

## CHAPTER 6

### DRIVE SYSTEM

#### CHAPTER OVERVIEW

| SECTION | TITLE                          |
|---------|--------------------------------|
| 6-1     | Equipment Description and Data |
| 6-2     | Troubleshooting                |
| 6-3     | On Aircraft Inspections        |
| 6-4     | Maintenance Procedures         |
| 6-5     | Maintenance Procedures (BENCH) |



## SECTION 6-1

### EQUIPMENT DESCRIPTION AND DATA

#### SECTION OVERVIEW

| PARAGRAPH | TITLE                                                                    |
|-----------|--------------------------------------------------------------------------|
| 6-1-1     | Transmission System Description and Data                                 |
| 6-1-2     | Main Transmission Lubrication System Description and Data                |
| 6-1-3     | Main Transmission Warning and Indicating System Description and Data     |
| 6-1-4     | Main Transmission and Gear Box Chip Detector System Description and Data |
| 6-1-5     | Rotor Brake System Description and Data                                  |



## 6-1-1 TRANSMISSION SYSTEM DESCRIPTION AND DATA.

### 6-1-1.1 Transmission System.

The Transmission System carries engine torque to the main rotor and tail rotor (Figure 6-1-1-1). The Transmission System consists of:

- a. Chip Detector System
- b. Drive Shafts
- c. Intermediate Gear Box
- d. Hot Oil Warning System
- e. Low Oil Pressure Warning System
- f. Main Transmission With Oil Cooler
- g. Oil Pressure Indicating System
- h. Oil Temperature Indicating System
- i. Tail Gear Box

The main transmission drives the main rotor, tail rotor, main transmission oil cooler fan, No. 1 and No. 2 hydraulic pump modules, and No. 1 and No. 2 generators.

### 6-1-1.2 Transmission System Operation.

The helicopter's power to the transmission begins at the front end of the engines. The engine input drive shaft, turning at 20,900 rpm (100% main rotor speed (Nr)) provides the power to the input module, which then drives the main module and accessory modules (Figure 6-1-1-2). The input module reduces engine input rpm to 5,750 rpm and also allows the drive angle to be changed from the engine to the main module. The main module then provides reduction for the main rotor head down to 258 rpm and a reduction for the tail drive and oil cooler to 4,116 rpm. The intermediate gear box receiving the tail drive shaft rpm then provides a reduction to 3,319 rpm plus changes the angle of drive about 58°. The tail gear box provides the remaining gear reduction for the tail rotor to 1,190 rpm and a 105° change in drive direction. During

operation of the main transmission, the hydraulic pump modules are driven at 7,188 rpm and the generators at 11,809 rpm.

### 6-1-1.3 Main Transmission.

The main transmission is mounted on the main fuselage with a built in 3° forward tilt. It consists of five modules: a main module, two input modules, and two accessory modules (Figure 6-1-1-3). It mounts and drives the main rotor head, changes the angle of drive from the engines, reduces engine rpm, and drives the tail rotor drive shaft along with the accessory modules. Both input modules and both accessory modules are interchangeable with one another and are replaceable without removing any other major components.

#### 6-1-1.3.1 Main Module.

The main module is mounted on top of the cabin fuselage. The main module supports and drives the main rotor head. It also drives the Tail Rotor System.

#### 6-1-1.3.2 Input Module.

The two input modules are mounted on the left and right front side of the main module (Figure 6-1-1-3). They connect the main module to the engines by shafting and gears. Each input module is identical and directly interchangeable. Over running clutches (free-wheeling units) disengage a non operating engine from the transmission, but not the accessory module.

#### 6-1-1.3.3 Accessory Module.

An accessory module is mounted on the front of each input module (Figure 6-1-1-3). The accessory module drives the electrical generators and hydraulic pump modules. A rotor speed sensor is mounted on the right accessory module and an oil pressure switch is mounted in the left accessory module. An additional rotor speed sensor is mounted on the left accessory module. Each basic module is identical and directly interchangeable.

#### 6-1-1.4 Intermediate Gear Box.

The intermediate gear box is mounted at the base of the pylon (Figure 6-1-1-4 and Figure 6-1-1-5). The intermediate gear box carries main transmission torque to the tail gear box and changes the angles of the drive about 58°. The intermediate gear box also reduces tail drive shaft input speed of 4,110 rpm to 3,319 rpm pylon shaft output speed. The gear box is divided into three sections, the input assembly and flange, the center housing, and the output assembly and flange.

#### 6-1-1.5 Tail Gear Box.

The tail gear box, mounted at the top of the pylon, holds the tail rotor head and changes the direction of drive 105° (Figure 6-1-1-6). The tail gear box is divided into three sections: input assembly, center housing assembly, and output assembly. The tail gear box includes an input linkage, bearing, and pitch control shaft, for tail rotor pitch controls. The tail rotor servo mounts on and in the tail gear box. The tail rotor blade deice slipring can be installed on the tail gear box.

#### 6-1-1.6 Gust Lock.

A manually-operated gust lock, at the tail takeoff flange, permits the main rotor and tail rotor to be locked when the helicopter is parked or stored. The gust lock is operated by pressing the handle release

button, in the cabin, and moving the handle in or out (Figure 6-1-1-7). The gust lock lever meshes with teeth on the tail takeoff flange. A switch in the rod assembly, between the handle and lock lever, turns on the GUST LOCK capsule on the caution/advisory panel.

#### 6-1-1.7 Tail Rotor Drive Shaft.

The tail rotor drive shaft runs from the tail takeoff flange on the rear of the main transmission, to the intermediate gear box. It then runs up from the intermediate gear box to the tail gear box (Figure 6-1-1-8). The drive shaft carries engine torque to the tail rotor and drives the oil cooler blower. The aluminum tail drive shaft is made up of four sections containing seven shafts. Each shaft is dynamically balanced. The sections are joined together by flexible couplings. The first three sections of shafting, from the rear of the oil cooler, are directly interchangeable. The rear end of these three shafts are held by special shock absorbing bearings. Alignment is not necessary after the replacement of any or all sections of the drive shaft.

Multiple disc flexible couplings are used to carry torque and allow for minor misalignment of tail drive shaft components. Along with the special shock absorbing bearings, the couplings also allow the drive shaft to remain in alignment as the airframe flexes in flight.

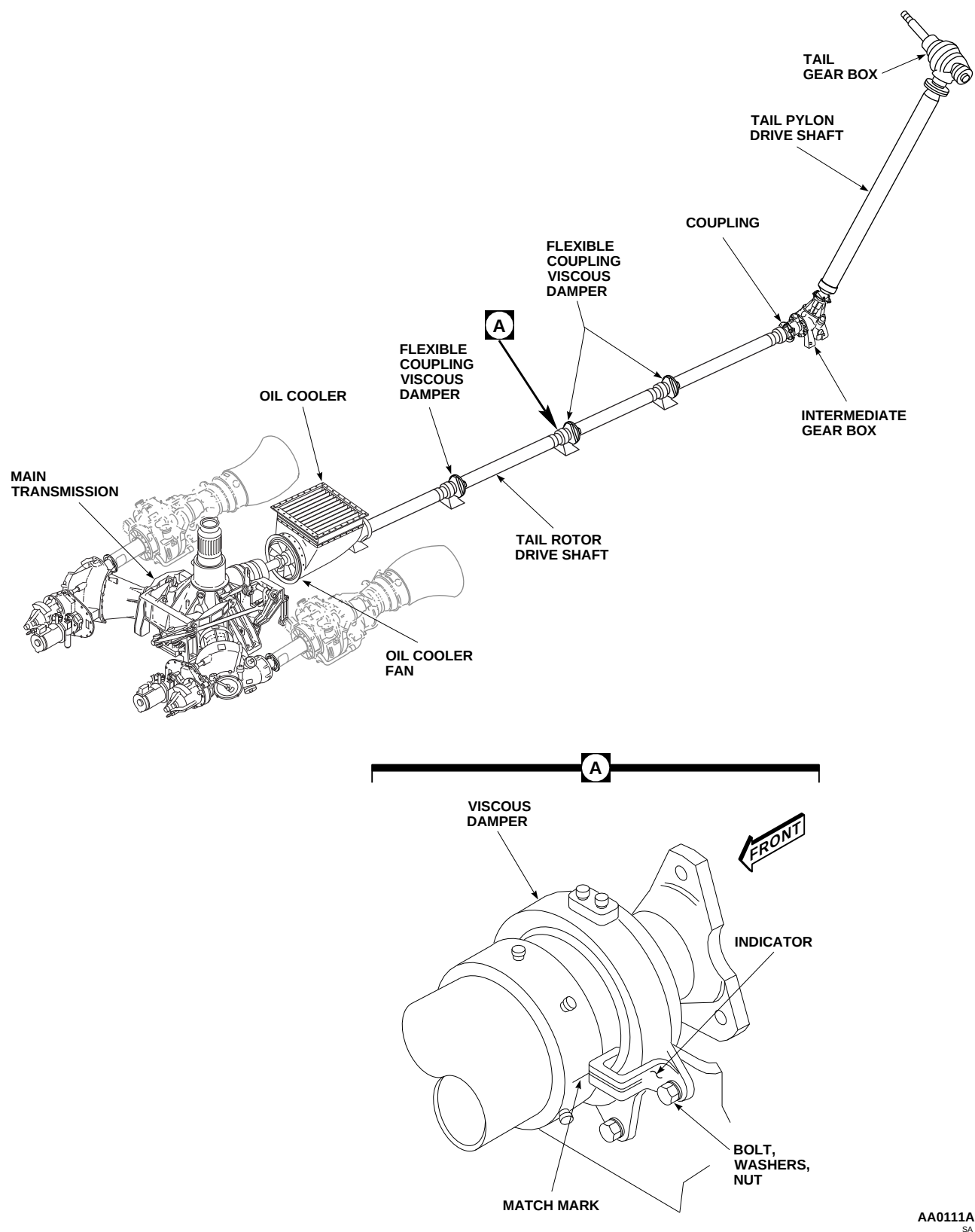
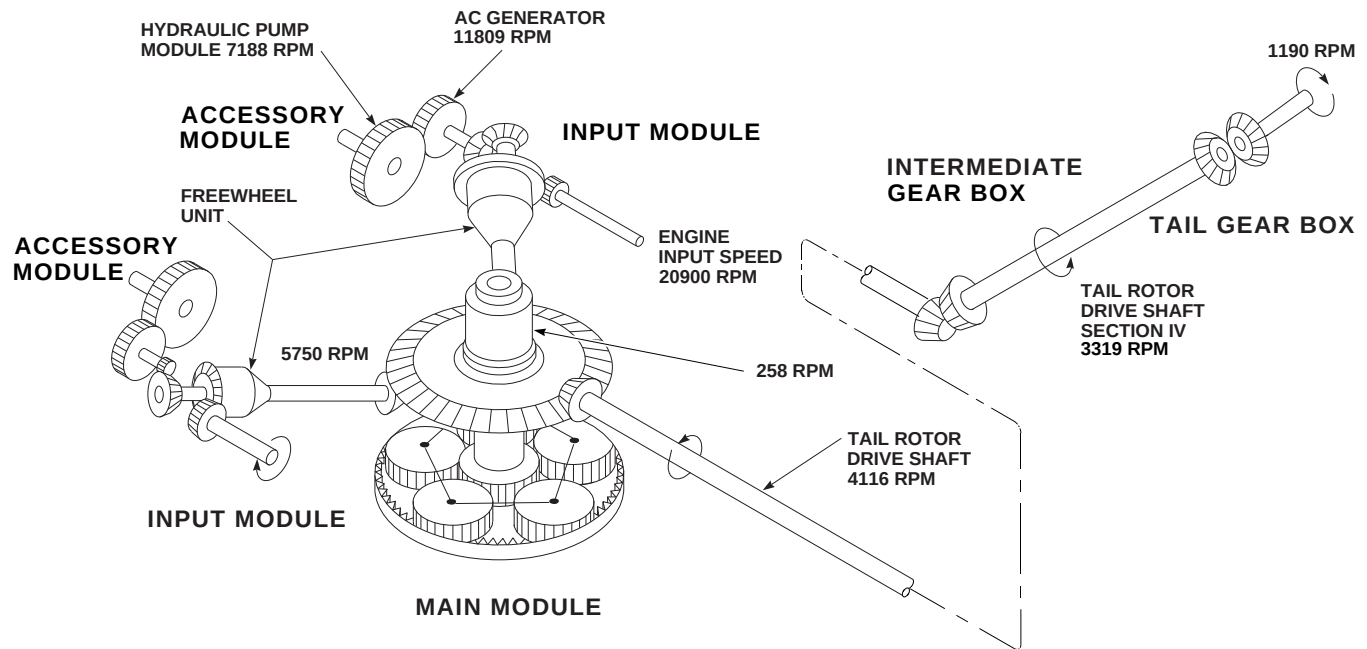


FIGURE 6-1-1-1. TRANSMISSION SYSTEM POWERTRAIN



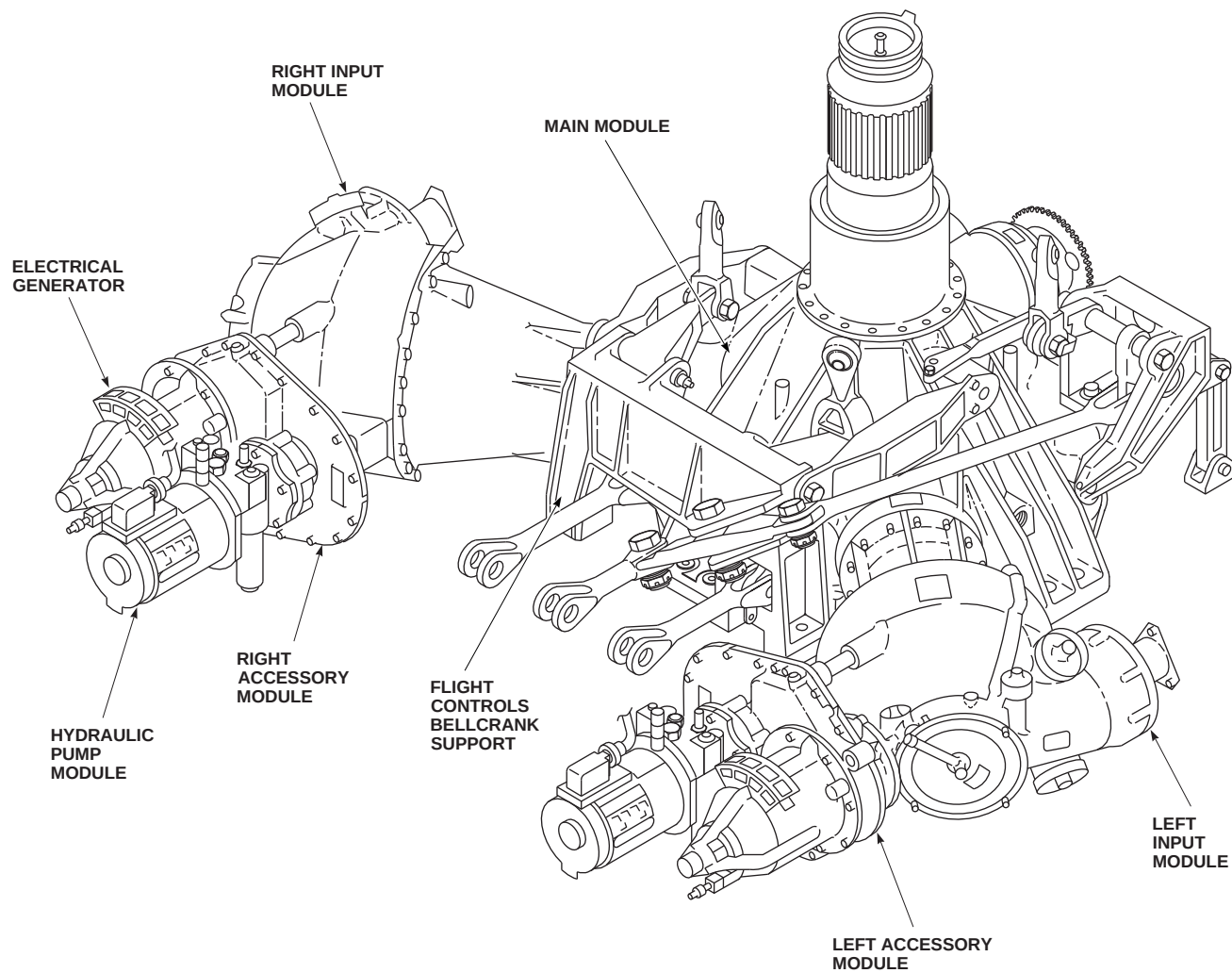
|                                | RPM'S |      | REDUCTION |
|--------------------------------|-------|------|-----------|
|                                | IN    | OUT  |           |
| MAIN TRANSMISSION INPUT MODULE | 20900 | 5750 | 3.63:1    |
| MAIN TRANSMISSION MAIN MODULE  | 5750  | 258  | 22.89:1   |
| INTERMEDIATE GEAR BOX          | 4116  | 3319 | 1.24      |
| TAIL GEAR BOX                  | 3319  | 1190 | 2.79      |

**NOTE**

ALL RPM CALLOUTS AT 100%.

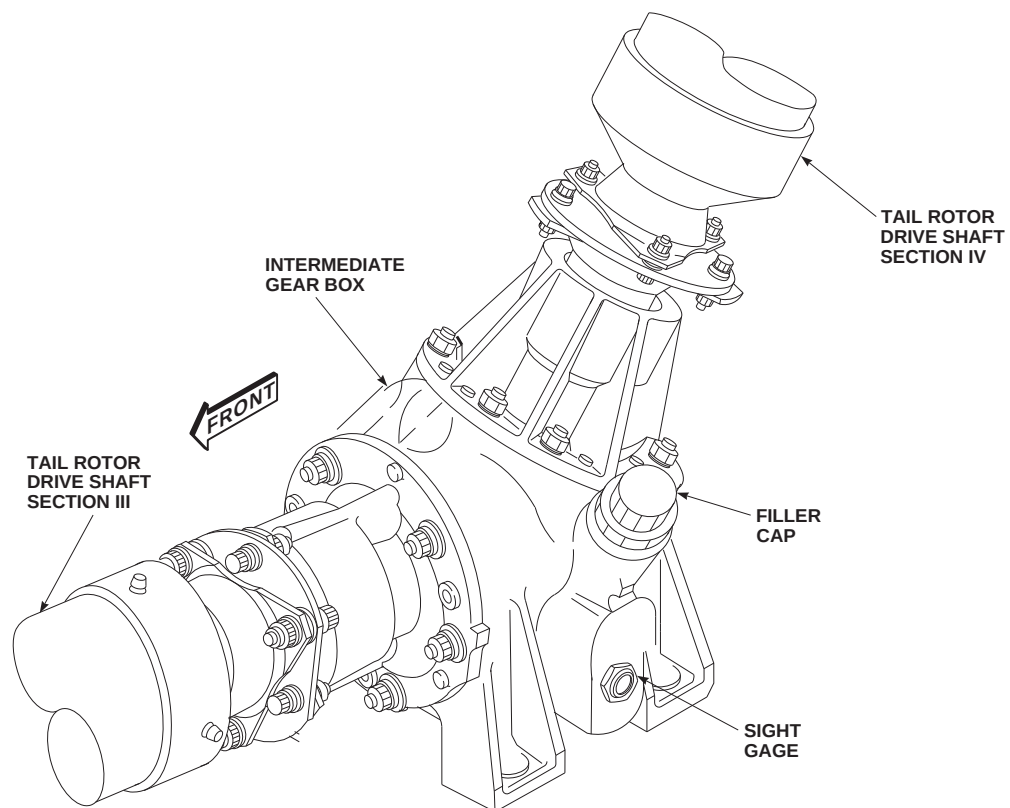
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SA**FIGURE 6-1-1-2. TRANSMISSION GEAR RATIOS**





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FIGURE 6-1-1-3. MAIN TRANSMISSION

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SA**FIGURE 6-1-1-4. INTERMEDIATE GEAR BOX INSTALLED**

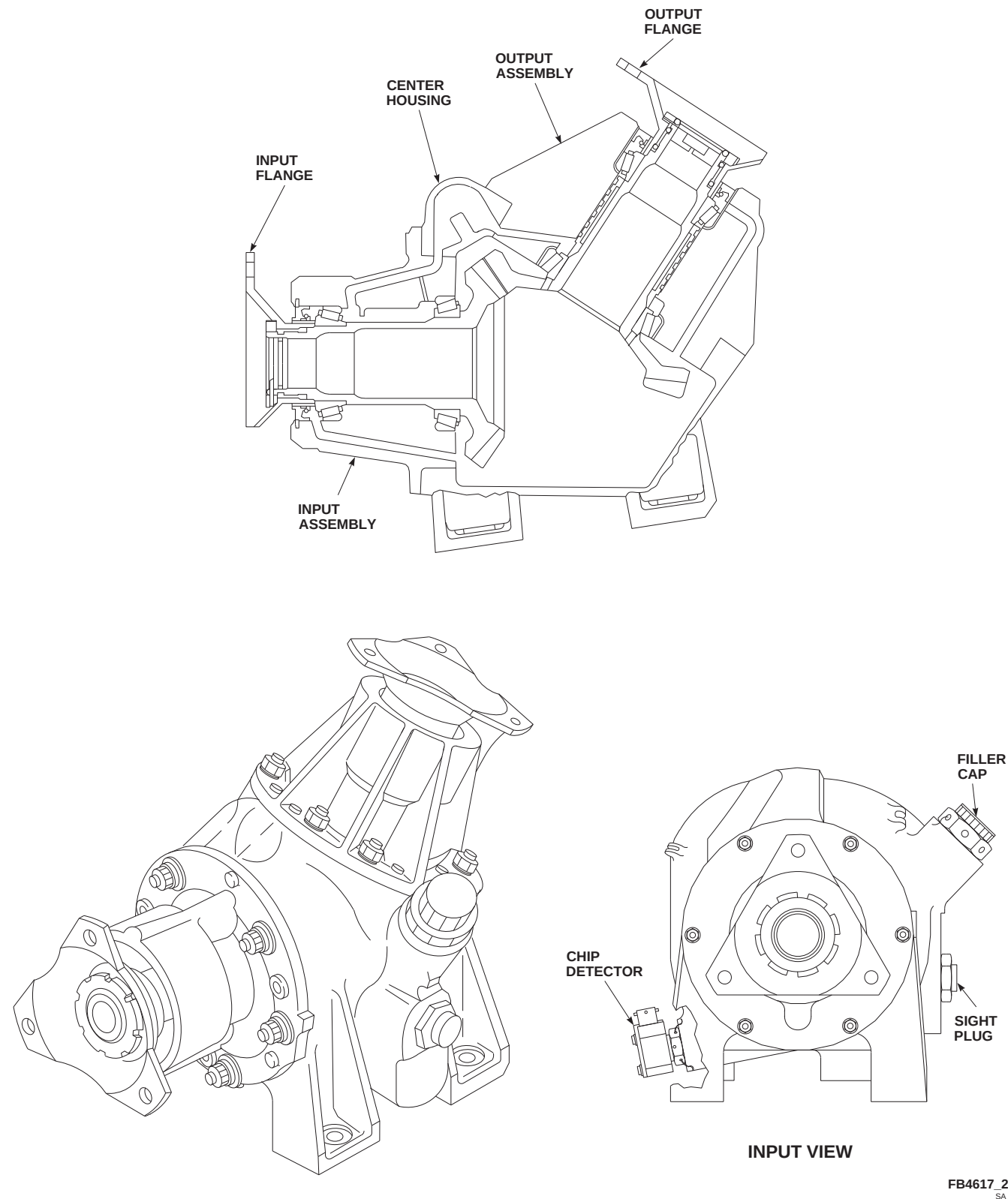
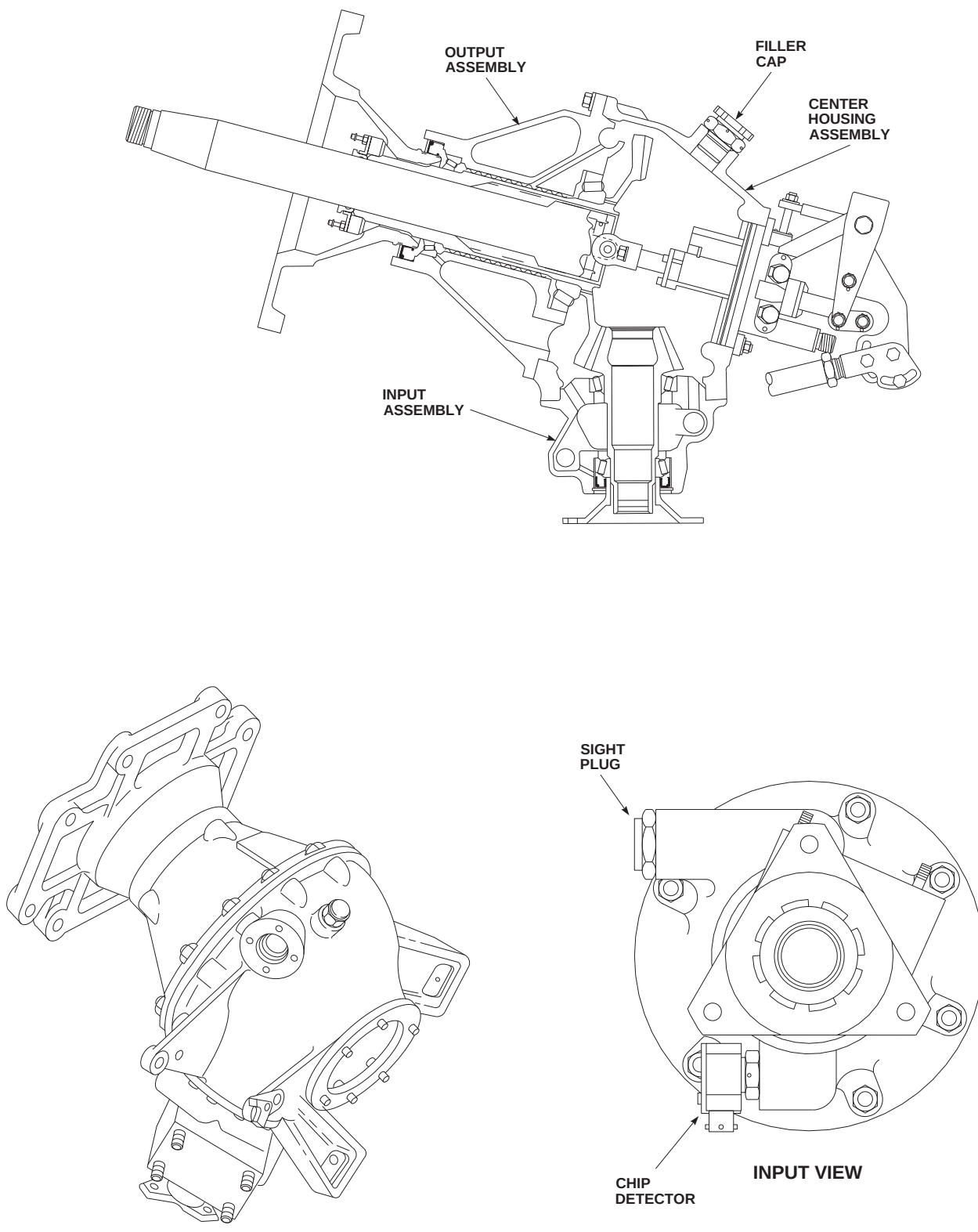
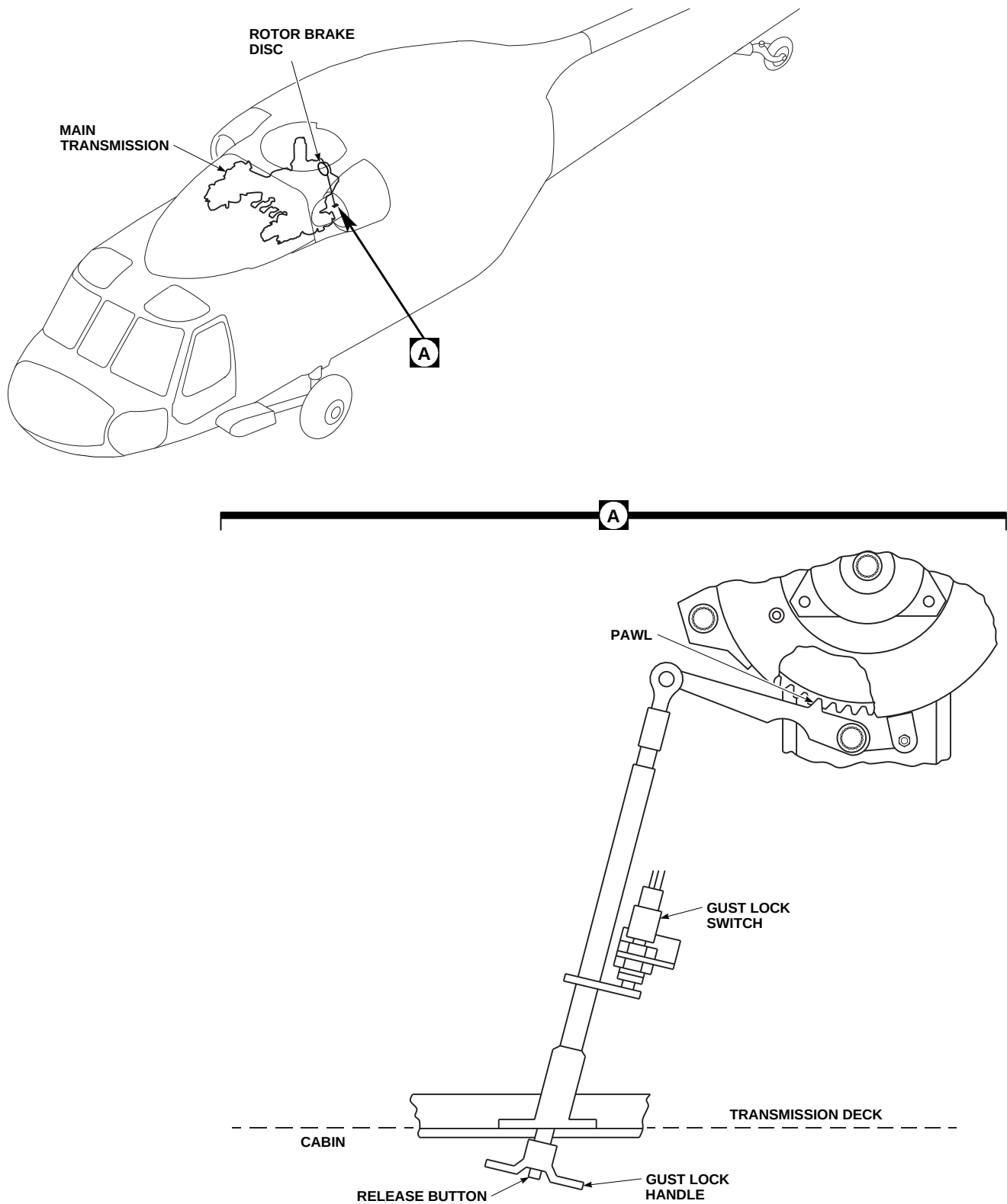


FIGURE 6-1-1-5. INTERMEDIATE GEAR BOX

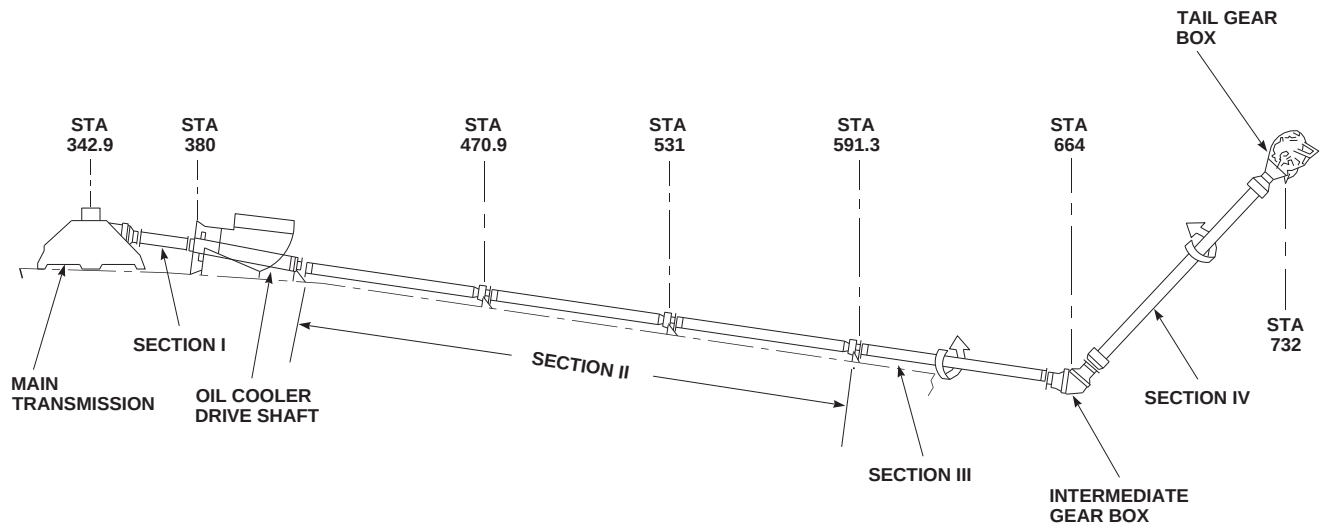
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SA**FIGURE 6-1-1-6. TAIL GEAR BOX**



GUST LOCK SYSTEM COMPONENTS

FIGURE 6-1-1-7. GUST LOCK

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ROTATION OF ALL SHAFTS

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FIGURE 6-1-1-8. TAIL ROTOR DRIVE SHAFT

## 6-1-2 MAIN TRANSMISSION LUBRICATION SYSTEM DESCRIPTION AND DATA.

### 6-1-2.1 Main Transmission Lubrication System.

The Main Transmission Lubrication System has a Wet Sump Lubrication System that cools and filters the oil to all the gears and bearings (Figure 6-1-2-1). The No. 1 and No. 2 generators also receive oil for cooling by way of internal lubrication lines. The oil is pumped through internally cored oil lines, except for the oil cooler inlet and outlet lines. The main transmission has an oil capacity of about seven gallons. A dipstick is used for checking oil quantity. When the oil level reaches the ADD mark, the Main Transmission Lubrication System is two quarts low.

The Main Transmission Lubrication System consists of:

- a. Two Pressure and Scavenge, Vane-Type Lubricating Pumps
- b. Pressure Regulating and Bypass Valves
- c. Two-Stage Oil Filter
- d. Oil Cooler and Blower
- e. Warning System
- f. Indicating System

### 6-1-2.2 Lubrication Pumps.

The Main Transmission Lubrication System pumps are combination pressure and scavenge vane-type operating in parallel. The pressure side of the pumps supplies oil at 15 gpm at a pressure of  $52.5 \pm 2.55$  pounds psi. The scavenge side returns oil at a rate of 7 gpm at a pressure of  $52.5 \pm 2.55$  psi to the sump. The pressure side is regulated by an adjustable pressure regulating valve. As pressure exceeds 55 psi, the bypass valve starts to open and extra oil is bypassed back to the inlet side of the pump.

### 6-1-2.3 Two-Stage Oil Filter.

The two-stage oil filter, at the right rear section of the sump, protects the Main Transmission Lubrica-

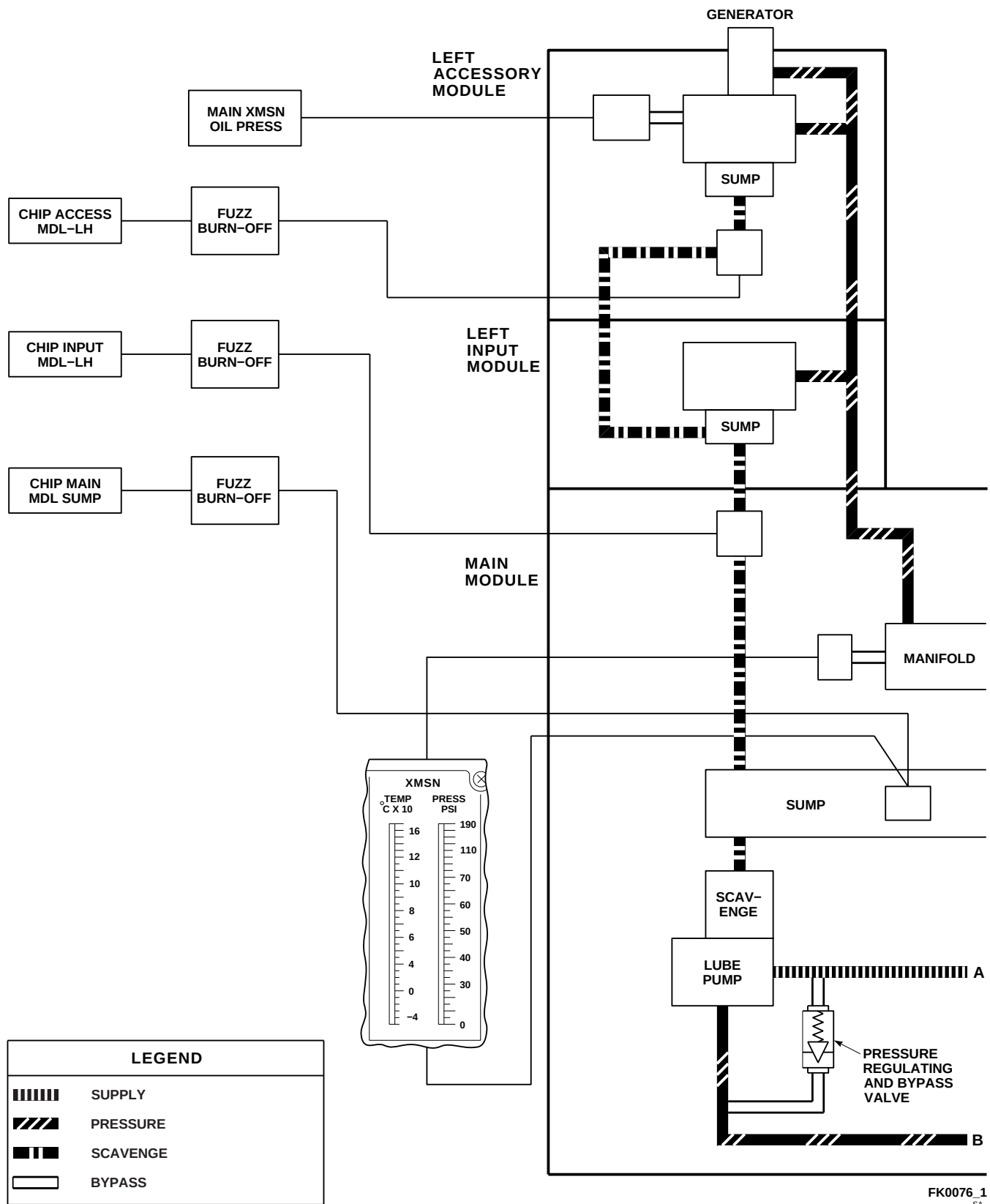
tion System by removing lubricant contaminants. The filter has two separate elements, a 3 micrometer (micron) first stage filter and a 75 micron second stage filter. With filter, 70351-08143-101, the filter has two separate elements, a 46 micron first stage filter and a 75 micron second stage filter. Both filters are the paper throwaway type. When the first stage filter starts to clog and pressure drops  $12 \pm 3$  psi, a red button extends  $\frac{3}{16}$ -inch from the bottom of the filter bowl. It cannot be reset until the filter is replaced. A thermal lockout prevents the indicator from extending when the temperature goes below 60.8°F (16°C) and the oil is cold and thick. The first stage filter will bypass at a pressure drop of about  $20 \pm 4$  psi. The second stage filter will protect the System to a drop of  $35 \pm 5$  psi, and then the second filter goes into bypass.

### 6-1-2.4 Oil Cooler and Fan.

The oil cooler and fan, in the rear of the main rotor pylon, consists of a radiator, duct, fan, and shafting. The fan, driven by the tail rotor drive shaft, forces air through the radiator. Hot oil from the main module sump is pumped into the radiator. A thermostatic control valve within the radiator allows cold oil (less than 158°F (70°C)) to bypass the radiator. Also, if the radiator becomes clogged, the oil will bypass. Oil is routed from the oil cooler to the main module manifold to be divided between the lubrication jets in the main module and the oil passages to the input modules, accessory modules, and generators.

### 6-1-2.5 Spline Wear Indicators.

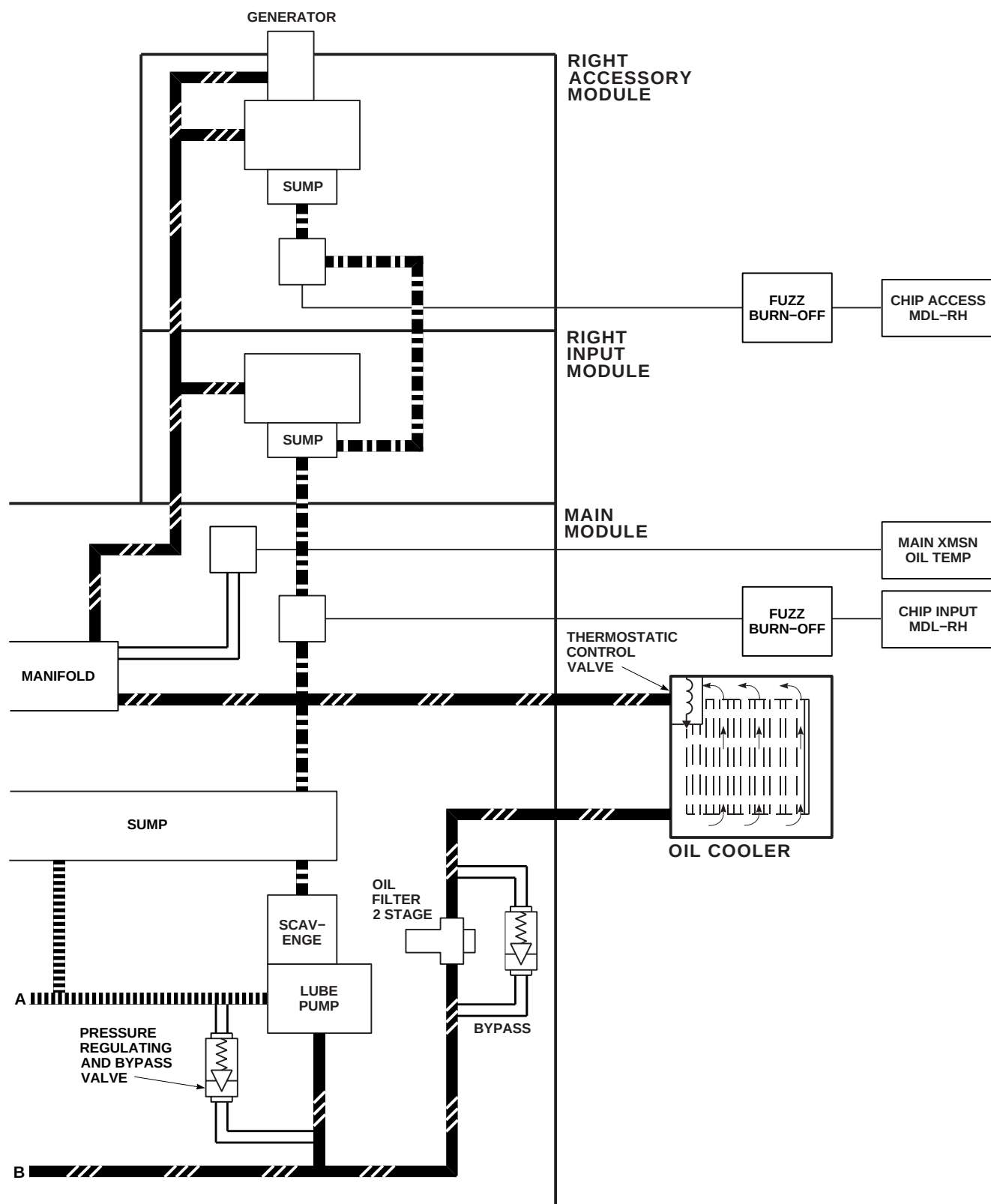
Spline wear indicators are installed on the viscous damper at station 531. The viscous damper has a scribed line on the tail rotor drive shaft which is lined up with an indicator mounted on the viscous damper. On helicopters with oil cooler fan, 70361-03005-103 through 70361-03005-106, installed, the oil cooler fan has a scribed mark on the fan blade which is compared to a wear indicator on fan housing. On helicopters with oil cooler fan, 70361-03005-107, installed, spline wear inspection is not required.



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**FIGURE 6-1-2-1. MAIN TRANSMISSION LUBRICATION SYSTEM BLOCK DIAGRAM (SHEET 1 OF 2)**





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FIGURE 6-1-2-1. MAIN TRANSMISSION LUBRICATION SYSTEM BLOCK DIAGRAM  
(SHEET 2 OF 2)



6-1-3 MAIN TRANSMISSION WARNING AND INDICATING SYSTEM  
 DESCRIPTION AND DATA.

6-1-3.1 Main Transmission Warning and Indicating System.

The Transmission Warning and Indicating Systems indicate possible troubles in the Transmission System. The Warning and Indicating Systems consists of:

- a. Oil Pressure Indication
- b. Oil Temperature Indication
- c. Chip Detectors throughout the Main Transmission

6-1-3.2 Main Transmission Oil Temperature and Pressure Warning Indicating System.

When the temperature of the oil entering the manifold is over 241.7°F ± 40.1°F (116.5°C ± 4.5°C), the sensor lights the MAIN XMSN OIL TEMP light in the caution/advisory panel (Figure 6-1-3-1). The Oil Pressure Warning System has an oil pressure switch in the left accessory module connected to the MAIN XMSN OIL PRESS light in the caution/advisory panel. When the oil pressure in the left accessory module falls below 14 pounds psi, the switch turns on the MAIN XMSN OIL PRESS capsule.

6-1-3.3 Main Transmission Oil Temperature and Pressure Indicating Systems.

The Oil Temperature Indicating System has an oil temperature sensor and MAIN XMSN OIL TEMP indicator in the Central Display Unit (CDU) (Figure 6-1-3-1). The MAIN XMSN OIL TEMP

indicator gives a main transmission temperature. Indicator ranges are shown in TABLE NO. 6-1-3-1.

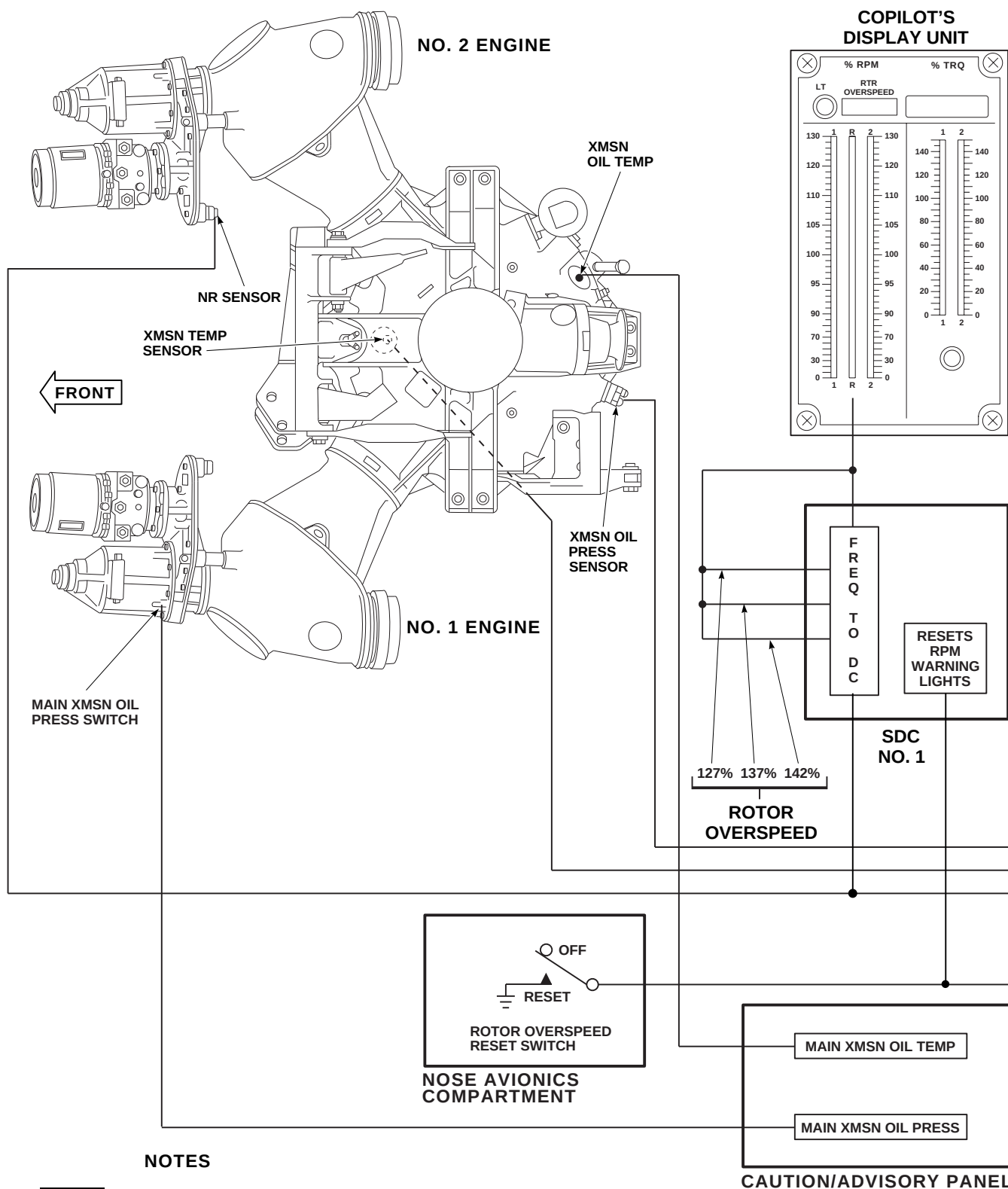
TABLE NO. 6-1-3-1. XMSN TEMP  
 Ranges

|       |                           |
|-------|---------------------------|
| GREEN | 122 - 248°F (50 - 120°C)  |
| AMBER | 248 - 284°F (120 - 140°C) |
| RED   | 284 - 338°F (140 - 170°C) |

The Oil Pressure Indicating System has an oil pressure sensor and MAIN XMSN OIL PRESS indicator in the central display unit. The MAIN XMSN OIL PRESS indicator gives main transmission oil pressure in psi. Indicator ranges are shown in TABLE NO. 6-1-3-2.

TABLE NO. 6-1-3-2. XMSN PRESS  
 Ranges

|       |               |
|-------|---------------|
| GREEN | 30 - 65 psi   |
| AMBER | 20 - 30 psi   |
| AMBER | 65 - 130 psi  |
| RED   | 130 - 190 psi |

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**FIGURE 6-1-3-1. MAIN TRANSMISSION OIL TEMPERATURE, PRESSURE WARNING AND INDICATING SYSTEMS BLOCK DIAGRAM (SHEET 1 OF 2)**

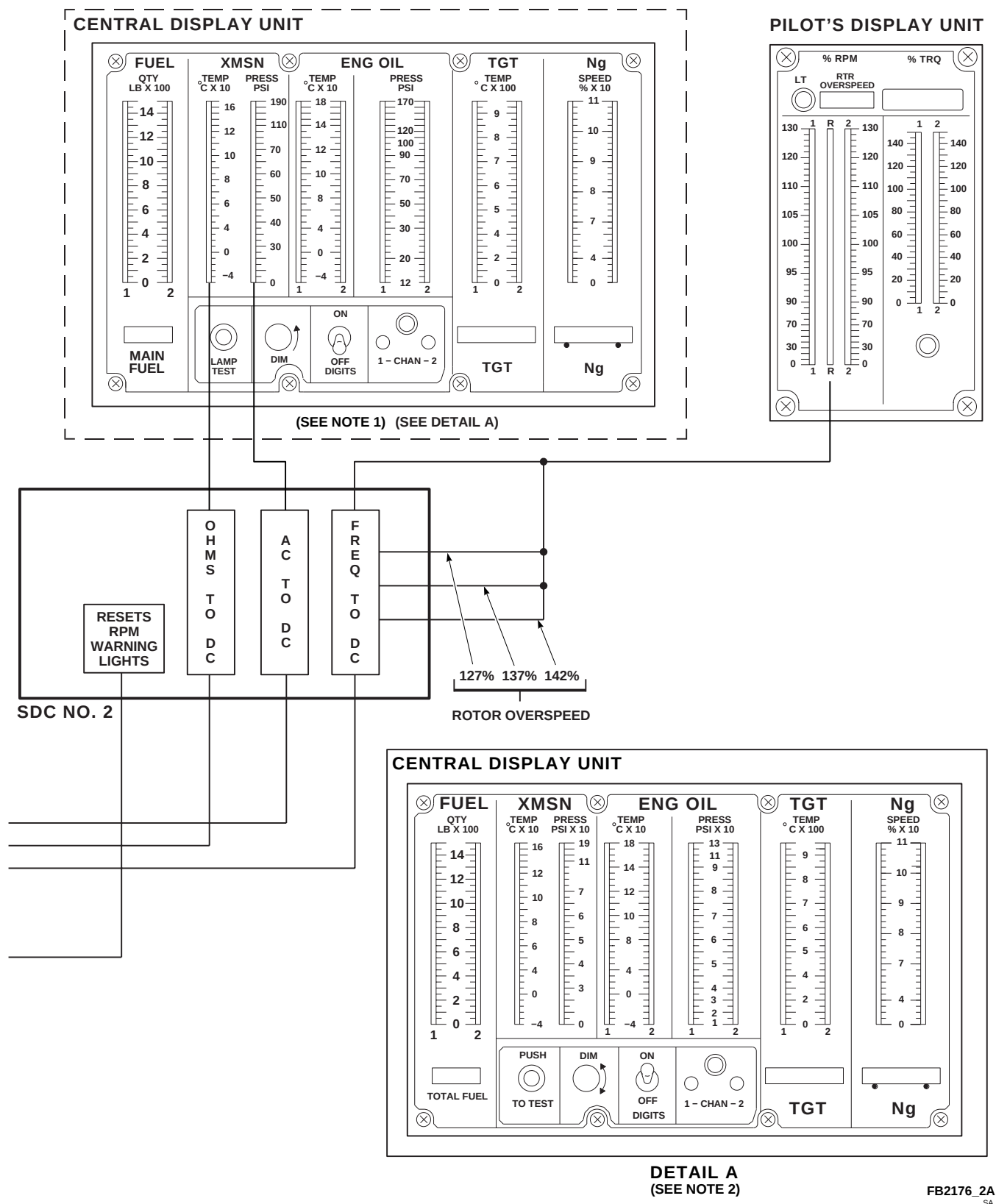


FIGURE 6-1-3-1. MAIN TRANSMISSION OIL TEMPERATURE, PRESSURE WARNING AND INDICATING SYSTEMS BLOCK DIAGRAM (SHEET 2 OF 2)



## 6-1-4 MAIN TRANSMISSION AND GEAR BOX CHIP DETECTOR SYSTEM DESCRIPTION AND DATA.

### 6-1-4.1 Main Module Sump Chip Detector.

One chip detector, mounted on the sump assembly, constantly monitors lubricating oil for possible metal contamination (Figure 6-1-4-1). Any metal chips that accumulate within the chip detector gaps close an electrical circuit that lights the CHIP MAIN MDL SUMP capsule on the caution/advisory panel. The chip detector has a fuzz-suppressing feature that prevents minute metal particles (fuzz) from activating the chip capsule by burning them off. Refer to PARA 8-1-7 for information on latching and time-delay circuits.

### 6-1-4.2 Accessory Modules Chip Detectors.

Two chip detectors, one mounted on each accessory module, constantly monitor lubricating oil for possible metal contamination (Figure 6-1-4-1). Any metal chips that accumulate within chip detector gaps close an electrical circuit that lights either CHIP ACCESS MDL - LH or CHIP ACCESS MDL - RH capsule on the caution/advisory panel (Figure 6-1-4-1 and Figure 6-1-4-2). Each chip detector has a fuzz-suppressing feature that prevents minute metal particles (fuzz) from activating the chip capsule by burning them off. For information on latching circuits, refer to PARA 8-1-7.

### 6-1-4.3 Input Modules Chip Detectors.

Two chip detectors, on the sump assembly, constantly monitor lubricating oil for possible metal contamination (Figure 6-1-4-1). Any metal chips that accumulate within chip detector gaps close an electrical circuit that lights either CHIP INPUT MDL - LH or CHIP INPUT MDL - RH capsule on the caution/advisory panel (Figure 6-1-4-1 and Figure 6-1-4-2). Each chip detector has a fuzz-suppressing feature that prevents minute metal particles (fuzz) from activating the chip capsule by burning them off. For information on latching circuits, refer to PARA 8-1-7.

### 6-1-4.4 Intermediate Gear Box Chip Detector/Temperature Sensor.

One chip detector in the gear box monitors possible metal contamination. Any metal chips that accumulate on the chip detector plug close an electrical circuit and light the CHIP INT XMSN light. The chip detector has a burn-off circuit that burns off minute metal particles (fuzz) to prevent unnecessary lighting of the caution light. The chip detector also contains a normally open bimetal temperature switch. When gear box oil temperature reaches 284°F (140°C), the switch closes and causes the fuzz burn capacitor to discharge. This prevents arcing within the gear box when gear box oil is hot. When the temperature switch closes, the INT XMSN OIL TEMP light on the caution/advisory panel goes on. The magnetic capability of the chip detector is retained. The chip detector is self-sealing to permit removal for inspection without loss of oil.

### 6-1-4.5 Tail Gear Box Chip Detector/Temperature Sensor.

One chip detector in the gear box monitors possible metal contamination. Any metal chips that accumulate on the chip detector plug close an electrical circuit and light the CHIP TAIL XMSN light. The chip detector has a burn-off circuit that burns off minute metal particles (fuzz) to prevent unnecessary lighting of the caution light. The chip detector also contains a normally open bimetal temperature switch. When gear box oil temperature reaches 284°F (140°C), the switch closes and causes the fuzz burn capacitor to discharge. This prevents arcing within the gear box when gear box oil is hot. When the temperature switch closes, the TAIL XMSN OIL TEMP light on the caution/advisory panel goes on. The magnetic capability of

the chip detector is retained. The chip detector is self-sealing to permit removal for inspection without loss of oil.

#### **6-1-4.6 Chip Detector With Built-In-Test (BIT) Circuitry.**

The chip detector with Built-In-Test (BIT) circuitry is a self-test used to check each chip detector and its wiring. The BIT feature operates upon the power up of helicopter. The BIT circuitry, inside

the pod of each chip detector, simulates fine metal particles, chips, and fuzz, which turns on the appropriate warning capsule on the caution/advisory panel. The appropriate warning capsule on the caution/advisory panel remains illuminated for approximately 45 to 75 seconds except for the main module sump warning capsule. The CHIP MAIN MDL SUMP capsule shall go on after a 30 second time delay and remain on for an additional 30 seconds.



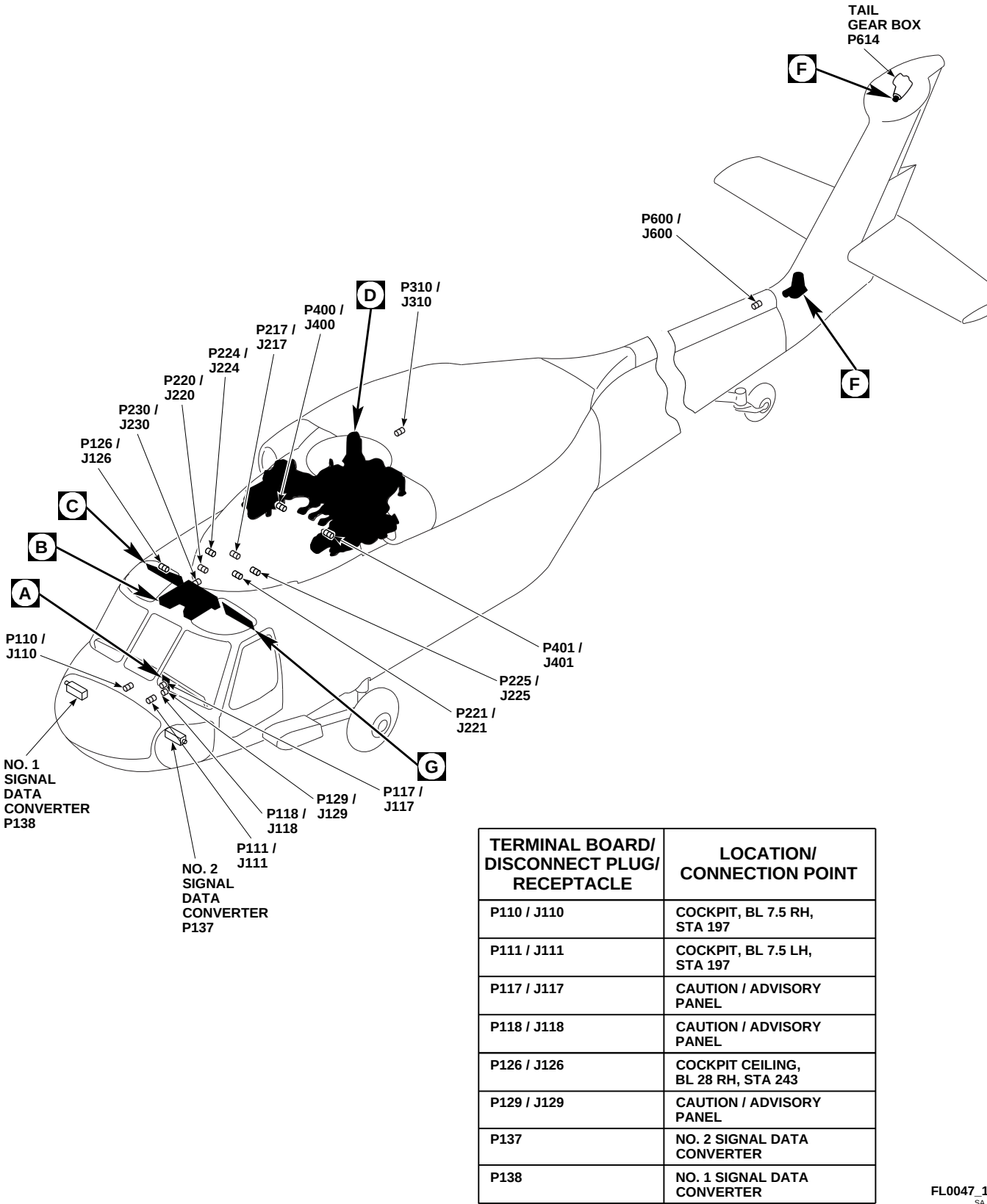
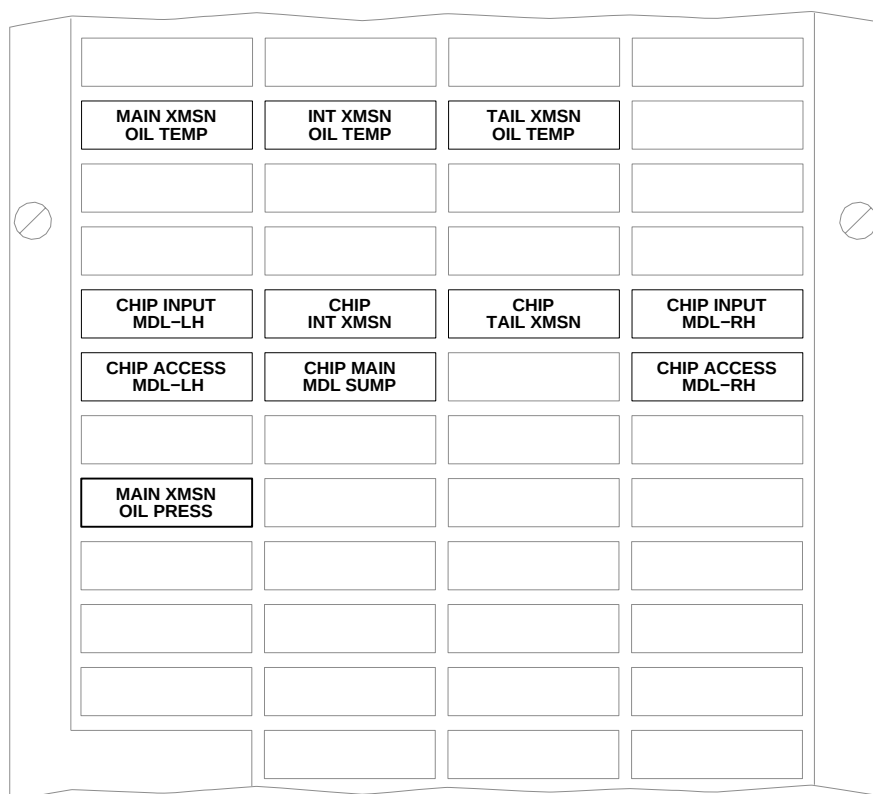


FIGURE 6-1-4-1. TRANSMISSION CHIP DETECTOR, INSTRUMENTS, AND OIL WARNING SYSTEMS LOCATION DIAGRAM (SHEET 1 OF 5)



CAUTION/ADVISORY PANEL

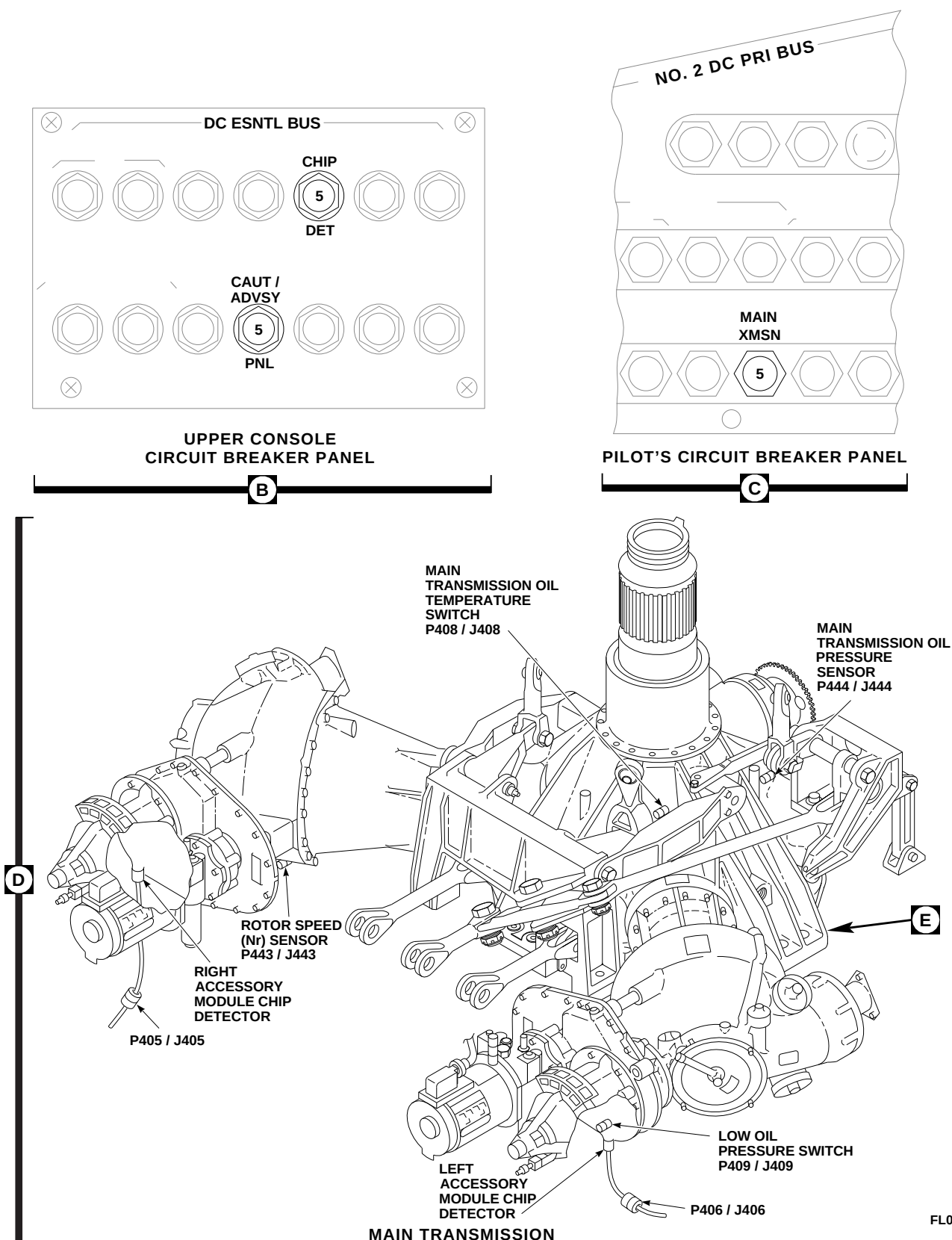
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| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT                               |
|---------------------------------------------------|-------------------------------------------------------------|
| P217 / J217                                       | CABIN CEILING, BL 0,<br>STA 284                             |
| P220 / J220                                       | MAIN ROTOR PYLON DECK<br>BL 3 RH, STA 243                   |
| P221 / J221                                       | JUNCTION BOX MAIN<br>ROTOR PYLON DECK<br>BL 3 LH, STA 245.5 |
| P224 / J224                                       | JUNCTION BOX MAIN<br>ROTOR PYLON DECK                       |
| P225 / J225                                       | JUNCTION BOX MAIN<br>ROTOR PYLON DECK<br>BL 3 LH, STA 244   |
| P230 / J230                                       | CABIN CEILING, BL 8 RH,<br>STA 247                          |
| P310 / J310                                       | CABIN CEILING, BL 17 RH,<br>STA 387                         |
| P400 / J400                                       | MAIN ROTOR PYLON DECK<br>BL 24 RH, STA 318                  |
| P401 / J401                                       | MAIN ROTOR PYLON DECK<br>BL 24 LH, STA 318                  |
| P403 / J403                                       | CHIP DETECTOR LEFT<br>INPUT MODULE                          |

| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT                |
|---------------------------------------------------|----------------------------------------------|
| P404 / J404                                       | CHIP DETECTOR RIGHT<br>INPUT MODULE          |
| P405 / J405                                       | CHIP DETECTOR RIGHT<br>ACCESSORY MODULE      |
| P406 / J406                                       | CHIP DETECTOR LEFT<br>ACCESSORY MODULE       |
| P408 / J408                                       | MAIN TRANSMISSION OIL<br>TEMPERATURE SWITCH  |
| P409 / J409                                       | MAIN TRANSMISSION LOW<br>OIL PRESSURE SWITCH |
| P412 / J412                                       | MAIN TRANSMISSION SUMP<br>TEMPERATURE SENSOR |
| P443 / J443                                       | Nr SPEED SENSOR                              |
| P444 / J444                                       | MAIN TRANSMISSION OIL<br>PRESSURE SENSOR     |
| P600 / J600                                       | TAIL CONE, BL 0, STA 600                     |
| P613                                              | CHIP DETECTOR<br>INTERMEDIATE GEAR BOX       |
| P614                                              | CHIP DETECTOR TAIL<br>GEAR BOX               |

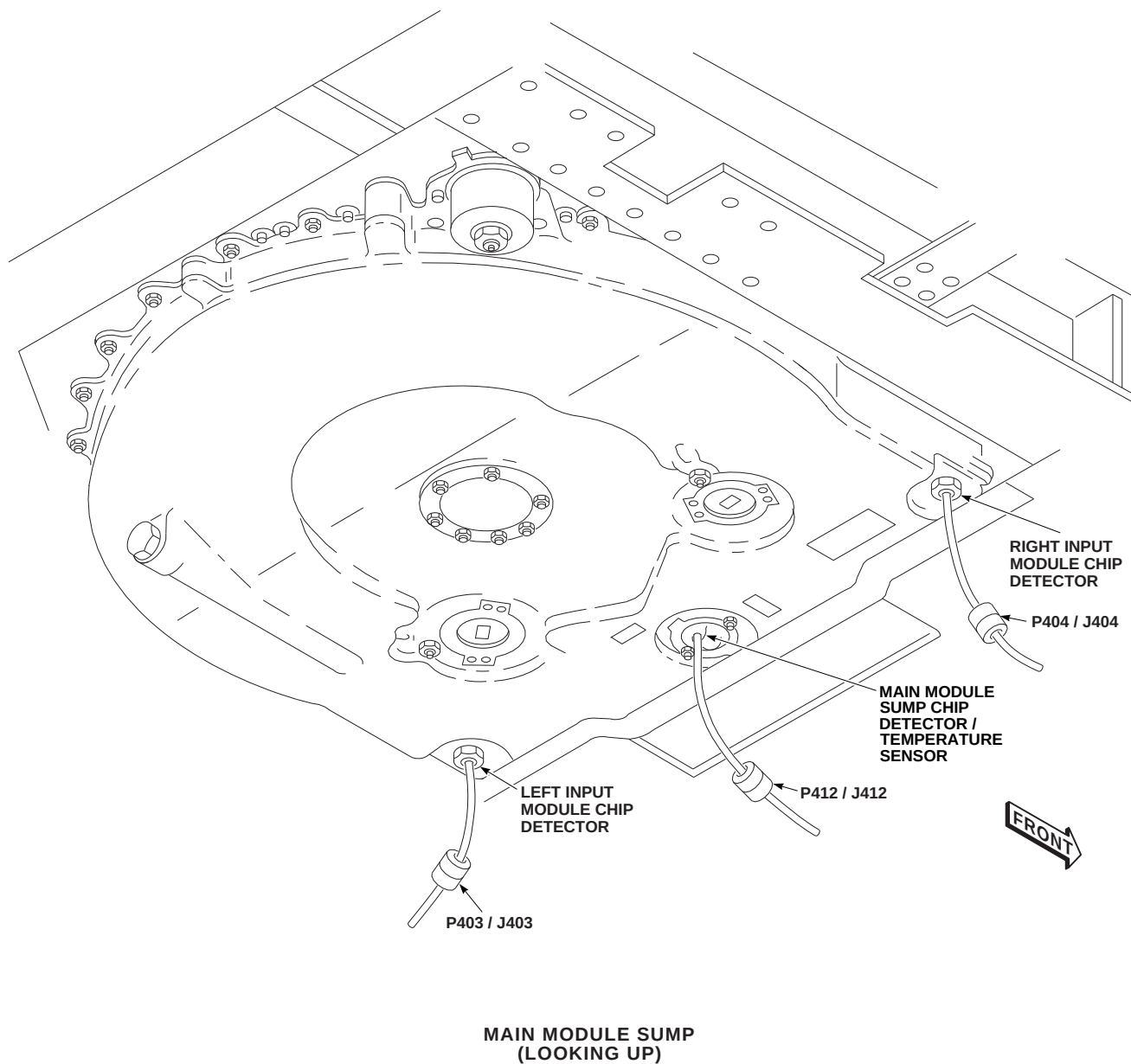
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FIGURE 6-1-4-1. TRANSMISSION CHIP DETECTOR, INSTRUMENTS, AND OIL WARNING SYSTEMS LOCATION DIAGRAM (SHEET 2 OF 5)

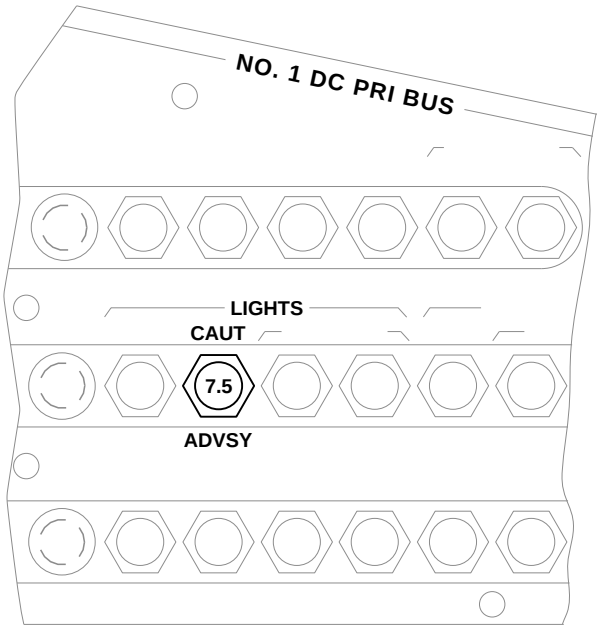
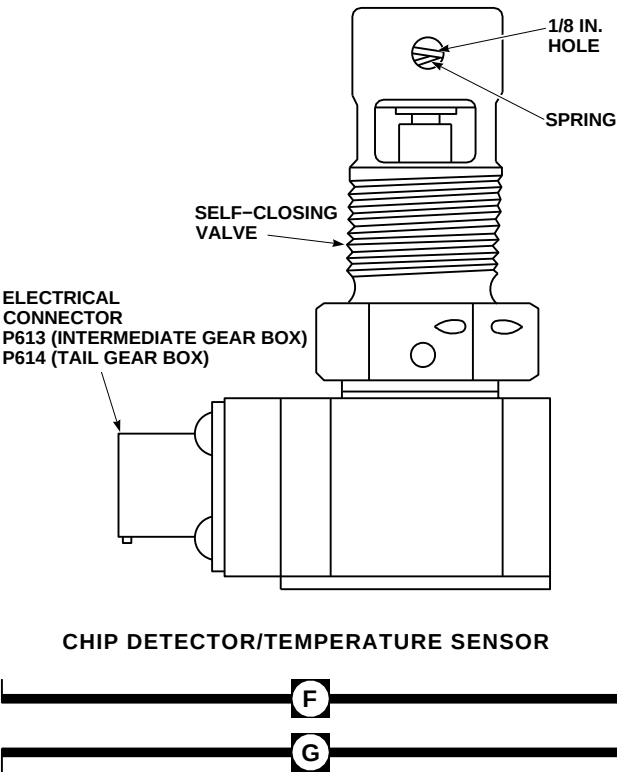


**FIGURE 6-1-4.1. TRANSMISSION CHIP DETECTOR, INSTRUMENTS, AND OIL WARNING SYSTEMS LOCATION DIAGRAM (SHEET 3 OF 5)**

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**FIGURE 6-1-4-1. TRANSMISSION CHIP DETECTOR, INSTRUMENTS, AND OIL WARNING SYSTEMS LOCATION DIAGRAM (SHEET 4 OF 5)**



COPLOT'S CIRCUIT BREAKER PANEL

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FIGURE 6-1-4-1. TRANSMISSION CHIP DETECTOR, INSTRUMENTS, AND OIL WARNING SYSTEMS LOCATION DIAGRAM (SHEET 5 OF 5)

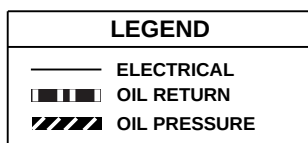
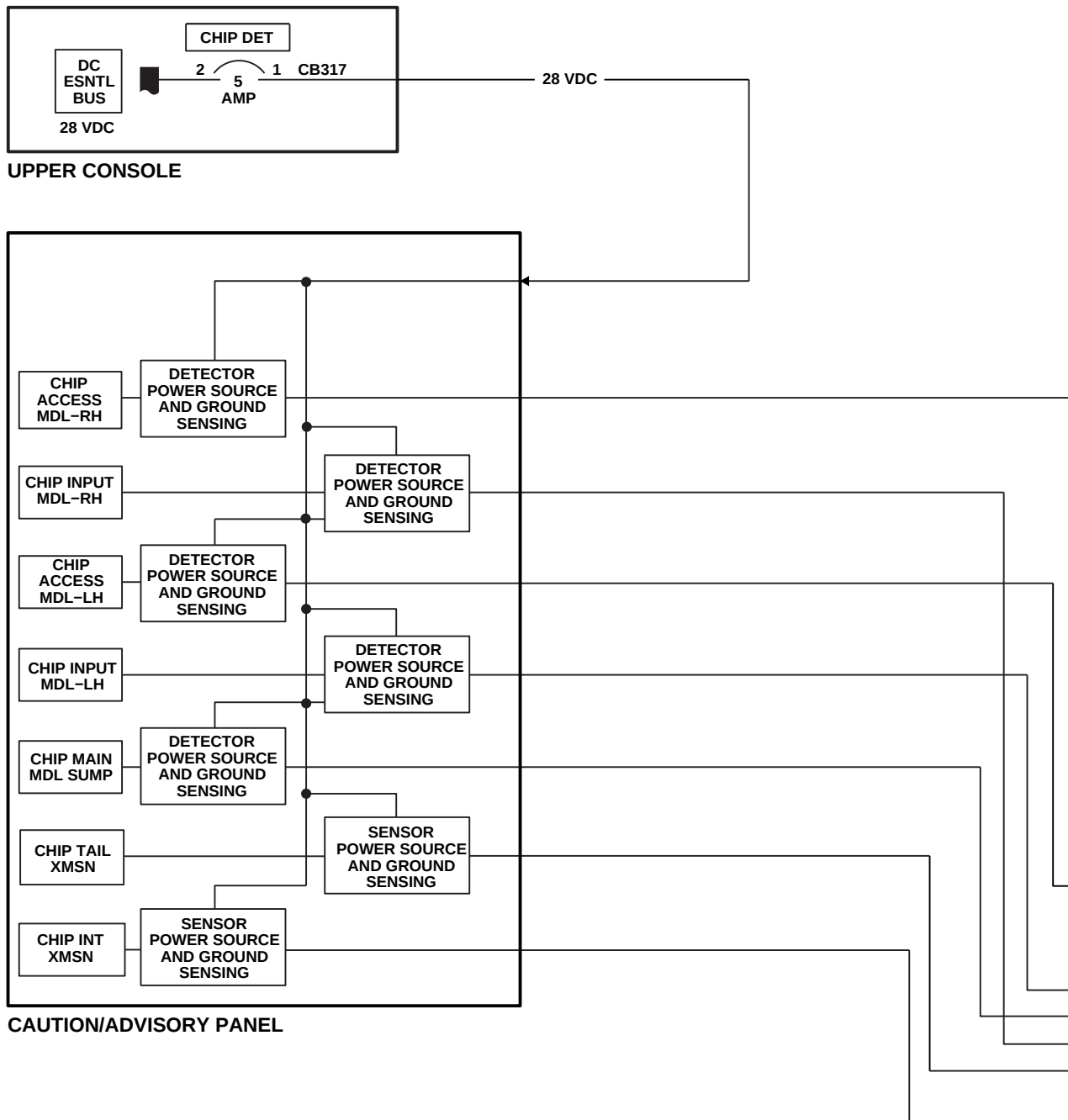
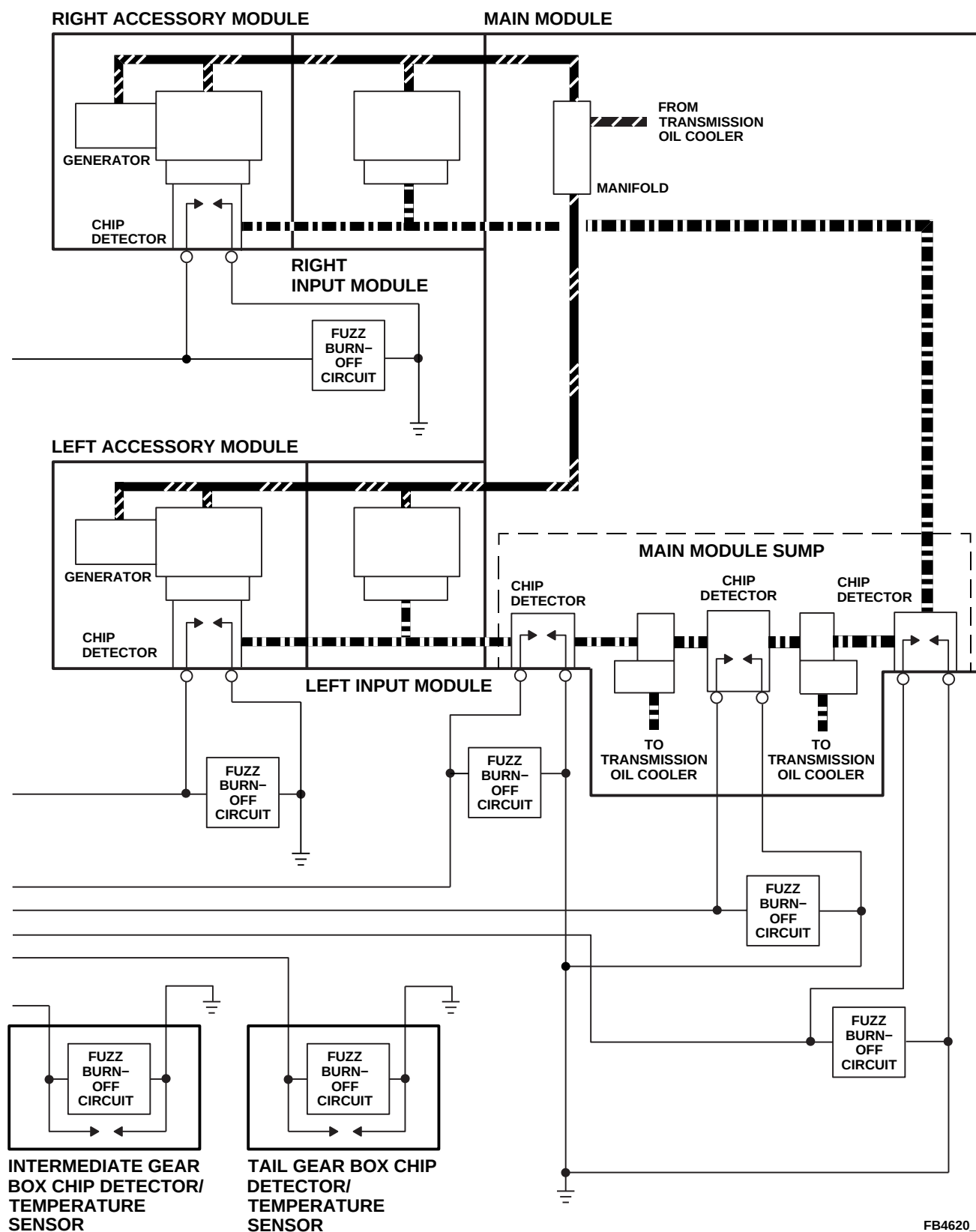
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FIGURE 6-1-4-2. MAIN TRANSMISSION CHIP DETECTOR SYSTEM BLOCK DIAGRAM  
(SHEET 1 OF 2)



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FIGURE 6-1-4-2. MAIN TRANSMISSION CHIP DETECTOR SYSTEM BLOCK DIAGRAM  
(SHEET 2 OF 2)





## 6-1-5 ROTOR BRAKE SYSTEM DESCRIPTION AND DATA.

### 6-1-5.1 System Description.

The Rotor Brake System uses hydraulic fluid pressure to stop and hold the rotor blades for extended periods. The Rotor Brake System is also used during manual blade fold operations.

### 6-1-5.2 Principles of Operation.

Pulling the master cylinder pump handle down and forward provides hydraulic fluid pressure through tubing to the rotor brake assembly. The rotor brake pressure gauge on the master cylinder visually indicates hydraulic fluid pressure. Normal Rotor Brake System operating pressure is 450 to 690 psi. Pressure-activated brake linings grip both sides of the rotor brake disc on the tail takeoff shaft, preventing the blades from turning. An accumulator mounted to the rear of the main gear box maintains constant hydraulic fluid pressure to the brake linings.

With  $7 \pm 8$  psi applied to the rotor brake pressure switch, the switch contacts close, turning on the ROTOR BRAKE ON advisory legends on both inboard MFDs and de-energizes a ground idle rotor brake solenoid in the engine control quadrant. The rotor brake solenoid locks the engine power levers in the ground idle position through a locking arm linked to the solenoid. The manifold relief valve allows excess hydraulic fluid pressure to return to the reservoir. The Rotor Brake System servicing selector valve provides hydraulic fluid to the master cylinder. The manifold check valve (shuttle valve) allows hydraulic fluid flow in one direction only, from the servicing selector valve to the master cylinder.

Electrical power of 28 VDC is supplied by DC ESNTL BUS through the ROTOR BRAKE circuit breaker, on the copilot's auxiliary circuit breaker panel, to the rotor brake pressure switch.

Hydraulic fluid pressure closes the rotor brake pressure switch, providing electrical power to the No. 1 and No. 2 Data Concentrator Units (DCUs) to activate ROTOR BRAKE ON legend on both inboard MFDs.

When the Rotor Brake System is off, the gust lock switch can operate an actuator to mechanically prevent the rotor head from turning. The rotor brake and gust lock cannot be engaged at the same time.

### 6-1-5.3 Component Description.

#### 6-1-5.3.1 Master Cylinder.

The master cylinder has an external lock on the cylinder body and an internal valve and piston. Pushing the pump handle down and forward provides hydraulic fluid pressure. The bleeder valve can be used to purge air from the master cylinder. The rotor brake pressure gauge visually indicates hydraulic fluid pressure.

#### 6-1-5.3.2 Brake Assembly.

The brake assembly consists of a spark shield, six wear indicator pins, six pistons and brake linings, and a bleeder valve.

#### 6-1-5.3.3 Accumulator.

The accumulator, by absorbing excessive pressure surges, maintains constant hydraulic fluid pressure to the Rotor Brake System. The accumulator has a bleed port for purging air.

#### 6-1-5.3.4 Pistons and Brake Linings.

Hydraulic fluid pressure forces three pistons and self-adjusting brake linings to grip the rotor brake disc on the tail takeoff shaft. This braking action on the rotor brake disc prevents the rotor head from turning.

#### 6-1-5.3.5 Wear Indicator Pins.

Six wear indicator pins, three on each side of the brake assembly, are used to indicate uneven wear on each brake lining. Brake lining wear is indicated by the  $\frac{1}{4}$ -inch wear indicator pins extending less than  $\frac{3}{16}$ -inch from housing. A spark shield is used to cover the assembly.

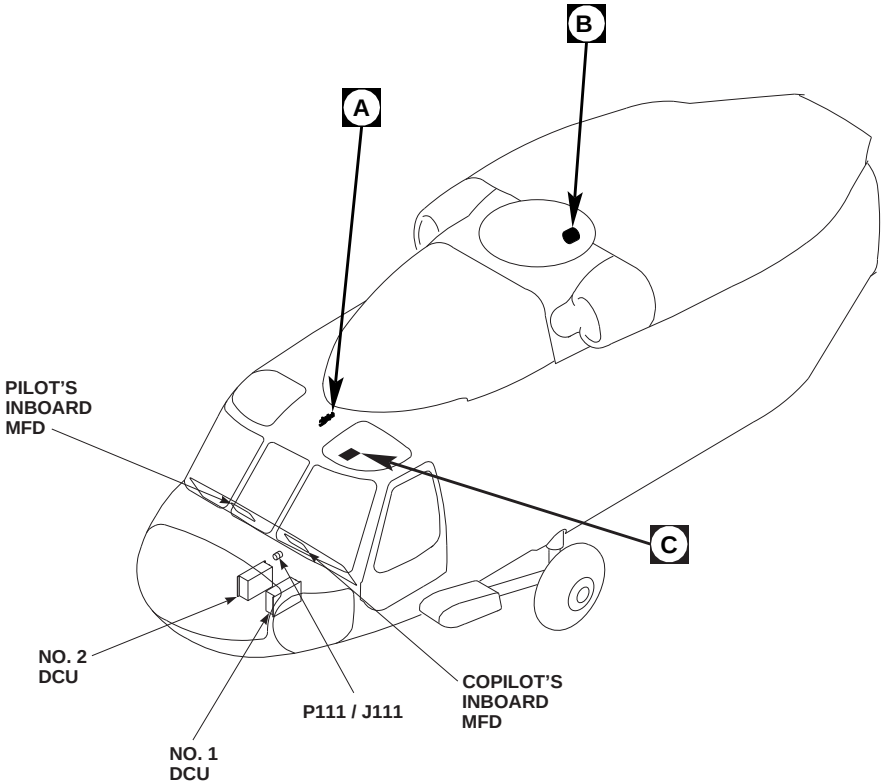
**6-1-5.3.6 Manifold and Valve Assembly.**

The manifold and valve assembly consists of a manifold with a shuttle valve and a pressure relief valve installed. The shuttle valve allows hydraulic fluid to enter from the servicing selector valve and prevents reverse hydraulic fluid flow. The pressure relief valve allows fluid flow to the reservoir with a high-pressure condition.

Excessive hydraulic fluid pressure at the rotor brake is prevented by a manifold relief valve. Hydraulic fluid passed by the manifold relief valve is returned to the reservoir. The 1.3 quart capacity reservoir has a handpump for servicing the Hydraulic System.

**6-1-5.3.7 Reservoir.**

The reservoir supplies the rotor brake master cylinder with hydraulic fluid through the manifold.



| TERMINAL BOARD /<br>DISCONNECT PLUG /<br>RECEPTACLE | LOCATION /<br>CONNECTION POINT               |
|-----------------------------------------------------|----------------------------------------------|
| P111 / J111                                         | COCKPIT, BL 0, STA 197                       |
| P129 / J129                                         | COPILOT'S AUXILIARY<br>CIRCUIT BREAKER PANEL |
| P166 / J166                                         | ENGINE CONTROL<br>QUADRANT                   |
| P221 / J221                                         | LEFT MAIN DISCONNECT                         |
| P410                                                | ROTOR BRAKE PRESSURE<br>SWITCH               |
| P1652 / J1                                          | NO. 1 DCU                                    |
| P1662 / J1                                          | NO. 2 DCU                                    |

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FIGURE 6-1-5-1. ROTOR BRAKE SYSTEM LOCATION DIAGRAM (SHEET 1 OF 2)

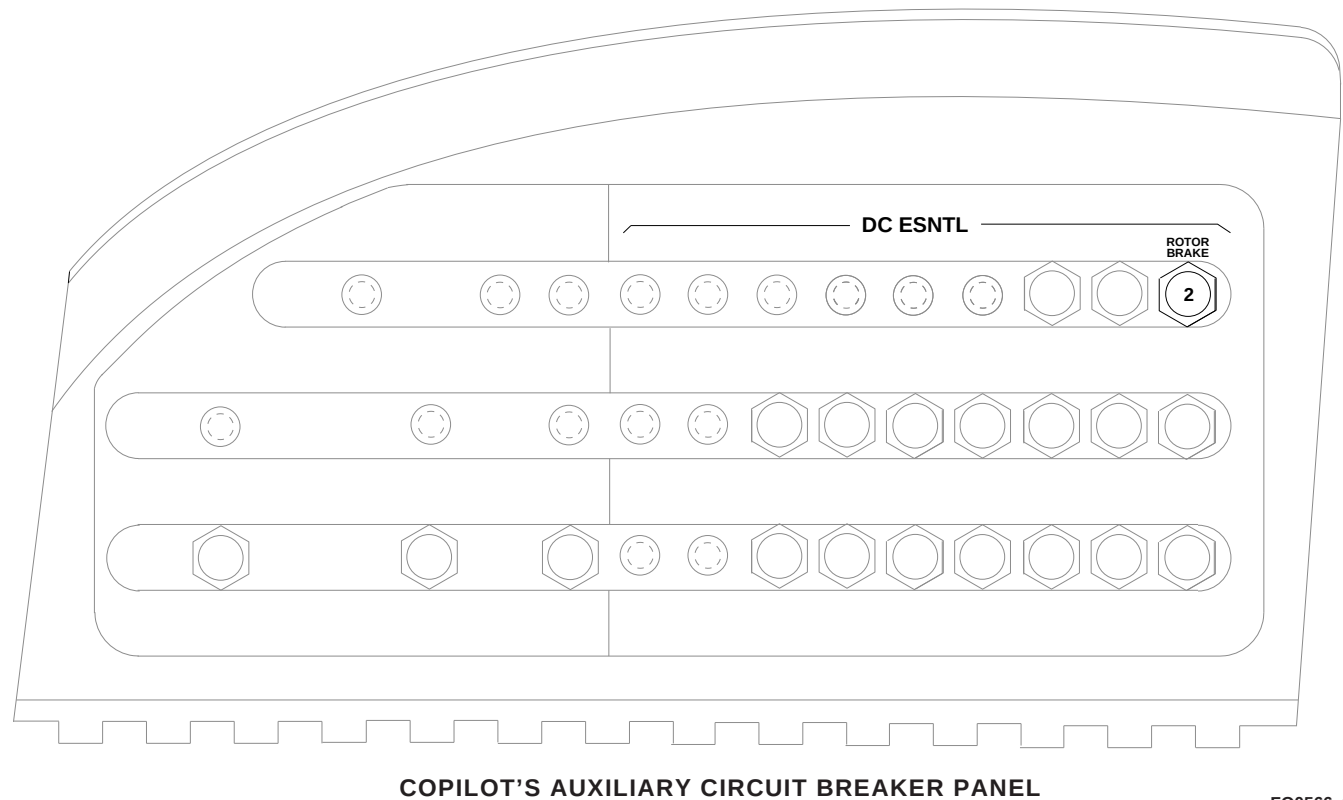
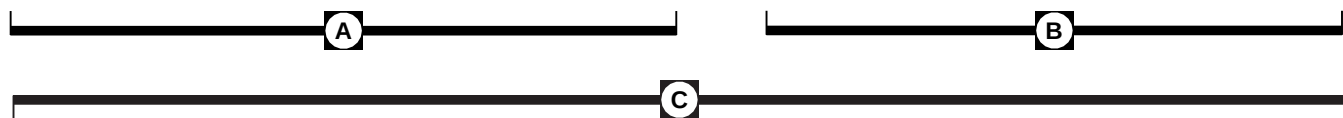
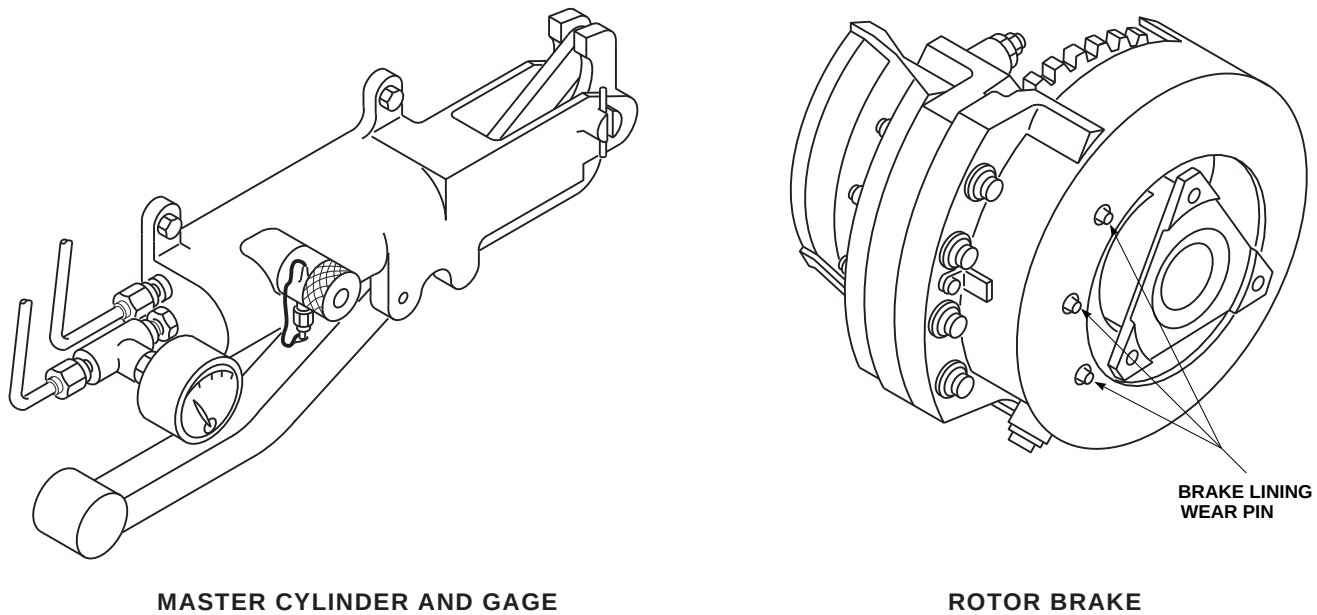
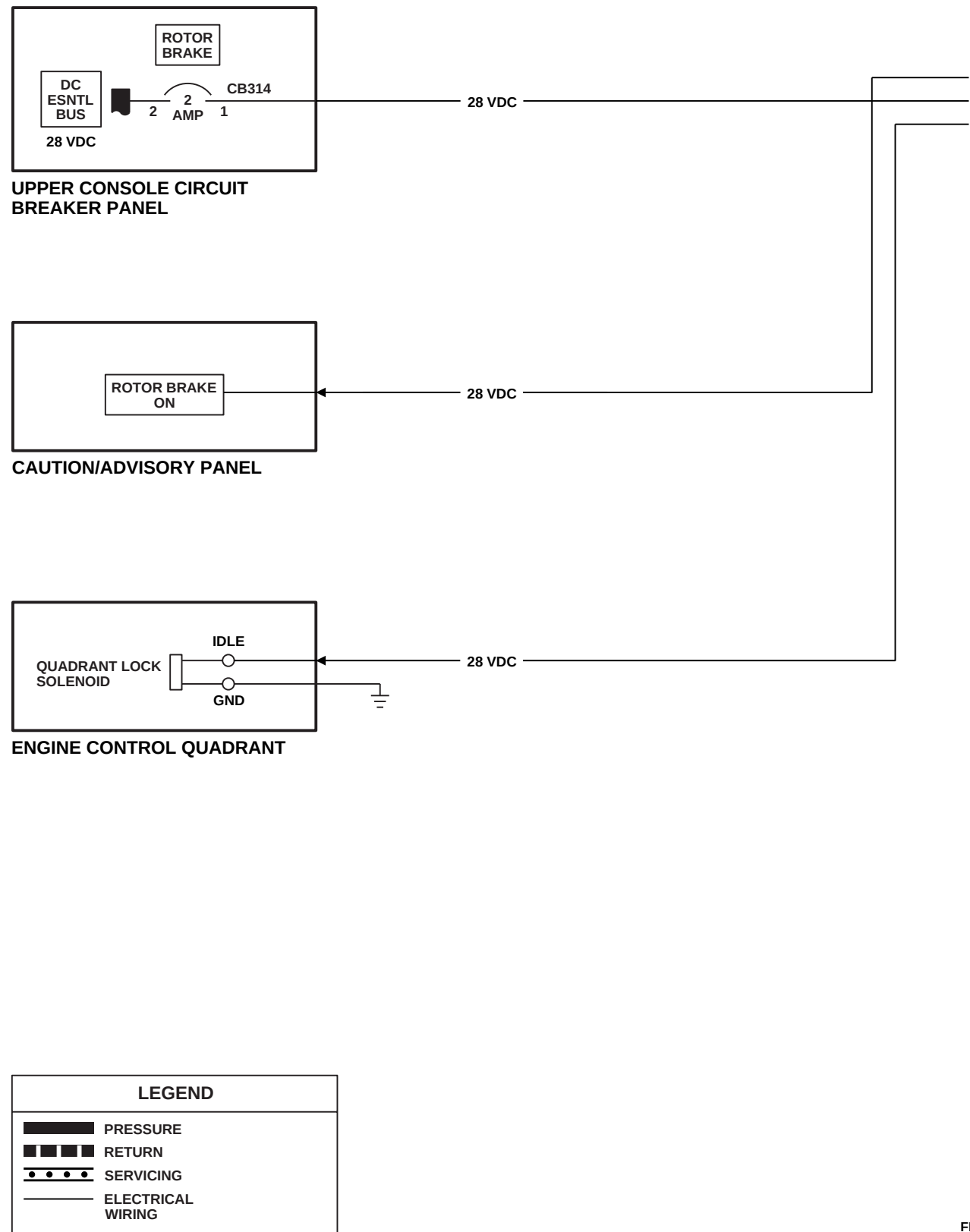


FIGURE 6-1-5-1. ROTOR BRAKE SYSTEM LOCATION DIAGRAM (SHEET 2 OF 2)

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FIGURE 6-1-5-2. ROTOR BRAKE SYSTEM BLOCK DIAGRAM (SHEET 1 OF 2)

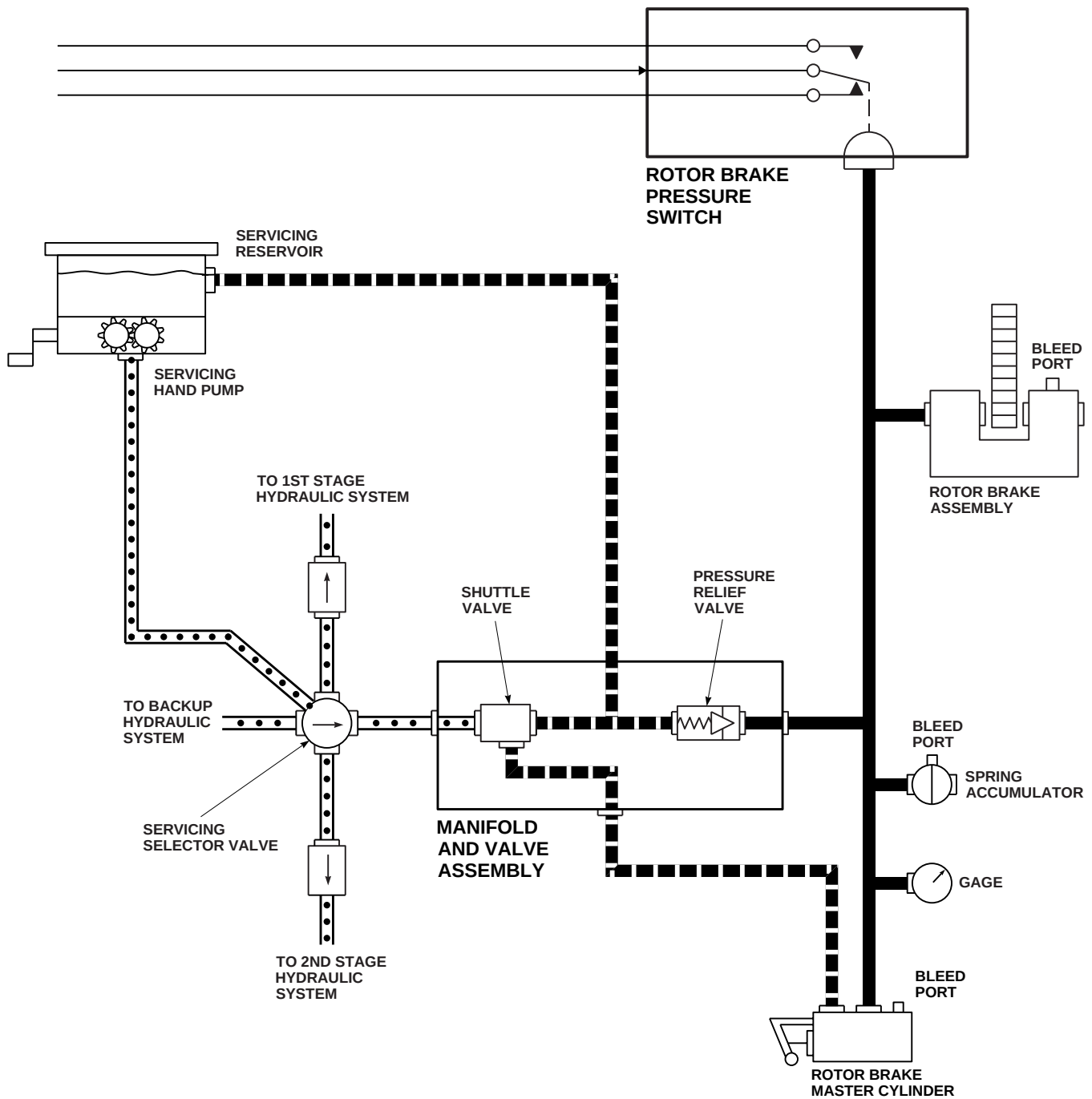
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FIGURE 6-1-5-2. ROTOR BRAKE SYSTEM BLOCK DIAGRAM (SHEET 2 OF 2)

## SECTION 6-2

# TROUBLESHOOTING

### SECTION OVERVIEW

| PARAGRAPH | TITLE                                                            |
|-----------|------------------------------------------------------------------|
| 6-2-1     | Gust Lock System                                                 |
| 6-2-2     | Transmission Chip Detector, Instruments, and Oil Warning Systems |
| 6-2-3     | Rotor Brake System                                               |





6-2-1 GUST LOCK SYSTEM.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer Toolkit

References

- PARA 1-6-16

Equipment Conditions

- External Power Available or APU Operational

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

6-2-1.1 Operational/Troubleshooting Procedure.

- Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- Turn on electrical power (PARA 1-6-16).
- Press button on gust lock handle and push handle up.

RESULT: GUST LOCK capsule on caution/ advisory panel shall go on.

- If result is not as specified, go to TABLE NO. 6-2-1-1.

- Press button on gust lock handle and pull handle down.

RESULT: GUST LOCK capsule on caution/ advisory panel shall go off.

- If result is not as specified, replace gust lock switch (PARA 6-4-9).

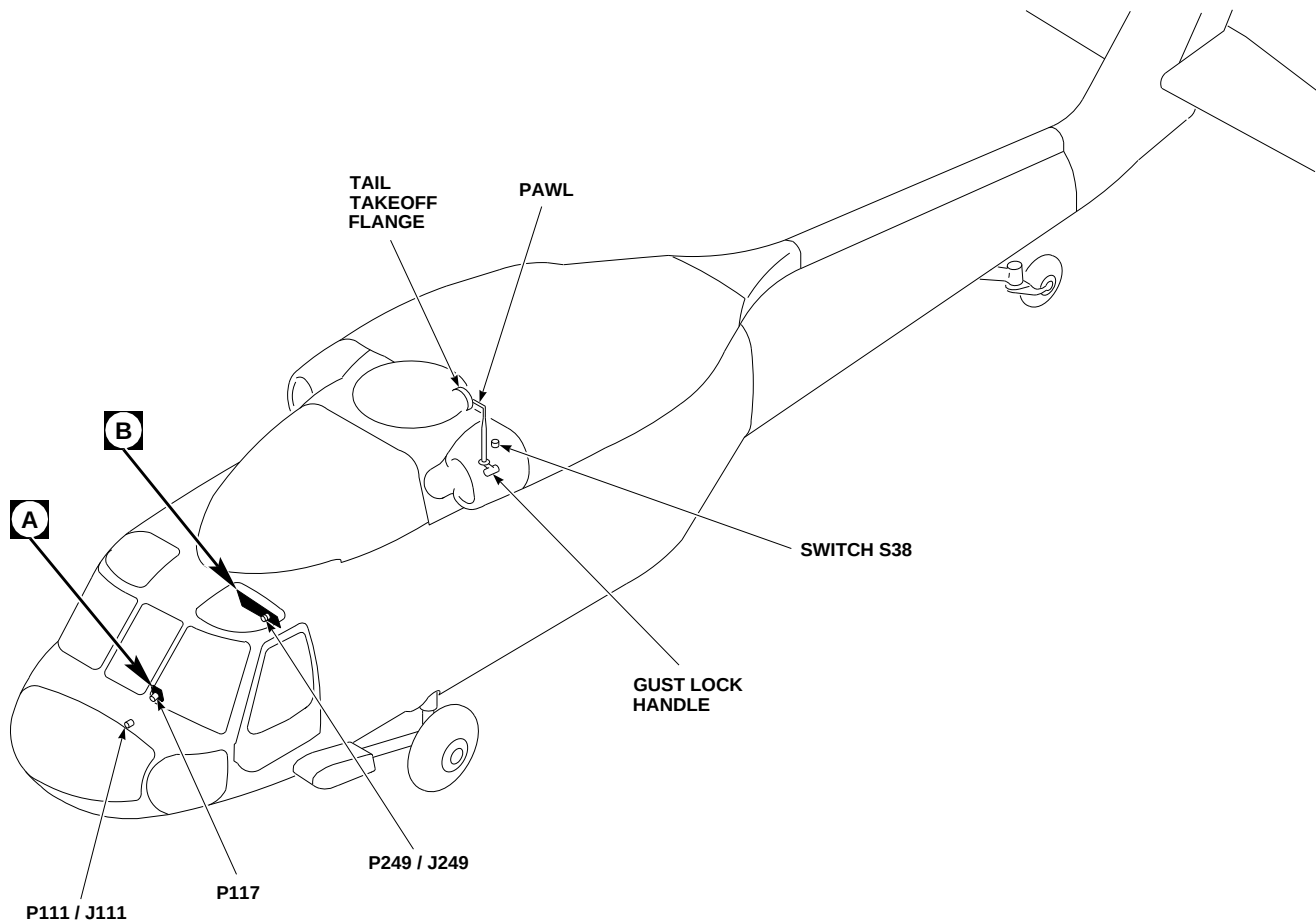
- Press button on handle and push handle up.
- Turn off electrical power (PARA 1-6-16).

TABLE NO. 6-2-1-1. GUST LOCK Capsule Does Not Go On.

| TEST OR INSPECTION                                                             | CORRECTIVE ACTION                                                                                                                                             |
|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST. Go to 2. |                                                                                                                                                               |
| 2. Check that GUST LOCK capsule goes on.                                       |                                                                                                                                                               |
|                                                                                | Step 1. If capsule goes on, go to 3.<br>Step 2. If capsule does not go on, troubleshoot caution/advisory warning system (PARA 8-2-4). Go to 14.               |
| 3. Check pawl for proper engagement.                                           |                                                                                                                                                               |
|                                                                                | Step 1. If pawl has proper engagement, go to 4.<br>Step 2. If pawl does not have proper engagement, go to 12.                                                 |
| 4. Push in the pushbutton on gust lock switch. Go to 5.                        |                                                                                                                                                               |
| 5. Check that GUST LOCK capsule goes on.                                       |                                                                                                                                                               |
|                                                                                | Step 1. If capsule goes on, adjust switch (PARA 6-4-9). Go to 14.<br>Step 2. If capsule does not go on, go to 6.                                              |
| 6. Check helicopter configuration.                                             |                                                                                                                                                               |
|                                                                                | Step 1. <b>EMEP</b> , go to 7.<br>Step 2. <b>W/O EMEP</b> , go to 9.                                                                                          |
| <b>EMEP</b> 7. Remove and check pin filtered adapter P117 (PARA 9-4-98).       |                                                                                                                                                               |
|                                                                                | Step 1. If pin filtered adapter is good, go to 8.<br>Step 2. If pin filtered adapter is not good, replace pin filtered adapter (PARA 9-4-98). Go to 14.       |
| <b>EMEP</b> 8. Install pin filtered adapter (PARA 9-4-98). Go to 9.            |                                                                                                                                                               |
| 9. Check for 28 vdc between P117-22 and ground.                                |                                                                                                                                                               |
|                                                                                | Step 1. If voltage is as specified, troubleshoot caution/advisory warning system (PARA 8-2-4). Go to 14.<br>Step 2. If voltage is not as specified, go to 10. |

**TABLE NO. 6-2-1-1. GUST LOCK Capsule Does Not Go On. (Cont)**

| TEST OR INSPECTION                                                                                                                                                | CORRECTIVE ACTION                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>10. Check for 28 vdc between LIGHTS ADVSY circuit breaker terminal 1 and ground.</b>                                                                           | <p>Step 1. If voltage is as specified, go to <b>11</b>.</p> <p>Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1).<br/>Go to <b>14</b>.</p>                                                   |
| <b>11. Check continuity between LIGHTS ADVSY circuit breaker terminal 1 and GUST LOCK switch contact NC, and between P117-22 and GUST LOCK switch contact NO.</b> | <p>Step 1. If continuity is present, replace GUST LOCK switch (PARA 6-4-9).<br/>Go to <b>14</b>.</p> <p>Step 2. If continuity is not present, repair/replace wiring as required (Appendix F).<br/>Go to <b>14</b>.</p>  |
| <b>12. Turn rotor head slightly to engage pawl. Go to 13.</b>                                                                                                     |                                                                                                                                                                                                                         |
| <b>13. Check pawl for proper engagement and that GUST LOCK capsule goes on.</b>                                                                                   | <p>Step 1. If pawl has proper engagement and capsule goes on, system is operational.<br/>Go to <b>14</b>.</p> <p>Step 2. If pawl does not have proper engagement, adjust rod end (PARA 6-4-9).<br/>Go to <b>14</b>.</p> |
| <b>14. Procedure completed.</b>                                                                                                                                   |                                                                                                                                                                                                                         |



| TERMINAL BOARD/<br>DISCONNECT PLUG/<br>RECEPTACLE | LOCATION/<br>CONNECTION POINT                    |
|---------------------------------------------------|--------------------------------------------------|
| P111 / J111                                       | COCKPIT, BL 7.5, STA 197                         |
| P117                                              | CAUTION / ADVISORY<br>PANEL                      |
| P249 / J249                                       | BEHIND COPILOT'S CIRCUIT<br>BREAKER PANEL, BL 23 |

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FIGURE 6-2-1-1. GUST LOCK SYSTEM LOCATION DIAGRAM (SHEET 1 OF 2)

#### 6-2-1-4

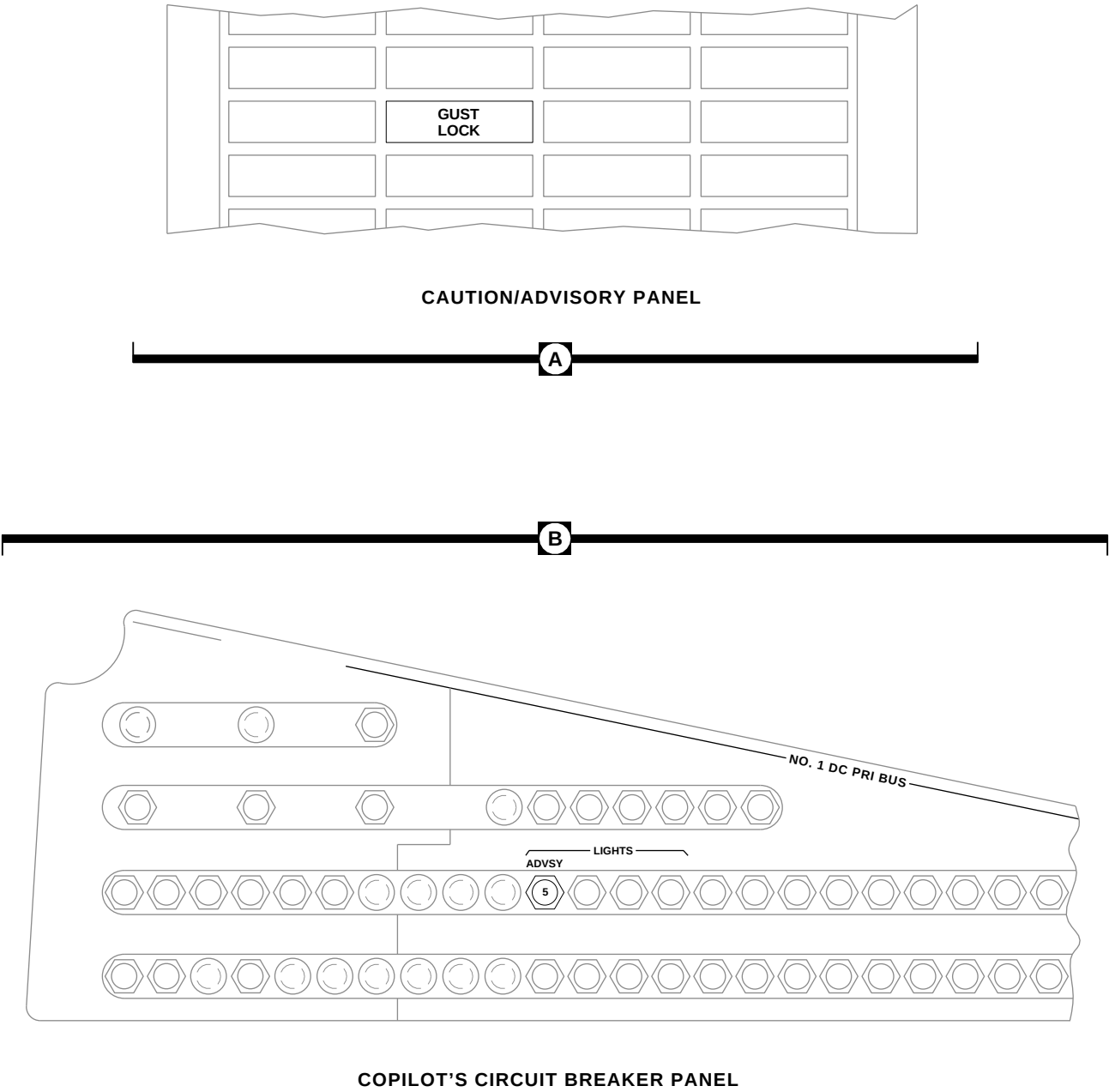


FIGURE 6-2-1-1. GUST LOCK SYSTEM LOCATION DIAGRAM (SHEET 2 OF 2)

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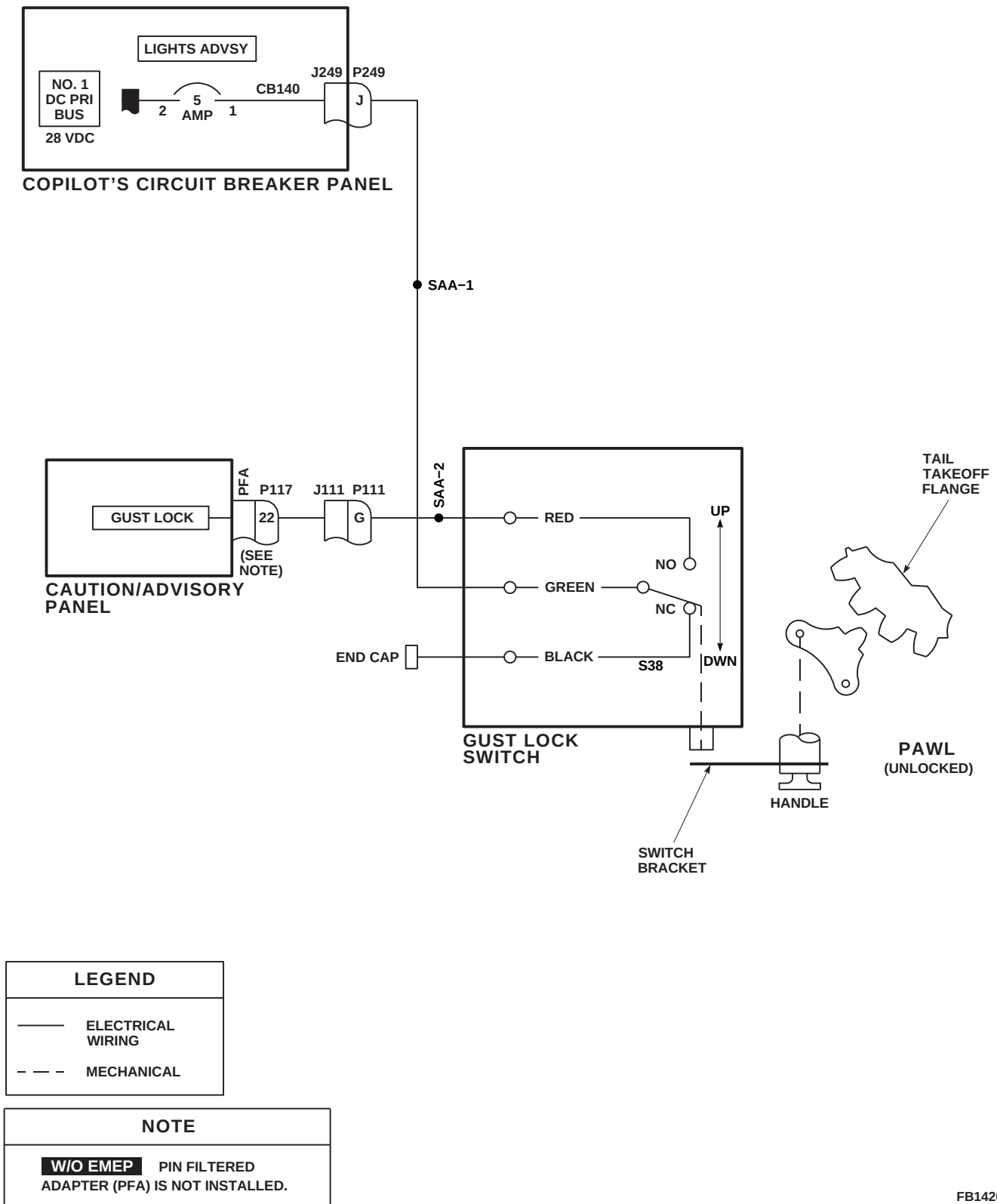


FIGURE 6-2-1-2. GUST LOCK SYSTEM SCHEMATIC DIAGRAM

## 6-2-2 TRANSMISSION CHIP DETECTOR, INSTRUMENTS, AND OIL WARNING SYSTEMS.

### INITIAL SETUP

#### Test Equipment

- Multimeter, Fluke 45

#### Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer Toolkit
- Insulated Jumper Wire

#### Materials/Parts

- Machinery Towel, Item 211, Appendix D

#### References

- Appendix D
- Appropriate Flight Manual
- Appropriate MTF
- PARA 1-3-12
- PARA 1-3-13
- PARA 1-3-14
- PARA 1-6-16
- PARA 6-3-11
- PARA 6-3-17
- PARA 6-4-15
- PARA 6-4-17
- PARA 6-4-18
- PARA 6-4-37

- PARA 6-4-59

#### Equipment Conditions

- External Power Available and APU Operational
- Intermediate Gear Box Serviced
- Main Gear Box Serviced
- Tail Gear Box Serviced
- Upper Console BATT Switch Set to OFF

#### NOTE

- If a circuit breaker pops out during operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).
- When a chip detector is installed after an operational/troubleshooting procedure, place upper console EXT PWR or GENERATORS APU switch to OFF, then ON, before removing next chip detector.

#### Special Instructions

If any of the helicopters have any new, self-test chip detectors, the Improved Chip Detector section must be performed for those positions. For all other positions with the old-style chip detectors, the Chip Detector section must be performed.

The Transmission Chip Detector, Instruments, and Oil Warning Systems operational/troubleshooting procedure is divided as follows:

6-2-2.1 Improved Chip Detector

6-2-2.2 Chip Detector

6-2-2.2.1 Left Accessory Module Chip Detector

6-2-2.2.2 Left Input Module Chip Detector

6-2-2.2.3 Right Accessory Module Chip Detector

6-2-2.2.4 Right Input Module Chip Detector

6-2-2.2.5 Main Module Sump Chip Detector/Temperature Sensor

6-2-2.2.6 Intermediate Gear Box Chip Detector/Temperature Sensor

6-2-2.2.7 Tail Gear Box Chip Detector/Temperature Sensor

6-2-2.3 Instruments and Oil Warning

6-2-2.4 Shutdown

### 6-2-2.1 Improved Chip Detector.

#### NOTE

If helicopter has any new, self-test chip detectors, the following Improved Chip Detector section must be performed for those positions.

- a. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- b. Turn on electrical power (PARA 1-6-16).

RESULT: The following caution/advisory panel capsules shall be on for 45 to 70 seconds:

CHIP ACCESS MDL-LH,  
CHIP INPUT MDL-LH,  
CHIP ACCESS MDL-RH,  
CHIP INPUT MDL-RH,  
CHIP TAIL XMSN, and  
CHIP INT XMSN.

The following caution/advisory panel capsules shall be on after a 30 second delay and shall remain on for an additional 30 seconds:

CHIP MAIN MDL SUMP.

- If CHIP ACCESS MDL-LH capsule is not as specified, go to TABLE NO. 6-2-2-1.
- If CHIP ACCESS MDL-LH capsule goes on, but CHIP DET circuit breaker (DC ESNTL BUS - copilot's side upper console circuit breaker panel) pops out, go to TABLE NO. 6-2-2-2.

- If CHIP INPUT MDL-LH capsule is not as specified, go to TABLE NO. 6-2-2-3.
- If CHIP INPUT MDL-LH capsule goes on, but CHIP DET circuit breaker (DC ESNTL BUS - copilot's side upper console circuit breaker panel) pops out, go to TABLE NO. 6-2-2-2.
- If CHIP ACCESS MDL-RH capsule is not as specified, go to TABLE NO. 6-2-2-4.
- If CHIP ACCESS MDL-RH capsule goes on, but CHIP DET circuit breaker (DC ESNTL BUS - copilot's side upper console circuit breaker panel) pops out, go to TABLE NO. 6-2-2-2.
- If CHIP INPUT MDL-RH capsule is not as specified, go to TABLE NO. 6-2-2-5.
- If CHIP INPUT MDL-RH capsule goes on, but CHIP DET circuit breaker (DC ESNTL BUS - copilot's side upper console circuit breaker panel) pops out, go to TABLE NO. 6-2-2-2.
- If CHIP MAIN MDL SUMP capsule is not as specified, go to TABLE NO. 6-2-2-6.
- If CHIP MAIN MDL SUMP capsule goes on before 30 seconds, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).
- If CHIP MAIN MDL SUMP capsule goes on, but CHIP DET circuit breaker (DC



ESNTL BUS - copilot's side upper console circuit breaker panel) pops out, go to TABLE NO. 6-2-2-2.

- If CHIP INT XMSN capsule is not as specified, go to TABLE NO. 6-2-2-7.
- If CHIP INT XMSN capsule goes on, but CHIP DET circuit breaker (DC ESNTL BUS - copilot's side upper console circuit breaker panel) pops out, go to TABLE NO. 6-2-2-2.
- If CHIP TAIL XMSN capsule is not as specified, go to TABLE NO. 6-2-2-8.
- If CHIP TAIL XMSN capsule goes on, but CHIP DET circuit breaker (DC ESNTL BUS - copilot's side upper console circuit breaker panel) pops out, go to TABLE NO. 6-2-2-2.

c. Turn off electrical power (PARA 1-6-16).

RESULT: CHIP ACCESS MDL-LH,  
CHIP INPUT MDL-LH,  
CHIP ACCESS MDL-RH,  
CHIP INPUT MDL-RH,  
CHIP MAIN MDL SUMP, and  
CHIP INT XMSN capsules on  
caution/advisory panel shall go off  
after five seconds.

- If CHIP ACCESS MDL-LH is not as specified, disconnect P406.
  - (a) If CHIP ACCESS MDL-LH capsule goes off, replace chip detector (PARA 6-4-15).
  - (b) If CHIP ACCESS MDL-LH capsule remains on, disconnect P117.

1 **EMEP** Remove and check Pin Filter Adapter (PFA) (PARA 9-4-98).◀

2 **EMEP** Install PFA (PARA 9-4-98).◀

3 If CHIP ACCESS MDL-LH capsule goes off, repair/replace wiring between P117 and P406 (Appendix F).

4 If CHIP ACCESS MDL-LH capsule remains on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).

- If CHIP INPUT MDL-LH is not as specified, disconnect P403.
  - (a) If CHIP INPUT MDL-LH capsule goes off, replace chip detector (PARA 6-4-17).
  - (b) If CHIP INPUT MDL-LH capsule remains on, disconnect P117.

1 **EMEP** Remove and check PFA (PARA 9-4-98).◀

2 **EMEP** Install PFA (PARA 9-4-98).◀

3 If CHIP INPUT MDL-LH capsule goes off, repair/replace wiring between P117 and P403 (Appendix F).

4 If CHIP INPUT MDL-LH capsule remains on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).

- If CHIP ACCESS MDL-RH is not as specified, disconnect P405.
  - (a) If CHIP ACCESS MDL-RH capsule goes off, replace chip detector (PARA 6-4-15).
  - (b) If CHIP ACCESS MDL-RH capsule remains on, disconnect P117.

1 **EMEP** Remove and check PFA (PARA 9-4-98).◀

2 **EMEP** Install PFA (PARA 9-4-98).◀

3 If CHIP ACCESS MDL-RH capsule goes off, repair/replace wiring between P117 and P405 (Appendix F).

- 4 If CHIP ACCESS MDL-RH capsule remains on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).
- If CHIP INPUT MDL-RH is not as specified, disconnect P404.
  - (a) If CHIP INPUT MDL-RH capsule goes off, replace chip detector (PARA 6-4-17).
  - (b) If CHIP INPUT MDL-RH capsule remains on, disconnect P117.
    - 1 **EMEP** Remove and check PFA (PARA 9-4-98).■
    - 2 **EMEP** Install PFA (PARA 9-4-98).■
    - 3 If CHIP INPUT MDL-RH capsule goes off, repair/replace wiring between P117 and P404 (Appendix F).
    - 4 If CHIP INPUT MDL-RH capsule remains on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).
- If CHIP MAIN MDL SUMP is not as specified, disconnect P412.
  - (a) If CHIP MAIN MDL SUMP capsule goes off, replace chip detector (PARA 6-4-18).
  - (b) If CHIP MAIN MDL SUMP capsule remains on, disconnect P117.
    - 1 **EMEP** Remove and check PFA (PARA 9-4-98).■
    - 2 **EMEP** Install PFA (PARA 9-4-98).■
    - 3 If CHIP MAIN MDL SUMP capsule goes off, repair/replace wiring between P117 and P412 (Appendix F).

- 4 If CHIP MAIN MDL SUMP capsule remains on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).
- If CHIP INT XMSN is not as specified, visually check sensor for chips.
  - (a) If chips are found, perform a serviceability check of intermediate gear box (PARA 6-3-12).
  - (b) If chips are not found, disconnect P613.
    - 1 If CHIP INT XMSN capsule goes off, replace chip detector/temperature sensor (PARA 6-4-37).
    - 2 If CHIP INT XMSN capsule remains on, disconnect P117.
      - a **EMEP** Remove and check PFA (PARA 9-4-98).■
      - b **EMEP** Install PFA (PARA 9-4-98).■
      - c If CHIP INT XMSN capsule goes off, repair/replace wiring between P117 and P613 (Appendix F).
      - d If CHIP INT XMSN capsule remains on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).
- If CHIP TAIL XMSN is not as specified, check sensor for chips.
  - (a) If chips are found, perform a serviceability check of tail gear box (PARA 6-3-18).
  - (b) If chips are not found, disconnect P614.

- 1 If CHIP TAIL XMSN capsule goes off, replace chip detector/temperature sensor (PARA 6-4-59).
- 2 If CHIP TAIL XMSN capsule remains on, disconnect P117.
  - a **EMEP** Remove and check PFA (PARA 9-4-98).
  - b **EMEP** Install PFA (PARA 9-4-98).
  - c If CHIP TAIL XMSN capsule goes off, repair/replace wiring between P117 and P614 (Appendix F).
  - d If CHIP TAIL XMSN capsule remains on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).
- d. Check main transmission for proper oil level and service as necessary (PARA 1-3-12).
- e. Check intermediate gear box for proper oil level and service as necessary (PARA 1-3-13).
- f. Check tail gear box for proper oil level and service as necessary (PARA 1-3-14).
- g. If no other check is to be done, go to Shutdown.

### 6-2-2.2 Chip Detector.

#### NOTE

If the helicopter has any new, self-test chip detectors, perform the Improved Chip Detector section instead of this section for the self-test chip detectors in those positions.

#### 6-2-2.2.1 Left Accessory Module Chip Detector.

#### NOTE

- Chip capsules may momentarily flash as detectors are removed or installed in gear box.

- When grounding chip detector with an insulated screwdriver, a spark will be noticed. The spark indicates that the fuzz burn-off circuit is working.

- a. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.

#### NOTE

In the following step, do not disconnect electrical connector when removing chip detector.

- b. Remove left accessory module chip detector (PARA 6-4-15) and inspect for chips.
- c. Using machinery towel, Item 211, Appendix D, clean all debris and oil off of chip detector.
- d. Turn on electrical power (PARA 1-6-16).

RESULT: CHIP ACCESS MDL-LH capsule on caution/advisory panel shall be off.

- If result is not as specified, disconnect P406.
  - (a) If CHIP ACCESS MDL-LH capsule goes off, replace chip detector (PARA 6-4-15).
  - (b) If CHIP ACCESS MDL-LH capsule remains on, disconnect P117.

- 1 **EMEP** Remove and check PFA (PARA 9-4-98).

- 2 **EMEP** Install PFA (PARA 9-4-98).

- 3 If CHIP ACCESS MDL-LH capsule goes off, repair/replace wiring between P117 and P406 (Appendix F).

- 4 If CHIP ACCESS MDL-LH capsule remains on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).

- e. Ground chip detector by placing a clean screwdriver with an insulated handle between each gap.

RESULT: CHIP ACCESS MDL-LH capsule on caution/advisory panel shall go on.

- If result is not as specified, go to TABLE NO. 6-2-2-1.
- If capsule goes on, but CHIP DET circuit breaker (DC ESNTL BUS - copilot's side upper console circuit breaker panel) pops out, go to TABLE NO. 6-2-2-2.

- f. Turn off electrical power (PARA 1-6-16).

- g. Install left accessory module chip detector (PARA 6-4-15).

- h. If no other check is to be done, go to Shutdown.

#### 6-2-2.2.2 Left Input Module Chip Detector.

##### NOTE

- Chip capsules may momentarily flash as detectors are removed or installed in gear box.
- When grounding chip detector with an insulated screwdriver, a spark will be noticed. The spark indicates that the fuzz burn-off circuit is working.

- a. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.

##### NOTE

In the following step, do not disconnect electrical connector when removing chip detector.

- b. Remove left input module chip detector (PARA 6-4-17) and inspect for chips.

- c. Using machinery towel, Item 211, Appendix D, clean all debris and oil off of chip detector.

- d. Turn on electrical power (PARA 1-6-16).

RESULT: CHIP INPUT MDL-LH capsule on caution/advisory panel shall be off.

- If result is not as specified, disconnect P403.
  - (a) If CHIP INPUT MDL-LH capsule goes off, replace chip detector (PARA 6-4-17).
  - (b) If CHIP INPUT MDL-LH capsule remains on, disconnect P117.

1 **EMEP** Remove and check PFA (PARA 9-4-98).

2 **EMEP** Install PFA (PARA 9-4-98).

3 If CHIP INPUT MDL-LH capsule goes off, repair/replace wiring between P117 and P403 (Appendix F).

4 If CHIP INPUT MDL-LH capsule remains on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).

- e. Ground chip detector by placing a clean screwdriver with an insulated handle between each gap.

RESULT: CHIP INPUT MDL-LH capsule on caution/advisory panel shall go on.

- If result is not as specified, go to TABLE NO. 6-2-2-3.
- If capsule goes on, but CHIP DET circuit breaker (DC ESNTL BUS - copilot's side upper console circuit breaker panel) pops out, go to TABLE NO. 6-2-2-2.

- f. Turn off electrical power (PARA 1-6-16).

g. Install left input module chip detector (PARA 6-4-17).

h. If no other check is to be done, go to Shutdown.

### 6-2-2.2.3 Right Accessory Module Chip Detector.

#### NOTE

- Chip capsules may momentarily flash as detectors are removed or installed in gear box.
- When grounding chip detector with an insulated screwdriver, a spark will be noticed. The spark indicates that the fuzz burn-off circuit is working.

a. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.

#### NOTE

In the following step, do not disconnect electrical connector when removing chip detector.

- b. Remove right accessory module chip detector (PARA 6-4-15) and inspect for chips.
- c. Using machinery towel, Item 211, Appendix D, clean all debris and oil off of chip detector.
- d. Turn on electrical power (PARA 1-6-16).

RESULT: CHIP ACCESS MDL-RH capsule on caution/advisory panel shall be off.

- If result is not as specified, disconnect P405.
  - (a) If CHIP ACCESS MDL-RH capsule goes off, replace chip detector (PARA 6-4-15).
  - (b) If CHIP ACCESS MDL-RH capsule remains on, disconnect P117.

1 **EMEP** Remove and check PFA (PARA 9-4-98).

2 **EMEP** Install PFA (PARA 9-4-98).

3 If CHIP ACCESS MDL-RH capsule goes off, repair/replace wiring between P117 and P405 (Appendix F).

4 If CHIP ACCESS MDL-RH capsule remains on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).

e. Ground chip detector by placing a clean screwdriver with an insulated handle between each gap.

RESULT: CHIP ACCESS MDL-RH capsule on caution/advisory panel shall go on.

- If result is not as specified, go to TABLE NO. 6-2-2-4.
- If capsule goes on, but CHIP DET circuit breaker (DC ESNTL BUS - copilot's side upper console circuit breaker panel) pops out, go to TABLE NO. 6-2-2-2.

f. Turn off electrical power (PARA 1-6-16).

g. Install right accessory module chip detector (PARA 6-4-15).

h. If no other check is to be done, go to Shutdown.

### 6-2-2.2.4 Right Input Module Chip Detector.

#### NOTE

- Chip capsules may momentarily flash as detectors are removed or installed in gear box.
- When grounding chip detector with an insulated screwdriver, a spark will be noticed. The spark indicates that the fuzz burn-off circuit is working.

- a. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.

### NOTE

In the following step, do not disconnect electrical connector when removing chip detector.

- b. Remove right input module chip detector (PARA 6-4-17) and inspect for chips.
- c. Using machinery towel, Item 211, Appendix D, clean all debris and oil off of chip detector.
- d. Turn on electrical power (PARA 1-6-16).

RESULT: CHIP INPUT MDL-RH capsule on caution/advisory panel shall be off.

- If result is not as specified, disconnect P404.
  - (a) If CHIP INPUT MDL-RH capsule goes off, replace chip detector (PARA 6-4-17).
  - (b) If CHIP INPUT MDL-RH capsule remains on, disconnect P117.
    - 1 **EMEP** Remove and check PFA (PARA 9-4-98).
    - 2 **EMEP** Install PFA (PARA 9-4-98).
    - 3 If CHIP INPUT MDL-RH capsule goes off, repair/replace wiring between P117 and P404 (Appendix F).
    - 4 If CHIP INPUT MDL-RH capsule remains on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).
- e. Ground chip detector by placing a clean screwdriver with an insulated handle between each gap.

RESULT: CHIP INPUT MDL-RH capsule on caution/advisory panel shall go on.

- If result is not as specified, go to TABLE NO. 6-2-2-5.
  - If capsule goes on, but CHIP DET circuit breaker (DC ESNTL BUS - copilot's side upper console circuit breaker panel) pops out, go to TABLE NO. 6-2-2-2.
- f. Turn off electrical power (PARA 1-6-16).
  - g. Install right input module chip detector (PARA 6-4-17).
  - h. Reconnect P404 to chip detector.
  - i. If no other check is to be done, go to Shutdown.

### 6-2-2.2.5 Main Module Sump Chip Detector/ Temperature Sensor.

### NOTE

- Chip capsules may momentarily flash as detectors are removed or installed in gear box.
  - When grounding chip detector with an insulated screwdriver, a spark will be noticed. The spark indicates that the fuzz burn-off circuit is working.
- a. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.

### NOTE

In the following step, do not disconnect electrical connector when removing chip detector/temperature sensor.

- b. Remove main module sump chip detector/temperature sensor and remove strainer from detector (PARA 6-4-18), and inspect for chips.

- c. Using machinery towel, Item 211, Appendix D, clean all debris and oil off of chip detector/temperature sensor.
- d. Turn on electrical power (PARA 1-6-16).

RESULT: CHIP MAIN MDL SUMP capsule on caution/advisory panel shall be off.

- If result is not as specified, disconnect P412.
  - (a) If CHIP MAIN MDL SUMP capsule goes off, replace chip detector (PARA 6-4-18).
  - (b) If CHIP MAIN MDL SUMP capsule remains on, disconnect P117.
    - 1 **EMEP** Remove and check PFA (PARA 9-4-98).
    - 2 **EMEP** Install PFA (PARA 9-4-98).
    - 3 If CHIP MAIN MDL SUMP capsule goes off, repair/replace wiring between P117 and P412 (Appendix F).
    - 4 If CHIP MAIN MDL SUMP capsule remains on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).

- e. Ground chip detector by placing a clean screwdriver with an insulated handle between each gap for at least 30 seconds.

RESULT: CHIP MAIN MDL SUMP capsule on caution/advisory panel shall go on 30 seconds after the detector was grounded.

- If result is not as specified, go to TABLE NO. 6-2-2-6.
- If capsule goes on before 30 seconds, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).

- If capsule goes on, but CHIP DET circuit breaker (DC ESNTL BUS - copilot's side upper console circuit breaker panel) pops out, go to TABLE NO. 6-2-2-2.

- f. Turn off electrical power (PARA 1-6-16).
- g. Install main module sump chip detector/temperature sensor (PARA 6-4-18).
- h. Check main transmission for proper oil level and service as necessary (PARA 1-3-12).
- i. If no other check is to be done, go to Shutdown.

#### 6-2-2.2.6 Intermediate Gear Box Chip Detector/Temperature Sensor.

##### NOTE

- Chip capsules may momentarily flash as detectors are removed or installed in gear box.
- When grounding chip detector with an insulated screwdriver, a spark will be noticed. The spark indicates that the fuzz burn-off circuit is working.

- a. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.

##### NOTE

In the following step, do not disconnect electrical connector when removing chip detector/temperature sensor.

- b. Remove intermediate gear box chip detector/temperature sensor (PARA 6-4-37) and inspect for chips.
- c. Using machinery towel, Item 211, Appendix D, clean all debris and oil off of chip detector/temperature sensor.
- d. Turn on electrical power (PARA 1-6-16).



RESULT: CHIP INT XMSN capsule on caution/advisory panel shall be off.

- If result is not as specified, visually check sensor for chips.

(a) If chips are found, perform a serviceability check of intermediate gear box (PARA 6-3-12).

(b) If chips are not found, disconnect P613.

1 If CHIP INT XMSN capsule goes off, replace chip detector/temperature sensor (PARA 6-4-37).

2 If CHIP INT XMSN capsule remains on, disconnect P117.

(a) **EMEP** Remove and check PFA (PARA 9-4-98).

(b) **EMEP** Install PFA (PARA 9-4-98).

(c) If CHIP INT XMSN capsule goes off, repair/replace wiring between P117 and P613 (Appendix F).

(d) If CHIP INT XMSN capsule remains on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).

e. Ground chip detector/temperature sensor by placing a clean screwdriver with an insulated handle between each gap.

RESULT: CHIP INT XMSN capsule on caution/advisory panel shall go on.

- If result is not as specified, go to TABLE NO. 6-2-2-7.
- If capsule goes on, but CHIP DET circuit breaker (DC ESNTL BUS - copilot's side

upper console circuit breaker panel) pops out, go to TABLE NO. 6-2-2-2.

f. Turn off electrical power (PARA 1-6-16).

g. Install intermediate gear box chip detector/temperature sensor (PARA 6-4-37).

h. Check intermediate gear box for proper oil level and service as necessary (PARA 1-3-13).

i. If no other check is to be done, go to Shutdown.

### 6-2-2.2.7 Tail Gear Box Chip Detector/ Temperature Sensor.

#### NOTE

- Chip capsules may momentarily flash as detectors are removed or installed in gear box.
  - When grounding chip detector with an insulated screwdriver, a spark will be noticed. The spark indicates that the fuzz burn-off circuit is working.
- a. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.

#### NOTE

In the following step, do not disconnect electrical connector when removing chip detector/temperature sensor.

- b. Remove tail gear box chip detector/temperature sensor (PARA 6-4-59) and inspect for chips.
- c. Using machinery towel, Item 211, Appendix D, clean all debris and oil off of chip detector/temperature sensor.
- d. Turn on electrical power (PARA 1-6-16).

RESULT: CHIP TAIL XMSN capsule on caution/advisory panel shall be off.

- If result is not as specified, check sensor for chips.



- (a) If chips are found, perform a service-ability check of tail gear box (PARA 6-3-18).
- (b) If chips are not found, disconnect P614.
 

1

 If CHIP TAIL XMSN capsule goes off, replace chip detector/temperature sensor (PARA 6-4-59).

2

 If CHIP TAIL XMSN capsule remains on, disconnect P117.

(a)

EMEP

 Remove and check PFA (PARA 9-4-98).

(b)

EMEP

 Install PFA (PARA 9-4-98).

(c)

 If CHIP TAIL XMSN capsule goes off, repair/replace wiring between P117 and P614 (Appendix F).

(d)

 If CHIP TAIL XMSN capsule remains on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4).
- e. Ground chip detector/temperature sensor by placing a screwdriver with an insulated handle between each gap.

RESULT: CHIP TAIL XMSN capsule on caution/advisory panel shall go on.

- If result is not as specified, go to TABLE NO. 6-2-2-8.
- If capsule goes on, but CHIP DET circuit breaker (DC ESNTL BUS - copilot's side upper console circuit breaker panel) pops out, go to TABLE NO. 6-2-2-2.

- f. Turn off electrical power (PARA 1-6-16).

- g. Install tail gear box chip detector/temperature sensor (PARA 6-4-59).
- h. Check tail gear box for proper oil level and service as necessary (PARA 1-3-14).
- i. If no other check is to be done, go to Shutdown.

6-2-2.3 Instruments and Oil Warning.

- a. Check oil filter bypass button.
 

RESULT: Oil filter bypass button shall not be extended.
 
  - If result is not as specified, go to TABLE NO. 6-2-2-26.
- b. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- c. Start APU (Appropriate Flight Manual).
 

RESULT: MAIN XMSN OIL PRESS capsule on caution/advisory panel shall be on.
 
  - If result is not as specified, go to TABLE NO. 6-2-2-9.
- d. With qualified personnel at controls, operate No. 1 and No. 2 engines with main rotor turning at 100% Nr until main transmission instruments have stabilized (Appropriate Flight Manual).

|                            |                   |
|----------------------------|-------------------|
| RESULT:                    |                   |
| CAUTION/<br>ADVISORY PANEL | CAPSULE<br>STATUS |
| MAIN XMSN OIL<br>TEMP      | OFF               |
| MAIN XMSN OIL<br>PRESS     | OFF               |
| INT XMSN OIL TEMP          | OFF               |

**CAUTION/  
ADVISORY PANEL**TAIL XMSN OIL  
TEMP**CAPSULE  
STATUS**

OFF

(c) Is not steady, go to TABLE NO. 6-2-2-14.

(d) Has no indication, go to TABLE NO. 6-2-2-14.

**CENTRAL DISPLAY  
UNIT**

XMSN TEMP

**INDICATION**

(Refer to Appropriate MTF for normal indications)

(e) Suddenly drops (more than 10 psi) without fluctuations, go to TABLE NO. 6-2-2-27.

XMSN PRESS

(Refer to Appropriate MTF for normal indications)

## • IF XMSN TEMP:

(a) Is too high (MAIN XMSN OIL TEMP capsule goes on), go to TABLE NO. 6-2-2-11.

(b) Is not steady, go to TABLE NO. 6-2-2-11.

(c) Has no indication, go to TABLE NO. 6-2-2-15.

%RPM RTR

100%

- If these caution/advisory panel capsules are on:

## • If %RPM RTR:

**CAPSULE****GO TO**MAIN XMSN OIL  
PRESS

TABLE NO. 6-2-2-10.

(a) Is not steady, go to TABLE NO. 6-2-2-16.

MAIN XMSN OIL  
TEMP

TABLE NO. 6-2-2-11.

(b) Has no indication, make sure P443 is connected to main gear box.

INT XMSN OIL TEMP

TABLE NO. 6-2-2-12.

1 If P443 is connected, go to TABLE NO. 6-2-2-16.TAIL XMSN OIL  
TEMP

TABLE NO. 6-2-2-13.

2 If P443 is not connected, connect P443 to main gear box.

- If any RTR OVERSPEED capsule goes on, perform a special inspection (PARA 1-7-2).

## • If XMSN PRESS:

e. Shut down No. 1 and No. 2 engines (Appropriate Flight Manual).

(a) Is too low (MAIN XMSN OIL PRESS capsule is on), go to TABLE NO. 6-2-2-10.

f. Shut down APU (Appropriate Flight Manual).

(b) Is too high, go to TABLE NO. 6-2-2-14.

g. Turn on electrical power (PARA 1-6-16).

h. Install an insulated jumper wire between P408-A and P408-C.

RESULT: MAIN XMSN OIL TEMP capsule on caution/advisory panel shall go on.

- If result is not as specified, go to TABLE NO. 6-2-2-17.

- i. Remove jumper wire from P408 and connect to main gear box.
- j. Install an insulated jumper wire between P613-B and P613-C (intermediate gear box chip detector/temperature sensor electrical connector).

RESULT: INT XMSN OIL TEMP capsule on caution/advisory panel shall go on.

- If result is not as specified, go to TABLE NO. 6-2-2-18.

- k. Remove jumper wire from P613.
- l. Remove intermediate gear box chip detector/temperature sensor (PARA 6-4-37).
- m. Connect P613 to chip detector/temperature sensor.
- n. Connect an insulated jumper wire between spring and valve housing.

RESULT: INT XMSN OIL TEMP capsule on caution/advisory panel shall go on.

- If result is not as specified, replace intermediate gear box chip detector/temperature sensor (PARA 6-4-37).

- o. Install intermediate gear box chip detector/temperature sensor (PARA 6-4-37).

- p. Install an insulated jumper wire between P614-B and P614-C (tail gear box chip detector/temperature sensor electrical connector).

RESULT: TAIL XMSN OIL TEMP capsule on caution/advisory panel shall go on.

- If result is not as specified, go to TABLE NO. 6-2-2-19.

- q. Remove jumper wire from P614.
- r. Remove tail gear box chip detector/temperature sensor (PARA 6-4-59).
- s. Connect P614 to tail gear box chip detector/temperature sensor.
- t. Connect an insulated jumper wire between spring and housing valve.

RESULT: TAIL XMSN OIL TEMP capsule on caution/advisory panel shall go on.

- If result is not as specified, replace tail gear box chip detector/temperature sensor (PARA 6-4-59).

- u. Install tail gear box chip detector/temperature sensor (PARA 6-4-59).
- v. If no other check is to be done, go to Shutdown.

6-2-2.4 Shutdown.

Turn off electrical power (PARA 1-6-16).

TABLE NO. 6-2-2-1. CHIP ACCESS MDL-LH Capsule Does Not Go On.

| TEST OR INSPECTION |
|--------------------|
| CORRECTIVE ACTION  |

- 1. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST. Go to 2.

TABLE NO. 6-2-2-1. CHIP ACCESS MDL-LH Capsule Does Not Go On. (Cont)

---

**TEST OR INSPECTION  
CORRECTIVE ACTION**

---

**2. Check that CHIP ACCESS MDL-LH capsule goes on.**

Step 1. If capsule goes on, go to **3**.

Step 2. If capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to **16**.

**3. Install an insulated jumper wire between J117-14 and ground. Go to 4.****4. Check that CHIP ACCESS MDL-LH capsule goes on.**

Step 1. If capsule goes on, go to **5**.

Step 2. If capsule does not go on, go to **10**.

**5. Remove jumper wire. Go to 6.****6. Check helicopter configuration.**

Step 1. **EMEP**, go to **7**.

Step 2. **W/O EMEP**, go to **9**.

**7. Remove and check PFA P117 (PARA 9-4-98).**

Step 1. If PFA is good, go to **8**.

Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to **16**.

**8. Install PFA (PARA 9-4-98). Go to 9.****9. Repair/replace wiring between P117-14 and left accessory module chip detector (Appendix F). Go to 16.****10. Remove jumper wire. Go to 11.****11. Check helicopter configuration.**

Step 1. **EMEP**, go to **12**.

Step 2. **W/O EMEP**, go to **14**.

**12. Remove and check PFA P118 (PARA 9-4-98).**

Step 1. If PFA is good, go to **13**.

Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to **16**.

TABLE NO. 6-2-2-1. CHIP ACCESS MDL-LH Capsule Does Not Go On. (Cont)

|             | TEST OR INSPECTION<br>CORRECTIVE ACTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>EMEP</b> | <p>13. Install PFA (PARA 9-4-98). Go to 14.</p> <p>14. Check for 28 VDC between P118-f and ground.</p> <p>Step 1. If voltage is as specified, replace caution/advisory panel (PARA 8-4-19).<br/>Go to 16.</p> <p>Step 2. If voltage is not as specified, go to 15.</p> <p>15. Check for 28 VDC between terminal 1 (CHIP DET circuit breaker, DC ESNTL BUS - copilot's side upper console circuit breaker panel) and ground.</p> <p>Step 1. If voltage is as specified, repair/replace wiring between P118-f and terminal 1 (Appendix F). Go to 16.</p> <p>Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1).<br/>Go to 16.</p> <p>16. Procedure completed.</p> |  |

TABLE NO. 6-2-2-2. CHIP DET Circuit Breaker Pops When A Chip Detector Light Goes On.

|  | TEST OR INSPECTION<br>CORRECTIVE ACTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <p>1. Check that CHIP ACCESS MDL-RH capsule is on.</p> <p>Step 1. If capsule is on, troubleshoot for CHIP DET circuit breaker (DC ESNTL BUS - copilot's side upper console circuit breaker panel) popping (PARA 9-2-3).</p> <p>Step 2. If trouble remains, go to TABLE NO. 6-2-2-20.</p> <p>Step 3. If capsule is not on, go to 2.</p> <p>2. Check that CHIP ACCESS MDL-LH capsule is on.</p> <p>Step 1. If capsule is on, troubleshoot for CHIP DET circuit breaker (DC ESNTL BUS - copilot's side upper console circuit breaker panel) popping (PARA 9-2-3).</p> <p>Step 2. If trouble remains, go to TABLE NO. 6-2-2-21.</p> <p>Step 3. If capsule is not on, go to 3.</p> |  |

**TABLE NO. 6-2-2-2. CHIP DET Circuit Breaker Pops When A Chip Detector Light Goes On. (Cont)**

| TEST OR INSPECTION | CORRECTIVE ACTION |
|--------------------|-------------------|
|--------------------|-------------------|

**3. Check that CHIP INPUT MDL-LH capsule is on.**

Step 1. If capsule is on, troubleshoot for CHIP DET circuit breaker (DC ESNTL BUS - copilot's side upper console circuit breaker panel) popping (PARA 9-2-3).

Step 2. If trouble remains, go to **TABLE NO. 6-2-2-22**.

Step 3. If capsule is not on, go to **4**.

**4. Check that CHIP INPUT MDL-RH capsule is on.**

Step 1. If capsule is on, troubleshoot for CHIP DET circuit breaker (DC ESNTL BUS - copilot's side upper console circuit breaker panel) popping (PARA 9-2-3).

Step 2. If trouble remains, go to **TABLE NO. 6-2-2-23**.

Step 3. If capsule is not on, go to **5**.

**5. CHIP MAIN MDL SUMP capsule is on. Go to 6.****6. Troubleshoot for CHIP DET circuit breaker popping (PARA 9-2-3).****7. If trouble remains, go to TABLE NO. 6-2-2-24. Then go to 8.****8. Procedure completed.****TABLE NO. 6-2-2-3. CHIP INPUT MDL-LH Capsule Does Not Go On.**

| TEST OR INSPECTION | CORRECTIVE ACTION |
|--------------------|-------------------|
|--------------------|-------------------|

**1. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST. Go to 2.****2. Check that CHIP INPUT MDL-LH capsule goes on.**

Step 1. If capsule goes on, go to **3**.

Step 2. If capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to **12**.

**3. Install an insulated jumper wire between J117-13 and ground. Go to 4.**

TABLE NO. 6-2-2-3. CHIP INPUT MDL-LH Capsule Does Not Go On. (Cont)

| TEST OR INSPECTION<br>CORRECTIVE ACTION                                                              |                                             |
|------------------------------------------------------------------------------------------------------|---------------------------------------------|
| 4. Check that CHIP INPUT MDL-LH capsule goes on.                                                     |                                             |
| Step 1. If capsule goes on, go to 5.                                                                 |                                             |
| Step 2. If capsule does not go on, go to 10.                                                         |                                             |
| 5. Remove jumper wire. Go to 6.                                                                      |                                             |
| 6. Check helicopter configuration.                                                                   |                                             |
| Step 1. <b>EMEP</b> , go to 7.                                                                       |                                             |
| Step 2. <b>W/O EMEP</b> , go to 9.                                                                   |                                             |
| <b>EMEP</b>                                                                                          | 7. Remove and check PFA P117 (PARA 9-4-98). |
| Step 1. If PFA is good, go to 8.                                                                     |                                             |
| Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to 12.                                     |                                             |
| <b>EMEP</b>                                                                                          | 8. Install PFA (PARA 9-4-98). Go to 9.      |
| 9. Repair/replace wiring between P117-13 and left input module chip detector (Appendix F). Go to 12. |                                             |
| 10. Remove jumper wire. Go to 11.                                                                    |                                             |
| 11. Replace caution/advisory panel (PARA 8-4-19). Go to 12.                                          |                                             |
| 12. Procedure completed.                                                                             |                                             |

TABLE NO. 6-2-2-4. CHIP ACCESS MDL-RH Capsule Does Not Go On.

| TEST OR INSPECTION<br>CORRECTIVE ACTION                                        |  |
|--------------------------------------------------------------------------------|--|
| 1. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST. Go to 2. |  |
| 2. Check that CHIP ACCESS MDL-RH capsule goes on.                              |  |
| Step 1. If capsule goes on, go to 3.                                           |  |

TABLE NO. 6-2-2-4. CHIP ACCESS MDL-RH Capsule Does Not Go On. (Cont)

---

**TEST OR INSPECTION  
CORRECTIVE ACTION**

---

Step 2. If capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 12.

3. Install an insulated jumper wire between J117-62 and ground. Go to 4.

4. Check that CHIP ACCESS MDL-RH capsule goes on.

Step 1. If capsule goes on, go to 5.

Step 2. If capsule does not go on, go to 10.

5. Remove jumper wire. Go to 6.

6. Check helicopter configuration.

Step 1. **EMEP**, go to 7.

Step 2. **W/O EMEP**, go to 9.

**EMEP**

7. Remove and check PFA P117 (PARA 9-4-98).

Step 1. If PFA is good, go to 8.

Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to 12.

**EMEP**

8. Install PFA (PARA 9-4-98). Go to 9.

9. Repair/replace wiring between P117-62 and right accessory module chip detector (Appendix F). Go to 12.

10. Remove jumper wire. Go to 11.

11. Replace caution/advisory panel (PARA 8-4-19). Go to 12.

12. Procedure completed.

---

TABLE NO. 6-2-2-5. CHIP INPUT MDL-RH Capsule Does Not Go On.

---

**TEST OR INSPECTION  
CORRECTIVE ACTION**

---

1. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST. Go to 2.



TABLE NO. 6-2-2-5. CHIP INPUT MDL-RH Capsule Does Not Go On. (Cont)

|      | TEST OR INSPECTION<br>CORRECTIVE ACTION                                                                 |  |
|------|---------------------------------------------------------------------------------------------------------|--|
|      | 2. Check that CHIP INPUT MDL-RH capsule goes on.                                                        |  |
|      | Step 1. If capsule goes on, go to 3.                                                                    |  |
|      | Step 2. If capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 12. |  |
|      | 3. Install an insulated jumper wire between J117-61 and ground. Go to 4.                                |  |
|      | 4. Check that CHIP INPUT MDL-RH capsule goes on.                                                        |  |
|      | Step 1. If capsule goes on, go to 5.                                                                    |  |
|      | Step 2. If capsule does not go on, go to 10.                                                            |  |
|      | 5. Remove jumper wire. Go to 6.                                                                         |  |
|      | 6. Check helicopter configuration.                                                                      |  |
|      | Step 1. EMEP , go to 7.                                                                                 |  |
|      | Step 2. W/O EMEP , go to 9.                                                                             |  |
| EMEP | 7. Remove and check PFA P117 (PARA 9-4-98).                                                             |  |
|      | Step 1. If PFA is good, go to 8.                                                                        |  |
|      | Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to 12.                                        |  |
| EMEP | 8. Install PFA (PARA 9-4-98). Go to 9.                                                                  |  |
|      | 9. Repair/replace wiring between P117-61 and right input module chip detector (Appendix F). Go to 12.   |  |
|      | 10. Remove jumper wire. Go to 11.                                                                       |  |
|      | 11. Replace caution/advisory panel (PARA 8-4-19). Go to 12.                                             |  |
|      | 12. Procedure completed.                                                                                |  |

TABLE NO. 6-2-2-6. CHIP MAIN MDL SUMP Capsule Does Not Go On.

---

| TEST OR INSPECTION | CORRECTIVE ACTION |
|--------------------|-------------------|
|--------------------|-------------------|

---

1. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST. Go to 2.

2. Check that CHIP MAIN MDL SUMP capsule goes on.

Step 1. If capsule goes on, go to 3.

Step 2. If capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 13.

3. Install an insulated jumper wire between J117-30 and ground. Go to 4.

4. Wait 30 seconds. Go to 5.

5. Check that CHIP MAIN MDL SUMP capsule goes on.

Step 1. If capsule goes on, go to 6.

Step 2. If capsule does not go on, go to 11.

6. Remove jumper wire. Go to 7.

7. Check helicopter configuration.

Step 1. **EMEP**, go to 8.

Step 2. **W/O EMEP**, go to 10.

**EMEP** 8. Remove and check PFA P117 (PARA 9-4-98).

Step 1. If PFA is good, go to 9.

Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to 13.

**EMEP** 9. Install PFA (PARA 9-4-98). Go to 10.

10. Repair/replace wiring between P117-30 and main module sump chip detector (Appendix F). Go to 13.

11. Remove jumper wire. Go to 12.

12. Replace caution/advisory panel (PARA 8-4-19). Go to 13.

13. Procedure completed.

---

TABLE NO. 6-2-2-7. CHIP INT XMSN Capsule Does Not Go On.

| TEST OR INSPECTION | CORRECTIVE ACTION                                                                                        |
|--------------------|----------------------------------------------------------------------------------------------------------|
|                    | 1. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST. Go to 2.                           |
|                    | 2. Check that CHIP INT XMSN capsule goes on.                                                             |
|                    | Step 1. If capsule goes on, go to 3.                                                                     |
|                    | Step 2. If capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 12.  |
|                    | 3. Install an insulated jumper wire between J117-29 and ground. Go to 4.                                 |
|                    | 4. Check that CHIP INT XMSN capsule goes on.                                                             |
|                    | Step 1. If capsule goes on, go to 5.                                                                     |
|                    | Step 2. If capsule does not go on, go to 10.                                                             |
|                    | 5. Remove jumper wire. Go to 6.                                                                          |
|                    | 6. Check helicopter configuration.                                                                       |
|                    | Step 1. <b>EMEP</b> , go to 7.                                                                           |
|                    | Step 2. <b>W/O EMEP</b> , go to 9.                                                                       |
| <b>EMEP</b>        | 7. Remove and check PFA P117 (PARA 9-4-98). <span style="float: right;"> </span>                         |
|                    | Step 1. If PFA is good, go to 8. <span style="float: right;"> </span>                                    |
|                    | Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to 12. <span style="float: right;"> </span>    |
| <b>EMEP</b>        | 8. Install PFA (PARA 9-4-98). Go to 9. <span style="float: right;"> </span>                              |
|                    | 9. Repair/replace wiring between P117-29 and intermediate gear box chip detector (Appendix F). Go to 12. |
|                    | 10. Remove jumper wire. Go to 11.                                                                        |
|                    | 11. Replace caution/advisory panel (PARA 8-4-19). Go to 12.                                              |
|                    | 12. Procedure completed.                                                                                 |

TABLE NO. 6-2-2-8. CHIP TAIL XMSN Capsule Does Not Go On.

---

| TEST OR INSPECTION | CORRECTIVE ACTION |
|--------------------|-------------------|
|--------------------|-------------------|

---

1. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST. Go to 2.

2. Check that CHIP TAIL XMSN capsule goes on.

Step 1. If capsule goes on, go to 3.

Step 2. If capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 12.

3. Install an insulated jumper wire between J117-45 and ground. Go to 4.

4. Check that CHIP TAIL XMSN capsule goes on.

Step 1. If capsule goes on, go to 5.

Step 2. If capsule does not go on, go to 10.

5. Remove jumper wire. Go to 6.

6. Check helicopter configuration.

Step 1. **EMEP**, go to 7.

Step 2. **W/O EMEP**, go to 9.

**EMEP**

7. Remove and check PFA P117 (PARA 9-4-98).

Step 1. If PFA is good, go to 8.

Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to 12.

**EMEP**

8. Install PFA (PARA 9-4-98). Go to 9.

9. Repair/replace wiring between P117-45 and tail gear box chip detector (Appendix F). Go to 12.

10. Remove jumper wire. Go to 11.

11. Replace caution/advisory panel (PARA 8-4-19). Go to 12.

12. Procedure completed.

---

TABLE NO. 6-2-2-9. MAIN XMSN OIL PRESS Capsule Does Not Go On.

|             | TEST OR INSPECTION<br>CORRECTIVE ACTION                                                                                                                                                                   |  |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|             | 1. Place and hold caution/advisory panel BRT/DIM-TEST switch to TEST. Go to 2.                                                                                                                            |  |
|             | 2. Check that MAIN XMSN OIL PRESS capsule goes on.<br><br>Step 1. If capsule goes on, go to 3.<br>Step 2. If capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 16. |  |
|             | 3. Install an insulated jumper wire between P409-2 and P409-3. Go to 4.                                                                                                                                   |  |
|             | 4. Check that MAIN XMSN OIL PRESS capsule goes on.<br><br>Step 1. If capsule goes on, go to 5.<br>Step 2. If capsule does not go on, go to 7.                                                             |  |
|             | 5. Remove jumper wire. Go to 6.                                                                                                                                                                           |  |
|             | 6. Replace low oil pressure switch (PARA 6-4-28). Go to 16.                                                                                                                                               |  |
|             | 7. Check helicopter configuration.<br><br>Step 1. <b>EMEP</b> , go to 8.<br>Step 2. <b>W/O EMEP</b> , go to 10.                                                                                           |  |
| <b>EMEP</b> | 8. Remove and check PFA P117 (PARA 9-4-98).<br><br>Step 1. If PFA is good, go to 9.<br>Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to 16.                                                   |  |
| <b>EMEP</b> | 9. Install PFA (PARA 9-4-98). Go to 10.                                                                                                                                                                   |  |
|             | 10. Check for 28 VDC between P117-16 and ground.<br><br>Step 1. If voltage is as specified, go to 11.<br>Step 2. If voltage is not as specified, go to 13.                                                |  |
|             | 11. Remove jumper wire. Go to 12.                                                                                                                                                                         |  |

TABLE NO. 6-2-2-9. MAIN XMSN OIL PRESS Capsule Does Not Go On. (Cont)

| TEST OR INSPECTION                                                                                                                | CORRECTIVE ACTION                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12. Troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 16.                                                          |                                                                                                                                                                                                                       |
| 13. Remove jumper wire. Go to 14.                                                                                                 |                                                                                                                                                                                                                       |
| 14. Check for 28 VDC between P409-3 and ground.                                                                                   | <p>Step 1. If voltage is as specified, repair/replace wiring between P409-3 and P117-16 (Appendix F). Go to 16.</p> <p>Step 2. If voltage is not as specified, go to 15.</p>                                          |
| 15. Check for 28 VDC between terminal 1 (MAIN XMSN circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel) and ground. | <p>Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and P409-2 (Appendix F). Go to 16.</p> <p>Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 16.</p> |
| 16. Procedure completed.                                                                                                          |                                                                                                                                                                                                                       |

TABLE NO. 6-2-2-10. XMSN PRESS Indication Is Too Low (MAIN XMSN OIL PRESS Capsule Goes On).

| TEST OR INSPECTION                                                                          | CORRECTIVE ACTION                                                                                                                                 |
|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Check main transmission oil for proper oil level using dipstick.                         | <p>Step 1. If oil level is good, go to 2.</p> <p>Step 2. If oil level is not good, properly service main transmission (PARA 1-3-12). Go to 4.</p> |
| 2. Inspect screens and replace oil filter (PARA 6-4-19). Go to 3.                           |                                                                                                                                                   |
| 3. If pressure is not good, replace both pressure regulating valves (PARA 6-4-22). Go to 4. |                                                                                                                                                   |
| 4. Procedure completed.                                                                     |                                                                                                                                                   |

TABLE NO. 6-2-2-11. XMSN TEMP Indication Is Too High Or Is Not Steady (MAIN XMSN OIL TEMP Capsule Goes On).

| TEST OR INSPECTION                                                                        | CORRECTIVE ACTION                                                                                                                                                               |
|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Check main module and other modules for any discoloration.                             | Step 1. If discoloration is present, replace affected module (PARA 6-4-25 or PARA 6-4-28). Go to 6.<br>Step 2. If modules are good, go to 2.                                    |
| 2. Check main gear box for proper oil level.                                              | Step 1. If oil level is good, go to 3.<br>Step 2. If oil level is not good, service main gear box (PARA 1-3-12). Go to 6.                                                       |
| 3. Troubleshoot XMSN TEMP scale on central display unit (PARA 8-2-3). Go to 4.            |                                                                                                                                                                                 |
| 4. If Instrument Display System is good, go to 5.                                         |                                                                                                                                                                                 |
| 5. Remove thermostatic valve from transmission oil cooler and check for proper operation. | Step 1. If operation is proper, replace transmission oil cooler (PARA 6-4-50). Go to 6.<br>Step 2. If operation is improper, replace thermostatic valve (PARA 6-4-48). Go to 6. |
| 6. Procedure completed.                                                                   |                                                                                                                                                                                 |

TABLE NO. 6-2-2-12. INT XMSN OIL TEMP Capsule Goes On.

| TEST OR INSPECTION                                                                      | CORRECTIVE ACTION                                                                                                                                                           |
|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Visually check intermediate gear box for signs of paint blistering or discoloration. | Step 1. If blistering or discoloration is present, perform a serviceability check of intermediate gear box (PARA 6-3-12). Go to 8.<br>Step 2. If gear box is good, go to 2. |

TABLE NO. 6-2-2-12. INT XMSN OIL TEMP Capsule Goes On. (Cont)

| TEST OR INSPECTION                                                                 | CORRECTIVE ACTION                                                                                                                                           |
|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2. Visually check intermediate gear box for proper oil level.                      | <p>Step 1. If oil level is good, go to 3.</p> <p>Step 2. If oil level is not good, properly service intermediate gear box (PARA 1-3-13). Go to 8.</p>       |
| 3. Disconnect P613 from intermediate gear box. Go to 4.                            |                                                                                                                                                             |
| 4. Check that INT XMSN OIL TEMP capsule goes off.                                  | <p>Step 1. If capsule goes off, go to 5.</p> <p>Step 2. If capsule does not go off, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 8.</p> |
| 5. Replace intermediate gear box oil temperature sensor (PARA 6-4-37). Go to 6.    |                                                                                                                                                             |
| 6. Reconnect P613 to sensor. Go to 7.                                              |                                                                                                                                                             |
| 7. Perform a serviceability check of intermediate gear box (PARA 6-3-12). Go to 8. |                                                                                                                                                             |
| 8. Procedure completed.                                                            |                                                                                                                                                             |

TABLE NO. 6-2-2-13. TAIL XMSN OIL TEMP Capsule Goes On.

| TEST OR INSPECTION                                                                    | CORRECTIVE ACTION                                                                                                                                                                    |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Visually check tail rotor gear box for signs of paint blistering or discoloration. | <p>Step 1. If blistering or discoloration is present, perform a serviceability check of tail rotor gear box (PARA 6-3-12). Go to 8.</p> <p>Step 2. If gear box is good, go to 2.</p> |
| 2. Visually check tail rotor gear box for proper oil level.                           | <p>Step 1. If oil level is good, go to 3.</p>                                                                                                                                        |



**TABLE NO. 6-2-2-13. TAIL XMSN OIL TEMP Capsule Goes On. (Cont)**

| <b>TEST OR INSPECTION</b> | <b>CORRECTIVE ACTION</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>Step 2. If oil level is not good, properly service tail rotor gear box (PARA 1-3-14).<br/>Go to <b>8</b>.</p> <p><b>3. Disconnect P614 from tail gear box. Go to 4.</b></p> <p><b>4. Check that TAIL XMSN OIL TEMP capsule goes off.</b></p> <p>Step 1. If capsule goes off, go to <b>5</b>.<br/>Step 2. If capsule is on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to <b>8</b>.</p> <p><b>5. Replace tail gear box chip detector (PARA 6-4-59).</b></p> <p><b>6. Reconnect P614 to chip detector. Go to 7.</b></p> <p><b>7. Perform a serviceability check of intermediate gear box (PARA 6-3-18). Go to 8. █</b></p> <p><b>8. Procedure completed.</b></p> |

**TABLE NO. 6-2-2-14. XMSN PRESS Has No Indication, Not Steady, Or Too High.**

| <b>TEST OR INSPECTION</b> | <b>CORRECTIVE ACTION</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p><b>1. Check main module oil filter and screens between input and accessory modules for possible contamination.</b></p> <p>Step 1. If contamination is present, perform a serviceability check of affected modules or inputs (PARA 6-3-5). Go to <b>9</b>.<br/>Step 2. If contamination is not present, go to <b>2</b>.</p> <p><b>2. Check main transmission oil for proper level using dipstick.</b></p> <p>Step 1. If level is good, go to <b>3</b>.<br/>Step 2. If level is not good, properly service main transmission (PARA 1-3-12).<br/>Go to <b>9</b>.<br/>Step 3. If oil level is high, drain oil until oil is at proper level (PARA 1-3-12).<br/>Go to <b>9</b>.</p> |

TABLE NO. 6-2-2-14. XMSN PRESS Has No Indication, Not Steady, Or Too High. (Cont)

---

**TEST OR INSPECTION  
CORRECTIVE ACTION**

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3. Troubleshoot XMSN PRESS scale (central display unit) (PARA 8-2-3). Go to 4.

4. If Instrument Display System is good, go to 5.

5. Check helicopter configuration.

Step 1. **EMEP** , go to 6.

Step 2. **W/O EMEP** , go to 8.

**EMEP**

6. Remove and check PFA P137 (PARA 9-4-98).

Step 1. If PFA is good, go to 7.

Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to 9.

**EMEP**

7. Install PFA (PARA 9-4-98). Go to 8.

8. Check continuity between:

P137-T and P444-3

P137-W and P444-2

P137-j and P444-1

Step 1. If continuity is present, replace transmission oil pressure sensor (PARA 6-4-21). Go to 9.

Step 2. If trouble remains, replace pressure regulating valves (PARA 6-4-22).  
Go to 9.

Step 3. If trouble still remains, replace packings on both oil pumps (PARA 6-4-20).  
Go to 9.

Step 4. If continuity is not present, repair/replace wiring as required (Appendix F).  
Go to 8.

9. Procedure completed.

---

TABLE NO. 6-2-2-15. XMSN TEMP Has No Indication.

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**TEST OR INSPECTION  
CORRECTIVE ACTION**

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1. Make sure connector P412 is connected to main gear box.

TABLE NO. 6-2-2-15. XMSN TEMP Has No Indication. (Cont)

|             | TEST OR INSPECTION<br>CORRECTIVE ACTION                                                                                                                                                                                                                                          |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | <p>Step 1. If receptacle is connected, go to 2.</p> <p>Step 2. If receptacle is not connected, connect receptacle to main gear box. Go to 8.</p>                                                                                                                                 |
|             | 2. Troubleshoot XMSN TEMP scale (central display unit) (PARA 8-2-3). Go to 3.                                                                                                                                                                                                    |
|             | 3. If Instrument Display System is good, go to 4.                                                                                                                                                                                                                                |
|             | 4. Check helicopter configuration.                                                                                                                                                                                                                                               |
|             | <p>Step 1. <b>EMEP</b> , go to 5.</p> <p>Step 2. <b>W/O EMEP</b> , go to 7.</p>                                                                                                                                                                                                  |
| <b>EMEP</b> | <p>5. Remove and check PFA P137 (PARA 9-4-98). <span style="float: right;"> </span></p> <p>Step 1. If PFA is good, go to 6. <span style="float: right;"> </span></p> <p>Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to 8. <span style="float: right;"> </span></p> |
| <b>EMEP</b> | 6. Install PFA (PARA 9-4-98). Go to 7. <span style="float: right;"> </span>                                                                                                                                                                                                      |
|             | <p>7. Check continuity between:</p> <p style="padding-left: 40px;">P137-J and P412-C</p> <p style="padding-left: 40px;">P137-K and P412-D</p>                                                                                                                                    |
|             | <p>Step 1. If continuity is present, replace transmission oil strainer (PARA 6-4-18).<br/>Go to 8.</p> <p>Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to 8.</p>                                                                                 |
|             | 8. Procedure completed.                                                                                                                                                                                                                                                          |

TABLE NO. 6-2-2-16. %RPM RTR Indication Is Not Steady Or Has No Indication.

|  | TEST OR INSPECTION<br>CORRECTIVE ACTION                             |
|--|---------------------------------------------------------------------|
|  | 1. Troubleshoot %RPM RTR scale (central display unit) (PARA 8-2-3). |
|  | 2. If Instrument Display System is good, go to 3.                   |
|  | 3. Check helicopter configuration.                                  |

TABLE NO. 6-2-2-16. %RPM RTR Indication Is Not Steady Or Has No Indication. (Cont)

---

| TEST OR INSPECTION | CORRECTIVE ACTION |
|--------------------|-------------------|
|--------------------|-------------------|

---

- Step 1. **EMEP** , go to 4.  
 Step 2. **W/O EMEP** , go to 6.

- |             |                                                       |
|-------------|-------------------------------------------------------|
| <b>EMEP</b> | 4. Remove and check PFAs P137 and P138 (PARA 9-4-98). |
|-------------|-------------------------------------------------------|

- Step 1. If PFAs are good, go to 5.  
 Step 2. If PFAs are not good, replace PFA as required (PARA 9-4-98). Go to 7.

- |             |                                         |
|-------------|-----------------------------------------|
| <b>EMEP</b> | 5. Install PFAs (PARA 9-4-98). Go to 6. |
|-------------|-----------------------------------------|

6. Check continuity between:  
     P443-1 and P137-B  
     P443-1 and P138-B  
     P443-2 and P137-A  
     P443-2 and P138-A

- Step 1. If continuity is present, replace Nr speed sensor (PARA 6-4-28). Go to 7.  
 Step 2. If continuity is not present, repair/replace wiring as required (Appendix F).  
 Go to 7.

7. Procedure completed.
- 

TABLE NO. 6-2-2-17. MAIN XMSN OIL TEMP Capsule Does Not Go On.

---

| TEST OR INSPECTION | CORRECTIVE ACTION |
|--------------------|-------------------|
|--------------------|-------------------|

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1. Place and hold BRT/DIM-TEST switch (caution/advisory panel) to TEST.  
 Go to 2.

2. Check that MAIN XMSN OIL TEMP capsule goes on.

- Step 1. If capsule goes on, go to 3.  
 Step 2. If capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 16.

3. Install an insulated jumper wire between P408-A and P408-C. Go to 4.

TABLE NO. 6-2-2-17. MAIN XMSN OIL TEMP Capsule Does Not Go On. (Cont)

|      | TEST OR INSPECTION<br>CORRECTIVE ACTION                                                                                                                           |  |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|      | 4. Check that MAIN XMSN OIL TEMP capsule goes on.                                                                                                                 |  |
|      | Step 1. If capsule goes on, go to 5.<br>Step 2. If capsule does not go on, go to 7.                                                                               |  |
|      | 5. Remove jumper wire. Go to 6.                                                                                                                                   |  |
|      | 6. Replace transmission oil temperature switch (PARA 6-4-24) Go to 16.                                                                                            |  |
|      | 7. Check helicopter configuration.                                                                                                                                |  |
|      | Step 1. EMEP , go to 8.<br>Step 2. W/O EMEP , go to 10.                                                                                                           |  |
| EMEP | 8. Remove and check PFA P117 (PARA 9-4-98).                                                                                                                       |  |
|      | Step 1. If PFA is good, go to 9.<br>Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to 16.                                                              |  |
| EMEP | 9. Install PFA (PARA 9-4-98). Go to 10.                                                                                                                           |  |
|      | 10. Check for 28 VDC between P117-10 and ground.                                                                                                                  |  |
|      | Step 1. If voltage is as specified, go to 11.<br>Step 2. If voltage is not as specified, go to 13.                                                                |  |
|      | 11. Remove jumper wire. Go to 12.                                                                                                                                 |  |
|      | 12. Troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 16.                                                                                          |  |
|      | 13. Remove jumper wire. Go to 14.                                                                                                                                 |  |
|      | 14. Check for 28 VDC between P408-A and ground.                                                                                                                   |  |
|      | Step 1. If voltage is as specified, repair/replace wiring between P408-C and P117-10 (Appendix F). Go to 16.<br>Step 2. If voltage is not as specified, go to 15. |  |
|      | 15. Check for 28 VDC between terminal 1 (MAIN XMSN circuit breaker, NO. 2 DC PRI BUS - pilot's circuit breaker panel) and ground.                                 |  |

TABLE NO. 6-2-2-17. MAIN XMSN OIL TEMP Capsule Does Not Go On. (Cont)

| TEST OR INSPECTION              | CORRECTIVE ACTION                                                                                                                                                                                                                   |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | <p>Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and P408-A (Appendix F). Go to <b>16</b>.</p> <p>Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to <b>16</b>.</p> |
| <b>16. Procedure completed.</b> |                                                                                                                                                                                                                                     |



TABLE NO. 6-2-2-18. INT XMSN OIL TEMP Capsule Does Not Go On.

| TEST OR INSPECTION | CORRECTIVE ACTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | <p>1. Place and hold BRT/DIM-TEST switch (caution/advisory panel) to TEST. Go to 2.</p> <p>2. Check that INT XMSN OIL TEMP capsule goes on.</p> <p>Step 1. If capsule goes on, go to 3.</p> <p>Step 2. If capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 17.</p> <p>3. Install an insulated jumper wire between P613-B and P613-C. Go to 4.</p> <p>4. Check that INT XMSN OIL TEMP capsule goes on.</p> <p>Step 1. If capsule goes on, go to 5.</p> <p>Step 2. If capsule does not go on, go to 7.</p> <p>5. Remove jumper wire. Go to 6.</p> <p>6. Replace intermediate gear box chip detector/temperature sensor (PARA 6-4-37). Go to 17.</p> <p>7. Remove jumper wire. Go to 8.</p> <p>8. Check helicopter configuration.</p> <p>Step 1. <b>EMEP</b> , go to 9.</p> <p>Step 2. <b>W/O EMEP</b> , go to 11.</p> |

TABLE NO. 6-2-2-18. INT XMSN OIL TEMP Capsule Does Not Go On. (Cont)

| TEST OR INSPECTION<br>CORRECTIVE ACTION |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EMEP</b>                             | <p><b>9. Remove and check PFA P117 (PARA 9-4-98).</b></p> <p>Step 1. If PFA is good, go to <b>10</b>.</p> <p>Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to <b>17</b>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>EMEP</b>                             | <p><b>10. Install PFA (PARA 9-4-98). Go to 11.</b></p> <p><b>11. Check for 28 VDC between P118-f and ground.</b></p> <p>Step 1. If voltage is as specified, go to <b>12</b>.</p> <p>Step 2. If voltage is not as specified, go to <b>16</b>.</p> <p><b>12. Check helicopter configuration.</b></p> <p>Step 1. <b>EMEP</b> , go to <b>13</b>.</p> <p>Step 2. <b>W/O EMEP</b> , go to <b>15</b>.</p>                                                                                                                                                                                                                                                                                                                                                                            |
| <b>EMEP</b>                             | <p><b>13. Remove and check PFA P117 (PARA 9-4-98).</b></p> <p>Step 1. If PFA is good, go to <b>14</b>.</p> <p>Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to <b>17</b>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>EMEP</b>                             | <p><b>14. Install PFA (PARA 9-4-98). Go to 15.</b></p> <p><b>15. Check continuity between:</b><br/> <b>P613-B and ground</b><br/> <b>P613-C and P117-26</b></p> <p>Step 1. If continuity is present, replace caution/advisory panel (PARA 8-4-19). Go to <b>17</b>.</p> <p>Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to <b>17</b>.</p> <p><b>16. Check for 28 VDC between terminal 1 (CHIP DET circuit breaker, DC ESNTL BUS - copilot's side upper console circuit breaker panel) and ground.</b></p> <p>Step 1. If voltage is as specified, repair/replace wiring between terminal 1 and P118-f (Appendix F). Go to <b>17</b>.</p> <p>Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to <b>17</b>.</p> |

TABLE NO. 6-2-2-18. INT XMSN OIL TEMP Capsule Does Not Go On. (Cont)

| TEST OR INSPECTION       | CORRECTIVE ACTION |
|--------------------------|-------------------|
| 17. Procedure completed. |                   |

TABLE NO. 6-2-2-19. TAIL XMSN OIL TEMP Capsule Does Not Go On.

| TEST OR INSPECTION                                                                  | CORRECTIVE ACTION                                                                                                                                          |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Place and hold BRT/DIM-TEST switch (caution/advisory panel) to TEST.<br>Go to 2. |                                                                                                                                                            |
| 2. Check that TAIL XMSN OIL TEMP capsule goes on.                                   | <p>Step 1. If capsule goes on, go to 3.</p> <p>Step 2. If capsule does not go on, troubleshoot Caution/Advisory Warning System (PARA 8-2-4). Go to 12.</p> |
| 3. Install an insulated jumper wire between P614-B and P614-C. Go to 4.             |                                                                                                                                                            |
| 4. Check that TAIL XMSN OIL TEMP capsule goes on.                                   | <p>Step 1. If capsule goes on, go to 5.</p> <p>Step 2. If capsule does not go on, go to 7.</p>                                                             |
| 5. Remove jumper wire. Go to 6.                                                     |                                                                                                                                                            |
| 6. Replace tail gear box chip detector/temperature sensor (PARA 6-4-59). Go to 12.  |                                                                                                                                                            |
| 7. Remove jumper wire. Go to 8.                                                     |                                                                                                                                                            |
| 8. Check helicopter configuration.                                                  | <p>Step 1. <b>EMEP</b> , go to 9.</p> <p>Step 2. <b>W/O EMEP</b> , go to 11.</p>                                                                           |
| 9. Remove and check PFA P117 (PARA 9-4-98).                                         | <p>Step 1. If PFA is good, go to 10.</p> <p>Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to 12.</p>                                           |



TABLE NO. 6-2-2-19. TAIL XMSN OIL TEMP Capsule Does Not Go On. (Cont)

| TEST OR INSPECTION<br>CORRECTIVE ACTION |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EMEP</b>                             | <p><b>10. Install PFA (PARA 9-4-98). Go to 11.</b></p> <p><b>11. Check continuity between:</b><br/> <b>P614-B and ground</b><br/> <b>P614-C and P117-42</b></p> <p>Step 1. If continuity is present, replace caution/advisory panel (PARA 8-4-19).<br/> Go to <b>12.</b></p> <p>Step 2. If continuity is not present, repair/replace wiring as required (Appendix F).<br/> Go to <b>12.</b></p> <p><b>12. Procedure completed.</b></p> |

**NOTE**

If CHIP DET circuit breaker has popped, go to TABLE NO. 6-2-2-2.

TABLE NO. 6-2-2-20. CHIP ACCESS MDL-RH Capsule Goes On.

| TEST OR INSPECTION<br>CORRECTIVE ACTION |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | <p><b>1. Remove chip detector from right accessory module (PARA 6-4-15). Go to 2.</b></p> <p><b>2. Check chip detector.</b></p> <p>Step 1. If small amounts of fuzz are found, go to <b>TABLE NO. 6-2-2-25.</b></p> <p>Step 2. If chips are found, perform a serviceability check of right accessory module (PARA 6-3-9). Go to <b>9.</b></p> <p>Step 3. If no chips or fuzz are found, go to <b>3.</b></p> <p><b>3. Disconnect P117 (caution/advisory panel). Go to 4.</b></p> <p><b>4. Check that CHIP ACCESS MDL-RH capsule goes off.</b></p> <p>Step 1. If capsule is off, go to <b>5.</b></p> <p>Step 2. If capsule is on, replace caution/advisory panel (PARA 8-4-19). Go to <b>9.</b></p> <p><b>5. Check helicopter configuration.</b></p> |

TABLE NO. 6-2-2-20. CHIP ACCESS MDL-RH Capsule Goes On. (Cont)

---

**TEST OR INSPECTION  
CORRECTIVE ACTION**

---

- Step 1. **EMEP** , go to 6.  
Step 2. **W/O EMEP** , go to 8.

**| EMEP**

- 6. Remove and check PFA P117 (PARA 9-4-98).**

**|**  
**|**

- Step 1. If PFA is good, go to 7.  
Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to 9.

**| EMEP**

- 7. Install PFA (PARA 9-4-98). Go to 8.**

- 8. Repair/replace wiring between P117-62 and P405-A (Appendix F). Go to 9.**

- 9. Procedure completed.**
- 

**NOTE**

If CHIP DET circuit breaker has popped, go to TABLE NO. 6-2-2-2.

TABLE NO. 6-2-2-21. CHIP ACCESS MDL-LH Capsule Goes On.

---

**TEST OR INSPECTION  
CORRECTIVE ACTION**

---

- 1. Remove chip detector from left accessory module (PARA 6-4-15). Go to 2.**

- 2. Check chip detector.**

- Step 1. If small amounts of fuzz are found, go to **TABLE NO. 6-2-2-25**.  
Step 2. If no fuzz or chips are found, go to 3.

- 3. Disconnect P117 (caution/advisory panel). Go to 4.**

- 4. Check that CHIP ACCESS MDL-LH capsule goes off.**

- Step 1. If capsule is off, go to 5.  
Step 2. If capsule is on, replace caution/advisory panel (PARA 8-4-19). Go to 9.

- 5. Check helicopter configuration.**

TABLE NO. 6-2-2-21. CHIP ACCESS MDL-LH Capsule Goes On. (Cont)

| TEST OR INSPECTION<br>CORRECTIVE ACTION                              |                                                                                                                                                         |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Step 1. <b>EMEP</b> , go to 6.<br>Step 2. <b>W/O EMEP</b> , go to 8. |                                                                                                                                                         |
| <b>EMEP</b>                                                          | 6. Remove and check PFA P117 (PARA 9-4-98).<br><br>Step 1. If PFA is good, go to 7.<br>Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to 9.  |
| <b>EMEP</b>                                                          | 7. Install PFA (PARA 9-4-98). Go to 8.<br><br>8. Repair/replace wiring between P117-14 and P406-A (Appendix F). Go to 9.<br><br>9. Procedure completed. |

NOTE

If CHIP DET circuit breaker has popped, go to TABLE NO. 6-2-2-2.

TABLE NO. 6-2-2-22. CHIP INPUT MDL-LH Capsule Goes On.

| TEST OR INSPECTION<br>CORRECTIVE ACTION                                                                                                                                                                                                                                            |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 1. Remove left input module chip detector from main sump (PARA 6-4-17).<br>Go to 2.                                                                                                                                                                                                |  |
| 2. Check chip detector.<br><br>Step 1. If small amounts of fuzz are found, go to <b>TABLE NO. 6-2-2-25</b> .<br>Step 2. If chips are found, perform a serviceability check of left input module (PARA 6-3-8). Go to <b>10</b> .<br>Step 3. If no chips or fuzz are found, go to 3. |  |
| 3. Visually check oil filter screen for chip(s).<br><br>Step 1. If chips are found, perform a serviceability check of left input module (PARA 6-3-8). Go to <b>10</b> .<br>Step 2. If chips are not found, go to 4.                                                                |  |
| 4. Disconnect P117 (caution/advisory panel). Go to 5.                                                                                                                                                                                                                              |  |

TABLE NO. 6-2-2-22. CHIP INPUT MDL-LH Capsule Goes On. (Cont)

---

**TEST OR INSPECTION  
CORRECTIVE ACTION**

---

**5. Check that CHIP INPUT MDL-LH capsule goes off.**Step 1. If capsule is off, go to **6**.Step 2. If capsule is on, replace caution/advisory panel (PARA 8-4-19). Go to **10**.**6. Check helicopter configuration.**Step 1. **EMEP**, go to **7**.Step 2. **W/O EMEP**, go to **9**.**EMEP****7. Remove and check PFA P117 (PARA 9-4-98).**Step 1. If PFA is good, go to **8**.Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to **10**.**EMEP****8. Install PFA (PARA 9-4-98). Go to 9.****9. Repair/replace wiring between P117-13 and P403-A (Appendix F). Go to 10.****10. Procedure completed.****NOTE**

If CHIP DET circuit breaker has popped, go to TABLE NO. 6-2-2-2.

TABLE NO. 6-2-2-23. CHIP INPUT MDL-RH Capsule Goes On.

---

**TEST OR INSPECTION  
CORRECTIVE ACTION**

---

**1. Remove right input module chip detector from main sump (PARA 6-4-17).  
Go to 2.****2. Check chip detector.**Step 1. If small amounts of fuzz are found, go to **TABLE NO. 6-2-2-25**.Step 2. If chips are found, perform a serviceability check of right input module (PARA 6-3-8). Go to **10**.

TABLE NO. 6-2-2-23. CHIP INPUT MDL-RH Capsule Goes On. (Cont)

| TEST OR INSPECTION | CORRECTIVE ACTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | <p>Step 3. If no chips or fuzz are found, go to 3.</p> <p><b>3. Visually check oil filter screen for chip(s).</b></p> <p>Step 1. If chips are found, perform a serviceability check of right input module (PARA 6-3-8). Go to 10.</p> <p>Step 2. If chips are not found, go to 4.</p> <p><b>4. Disconnect P117 (caution/advisory panel). Go to 5.</b></p> <p><b>5. Check that CHIP INPUT MDL-RH capsule goes off.</b></p> <p>Step 1. If capsule is off, go to 6.</p> <p>Step 2. If capsule is on, replace caution/advisory panel (PARA 8-4-19). Go to 10.</p> <p><b>6. Check helicopter configuration.</b></p> <p>Step 1. <b>EMEP</b>, go to 7.</p> <p>Step 2. <b>W/O EMEP</b>, go to 9.</p> <p><b>EMEP</b> <b>7. Remove and check PFA P117 (PARA 9-4-98).</b></p> <p>Step 1. If PFA is good, go to 8.</p> <p>Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to 10.</p> <p><b>EMEP</b> <b>8. Install PFA (PARA 9-4-98). Go to 9.</b></p> <p><b>9. Repair/replace wiring between P117-61 and P404-A (Appendix F). Go to 10.</b></p> <p><b>10. Procedure completed.</b></p> |

**NOTE**

If CHIP DET circuit breaker has popped, go to TABLE NO. 6-2-2-2.

**TABLE NO. 6-2-2-24. CHIP MAIN MDL SUMP Capsule Is On.**

---

**TEST OR INSPECTION  
CORRECTIVE ACTION**

---

**1. Remove chip detector from main module sump (PARA 6-4-18). Go to 2.**

**2. Check chip detector.**

Step 1. If small amounts of fuzz are found, go to **TABLE NO. 6-2-2-25**.

Step 2. If chips are found, perform a serviceability check of main module (PARA 6-3-5). Go to **9**.

Step 3. If no chips or fuzz are found, go to **3**.

**3. Disconnect P117 (caution/advisory panel). Go to 4.**

**4. Check that CHIP MAIN MDL SUMP capsule goes off.**

Step 1. If capsule is off, go to **5**.

Step 2. If capsule is on, replace caution/advisory panel (PARA 8-4-19). Go to **9**.

**5. Check helicopter configuration.**

Step 1. **EMEP**, go to **6**.

Step 2. **W/O EMEP**, go to **8**.

**6. Remove and check PFA P117 (PARA 9-4-98).**

Step 1. If PFA is good, go to **7**.

Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to **9**.

**7. Install PFA (PARA 9-4-98). Go to 8.**

**8. Repair/replace wiring between P117-30 and P412-A (Appendix F). Go to 9.**

**9. Procedure completed.**

---

**TABLE NO. 6-2-2-25. Chip Detector Does Not Burn Off Fuzz.**

| TEST OR INSPECTION<br>CORRECTIVE ACTION                                                                                                                                                                                  |          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>1. Is right accessory module chip detector good?</b>                                                                                                                                                                  |          |
| Step 1. If chip detector is good, go to <b>2</b> .<br>Step 2. If chip detector is not good, go to <b>9</b> .                                                                                                             |          |
| <b>2. Is left accessory module chip detector good?</b>                                                                                                                                                                   |          |
| Step 1. If chip detector is good, go to <b>3</b> .<br>Step 2. If chip detector is not good, go to <b>8</b> .                                                                                                             |          |
| <b>3. Is left input module chip detector good?</b>                                                                                                                                                                       |          |
| Step 1. If chip detector is good, go to <b>4</b> .<br>Step 2. If chip detector is not good, go to <b>7</b> .                                                                                                             |          |
| <b>4. Is right input module chip detector good?</b>                                                                                                                                                                      |          |
| Step 1. If chip detector is good, go to <b>5</b> .<br>Step 2. If chip detector is not good, replace chip detector (PARA 6-4-17). Go to <b>6</b> .                                                                        |          |
| <b>5. Check for 28 VDC between P412-A and ground.</b>                                                                                                                                                                    | <b>I</b> |
| Step 1. If voltage is as specified, replace main module sump chip detector (PARA 6-4-18). Go to <b>10</b> .<br>Step 2. If voltage is not as specified, replace caution/advisory panel (PARA 8-4-19). Go to <b>10</b> .   |          |
| <b>6. Check for 28 VDC between P404-A and ground.</b>                                                                                                                                                                    | <b>I</b> |
| Step 1. If voltage is as specified, replace right input module chip detector (PARA 6-4-17). Go to <b>10</b> .<br>Step 2. If voltage is not as specified, replace caution/advisory panel (PARA 8-4-19). Go to <b>10</b> . |          |
| <b>7. Check for 28 VDC between P403-A and ground.</b>                                                                                                                                                                    | <b>I</b> |
| Step 1. If voltage is as specified, replace left input module chip detector (PARA 6-4-17). Go to <b>10</b> .<br>Step 2. If voltage is not as specified, replace caution/advisory panel (PARA 8-4-19). Go to <b>10</b> .  |          |

TABLE NO. 6-2-2-25. Chip Detector Does Not Burn Off Fuzz. (Cont)

| TEST OR INSPECTION | CORRECTIVE ACTION                                                                                                                                                                                                                                                                                  |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | <p><b>8. Check for 28 VDC between P406-A and ground.</b></p> <p>Step 1. If voltage is as specified, replace left accessory module chip detector (PARA 6-4-15). Go to <b>10</b>.</p> <p>Step 2. If voltage is not as specified, replace caution/advisory panel (PARA 8-4-19). Go to <b>10</b>.</p>  |
|                    | <p><b>9. Check for 28 VDC between P405-A and ground.</b></p> <p>Step 1. If voltage is as specified, replace right accessory module chip detector (PARA 6-4-15). Go to <b>10</b>.</p> <p>Step 2. If voltage is not as specified, replace caution/advisory panel (PARA 8-4-19). Go to <b>10</b>.</p> |
|                    | <p><b>10. Procedure completed.</b></p>                                                                                                                                                                                                                                                             |

TABLE NO. 6-2-2-26. Oil Filter Bypass Button Extended.

| TEST OR INSPECTION | CORRECTIVE ACTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | <p><b>1. Refer to maintenance log book.</b></p> <p>Step 1. If first stage filter element has been replaced more than three times within 200 hours with no oil changes, replace filter bowl assembly (PARA 6-4-19). Go to <b>2</b>.</p> <p>Step 2. If maintenance has been performed on the main transmission that caused oil to be drained, exposed to a non-purified container, or exposed to dust, replace filter bowl assembly (PARA 6-4-19). Go to <b>2</b>.</p> <p>Step 3. If the above steps have not occurred, perform a serviceability check (PARA 6-3-5). Go to <b>2</b>.</p> |
|                    | <p><b>2. Procedure completed.</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |



**TABLE NO. 6-2-2-27. Pressure Suddenly Drops More Than 10 PSI Without Fluctuations.**

| <b>TEST OR INSPECTION<br/>CORRECTIVE ACTION</b>                                                                                                                                            | <b>NOTE</b>                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                            | If situation permits, verify that pressure drop is not an instrumentation problem using a direct reading pressure gage.                                                                                                                                                          |
| <b>1. Inspect sump housing for oil leakage and/or cracks in the core wall (circular depression in the sump housing between pumps and filter bowl).</b>                                     | <p>Step 1. If oil leakage and/or cracks in the core wall are visible, replace main module. (PARA 6-4-1) Go to <b>5</b>.</p> <p>Step 2. If oil leakage and/or cracks in the core wall are not visible, go to <b>2</b>.</p>                                                        |
| <b>2. Is pressure within or below green range after sudden pressure drop?</b>                                                                                                              | <p>Step 1. Check chip detectors, strainers and screens for contamination in main, input, and accessory modules (PARA 6-3-6).</p> <p>Step 2. If pressure is within green range, go to <b>3</b>.</p> <p>Step 3. If pressure is below green range, go to <b>TABLE 6-2-2-10</b>.</p> |
|                                                                                                                                                                                            | <p><b>NOTE</b></p> <p>Stop running system if pressure drops below green range.</p>                                                                                                                                                                                               |
| <b>3. Perform a one-hour ground run (Appropriate Flight Manual) and observe XMSN PRESS. Record pressure every 5 minutes and any sudden pressure drops. Did pressure drop below 30 psi?</b> | <p>Step 1. If pressure has dropped below 30 psi, replace main module (PARA 6-4-1). Go to <b>5</b>.</p> <p>Step 2. If pressure has not dropped below 30 psi, go to <b>4</b>.</p>                                                                                                  |
| <b>4. Inspect sump housing for oil leakage and/or cracks in the core wall.</b>                                                                                                             | <p>Step 1. If oil leakage and/or cracks in the core wall are visible, replace main module (PARA 6-4-1). Go to <b>5</b>.</p> <p>Step 2. If oil leakage and/or cracks in the core wall are not visible. Go to <b>TABLE 6-2-2-10</b>.</p>                                           |
| <b>5. Procedure completed.</b>                                                                                                                                                             |                                                                                                                                                                                                                                                                                  |

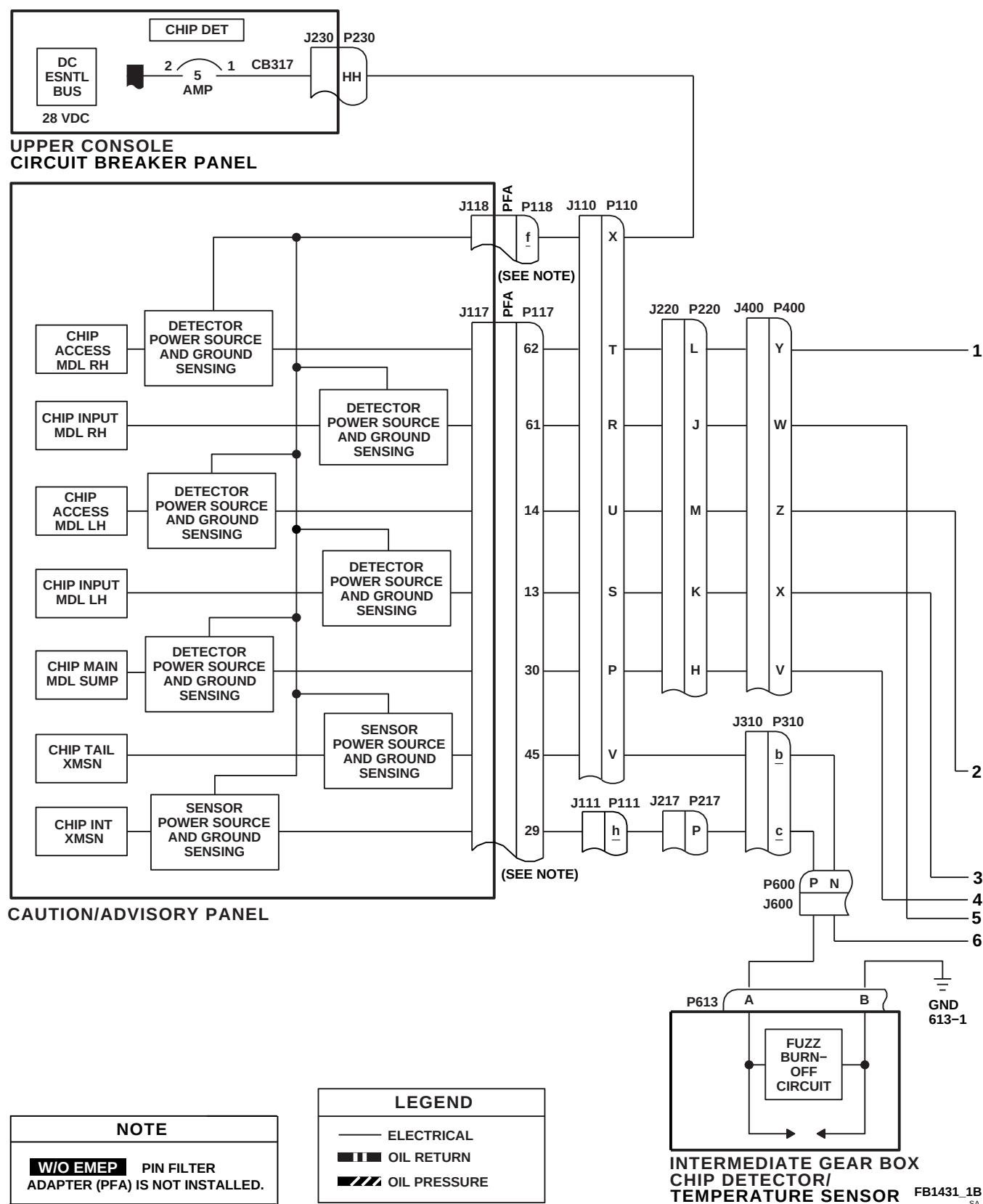


FIGURE 6-2-2-1. TRANSMISSION CHIP DETECTOR SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 2)

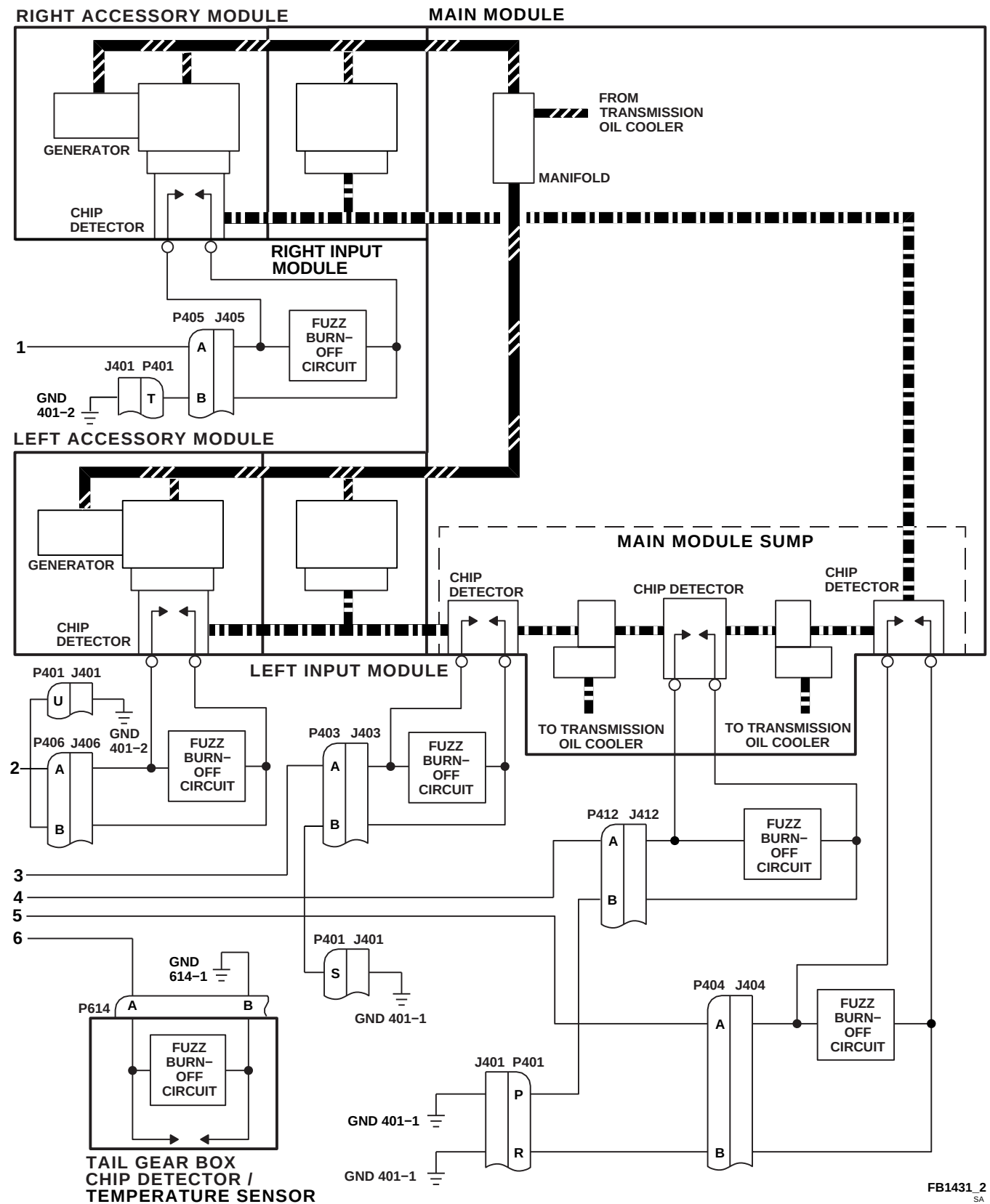
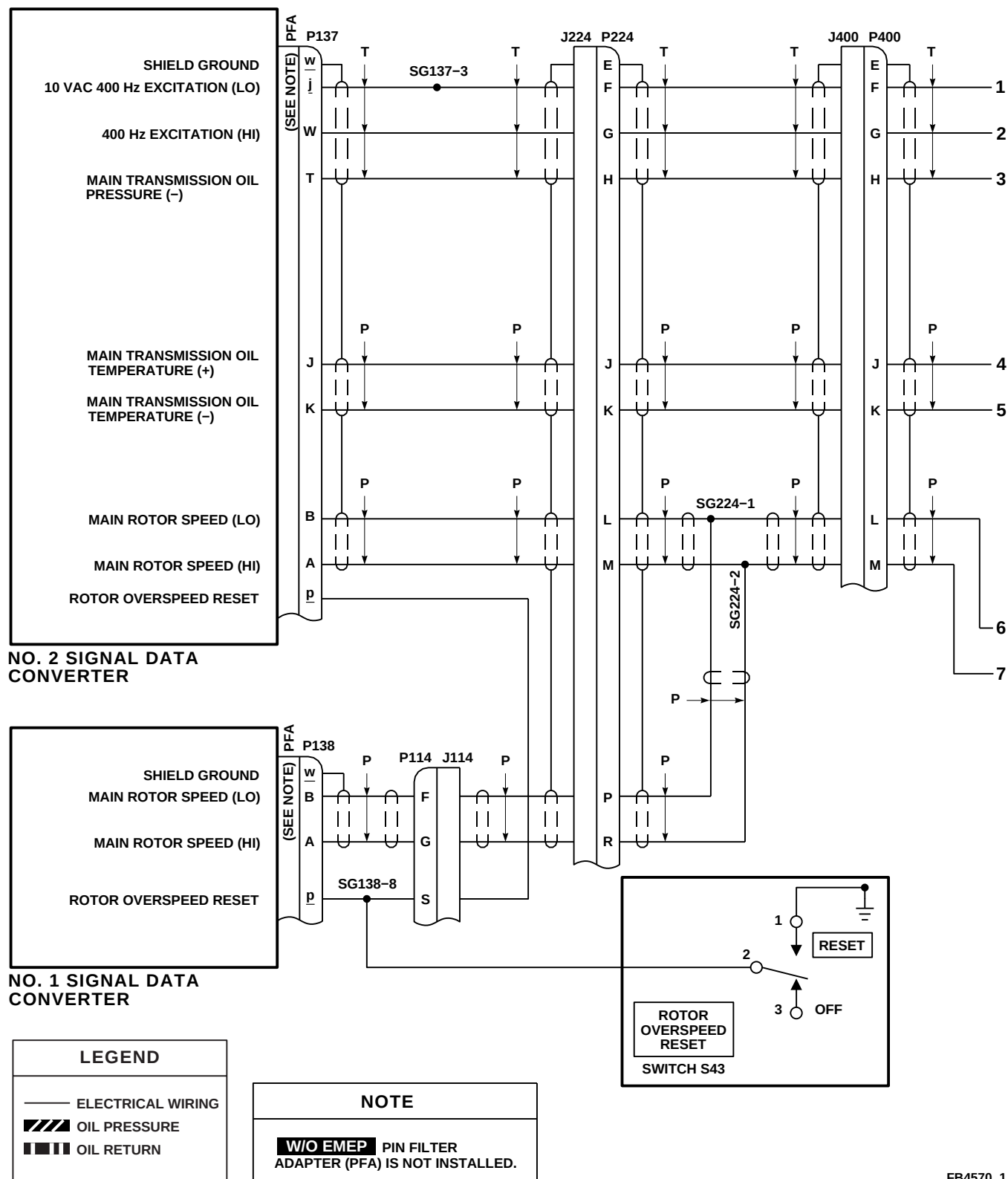


FIGURE 6-2-2-1. TRANSMISSION CHIP DETECTOR SYSTEM SCHEMATIC DIAGRAM  
(SHEET 2 OF 2)



FB4570\_1  
SA

FIGURE 6-2-2-2. TRANSMISSION INSTRUMENT SYSTEM SCHEMATIC DIAGRAM  
(SHEET 1 OF 2)

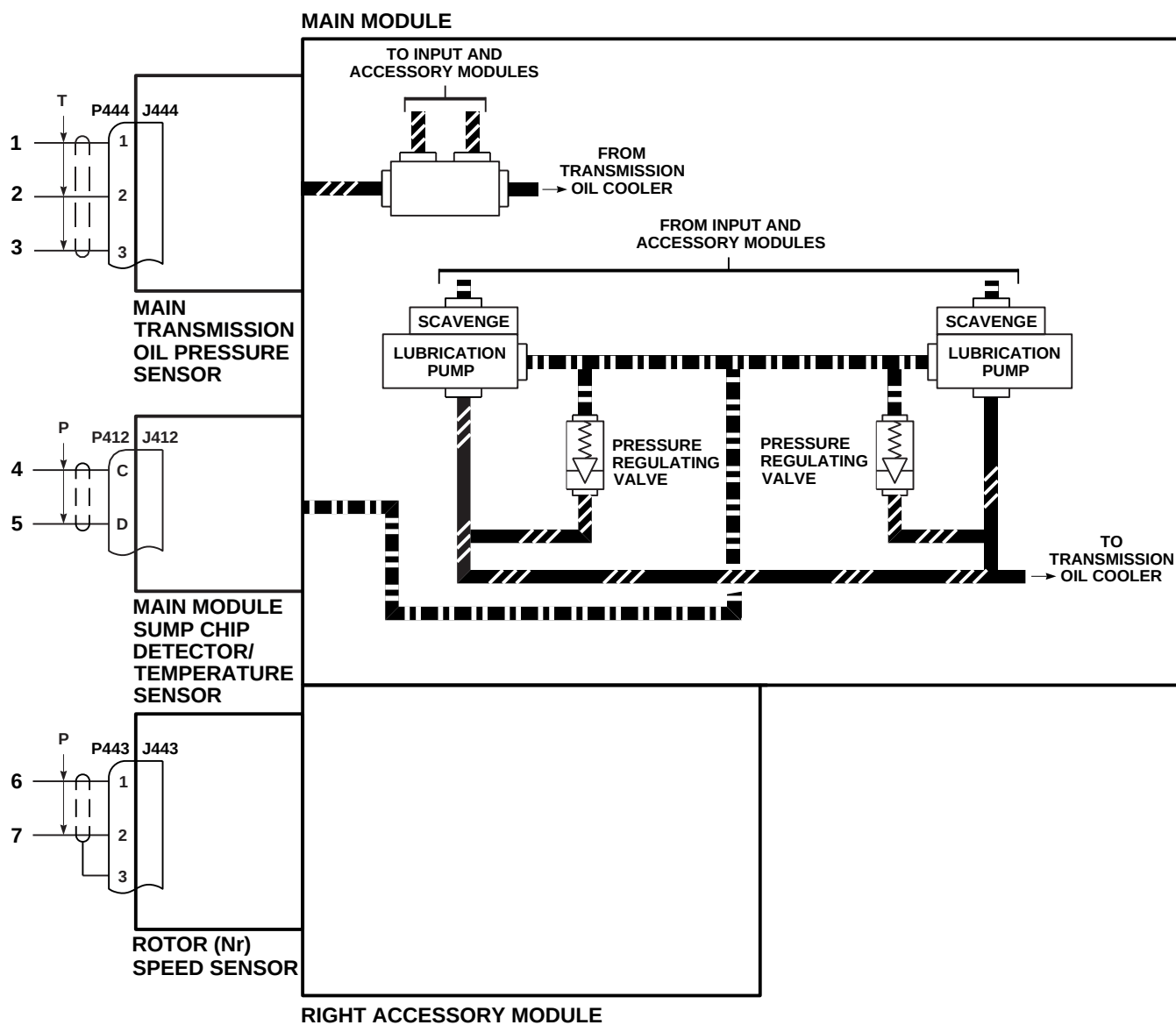
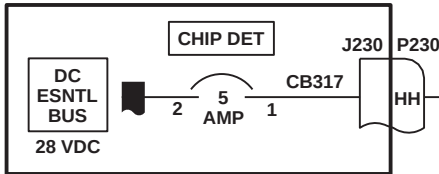
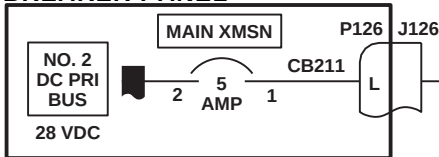


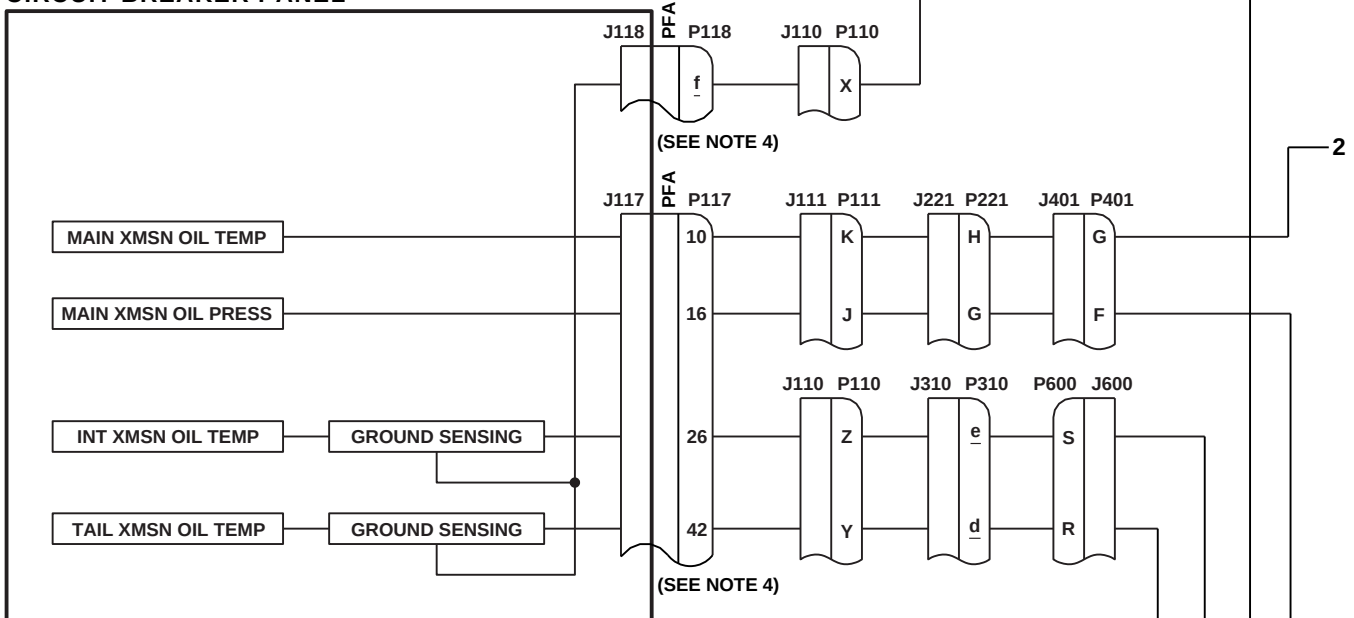
FIGURE 6-2-2-2. TRANSMISSION INSTRUMENT SYSTEM SCHEMATIC DIAGRAM  
(SHEET 2 OF 2)

FB4570\_2  
SA

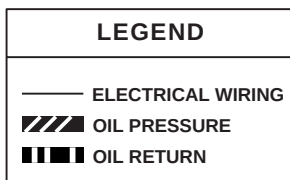
### PILOT'S CIRCUIT BREAKER PANEL



### UPPER CONSOLE CIRCUIT BREAKER PANEL



### CAUTION/ADVISORY PANEL



### NOTES

1. SWITCH CLOSES WHEN TEMPERATURE IS ABOVE 273°F TO 295°F (134°C TO 146°C).
2. SWITCH CLOSES WHEN OIL PRESSURE IS BELOW 12 TO 16 PSI.
3. SWITCH CLOSES WHEN TEMPERATURE IS ABOVE 233°F TO 250°F (112°C TO 121°C).
4. **W/O EMEP** PIN FILTER ADAPTER (PFA) IS NOT INSTALLED.

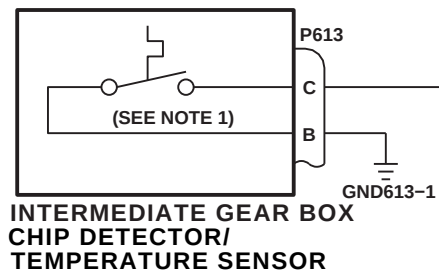
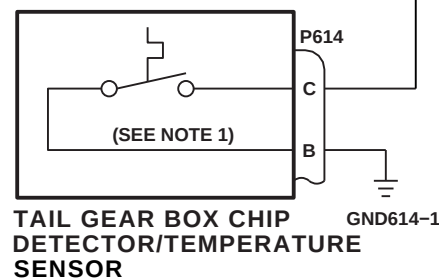
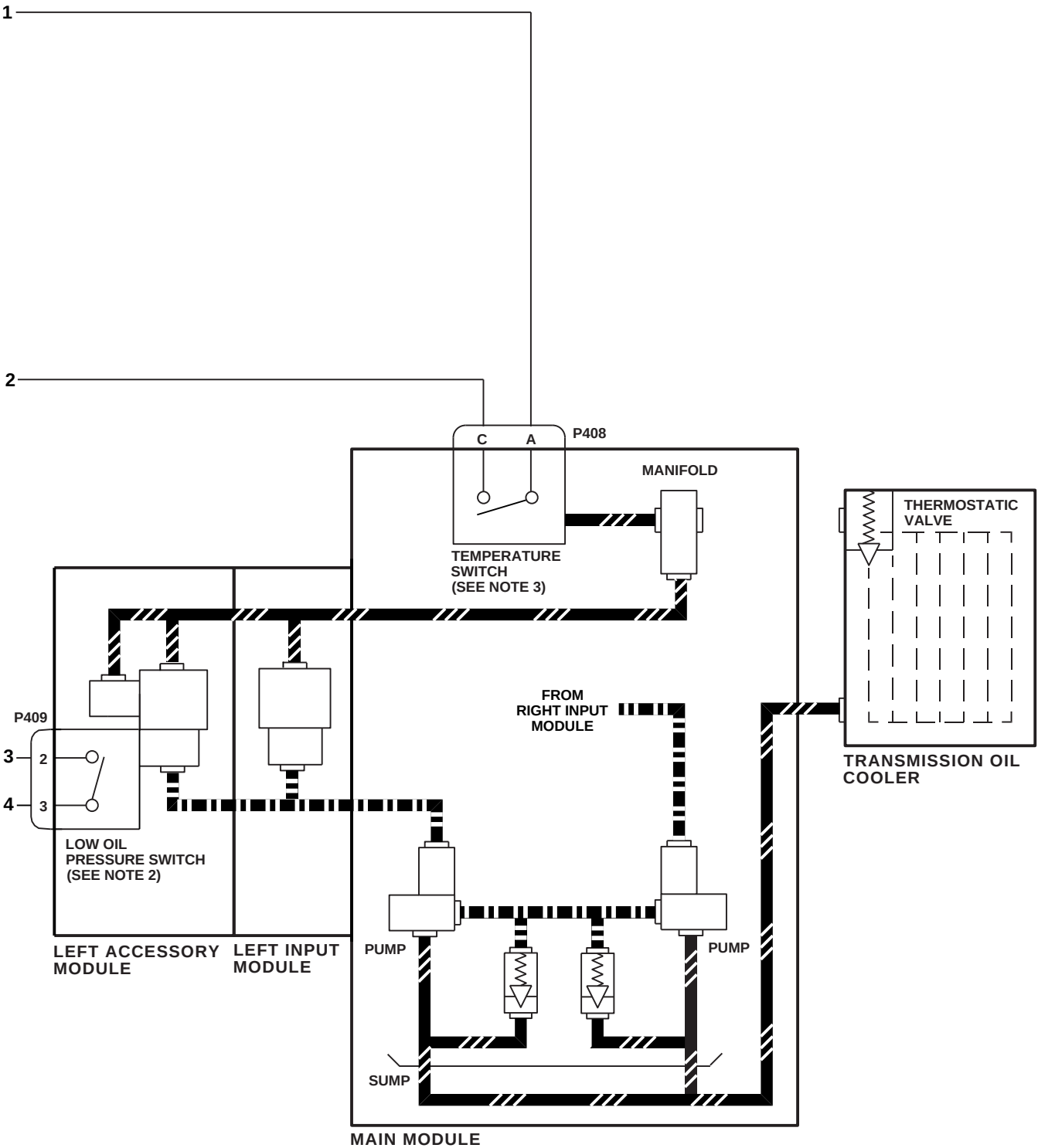
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FIGURE 6-2-2-3. TRANSMISSION OIL WARNING SYSTEM SCHEMATIC DIAGRAM  
(SHEET 1 OF 2)



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FIGURE 6-2-2-3. TRANSMISSION OIL WARNING SYSTEM SCHEMATIC DIAGRAM  
(SHEET 2 OF 2)





6-2-3 ROTOR BRAKE SYSTEM.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer Toolkit

Materials/Parts

- Hydraulic Fluid, Item 155, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D
- Appropriate Flight Manual
- PARA 1-6-16
- PARA 1-6-18
- PARA 2-4-113

- PARA 6-4-68

Equipment Conditions

- External Power Available And/Or APU Operational
- Gust Lock System Operational
- Rotor Brake System Serviced

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

Special Instructions

The Rotor Brake System operational/troubleshooting procedure is subdivided as follows:

- (1) Rotor Brake Static Check
- (2) Rotor Brake Break-In Procedure
- (3) Rotor Brake Dynamic Check
- (4) Rotor Brake Seal Verification Check
- (5) Shutdown

6-2-3.1 Rotor Brake Static Check.

- Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
- Turn on electrical power (PARA 1-6-16).
- Engage and disengage rotor brake at least three times, leaving the brake disengaged (PARA 1-6-18).
- Measure the gap between brake pucks and rotor disc on each side of disc.

RESULT:

- (1) Gap between each brake puck and rotor disc shall be  $0.055 \pm 0.020$  inch.
- (2) Total of six gaps when added together shall not exceed 0.330 inch.
  - If result is not as specified, replace rotor brake linings (pucks) (PARA 6-4-71).
- e. Engage rotor brake (PARA 1-6-18).

RESULT: ROTOR BRAKE ON capsule on caution/advisory panel shall go on.

- If result is not as specified, go to TABLE NO. 6-2-3-1.

- f. Inspect rotor brake, brake master cylinder, manifold and valve, accumulator, pressure switch, and tubing for evidence of leakage.

RESULT: No leakage shall be evident.

- If result is not as specified, repair/replace components, as required.

- g. Inspect rotor brake housing and disc for evidence of warping, scoring, corrosion, or cracks.

RESULT: There shall be no warping, scoring, corrosion, or cracks.

- If result is not as specified, repair/replace components as required.

- h. Check wear pin extension.

RESULT: The ¼ inch diameter portion of all pins are ⅜ inch or greater from housing.

- If result is not as specified, replace brake linings (PARA 6-4-71).

- i. Disengage rotor brake and move handle into detent on master cylinder (PARA 1-6-18).

RESULT:

- (1) ROTOR BRAKE ON capsule on caution/advisory panel shall go off.
- (2) Handle shall engage in master cylinder detent.
- (3) Pressure gauge shall indicate zero.

- If result (1) is not as specified, replace rotor brake pressure switch (PARA 6-4-63).
- If result (2) is not as specified, replace master cylinder (PARA 6-4-60).

- If result (3) is not as specified, go to TABLE NO. 6-2-3-2.

- j. Engage rotor brake (PARA 1-6-18).

RESULT:

- (1) ROTOR BRAKE ON capsule on caution/advisory panel shall go on.
- (2) Master cylinder handle shall lock at end of travel.
- (3) Pressure gauge shall indicate  $570 \pm 120$  psi.

- If result (1) is not as specified, go to TABLE NO. 6-2-3-1.

- If result (2) is not as specified, replace master cylinder (PARA 6-4-60).

- If pressure in result (3) is less than 450 psi, go to TABLE NO. 6-2-3-3.

- If pressure in result (3) is greater than 690 psi, replace rotor brake manifold (PARA 6-4-62).

- k. Pull down and push engine control quadrant power levers toward FLY.

RESULT: Levers shall not move past IDLE.

- If result is not as specified, replace rotor brake solenoid (PARA 4-4-24).

- l. Disengage rotor brake (PARA 1-6-18).

- m. Move power control levers to FLY.

RESULT: Power control levers shall go to FLY.

- If result is not as specified, go to TABLE NO. 6-2-3-4.

- n. Turn off electrical power (PARA 1-6-16).

### 6-2-3.2 Rotor Brake Break-In Procedure.

#### NOTE

This Rotor Brake Break-In Procedure is to ensure proper mating of components. It is required only when component, rotor brake disc, linings, or housings are changed, or are serviced due to rotor brake slippage.

- a. Engage rotor brake (PARA 1-6-18).
- b. Move helicopter to a safe area for engine startup and rotor engagement.
- c. Disengage rotor brake (PARA 1-6-18).

#### WARNING

A new rotor brake which has not been broken in may allow the rotor head to slip with two engines running at idle. Ensure ground crew is aware of this possibility and remains clear.

#### NOTE

One engine may be used to perform Rotor Brake Break-In Procedure in order to limit engine start-ups.

- d. With qualified personnel at controls, start No. 1 and/or No. 2 engine(s) and advance power control lever(s) to IDLE (Appropriate Flight Manual).
- e. Shut down No. 1 and/or No. 2 engine (Appropriate Flight Manual).
- f. When rotor speed is 40%, slowly engage rotor brake (PARA 1-6-18).

RESULT:

- (1) Braking pressure shall be  $165 \pm 15$  psi.
- (2) Blade rotation shall stop in  $15 \pm 3$  seconds.

- If result (1) is not as specified, bleed Rotor Brake System (PARA 6-4-68).
- If rotor head stops in less than 12 seconds, perform a sudden stoppage inspection (PARA 1-7-50).
- If rotor head stops longer than 18 seconds, go to TABLE NO. 6-2-3-5.
- If rotor does not appear to be slowing down, disengage the rotor brake and allow rotor to coast down (PARA 1-6-18).

Inspect rotor brake for proper adjustment (PARA 6-4-70).

#### CAUTION

Reengaging rotor brake before cooldown could cause rotor brake disc to warp (PARA 1-6-18).

- g. Allow a 15 minute cooldown period before reengaging rotor brake (PARA 1-6-18).
- h. Disengage rotor brake (PARA 1-6-18).
- i. Manually turn rotor.

RESULT: Rotor shall turn.

- If result is as specified, go to TABLE NO. 6-2-3-5.
- j. Repeat Rotor Brake Break-In Procedure one more time to complete two break-in cycles.
- k. After break-in procedure is complete, perform Rotor Brake Dynamic Check.

### 6-2-3.3 Rotor Brake Dynamic Check.

- a. With qualified personnel at controls, start No. 1 and No. 2 engines and operate at IDLE speed (Appropriate Flight Manual).
- b. Shut down No. 1 and No. 2 engines (Appropriate Flight Manual).

- c. When rotor speed is 40%, slowly engage rotor brake (PARA 1-6-18).

**RESULT:**

- (1) Braking pressure shall be  $165 \pm 15$  psi.
  - (2) Blade rotation shall stop in  $15 \pm 3$  seconds.
- If result (1) is not as specified, bleed Rotor Brake System (PARA 6-4-68).
  - If rotor head stops in less than 12 seconds, perform a sudden stoppage inspection (PARA 1-7-50).
  - If rotor head stops longer than 18 seconds, go to TABLE NO. 6-2-3-5.
  - If rotor does not appear to be slowing down, disengage the rotor brake and allow rotor to coast down (PARA 1-6-18).

Inspect rotor brake for proper adjustment (PARA 6-4-70).

- e. Remove oil cooler compartment access doors (PARA 2-4-113).
- f. Remove bleed screw and washer from rotor brake bleeder valve (Figure 6-2-3-2, Detail C).

**NOTE**

As an alternate to an in-line pressure gage, the master cylinder pressure gage may also be used to monitor the pressure being applied to the Rotor Brake System.

- g. Attach pressure gage and hydraulic hand pump to bleeder valve on rotor brake.
- h. Using hydraulic hand pump, apply pressure to the Rotor Brake System at a rate of approximately 1 stroke every 5 seconds.

**CAUTION**

Pressure applied should not exceed 300 psi.

- i. While pressure is being applied, monitor fluid level of servicing reservoir for volume increase.

**RESULT:** Master cylinder servicing reservoir shall increase in volume.

- If no increase in volume in servicing reservoir is seen, replace master cylinder (PARA 6-4-60). Repeat steps h. and i. on bench with new master cylinder prior to installing.

- j. Disconnect hydraulic hand pump and pressure gage from bleeder valve on rotor brake and close bleeder valve.
- k. Install rotor brake washer and bleed screw.
- l. Install oil cooler compartment access doors (PARA 2-4-113).
- m. Bleed Rotor Brake System (PARA 6-4-68).

### 6-2-3.4 Rotor Brake Seal Verification Check.

**NOTE**

- All rotor brake components must be installed prior to performing check.
  - Make sure Rotor Brake System is filled and bled per PARA 6-4-68 with hydraulic fluid, Item 155, Appendix D, which meets system requirements for cleanliness.
- a. Turn off all electrical power (PARA 1-6-16).
  - b. Cover lower console with machinery towel, Item 211, Appendix D.
  - c. Engage gust lock and disengage rotor brake, making sure the master cylinder handle is in the detent position (stowed) (PARA 1-6-18).
  - d. Open main rotor pylon sliding cover.

- (1) If rotor brake master cylinder passes the seal-verification check, but the same cockpit indications are seen, replace rotor brake master cylinder (PARA 6-4-60).

(2) If rotor brake master cylinder has been replaced, but the same cockpit indications are seen, replace rotor brake pressure switch (PARA 6-4-63).

(3) If rotor brake master cylinder and rotor brake pressure switch have been re-

placed, but the same cockpit indications are seen, troubleshoot Rotor Brake System using all procedures in this PARA.

**6-2-3.5 Shutdown.**

  - a. Turn off electrical power (PARA 1-6-16).
  - b. Engage rotor brake after 15 minute cooldown (PARA 1-6-18).

TABLE NO. 6-2-3-1. ROTOR BRAKE ON Caution/Advisory Capsule Does Not Go On.

| TEST OR INSPECTION                                                           | CORRECTIVE ACTION                                                                                                                                                                                                        |
|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Check for 28 VDC between J280-J and gnd.                                  | Step 1. If voltage is as specified, go to 2.<br>Step 2. If voltage is not as specified, go to 3.                                                                                                                         |
| 2. Check for 28 VDC between P410-1 and gnd.                                  | Step 1. If voltage is as specified, go to 4.<br>Step 2. If voltage is not as specified, repair/replace wiring between P410-1 and P280-J (Appendix F). Go to 9.                                                           |
| 3. Check for 28 VDC between ROTOR BRAKE circuit breaker terminal 1 and gnd.  | Step 1. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and J280-J (Appendix F). Go to 9.<br>Step 2. If voltage is not as specified, replace circuit breaker (PARA 9-4-1). Go to 9. |
| 4. Check helicopter configuration.                                           | Step 1. <b>EMEP</b> , go to 5.<br>Step 2. <b>W/O EMEP</b> , go to 7.                                                                                                                                                     |
| <b>EMEP</b> 5. Remove and check Pin Filter Adapter (PFA) P118 (PARA 9-4-98). | Step 1. If PFA is good, go to 6.<br>Step 2. If PFA is not good, replace PFA (PARA 9-4-98). Go to 9.                                                                                                                      |
| <b>EMEP</b> 6. Install PFA (PARA 9-4-98). Go to 7.                           |                                                                                                                                                                                                                          |

TABLE NO. 6-2-3-1. ROTOR BRAKE ON Caution/Advisory Capsule Does Not Go On. (Cont)

| TEST OR INSPECTION | CORRECTIVE ACTION |
|--------------------|-------------------|
|--------------------|-------------------|

|                                             |  |
|---------------------------------------------|--|
| 7. Check for 28 VDC between P118-Z and gnd. |  |
|---------------------------------------------|--|

|                                                                                                       |  |
|-------------------------------------------------------------------------------------------------------|--|
| <p>Step 1. If voltage is as specified, replace caution/advisory panel (PARA 8-4-19).<br/>Go to 9.</p> |  |
|-------------------------------------------------------------------------------------------------------|--|

|                                                         |  |
|---------------------------------------------------------|--|
| <p>Step 2. If voltage is not as specified, go to 8.</p> |  |
|---------------------------------------------------------|--|

|                                                |  |
|------------------------------------------------|--|
| 8. Check continuity between P118-Z and P410-3. |  |
|------------------------------------------------|--|

|                                                                                                          |  |
|----------------------------------------------------------------------------------------------------------|--|
| <p>Step 1. If continuity is present, replace rotor brake pressure switch (PARA 6-4-63).<br/>Go to 9.</p> |  |
|----------------------------------------------------------------------------------------------------------|--|

|                                                                                                                     |  |
|---------------------------------------------------------------------------------------------------------------------|--|
| <p>Step 2. If continuity is not present, repair/replace wiring between P118-Z and P410-3 (Appendix F). Go to 9.</p> |  |
|---------------------------------------------------------------------------------------------------------------------|--|

|                         |  |
|-------------------------|--|
| 9. Procedure completed. |  |
|-------------------------|--|

TABLE NO. 6-2-3-2. Pressure Gauge Does Not Go To Zero.

| TEST OR INSPECTION | CORRECTIVE ACTION |
|--------------------|-------------------|
|--------------------|-------------------|

|                                            |  |
|--------------------------------------------|--|
| 1. Bleed Rotor Brake System (PARA 6-4-68). |  |
|--------------------------------------------|--|

|                                                                                       |  |
|---------------------------------------------------------------------------------------|--|
| <p>Step 1. Repeat operational/troubleshooting procedure, (PARA 6-2-3.1, step i.).</p> |  |
|---------------------------------------------------------------------------------------|--|

|                                             |  |
|---------------------------------------------|--|
| <p>Step 2. If trouble remains, go to 2.</p> |  |
|---------------------------------------------|--|

|                                                      |  |
|------------------------------------------------------|--|
| 2. Replace rotor brake pressure gauge (PARA 6-4-64). |  |
|------------------------------------------------------|--|

|                                             |  |
|---------------------------------------------|--|
| <p>Step 1. If trouble remains, go to 3.</p> |  |
|---------------------------------------------|--|

|                                                       |  |
|-------------------------------------------------------|--|
| 3. Replace rotor brake master cylinder (PARA 6-4-60). |  |
|-------------------------------------------------------|--|

|                                             |  |
|---------------------------------------------|--|
| <p>Step 1. If trouble remains, go to 4.</p> |  |
|---------------------------------------------|--|

|                                                          |  |
|----------------------------------------------------------|--|
| 4. Replace rotor brake manifold and valve (PARA 6-4-62). |  |
|----------------------------------------------------------|--|

|                                             |  |
|---------------------------------------------|--|
| <p>Step 1. If trouble remains, go to 5.</p> |  |
|---------------------------------------------|--|

|                                                |  |
|------------------------------------------------|--|
| 5. Replace rotor brake (PARA 6-4-70). Go to 6. |  |
|------------------------------------------------|--|

**TABLE NO. 6-2-3-2. Pressure Gauge Does Not Go To Zero. (Cont)**

| TEST OR INSPECTION      | CORRECTIVE ACTION |
|-------------------------|-------------------|
| 6. Procedure completed. |                   |

**TABLE NO. 6-2-3-3. Pressure Gauge Does Not Indicate At Least 450 PSI.**

## TEST OR INSPECTION CORRECTIVE ACTION

---

- 1. Check all rotor brake lines and fittings for leaks.**  
  
Step 1. If leaks are found, replace lines and fittings (PARA 6-4-67). Go to 7.  
Step 2. If no leaks are found, go to 2.
- 2. Bleed Rotor Brake System (PARA 6-4-68).**  
  
Step 1. Perform Rotor Brake Static Check (PARA 6-2-3.1).  
Step 2. If trouble remains, go to 3.
- 3. Replace rotor brake pressure gauge (PARA 6-4-64).**  
  
Step 1. If trouble remains, go to 4.
- 4. Replace rotor brake master cylinder (PARA 6-4-60).**  
  
Step 1. If trouble remains, go to 5.
- 5. Replace rotor brake manifold and valve (PARA 6-4-62).**  
  
Step 1. If trouble remains, go to 6.
- 6. Replace rotor brake (PARA 6-4-70). Go to 7.**
- 7. Procedure completed.**

**TABLE NO. 6-2-3-4. Power Control Levers Do Not Move Past IDLE Position With Rotor Brake Off.**

| TEST OR INSPECTION<br>CORRECTIVE ACTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>1. Check for 28 VDC between J166-1 and J166-3.</b></p> <p>Step 1. If voltage is as specified, troubleshoot engine quadrant (PARA 4-2-9).<br/>Go to <b>4.</b></p> <p>Step 2. If voltage is not as specified, go to <b>2.</b></p> <p><b>2. Check continuity between P166-3 and gnd.</b></p> <p>Step 1. If continuity is present, go to <b>3.</b></p> <p>Step 2. If continuity is not present, repair/replace wiring between P166-3 and GND166-1 (Appendix F). Go to <b>4.</b></p> <p><b>3. Check continuity between P166-1 and P410-2.</b></p> <p>Step 1. If continuity is present, replace rotor brake pressure switch (PARA 6-4-63).<br/>Go to <b>4.</b></p> <p>Step 2. If continuity is not present, repair/replace wiring between P166-1 and P410-2 (Appendix F). Go to <b>4.</b></p> <p><b>4. Procedure completed.</b></p> |

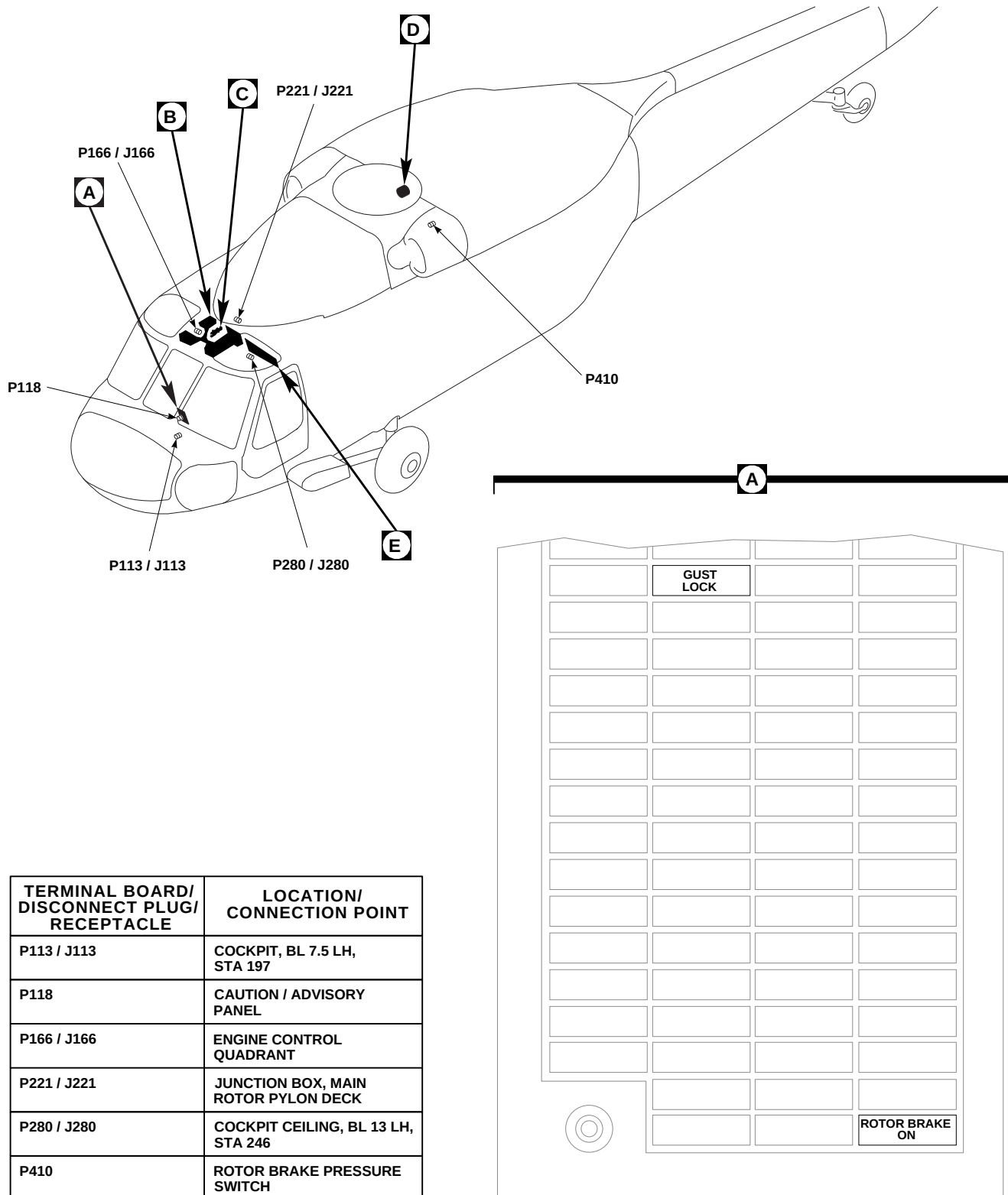
**TABLE NO. 6-2-3-5. Rotor Brake Does Not Function Properly.**

| TEST OR INSPECTION<br>CORRECTIVE ACTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>1. Check fluid level in hydraulic reservoir.</b></p> <p>Step 1. If fluid level is at or above designated minimum, go to <b>2.</b></p> <p>Step 2. If fluid level is below designated minimum, service reservoir (PARA 1-3-8).<br/>Go to <b>9.</b></p> <p><b>2. Check for hydraulic fluid on discs and linings.</b></p> <p>Step 1. If fluid is on discs or linings, clean/replace (PARA 6-4-71). Go to <b>9.</b></p> <p>Step 2. If no fluid is present, go to <b>3.</b></p> <p><b>3. Check if puck clearance total exceeds 0.330 inches.</b></p> |



**TABLE NO. 6-2-3-5. Rotor Brake Does Not Function Properly. (Cont)**

| TEST OR INSPECTION                                                                                           | CORRECTIVE ACTION                                                                                          |
|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Step 1. If total clearance is 0.330 inches or less, go to <b>4</b> .                                         | Step 2. If total clearance is greater than 0.330 inches, replace linings (PARA 6-4-71).                    |
| Go to <b>9</b> .                                                                                             |                                                                                                            |
| <b>4. Perform Rotor Brake Static Check (PARA 6-2-3.1).</b>                                                   |                                                                                                            |
| Step 1. If linings are good, go to <b>5</b> .                                                                | Step 2. If linings are defective, replace linings (PARA 6-4-71). Go to <b>9</b> .                          |
| <b>5. Engage rotor brake lever (PARA 1-6-18). Go to 6.</b>                                                   |                                                                                                            |
| <b>6. Does pressure gauge read between 450-690 psi?</b>                                                      |                                                                                                            |
| Step 1. If pressure is as specified, go to <b>7</b> .                                                        | Step 2. If pressure is not as specified, replace gauge (PARA 6-4-64). Go to <b>9</b> .                     |
| Step 3. If trouble remains, replace master cylinder (PARA 6-4-60). Go to <b>9</b> .                          |                                                                                                            |
| <b>7. Bleed Rotor Brake Hydraulic System (PARA 7-4-65). Go to 8.</b>                                         |                                                                                                            |
| <b>8. Does rotor brake function correctly?</b>                                                               |                                                                                                            |
| Step 1. If rotor brake functions correctly, continue operational/troubleshooting procedure. Go to <b>9</b> . | Step 2. If rotor brake does not function as specified, replace rotor brake (PARA 6-4-70). Go to <b>9</b> . |
| <b>9. Procedure completed.</b>                                                                               |                                                                                                            |



CAUTION/ADVISORY PANEL

FB1567\_1A  
SA

FIGURE 6-2-3-1. ROTOR BRAKE SYSTEM LOCATION DIAGRAM (SHEET 1 OF 3)

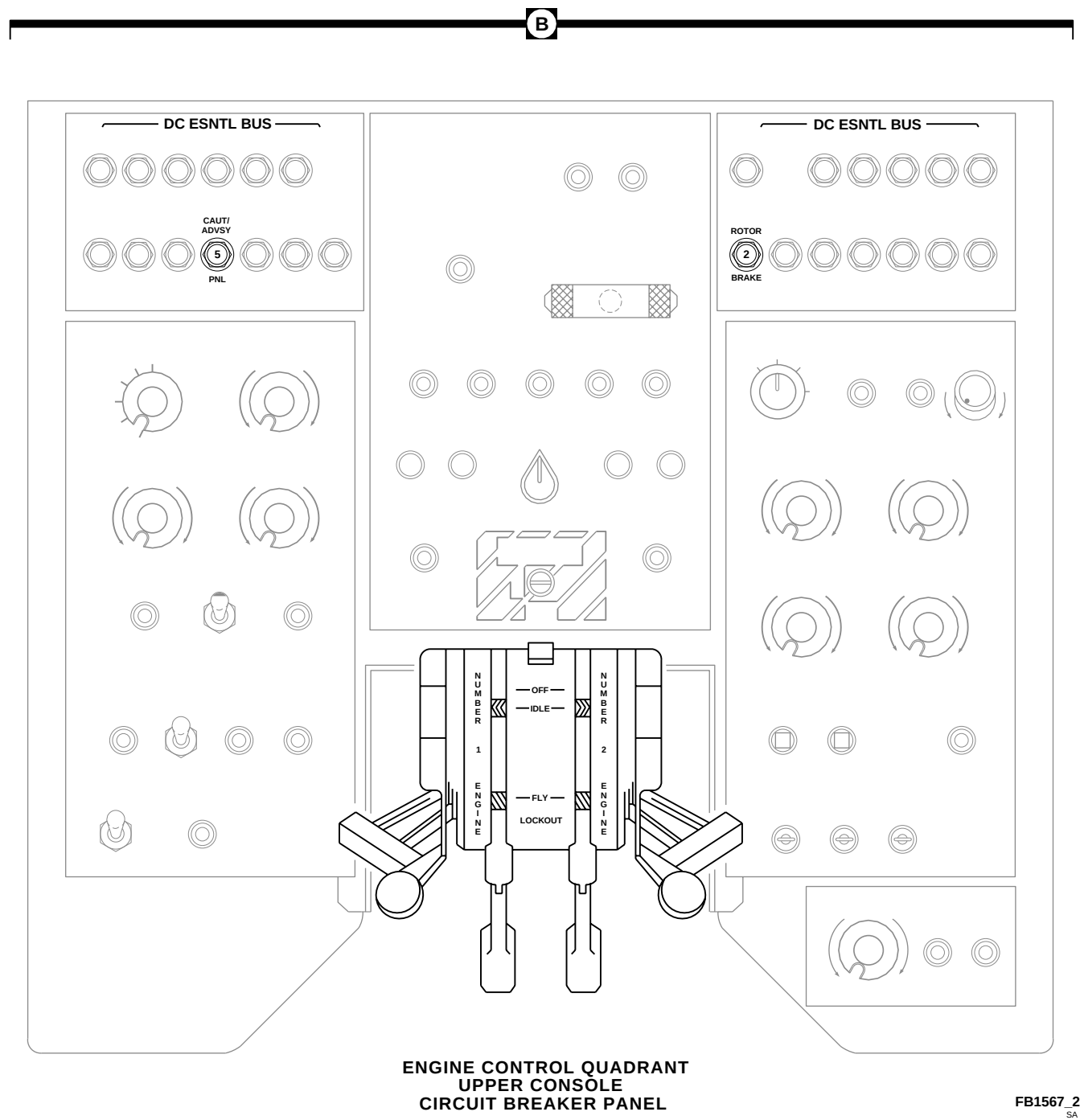
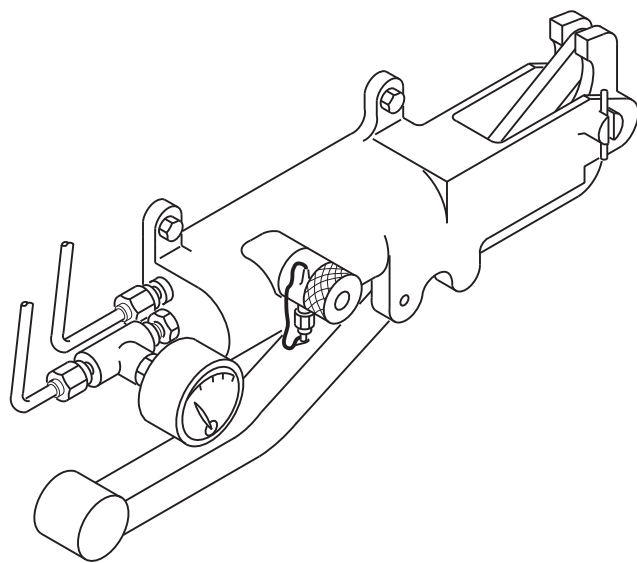
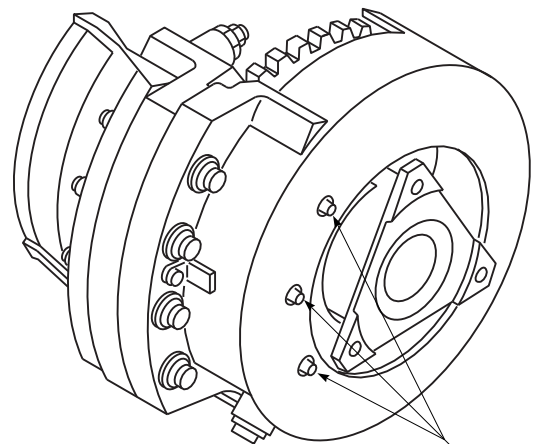


FIGURE 6-2-3-1. ROTOR BRAKE SYSTEM LOCATION DIAGRAM (SHEET 2 OF 3)

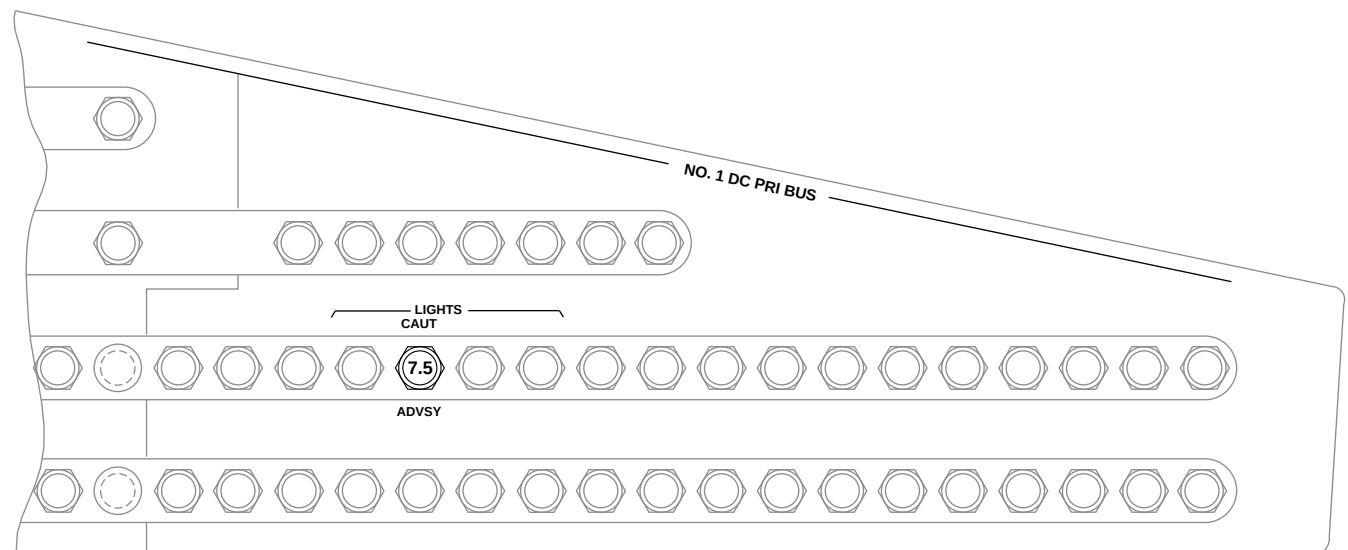


MASTER CYLINDER AND GAGE



BRAKE LINING  
WEAR PIN

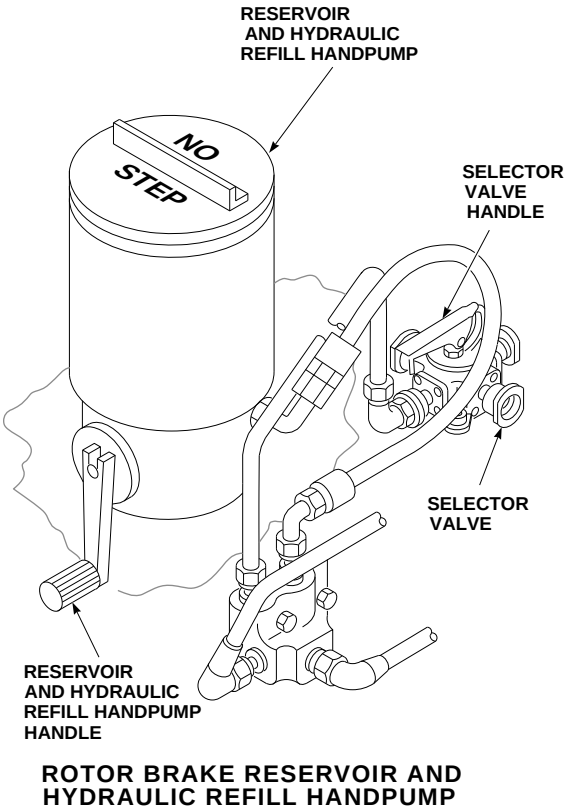
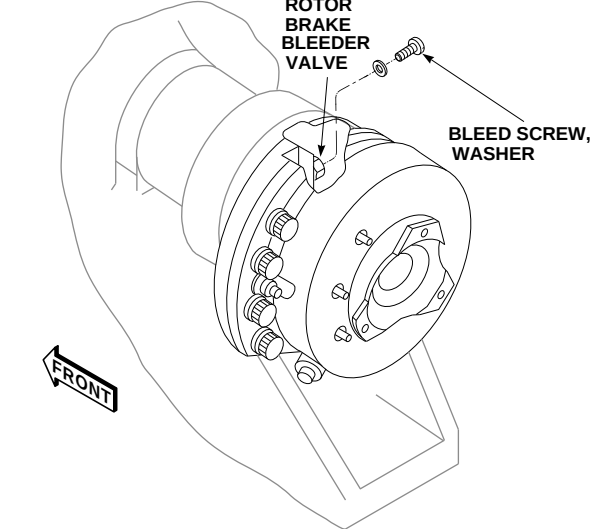
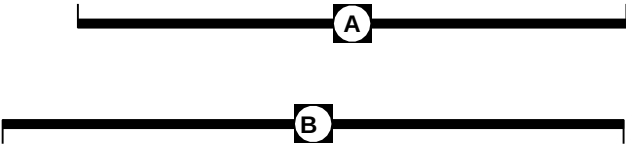
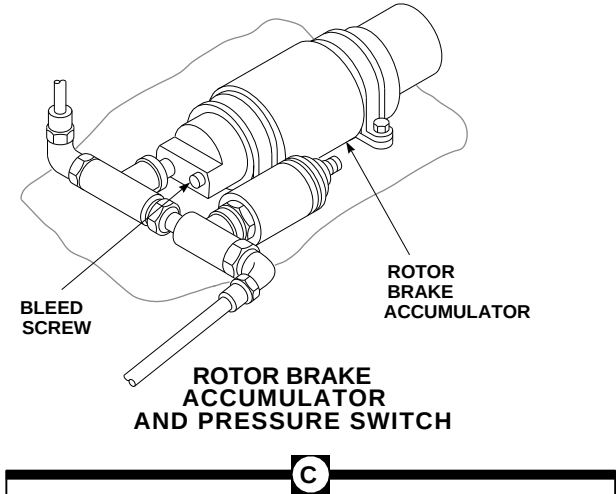
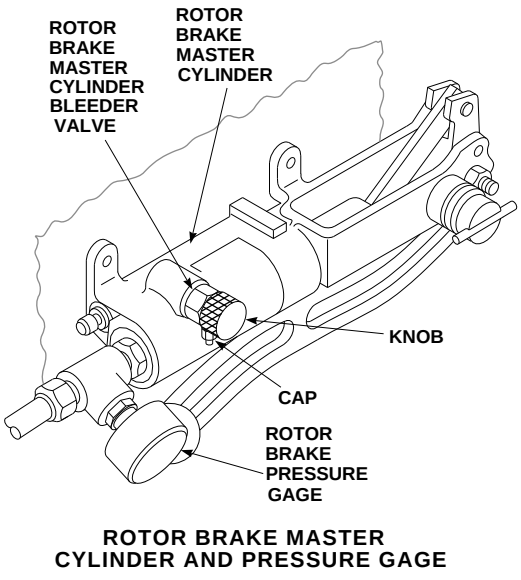
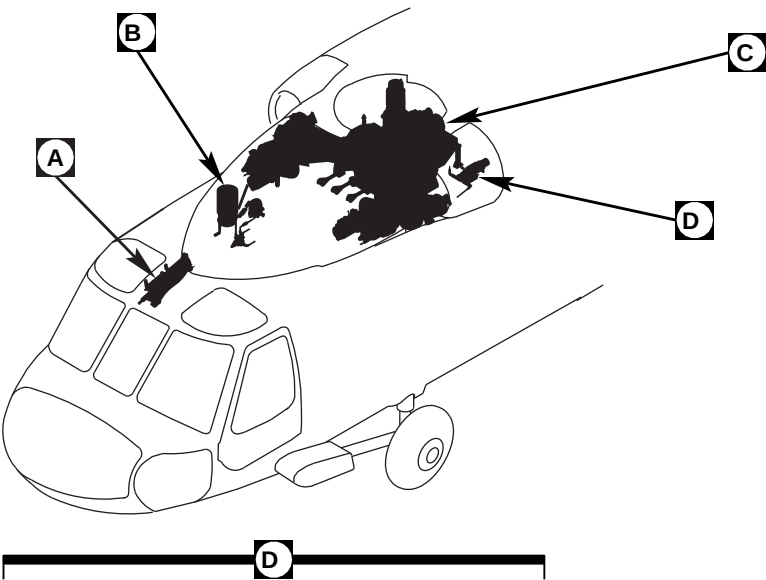
ROTOR BRAKE



COPLOT'S CIRCUIT BREAKER PANEL

FB1567\_3  
SA

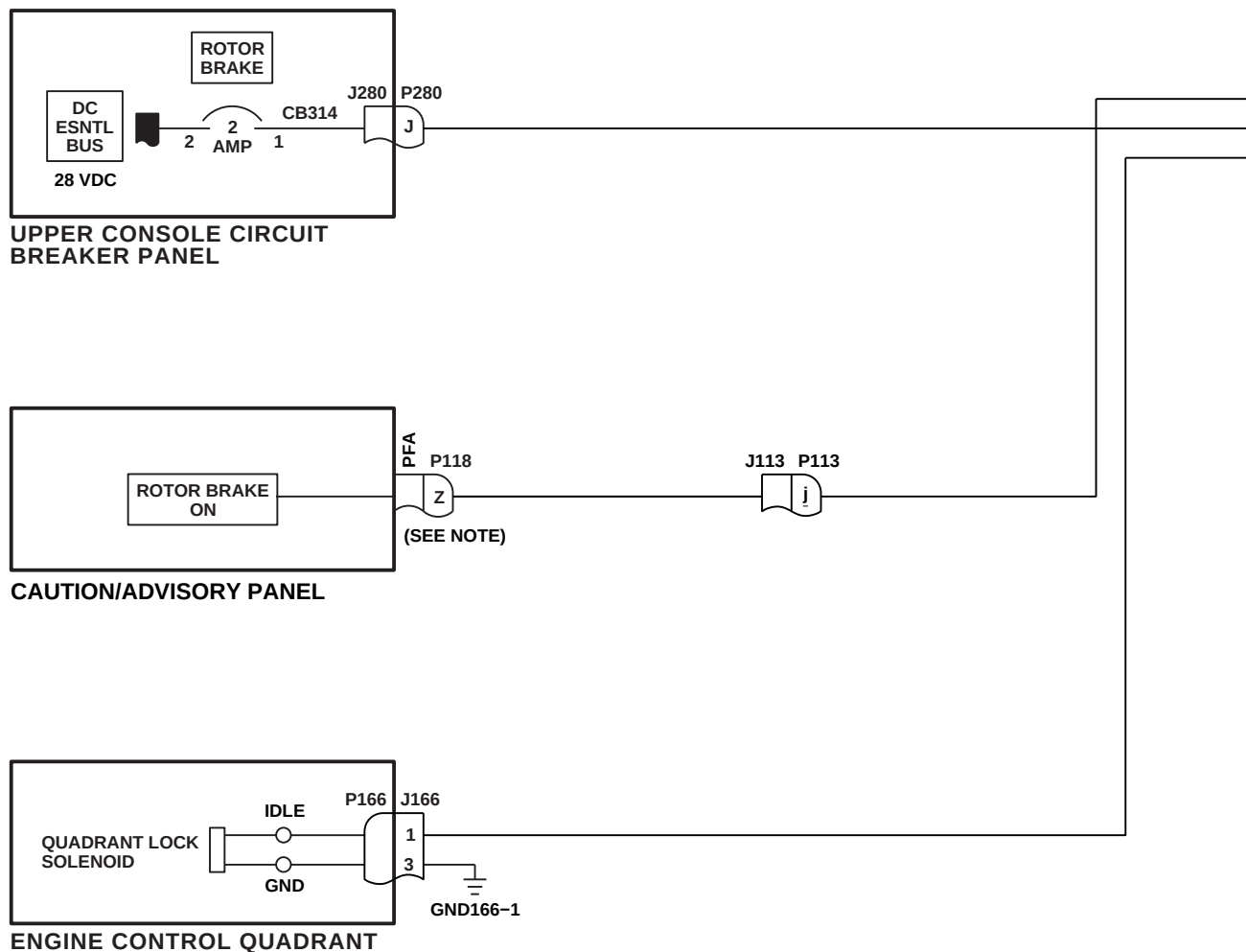
FIGURE 6-2-3-1. ROTOR BRAKE SYSTEM LOCATION DIAGRAM (SHEET 3 OF 3)



ROTOR BRAKE ASSEMBLY

ROTOR BRAKE RESERVOIR AND HYDRAULIC REFILL HANDPUMP

FIGURE 6-2-3-2. ROTOR BRAKE SYSTEM BLEED DIAGRAM



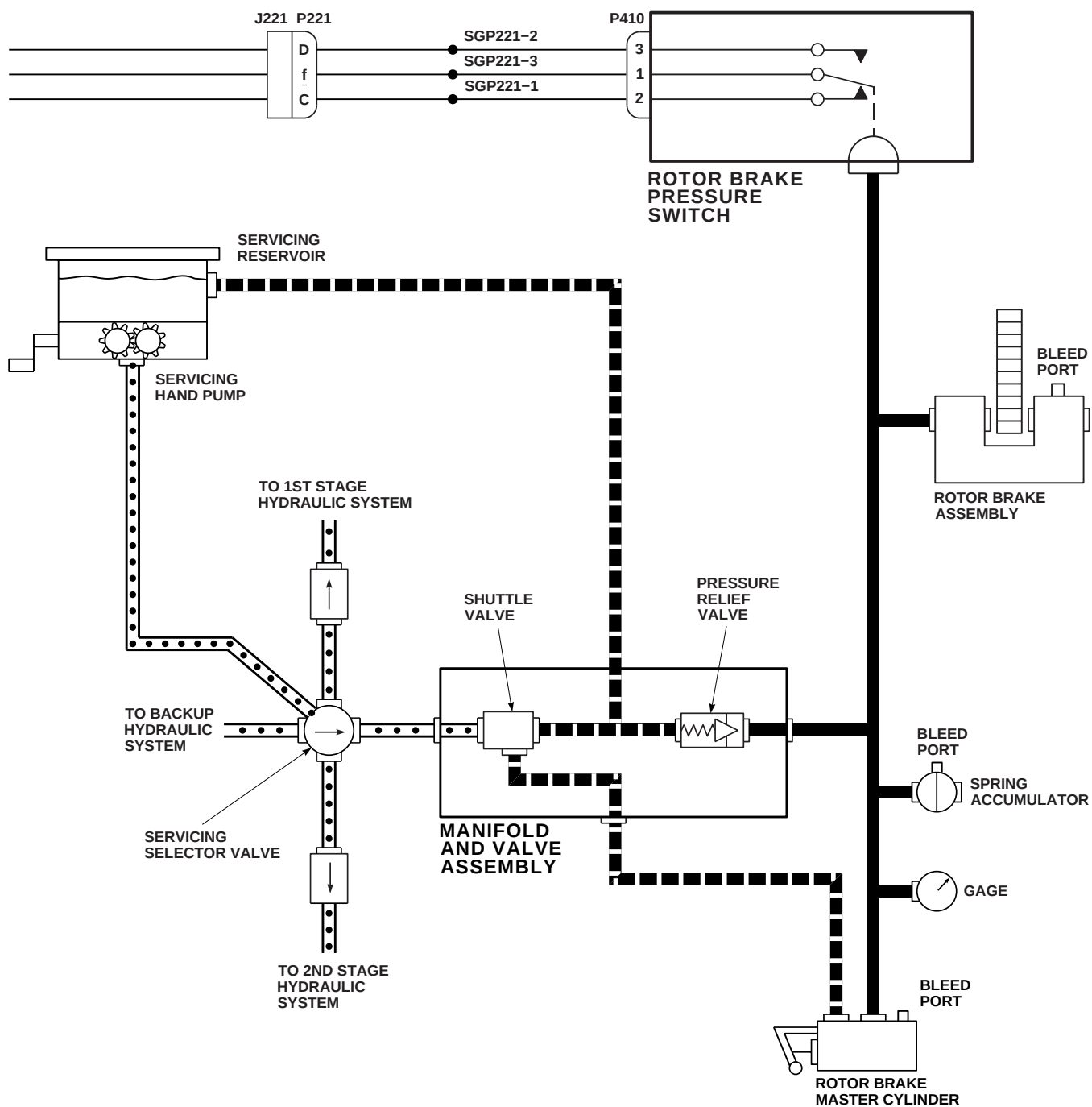
| LEGEND |                   |
|--------|-------------------|
|        | PRESSURE          |
|        | RETURN            |
|        | SERVICING         |
|        | ELECTRICAL WIRING |

**NOTE**

**W/O EMEP** PIN FILTER  
ADAPTER (PFA) IS NOT INSTALLED.

FB1568\_1C  
SA

FIGURE 6-2-3-3. ROTOR BRAKE SYSTEM SCHEMATIC DIAGRAM (SHEET 1 OF 2)



FB1568\_2A  
SA

FIGURE 6-2-3-3. ROTOR BRAKE SYSTEM SCHEMATIC DIAGRAM (SHEET 2 OF 2)





## SECTION 6-3

### ON AIRCRAFT INSPECTIONS

#### SECTION OVERVIEW

| PARAGRAPH | TITLE                                                                       |
|-----------|-----------------------------------------------------------------------------|
| 6-3-1     | Main Transmission Mounting Bolt Inspection                                  |
| 6-3-2     | Main Transmission Dowel Pin Inspection                                      |
| 6-3-3     | Main Rotor Shaft Inspections                                                |
| 6-3-4     | Main Transmission Functional Test                                           |
| 6-3-5     | Main Transmission Serviceability Inspection                                 |
| 6-3-6     | Main Transmission Chip Detectors Particle Inspections                       |
| 6-3-7     | Main Transmission Lubrication System Inspection                             |
| 6-3-8     | Input Module Inspections                                                    |
| 6-3-9     | Accessory Module Inspection                                                 |
| 6-3-10    | Intermediate Gear Box Inspections                                           |
| 6-3-11    | Intermediate Gear Box Chip Detector/Temperature Sensor Particle Inspections |
| 6-3-12    | Intermediate Gear Box Serviceability Inspection                             |
| 6-3-13    | Tail Rotor Drive Shaft Inspection                                           |
| 6-3-14    | Tail Rotor Drive Shaft Viscous Damper Bearing Assembly Inspection           |
| 6-3-15    | Oil Cooler Inspections                                                      |
| 6-3-16    | Tail Gear Box Inspections                                                   |
| 6-3-17    | Tail Gear Box Chip Detector/Temperature Sensor Particle Inspections         |
| 6-3-18    | Tail Gear Box Serviceability Inspection                                     |
| 6-3-19    | Main Transmission Housing Interference Inspection                           |

**6-3-1/(6-3-2 Blank)**



6-3-1 MAIN TRANSMISSION MOUNTING BOLT INSPECTION.

PARAGRAPH BREAKDOWN

|         |                                              | PAGE    |
|---------|----------------------------------------------|---------|
| 6-3-1.1 | Main Transmission Mounting Bolt Torque Check | 6-3-1-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Mounting Bolt Adapter, Locally-Made, Figure H-46, Appendix H
- Nonmetallic Scraper, Locally-Made
- Powertrain Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 100 - 500 ft lbs
- Vise, Soft-Jawed

Materials/Parts

- Antiseize Compound, Item 74 Appendix D
- Cap Plug, NAS813-16

- Corrosion Preventive Compound, Item 113, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Masking Tape, 3-inches, Item 215, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- Appendix H
- PARA 1-7-3
- PARA 2-4-70

6-3-1.1 Main Transmission Mounting Bolt Torque Check.

- a. Open main rotor pylon sliding cover.
- b. Open oil cooler compartment access doors.
- c. Remove main transmission drip pan (PARA 2-4-70).

**CAUTION**

Damage to main transmission will result if metal scrapers are used on surfaces. Do not use metallic scrapers to remove sealing compound or other debris.

- d. Using nonmetallic scraper, remove sealing compound from bolts securing main transmission.
- e. Locally make mounting bolt adapter (Figure H-46, Appendix H).

**WARNING****FSCAP**

Verification of minimum torque of bolts is a critical characteristic.

**NOTE**

- The use of mounting bolt adapter will be required to check torque on two front mounting bolts.
  - Torque value will change due to added length of adapter. Make sure to compute torque requirements using nonconcentric torque formula.
- f. **CHECK EIGHT MAIN TRANSMISSION MOUNTING BOLTS FOR TORQUE OF 154 - 170 FOOT-POUNDS** (Figure 6-3-1-1, Detail A). If torque is 154 - 170 foot-pounds, torque check is complete, proceed to step m. If bolt is loose or moves during check, proceed to step g. and continue with inspection.
  - g. If a bolt is loose or moves during check, remove that bolt and its corresponding barrel-nut as follows:
    - (1) From inside main transmission well, using nonmetallic scraper, remove sealing compound from around cap plug in airframe or forward bellcrank support beam. Remove cap plug.

- (2) Remove barrel-nut and retainer from airframe or forward bellcrank support beam.

- h. Check breakaway torque of removed barrel-nut(s) as follows:

- (1) Place barrel-nut in soft-jawed vise.
- (2) Install bolt into barrel-nut until two threads extend beyond barrel-nut.
- (3) Using torque wrench, back out bolt. **MINIMUM BREAKAWAY TORQUE IS 32 INCH-POUNDS.** If breakaway torque is less than 32 inch-pounds, replace barrel-nut.

**CAUTION**

Damage to recess in airframe or forward bellcrank support beam will result if barrel-nut is incorrectly installed. To prevent damage to recess, make sure barrel-nut is installed with barrel (round side) toward bolthead.

**NOTE**

Make sure that only barrel-nuts, RMLH2577, are used. These barrel-nuts are identified with the letters "EN" or "SPS" stamped on an end (additional marking may also include "LH2577").

- i. Install serviceable barrel-nut and retainer as follows (Figure 6-3-1-1, Detail A):
  - (1) Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to barrel-nut and retainer.
  - (2) Install barrel-nut and retainer in airframe or forward bellcrank support beam while primer is still wet.
  - (3) Apply a 1/4-inch diameter bead of sealing compound, Item 292, Appendix D, under lip of cap plug.
  - (4) Install cap plug in airframe or forward bellcrank support beam.

- (5) Apply a 1/4-inch diameter bead of sealing compound, Item 292, Appendix D, around interface of cap plug and airframe or forward bellcrank support beam.
- (6) Apply sealing compound, Item 292, Appendix D, to entire cap plug. Make sure cap plug is entirely sealed.



Damage to equipment will result if epoxy primer is applied to threads of mounting bolts. Epoxy primer will make removal of bolts difficult and could damage barrel-nuts. Do not allow epoxy primer to contact bolt threads.

- j. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to shank of mounting bolts.
- k. Using antiseize compound, Item 74, Appendix D, apply a light coat on mounting bolt threads. Install bolt.



#### FSCAP

Verification of minimum torque of bolts is a critical characteristic.

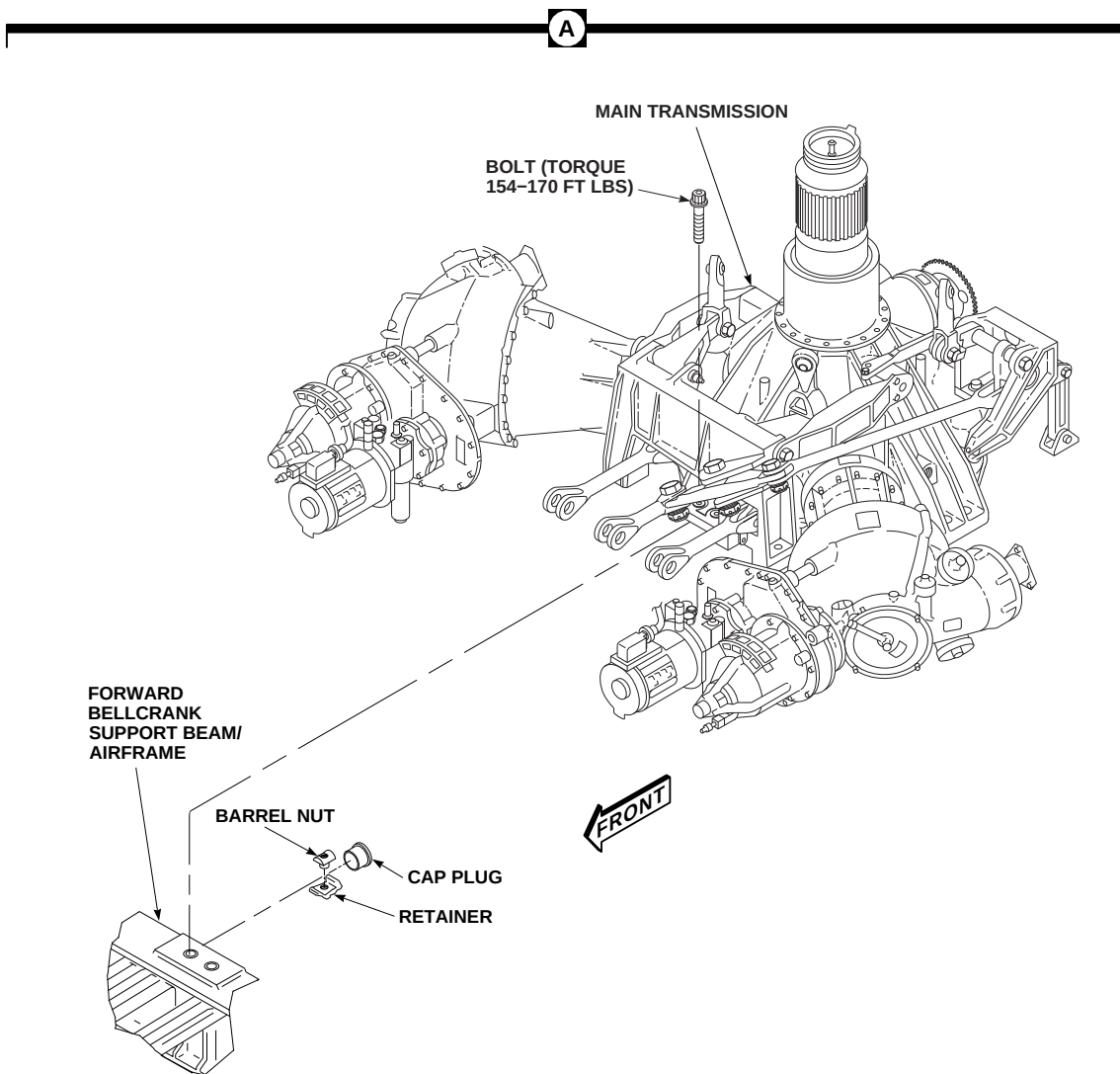
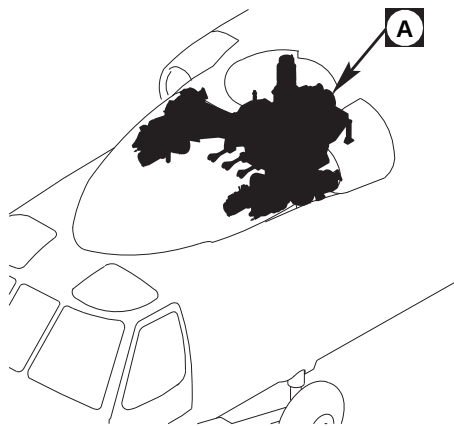
#### NOTE

- The use of mounting bolt adapter will be required to torque the two front mounting bolts.

- Torque value will change due to added length of adapter. Make sure to compute torque requirements using nonconcentric torque formula.

#### l. TORQUE BOLT TO 154 - 170 FOOT-POUNDS.

- m. Using 3-inch masking tape, Item 215, Appendix D, mask off adjacent areas from main transmission mounting bolt attach areas to prevent overspray.
- n. Using corrosion preventive compound, Item 113, Appendix D, apply to entire area where sealing compound was removed.
- o. Allow corrosion preventive compound to cure for 10 to 30 minutes.
- p. Using corrosion preventive compound, Item 113, Appendix D, apply a second coat.
- q. If any main transmission mounting bolt is replaced, a torque check is required 9 to 11 flight hours after replacement (PARA 1-7-3).
- r. Inspect torque every three days, and using corrosion preventive compound, Item 113, Appendix D, reapply two coats as necessary until bolt torque stabilizes.
- s. Make sure area is clean and free of foreign material.
- t. Install main transmission drip pan (PARA 2-4-70).
- u. Close oil cooler compartment access doors.
- v. Close main rotor pylon sliding cover.



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FIGURE 6-3-1-1. INSPECT MAIN TRANSMISSION MOUNTING BOLTS

6-3-2 MAIN TRANSMISSION DOWEL PIN INSPECTION.

PARAGRAPH BREAKDOWN

|                                                | PAGE    |
|------------------------------------------------|---------|
| 6-3-2.1 Main Transmission Dowel Pin Inspection | 6-3-2-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Feeler Gage
- Torque Wrench, 0 - 30 in. lbs

References

- PARA 6-4-1
- PARA 11-4-99
- PARA 11-4-100
- PARA 11-4-102

6-3-2.1 Main Transmission Dowel Pin Inspection.

- Open main rotor pylon sliding cover.
- Open engine cowlings.
- Open left oil cooler compartment access door.
-  Inspect main transmission flight control mount dowel pins as follows:◀

NOTE

- On main transmissions, 70351-08100-053 and 70351-08100-054, with slip-page stripe, axial movement will reveal an unpainted portion of the dowel pin. Radial movement will indicate a break in the black stripe between dowel pin head and adjoining white painted surface.
- On dowel pins with sealing compound, axial movement will be indicated by a bulging or bubbling of sealing compound surface. Radial movement will be

indicated by clockwise or counterclockwise ripples on cracking of sealing compound surface.

- Inspect right and left tie rod dowel pins for evidence of axial or radial movement (Figure 6-3-2-1, Sheet 1, and Detail A). **NONE ALLOWED. TIE ROD DOWEL PINS MUST BE FLUSH WITH, OR BELOW SURFACE OF TIE RODS.** If movement is found, or if dowel pin is beyond limits, replace main transmission (PARA 6-4-1).
- Inspect left vertical dowel pin for axial or radial movement (Figure 6-3-2-1, Sheet 2, Detail D). **NONE ALLOWED. LEFT VERTICAL DOWEL PIN MUST PROTRUDE NO MORE THAN 0.090-INCH ABOVE SURFACE OF AFT LONGITUDINAL BELLCRANK SUPPORT ROD.** If movement is found, or if dowel pin is beyond limits, replace main transmission (PARA 6-4-1).
- Inspect left aft dowel pin for evidence of axial or radial movement (Figure 6-3-2-

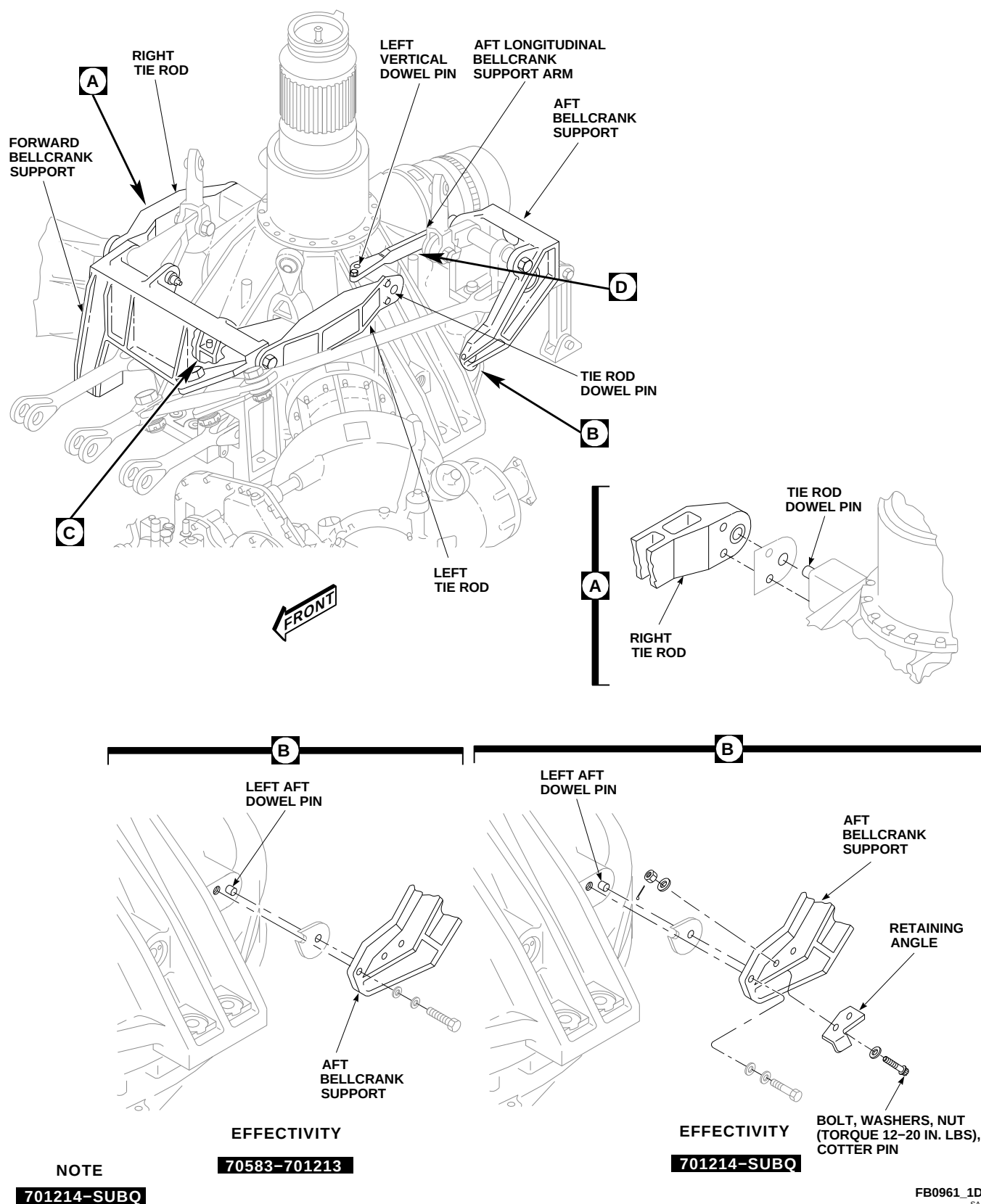
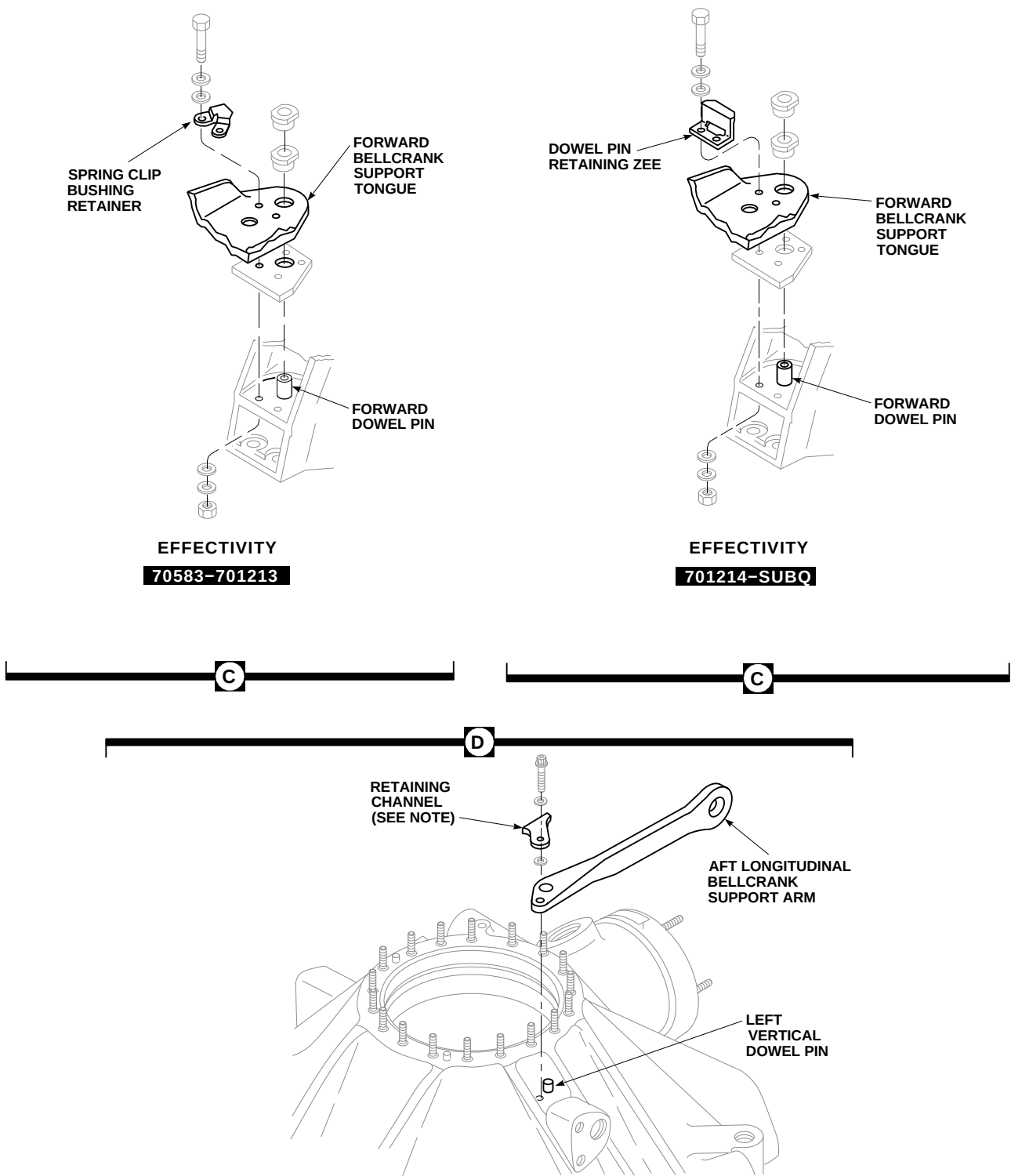


FIGURE 6-3-2-1. INSPECT MAIN TRANSMISSION DOWEL PINS (SHEET 1 OF 2)





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FIGURE 6-3-2-1. INSPECT MAIN TRANSMISSION DOWEL PINS (SHEET 2 OF 2)

1, Sheet 1, Detail B). **NONE ALLOWED.** **LEFT AFT DOWEL PIN, 70351-08 404-102, 70351-08405-102, OR NAS607-12-14P, MUST BE FLUSH WITH, OR BELOW SURFACE OF LEG OF AFT BELLCRANK SUPPORT. LEFT AFT DOWEL PIN, 70351-08404-104, MUST PROTRUDE NO MORE THAN 0.248-INCH ABOVE SURFACE OF LEG OF AFT BELLCRANK SUPPORT.** If movement is found, or if dowel pin is beyond limits, replace main transmission (PARA 6-4-1).

- (4) Inspect forward dowel pin for evidence of axial or radial movement (Figure 6-3-2-1, Sheet 2, Detail C). **NONE ALLOWED. FORWARD DOWEL PIN MUST PROTRUDE NO MORE THAN 0.938-INCH ABOVE SURFACE OF FORWARD BELLCRANK SUPPORT TONGUE.** If movement is found, or if dowel pin is beyond limits, replace main transmission (PARA 6-4-1).

e. **701214-SUBQ** Inspect main transmission flight control mount dowel pins as follows:■

- (1) Inspect bolts securing right and left tie rods for damage and security (Figure 6-3-2-1, Sheet 1). **NONE ALLOWED.** If tie rod attachment bolts are damaged or loose, perform the following:
  - (a) Remove right or left tie rod (PARA 11-4-99).
  - (b) Inspect tie rod dowel pin for axial or radial movement. **NONE ALLOWED. TIE ROD DOWEL PINS MUST BE FLUSH WITH, OR BELOW SURFACE OF TIE RODS.** If movement is found, or if dowel pin is beyond limits, replace main transmission (PARA 6-4-1).
  - (c) Install right or left tie rod (PARA 11-4-99).
- (2) Inspect retaining channel for deformation and attachment bolts for damage and

security (Figure 6-3-2-1, Sheet 2, Detail D). **NO DEFORMATION OF RETAINING CHANNEL ALLOWED. NO DAMAGE OR LOOSENESS OF ATTACHMENT BOLTS ALLOWED.** If retaining channel is deformed, or attachment bolts are damaged or loose, perform the following:

- (a) Remove retaining channel from aft longitudinal bellcrank support arm (PARA 11-4-102).
  - (b) Inspect left vertical dowel pin for axial or radial movement. **NONE ALLOWED. LEFT VERTICAL DOWEL PIN MUST PROTRUDE NO MORE THAN 0.090-INCH ABOVE SURFACE OF AFT LONGITUDINAL BELLCRANK SUPPORT ROD.** If movement is found, or if dowel pin is beyond limits, replace main transmission (PARA 6-4-1).
  - (c) Install retaining channel on aft longitudinal bellcrank support arm (PARA 11-4-102).
- (3) Inspect retaining angle for deformation and attachment bolts for damage and security (Figure 6-3-2-1, Sheet 1, Detail B). **NO DEFORMATION OF RETAINING ANGLE ALLOWED. NO DAMAGE OR LOOSENESS OF ATTACHMENT BOLTS ALLOWED.** If retaining angle is deformed, or attachment bolts are damaged or loose, perform the following:
    - (a) Remove bolts, washers, and nuts securing retaining angle to leg of aft bellcrank support. Remove retaining angle.
    - (b) Inspect left aft dowel pin for evidence of axial or radial movement. **NONE ALLOWED. LEFT AFT DOWEL PIN, 70351-08404-102, 70351-08405-102, OR NAS607-12-14P, MUST BE FLUSH WITH,**

OR BELOW SURFACE OF LEG OF AFT BELLCRANK SUPPORT. LEFT AFT DOWEL PIN, 70351-08404-104, MUST PROTRUDE NO MORE THAN 0.128-INCH ABOVE SURFACE OF LEG OF AFT BELLCRANK SUPPORT. If movement is found, or if dowel pin is beyond limits, replace main transmission (PARA 6-4-1).

**WARNING**

**FSCAP**

Verification of retaining angle installation is a critical characteristic.

**NOTE**

Bolts securing retaining angle to leg of aft bellcrank support are installed with boltheads facing down.

- (c) Install retaining angle to leg of aft bellcrank support and secure with bolts, washers, and nuts.
- (d) Using feeler gage, measure clearance between left aft dowel pin and retaining angle. **MINIMUM CLEARANCE IS 0.010-INCH.** If clearance is less than 0.010-inch, reseat left aft dowel pin (PARA 6-4-1).

- (e) **TORQUE NUTS TO 12 - 20 INCH-POUNDS.**
- (f) Install cotter pin.
- (4) Inspect dowel pin retaining zee for deformation and attachment bolts for damage and security (Figure 6-3-2-1, Sheet 2, Detail C). **NO DEFORMATION OF DOWEL PIN RETAINING ZEE ALLOWED. NO DAMAGE OR LOOSENESS OF ATTACHMENT BOLTS ALLOWED.** If dowel pin retaining zee is deformed, or attachment bolts are damaged or loose, perform the following:
  - (a) Remove forward dowel pin retaining zee (PARA 11-4-100).
  - (b) Inspect forward dowel pin for evidence of axial or radial movement. **NONE ALLOWED. FORWARD DOWEL PIN MUST PROTRUDE NO MORE THAN 0.938-INCH ABOVE SURFACE OF FORWARD BELLCRANK SUPPORT TONGUE.** If movement is found, or if dowel pin is beyond limits, replace main transmission (PARA 6-4-1).
  - (c) Install forward dowel pin retaining zee (PARA 11-4-100).
- f. Close left oil cooler compartment access door.
- g. Close engine cowlings.
- h. Close main rotor pylon sliding cover.



6-3-3 MAIN ROTOR SHAFT INSPECTIONS.

PARAGRAPH BREAKDOWN

|         |                                                               | PAGE    |         |                             | PAGE    |
|---------|---------------------------------------------------------------|---------|---------|-----------------------------|---------|
| 6-3-3.1 | Swashplate Guide and Main Rotor Shaft Seal Leakage Inspection | 6-3-3-1 | 6-3-3.2 | Main Rotor Shaft Inspection | 6-3-3-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Powertrain Repairer’s Toolkit

References

- PARA 6-4-1
- PARA 6-4-5

6-3-3.1 Swashplate Guide and Main Rotor Shaft Seal Leakage Inspection.

NOTE

- Checking amount of main rotor shaft seal leakage can prevent premature seal replacement, before leakage is over recommended limits.
- Allow at least 2 hours after flight for oil to drain back into sump before checking level.
- a. Measure seal leakage by observing drop in oil level on oil level gage (dipstick).
- b. **IF LEAKAGE IS MORE THAN 1/16-INCH PER FLIGHT HOUR, REPLACE MAIN ROTOR SHAFT SEAL (PARA 6-4-5).**

6-3-3.2 Main Rotor Shaft Inspection.

NOTE

- This inspection applies only to helicopters with main transmission, 70351-08000.
- a. Inspect main rotor shaft for radial and axial play measured just above swashplate guide.
    - (1) **MAXIMUM RADIAL PLAY IS 0.015 INCH.**
    - (2) **MAXIMUM AXIAL PLAY IS 0.065 INCH.**
  - b. If shaft is beyond limits, replace main transmission (PARA 6-4-1).



6-3-4 MAIN TRANSMISSION FUNCTIONAL TEST.

PARAGRAPH BREAKDOWN

|                                           | PAGE    |
|-------------------------------------------|---------|
| 6-3-4.1 Main Transmission Functional Test | 6-3-4-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Powertrain Repairer’s Toolkit

References

- Appropriate MTF
- PARA 1-3-12
- PARA 6-3-5

6-3-4.1 Main Transmission Functional Test.



FSCAP

Injury to personnel and damage to equipment will result if checks are not performed. These checks are mandatory and must be conducted for safety of flight.

NOTE

Functional test is required to make sure replacement main transmission or replacement module and its components are installed properly, and that lubrication system is working properly.

- a. Prepare helicopter for run up. Make sure helicopter is serviced.



Damage to equipment will result if instrument readings are not carefully

monitored. If any abnormal readings are indicated, shut down helicopter immediately and investigate per PARA 6-3-5.

- b. With qualified personnel, conduct a normal preflight warm-up, and perform the following:
  - (1) Inspect main transmission for leakage.
  - (2) Check main gear box oil temperature and oil pressure for normal operation limits.
  - (3) Record gage readings when they stabilize.
- c. If all checks are normal, with qualified personnel, hover helicopter several feet above ground for 30 minutes. Continuously check oil pressure, oil temperature, and chip indicator for abnormal conditions.
- d. After landing, inspect main transmission for leakage.
- e. Inspect input module chip detector and main module sump chip detector/temperature sensor screens.

**NOTE**

Allow about 2 hours for oil to drain back before checking.

- f. Check oil level on oil level gage (dipstick) and, if necessary, service main transmission (PARA 1-3-12).

**NOTE**

Maintenance test flight is not required after functional test for accessory

module, oil cooler, or any seal replacement.

- g. Perform maintenance test flight after functional test when replacing input or main modules (Appropriate MTF).



6-3-5 MAIN TRANSMISSION SERVICEABILITY INSPECTION.

PARAGRAPH BREAKDOWN

|                                                                                      | PAGE                                                            |
|--------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| 6-3-5.1 Main Transmission Serviceability Inspection                                  | 6-3-5-1                                                         |
| <hr/>                                                                                |                                                                 |
| INITIAL SETUP                                                                        | <ul style="list-style-type: none"> <li>• PARA 6-3-6</li> </ul>  |
| Tools                                                                                | <ul style="list-style-type: none"> <li>• PARA 6-4-1</li> </ul>  |
| <ul style="list-style-type: none"> <li>• Aircraft Mechanic’s Toolkit</li> </ul>      | <ul style="list-style-type: none"> <li>• PARA 6-4-15</li> </ul> |
| <ul style="list-style-type: none"> <li>• Container</li> </ul>                        | <ul style="list-style-type: none"> <li>• PARA 6-4-16</li> </ul> |
| <ul style="list-style-type: none"> <li>• Powertrain Repairer’s Toolkit</li> </ul>    | <ul style="list-style-type: none"> <li>• PARA 6-4-17</li> </ul> |
| Materials/Parts                                                                      | <ul style="list-style-type: none"> <li>• PARA 6-4-18</li> </ul> |
| <ul style="list-style-type: none"> <li>• Cheesecloth, Item 90, Appendix D</li> </ul> | <ul style="list-style-type: none"> <li>• PARA 6-4-19</li> </ul> |
| References                                                                           | <ul style="list-style-type: none"> <li>• PARA 6-4-25</li> </ul> |
| <ul style="list-style-type: none"> <li>• Appendix D</li> </ul>                       | <ul style="list-style-type: none"> <li>• PARA 6-4-28</li> </ul> |
| <ul style="list-style-type: none"> <li>• PARA 1-3-12</li> </ul>                      | <ul style="list-style-type: none"> <li>• PARA 6-4-47</li> </ul> |
| <ul style="list-style-type: none"> <li>• PARA 6-2-2</li> </ul>                       | <ul style="list-style-type: none"> <li>• PARA 6-4-49</li> </ul> |
| <hr/>                                                                                |                                                                 |

6-3-5.1 Main Transmission Serviceability Inspection.

6-3-5.1.1 Pressure Indicator Extended.

- a. If pressure indicator is extended and main transmission oil filter, 70351-08143-101, is installed, perform serviceability inspection (PARA 6-3-5.1.2).
- b. If pressure indicator is extended and main transmission oil filter, 70351-08143-104 or 70351-38801-101, is installed, perform the following:

NOTE

- Only main module sump chip detector/temperature sensor and input module chip detectors have strainers. All main transmission chip detectors require inspection.
- If found, collect and package any debris and particles, and label them for comparison after main transmission run up.
  - (1) Inspect chip detectors and strainers, if applicable, for particles (PARA 6-3-6).

(2) If no chips are found in chip detectors or strainers, serviceability inspection is not required. Perform the following:

- (a) Replace first stage filter element (PARA 6-4-19).
- (b) Check second stage filter element for contamination. If contamination is found, replace second stage filter element (PARA 6-4-19).

(3) If chips are found on chip detectors or strainers, perform serviceability inspection (PARA 6-3-5.1.2).

(4) If pressure indicator continues to pop after all filter elements have been replaced, troubleshoot transmission chip detector system (PARA 6-2-2).

(4) Remove, clean, and install input module chip detectors and strainers for both input modules (PARA 6-4-17).

(5) Remove, clean, and install main module sump chip detector/temperature sensor and strainer from sump (PARA 6-4-18).

c. If no metal particles are found, reconnect oil inlet line to oil cooler radiator (PARA 6-4-49).

#### NOTE

Make sure replacement of oil filter is entered in helicopter logbook.

d. Replace first stage filter element (PARA 6-4-19). Check second stage filter element for contamination. If contamination is found, replace second stage filter element (PARA 6-4-19).

e. Service main transmission (PARA 1-3-12).

f. Ground run helicopter for 30 minutes, and then hover for 30 minutes.

g. After ground run and hover, perform the following:

(1) Inspect chip detector and strainer, if installed, of affected module (PARA 6-3-6).

#### NOTE

If module is being replaced due to an increase in particles, make sure all foreign material from chip detectors, strainers, screens, and/or filters is carefully packaged and returned with module.

(a) If amount of particles has increased, replace affected module (PARA 6-4-1 (for main module), PARA 6-4-25 (for input module), or PARA 6-4-28 (for accessory module)).

(b) If amount of particles has decreased and is within limit, return main transmission to service.

### 6-3-5.1.2 Serviceability Inspection.

#### NOTE

Serviceability inspection is required if:

- (1) pressure indicator is extended and filter, 70351-08143-101, is installed, or
- (2) particles are found on any chip detector or chip detector oil screen.

a. Disconnect oil inlet line from oil cooler radiator (PARA 6-4-49), and drain oil through clean cheesecloth, Item 90, Appendix D, into suitable container.

b. If any metal particles are found, perform the following:

- (1) Replace oil cooler radiator (PARA 6-4-47).
- (2) Drain main transmission oil through clean cheesecloth, Item 90, Appendix D, into suitable container.
- (3) Remove, clean, and install accessory module chip detectors and sleeves (PARA 6-4-15), and accessory module oil screens from both accessory modules (PARA 6-4-16).

- (2) If particles were found in oil cooler inlet line, inspect all oil screens and strainers (PARA 6-4-16, PARA 6-4-17, and PARA 6-4-18).

**NOTE**

If module is being replaced due to an increase in particles, make sure all foreign material from chip detectors, strainers, screens, and/or filters is carefully packaged and returned with module.

- (a) If amount of particles has increased, replace affected module (PARA

6-4-1 (for main module), PARA 6-4-25 (for input module), or PARA 6-4-28 (for accessory module)).

- (b) If amount of particles has decreased and is within limit, return main transmission to service.

- h. If necessary, service main transmission (PARA 1-3-12).

- i. Return helicopter to service.



6-3-6 MAIN TRANSMISSION CHIP DETECTORS PARTICLE INSPECTIONS.

PARAGRAPH BREAKDOWN

|         |                                                       | PAGE    |
|---------|-------------------------------------------------------|---------|
| 6-3-6.1 | General                                               | 6-3-6-1 |
| 6-3-6.2 | Main Transmission Chip Detectors Particle Inspections | 6-3-6-2 |

INITIAL SETUP

- PARA 6-4-1

Tools

- Aircraft Mechanic’s Toolkit
- Container
- Powertrain Repairer’s Toolkit

- PARA 6-4-15
- PARA 6-4-17
- PARA 6-4-18

Materials/Parts

- Cheesecloth, Item 90, Appendix D

- PARA 6-4-25
- PARA 6-4-28

References

- Appendix D
- PARA 6-3-5

- PARA 9-4-16

6-3-6.1 General.

- When CHIP INPUT MDL-LH, CHIP ACCESS MDL-LH, CHIP MAIN MDL SUMP, CHIP INPUT MDL-RH, or CHIP ACCESS MDL-RH light on caution/advisory panel goes on, it indicates which chip detector has been activated.
- Chip detectors and screens in main transmission isolate problems to a particular module, and avoid contaminating lubrication system. If particles are found, it does not necessarily

mean affected module must be replaced. Quantity, size, type, and source of the particles, along with history of main transmission, must be considered. Also, main transmission time, previous malfunctions, operating conditions, temperatures, and pressure should be considered.

- Particles found may be magnetic (steel), nonmagnetic (aluminum, bronze, cadmium, lead, magnesium, or silver), or nonmetallic (dust, baked resin/paint, lint, paint, paper, phenolic, rubber, or wood).

### 6-3-6.2 Main Transmission Chip Detectors Particle Inspections.

- a. As required, remove chip detector and strainer, if installed (PARA 6-4-15, PARA 6-4-17, or PARA 6-4-18).



Main transmission oil filter normally keeps oil clean. Metallic-black oil discoloration found in main transmission is indicative of high rate of fretting and/or wear. Prior to next flight, contact Sikorsky Aircraft Corporation ILS.

- b. Inspect particles. If particles are easily identified as fragments from gears, nuts, bearings, etc., replace applicable module (PARA 6-4-1 (for main module), PARA 6-4-25 (for input module), or PARA 6-4-28 (for accessory module)). If replacement of module is questionable, contact Sikorsky Aircraft Corporation ILS.
- c. Count and measure particles. If 10 or more particles are found, or if any one particle is larger than 0.050-inch diameter or 0.032-inch thick  $\times$  0.125-inch long, replace applicable

module (PARA 6-4-1 (for main module), PARA 6-4-25 (for input module), or PARA 6-4-28 (for accessory module)).

- d. If a module is replaced due to contamination, perform the following:
- (1) Foreign material from screens and filters must be carefully packaged and returned with the module.
  - (2) Main transmission oil must be drained and strained with cheesecloth, Item 90, Appendix D, into suitable container.
  - (3) If necessary, any particles found in oil should be combined with particles from screens and filters for evaluation and returned with module.
  - (4) Package cheesecloth, Item 90, Appendix D, put oil in sealed container, and return both with module.
- e. For appropriate action after determining particle material, size, and quantity, refer to Table 6-3-6-1.
- f. If required, install chip detectors and strainers (PARA 6-4-15, PARA 6-4-17, or PARA 6-4-18).

**Table 6-3-6-1. Particle Classification**

| <b>Material</b>                                                     | <b>Type-Quantity Size</b>                                                                                                   | <b>Action Required</b>                                                                                                                                                                                                                        |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Magnetic (Steel)                                                    | Fuzz, fine hair-like particles.                                                                                             | Acceptable. Return main transmission to service. No main transmission serviceability inspection is required.                                                                                                                                  |
|                                                                     | No more than 10 particles, none over 0.050-inch diameter or over 0.032-inch thick × 0.125-inch long.                        | Perform main transmission serviceability inspection (PARA 6-3-5). Record findings on log card. Refer to Table 6-3-6-2.                                                                                                                        |
|                                                                     | Splinters or granular particles.                                                                                            | Perform main transmission serviceability inspection (PARA 6-3-5).                                                                                                                                                                             |
|                                                                     | Any quantity of chips beyond the above dimensions.                                                                          | Replace applicable module (PARA 6-4-1 (for main module), PARA 6-4-25 (for input module), or PARA 6-4-28 (for accessory module)).                                                                                                              |
|                                                                     | Flathead #8 machine screws.                                                                                                 | Remove screws. Replace main generator (PARA 9-4-16).                                                                                                                                                                                          |
| Nonmagnetic (Aluminum, Bronze, Cadmium, Lead, Magnesium, or Silver) | Pieces of small coil spring. Overall length of unstretched spring from lower main rotor shaft seal, approximately 9-inches. | Performance of main rotor shaft seal will not be seriously affected by loss of spring. Measure unstretched spring (9-inches) to make sure entire spring has been recovered. Perform main transmission serviceability inspection (PARA 6-3-5). |
|                                                                     | Thin flakes not over 0.050-inch diameter or over 0.032-inch thick × 0.125-inch long. No more than 10 flakes permitted.      | Acceptable. Return main transmission to service. No main transmission serviceability inspection is required.                                                                                                                                  |
|                                                                     | More than 10 flakes, but less than 25.                                                                                      | Perform main transmission serviceability inspection (PARA 6-3-5). Record findings on log card. Refer to Table 6-3-6-2.                                                                                                                        |

Table 6-3-6-1. Particle Classification (Cont)

| Material                                                                             | Type-Quantity Size                                                                                                         | Action Required                                                                                                                                                                       |
|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nonmetallic (Dust, Baked Resin/Paint, Lint, Paint, Paper, Phenolic, Rubber, or Wood) | Flakes of aluminum material that appear to be shim-like, and gear box reported to have drop in pressure of 10 psi or more. | Replace main module (PARA 6-4-1).                                                                                                                                                     |
|                                                                                      | More than 25 flakes.                                                                                                       | Replace module (PARA 6-4-1 (for main module), PARA 6-4-25 (for input module), or PARA 6-4-28 (for accessory module)).                                                                 |
|                                                                                      | Flakes not over 0.050-inch diameter or over 0.032-inch thick $\times$ 0.125-inch long. No more than 10 flakes permitted.   | Acceptable. Return main transmission to service. No main transmission serviceability inspection is required.                                                                          |
|                                                                                      | Less than a total area of $\frac{1}{2}$ -square-inch.                                                                      | Perform main transmission serviceability inspection (PARA 6-3-5). Refer to Table 6-3-6-2.                                                                                             |
|                                                                                      | More than a total area of $\frac{1}{2}$ -square-inch.                                                                      | If contamination is sufficient to partially block oil strainer, replace module (PARA 6-4-1 (for main module), PARA 6-4-25 (for input module), or PARA 6-4-28 (for accessory module)). |



Table 6-3-6-1. Particle Classification (Cont)

| Material             | Type-Quantity Size                  | Action Required                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nonmetallic (Rubber) | More than an area of ¼-square-inch. | Conduct visual leakage inspection of all external oil seal areas and main transmission housing attachment joints. Evaluate indicator oil pressure readings. If low oil pressure warning light has been illuminated, and if leakage source can not be corrected or identified, remove applicable module (PARA 6-4-1 (for main module), PARA 6-4-25 (for input module), or PARA 6-4-28 (for accessory module)). |

NOTE

If continuing chip indications are being found, the following limitations in Table 6-3-6-2 apply.



Table 6-3-6-2. Particle Classification - Continuing Indications

| Material                                                            | Type-Quantity<br>Size                                                                                             | Time Interval  | Number of<br>Indications | Action Required                                                                                                                  |
|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Magnetic (Steel)                                                    | More than 2 flakes or granular particles over 0.005-inch thick or 0.020-inch diameter. Noted at each indication.  | 10 hour period | Three                    | Replace applicable module (PARA 6-4-1 (for main module), PARA 6-4-25 (for input module), or PARA 6-4-28 (for accessory module)). |
|                                                                     |                                                                                                                   | 25 hour period | Five                     | Replace applicable module (PARA 6-4-1 (for main module), PARA 6-4-25 (for input module), or PARA 6-4-28 (for accessory module)). |
|                                                                     |                                                                                                                   | 50 hour period | Nine                     | Replace applicable module (PARA 6-4-1 (for main module), PARA 6-4-25 (for input module), or PARA 6-4-28 (for accessory module)). |
| Nonmagnetic (Aluminum, Bronze, Cadmium, Lead, Magnesium, or Silver) | More than 10 flakes or granular particles over 0.032-inch thick or 0.125-inch diameter. Noted at each indication. | 10 hour period | Three                    | Replace applicable module (PARA 6-4-1 (for main module), PARA 6-4-25 (for input module), or PARA 6-4-28 (for accessory module)). |

Table 6-3-6-2. Particle Classification - Continuing Indications (Cont)

| Material | Type-Quantity<br>Size                                                                                                      | Time Interval | Number of<br>Indications | Action Required                   |
|----------|----------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------|-----------------------------------|
|          | Flakes of aluminum material that appear to be shim-like, and gear box reported to have drop in pressure of 10 psi or more. |               |                          | Replace main module (PARA 6-4-1). |

NOTE

To make use of time intervals and number of indications, chip light history must be entered on log cards.

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6-3-7    MAIN TRANSMISSION LUBRICATION SYSTEM INSPECTION.

PARAGRAPH BREAKDOWN

|         |                                            | PAGE    |
|---------|--------------------------------------------|---------|
| 6-3-7.1 | Main Transmission Oil<br>Filter Inspection | 6-3-7-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 6-3-5

6-3-7.1    Main Transmission Oil Filter Inspection.

Inspect pressure indicator on filter housing. If pressure indicator is popped, refer to PARA 6-3-5.



6-3-8 INPUT MODULE INSPECTIONS.

PARAGRAPH BREAKDOWN

|         |                                     | PAGE    |         |                               | PAGE    |
|---------|-------------------------------------|---------|---------|-------------------------------|---------|
| 6-3-8.1 | Elastomeric Gimbal Inspection       | 6-3-8-1 | 6-3-8.3 | Input Seal Leakage Inspection | 6-3-8-3 |
| 6-3-8.2 | Input Module/Quill Shaft Inspection | 6-3-8-3 | 6-3-8.4 | Input Pinion Inspection       | 6-3-8-3 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Dial Indicator
- Feeler Gage, 0.005-inch
- Powertrain Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 1A, Appendix D
- Abrasive Wheel, Item 22, Appendix D

- Corrosion Resistant Coating, Item 118, Appendix D

References

- Appendix D
- PARA 6-4-12
- PARA 6-4-25
- PARA 6-4-26

6-3-8.1 Elastomeric Gimbal Inspection.

- Inspect elastomeric gimbal, 70361-08001-101, as follows:
  - USING FINGERS ONLY, CHECK EIGHT ALLEN-HEAD BOLTS FOR TIGHTNESS.**
  - If 1 to 4 bolts are loose, using 0.005-inch feeler gage, inspect for bond separation (Figure 6-3-8-1).
  - If more than 4 bolts are loose, replace elastomeric gimbal (PARA 6-4-25).
  - Check elastomeric gimbal for nicks, scratches, or gouges. **DAMAGE UP TO**

0.015-INCH DEEP CAN BE BLENDED OUT. DAMAGE MUST BE LESS THAN 15% OF CIRCUMFERENCE OF ELASTOMERIC GIMBAL.

- Using abrasive cloth, Item 1A, Appendix D, or abrasive wheel, Item 22, Appendix D, as needed, blend out damage. Blend out all reworked areas to a ½-inch radius.
  - Using corrosion resistant coating, Item 118, Appendix D, touch up reworked areas.
- Inspect elastomeric gimbal, 70361-08001-102, as follows:

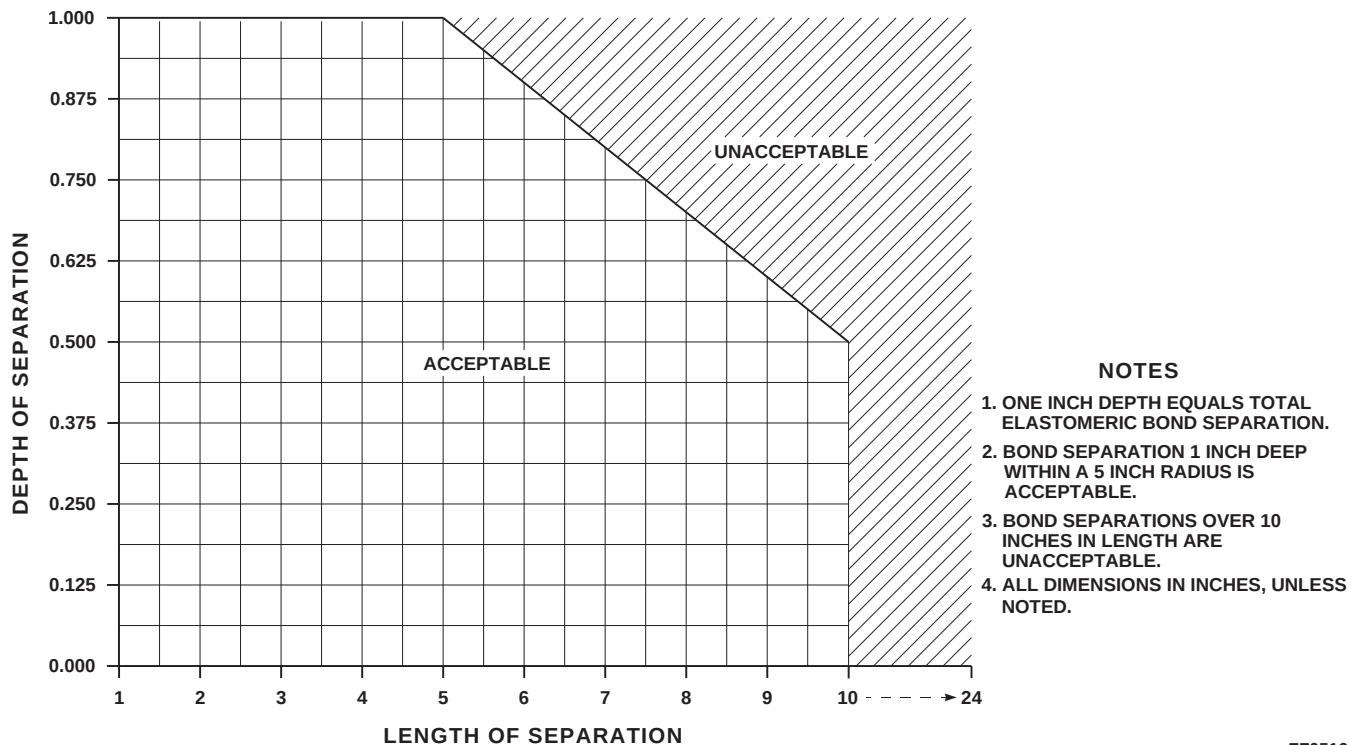
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FIGURE 6-3-8-1. INSPECT ELASTOMERIC GIMBAL

- (1) Visually inspect for loose or broken bolts.
  - (2) If 1 to 4 bolts are loose, **RETORQUE LOOSE BOLTS TO 57 - 63 INCH-POUNDS.**
  - (3) If any bolts are broken, replace elastomeric gimbal (PARA 6-4-25).
  - (4) If loose bolts are found, using 0.005-inch feeler gage, inspect for bond separation (Figure 6-3-8-1).
  - (5) Check elastomeric gimbal for nicks, scratches, or gouges. **DAMAGE UP TO 0.015-INCH DEEP CAN BE BLENDED OUT. DAMAGE MUST BE LESS THAN 15% OF CIRCUMFERENCE OF ELASTOMERIC GIMBAL.**
  - (6) Using abrasive cloth, Item 1A, Appendix D, or abrasive wheel, Item 22, Appendix D, as needed, blend out damage. Blend out all reworked areas to a ½-inch radius. Using corrosion resistant coating, Item 118, Appendix D, touch up reworked areas.
- c. Inspect elastomeric gimbal, 70361-08001-103 as follows:
- (1) Visually inspect for loose bolts. **RETORQUE ANY LOOSE BOLT TO 81 - 89 INCH-POUNDS.**
  - (2) If any bolts are broken, replace elastomeric gimbal (PARA 6-4-25).
  - (3) If loose bolts are found, using 0.005-inch feeler gage, inspect for bond separation (Figure 6-3-8-1).
  - (4) Check elastomeric gimbal for nicks, scratches or gouges. **DAMAGE UP TO 0.015-INCH DEEP CAN BE BLENDED OUT. DAMAGE MUST BE LESS THAN 15% OF CIRCUMFERENCE OF ELASTOMERIC GIMBAL.**
  - (5) Using abrasive cloth, Item 1A, Appendix D, or abrasive wheel, Item 22, Appendix D, as needed, blend out damage. Blend out all reworked areas to a ½-inch radius. Using corrosion resistant coating, Item



118, Appendix D, touch up reworked areas.

### 6-3-8.2 Input Module/Quill Shaft Inspection.

- a. Check input module housing surface for paint discoloration. Discoloration from original aluminized-gray to a flat tone of gray is acceptable. **IF ANY PAINT HAS DISCOLORED TO A BROWN TONE, INDICATING POSSIBLE EXCESSIVE HEAT OR OIL LEAKAGE, REPLACE INPUT MODULE** (PARA 6-4-25).
- b. Visually inspect input module housing surface for nicks, gouges, scratches, and corrosion (PARA 6-4-12).
- c. Inspect for axial movement of input pinion (PARA 6-3-8.4).
- d. Visually inspect quill shaft and splines for damage. **NONE ALLOWED.** If damaged, replace quill shaft.

### 6-3-8.3 Input Seal Leakage Inspection.

#### NOTE

- Checking amount of input seal leakage can prevent premature seal replacement,

before leakage is over recommended limits.

- Allow at least 2 hours after flight for oil to drain back into sump before checking level.

Measure seal leakage by observing drop in oil level on oil level gage (dipstick). **IF LEAKAGE IS MORE THAN 1/8-INCH AFTER ANY ONE FLIGHT, REPLACE INPUT MODULE SEAL** (PARA 6-4-26).

### 6-3-8.4 Input Pinion Inspection.

#### NOTE

This procedure is to be performed whenever input module input flange, input module, elastomeric gimbal, output shaft, or engine is removed or replaced.

- a. Press input flange forward toward input module to remove any axial play in pinion shaft.
- b. Move input flange to rear and measure axial play using dial indicator. **MAXIMUM ALLOWABLE AXIAL PLAY IS 0.030-INCH.** If limit is exceeded, replace input module (PARA 6-4-25).



6-3-9 ACCESSORY MODULE INSPECTION.

PARAGRAPH BREAKDOWN

|                            | PAGE    |
|----------------------------|---------|
| 6-3-9.1 Leakage Inspection | 6-3-9-1 |

INITIAL SETUP

References

Tools

- PARA 6-4-28

- Aircraft Mechanic’s Toolkit
- Powertrain Repairer’s Toolkit

6-3-9.1 Leakage Inspection.

- Visually inspect accessory module for oil leaks, paying close attention to area adjacent generator mounting pad. **NONE ALLOWED.** If leaks are found, replace accessory module (PARA 6-4-28).
- If any of the following conditions exist, remove accessory module and inspect oil screens (PARA 6-4-28):
  - Increase of main gear box oil temperature above 212°F (100°C).

- Main gear box oil pressure fluctuates below 30 psi.
- Accessory module chip detector light (CHIP ACCESS MDL-LH or CHIP ACCESS MDL-RH) on the caution/advisory panel is on.
- Accessory and/or input module housing paint discolored from aluminized-gray to brown.



6-3-10 INTERMEDIATE GEAR BOX INSPECTIONS.

PARAGRAPH BREAKDOWN

|          |                                                      | PAGE     |          |                                                          | PAGE     |
|----------|------------------------------------------------------|----------|----------|----------------------------------------------------------|----------|
| 6-3-10.1 | Intermediate Gear Box Input Seal Leakage Inspection  | 6-3-10-1 | 6-3-10.3 | Intermediate Gear Box Sight Plug Inspection              | 6-3-10-2 |
| 6-3-10.2 | Intermediate Gear Box Output Seal Leakage Inspection | 6-3-10-2 | 6-3-10.4 | Intermediate Gear Box Input and Output Flange Inspection | 6-3-10-2 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Powertrain Repairer’s Toolkit
- Tail Drive Flange Wrench, 70700-20687-102
- Torque Reactor, 70700-20688-041

Materials/Parts

- Abrasive Cloth, Item 6, Appendix D
- Abrasive Cloth, Item 10, Appendix D

References

- Appendix D
- PARA 6-4-32
- PARA 6-4-33
- PARA 6-4-34

6-3-10.1 Intermediate Gear Box Input Seal Leakage Inspection.



Operation with excess leakage may result in gear box replacement. One-hour leak rate must not be used for operation greater than one hour. Leak rates are determined by amount of leakage that can occur in specified amount of time without affecting gear box performance. If flights of longer dura-

tion are anticipated, use rates for greater hours.

NOTE

- Checking amount of input seal leakage can prevent premature seal replacement, before leakage is over recommended limits.
- Allow at least 45 minutes after flight for oil to drain back into sump before checking level.
- If operation falls between intervals shown in Table 6-3-10-1, use next

higher time interval. Refer to notes in Table 6-3-10-1 for examples.

Measure seal leakage by observing drop in oil level on intermediate gear box sight plug (Table 6-3-10-1). **LEAKAGE MUST NOT BE MORE THAN 1/8-INCH AFTER ANY ONE FLIGHT.** If leakage exceeds limit, replace input seal (PARA 6-4-32).

### 6-3-10.2 Intermediate Gear Box Output Seal Leakage Inspection.

#### WARNING

Operation with excess leakage may result in gear box replacement. One-hour leak rate must not be used for operation greater than one hour. Leak rates are determined by amount of leakage that can occur in specified amount of time without affecting gear box performance. If flights of longer duration are anticipated, use rates for greater hours.

#### NOTE

- Checking amount of output seal leakage can prevent premature seal replacement before leakage is over recommended limits.
- Allow at least 45 minutes after flight for oil to drain back into sump before checking level.
- If operation falls between intervals shown in Table 6-3-10-1, use next higher time interval. Refer to notes in Table 6-3-10-1 for examples.

Measure seal leakage by observing drop in oil level on intermediate gear box sight plug (Table 6-3-10-1). **LEAKAGE MUST NOT BE MORE THAN 1/8-INCH AFTER ANY ONE FLIGHT. IF LEAKAGE EXCEEDS LIMIT, REPLACE INTERMEDIATE GEAR BOX OUTPUT SEAL (PARA 6-4-33).**

### 6-3-10.3 Intermediate Gear Box Sight Plug Inspection.

- Inspect intermediate gear box sight plug for cracks. **NONE ALLOWED.** If cracked, replace intermediate gear box sight plug (PARA 6-4-34).
- Check oil level on intermediate gear box sight plug. **OIL LEVEL MUST BE VISIBLE THROUGH LENS, EVEN WITH CRAZING OR STAINING.** If oil level can not be seen, replace intermediate gear box sight plug (PARA 6-4-34).
- Inspect intermediate gear box sight plug for leaks. **NONE ALLOWED.** If leaks found replace intermediate gear box sight plug (PARA 6-4-34).

### 6-3-10.4 Intermediate Gear Box Input and Output Flange Inspection.

#### NOTE

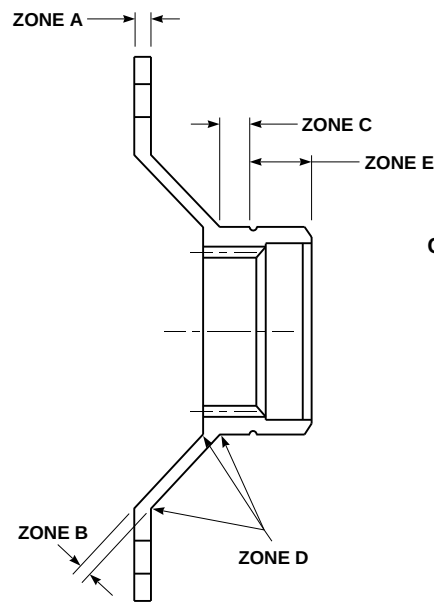
Rework of intermediate gear box input flange may be done on helicopter.

- If necessary, remove damaged intermediate gear box input flange (PARA 6-4-32) and/or intermediate gear box output flange (PARA 6-4-33).

#### NOTE

Wear pattern caused by seal is not cause for replacement, unless wear causes seal to leak.

- Inspect chrome-plated seal journal area of flange for nicks or scratches. **NONE ALLOWED.** If damage is found, replace intermediate gear box input flange (PARA 6-4-32) and/or intermediate gear box output flange (PARA 6-4-33).
- Fine scratches may be blended smooth using abrasive cloth, Item 10, Appendix D.
- Check present dimension of damaged area (Figure 6-3-10-1).
- Check depth of damaged area to determine whether intermediate gear box input flange



| COMPONENT | ZONE | DIMENSION BEFORE REPAIR | MINIMUM DIMENSION AFTER REPAIR |
|-----------|------|-------------------------|--------------------------------|
| LUG       | A    | 0.175 (±0.005)          | 0.160 (SEE NOTE 3)             |
| WEB       | B    | 0.105 (±0.005)          | 0.090                          |
| OD        | C    | 2.270 (±0.010)          | 2.250                          |
| RADIUS    | D    | 0.125 (±0.010)          | 0.105                          |
| LINER     | E    | 2.2505 (±0.0015)        | 2.248 (SEE NOTE 2)             |

NOTES

- ALL DIMENSIONS ARE IN INCHES, UNLESS NOTED.
- NO SCRATCHES OR NICKS ALLOWED IN CHROME-PLATED LINER.
- NO REMOVAL OF MATERIAL ALLOWED IN WASHER CONTACT AREA.

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FIGURE 6-3-10-1. REPAIR INTERMEDIATE GEAR BOX INPUT AND OUTPUT FLANGE

minimum dimension after repair can be maintained.

- If damage can be repaired, using abrasive cloth, Item 6, Appendix D, blend smooth. Remove only what is necessary, so flange will stay within limits.
- Using abrasive cloth, Item 10, Appendix D, blend reworked area to remove all traces of abrasion marks and mark surface smoothness.
- Repair nicks, scratches, or scoring by polishing as follows:
  - Polish only in damaged area. Do not polish circumference, unless damage extends around it.
  - Polish with blend radius of 1/2-inch or greater.
- If removed, install intermediate gear box input flange (PARA 6-4-32) and/or intermediate gear box output flange (PARA 6-4-33).

- If intermediate gear box input or output flange was not removed, perform torque check as follows:
  - Remove cotter pin from retaining nut and install torque reactor.

NOTE

Check torque in tightening direction only.

- Using torque reactor and tail drive flange wrench, check torque of retaining nut. **IF TORQUE IS 214 - 236 FOOT-POUNDS, INSTALL NEW COTTER PIN. IF REQUIRED TO ALIGN COTTER PIN HOLE, TURN RETAINING NUT IN TIGHTENING DIRECTION.** Torque check is complete.
- IF TORQUE IS LESS THAN 214 FOOT-POUNDS, INSPECT RETAINING NUT AND FLANGE FOR BINDING AND CORROSION. REPLACE RETAINING NUT IF:**

- (a) PITS ARE 0.030-INCH DEEP.
- (b) PITTED MORE THAN 10% OF MATING SURFACE.
- (c) THREADS SHOW SIGNS OF CORROSION.
- (d) RETAINING NUT CAN NOT BE FULLY RUN ON BY HAND.
- (4) IF RETAINING NUT IS NOT BOUND OR CORRODED, USING TORQUE REACTOR AND TAIL DRIVE FLANGE WRENCH, TORQUE RETAINING NUT TO 214 - 236 FOOT-POUNDS. INSTALL COTTER PIN. IF REQUIRED TO ALIGN COTTER PIN HOLE, TURN RETAINING NUT IN TIGHTENING DIRECTION.

Table 6-3-10-1. Intermediate Gear Box Input and Output Seals Allowable Leakage Limits

| Hours of Operation             | Allowable Leakage                                          |
|--------------------------------|------------------------------------------------------------|
| One Hour (see Note 1)          | $\frac{1}{8}$ -inch = 3.38 fluid ounces = 100 milliliters  |
| Five Hours (see Notes 1 and 2) | $\frac{1}{4}$ -inch = 6.76 fluid ounces = 200 milliliters  |
| Ten Hours (see Note 2)         | $\frac{1}{2}$ -inch = 13.52 fluid ounces = 400 milliliters |

## NOTES

1. If flight is between one and five hours, use five hours allowable leakage rate.
2. If flight is between five and ten hours, use ten hours allowable leakage rate.
3. Anytime flight hour exceeds given interval, leakage rate per hour is 0.05-inch, which is equivalent to 3.2 ounces per hour (for main gear box) and 1.3 ounces per hour (for intermediate and tail gear boxes).
4. Example: Leakage at  $\frac{1}{8}$ -inch per hour is experienced and a subsequent flight of 5 continuous flight hours is planned.  $\frac{5}{8}$ -inch of oil will be expected to be lost which exceeds allowable leak rate. Since subsequent servicing after 5-hour flight may indicate that gearbox ran low on oil requiring gearbox replacement, seal replacement is required to prevent leakage.



6-3-11 INTERMEDIATE GEAR BOX CHIP DETECTOR/TEMPERATURE SENSOR PARTICLE INSPECTIONS.

PARAGRAPH BREAKDOWN

|          |                                                                                    | PAGE     |
|----------|------------------------------------------------------------------------------------|----------|
| 6-3-11.1 | General                                                                            | 6-3-11-1 |
| 6-3-11.2 | Intermediate Gear Box Chip Detector/<br>Temperature Sensor<br>Particle Inspections | 6-3-11-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container

Materials/Parts

- Cheesecloth, Item 90, Appendix D

References

- Appendix D
- PARA 6-3-12
- PARA 6-4-31
- PARA 6-4-37

6-3-11.1 General.

- When CHIP INT XMSN light on caution/ advisory panel goes on, it indicates chip detector sensor in intermediate gear box has activated.
- Metal particles found on chip detector may indicate an internal part of intermediate gear box has malfunctioned. If particles are found, it does not necessarily mean intermediate gear box must be replaced. Quantity, size, type, and source of particles, along with history of the intermediate gear box, must be considered. Also intermediate gear box time, previous malfunctions, and operating conditions should be considered.
- Particles found may be magnetic (steel), nonmagnetic (aluminum, bronze, cadmium, lead, magnesium, or silver), or nonmetallic

(dust, baked resin/paint, lint, paint, paper, phenolic, rubber, or wood).

6-3-11.2 Intermediate Gear Box Chip Detector/Temperature Sensor Particle Inspections.

- Remove chip detector (PARA 6-4-37).

NOTE

Intermediate gear boxes do not have oil filters and discoloration is possible.

- Inspect particles. If particles are easily identified as fragments from gears, nuts, bearings, etc., replace intermediate gear box (PARA 6-4-31) . If replacement of intermediate gear box is questionable, contact Sikorsky Aircraft Corporation ILS.

- c. Count and measure particles. If 10 or more particles are found, or if any one particle is larger than 0.050-inch diameter or 0.032-inch  $\times$  0.125-inch long, replace intermediate gear box (PARA 6-4-31).
- d. If a intermediate gear box is replaced due to contamination, perform the following:
  - (1) Foreign material from screens and filters must be carefully packaged and returned with the intermediate gear box.
  - (2) Intermediate gear box oil must be drained and strained with cheesecloth, Item 90, Appendix D, into suitable container.
  - (3) If necessary, any particles found in oil should be combined with particles from screens and filters for evaluation and returned with intermediate gear box.
  - (4) Package cheesecloth, Item 90, Appendix D, put oil in sealed container, and return both with intermediate gear box.
- e. For appropriate action after determining particle material, size, and quantity, refer to Table 6-3-11-1.
- f. Install chip detector (PARA 6-4-37).

**Table 6-3-11-1. Particle Classification**

| <b>Material</b>                                                                       | <b>Type-Quantity Size</b>                                                                                                     | <b>Action Required</b>                                                                                                       |
|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Magnetic (Steel)                                                                      | Fuzz, fine hair-like particles.                                                                                               | Acceptable. Return intermediate gear box to service. No intermediate gear box serviceability inspection is required.         |
|                                                                                       | No more than 10 particles, none over 0.050-inch diameter or over 0.032-inch thick $\times$ 0.125-inch long.                   | Perform intermediate gear box serviceability inspection (PARA 6-3-12). Record findings on log card. Refer to Table 6-3-11-2. |
|                                                                                       | Splinters or granular particles.                                                                                              | Perform intermediate gear box serviceability inspection (PARA 6-3-12).                                                       |
|                                                                                       | Any quantity of chips beyond the above dimensions.                                                                            | Replace intermediate gear box (PARA 6-4-31).                                                                                 |
| Nonmagnetic (Aluminum, Bronze, Cadmium, Lead, Magnesium, or Silver)                   | Thin flakes not over 0.050-inch diameter or over 0.032-inch thick $\times$ 0.125-inch long. No more than 10 flakes permitted. | Acceptable. Return intermediate gear box to service. No intermediate gear box serviceability inspection is required.         |
|                                                                                       | More than 10 flakes, but less than 25.                                                                                        | Perform intermediate gear box serviceability inspection (PARA 6-3-12). Record findings on log card. Refer to Table 6-3-11-2. |
|                                                                                       | More than 25 flakes.                                                                                                          | Replace intermediate gear box (PARA 6-4-31).                                                                                 |
| Nonmetallic (Dust, Baked Resin/ Paint, Lint, Paint, Paper, Phenolic, Rubber, or Wood) | Flakes not over 0.050-inch diameter or over 0.032-inch thick $\times$ 0.125-inch long. No more than 10 flakes permitted.      | Acceptable. Return intermediate gear box to service. No intermediate gear box serviceability inspection is required.         |
|                                                                                       | Less than a total area of $\frac{1}{2}$ -square-inch.                                                                         | Perform intermediate gear box serviceability inspection (PARA 6-3-12). Refer to Table 6-3-11-2.                              |
|                                                                                       | More than a total area of $\frac{1}{2}$ -square-inch.                                                                         | If contamination is sufficient to partially block oil strainer, replace intermediate gear box (PARA 6-4-31).                 |

Table 6-3-11-1. Particle Classification (Cont)

| Material             | Type-Quantity Size                    | Action Required                                                                                                                                                                                                                                                                                                                   |
|----------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nonmetallic (Rubber) | More than an area of 1/4-square-inch. | Conduct visual leakage inspection of all external oil seal areas and intermediate gear box housing attachment joints. Evaluate indicator oil pressure readings. If the low oil pressure warning light has been illuminated, and if leakage source can not be corrected or identified, remove intermediate gear box (PARA 6-4-31). |

## NOTE

If continuing chip indications are being found, the following limitations in Table 6-3-11-2 apply.

---

**Table 6-3-11-2. Particle Classification - Continuing Indications**

| <b>Material</b>                                                     | <b>Type-Quantity<br/>Size</b>                                                                                     | <b>Time Interval</b> | <b>Number of<br/>Indications</b> | <b>Action Required</b>                       |
|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------|----------------------------------|----------------------------------------------|
| Magnetic (Steel)                                                    | More than 2 flakes or granular particles over 0.005-inch thick or 0.020-inch diameter. Note at each indication.   | 10 hour period       | Three                            | Replace intermediate gear box (PARA 6-4-31). |
|                                                                     |                                                                                                                   | 25 hour period       | Five                             | Replace intermediate gear box (PARA 6-4-31). |
|                                                                     |                                                                                                                   | 50 hour period       | Nine                             | Replace intermediate gear box (PARA 6-4-31). |
| Nonmagnetic (Aluminum, Bronze, Cadmium, Lead, Magnesium, or Silver) | More than 10 flakes or granular particles over 0.032-inch thick or 0.125-inch diameter. Noted at each indication. | 10 hour period       | Three                            | Replace intermediate gear box (PARA 6-4-31). |

**NOTE**

To make use of time interval and number of indications, chip light history must be entered on log cards.



6-3-12 INTERMEDIATE GEAR BOX SERVICEABILITY INSPECTION.

PARAGRAPH BREAKDOWN

|                                                          | PAGE     |
|----------------------------------------------------------|----------|
| 6-3-12.1 Intermediate Gear Box Serviceability Inspection | 6-3-12-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 2-Quart
- Drain Line, H538A-3
- Powertrain Repairer’s Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D

References

- Appendix D
- PARA 1-3-13
- PARA 6-4-31
- PARA 6-4-37

6-3-12.1 Intermediate Gear Box Serviceability Inspection.

NOTE

Perform this procedure if high metal count is detected during oil sampling, or if particles were found on intermediate gear box chip detector/temperature sensor.

- Drain intermediate gear box oil through cheesecloth as follows:
  - Remove intermediate gear box chip detector/temperature sensor (PARA 6-4-37).

NOTE

Position 2-quart container and cheesecloth as necessary so oil will be strained though cheesecloth when drained.

- Position 2-quart container and clean cheesecloth, Item 90, Appendix D. Place one end of drain line into container. Insert other end into self-closing valve by pushing in and turning clockwise.
- Remove intermediate gear box oil filler plug and drain oil through cheesecloth into container.
- Remove drain line.
- If required, inspect and clean intermediate gear box chip detector/temperature sensor (PARA 6-4-37).
- Install intermediate gear box chip detector/temperature sensor (PARA 6-4-37).
- Service intermediate gear box (PARA 1-3-13).
- Ground run helicopter for 30 minutes, and then hover for 30 minutes.

- f. After ground run and hover, perform the following:
  - (1) Inspect intermediate gear box chip detector/temperature sensor (PARA 6-4-37).
  - (2) If amount of metal particles has increased, replace intermediate gear box (PARA 6-4-31).
  - (3) If amount of particles has decreased and is within limits, serviceability inspection is complete.
- g. If necessary, service intermediate gear box (PARA 1-3-13).
- h. Return helicopter to service.



6-3-13 TAIL ROTOR DRIVE SHAFT INSPECTION.

PARAGRAPH BREAKDOWN

|                                         | PAGE     |
|-----------------------------------------|----------|
| 6-3-13.1 Inspect Tail Rotor Drive Shaft | 6-3-13-1 |

INITIAL SETUP

- PARA 6-4-40

Tools

- Aircraft Mechanic’s Toolkit

- PARA 6-4-41

References

- PARA 6-4-38
- PARA 6-4-39

- PARA 6-4-50

6-3-13.1 Inspect Tail Rotor Drive Shaft.

NOTE

Inspection of all tail rotor drive shaft sections is the same, except as noted.

drive shaft tube to flange and stub shaft is not cause for rejection. Replace tail rotor drive shaft if corrosion between drive shaft tube and flange or between drive shaft tube and stub shaft is present, or if fasteners are loose.

6-3-13.1.1. Inspect.

NOTE

The inspections in step a. and step b. are performed during the 500 hour inspection when the flexible coupling is disconnected.

- Visually inspect tail rotor drive shaft for cracks. **NONE ALLOWED.** If cracks are found, replace applicable section(s) of tail rotor drive shaft (PARA 6-4-38 (for section I), PARA 6-4-39 (for section II), PARA 6-4-40 (for section III), or PARA 6-4-41 (for section IV)).

- Visually inspect tail rotor drive shaft for corrosion between drive shaft tube and stub shaft, and drive shaft tube and flange. Corrosion between drive shaft tube and stub shaft, and drive shaft tube and flange is identified by flaking or lifting of paint, crack, or presence of white powder on shaft. **NONE ALLOWED.** If corrosion is found, replace applicable section(s) of tail rotor drive shaft (PARA 6-4-38 (for section I), PARA 6-4-39 (for section II), PARA 6-4-40 (for section III), or PARA 6-4-41 (for section IV)).
- Inspect tail rotor drive shaft section II, III, and IV flange boltholes for elongation. **NONE ALLOWED.** If elongated, replace applicable section(s) of tail rotor drive shaft (PARA 6-4-39 (for section II), PARA 6-4-40 (for section III), or PARA 6-4-41 (for section IV)).

NOTE

Light surface corrosion on inside collars of fasteners used to connect tail rotor

- d. Inspect oil cooler fan exit flange and oil cooler drive shaft flange boltholes for elongation. **NONE ALLOWED.** If oil cooler fan exit flange boltholes are elongated, replace oil cooler or fan

(PARA 6-4-50). If drive shaft boltholes are elongated, replace oil cooler or drive shaft (PARA 6-4-50).

6-3-14 TAIL ROTOR DRIVE SHAFT VISCOUS DAMPER BEARING ASSEMBLY INSPECTION.

PARAGRAPH BREAKDOWN

|          |                                                                           | PAGE     |
|----------|---------------------------------------------------------------------------|----------|
| 6-3-14.1 | Inspect Tail Rotor Drive Shaft Viscous Damper Bearing Assembly Inspection | 6-3-14-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Feeler Gage, 0.002-inch
- Flashlight
- Powertrain Repairer’s Toolkit

References

- PARA 1-3-16
- PARA 6-4-39
- PARA 6-4-44
- PARA 6-4-45
- PARA 6-4-50

6-3-14.1 Inspect Tail Rotor Drive Shaft Viscous Damper Bearing Assembly Inspection.

tube assembly. If disbonding is found, perform the following:

NOTE

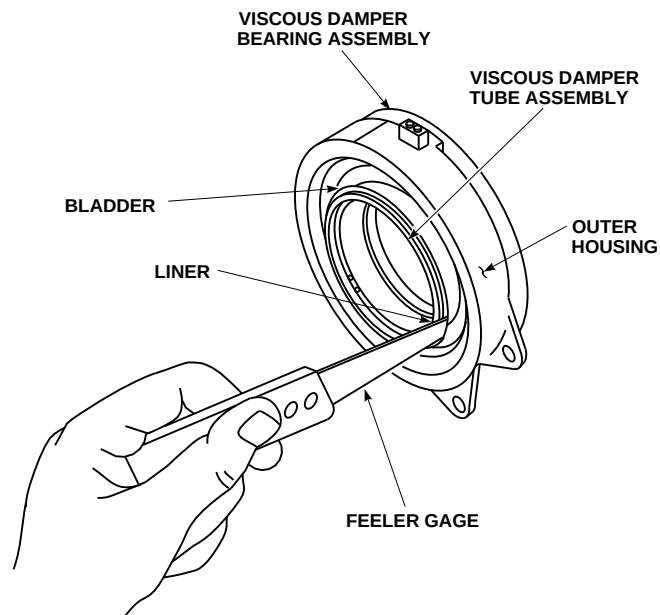
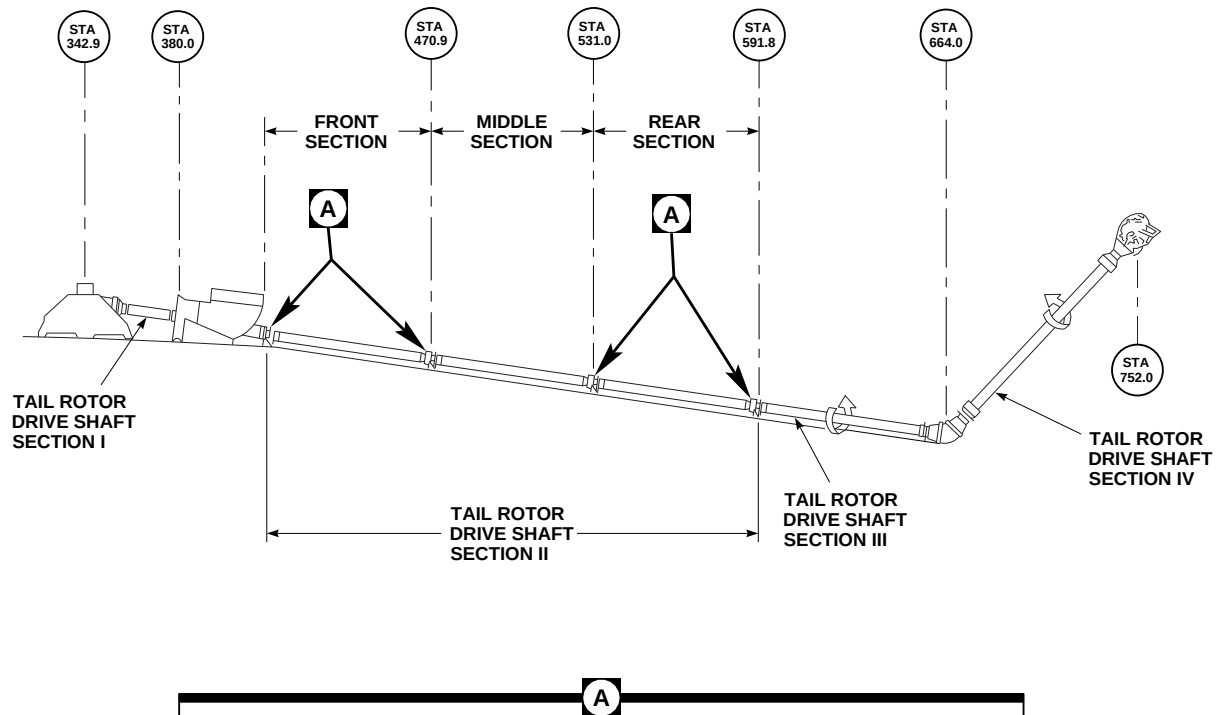
Variations in tolerance may allow movement of viscous damper tube assembly within outer housing. Axial and radial movement inside outer housing is not cause for rejection/replacement, provided the viscous damper tube assembly does not become cocked inside outer housing and does not move past snapping.

- Visually inspect bearing, viscous damper tube assembly, tube housing, and bearing mount area for indications of excessive temperature.
- Visually inspect bearing for evidence of disbonding between liner and viscous damper



Damage to bladder will result if feeler gage is not used carefully. Use care when inserting feeler gage.

- Using 0.002-inch feeler gage, carefully insert and measure depth of separation on both sides of tail rotor drive shaft (Figure 6-3-14-1, Detail A).
- MAXIMUM ALLOWABLE SEPARATION IS 0.200-INCH DEEP FOR 40% OF THE CIRCUMFERENCE ON ONE OR BOTH SIDES.** If separation exceeds limit, replace liner and viscous damper tube assembly (PARA 6-4-45).



**NOTE**  
BEARING SHOWN REMOVED.

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**FIGURE 6-3-14-1. INSPECT TAIL ROTOR DRIVE SHAFT VISCIOUS DAMPER BEARING ASSEMBLY**

**6-3-14-2**

- c. Inspect both sides of bearing to make sure that bearing seals are intact and properly positioned.

**NOTE**

- A small amount of grease purged from bearing, with no accompanying discoloration or damage, is not cause for rejection/replacement.
  - Puddling of grease beneath new bearing should not occur after 10 flight hours. Purged grease can sling out onto seals, adjacent components, cowlings, or anywhere within 360° plane of bearing rotation. Majority of grease purging should lessen within 100 flight hours. A thin film of grease uniformly around full 360° of bearing seal is acceptable for life of bearing. In order to properly determine condition of bearing, make sure to document amount of grease loss.
  - An assembly compound is used between liner and bearing. This assembly compound darkens with age and appears like burned grease. Assembly compound will appear as a thin bead at outside diameter of bearing. Assembly compound may or may not be visible. Do not replace bearing if assembly compound is visible.
- d. Inspect bearing for grease seepage. If a large amount of grease is showing, perform the following:
    - (1) Depending on bearing to be inspected, disconnect oil cooler shaft (PARA 6-4-50), or applicable tail rotor drive shaft

section from tail rotor drive shaft section II (PARA 6-4-39).

- (2) Rotate bearing by hand. **BEARING MUST ROTATE SMOOTHLY.** If bearing ratchets or feels rough when rotated, replace bearing (PARA 6-4-44).
  - (3) If bearing show a large amount of grease seepage accompanied by heat discoloration, replace bearing (PARA 6-4-44).
  - (4) As required, connect oil cooler shaft (PARA 6-4-50), or applicable tail rotor drive shaft to tail rotor drive shaft section II (PARA 6-4-39).
- e. Check bearing for leakage. **NONE ALLOWED.** If leakage is suspected, replace viscous damper tube assembly (PARA 6-4-45). If viscous damper tube assembly is transparent, check for leakage as follows:
    - (1) Using flashlight, hold behind bearing and check for air bubbles in viscous damper tube assembly.
    - (2) **BEARING WITH NO AIR BUBBLES SHOULD BE CHECKED FOR FLUID.**
    - (3) **IF BEARING HAS AN AIR BUBBLE LESS THAN 1-INCH LONG, SERVICE MAY BE CONTINUED WITHOUT RE-SERVICING.**
    - (4) **IF BEARING HAS AN AIR BUBBLE 1 TO 2-INCHES LONG, SERVICE BEARING** (PARA 1-3-16).
    - (5) **IF BEARING HAS AN AIR BUBBLE OVER 2-INCHES LONG, REPLACE VISCOUS DAMPER TUBE ASSEMBLY** (PARA 6-4-45).



6-3-15 OIL COOLER INSPECTIONS.

PARAGRAPH BREAKDOWN

|          |                                                       | PAGE     |          |                                   | PAGE     |
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| 6-3-15.1 | Exit Flange Retaining Nut Torque Check and Inspection | 6-3-15-2 | 6-3-15.3 | Oil Cooler Spline Wear Inspection | 6-3-15-4 |
| 6-3-15.2 | Oil Cooler Drive Shaft Bearing Inspection             | 6-3-15-3 |          |                                   |          |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Borescope Kit
- Flashlight
- Inspection Mirror
- Powertrain Repairer’s Toolkit
- Ruler, 6-inch (or similar straight edge)
- Scribe (or equivalent)
- Spring Scale
- Torque Adapter
- Torque Reactor, 70700-20688-041

- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 100 - 500 ft lbs

Materials/Parts

- Corrosion Preventive Compound, Item 113, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Tiedown Strap, Item 336, Appendix D

References

- Appendix D
- PARA 6-4-43
- PARA 6-4-47
- PARA 6-4-50

### 6-3-15.1 Exit Flange Retaining Nut Torque Check and Inspection.

#### WARNING

Injury to personnel and damage to equipment will result if tail rotor is not secured when drive shaft is disconnected. Secure tail rotor to prevent shafts and rotor from turning.

#### NOTE

This procedure applies only to helicopters with oil cooler fans, 70361-03005-103 through 70361-03005-106, installed.

- a. Remove oil cooler (PARA 6-4-50).
- b. Remove oil cooler radiator (PARA 6-4-47).
- c. Match mark oil cooler drive shaft input flange with oil cooler fan drive shaft exit flange prior to disconnecting shafts.

#### NOTE

It is not required to completely remove oil cooler drive shaft from oil cooler duct since this would require removing the viscous damper bearing assembly. Disconnecting oil cooler drive shaft will provide access to fan exit flange retaining nut.

- d. Disconnect oil cooler drive shaft from oil cooler fan exit flange and slide shaft aft through oil cooler duct (PARA 6-4-50).

#### WARNING

Injury to personnel and damage to equipment will result if fan exit flange retaining nut is not torqued properly. Loosening or removal of retaining nut

may cause improper torque to be present and is not allowed. Only inspect and repair surfaces that do not require removal or loosening of fan exit flange retaining nut.

- e. Visually inspect area around oil cooler fan exit flange retaining nut, inside diameter (ID) of oil cooler fan drive shaft, and ID of oil cooler drive shaft for red oxide residue. **NONE ALLOWED.** Red oxide residue may be an indication of fretting of internal splines. If evidence of fretting or wear (red oxide) is found, replace oil cooler (PARA 6-4-50).
- f. Inspect exit flange, retaining nut, and mating surface of oil cooler drive shaft for corrosion. **IF PITS ARE 0.030-INCH DEEP, OR IF PITTING COVERS MORE THAN 10% OF OIL COOLER DRIVE SHAFT MATING SURFACE, REPLACE OIL COOLER FAN.**
- g. Inspect oil cooler fan shaft inside diameter (PARA 6-4-50).
- h. Remove corrosion from exposed surfaces of fan exit flange and/or retaining nut.
- i. Remove cotter pin from oil cooler fan drive shaft exit flange retaining nut.

#### WARNING

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- Verification of minimum torque of oil cooler fan exit flange retaining nut is a critical characteristic.
- Injury to personnel and damage to equipment will result if fan exit flange retaining nut is not torqued properly. Loosening or removal of retaining nut may cause improper torque to be present and is not allowed. Check torque of fan exit flange retaining nut in tightening direction only.



### NOTE

Torque reactor will be attached to oil cooler fan input flange to react against torquing of oil cooler fan exit flange retaining nut.

- j. Check and record torque of oil cooler fan exit flange retaining nut using torque wrench and torque adapter. Torque is to be checked in a tightening direction, while holding oil cooler fan shaft with torque reactor, to the following specified torque or until nut moves indicating looseness, whichever occurs first. **TORQUE SHALL BE 214 - 236 FOOT-POUNDS.**
- k. **IF RETAINING NUT TORQUE IS 214 - 236 FOOT-POUNDS, AND NO RED OXIDE RESIDUE IS FOUND,** install new cotter pin. If required, align cotter pin hole by turning nut in tightening direction. Torque check is complete.
- l. **IF RETAINING NUT TORQUE IS 180 - 213 FOOT-POUNDS, AND NO RED OXIDE RESIDUE IS FOUND, TORQUE RETAINING NUT TO 214 - 236 FOOT-POUNDS.** If required, align cotter pin hole by turning nut in tightening direction. Install cotter pin. Torque check is complete.
- m. **IF RETAINING NUT TORQUE IS LESS THAN 180 FOOT-POUNDS, REPLACE OIL COOLER** (PARA 6-4-50).
- n. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to exposed surfaces of retaining nut.
- o. Using corrosion preventive compound, Item 113, Appendix D, apply to retaining nut and area around nut. Do not apply corrosion preventive compound to flange mating surfaces.

### CAUTION

Damage to oil cooler will result if match marks on flanges are not aligned correctly prior to installation. Make sure

match marks on oil cooler drive shaft input flange line up with corresponding match marks on oil cooler fan drive shaft exit flange.

- p. Connect oil cooler drive shaft input flange to oil cooler fan drive shaft exit flange making sure to line up match marks on flanges (PARA 6-4-50).
- q. Install oil cooler radiator (PARA 6-4-47).
- r. Install oil cooler (PARA 6-4-50).

### 6-3-15.2 Oil Cooler Drive Shaft Bearing Inspection.

### NOTE

If bearing fails inspection and is to be replaced, a drive shaft runout check must be performed (PARA 6-4-50) prior to replacing bearing.

- a. Turn main rotor blades as required to gain access to oil cooler.
- b. Engage gust lock.
- c. Open oil cooler compartment access doors.

### NOTE

- Movement of viscous damper tube assembly is not cause for replacement of bearing.
- Grease purged from bearing, with no accompanying discoloration or indication of damage to bearing, is not cause for rejection/replacement.
- d. On helicopters with sleeve adapter, 70361-03001-041, perform the following:
  - (1) Remove bolts and washers that secure access cover to rear of oil cooler duct (Figure 6-3-15-1, Detail B).
  - (2) Using inspection mirror and flashlight, inspect area inside of duct for debris. Use

flexible borescope to inspect area at forward end of bearing and between bearing and duct.

- e. On helicopters with sleeve adapter, 70361-03011-105, using slots in sleeve, inspect inside of duct for debris located at the forward end of bearing, and between bearing and duct.
- f. If debris is found at bottom of duct, but no debris is found at forward end of bearing or between viscous damper bearing assembly and duct, clean out debris. Inspection is complete.
- g. If debris is found at forward end of bearing or between bearing and duct, clean out debris as follows:
  - (1) On helicopters with sleeve adapter, 70361-03011-105, using scribe (or equivalent), clean out debris through sleeve slots.
  - (2) On helicopters with sleeve adapter, 70361-03011-041, remove oil cooler assembly and viscous damper bearing assembly (PARA 6-4-50). Clean out debris from inside duct.
  - (3) Check bearing for evidence of overheating (temperature discoloration) and debris damage. Pay particular attention to bearing seal. **NONE ALLOWED.** If evidence of overheating or debris damage is found, replace viscous damper bearing assembly (PARA 6-4-43).
  - (4) If removed, install viscous damper bearing assembly and oil cooler (PARA 6-4-50).
- h. Position flashlight inside of duct, pointing to rear at the bearing.
- i. Inspect bearing assembly for proper fluid level, indicated by bubbles in the fluid.

- j. Inspect bearing for fluid leakage.
- k. If bearing has any of the above deficiencies, replace bearing (PARA 6-4-43).
- l. On helicopters with sleeve adapter, 70361-03011-041, perform the following:
  - (1) Install access cover at rear of oil cooler duct with bolts and washers (Figure 6-3-15-1, Detail B).
  - (2) **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**
- m. Close oil cooler compartment access doors.

### 6-3-15.3 Oil Cooler Spline Wear Inspection.

#### NOTE

This procedure applies only to helicopters with oil cooler fans, 70361-03005-103 through 70361-03005-106, installed.

- a. Open tail rotor drive shaft covers for sections II and III.
- b. Open oil cooler compartment access doors.

#### WARNING

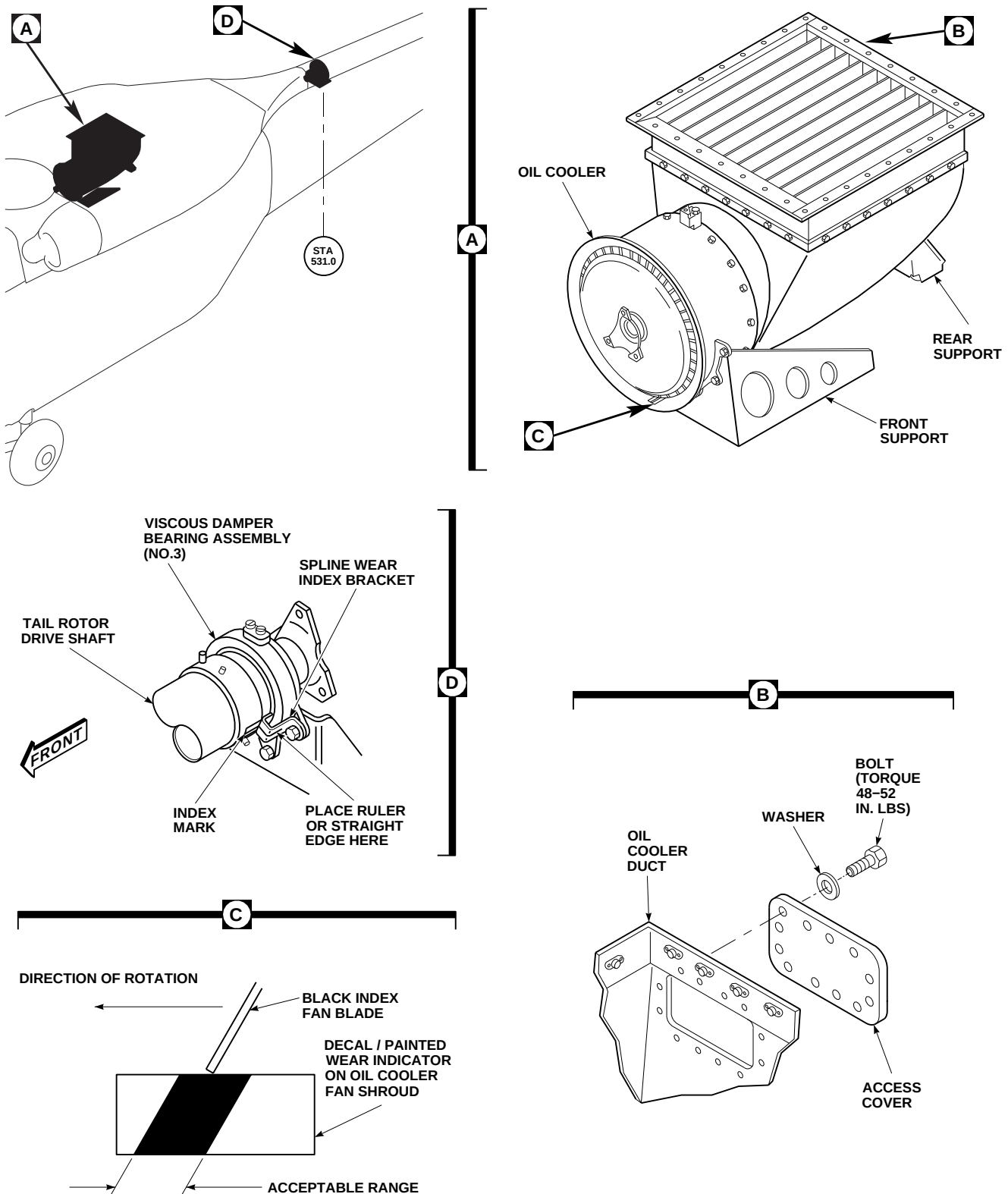
Injury to personnel and damage to equipment will result if spline wear inspection is not performed accurately. Excessive spline wear will cause loss of tail rotor drive. Make sure spline wear inspection is performed accurately.

- c. Engage gust lock.

**WARNING**

Injury to personnel and damage to equipment will result if spline wear inspection is performed improperly by rotating tail rotor in normal direction of rotation. Tail rotor must be rotated in direction opposite normal rotation to simulate in-flight loads. Failure to rotate tail rotor in direction opposite normal rotation will result in false spline wear readings. If tail rotor is rotated in the wrong direction, the entire inspection procedure must be repeated.

- d. Using tiedown strap, Item 336, Appendix D, and spring scale, apply a 20 pound load for five seconds to tail rotor blade leading edge. Make sure tail rotor is rotated in direction opposite normal rotation.
- e. Release gust lock with load applied.
- f. Slowly rotate tail rotor blade in opposite direction of rotation until index mark on tail rotor drive shaft lines up with spline wear index bracket slot on No. 3 viscous damper bearing assembly. Use a 6-inch ruler or similar straight edge to line up bracket slot with black index mark line (Figure 6-3-15-1, Detail D).
- g. If index mark on tail rotor drive shaft rotates past spline wear index bracket slot on No. 3 viscous damper bearing assembly, continue to rotate in opposite direction of normal rotation until index mark is aligned with slot.
- h. If tail rotor is inadvertently rotated in direction of normal rotation, repeat steps c. through g.
- i. Check position of black painted index blade on oil cooler. **BLACK PAINTED INDEX BLADE MUST BE WITHIN BLACK WEAR INDICATOR RANGE** (Figure 6-3-15-1, Detail C).
- j. If any portion of black painted index blade is not within black wear indicator range, perform the following:
  - (1) Repeat steps c. through i.
  - (2) If any portion of black painted index blade is still not within black wear indicator range, replace oil cooler (PARA 6-4-50).
- k. Make sure area is clean and free of foreign material.
- l. Close oil cooler compartment access doors.
- m. Close tail rotor drive shaft covers.



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FIGURE 6-3-15-1. OIL COOLER SPLINE WEAR INSPECTION

6-3-16 TAIL GEAR BOX INSPECTIONS.

PARAGRAPH BREAKDOWN

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| 6-3-16.1 | Tail Gear Box Input Seal Leakage Inspection     | 6-3-16-2 | 6-3-16.3 | Tail Gear Box Sight Plug Inspection   | 6-3-16-2 |
| 6-3-16.2 | Inboard Retention Plate Seal Leakage Inspection | 6-3-16-2 | 6-3-16.4 | Tail Gear Box Input Flange Inspection | 6-3-16-2 |

INITIAL SETUP

- PARA 6-4-54

Tools

- Aircraft Mechanic’s Toolkit

- PARA 6-4-55

References

- PARA 6-3-10
- PARA 6-4-53

### 6-3-16.1 Tail Gear Box Input Seal Leakage Inspection.

#### WARNING

Operation with excess leakage may result in gear box replacement. One hour leak rate shall not be used for operation greater than one hour. Leak rates are determined by amount of leakage that can occur in specified amount of time without affecting gear box performance. If flights of longer duration are anticipated, use rates for greater hours.

#### NOTE

- Checking amount of input seal leakage can prevent premature seal replacement, before leakage is over recommended limits.
- Allow at least 45 minutes after flight for oil to drain back into sump before checking level.
- If operation falls between intervals shown in Table 6-3-16-1, use next higher time interval. Refer to notes in Table 6-3-16-1 for examples.

Measure seal leakage by observing drop in oil level on tail gear box sight gage (Table 6-3-16-1). **IF LEAKAGE IS MORE THAN 1/8-INCH AFTER ANY ONE FLIGHT, REPLACE TAIL GEAR BOX INPUT SEAL (PARA 6-4-53).**

### 6-3-16.2 Inboard Retention Plate Seal Leakage Inspection.

#### WARNING

Operation with excess leakage may result in gear box replacement. One hour leak rate shall not be used for operation greater than one hour. Leak rates are determined by amount of leakage that

can occur in specified amount of time without affecting gear box performance. If flights of longer duration are anticipated, use rates for greater hours.

#### NOTE

- Checking amount of retention plate seal leakage can prevent premature seal replacement before leakage is over recommended limits.
- Allow at least 45 minutes after flight for oil to drain back into sump before checking level.
- If operation falls between intervals shown in Table 6-3-16-1, use next higher time interval. Refer to notes in Table 6-3-16-1 for examples.

Measure seal leakage by observing drop in oil level on tail gear box sight gage (Table 6-3-16-1). **IF LEAKAGE IS MORE THAN 1/8-INCH AFTER ANY ONE FLIGHT, REPLACE INBOARD RETENTION PLATE SEAL (PARA 6-4-54).**

### 6-3-16.3 Tail Gear Box Sight Plug Inspection.

- IF TAIL GEAR BOX SIGHT PLUG HAS ONE OR MORE CRACKS, REPLACE SIGHT PLUG (PARA 6-4-34).**
- IF OIL LEVEL CANNOT BE SEEN THROUGH LENS DUE TO CRAZING OR STAINING, REPLACE TAIL GEAR BOX SIGHT PLUG.**
- IF TAIL GEAR BOX SIGHT PLUG LEAKS, REPLACE SIGHT PLUG.**

### 6-3-16.4 Tail Gear Box Input Flange Inspection.

#### NOTE

Inspection of tail gear box input flange and intermediate gear box input and output flanges is identical.

Inspect tail gear box input flange (PARA 6-3-10).

**Table 6-3-16-1. Tail Gear Box Input Seal Allowable Leakage Limits**

| Hours of Operation             | Allowable Leakage                                          |
|--------------------------------|------------------------------------------------------------|
| One Hour (see note 1)          | $\frac{1}{8}$ -inch = 3.38 fluid ounces = 100 milliliters  |
| Five Hours (see notes 1 and 2) | $\frac{1}{4}$ -inch = 6.76 fluid ounces = 200 milliliters  |
| Ten Hours (see note 2)         | $\frac{1}{2}$ -inch = 13.52 fluid ounces = 400 milliliters |

**NOTES**

1. If flight is between one and five hours, use five hours allowable leakage rate.
2. If flight is between five and ten hours, use ten hours allowable leakage rate.
3. Anytime flight hour exceeds given interval, leakage rate per hour is 0.05-inch, which is equivalent to 3.2 ounces per hour (for main gear box) and 1.3 ounces per hour (for intermediate and tail gear boxes).
4. Example: Leakage at  $\frac{1}{8}$ -inch per hour is experienced and a subsequent flight of 5 continuous flight hours is planned.  $\frac{5}{8}$ -inch of oil will be expected to be lost which exceeds allowable leak rate. Since subsequent servicing after 5-hour flight may indicate that gearbox ran low on oil requiring gearbox replacement, seal replacement is required to prevent leakage.





6-3-17 TAIL GEAR BOX CHIP DETECTOR/TEMPERATURE SENSOR PARTICLE INSPECTIONS.

PARAGRAPH BREAKDOWN

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| 6-3-17.1 | General                                                             | 6-3-17-1 |
| 6-3-17.2 | Tail Gear Box Chip Detector/Temperature Sensor Particle Inspections | 6-3-17-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container

Materials/Parts

- Cheesecloth, Item 90, Appendix D

References

- Appendix D
- PARA 6-3-18
- PARA 6-4-52
- PARA 6-4-59

6-3-17.1 General.

- When the CHIP TAIL XMSN light on caution/advisory panel goes on, it indicates the tail gear box chip detector/temperature sensor in the tail gear box has activated.
- Metal particles found on the tail gear box chip detector/temperature sensor may indicate an internal part of the tail gear box has malfunctioned. If particles are found, it does not necessarily mean the tail gear box must be replaced. Quantity, size, type, and source of the particles, along with the history of the tail gear box, must be considered. Also, tail gear box time, previous malfunctions, and operating conditions should be considered.
- Particles found may be magnetic (steel), nonmagnetic (aluminum, bronze, cadmium, lead, magnesium, or silver), or nonmetallic

(dust, baked resin/paint, lint, paint, paper, phenolic, rubber, or wood).

6-3-17.2 Tail Gear Box Chip Detector/ Temperature Sensor Particle Inspections.

- Remove tail gear box chip detector/temperature sensor (PARA 6-4-59).

NOTE

Tail gear boxes do not have oil filters and discoloration is possible.

- Inspect particles. If particles are easily identified as fragments from gears, nuts, bearings, etc., replace tail gear box (PARA 6-4-52). If replacement of tail gear box is questionable, contact Sikorsky Aircraft Corporation ILS.
- Count and measure particles. If 10 or more particles are found, or if any one particle is

larger than 0.050-inch diameter or 0.032-inch  $\times$  0.125-inch long, replace tail gear box (PARA 6-4-52).

- d. If a tail gear box is replaced due to contamination, perform the following:
  - (1) Foreign material from screens and filters must be carefully packaged and returned with the tail gear box.
  - (2) Tail gear box oil must be drained and strained with cheesecloth, Item 90, Appendix D, into suitable container.
  - (3) If necessary, any particles found in oil should be combined with particles from

screens and filters for evaluation and returned with tail gear box.

- (4) Package cheesecloth, Item 90, Appendix D, put oil in sealed container, and return both with tail gear box.
- e. For appropriate action after determining particle material, size, and quantity, refer to Table 6-3-17-1.
- f. Install tail gear box chip detector/temperature sensor (PARA 6-4-59).

**Table 6-3-17-1. Particle Classification**

| <b>Material</b>                                                                       | <b>Type-Quantity Size</b>                                                                                                     | <b>Action Required</b>                                                                                               |
|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Magnetic (Steel)                                                                      | Fuzz, fine hair-like particles.                                                                                               | Acceptable. Return tail gear box to service. No tail gear box serviceability inspection is required.                 |
|                                                                                       | No more than 10 particles, none over 0.050-inch diameter or over 0.032-inch thick $\times$ 0.125-inch long.                   | Perform tail gear box serviceability inspection (PARA 6-3-18). Record findings on log card. Refer to Table 6-3-17-2. |
|                                                                                       | Splinters or granular particles.                                                                                              | Perform tail gear box serviceability inspection (PARA 6-3-18).                                                       |
|                                                                                       | Any quantity of chips beyond the above dimensions.                                                                            | Replace tail gear box (PARA 6-4-52).                                                                                 |
| Nonmagnetic (Aluminum, Bronze, Cadmium, Lead, Magnesium, or Silver)                   | Thin flakes not over 0.050-inch diameter or over 0.032-inch thick $\times$ 0.125-inch long. No more than 10 flakes permitted. | Acceptable. Return tail gear box to service. No tail gear box serviceability inspection is required.                 |
|                                                                                       | More than 10 flakes, but less than 25.                                                                                        | Perform tail gear box serviceability inspection (PARA 6-3-18). Record findings on log card. Refer to Table 6-3-17-2. |
|                                                                                       | More than 25 flakes.                                                                                                          | Replace tail gear box (PARA 6-4-52).                                                                                 |
| Nonmetallic (Dust, Baked Resin/ Paint, Lint, Paint, Paper, Phenolic, Rubber, or Wood) | Flakes not over 0.050-inch diameter or over 0.032-inch thick $\times$ 0.125-inch long. No more than 10 flakes permitted.      | Acceptable. Return tail gear box to service. No tail gear box serviceability inspection is required.                 |
|                                                                                       | Less than a total area of $\frac{1}{2}$ -square-inch.                                                                         | Perform tail gear box serviceability inspection (PARA 6-3-18). Refer to Table 6-3-17-2.                              |
|                                                                                       | More than a total area of $\frac{1}{2}$ -square-inch.                                                                         | If contamination is sufficient to partially block oil strainer, replace tail gear box (PARA 6-4-52).                 |

Table 6-3-17-1. Particle Classification (Cont)

| Material             | Type-Quantity Size                    | Action Required                                                                                                                                                                                                                                                                                                  |
|----------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nonmetallic (Rubber) | More than an area of 1/4-square-inch. | Conduct visual leakage inspection of all external oil seal areas and tail gear box housing attachment joints. Evaluate indicator oil pressure readings. If the low oil pressure warning light has been illuminated and if leakage source can not be corrected or identified, remove tail gear box (PARA 6-4-52). |

## NOTE

If continuing chip indications are being found, the following limitations in Table 6-3-17-2 apply.

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**Table 6-3-17-2. Particle Classification - Continuing Indications**

| <b>Material</b>                                                     | <b>Type-Quantity<br/>Size</b>                                                                                     | <b>Time Interval</b> | <b>Number of<br/>Indications</b> | <b>Action Required</b>               |
|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------|----------------------------------|--------------------------------------|
| Magnetic (Steel)                                                    | More than 2 flakes or granular particles over 0.005-inch thick or 0.020-inch diameter. Noted at each indication.  | 10 hour period       | Three                            | Replace tail gear box (PARA 6-4-52). |
|                                                                     |                                                                                                                   | 25 hour period       | Five                             | Replace tail gear box (PARA 6-4-52). |
|                                                                     |                                                                                                                   | 50 hour period       | Nine                             | Replace tail gear box (PARA 6-4-52). |
| Nonmagnetic (Aluminum, Bronze, Cadmium, Lead, Magnesium, or Silver) | More than 10 flakes or granular particles over 0.032-inch thick or 0.125-inch diameter. Noted at each indication. | 10 hour period       | Three                            | Replace tail gear box (PARA 6-4-52). |

**NOTE**

To make use of time interval and number of indications, chip light history must be entered on log cards.



6-3-18 TAIL GEAR BOX SERVICEABILITY INSPECTION.

PARAGRAPH BREAKDOWN

|                                                       | PAGE     |
|-------------------------------------------------------|----------|
| 6-3-18.1 Tail Gear Box Service-<br>ability Inspection | 6-3-18-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 2-Quart
- Drain Line, H538A-3

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Cloth, Cleaning, Item 102, Appendix D

References

- Appendix D
- PARA 1-3-14
- PARA 6-4-52
- PARA 6-4-59
- PARA 11-4-113

6-3-18.1 Tail Gear Box Serviceability Inspection.

NOTE

Perform this procedure if particles were found on tail gear box chip detector/temperature sensor.

- a. Drain tail gear box oil through cheesecloth as follows:
- (1) Remove tail gear box chip detector/temperature sensor (PARA 6-4-59).

NOTE

Position 2-quart container and cheesecloth as necessary so oil will be

strained through cheesecloth when drained.

- (2) Position 2-quart container and clean cheesecloth, Item 90, Appendix D. Place one end of drain line into container. Insert other end into self-closing valve by pushing in and turning clockwise.
- (3) Remove tail gear box oil filler plug and drain oil through cheesecloth into container.
- (4) Remove drain line.
- b. If required, inspect and clean tail gear box chip detector/temperature sensor (PARA 6-4-59).
- c. Install tail gear box chip detector/temperature sensor (PARA 6-4-59).

**CAUTION**

Damage to equipment will result if foreign objects enter tail gear box. Foreign objects may enter tail gear box when tail rotor servo is removed. Make sure that no foreign objects enter tail gear box while performing step d. through step g.

**NOTE**

Step d. through step g. are only required if excessive particle counts are found.

- d. Remove tail rotor servo (PARA 11-4-113).
- e. Check inside of tail gear box, especially inside diameter of input bevel pinion, for buildup of sludge or debris accumulation. Using clean, dry, cleaning cloth, Item 102, Appendix D, remove any sludge/debris accumulation.
- f. Flush tail gear box (PARA 1-3-14).
- g. Install tail rotor servo (PARA 11-4-113).
- h. Service tail gear box (PARA 1-3-14).
- i. Ground run helicopter for 30 minutes, and then hover for 30 minutes.
- j. After ground run and hover, perform the following:
  - (1) Inspect tail gear box chip detector/temperature sensor (PARA 6-4-59).
  - (2) If amount of metal particles has increased, replace tail gear box (PARA 6-4-52).
  - (3) If amount of particles has decreased and is within limits, serviceability inspection is complete.
- k. If necessary, service tail gear box (PARA 1-3-14).
- l. Return helicopter to service.



6-3-19    MAIN TRANSMISSION HOUSING INTERFERENCE INSPECTION.

PARAGRAPH BREAKDOWN

|          |                                                   | PAGE     |
|----------|---------------------------------------------------|----------|
| 6-3-19.1 | Main Transmission Housing Interference Inspection | 6-3-19-2 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Beaker (10 ml)
- Beaker (200 ml)
- Beaker (600 ml)
- Container, Plastic, 1-Pint
- Container, Plastic, 2-Gallon
- Flashlight
- Fluorescent Penetrant Inspection Kit
- Ink Marker
- Inspection Mirror
- Powertrain Repairer’s Toolkit
- Protection, Eye
- Radius Gage, 2-inch, Locally-Made
- Vessel(s), with equivalent measuring accuracy of Beakers (Alternate for Beakers)

Materials/Parts

- Abrasive Cloth, Item 6, Appendix D
- Abrasive Cloth, Item 10, Appendix D

- Acetic Acid, Item 24, Appendix D
- Applicator, Item 78, Appendix D
- Chromium Trioxide, Technical, Item 91, Appendix D
- Corrosion Preventive Compound, Item 108C, Appendix D
- Corrosion Preventive Compound, Item 113, Appendix D
- Distilled Water, Item 123, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Coating Kit, Item 126A, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Ethylene Glycol, Item 137, Appendix D
- Nitric Acid, Item 220, Appendix D
- Paint, Polyurethane, Item 224, Appendix D
- Paint Remover, Item 231, Appendix D
- Pressure Sensitive Adhesive Tape, Item 250, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 1-6-15

- PARA 1-6-16
- PARA 11-4-2
- PARA 11-4-96

### 6-3-19.1 Main Transmission Housing Interference Inspection.

- a. Using inspection mirror and flashlight, inspect main transmission housing right side support web for evidence of contact with forward bellcrank. **NONE ALLOWED.** Proceed as follows:
  - (1) If no evidence of contact is found, proceed to step b.
  - (2) If evidence of contact is found, proceed to step c.
- b. Check minimum clearance between main transmission housing and forward bellcrank as follows:

**CAUTION**

Damage to flight control components will result if hydraulic power is shut down with rigging pins still installed. Make sure all rigging pins are removed before shutting down hydraulic power.

**NOTE**

Backup pump should be used only as an alternate source to power hydraulic systems. External hydraulic power should be applied whenever available.

- (1) Connect external hydraulic power to backup system and connect electrical power (PARA 1-6-15 and PARA 1-6-16).
- (2) Install pilot's and copilot's lateral cyclic stick rigging pins, 70700-20380-066 (PARA 11-4-2).
- (3) Raise collective stick to full up position and apply full left pedal. Move cyclic stick full aft.

- (4) While holding full up collective, full aft cyclic, and full left pedal, **VERIFY FORWARD BELLCRANK IS NOT CONTACTING MAIN GEAR BOX HOUSING AND A MINIMUM OF 1/16-INCH CLEARANCE IS PRESENT.** Evaluate as follows:

- (a) If forward bellcrank is not contacting main transmission housing, and a minimum of 1/16-inch clearance is present, inspection is complete.
- (b) If forward bellcrank is contacting main transmission housing, and/or a minimum of 1/16-inch clearance is not present, proceed to step c.

- (5) Remove pilot's and copilot's lateral cyclic stick rigging pins, 70700-20380-066 (PARA 11-4-2).

- (6) Disconnect external hydraulic power and electrical power (PARA 1-6-15 and PARA 1-6-16).

- c. Rework main transmission housing as follows (Figure 6-3-19-1, Detail A):

- (1) Remove forward bellcrank (PARA 11-4-96).
- (2) Using paint remover, Item 231, Appendix D, remove paint from web area.
- (3) Locally make 2-inch radius gage.
- (4) Locate right input module to main transmission joint line. Measure back along main transmission housing web to a point 3.31 inches from joint line. Mark this point on web using ink marker.
- (5) Using ink marker and radius gage, mark guide line starting at previously marked point located 3.31 inches from joint line towards existing web cut out.

- (6) Using abrasive cloth, Item 6, Appendix D, carefully blend web area until there is approximately 0.010-inch of material remaining above guide line.
- (7) Using abrasive cloth, Item 10, Appendix D, carefully blend remaining 0.010-inch of material. Make sure surface is smooth and continuous with no mismatches.
- (8) Remove displaced metal caused by abrasive blending by using magnesium-etching procedure as follows:

CAUTION

- Magnesium-etchant measurements are to be as accurate as possible to values as can be measured in a milliliter beaker or vessel with equivalent mixing accuracy.
  - Magnesium-etchant formulas must be prepared and stored in unbreakable 1-pint plastic containers and safely stored in designated areas not subjected to extreme heat or cold.
  - Before preparing magnesium-etchant, make sure storage area does not contain premixed magnesium-etchant formula. The correct stock bottle for this repair must be marked MAGNESIUM-ETCHANT.
- (a) Prepare magnesium-etchant solution in clean unbreakable plastic container by mixing the following:

| MATERIAL                                    | QTY             |
|---------------------------------------------|-----------------|
| Ethylene Glycol,<br>Item 137, Appendix<br>D | 600 milliliters |

| MATERIAL                                 | QTY             |
|------------------------------------------|-----------------|
| Acetic Acid, Item<br>24, Appendix D      | 200 milliliters |
| Nitric Acid, Item<br>220, Appendix D     | 10 milliliters  |
| Distilled Water, Item<br>123, Appendix D | 190 milliliters |

- (b) Using dry-cleaning solvent, Item 124, Appendix D, clean area to be etched.
- (c) Using pressure sensitive adhesive tape, Item 250, Appendix D, mask area.
- (d) Pour small amount of magnesium-etchant from stock bottle into 1-pint plastic container.

CAUTION

Damage to components will result if magnesium-etchant is allowed to come in contact with areas not being etched. Extreme caution must be used to protect those areas not being etched.

- (e) Using applicator, Item 78, Appendix D, apply magnesium-etchant to reworked area. Swab area for about 10 - 15 seconds.
- (f) Neutralize magnesium-etchant immediately by flooding area with generous amount of clean water.

**WARNING**

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- (g) Using clean, dry, compressed air, blow dry area thoroughly.
  - (h) After use, do not return unused magnesium-etchant to stock bottle. Dispose of used magnesium-etchant per local instructions.
  - (i) Identify contents of stock bottle as MAGNESIUM-ETCHANT before placing bottle in storage area.
  - (j) Prepare technical chromium trioxide solution in clean 2-gallon plastic container by mixing 24 ounces of technical chromium trioxide, Item 91, Appendix D, with 1 gallon of water.
  - (k) Using technical chromium trioxide solution, treat bare surface of housing.
- (9) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect reworked area on main transmission housing for cracks (ASTM E1417, Type I, Method C, Level 4 Sensitivity).

- (10) Coat reworked area with corrosion preventive compound, Item 108C, Appendix D.
- (11) Using epoxy primer coating kit, Item 135, Appendix D (or equivalent), apply one coat of epoxy primer to reworked area within four hours of application of corrosion preventive compound.

**NOTE**

Housings coated with epoxy coating are glossy silver. Housings coated with polyurethane paint are matte surface medium-gray.

- (12) Using epoxy coating kit, Item 126A, Appendix D, apply one coat of epoxy coating, or one coat of polyurethane paint, Item 224, Appendix D, depending on original paint finish.
- (13) Apply one coat of corrosion preventive compound, Item 113, Appendix D. Allow corrosion preventive compound to cure for 30 minutes.
- (14) Apply second coat of corrosion preventive compound, Item 113, Appendix D.
- (15) Make sure area is clean and free of foreign material.
- (16) Install forward bellcrank (PARA 11-4-96).
- (17) Record brief description of repair in helicopter component historical records.
- (18) Check minimum clearance between main transmission housing and forward bellcrank per step b. If forward bellcrank still contacts main transmission, or there is less than  $\frac{1}{16}$ -inch clearance, contact Sikorsky Aircraft.

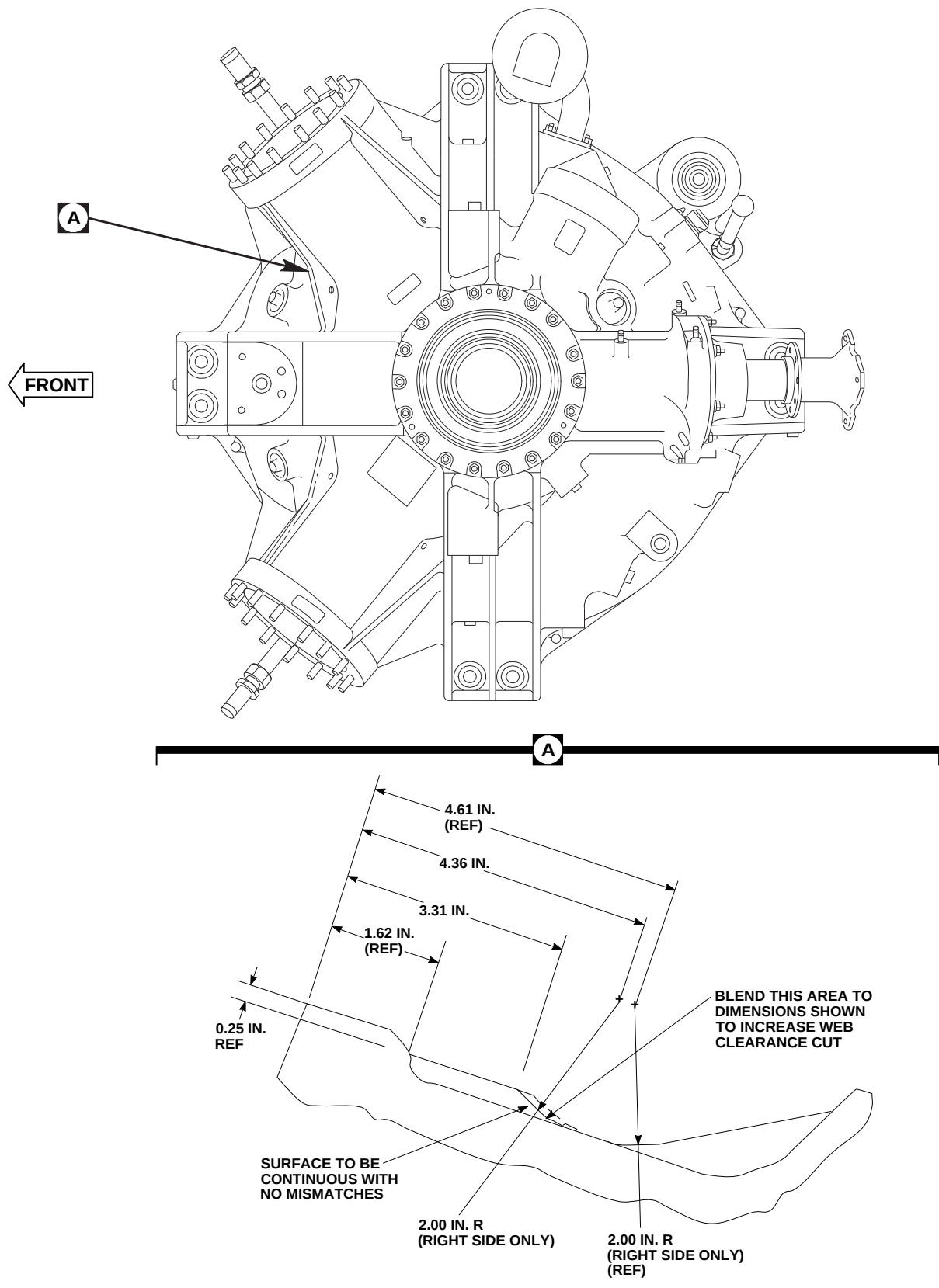


FIGURE 6-3-19-1. MAIN TRANSMISSION HOUSING INTERFERENCE INSPECTION



## SECTION 6-4

# MAINTENANCE PROCEDURES

## SECTION OVERVIEW

| PARAGRAPH | TITLE                                                                                |
|-----------|--------------------------------------------------------------------------------------|
| 6-4-1     | Main Transmission                                                                    |
| 6-4-2     | Main Transmission Tail Takeoff Flange                                                |
| 6-4-3     | Main Transmission Output Shaft Seal                                                  |
| 6-4-4     | Main Transmission Breather Plug                                                      |
| 6-4-5     | Swashplate Guide and Main Rotor Shaft Seal                                           |
| 6-4-6     | Main Rotor Shaft Plug                                                                |
| 6-4-7     | Main Rotor Shaft Inspect/Repair                                                      |
| 6-4-8     | Gust Lock Rod and Lever                                                              |
| 6-4-9     | Gust Lock Indicator Switch                                                           |
| 6-4-10    | Lever Bracket <b>RTR BRAKE</b>                                                       |
| 6-4-11    | Main Module Housing Repair                                                           |
| 6-4-12    | Input Module Housing Repair                                                          |
| 6-4-13    | Accessory Module Housing Repair                                                      |
| 6-4-14    | Main, Input, and Accessory Module Housing Water Entrapment Area Corrosion Protection |
| 6-4-15    | Accessory Module Chip Detector                                                       |
| 6-4-16    | Accessory Module Oil Screen                                                          |
| 6-4-17    | Input Module Chip Detector                                                           |
| 6-4-18    | Main Module Sump Chip Detector/Temperature Sensor                                    |
| 6-4-19    | Main Transmission Oil Filter                                                         |
| 6-4-20    | Main Transmission Oil Pump                                                           |

| PARAGRAPH | TITLE                                                  |
|-----------|--------------------------------------------------------|
| 6-4-21    | Main Transmission Oil Pressure Sensor                  |
| 6-4-22    | Main Transmission Oil Pump Pressure Regulating Valves  |
| 6-4-23    | Main Transmission Oil Gage Tube Assembly               |
| 6-4-24    | Main Transmission Oil Temperature Switch               |
| 6-4-25    | Input Module                                           |
| 6-4-26    | Input Module Oil Seal                                  |
| 6-4-27    | Input Module Input Flange                              |
| 6-4-28    | Accessory Module                                       |
| 6-4-29    | Accessory Module Spline Adapter                        |
| 6-4-30    | Accessory Module Shim                                  |
| 6-4-31    | Intermediate Gear Box                                  |
| 6-4-32    | Intermediate Gear Box Input Seal and Flange            |
| 6-4-33    | Intermediate Gear Box Output Seal and Flange           |
| 6-4-34    | Intermediate Gear Box Sight Plug                       |
| 6-4-35    | Intermediate Gear Box Oil Filler Plug                  |
| 6-4-36    | Intermediate Gear Box Housing Repair                   |
| 6-4-37    | Intermediate Gear Box Chip Detector/Temperature Sensor |
| 6-4-38    | Tail Rotor Drive Shaft Section I                       |
| 6-4-39    | Tail Rotor Drive Shaft Section II                      |
| 6-4-40    | Tail Rotor Drive Shaft Section III                     |
| 6-4-41    | Tail Rotor Drive Shaft Section IV                      |
| 6-4-42    | Tail Rotor Drive Shaft Flexible Coupling               |
| 6-4-43    | Tail Rotor Drive Shaft Viscous Damper Bearing Assembly |
| 6-4-44    | Tail Rotor Drive Shaft Bearing                         |
| 6-4-45    | Viscous Damper Tube Assembly                           |



| PARAGRAPH | TITLE                                          |
|-----------|------------------------------------------------|
| 6-4-46    | Tail Rotor Drive Shaft Repair                  |
| 6-4-47    | Oil Cooler Radiator                            |
| 6-4-48    | Oil Cooler Radiator Thermostatic Valve         |
| 6-4-49    | Oil Cooler Radiator Lines and Elbows           |
| 6-4-50    | Oil Cooler                                     |
| 6-4-51    | Oil Cooler Spline Wear Indicator System        |
| 6-4-52    | Tail Gear Box                                  |
| 6-4-53    | Tail Gear Box Input Seal and Flange            |
| 6-4-54    | Inboard Retention Plate and Seals              |
| 6-4-55    | Tail Gear Box Sight Plug                       |
| 6-4-56    | Tail Gear Box Oil Filler Plug                  |
| 6-4-57    | Tail Gear Box Housing Repair                   |
| 6-4-58    | Tail Gear Box Pulley Guards                    |
| 6-4-59    | Tail Gear Box Chip Detector/Temperature Sensor |
| 6-4-60    | Rotor Brake Master Cylinder                    |
| 6-4-61    | Rotor Brake Accumulator                        |
| 6-4-62    | Rotor Brake Manifold and Valve Assembly        |
| 6-4-63    | Rotor Brake Pressure Switch                    |
| 6-4-64    | Rotor Brake Pressure Gage                      |
| 6-4-65    | Rotor Brake Master Cylinder Bleeder Valve      |
| 6-4-66    | Rotor Brake Plunger                            |
| 6-4-67    | Rotor Brake System Lines                       |
| 6-4-68    | Bleed Rotor Brake System                       |
| 6-4-69    | Rotor Brake Tubing                             |
| 6-4-70    | Rotor Brake Assembly                           |

PARAGRAPH

TITLE

|        |                              |
|--------|------------------------------|
| 6-4-71 | Rotor Brake Linings (Pucks)  |
| 6-4-72 | Rotor Brake Bleeder Valve    |
| 6-4-73 | Rotor Brake Disc             |
| 6-4-74 | Rotor Brake Mounting Bracket |
| 6-4-75 | Rotor Speed (Nr) Sensor      |
| 6-4-76 | Low Oil Pressure Switch      |
| 6-4-77 | Main Rotor Shaft Lower Seal  |

6-4-1 MAIN TRANSMISSION.

PARAGRAPH BREAKDOWN

|         |                                | PAGE    |         |                  | PAGE     |
|---------|--------------------------------|---------|---------|------------------|----------|
| 6-4-1.1 | Replace Main Transmis-<br>sion | 6-4-1-3 | 6-4-1.2 | Reseat Dowel Pin | 6-4-1-54 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Allen Wrench, 1⁄8-inch
- Apex Ring(s), 70700-20322-101
- Brass Drift
- Caulking Gun
- Chocks
- Container, 1-Quart
- Crowfoot Attachment
- Feeler Gage
- Fluorescent Penetrant Inspection Kit
- Guide Straps
- Hoist, Capacity to Support Main Transmission with Main Rotor Installed (Alternate for Cabin Maintenance Crane)
- Hoisting Sling, 70700-20320-042
- Main Gear Box Adapter, 70700-20420-041
- Maintenance Trailer
- Mounting Bolt Adapter, Locally-Made, Figure H-46, Appendix H
- Nonmetallic Scraper, Locally-Made

- Powertrain Repairers’s Toolkit
- Protective Covering
- Rawhide Mallet, 2 - 5 pounds
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1600 in. lbs
- Torque Wrench, 100 - 500 ft. lbs
- Universal Hoist Sling Assembly, 70700-20320-045

Materials/Parts

- Bolt, NAS630-35
- Bolt, SS5209-10-34
- Brush, Paint, Item 86, Appendix D
- Cap Plug, NAS813-14
- Cap Plug, NAS813-16
- Cap Plug, NAS813-20
- Cloth, Cleaning, Item 102, Appendix D
- Corrosion Preventive Compound, Item 109, Appendix D
- Corrosion Preventive Compound, Item 111, Appendix D

- Corrosion Preventive Compound, Item 113, Appendix D
- Corrosion Preventive Compound, Item 114, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Coating Kit, Item 126A, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Insulation Sleeving
- Lockwire, Item 194, Appendix D
- Lockwire, Item 196, Appendix D
- Lockwire, Item 197, Appendix D
- Lubricant, Solid Film, Item 201, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D
- Lubricating Oil, Item 208, Appendix D
- Machinery Towel, Item 211, Appendix D
- Masking Tape, 3-inches, Item 215, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint, Polyurethane, Item 224, Appendix D
- Primer, Item 258, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 287, Appendix D
- Sealing Compound, Item 292, Appendix D
- Spiral Wrap, Item 315, Appendix D
- Tags
- Tape, Item 317C, Appendix D

- Tiedown Strap, Item 329, Appendix D
- Tiedown Strap, Item 334, Appendix D

#### References

- Appendix D
- Appendix H
- ASTM E1417
- PARA 1-3-12
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-3
- PARA 1-7-20
- PARA 2-4-69
- PARA 2-4-70
- PARA 2-4-106
- PARA 2-4-107
- PARA 2-4-113
- PARA 2-6-5
- PARA 4-4-6
- PARA 4-4-27
- PARA 5-4-22
- PARA 5-4-31
- PARA 6-2-2
- PARA 6-3-2
- PARA 6-3-4
- PARA 6-4-8
- PARA 6-4-15

- PARA 6-4-16
- PARA 6-4-17
- PARA 6-4-18
- PARA 6-4-25
- PARA 6-4-38
- PARA 11-3-13
- PARA 11-3-15
- PARA 11-3-16
- PARA 11-3-17
- PARA 11-3-18
- PARA 11-3-19
- PARA 11-3-20
- PARA 11-3-23
- PARA 11-4-1
- PARA 11-4-4
- PARA 12-4-5
- PARA 12-4-26
- PARA 13-4-4
- PARA 18-4-1

#### 6-4-1.1 Replace Main Transmission.

#### NOTE

##### 6-4-1.1.1 Remove.

- a. Make sure helicopter wheels are chocked.
- b. If necessary, install main gear box adapter on maintenance trailer.
- c. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).
- d. Open main rotor pylon sliding cover.
- e. Open engine cowlings.
- f. Remove main transmission drip pan (PARA 2-4-70).



Damage to cabin maintenance crane and helicopter will occur if cabin maintenance crane is used to remove main transmission with main rotor head and swashplate installed. Use hoisting sling and overhead hoist to remove main transmission with main rotor head and swashplate installed.

- Main transmission, 70351-08000, with all equipment (input modules, accessory modules, main generators, and hydraulic pump modules) installed, weighs about 1,200 pounds. Main transmission with all equipment, main rotor head, and swashplate installed, weighs about 2,400 pounds. Main module alone weighs about 750 pounds.
- Main transmissions, 70351-48000 and 70351-58000, with all equipment (input modules, accessory modules, main generators, and hydraulic pump modules) installed, weighs about 1,365 pounds. Main transmission with all equipment, main rotor head, and swashplate installed, weighs about 2,565 pounds. Main module alone weighs about 915 pounds.
- If an overhead hoist capable of lifting main transmission with main rotor head and swashplate installed is available, it is easier to remove main transmission (with main rotor head installed) and then remove main rotor head while on ground. If cabin maintenance crane will

be used, main rotor head and swashplate must be removed first.

- g. If cabin maintenance crane will be used, perform the following:

- (1) Erect cabin maintenance crane (PARA 18-4-1).

#### NOTE

Main rotor head will be removed when swashplate is removed.

- (2) Remove swashplate (PARA 5-4-22).
  - (3) If installed, remove fire extinguisher line from right firewall (PARA 12-4-5).
- h. If overhead hoist and hoisting sling will be used, perform the following:
- (1) If blade deice kit is installed, disconnect deice harness from main rotor deice junction box (PARA 12-4-26).
  - (2) Remove main rotor blades (PARA 5-4-31).
  - (3) If installed, remove fire extinguisher line from right firewall (PARA 12-4-5).
- i. Drain main transmission (PARA 1-3-12).
- j. Remove soundproofing panels, rear of main transmission drip pan, from cabin ceiling (PARA 2-4-69).
- k. Remove oil cooler compartment access doors (PARA 2-4-113).
- l. Remove main rotor pylon transmission cover (PARA 2-4-107).
- m. Remove left and right main rotor pylon air inlet fairings (PARA 2-4-106).
- n. Disconnect engine output shafts from input module flanges (PARA 4-4-6).
- o. Disconnect forward support tubes from elastomeric gimbals on input modules (PARA 4-4-6).

- p. Disconnect lines and wiring from main transmission as follows:

- (1) Disconnect hydraulic pressure line and hydraulic return line at quick-disconnects on hydraulic pump modules (Figure 6-4-1-1, Sheet 1, Detail A).
- (2) Disconnect electrical connectors from hydraulic pump modules as follows:
  - (a) Disconnect electrical connector, P447 (No. 1 hydraulic pump module), and P432 (No. 2 hydraulic pump module).
  - (b) Install protective caps and plugs on electrical connectors.
- (3) Disconnect drain lines from seal drain elbows.
- (4) Disconnect drain lines from hydraulic pump module bleed relief valves.
- (5) Disconnect electrical leads from main generators as follows (Figure 6-4-1-1, Sheet 1, Detail C or Detail D):
  - (a) If installed, remove screws and washers and slide terminal block cover forward on electrical leads, or pull rubber terminal covers back from terminals to expose terminal connections.
  - (b) Tag electrical leads. Remove nuts, washers, and lockwashers (if applicable) that secure electrical leads to terminals.

#### NOTE

Terminal block cover will not be reused.

- (c) If terminal block cover was installed, pull electrical leads through terminal block cover and remove terminal block cover (Figure 6-4-1-1, Sheet 1, Detail D).
- (d) Remove screw, washer, and clamp that secure electrical leads to end of main generator.

- (6) Tag and disconnect electrical connectors from main generators as follows (Figure 6-4-1-1, Sheet 1, Detail A). Install protective caps and plugs on electrical connectors as they are disconnected:

| No. 1 Main Generator | No. 2 Main Generator |
|----------------------|----------------------|
| P251 from J251       | P250 from J250       |
| P278 from J278       | P277 from J277       |

- (7) Disconnect electrical connectors under forward support tube as follows (Figure 6-4-1-1, Sheet 1, Detail B):
  - (a) Disconnect electrical connectors, P400 (right side), and P401 (left side), from brackets under forward support tube.
  - (b) Install protective caps and plugs on electrical connectors.
- q. Disconnect engine load-demand push/pull cable supports from accessory modules by removing bolts, washers, and nuts. Retain bolts, washers, and nuts with supports (Figure 6-4-1-1, Sheet 2, Detail F).
- r. Disconnect rod ends of collective bias tube from mixer (PARA 4-4-27). Reposition cables away from main transmission components.
- s. Remove heating and ventilation system bleed-air tube (PARA 13-4-4).
- t. Remove tail rotor drive shaft section I (PARA 6-4-38).
- u. Remove gust lock lever (PARA 6-4-8).
- v. Disconnect oil inlet and outlet lines and if installed, rotor brake hose from main transmission as follows:
  - (1) **RTR BRAKE** Tag and disconnect rotor brake hose from elbow on rotor brake (Figure 6-4-1-2, Detail A).

- (2) **RTR BRAKE** Install protective cap and plug on elbow and rotor brake hose.
- (3) Position 1-quart container under oil lines to catch fluid trapped in lines.
- (4) Disconnect oil inlet and oil outlet lines from fittings.
- (5) Install protective caps and plugs on oil inlet and oil outlet lines and fittings.
- (6) Remove screw and nut securing clamp and oil drain line to bracket on oil scupper. Disconnect oil drain line from oil scupper (Figure 6-4-1-2, Detail B).

- w. Attach and adjust hoisting sling as follows:

### WARNING

Injury to personnel and damage to equipment may result if two apex rings are not used when removing main transmission, 70351-48000 and 70351-58000, with all equipment (input modules, accessory modules, main generators, and hydraulic pump modules), main rotor head, and swashplate installed. Main transmission with all equipment, main rotor head, and swashplate installed, weighs about 2,565 pounds. Use of universal hoist sling assembly, 70700-20320-045, with a rated capacity of 2,500 pounds, can only be used with the addition of a second apex ring.

### CAUTION

Damage to cabin maintenance crane and helicopter will occur if cabin maintenance crane is used to remove main transmission with main rotor head and swashplate installed. Use hoisting sling and overhead hoist to remove main transmission with main rotor head and swashplate installed.

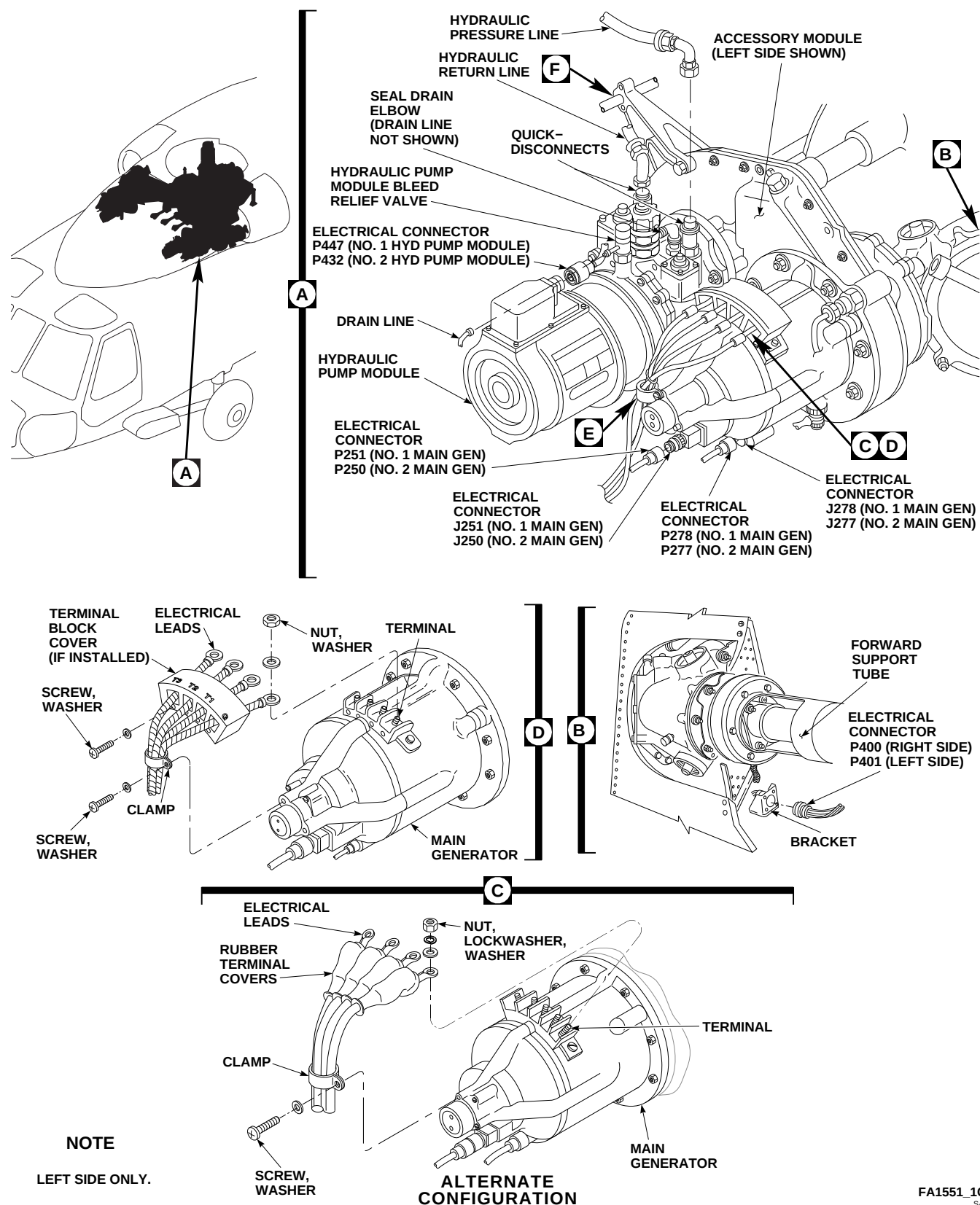
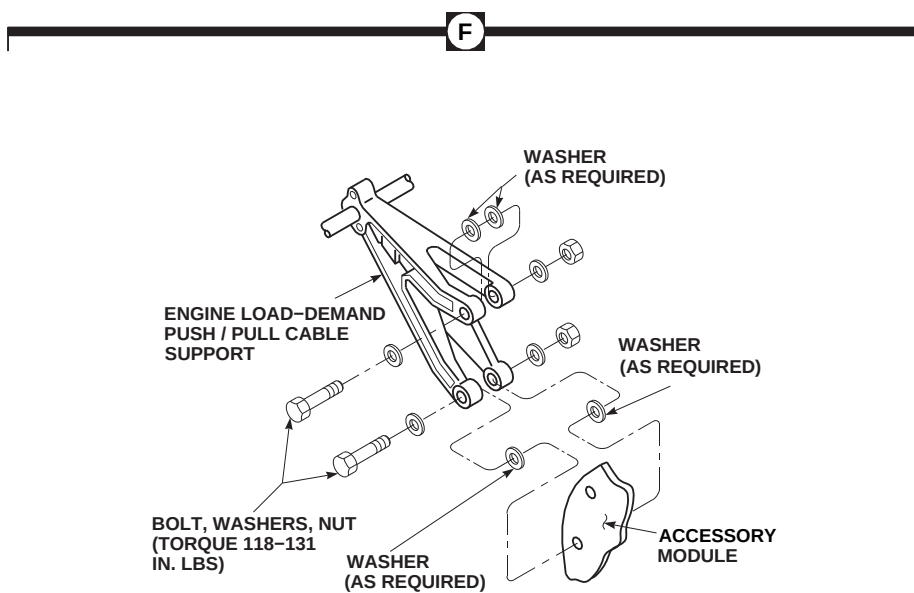
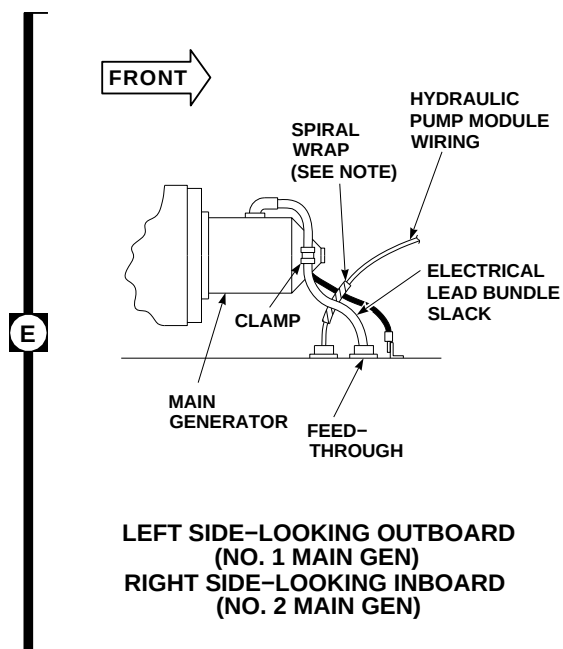
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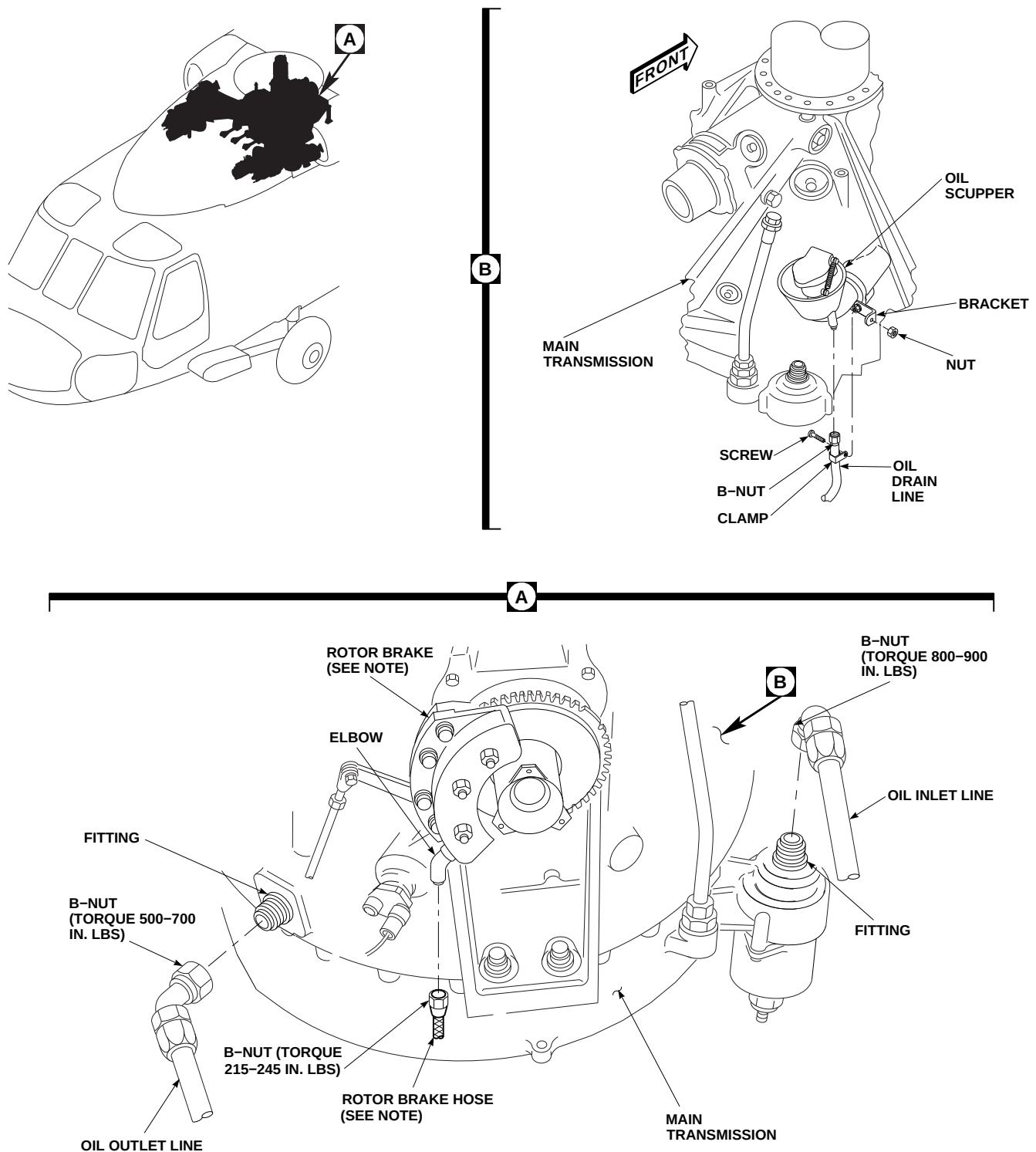
FIGURE 6-4-1-1. DISCONNECT/CONNECT LINES AND WIRING (SHEET 1 OF 2)





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FIGURE 6-4-1-1. DISCONNECT/CONNECT LINES AND WIRING (SHEET 2 OF 2)

**NOTE****RTR BRAKE****LOOKING TOWARD FRONT**FC3747A  
SA**FIGURE 6-4-1-2. DISCONNECT/CONNECT OIL INLET AND OUTLET LINES, AND ROTOR BRAKE HOSE**

**NOTE**

If removing main transmission with main rotor head installed, perform step w. (1). If removing main transmission with main rotor head and swashplate removed, perform step w. (2) (Figure 6-4-1-3).

- (1) If removing main transmission with main rotor head installed, attach and adjust hoisting sling as follows:

**CAUTION**

Damage to main rotor head components will result if hoisting sling is not properly installed. Make sure hoisting sling is not put around damper, damper reservoir, or other parts of main rotor head, other than main rotor hub arms.

- (a) Position one strap around each main rotor hub arm.
- (b) Attach two adjacent straps to one apex ring.
- (c) Attach other two adjacent straps to second apex ring.
- (d) Connect both apex rings (with straps attached) to hoist with enough capacity to lift main transmission and main rotor head (minimum of 3,000 pounds). Make sure apex rings are not interlaced.

**NOTE**

Five assistants are required to help assist and guide main transmission, with main rotor head installed, clear of helicopter.

- (e) Place one assistant at each corner of main transmission and another to operate hoist.

**CAUTION**

Damage to main rotor head components will result if hoisting sling is not properly installed. Weight distribution on hoisting sling must be evenly distributed on each main rotor hub arm so main transmission can be lifted straight up.

- (f) Adjust front straps taut and adjust rear straps to hang loosely.
  - (g) Operate hoist to put tension on hoisting sling for even distribution of weight.
  - (h) Main transmission should be level when hoisted. If not, lower hoist until tension is removed from hoisting sling and adjust front or rear straps as necessary.
  - (i) Repeat step w. (1) (g) and step w. (1) (h) until main transmission is level.
- (2) If removing main transmission with main rotor head and swashplate removed, attach and adjust hoisting sling as follows:

**CAUTION**

On helicopters with main transmission, 70351-48000 and 70351-58000, damage to hoisting sling will result if hoisting sling contacts studs on main transmission tail takeoff housing. Studs will cut into hoisting sling. Make sure hoisting sling does not contact studs.

- (a) Attach straps to input modules and around main transmission tail takeoff housing.
- (b) Attach straps to apex ring.
- (c) Attach apex ring (with straps attached) to cabin maintenance crane

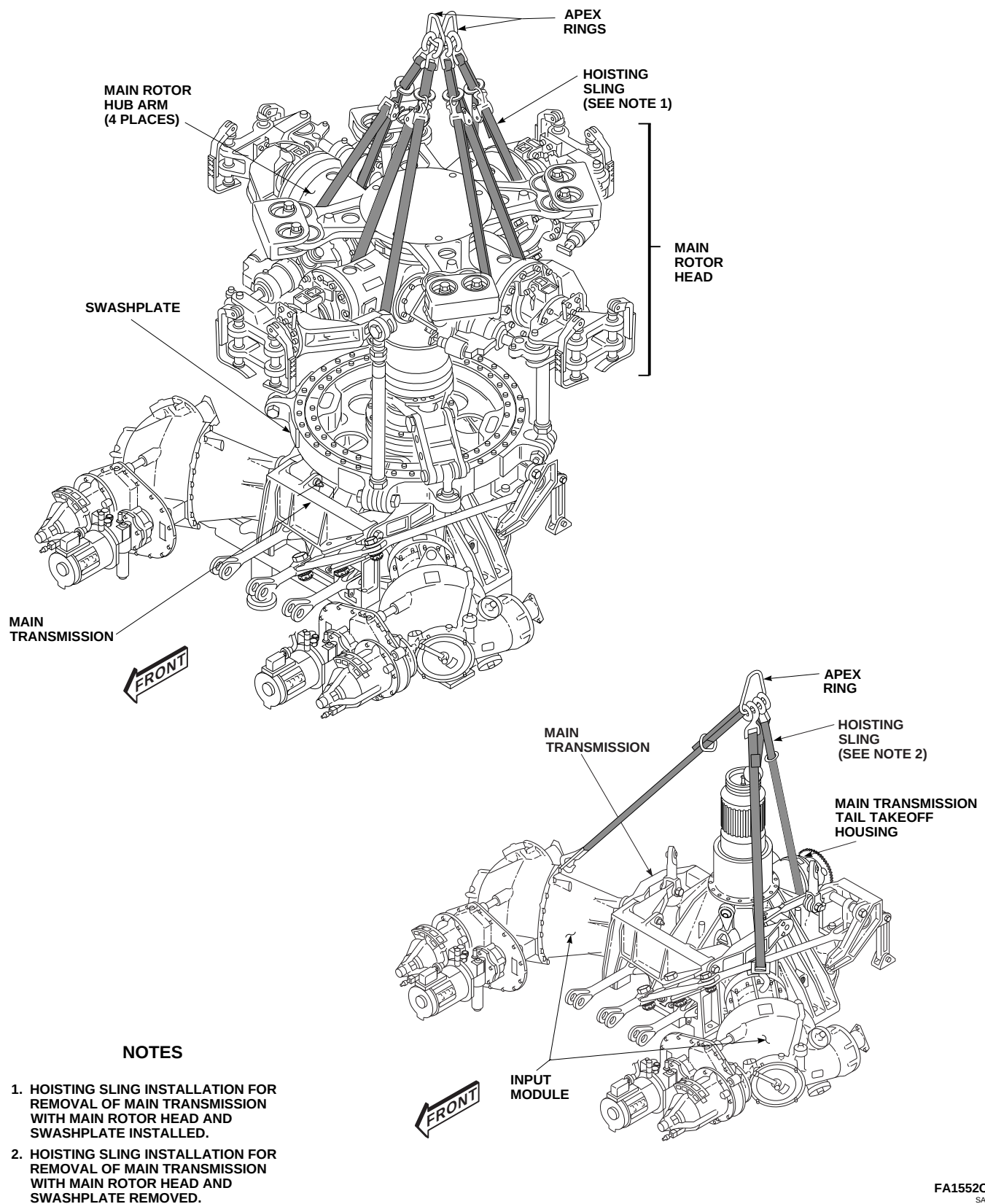
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FIGURE 6-4-1-3. ATTACH HOISTING SLING

or hoist with enough capacity to lift main transmission.

- (d) Adjust straps before lifting, for even distribution of weight.
- x. Disconnect forward and aft bellcrank supports and remove bolts securing main transmission as follows:

**WARNING**

**FSCAP**

Upper wall perimeter of main module housing is a critical area of main transmission. These surfaces must not be damaged during and after removal of main transmission. Use extreme caution when removing main transmission, and provide protective covering to these surfaces during removal.

- (1) Install protective covering on upper wall perimeter of main module housing.

**WARNING**

**FSCAP**

All self-retaining bolts, bellcranks, tie rods, connecting links, control rods, bellcrank supports, and aft tie rod support fitting contain critical surfaces which must not be damaged during and after removal or disconnection. Use extreme caution when removing or disconnecting, and provide protective covering after removal.

**CAUTION**

Damage to special self-retaining bolts, forward control rod, forward bellcrank, forward primary servo, aft control rod,

walking beam, aft primary servo, lateral control rod, lateral bellcrank, and lateral primary servo will result if special self-retaining bolts are removed incorrectly.

**NOTE**

To remove special self-retaining bolt, insert  $\frac{1}{8}$ -inch Allen wrench in hole in bolthead. Push on pin to release retaining balls at nut end of bolt. Keep pressure on pin while removing self-retaining bolt.

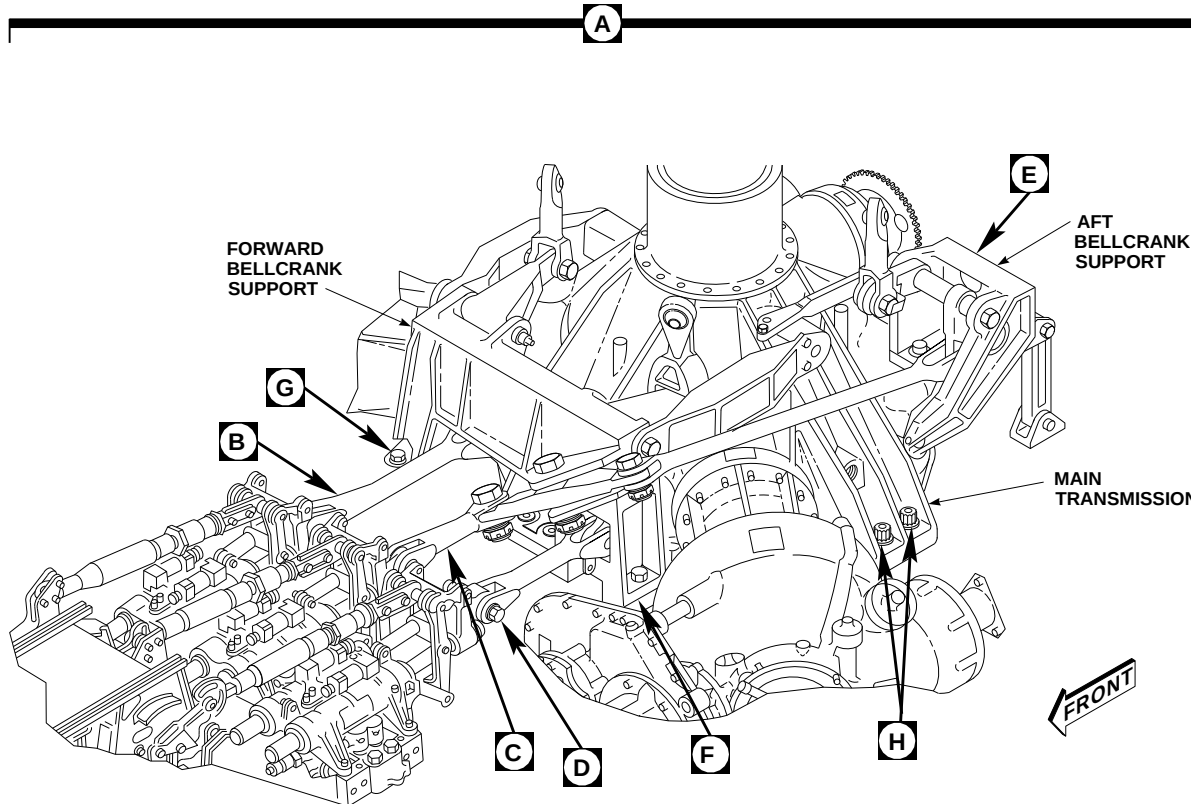
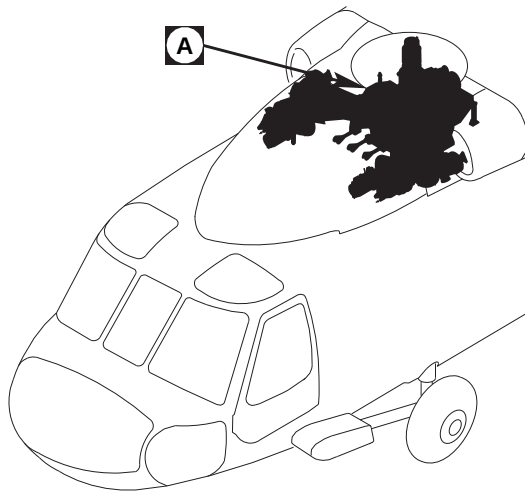
- (2) Disconnect forward, aft, and lateral control rods from forward, aft, and lateral primary servos by removing nuts, special self-retaining bolts, countersunk or flat washers, and additional washer(s), if installed (Figure 6-4-1-4, Sheet 2, Detail B, Detail C, and Detail D).

**WARNING**

**FSCAP**

Upper and lower ends of aft tie rod at aft bellcrank support are critical characteristic surfaces which must not be damaged during and after removal of main transmission. Use extreme caution when disconnecting.

- (3) Remove bolt, countersunk washer, washer, additional washer(s), if installed, and nut securing aft tie rod to aft tie rod support fitting (Figure 6-4-1-4, Sheet 2, Detail E).
- (4) Swing aft tie rod out. Remove bolts, countersunk washers, washer(s), and nuts securing aft tie rod support fitting to deck. Remove shims.
- (5) Temporarily install aft tie rod support fitting on lower end of aft tie rod and secure with bolt, countersunk washer, washer, additional washer(s), if installed, and nut.

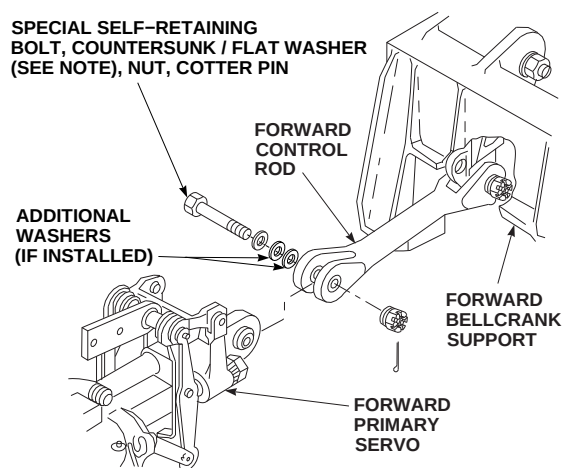
**NOTE**

ON HELICOPTERS WITH SWASHPLATE LINKAGE CONTROLS ASSEMBLY, 70400-08200-041 OR -042, A FLAT WASHER IS USED.

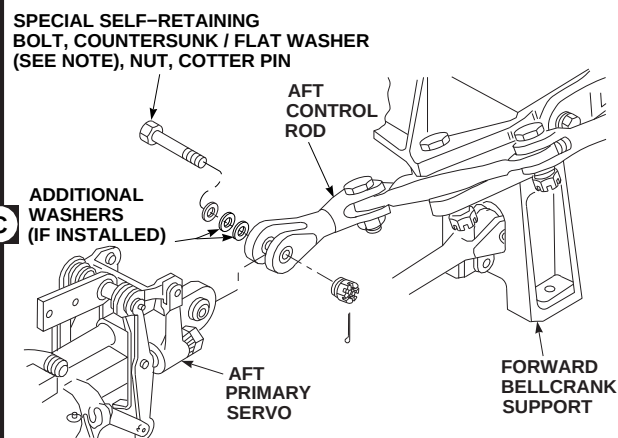
ON HELICOPTERS WITH SWASHPLATE LINKAGE CONTROLS ASSEMBLY, 70400-08200-043 OR -044, A COUNTERSUNK WASHER IS USED.

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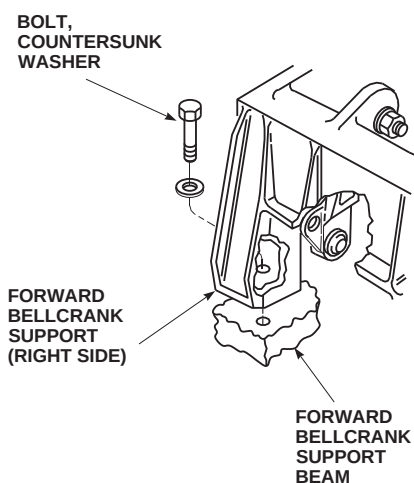
**FIGURE 6-4-1-4. DISCONNECT FORWARD AND AFT BELLCRANK SUPPORTS  
(SHEET 1 OF 3)**



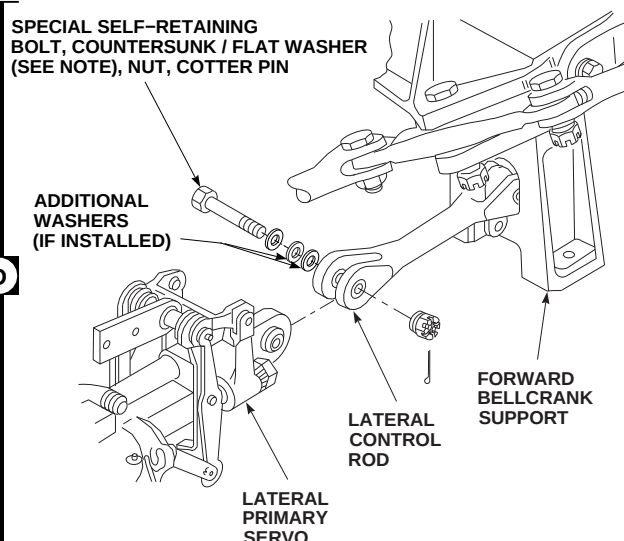
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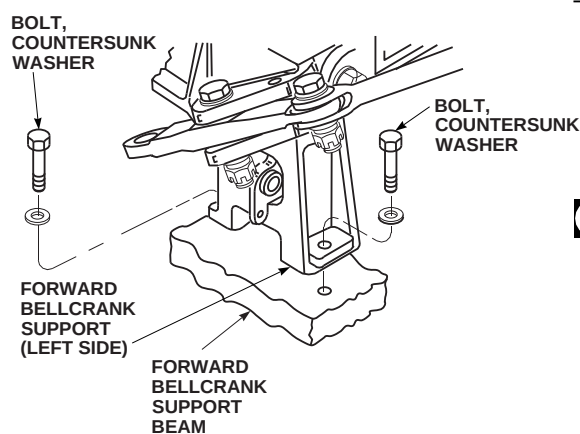
**C**



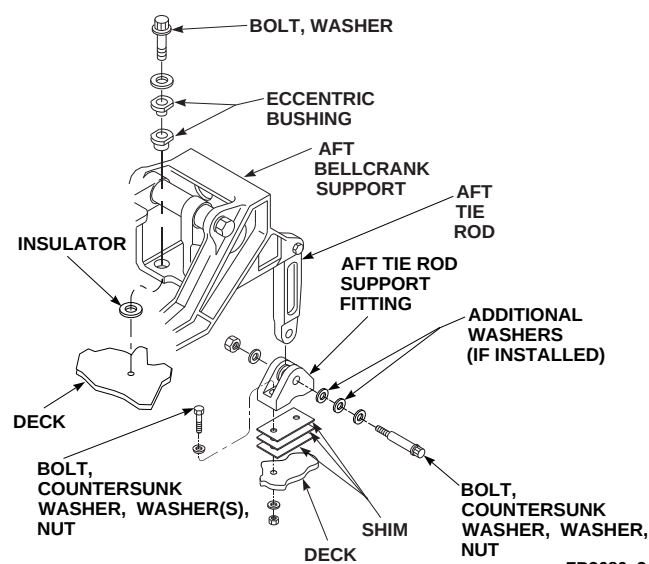
**G**



**D**



**F**

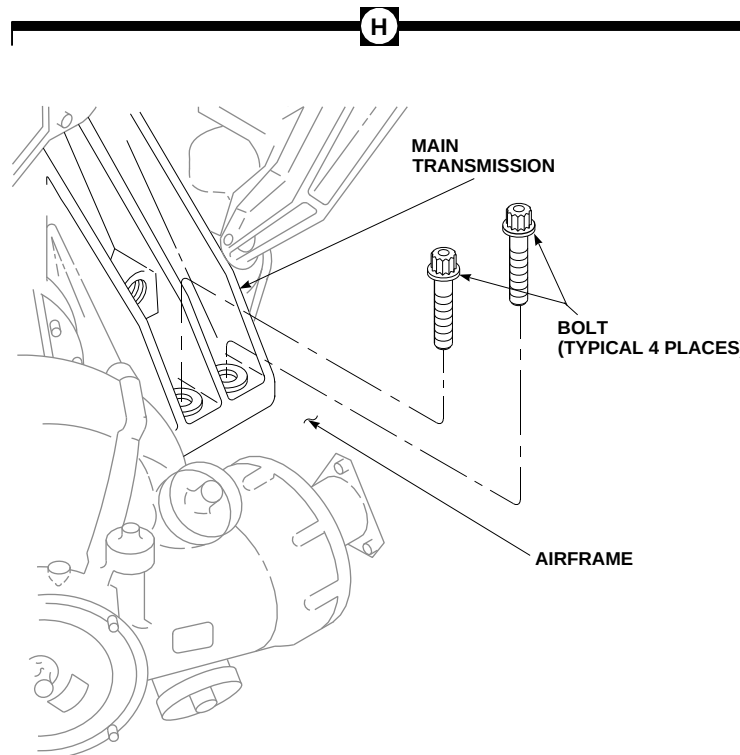


**E**

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**FIGURE 6-4-1-4. DISCONNECT FORWARD AND AFT BELLCRANK SUPPORTS (SHEET 2 OF 3)**



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**FIGURE 6-4-1-4. DISCONNECT FORWARD AND AFT BELLCRANK SUPPORTS  
(SHEET 3 OF 3)**

- (6) Remove bolt and washer securing aft bellcrank support and eccentric bushings to deck. Remove insulator.
- (7) Remove bolts and countersunk washers securing forward bellcrank support to forward bellcrank support beam (Figure 6-4-1-4, Sheet 2, Detail F and Detail G).
- (8) If not already done, install protective covering to removed components with critical characteristics.
- (9) Remove bolts securing main transmission to airframe and forward bellcrank support

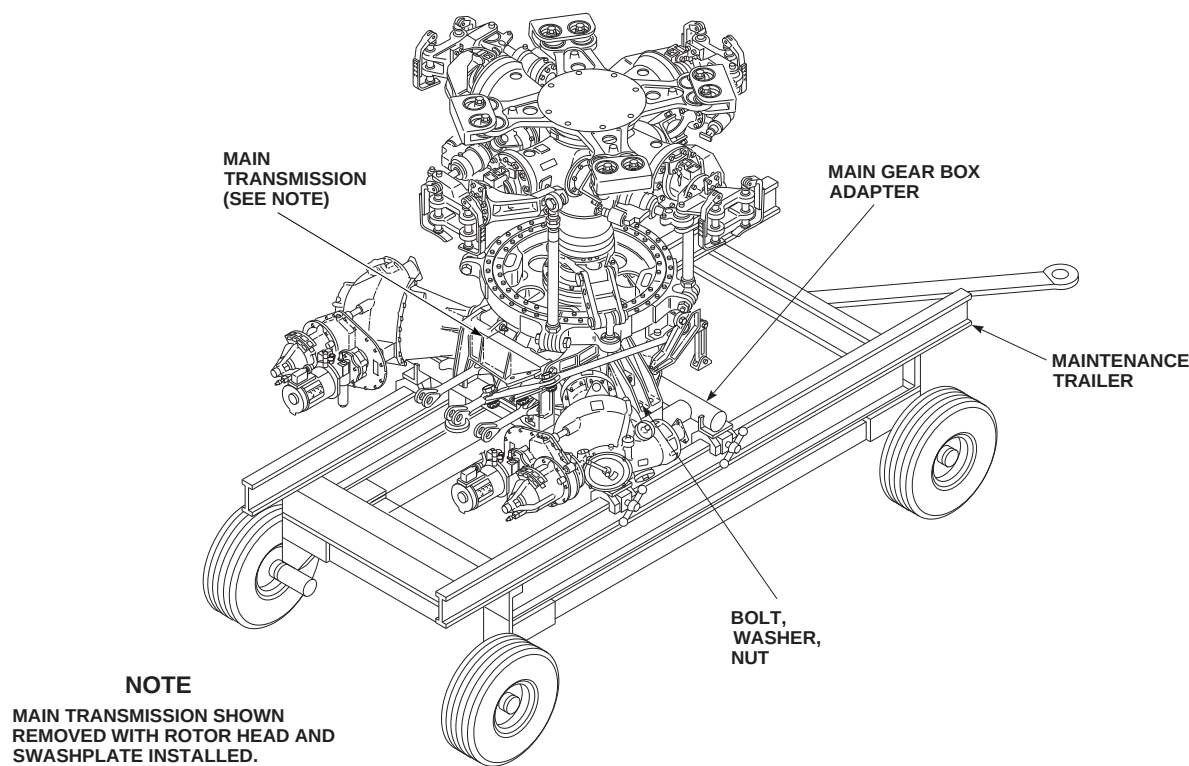
beam (Figure 6-4-1-4, Sheet 3, Detail H). If necessary, locally make mounting bolt adapter (Figure H-46, Appendix H), to remove two front bolts.

**CAUTION**

Damage to helicopter will result if all connections to main transmission are not removed prior to hoisting.

- y. Check main transmission to verify all attaching lines and electrical connections are removed/disconnected, and area is clear before hoisting.





**FIGURE 6-4-1-5. INSTALL MAIN TRANSMISSION ON MAIN GEAR BOX ADAPTER**

**NOTE**

- If main transmission is being removed without main rotor head installed, three assistants are required to help assist and guide main transmission clear of helicopter.
- If main transmission is being removed with main rotor head installed, five assistants are required to help assist and guide main transmission clear of helicopter.
- z. If main transmission is being removed without main rotor head installed, have an assistant on each side of main transmission, holding a guide strap to help guide main transmission, and another to operate hoist. If main transmission is being removed with main rotor head installed, have one assistant at each corner of main transmission and another to operate hoist.

**WARNING**

- Injury to personnel and damage to equipment will result if caution is not taken when hoisting main transmission. Keep area clear when hoisting main transmission.
- Hoist cable may break, causing injury to personnel or damage to equipment, if hoist is used to free a sticking component being removed from helicopter. Do not use too much force when removing a component.
- Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment if hook is allowed to contact pulley when removing a component from helicopter. Do not allow hook assembly to contact pulley.

- aa. Lift main transmission clear of helicopter. If main transmission is being removed with main rotor head installed, main transmission should be level when hoisted. If not, lower hoist until tension is removed from hoisting sling and adjust front or rear straps as necessary. Repeat step until main transmission is level when hoisted.
- ab. Lower main transmission onto main gear box adapter/maintenance trailer (Figure 6-4-1-5).
- ac. Install bolts, washers, and nuts securing main transmission to main gear box adapter.

#### 6-4-1.1.2 Main Transmission Teardown/Buildup.

##### NOTE

If same main transmission is to be installed and was removed with main rotor head installed, removal of main rotor head, swashplate, and main transmission harness is not required.

- a. If main transmission is to be replaced, remove the following:

##### WARNING

##### FSCAP

The upper wall perimeter of main module housing is a critical area of main transmission. These surfaces must not be damaged during teardown/buildup of main transmission. Use extreme caution when tearing down/building up main transmission. Keep protective covering on during teardown/buildup.

##### NOTE

Main rotor head will be removed when swashplate is removed.

- (1) If main transmission was removed with main rotor head installed, remove swashplate (PARA 5-4-22).

- (2) Remove main transmission harness as follows (Figure 6-4-1-6, Sheet 1, Detail A and Detail B, Sheet 2, Detail C, Detail D, and Detail E, Sheet 3, Detail F, and Sheet 4, Detail G, Detail H, and Detail J):

- (a) Tag and disconnect electrical connectors and install protective caps and plugs on electrical connectors as they are disconnected.

##### NOTE

To aid main transmission harness installation, note location of clamps and harness routing.

- (b) Remove screws, washers, spacers, and nuts securing clamps. Remove tape or tiedown straps securing main transmission harness. Remove insulation sleeving and spiral wrap from main transmission harness.
- (c) Remove main transmission harness.

##### WARNING

##### FSCAP

All self-retaining bolts, bellcranks, tie rods, connecting links, control rods, bellcrank supports, and aft tie rod support fitting contain critical surfaces which must not be damaged during and after removal or disconnection. Use extreme caution when removing or disconnecting, and provide protective covering after removal.

##### NOTE

If new main transmission is to be installed, forward and aft bellcrank supports must be removed.

- b. If necessary, remove forward bellcrank support as follows:

**CAUTION**

Damage to special self-retaining bolts, forward control rod, forward bellcrank, forward primary servo, aft control rod, walking beam, aft primary servo, lateral control rod, lateral bellcrank, and lateral primary servo will result if special self-retaining bolts are removed incorrectly.

**NOTE**

To remove special self-retaining bolt, insert 1/8-inch Allen wrench in hole in bolthead. Push on pin to release retaining balls at nut end of bolt. Keep pressure on pin while removing self-retaining bolt.

- (1) Remove nuts, special self-retaining bolts, countersunk or flat washers, and additional washer(s), if installed, securing forward, aft, and lateral control rods to forward bellcrank, walking beam, and lateral bellcrank on forward bellcrank support (Figure 6-4-1-6, Sheet 5, Detail M and Detail N, and Sheet 6, Detail Q).

**CAUTION**

Damage to special self-retaining bolts, long control rod, walking beam, and aft bellcrank will result if special self-retaining bolts are removed incorrectly.

**NOTE**

To remove special self-retaining bolt, insert 1/8-inch Allen wrench in hole in bolthead. Push on pin to release retaining balls at nut end of bolt. Keep pressure on pin while removing self-retaining bolt.

- (2) Remove nuts, special self-retaining bolts, countersunk or flat washers, and additional washer(s), if installed, securing long

control rod between walking beam and aft bellcrank (Figure 6-4-1-6, Sheet 5, Detail L).

- (3) Remove bolts, countersunk washers, and nylon washers securing left and right tie rods to main transmission.
  - (4) Remove bolts, countersunk washers, nylon washers, washers, and nuts securing forward bellcrank support tongue to main transmission. Remove eccentric bushings, and **70583-701213** spring clip bushing retainer **701214-SUBQ** dowel pin retaining zee from forward bellcrank support tongue (Figure 6-4-1-6, Sheet 6, Detail P).
  - (5) Loosen nuts securing left and right tie rods to forward bellcrank support enough so that tie rods can be slipped off tie rod dowel pins on main transmission (Figure 6-4-1-6, Sheet 5, Detail L).
  - (6) Remove forward bellcrank support with bellcranks, left and right tie rods, and walking beam attached. Remove insulators (Figure 6-4-1-6, Sheet 5, Detail L and Sheet 6, Detail P).
- c. Remove aft bellcrank support as follows (Figure 6-4-1-6, Sheet 5, Detail L):
- (1) Remove nut and eccentric bushings securing aft longitudinal bellcrank support arm to aft bellcrank support.
  - (2) Remove bolt, countersunk washer, **701214-SUBQ** retaining channel, and washer securing aft longitudinal bellcrank support arm to main transmission. Remove aft longitudinal bellcrank support arm.
  - (3) Remove bolt, countersunk washer, washer, additional washer(s), if installed, and nut securing aft tie rod to aft tie rod support fitting.
  - (4) **701214-SUBQ** Remove bolts, washers, and nuts securing retaining angle to aft bellcrank support.

- (5) Remove bolt, countersunk washer, and nylon washer securing leg of aft bellcrank support to main transmission.
- (6) Remove aft bellcrank support with insulator.
- d. If not already done, install protective covering to removed components with critical characteristics.
- e. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean primer from bolts, insulators, and areas on main transmission that mate with tie rods, forward bellcrank support, and aft bellcrank support.
- f. Using nonmetallic scraper, clean sealing compound from main transmission.
- g. Inspect left and right tie rods (PARA 11-3-16).
- h. Inspect aft bellcrank (PARA 11-3-20).
- i. Inspect lug surface areas of aft tie rod support fitting (PARA 11-3-19). After inspection, temporarily install aft tie rod support fitting on lower end of aft tie rod and secure with bolt, countersunk washer, washer, additional washer(s), if installed, and nut.
- j. Inspect long control rod (PARA 11-3-17).
- k. Inspect forward bellcrank support lug areas, top flange, radii, and support lug surfaces (PARA 11-3-15).
- l. Visually inspect bushings of forward bellcrank support for damaged or worn cadmium-plating. If cadmium-plating is damaged or worn, apply solid film lubricant, Item 201, Appendix D, for corrosion protection until part is replaced.
- m. Inspect main transmission dowel pins (PARA 6-3-2).
- n. Inspect lug areas of aft bellcrank support, top flange, and radius (PARA 11-3-18).
- o. Visually inspect bushings of aft bellcrank support for damaged or worn cadmium-plating. If cadmium-plating is damaged or worn, apply solid film lubricant, Item 201, Appendix D, for corrosion protection until part is replaced.
- p. Inspect special self-retaining bolts (PARA 11-3-13).
- q. Inspect lug and surface areas of aft longitudinal bellcrank support arm (PARA 11-3-23).
- r. Visually inspect main transmission and cabin area at STA 308.0 through STA 402.0.
- s. If new main module is being installed, the following items must be removed from old main module and installed on replacement main module:
  - (1) Remove left and right input module (PARA 6-4-25).
  - (2) Remove main transmission oil temperature switch as follows (Figure 6-4-1-6, Sheet 4, Detail K):
    - (a) Using nonmetallic scraper, remove sealing compound from around mating flange.
    - (b) Remove main transmission oil temperature switch and packing from mating flange of main transmission. Discard packing.
  - (3) Remove main transmission oil pressure sensor as follows:
 

Damage to threaded port will result if excessive torque is used to remove main transmission oil pressure sensor. Make sure all sealing compound is removed prior to removing main transmission oil pressure sensor.

    - (a) Using nonmetallic scraper, remove sealing compound from area around mating flange.

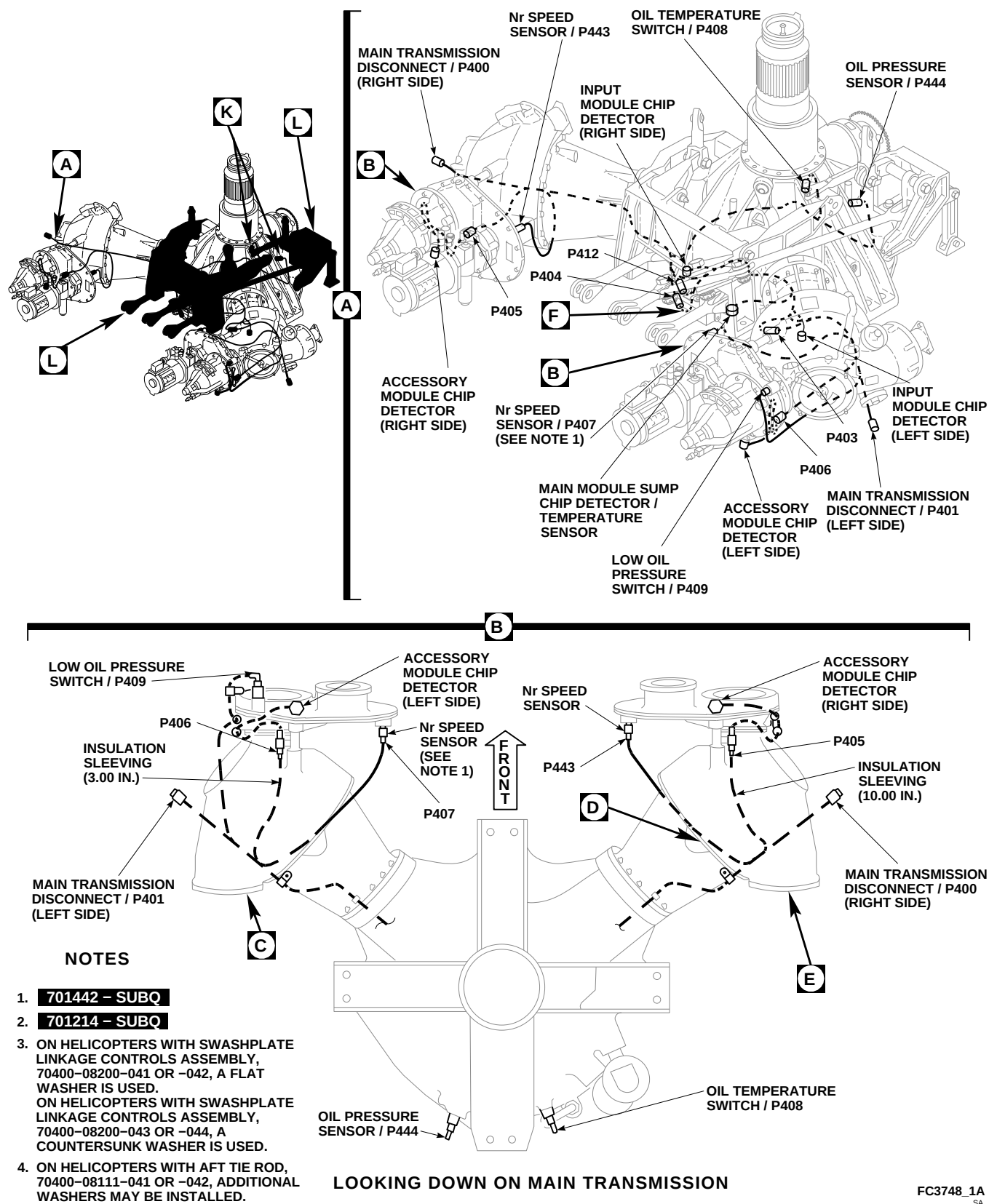
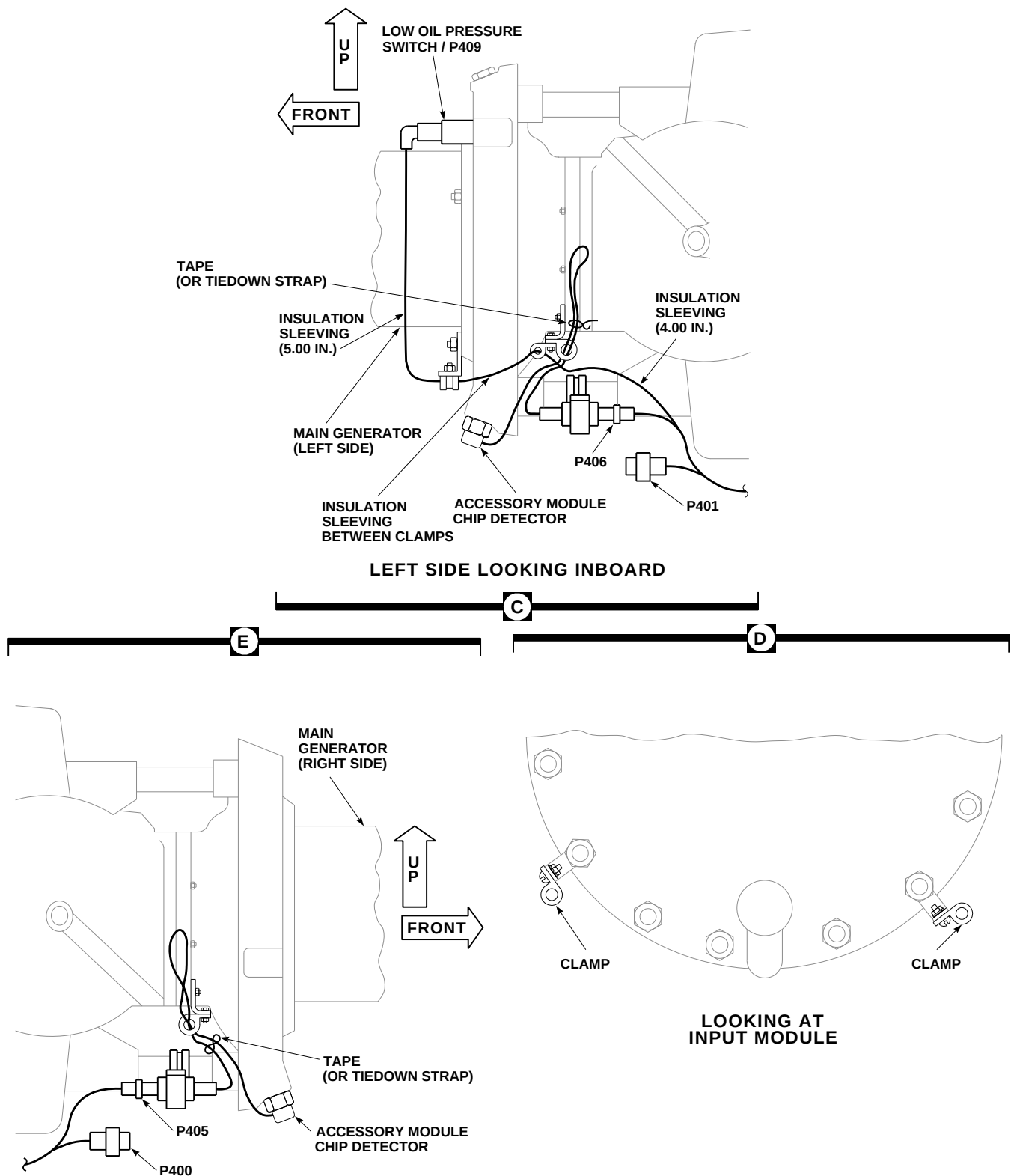


FIGURE 6-4-1-6. TEARDOWN/BUILDUP MAIN TRANSMISSION (SHEET 1 OF 6)

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SA



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FIGURE 6-4-1-6. TEARDOWN/BUILDUP MAIN TRANSMISSION (SHEET 2 OF 6)



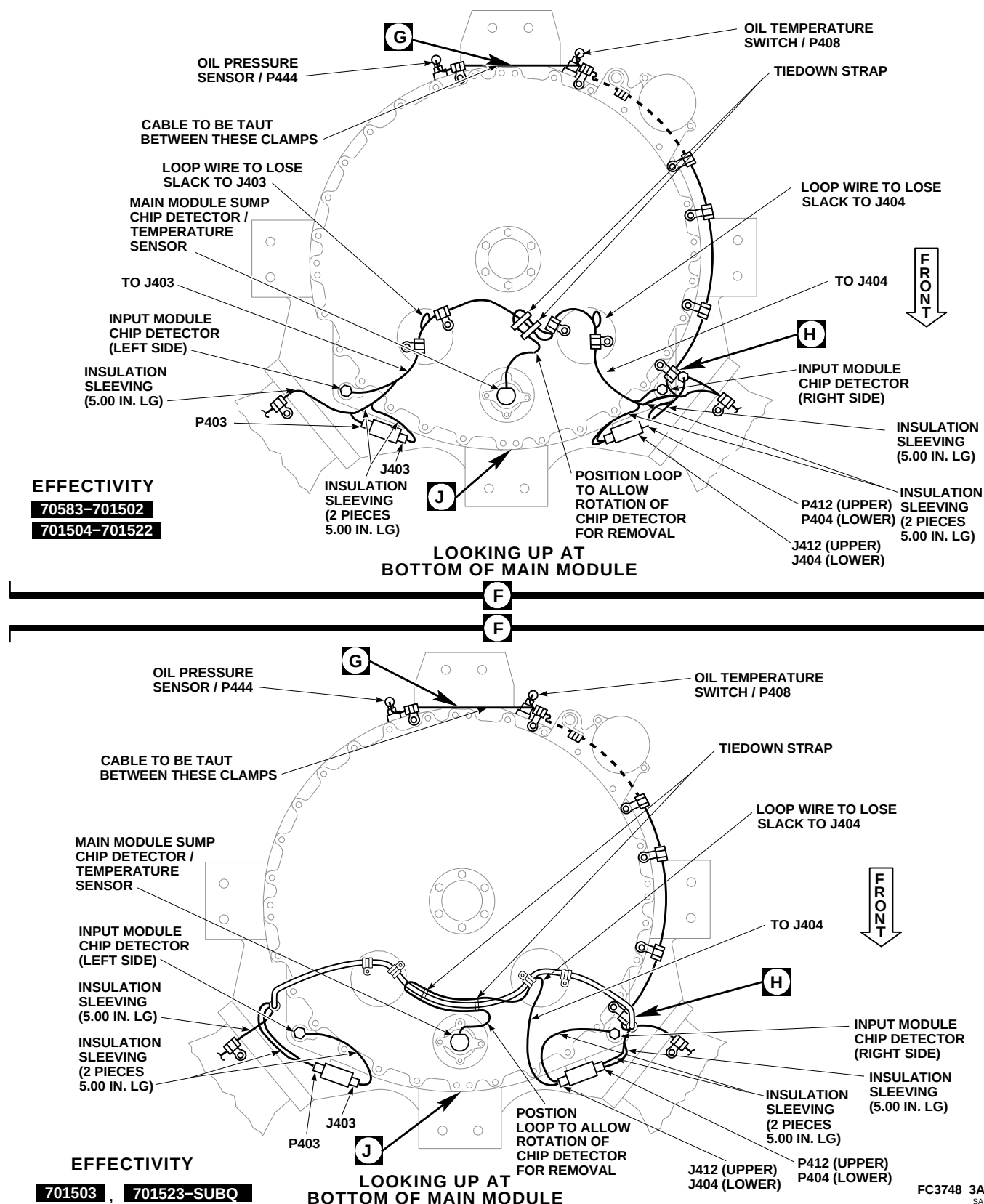
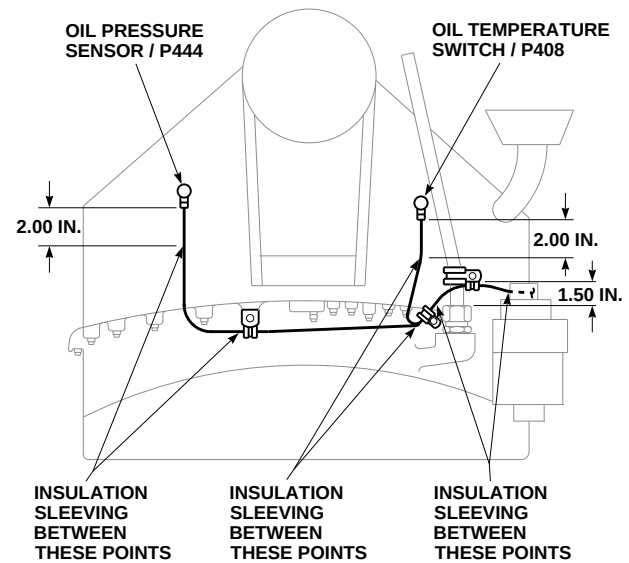
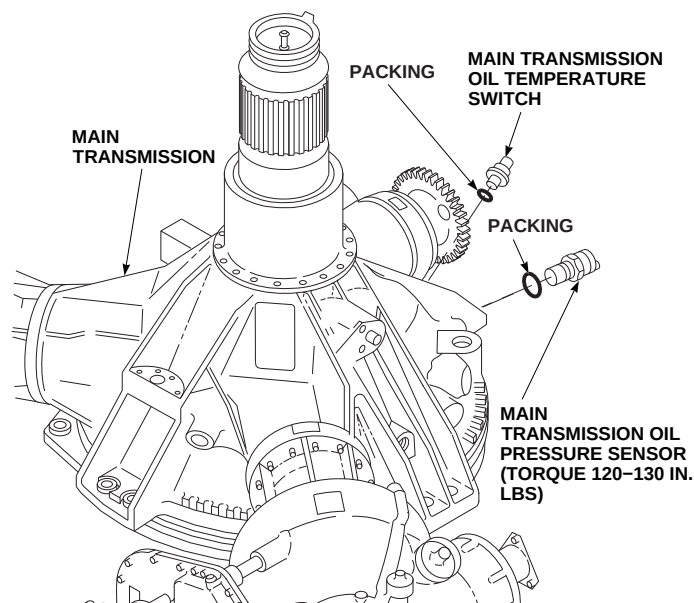


FIGURE 6-4-1-6. TEARDOWN/BUILDUP MAIN TRANSMISSION (SHEET 3 OF 6)

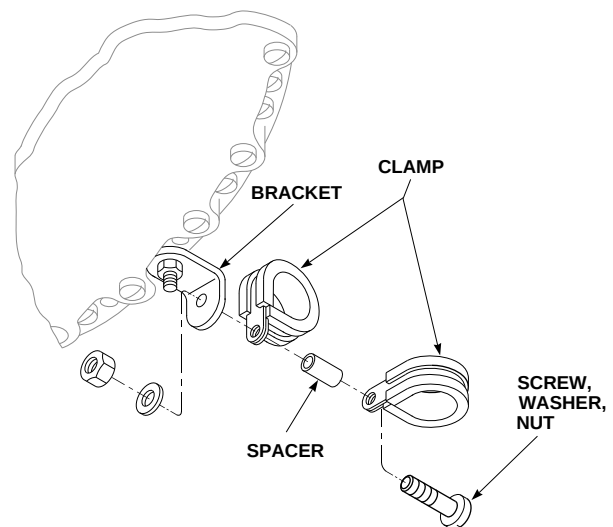
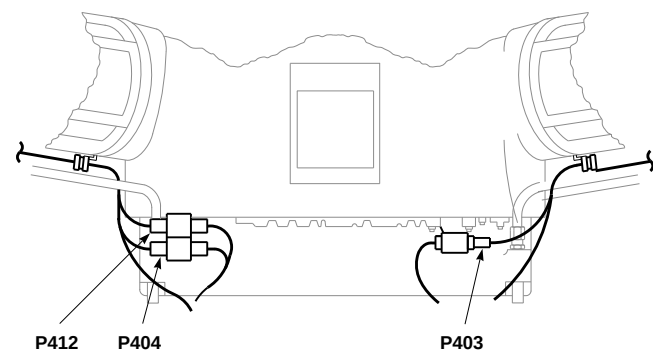


**K**

**G**

**J**

**H**



FRONT OF MAIN TRANSMISSION  
LOOKING AFT

LOOKING UP AT  
BOTTOM OF MAIN MODULE

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SA

FIGURE 6-4-1-6. TEARDOWN/BUILDUP MAIN TRANSMISSION (SHEET 4 OF 6)



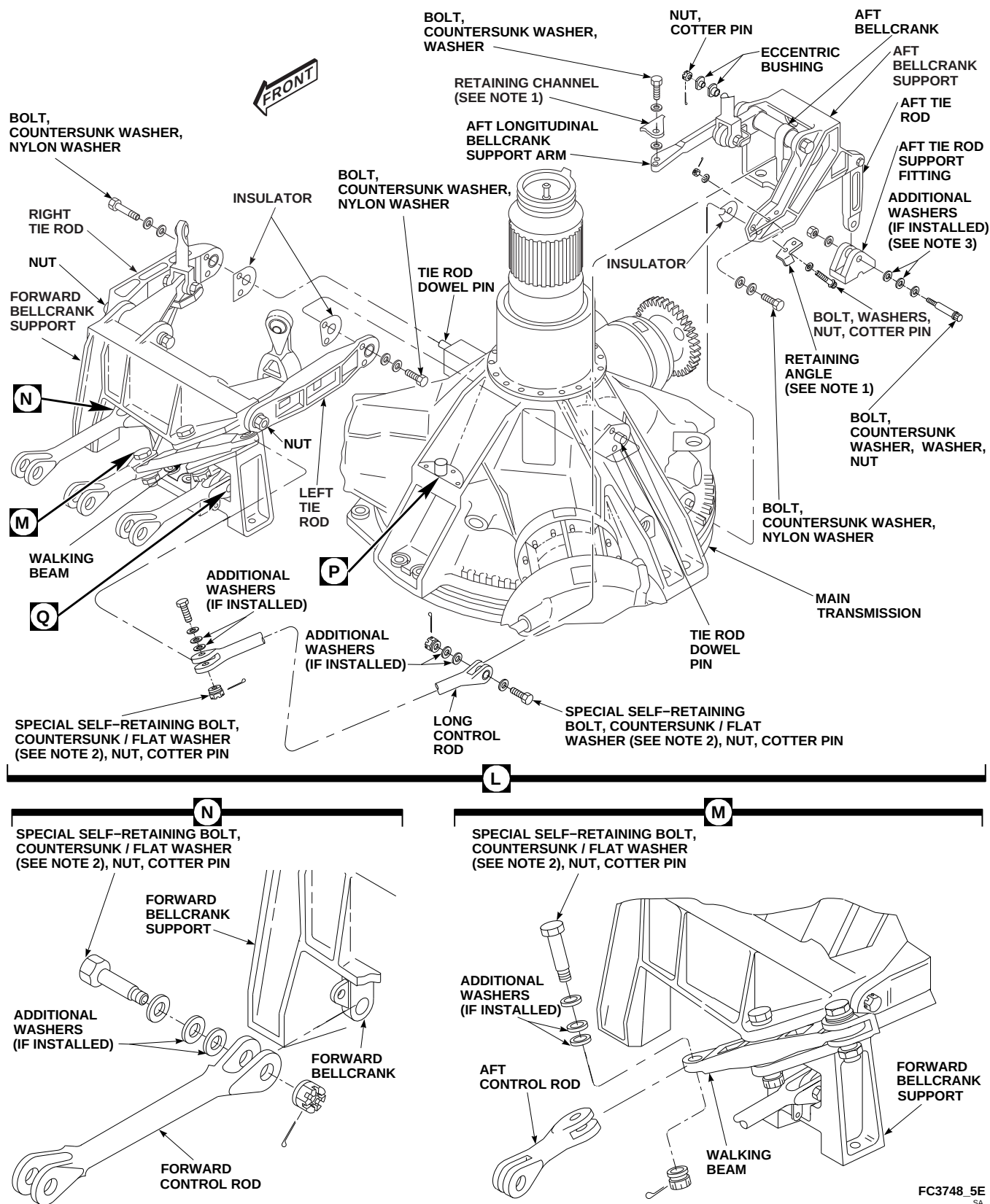


FIGURE 6-4-1-6. TEARDOWN/BUILDUP MAIN TRANSMISSION (SHEET 5 OF 6)

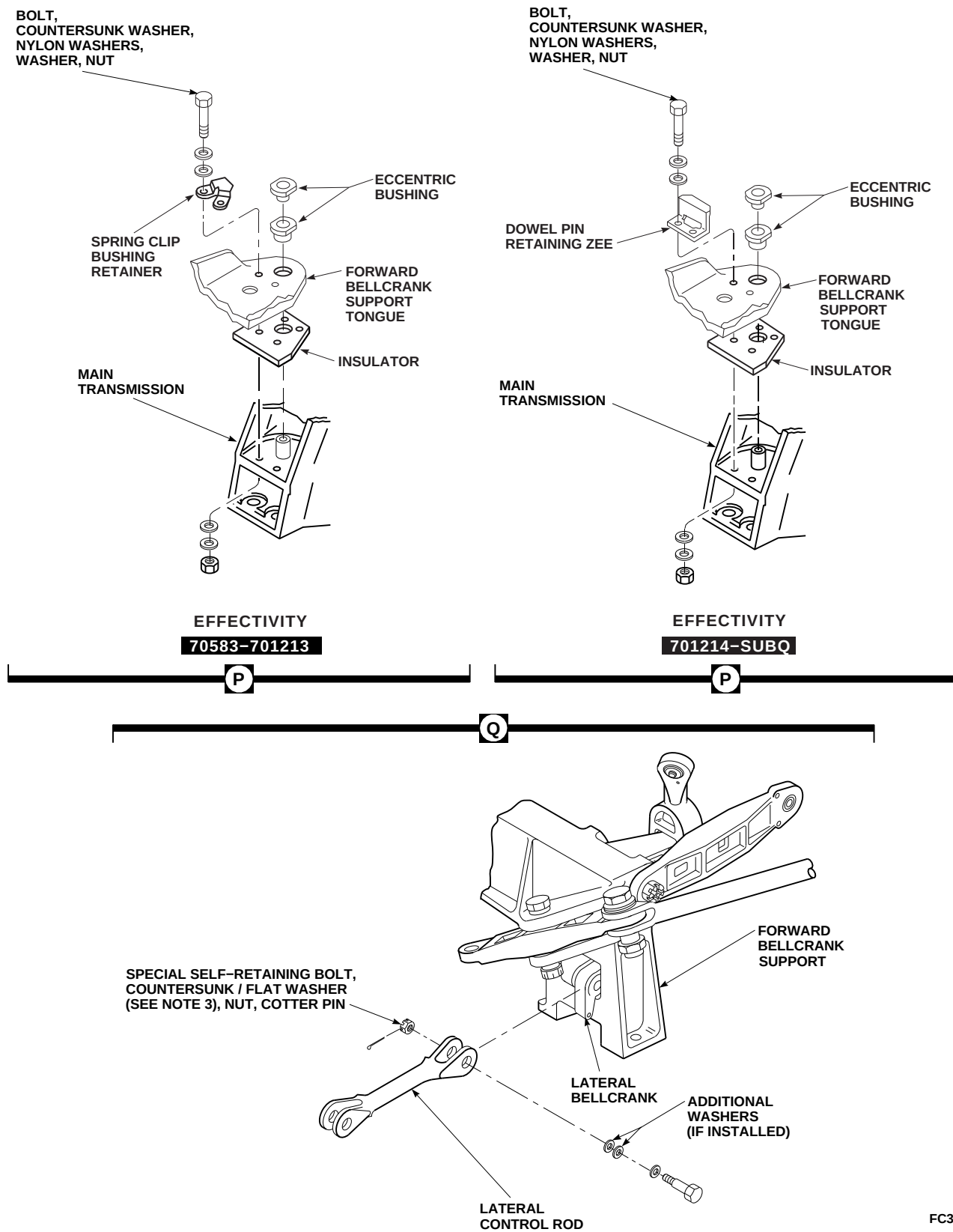


FIGURE 6-4-1-6. TEARDOWN/BUILDUP MAIN TRANSMISSION (SHEET 6 OF 6)

- (b) Position 1-quart container to catch residual oil.
- (c) Remove main transmission oil pressure sensor from mating flange of main transmission. Discard packing.
- (4) Install main transmission oil pressure sensor on replacement main module as follows:
  - (a) Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing.
  - (b) Install packing on main transmission oil pressure sensor.
  - (c) Install main transmission oil pressure sensor in mating flange of main transmission.
- (b) Install packing on main transmission oil temperature switch.
- (c) Install main transmission oil temperature switch in mating flange of main transmission.
- (d) Using lockwire, Item 197, Appendix D, lockwire main transmission oil temperature switch.
- (e) Using sealing compound, Item 292, Appendix D, apply to seal area around mating flange.
- (6) Install left and right input module on replacement main module (PARA 6-4-25).
- t. If necessary, install main transmission harness on replacement main module as follows (Figure 6-4-1-6, Sheet 1, Detail A and Detail B, Sheet 2, Detail C, Detail D, and Detail E, Sheet 3, Detail F, and Sheet 4, Detail G, Detail H, and Detail J):
  - (1) Remove protective caps and plugs from electrical connectors as they are connected.
  - (2) Inspect and treat electrical connectors as they are connected (PARA 1-7-20).



Damage to main transmission oil pressure sensor will result if overtorqued. Oil pressure readings will be affected. Do not overtorque main transmission oil pressure sensor.

- (d) **TORQUE MAIN TRANSMISSION OIL PRESSURE SENSOR TO 120 - 130 INCH-POUNDS.**
- (e) Using lockwire, Item 197, Appendix D, lockwire main transmission oil pressure sensor.
- (f) Using sealing compound, Item 292, Appendix D, apply to seal area around mating flange.
- (5) Install main transmission oil temperature switch on replacement main module as follows:
  - (a) Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing.

#### NOTE

Tiedown straps may be used as an alternate for tape.

- (3) Route and secure main transmission harness with clamps, screws, washers, nuts, and spacers. Install insulation sleeving, and spiral wrap, Item 315, Appendix D, as necessary. Secure slack in main transmission harness with tape, Item 317C, Appendix D, or tiedown straps, Item 329, Appendix D.

#### 6-4-1.1.3 Prepare to Install.

- a. Make sure helicopter wheels are chocked.
- b. Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).

- c. Inspect main transmission forward bridge drain hole (PARA 1-7-20).

**NOTE**

Main transmission must be installed prior to installing forward and aft bellcrank supports, and attaching flight controls.

- d. Check main transmission to make sure all parts removed from old main module have been installed on replacement main module, except for forward and aft bellcrank supports and attaching flight controls.

**NOTE**

Bolt, SS5209-10-34, is the preferred bolt. If bolts, SS5209-10-34, are not available, use new bolts, NAS630-35, as an alternative.

- e. Check to make sure bolts, SS5209-10-34, are being used. If bolts, NAS630-35, were being used, replace with bolts, SS5209-10-34.
- f. Check bolts and breakaway torque of barrel-nuts that secure main transmission as follows:
  - (1) Check bolt threaded area for damage.
  - (2) Using fluorescent penetrant inspection kit (ASTM E1417), fluorescent penetrant inspect bolt for cracks. **NONE ALLOWED.** If cracks are found, replace bolt.
  - (3) Install bolt into barrel-nut until two threads extend beyond barrel-nut.
  - (4) Using torque wrench, back out bolt. **MINIMUM BREAKAWAY TORQUE IS 32 INCH-POUNDS.** If breakaway torque is less than 32 inch-pounds, replace barrel-nut. Proceed to step g.

- g. If required, replace barrel-nut(s) as follows (Figure 6-4-1-7, Sheet 1, Detail B):

- (1) Using nonmetallic scraper, remove sealing compound from around cap plug in airframe or forward bellcrank support beam from inside main transmission well. Remove cap plug.
- (2) Remove barrel-nut and retainer from airframe or forward bellcrank support beam.

**NOTE**

Make sure that only barrel-nuts, RMLH2577, are used. These barrel-nuts are identified with letters "EN" or "SPS" stamped on an end (additional marking may also include "LH2577").

- (3) Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to barrel-nut and retainer.

**CAUTION**

Damage to recess in airframe or forward bellcrank support beam will result if barrel-nut is incorrectly installed. To prevent damage to recess, make sure barrel-nut is installed with barrel (round side) toward bolthead.

**NOTE**

Installation and sealing of cap plugs will be performed after main transmission and forward bellcrank support are installed.

- (4) Install barrel-nut and retainer in airframe or forward bellcrank support beam while epoxy primer is still wet.

**WARNING**

Injury to personnel and damage to equipment may result if two apex rings are not used when removing main transmission, 70351-48000 and 70351-58000, with all equipment (input modules, accessory modules, main generators, and hydraulic pump modules), main rotor head, and swashplate installed. Main transmission with all equipment, main rotor head, and swashplate installed, weighs about 2,565 pounds. Use of universal hoist sling assembly, 70700-20320-045, with a rated capacity of 2,500 pounds, can only be used with the addition of a second apex ring.

**CAUTION**

Damage to cabin maintenance crane and helicopter will occur if cabin maintenance crane is used to install main transmission with main rotor head and swashplate installed. Use hoisting sling and overhead hoist to install main transmission with main rotor head and swashplate installed.

**NOTE**

If installing main transmission, with main rotor head installed, perform step h. If installing main transmission with main rotor head and swashplate removed, perform step i. (Figure 6-4-1-3).

- h. If installing main transmission, with main rotor head installed, attach and adjust hoisting sling as follows:

**CAUTION**

Damage to main rotor head components will occur if hoisting sling is not

properly installed. Make sure hoisting sling is not put around damper, damper reservoir, or other parts of rotor head, other than main rotor hub arms.

- (1) Position one strap around each main rotor hub arm.
- (2) Attach two adjacent straps to one apex ring.
- (3) Attach other two adjacent straps to second apex ring.
- (4) Connect both apex rings (with straps attached) to hoist with enough capacity to lift main transmission and main rotor head (minimum of 3,000 pounds). Make sure apex rings are not interlaced.

**NOTE**

Five assistants are required to help assist and guide main transmission, with main rotor head installed, into helicopter.

- (5) Place one assistant at each corner of main transmission and another to operate hoist.

**CAUTION**

Damage to main rotor head components will result if hoisting sling is not properly installed. Weight distribution on hoisting sling must be evenly distributed on each main rotor hub arm so main transmission can be lifted straight up.

- (6) Adjust front straps taut and adjust rear straps to hang loosely.
- (7) Operate hoist to put tension on hoisting sling for even distribution of weight.
- (8) Main transmission should be level when hoisted. If not, lower hoist until tension is removed from hoisting sling and adjust front or rear straps as necessary.

- (9) Repeat step h. (7) and step h. (8) until main transmission is level.
- i. If installing main transmission with main rotor head and swashplate removed, attach and adjust hoisting sling as follows:

**CAUTION**

On helicopters with main transmission, 70351-48000 and 70351-58000, damage to hoisting sling will result if hoisting sling contacts studs on main transmission tail takeoff housing. Studs will cut into hoisting sling. Make sure hoisting sling does not contact studs.

- (1) Attach straps to input modules and around main transmission tail takeoff housing.
- (2) Attach straps to apex ring.
- (3) Attach apex ring (with straps attached) to hoist or cabin maintenance crane with enough capacity to lift main transmission.
- (4) Adjust straps before lifting, for even distribution of weight.

#### 6-4-1.1.4 Install.

- a. Before hoisting main transmission, line up barrel-nuts as follows:

**CAUTION**

Damage to recess in airframe or forward bellcrank support beam will result if barrel-nut is incorrectly installed. To prevent damage to recess, make sure barrel-nut is installed with barrel (round side) toward bolthead.

**NOTE**

Make sure that only barrel-nuts, RMLH2577, are used. These barrel-nuts are identified with letters "EN" or "SPS"

stamped on an end (additional marking may also include "LH2577").

- (1) Install mounting bolts into each barrel-nut two to three threads.
- (2) Line up mounting bolts perpendicular to mounting surface of airframe/forward bellcrank support beam.
- (3) Carefully remove bolts so as not to disturb position of barrel-nuts.

**NOTE**

Do not apply corrosion preventive compound to mounting feet of main transmission.

- b. Using corrosion preventive compound, Item 114, Appendix D, apply spray-coat on each mount pad on airframe.

**WARNING**

**FSCAP**

The upper wall perimeter of main module housing is a critical area of main transmission. These surfaces must not be damaged during and after installation of main transmission. Use extreme caution when installing main transmission. Keep protective covering on during installation.

**CAUTION**

- Damage to main transmission and adjacent components will result if main transmission is allowed to swing into components during installation. Carefully guide main transmission into place to clear all components on helicopter.
- Damage to cabin maintenance crane and helicopter will occur if cabin



maintenance crane is used to install main transmission with main rotor head and swashplate installed. Use hoisting sling and overhead hoist to install main transmission with main rotor head and swashplate installed.

### NOTE

- Main transmission, 70351-08000, with all equipment (input modules, accessory modules, main generators, and hydraulic pumps modules) installed, weighs about 1,200 pounds. Main transmission with all equipment, main rotor head, and swashplate installed, weighs about 2,400 pounds. Main module alone weighs about 750 pounds.
  - Main transmissions, 70351-48000 and 70351-58000, with all equipment (input modules, accessory modules, main generators, and hydraulic pump modules) installed, weighs about 1,365 pounds. Main transmission with all equipment, main rotor head, and swashplate installed, weighs about 2,565 pounds. Main module alone weighs about 915 pounds.
  - If main transmission is being installed without main rotor head installed, three assistants are required to help assist and guide main transmission into helicopter.
  - If main transmission is being installed with main rotor head installed, five assistants are required to help assist and guide main transmission into helicopter.
- c. If not installed, attach guide straps as necessary to main transmission.
- d. Remove bolts, washers, and nuts securing main transmission to main gear box adapter (Figure 6-4-1-5).

### WARNING

- Injury to personnel may result when applying sealing compound to underside of

mounting feet while main transmission is suspended. Use extreme care when applying sealing compound.

- Injury to personnel and damage to equipment will result if caution is not taken when hoisting main transmission. Keep area clear when hoisting main transmission.
  - Hoist cable may break, causing injury to personnel or damage to equipment, if hoist is used to free a sticking component being removed from main gear box adapter. Do not use too much force when removing a component.
  - Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment if hook is allowed to contact pulley when removing a component from main gear box adapter. Do not allow hook assembly to contact pulley.
- e. Apply sealing compound to mount pads or mounting feet as follows:

### NOTE

Make sure corrosion preventive compound has completely dried. Do not mix sealing compound and corrosion preventive compound.

- (1) Using caulking gun, apply sealing compound, Item 292, Appendix D, to mount pads on airframe over layer of corrosion preventive compound previously applied. Apply sealing compound in beads approximately  $\frac{1}{8}$ -inch thick as shown in Figure 6-4-1-7, Sheet 1, Detail C. Do not brush or spread sealing compound.
- (2) As an alternate method, perform the following:
  - (a) Lift main transmission enough to gain access to underside of mounting feet.

**WARNING**

Injury to personnel will result if hoisting sling or hoist allows main transmission to fall while personnel are working under main transmission. Do not work directly under main transmission when main transmission is supported only by hoisting sling and hoist.

**NOTE**

Do not move main gear box adapter and maintenance trailer from under main transmission while applying sealing compound.

- (b) Using paint brush, Item 86, Appendix D, apply thin brush-coat of sealing compound, Item 287 or 292, Appendix D, to underside of mounting feet on main transmission.
- (3) Using sealing compound, Item 287 or 292, Appendix D, apply bead around rim of boltholes on top and bottom sides of mounting feet.
- f. If main transmission is being installed without main rotor head installed, place an assistant on each side of main transmission, holding a guide strap to help guide main transmission, and another to operate hoist. If installing main transmission with main rotor head installed, place one assistant at each corner of main transmission and another to operate hoist.

**WARNING**

Injury to personnel and damage to equipment will result if caution is not taken when hoisting main transmission. Keep area clear when hoisting main transmission.

- g. Lift main transmission and lower into position on helicopter while sealing compound is still

wet. If installing main transmission with main rotor head installed, main transmission should be level when hoisted. If not, lower hoist until tension is removed from hoisting sling and adjust front or rear straps as necessary. Repeat step until main transmission is level when hoisted.

**CAUTION**

Damage to equipment will result if epoxy primer is applied to threads of bolts. Epoxy primer will make removal of bolts difficult and could damage barrel-nuts. Do not allow epoxy primer to contact bolt threads.

- h. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to shank of bolts.

**WARNING****FSCAP**

Verification of minimum torque of bolts is a critical characteristic.

**NOTE**

- Bolt, SS5209-10-34, is the preferred bolt. If bolts, SS5209-10-34, are not available, use new bolts, NAS630-35, as an alternative.
- It may be easier to line up front mounting holes first, insert bolts, and then line up rear holes and insert bolts.
- i. Install bolts securing main transmission to airframe and forward bellcrank support beam while primer is still wet (Figure 6-4-1-7, Sheet 2, Detail E).
- j. **TORQUE ALL EIGHT BOLTS SECURING MAIN TRANSMISSION TO 154 - 170 FOOT-POUNDS.**



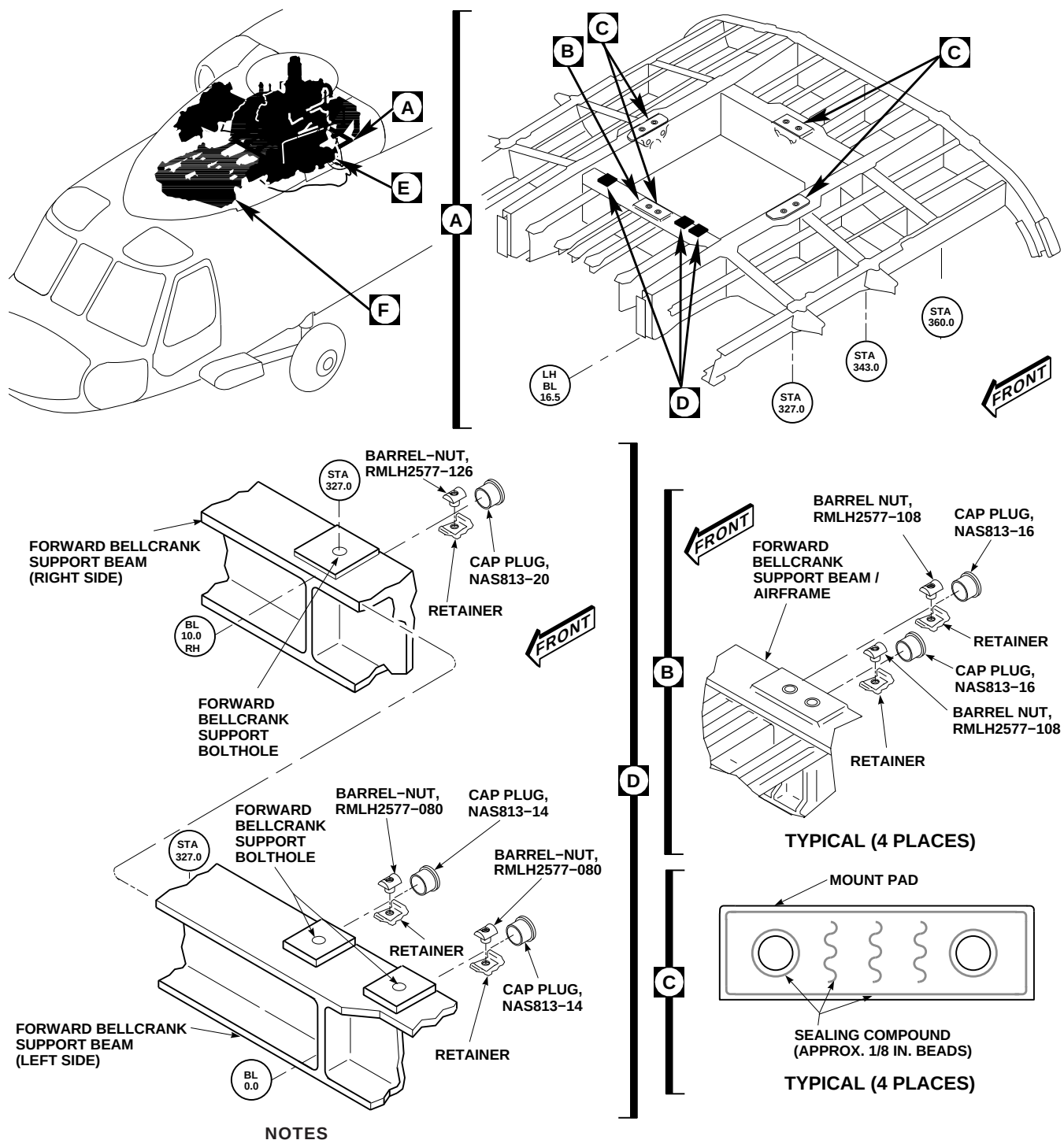


FIGURE 6-4-1-7. INSTALL MAIN TRANSMISSION AND FLIGHT CONTROLS (SHEET 1 OF 7)

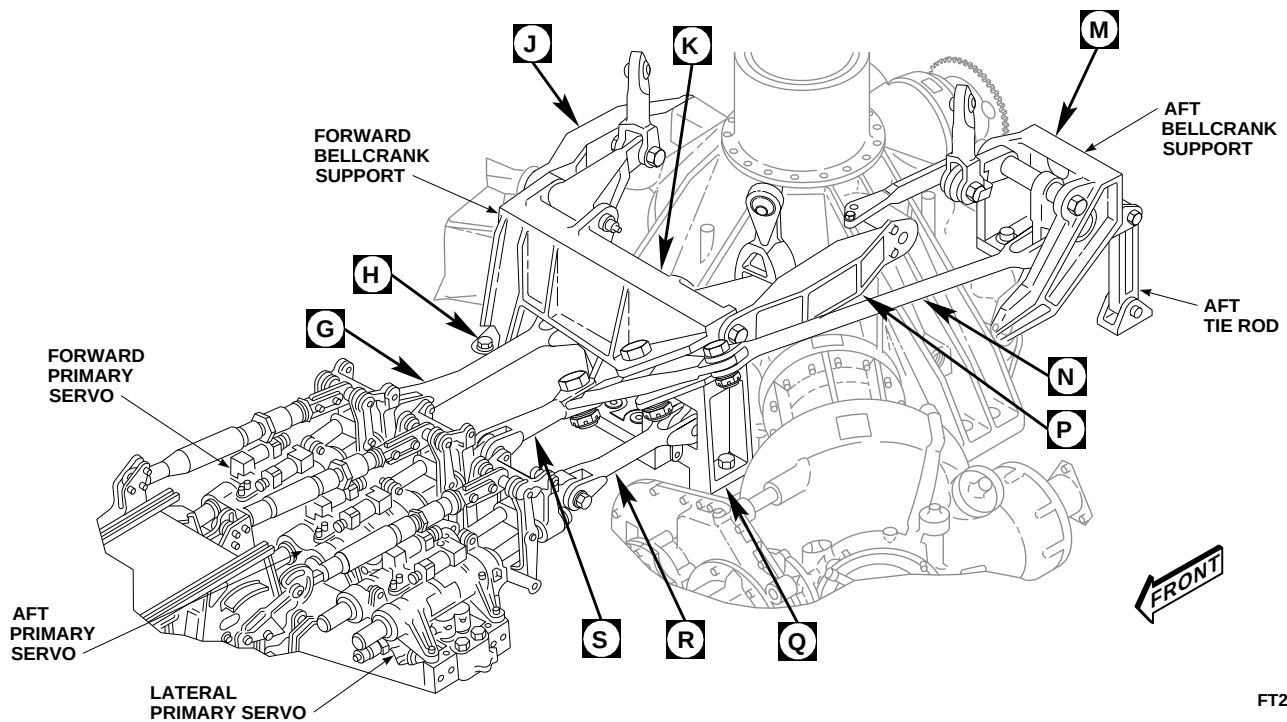
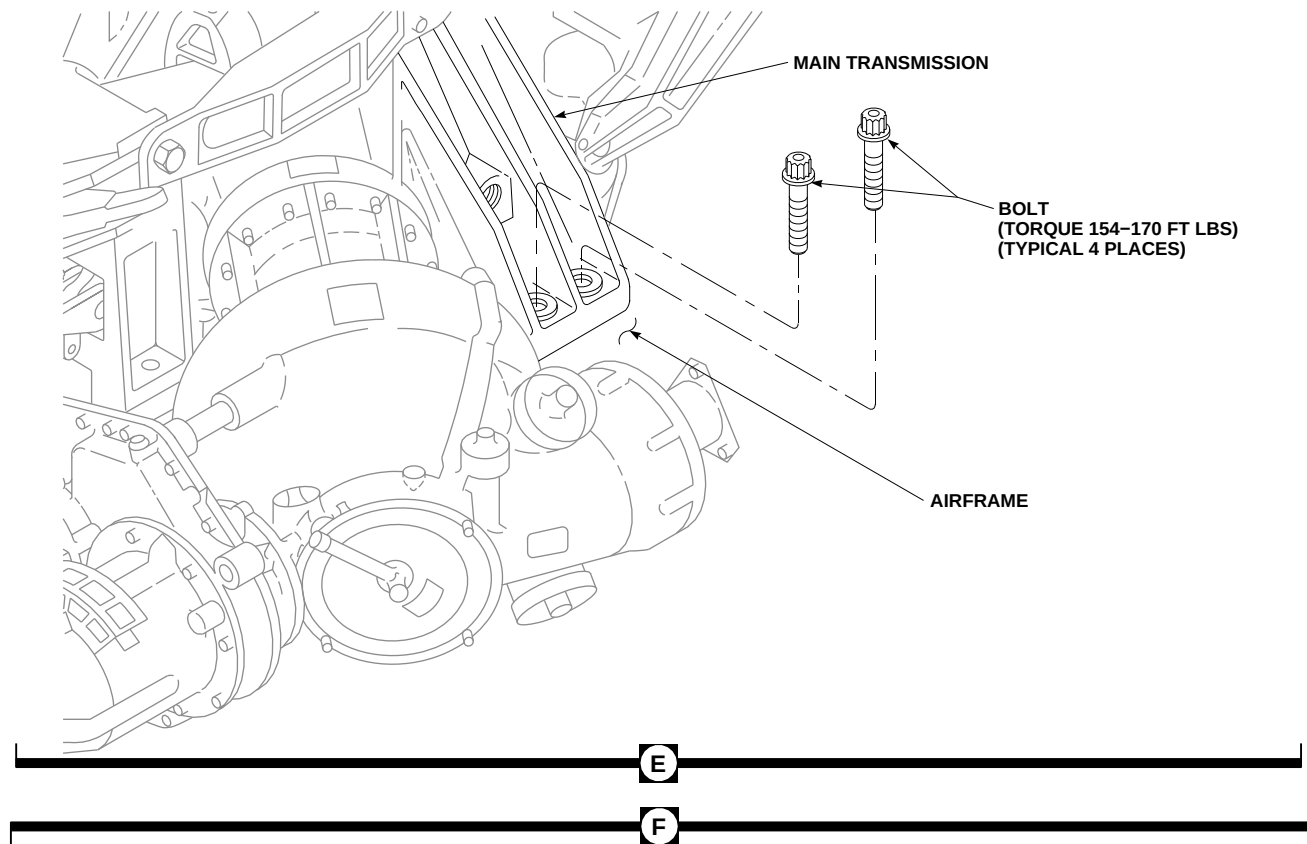
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FIGURE 6-4-1-7. INSTALL MAIN TRANSMISSION AND FLIGHT CONTROLS (SHEET 2 OF 7)

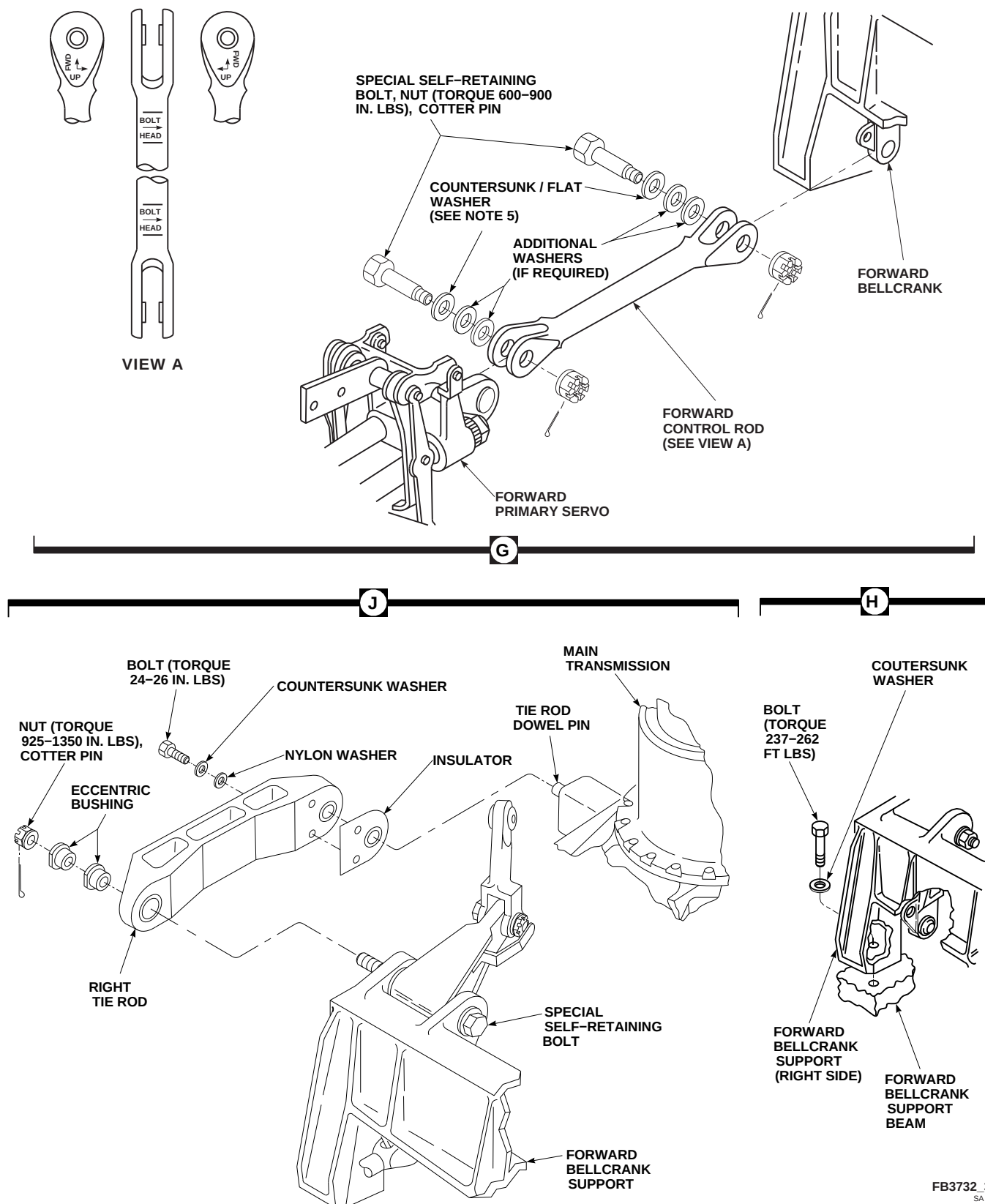
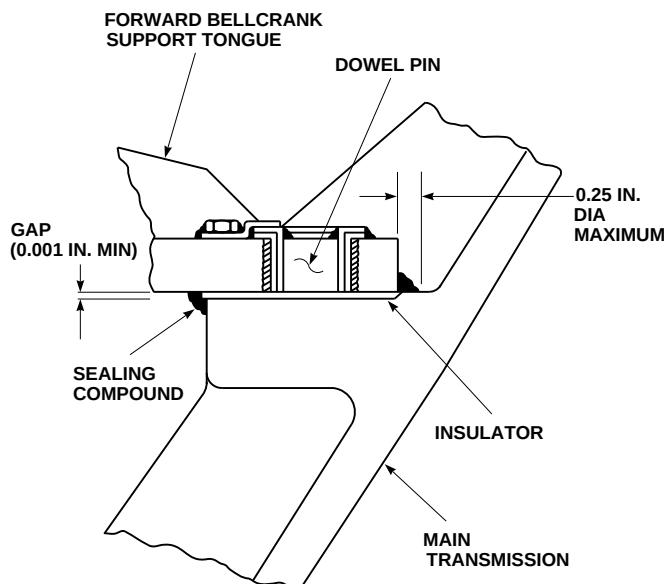
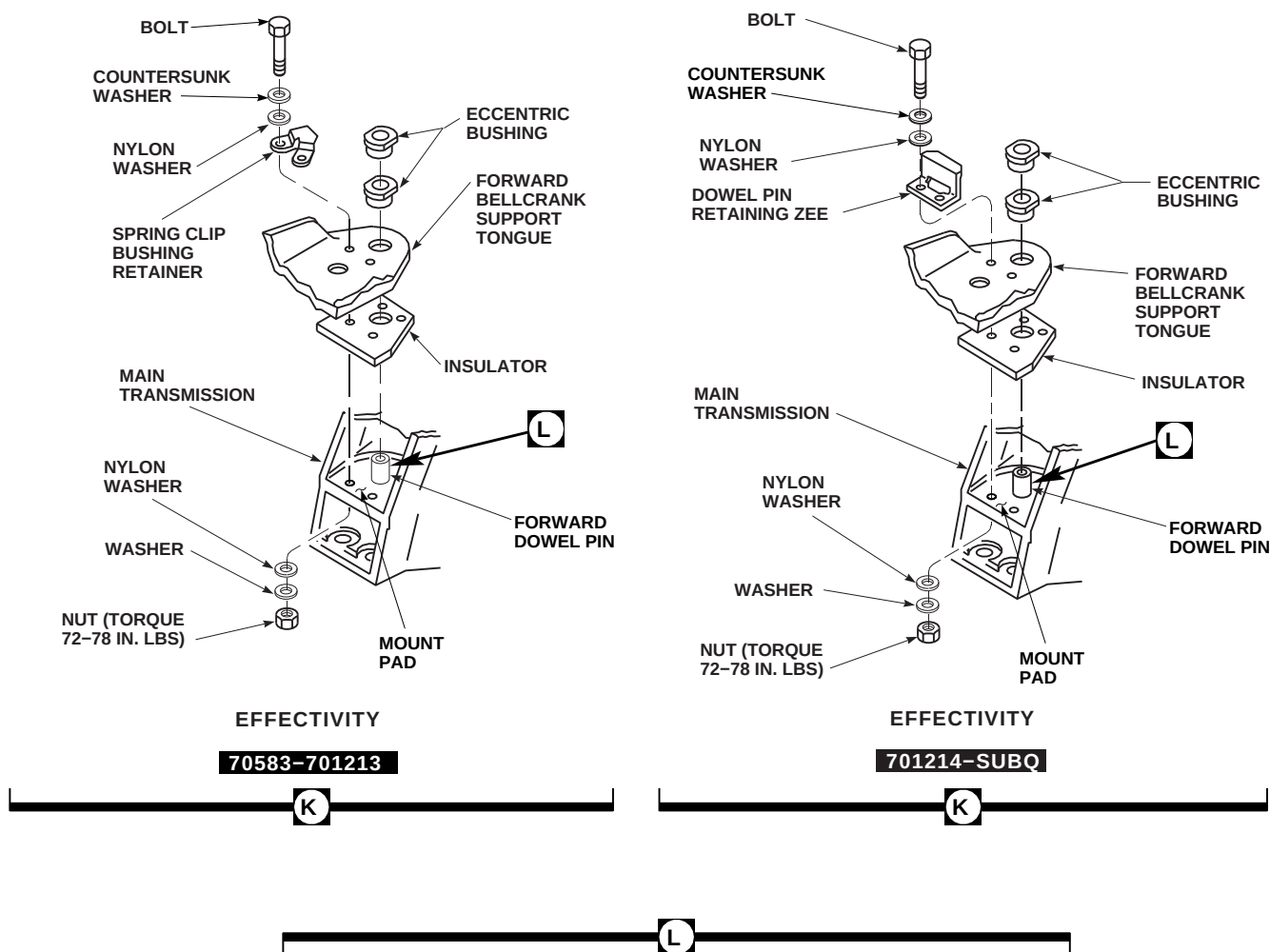


FIGURE 6-4-1-7. INSTALL MAIN TRANSMISSION AND FLIGHT CONTROLS (SHEET 3 OF 7)



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FIGURE 6-4-1-7. INSTALL MAIN TRANSMISSION AND FLIGHT CONTROLS (SHEET 4 OF 7)

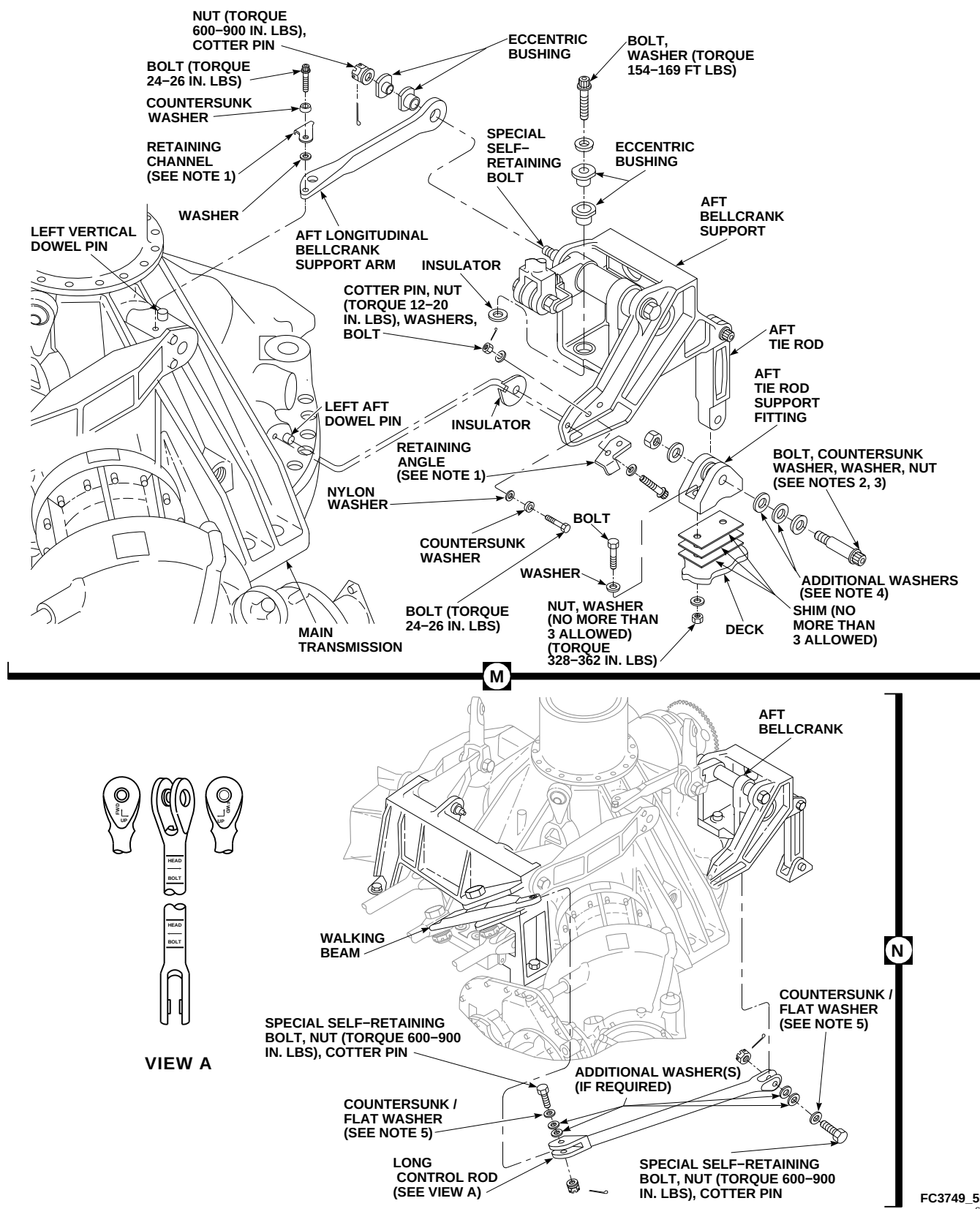
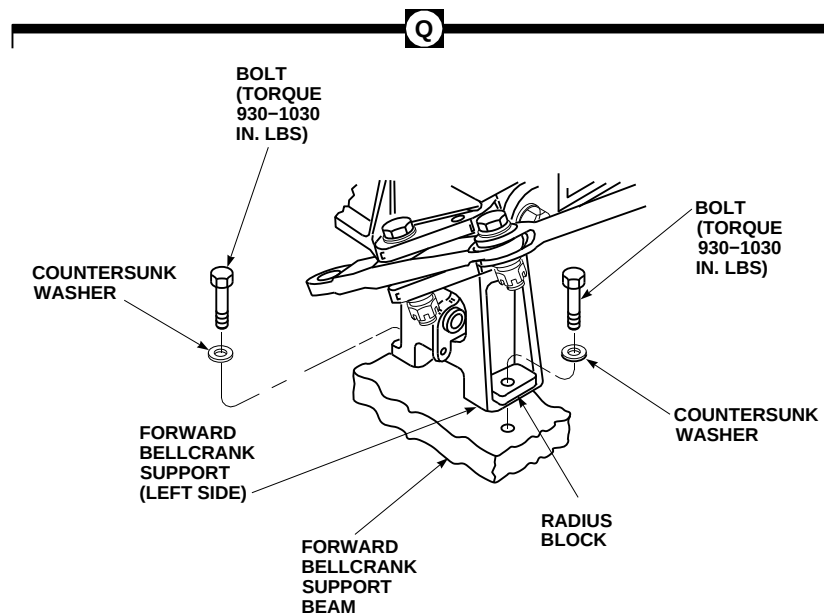
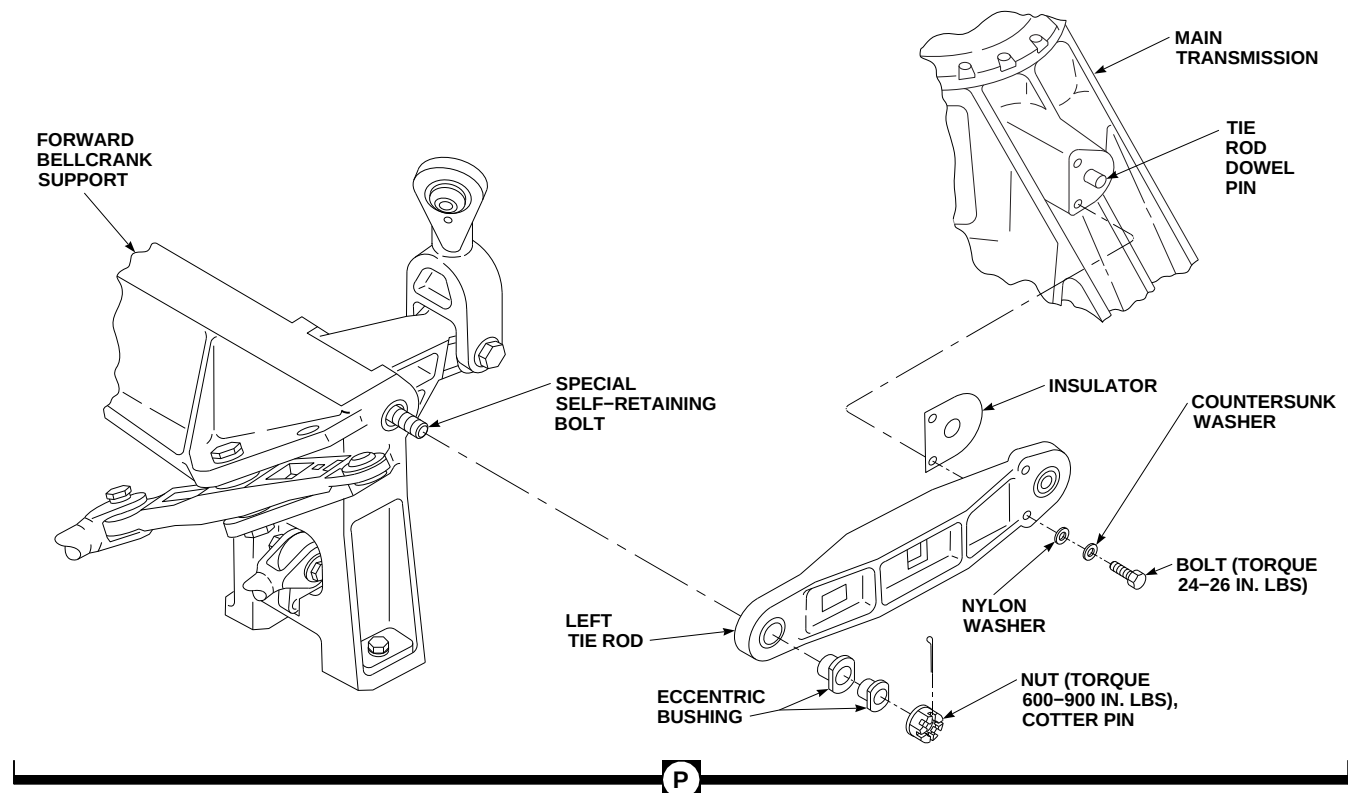
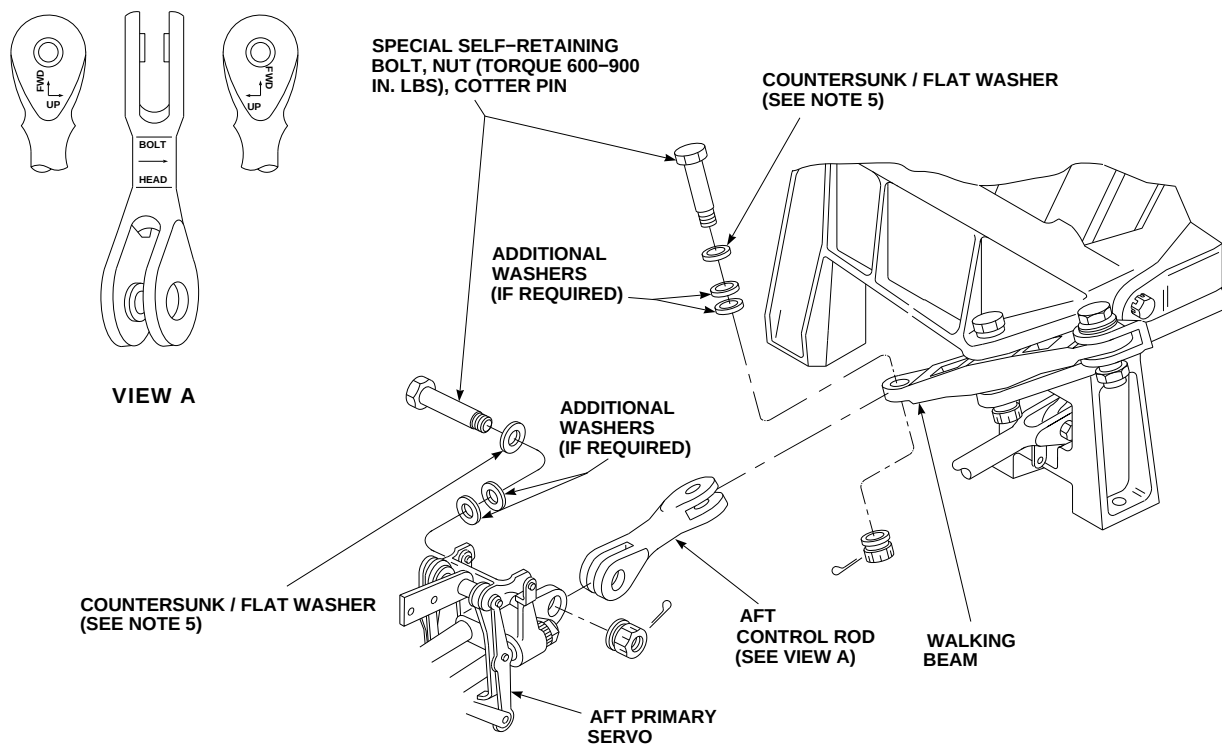
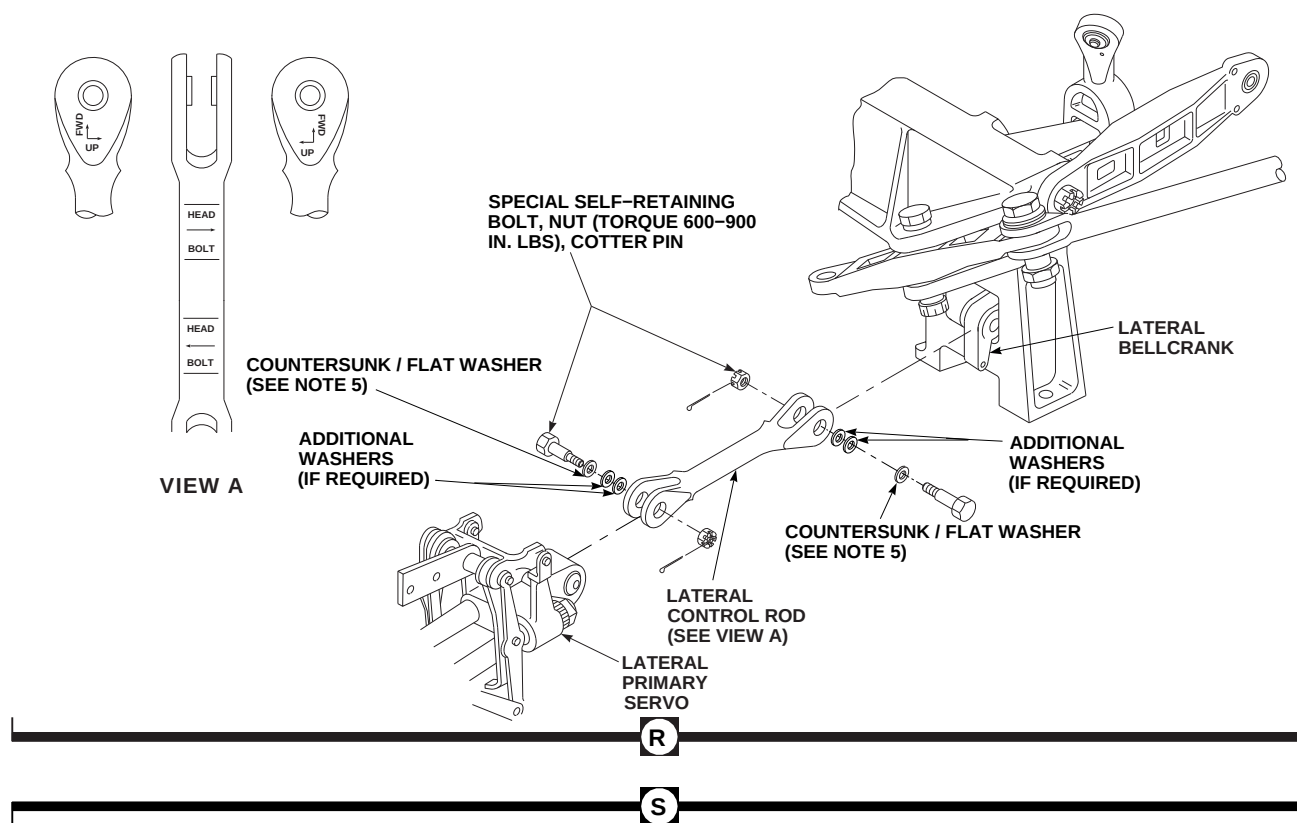


FIGURE 6-4-1-7. INSTALL MAIN TRANSMISSION AND FLIGHT CONTROLS (SHEET 5 OF 7)



FT2604\_6A  
SA

FIGURE 6-4-1-7. INSTALL MAIN TRANSMISSION AND FLIGHT CONTROLS (SHEET 6 OF 7)



FB3732\_7  
SA

FIGURE 6-4-1-7. INSTALL MAIN TRANSMISSION AND FLIGHT CONTROLS (SHEET 7 OF 7)



- k. Remove hoisting sling and guide straps.
- l. Using sealing compound, Item 287 or 292, Appendix D, apply no larger than 1/4-inch diameter bead, around mounting feet/mounting pad/airframe interface.
- m. Using 3-inch masking tape, Item 215, Appendix D, mask off adjacent areas of mounting feet to prevent overspray.
- n. Using corrosion preventive compound, Item 113, Appendix D, apply to entire area of mounting feet.
- o. Allow corrosion preventive compound to cure for 10 to 30 minutes.
- p. Using corrosion preventive compound, Item 113, Appendix D, apply second coat.
- q. Check helicopter logbook to make sure main transmission mounting bolt torque stabilization check is shown as due 9 to 11 flight hours after installation (PARA 1-7-3).
- r. Inspect torque every three days. Using corrosion preventive compound, Item 113, Appendix D, reapply two coats as necessary until bolt torque stabilizes.

### WARNING

- **702218 – SUBQ** Injury to personnel and damage to equipment will result if improved components are mixed with older parts. The improved components are as follows:
  - Lateral Bellcrank, 70400-08166-041
  - Right Tie Rod, 70400-08165-043
  - Aft Bellcrank Support, 70400-08163-043
  - Aft Tie Rod, 70400-08111-043
  - Aft Tie Rod Support Fitting, 70400-08112-046

- Do not mix these parts with older parts.

### FSCAP

- All self-retaining bolts between servo and swashplate bellcranks, tie rods, connecting links, control rods, bellcrank supports, and aft tie rod support fittings contain critical surfaces which must not be damaged during installation. Use extreme caution when installing critical parts.

### NOTE

Remove protective covering from components as they are installed.

- s. Install forward bellcrank support as follows:

### NOTE

Tie rods contain eccentric bushings which must be lined up after forward bellcrank support is installed.

- (1) If tie rods are bolted to forward bellcrank support, remove cotter pin and nut and remove tie rod from forward bellcrank support (Figure 6-4-1-7, Sheet 3, Detail J and Sheet 6, Detail P).
- (2) Check breakaway torque of bolts and barrel-nuts in forward bellcrank support beam that secure forward bellcrank support as follows (Figure 6-4-1-7, Sheet 1, Detail B and Detail D):

### WARNING

### FSCAP

Make sure correct bolts are used when installing forward bellcrank support. Bolts used on installation, 70400-08200-011, are shorter than those used on installation, 70400-08200-012/-013, and cannot be interchanged.





Damage to recess in airframe or forward bellcrank support beam will result if barrel-nut is incorrectly installed. To prevent damage to recess, make sure barrel-nuts are installed with barrel (round side) toward bolthead.

### NOTE

Make sure that only barrel-nuts, RMLH2577, are used. These barrel-nuts are identified with letters "EN" or "SPS" stamped on an end (additional marking may also include "LH2577").

- (a) Install bolt into barrel-nut until two threads extend beyond barrel-nut.
  - (b) Using torque wrench, back out bolt, noting breakaway torque.
  - (c) For right barrel-nut, **MINIMUM BREAKAWAY TORQUE IS 50 INCH-POUNDS**. If breakaway torque is less than 50 inch-pounds, replace barrel-nut (PARA 6-4-1.1.3, step g.).
  - (d) For left barrel-nuts, **MINIMUM BREAKAWAY TORQUE IS 18 INCH-POUNDS**. If breakaway torque is less than 18 inch-pounds, replace barrel-nut(s) (PARA 6-4-1.1.3, step g.).
  - (e) If barrel-nut(s) was replaced, recheck breakaway torque. **IF BREAKAWAY TORQUE IS STILL TOO LOW, REPLACE BOLT.**
- (3) Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean mount pad on forward bellcrank support tongue on main transmission. Wipe with clean machinery towel.

- (4) Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to forward dowel pin on main transmission.

### NOTE

Forward bellcrank support must be installed within two hours after application of sealing compound.

- (5) Using sealing compound, Item 287, Appendix D, apply thin coat to entire surface of mount pad area on main transmission.
- (6) Using sealing compound, Item 287, Appendix D, apply thin coat to bottom of insulator.
- (7) Install insulator over forward dowel pin and press onto mount pad area on main transmission. Align holes in insulator with holes in mount pad (Figure 6-4-1-7, Sheet 4, Detail K).
- (8) Check for minimum gap between forward bellcrank support tongue and main transmission as follows:
  - (a) Temporarily install forward bellcrank support on forward bellcrank support beam with bolts and countersunk washers. Make sure countersunk side of washer is toward bolthead. Tighten bolts until no relative motion is felt in forward bellcrank support.
  - (b) Using feeler gage, measure gap between forward bellcrank support tongue and main transmission. **MINIMUM GAP IS 0.001-INCH** (Figure 6-4-1-7, Sheet 4, Detail L).
  - (c) Remove bolts, countersunk washers, and forward bellcrank support from forward bellcrank support beam (Figure 6-4-1-7, Sheet 4, Detail K).
- (9) Using sealing compound, Item 287, Appendix D, apply thin coat to entire lower surface of forward bellcrank support

tongue (Figure 6-4-1-7, Sheet 4, Detail L).

- (10) Using sealing compound, Item 287, Appendix D, apply thin coat to entire upper surface of insulator.
- (11) Using sealing compound, Item 292, Appendix D, apply thin coat to mounting surface of forward bellcrank support beam and bottom of forward bellcrank support.
- (12) Using sealing compound, Item 292, Appendix D, apply fillet around forward bellcrank support dowel pin.
- (13) Position forward bellcrank support on forward bellcrank support beam and position forward bellcrank support tongue onto forward dowel pin on main transmission (Figure 6-4-1-7, Sheet 3, Detail H, Sheet 4, Detail K, and Sheet 6, Detail Q).

### WARNING

#### FSCAP

Make sure correct bolts are used when installing forward bellcrank support. Bolts used on installation, 70400-08200-011, are shorter than those used on installation, 70400-08200-012/-013, and cannot be interchanged.

- (14) Using primer, Item 258, Appendix D, apply coat to shank, threads, and area under boltheads.
- (15) Install bolts and countersunk washers in forward bellcrank support while primer is still wet. Make sure countersunk side of washer is toward bolthead (Figure 6-4-1-7, Sheet 3, Detail H and Sheet 6, Detail Q).

### WARNING

#### FSCAP

Verification of minimum torque of bolt is a critical characteristic.

- (16) On left side of forward bellcrank support, perform the following (Figure 6-4-1-7, Sheet 6, Detail Q):

- (a) **TORQUE BOLTS TO 930- 1,030 INCH-POUNDS.**

### WARNING

#### FSCAP

Verification of safety of bolts is a critical characteristic.

### CAUTION

To prevent damage to helicopter, do not use safety cable in this application. Use only lockwire that is specified in this step.

- (b) Using lockwire, Item 194, Appendix D, lockwire bolts to radius blocks.
- (c) Apply torque stripes (PARA 2-6-5).

### WARNING

#### FSCAP

Verification of minimum torque of bolt is a critical characteristic.

- (17) On right side of forward bellcrank support, **TORQUE BOLT TO 237 - 262**

**FOOT-POUNDS** (Figure 6-4-1-7, Sheet 3, Detail H).

- (18) Form fillet with sealing compound squeezeout around joint between forward bellcrank support and forward bellcrank support beam. Make sure sealing compound covers entire joint line, but is no larger than ¼-inch diameter.
- (19) Using sealing compound, Item 292, Appendix D, apply bead around bolts and countersunk washers securing forward bellcrank support.
- (20) Install and adjust eccentric bushing in forward bellcrank support tongue to fit over forward dowel pin on main transmission (Figure 6-4-1-7, Sheet 4, Detail K).
- (21) **70583-701213** Position spring clip bushing retainer on forward bellcrank support tongue as follows:■

## WARNING

### FSCAP

Verification of spring clip bushing retainer installation is a critical characteristic.

- (a) Using sealing compound, Item 292, Appendix D, apply bead around each bolthole of spring clip bushing retainer on top of forward bellcrank support tongue.
- (b) Position spring clip bushing retainer on forward bellcrank support tongue. Install bolts, countersunk washers, and nylon washers. Make sure countersunk side of washer is toward bolthead.
- (22) **701214-SUBQ** Position dowel pin retaining zee on forward bellcrank support tongue as follows:■

## WARNING

### FSCAP

Verification of dowel pin retaining zee installation is a critical characteristic.

### NOTE

Dowel pin retaining zee, 70400-08171-101, is used on main transmissions with tall forward dowel pin. Dowel pin retaining zee, 70400-08171-102, is used on main transmissions with short forward dowel pin.

- (a) Using sealing compound, Item 292, Appendix D, apply bead around each bolthole of dowel pin retaining zee on top of forward bellcrank support tongue.
- (b) Position dowel pin retaining zee on forward bellcrank support tongue. Install bolts, countersunk washers, and nylon washers. Make sure countersunk side of washer is toward bolthead.

### NOTE

Lower washers and nylon washers must be installed wet.

- (23) Using epoxy primer coating kit, Item 135, Appendix D, coat washers with epoxy primer.
- (24) Using sealing compound, Item 287, Appendix D, apply bead to one side of nylon washers.
- (25) Install nylon washers (with sealing compound side towards main transmission), washers, and nuts.
- (26) **TORQUE NUTS TO 72 - 78 INCH-POUNDS.**

**NOTE**

Do not use excess sealing compound to form fillet around forward bellcrank support tongue. Excessive sealing compound will hinder corrosion inspection.

- (27) Form fillet with sealing compound squeezeout around joint between main transmission and forward bellcrank support tongue. Make sure sealing compound covers entire joint line, but is no larger than 1/4-inch diameter (Figure 6-4-1-7, Sheet 4, Detail L).
  - (28) Using polyurethane paint, Item 224, Appendix D, apply to mounting area of upper and lower surfaces of forward bellcrank support tongue (PARA 2-6-5).
  - (29) Mask off adjacent area to prevent overspray, then using corrosion preventive compound, Item 113, Appendix D, apply two coats to mounting area of forward bellcrank support tongue and to water entrapment area behind mounting area. Allow to cure between coats.
- t. Install and seal cap plugs as follows (Figure 6-4-1-7, Sheet 1, Detail B and Detail D):

**NOTE**

Cap plug sizes vary depending on location. Cap plug part numbers and locations are as follows:

- Cap plug, NAS813-14 (two left side forward bellcrank support barrel-nut holes)
- Cap plug, NAS813-16 (eight main transmission barrel-nut holes)
- Cap plug, NAS813-20 (right side forward bellcrank support barrel-nut hole)

- (1) Using sealing compound, Item 292, Appendix D, apply 1/4-inch diameter bead under lip of cap plug.

- (2) Install cap plug in airframe or forward bellcrank support beam.

- (3) Using sealing compound, Item 292, Appendix D, apply 1/4-inch diameter bead around interface of cap plug and airframe or forward bellcrank support beam.

- (4) Using sealing compound, Item 292, Appendix D, apply to entire cap plug. Make sure cap plug is entirely sealed.

- u. Install left and right tie rods as follows (Figure 6-4-1-7, Sheet 3, Detail J and Sheet 6, Detail P):

- (1) Using primer, Item 258, Appendix D, apply coat to dowel pin and entire tie rod mount pad surface on main transmission, including fillets around threaded inserts and base of dowel pin. Make sure there are no gaps in coating.

- (2) Position insulator on main transmission mounting pads. Using primer, Item 258, Appendix D, apply coat to top surface of insulator and underside of tie rod. Make sure there are no gaps in coating.

- (3) Position tie rod on dowel pin. Turn eccentric bushings to align bolthole and position tie rod on special self-retaining bolt on forward bellcrank support.

**CAUTION**

If resistance or binding of retaining ball mechanism is observed when pin is actuated, do not apply corrosion preventive compound to free mechanism. Inner core of special self-retaining bolt is corroded and must be rejected.

- (4) Using cleaning cloth, Item 102, Appendix D, moistened with dry-cleaning solvent, Item 124, Appendix D, clean special self-retaining bolt.

- (5) Inspect special self-retaining bolt as follows:

- (a) Inspect special self-retaining bolt for evidence of corrosion in area of retaining balls, threads, cotter pin hole, retaining pin hole in threaded end of special self-retaining bolt. **NONE ALLOWED.** If corrosion is visible, replace bolt.
- (b) Visually inspect special self-retaining bolt for cracks in area of retaining balls and threads. **NONE ALLOWED.** If crack is visible, replace bolt.
- (c) If corrosion or cracks are found, replace special self-retaining bolt.
- (6) Release retaining balls by holding in retaining pin. Using corrosion preventive compound, Item 114, Appendix D, apply to special self-retaining bolt as follows:
  - (a) Work corrosion preventive compound through one retaining ball hole until excess is seen at opposite ball hole. Rotate bolt 180° and repeat procedure.
  - (b) While working retaining pin, using corrosion preventive compound, Item 114, Appendix D, spray into hole in bolthead and hole in end of special self-retaining bolt.
  - (c) Using cleaning cloth, Item 102, Appendix D, wipe off excess corrosion preventive compound from special self-retaining bolt prior to installation.
  - (d) Using corrosion preventive compound, Item 109, Appendix D, apply to exterior retaining ball area and threads.

**WARNING**

**FSCAP**

Verification of proper hardware installation, torque, safety installation, and

proof torque of special self-retaining bolts are critical characteristics.

- (7) Check special self-retaining bolt for free axial motion and that impedance feature is not restrained in part. If special self-retaining bolt binds and impedance feature is depressed, change washer stackup.
- (8) Install nut on tie rod and secure as follows:
  - (a) For left tie rod, **TORQUE NUT TO 600 - 900 INCH-POUNDS.**
  - (b) For right tie rod, **TORQUE NUT TO 925 - 1,350 INCH-POUNDS.**

**NOTE**

If unable to torque bolthead, use crowfoot attachment to torque nut, without disturbing cotter pin and without restraining bolthead.

- (c) For left tie rod, **APPLY TIGHTENING TORQUE (PROOF LOAD) OF 400 - 420 INCH-POUNDS IN TIGHTENING DIRECTION.** If bolt rotates, remove nut, adjust by adding washers (PARA 11-4-1), and repeat step u. (8).
- (d) For right tie rod, **APPLY TIGHTENING TORQUE (PROOF LOAD) OF 630 - 650 INCH-POUNDS IN TIGHTENING DIRECTION.** If bolt rotates, remove nut, adjust by adding washers (PARA 11-4-1), and repeat step u. (8).
- (e) Install cotter pin.
- (f) Apply torque stripe (PARA 2-6-5).
- (9) Using epoxy primer coating kit, Item 135, Appendix D, apply light coat of epoxy primer to bolthead, shank, and area under bolthead. Allow epoxy primer to dry.
- (10) Using corrosion preventive compound, Item 109, Appendix D, lubricate bolthead

and shank. Place countersunk washer and nylon washer on bolt. Make sure countersunk side of washer is toward bolthead.

- (11) Using epoxy primer coating kit, Item 135, Appendix D, apply light coat of epoxy primer to bolt threads. With threads still wet, secure tie rod to main transmission with bolts, countersunk washers, and nylon washers. Make sure countersunk side of washer is toward bolthead.
- (12) **TORQUE BOLTS TO 24 - 26 INCH-POUNDS.**
- (13) Using machinery towel, Item 211, Appendix D, wipe off excess epoxy primer.

#### NOTE

- Bolts securing left tie rod to main transmission must be lockwired to each other and to bolt of aft longitudinal bellcrank support arm before applying sealing compound.
- Lockwiring of bolt (securing aft longitudinal bellcrank support arm) to upper bolt on left tie rod, and application of sealing compound will be performed in step v. (24) and step v. (25).

- (14) For left tie rod, using lockwire, Item 197, Appendix D, lockwire left tie rod bolt-heads together. Do not lockwire to bolt of aft longitudinal bellcrank support arm at this time.
- (15) Using epoxy coating kit, Item 126A, Appendix D, cover exposed end of tie rod dowel pin with epoxy coating.
- (16) Using sealing compound, Item 292, Appendix D, apply thin coat around gap between main transmission mounting surface and tie rod.

- v. Install aft bellcrank support and aft longitudinal bellcrank support arm as follows (Figure 6-4-1-7, Sheet 5, Detail M):

- (1) Using primer, Item 258, Appendix D, apply light coat to area on main transmission that mates with aft bellcrank support.
- (2) Position insulators on aft bellcrank support and carefully position aft bellcrank support and insulators on main transmission while primer is still wet.
- (3) Using primer, Item 258, Appendix D, apply light coat to shank, threads, and area under bolthead.
- (4) Install and adjust eccentric bushings in aft bellcrank support. Install bolt and washer while primer is still wet.

#### WARNING

#### FSCAP

Verification of minimum torque and safety installation of bolts are critical characteristics.

- (5) **TORQUE BOLT TO 154 - 169 FOOT-POUNDS.**
- (6) Using machinery towel, Item 211, Appendix D, wipe off excess primer from bolt.
- (7) Using sealing compound, Item 292, Appendix D, apply bead around bolthead and washer.
- (8) Using epoxy primer coating kit, Item 135, Appendix D, apply light coat of epoxy primer to bolthead, shank, and area under bolthead. Allow epoxy primer to dry.
- (9) Using corrosion preventive compound, Item 109, Appendix D, lubricate bolthead and shank.
- (10) Using epoxy primer coating kit, Item 135, Appendix D, apply light coat of epoxy primer to bolt threads.
- (11) Secure leg of aft bellcrank support on left aft dowel pin to main transmission with



bolt, countersunk washer, and nylon washer. Install bolt while primer is still wet. Make sure countersunk side of washer faces bolthead.

- (12) **TORQUE BOLT TO 24 - 26 INCH-POUNDS.**
- (13) Using machinery towel, Item 211, Appendix D, wipe off excess epoxy primer from bolt.



To prevent damage to helicopter, do not use safety cable in this application. Use only lockwire that is specified in this step.

- (14) Using lockwire, Item 197, Appendix D, lockwire bolt to bolt installed in step v. (4).
- (15) Using sealing compound, Item 292, Appendix D, apply bead around bolt, countersunk washer, and nylon washer.
- (16) **701214-SUBQ** Install retaining angle as follows: 



**FSCAP**

Verification of retaining angle installation is a critical characteristic.

**NOTE**

Bolts securing retaining angle to leg of aft bellcrank support are installed with boltheads facing down.

- (a) Install retaining angle to leg of aft bellcrank support and secure with bolts, washers, and nuts.
- (b) Using feeler gage, measure clearance between left aft dowel pin and

retaining angle. **MINIMUM CLEARANCE IS 0.010-INCH.** If clearance is less than 0.010-inch, reseat left aft dowel pin (PARA 6-4-1.2).

- (c) **TORQUE NUTS TO 12 - 20 INCH-POUNDS.**
- (d) Install cotter pin.

**NOTE**

Do not allow primer to dry.

- (17) Using primer, Item 258, Appendix D, apply coat to left vertical dowel pin on main transmission.
- (18) Install aft longitudinal bellcrank support arm on left vertical dowel pin. Make sure aft longitudinal bellcrank support arm is fully seated and bolthole lines up with threaded hole in main transmission.
- (19) Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to bolt shank and bolthead. Allow epoxy primer to dry.
- (20) Using corrosion preventive compound, Item 111, Appendix D, apply compound to bolt shank and bolthead.

**NOTE**

Do not allow epoxy primer to dry.

- (21) Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to bolt threads.

**NOTE**

**701214-SUBQ** Retaining channel is mounted between countersunk washer and washer. 

- (22) Secure aft longitudinal bellcrank support arm to main transmission with bolt, countersunk washer, **701214-SUBQ** retain-

ing channel,  and washer. Make sure countersunk side of washer is toward bolt-head.

(23) **TORQUE BOLT TO 24 - 26 INCH-POUNDS.**

(24) Using lockwire, Item 194, Appendix D, lockwire bolt to left tie rod upper attachment bolt.

(25) Using sealing compound, Item 292, Appendix D, apply bead to boltheads and washers installed in rear of left tie rod and aft longitudinal bellcrank support arm.

**CAUTION**

If resistance or binding of retaining ball mechanism is observed when pin is actuated, do not apply corrosion preventive compound to free mechanism. Inner core of special self-retaining bolt is corroded and must be rejected.

(26) Using cleaning cloth, Item 102, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean special self-retaining bolt.

(27) Inspect special self-retaining bolt as follows:

(a) Inspect special self-retaining bolt for evidence of corrosion in area of retaining balls, threads, cotter pin hole, retaining pin hole in bolthead, and hole in threaded end of special self-retaining bolt. **NONE ALLOWED.** If corrosion is visible, replace bolt.

(b) Visually inspect special self-retaining bolt for cracks in area of retaining balls and threads. **NONE ALLOWED.** If crack is visible, replace bolt.

(c) If corrosion or cracks are found, replace special self-retaining bolt.

(28) Release retaining balls by holding in retaining pin. Using corrosion preventive compound, Item 114, Appendix D, apply to special self-retaining bolt as follows:

(a) Work corrosion preventive compound through one retaining ball hole until excess is seen at opposite ball hole. Rotate bolt 180° and repeat procedure.

(b) While working retaining pin, using corrosion preventive compound, Item 114, Appendix D, spray into hole in bolthead and hole in end of special self-retaining bolt.

(c) Using cleaning cloth, Item 102, Appendix D, wipe off excess corrosion preventive compound from special self-retaining bolt prior to installation.

(d) Using corrosion preventive compound, Item 109, Appendix D, apply to exterior retaining ball area and threads.

(29) Install eccentric bushings in aft longitudinal bellcrank support arm and aft bellcrank support. Turn eccentric bushings to align boltholes.

(30) Install nut on special self-retaining bolt.

(31) **TORQUE NUT TO 600 INCH-POUNDS. IF SLOT IN NUT DOES NOT LINE UP WITH COTTER PIN HOLE, SET TORQUE WRENCH TO 900 INCH-POUNDS AND TIGHTEN NUT UNTIL SLOT IN NUT AND COTTER PIN HOLE LINE UP. DO NOT EXCEED MAXIMUM TORQUE.**

(32) Install cotter pin.



# NOTE

If unable to torque bolthead, use crowfoot attachment to torque nut, without disturbing cotter pin and without restraining bolthead.

(33) **APPLY TIGHTENING TORQUE (PROOF LOAD) OF 400 - 420 INCH-POUNDS IN TIGHTENING DIRECTION.** If bolt rotates, perform the following:

- (a) Remove cotter pin, nut, special self-retaining bolt, and washer(s).
- (b) Install special self-retaining bolt and countersunk washer (under bolthead) through eccentric bushings and aft bellcrank. Make sure countersunk side of washer is toward bolthead. Add washers, as required (maximum of two additional washers under countersunk washer), to provide correct grip length and cotter pin position.
- (c) Check special self-retaining bolt for free axial motion and that impedance feature is not restrained in part. If special self-retaining bolt binds and impedance feature is depressed, change washer stackup.
- (d) Repeat step v. (30) through step v. (33).

(34) Apply torque stripe (PARA 2-6-5).

(35) Using sealing compound, Item 292, Appendix D, apply to exposed end of left vertical dowel pin, around gap between main transmission mounting surface and aft longitudinal bellcrank support arm, and around **701214-SUBQ** retaining channel, special self-retaining bolt, countersunk washer, and washer.

w. Install aft tie rod support fitting and connect aft tie rod as follows:

## WARNING

### FSCAP

Upper and lower ends of aft tie rod at aft bellcrank support are critical characteristic surfaces which must not be damaged during installation of main transmission. Use extreme caution when installing.

- (1) Swing aft tie rod support fitting down over deck. Using feeler gage, measure gap between aft tie rod support fitting and deck.
- (2) Remove bolt, countersunk washer, washer, additional washer(s), if installed, and nut securing aft tie rod to aft tie rod support fitting. Remove aft tie rod support fitting.
- (3) Peel shims to obtain thickness of 0.000 - 0.010-inch thicker than measured gap. No more than three shims may be used.
- (4) Using corrosion resistant coating, Item 118, Appendix D, treat shims.

## WARNING

### FSCAP

Make sure correct bolts are used when installing aft tie rod support fitting. Bolts used on installation, 70400-08200-011, are shorter than those used on installation, 70400-08200-012/-013, and cannot be interchanged.

- (5) Using primer, Item 258, Appendix D, apply light coat to shank, threads, and area under bolthead of aft tie rod support fitting bolts.

**NOTE**

A maximum of three washers, in any combination, may be installed between deck and nut to make sure nut is bottomed against deck and not bolt.

- (6) Install shims and aft tie rod support fitting to deck with bolts, countersunk washers, washer(s) (located between deck and nuts), and nuts while primer is still wet.
- (7) **TORQUE NUTS TO 328 - 362 INCH-POUNDS.**
- (8) Using machinery towel, Item 211, Appendix D, wipe off excess primer.

**NOTE**

Add washers, as necessary, to make sure nut is bottomed against aft tie rod and not on bolt.

- (9) Install bolt, countersunk washer, washer, and nut securing aft tie rod to aft tie rod support fitting. Make sure countersunk side of washer is toward bolthead. Add washers, as required, under countersunk washer to provide correct grip length and cotter pin position.
  - (10) On helicopters with aft tie rod, 70400-08111-043, **TORQUE NUTS TO 238 - 262 INCH-POUNDS.**
  - (11) On helicopters with aft tie rod, 70400-08111-041 or 70400-08111-042, **TORQUE NUTS TO 532 - 588 INCH-POUNDS.**
  - (12) Using sealing compound, Item 292, Appendix D, apply bead around bolts, washers, and nuts securing aft tie rod support fitting.
- x. After all corrosion preventive coatings are dry, perform the following:
- (1) Make sure all areas to be treated are clean and dry.

- (2) Using corrosion preventive compound, Item 113, Appendix D, apply two coats to flight control mount pads on main transmission.
- y. On main transmissions, 70351-08100-053 and 70351-08100-054, perform the following:
- (1) Paint exposed end of pin white (PARA 2-6-5).
  - (2) Paint 1/8-inch wide black strip across end of pin, extending onto adjoining flight control supports (PARA 2-6-5).
  - (3) Using epoxy coating kit, Item 126A, Appendix D, seal exposed end of dowel pins on main transmission with epoxy primer.
- z. Install long control rod as follows (Figure 6-4-1-7, Sheet 5, Detail N):

**CAUTION**

If resistance or binding of retaining ball mechanism is observed when pin is actuated, do not apply corrosion preventive compound to free mechanism. Inner core of special self-retaining bolt is corroded and must be rejected.

- (1) Using cleaning cloth, Item 102, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean special self-retaining bolt.
- (2) Inspect special self-retaining bolt as follows:
  - (a) Inspect special self-retaining bolt for evidence of corrosion in area of retaining balls, threads, cotter pin hole, retaining pin hole in bolthead, and hole in threaded end of special self-retaining bolt.
  - (b) Visually inspect special self-retaining bolt for cracks in area of retaining balls and threads.

- (c) If corrosion or cracks are found, replace special self-retaining bolt.
- (3) Release retaining balls by holding in retaining pin. Using corrosion preventive compound, Item 114, Appendix D, apply to special self-retaining bolt as follows:
  - (a) Work corrosion preventive compound through one retaining ball hole until excess is seen at opposite ball hole. Rotate bolt 180° and repeat procedure.
  - (b) While working retaining pin, using corrosion preventive compound, Item 114, Appendix D, spray into hole in bolthead and hole in end of special self-retaining bolt.
  - (c) Using cleaning cloth, Item 102, Appendix D, wipe off excess corrosion preventive compound from special self-retaining bolt prior to installation.
  - (d) Using corrosion preventive compound, Item 109, Appendix D, apply to exterior retaining ball area and threads.

**WARNING**

**FSCAP**

Verification of proper hardware installation, torque, safety installation, and proof torque of special self-retaining bolts are critical characteristics.

**NOTE**

Position long control rod according to identification plate on rod end (Figure 6-4-1-7, Sheet 5, Detail N, View A).

- (4) Install long control rod between walking beam and aft bellcrank.

**NOTE**

- Add washers, as necessary, to make sure nut is bottomed against long control rod and not on special self-retaining bolts.
- To install special self-retaining bolt, insert 1/8-inch Allen wrench in hole in bolthead. Push on pin to release retaining balls at nut end of bolt. Keep pressure on pin while installing self-retaining bolt.
- (5) Install special self-retaining bolts and countersunk or flat washers securing long control rod to walking beam and aft bellcrank. Make sure countersunk side of washer is toward bolthead. To provide correct grip length and cotter pin position, adjust as follows (Figure 6-4-1-7, Sheet 5, Detail N):
  - (a) On front end of long control rod, add maximum of two washers under countersunk washer, as required.
  - (b) On rear end of long control rod, add maximum of two washers under nut, as required.
- (6) Check special self-retaining bolt for free axial motion and that impedance feature is not restrained in part. If special self-retaining bolt binds and impedance feature is depressed, change washer stackup.
- (7) Install nuts.
- (8) **TORQUE NUTS TO 600 INCH-POUNDS. IF SLOT IN NUT DOES NOT LINE UP WITH COTTER PIN HOLE, SET TORQUE WRENCH TO 900 INCH-POUNDS AND TIGHTEN NUT UNTIL SLOT IN NUT AND COTTER PIN HOLE LINE UP. DO NOT EXCEED MAXIMUM TORQUE.**
- (9) Install cotter pin.

**NOTE**

If unable to torque bolthead, use crowfoot attachment to torque nut, without disturbing cotter pin and without restraining bolthead.

- (10) **APPLY TIGHTENING TORQUE (PROOF LOAD) OF 400 - 420 INCH-POUNDS IN TIGHTENING DIRECTION.** If bolt rotates, perform the following:

- (a) Remove cotter pin, nut, special self-retaining bolt, and washer(s).
- (b) Repeat step z. (5), adjusting washer thicknesses or adding washers.
- (c) Repeat step z. (6) through step z. (10) until bolt does not rotate.

- (11) Apply torque stripe (PARA 2-6-5).

- aa. Install each forward, lateral, and aft control rod as follows (Figure 6-4-1-7, Sheet 3, Detail G, and Sheet 7, Detail R and Detail S):

**WARNING**

- Injury to personnel and damage to equipment will result if incorrect control rod is installed. Forward and lateral primary servo output control rods are similar in length.
- Forward control rod is 12.670-inches from bolthole-center to bolthole-center.
- Lateral control rod is 11.700-inches from bolthole-center to bolthole-center.

**NOTE**

Installation of forward, aft, and lateral control rods is the same, except as noted.

**CAUTION**

If resistance or binding of retaining ball mechanism is observed when pin is

actuated, do not apply corrosion preventive compound to free mechanism. Inner core of special self-retaining bolt is corroded and must be rejected.

- (1) Using cleaning cloth, Item 102, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean special self-retaining bolt.

- (2) Inspect special self-retaining bolt as follows:

- (a) Inspect special self-retaining bolt for evidence of corrosion in area of retaining balls, threads, cotter pin hole, retaining pin hole in bolthead, and hole in threaded end of special self-retaining bolt. **NONE ALLOWED.** If corrosion is visible, replace bolt.

- (b) Visually inspect special self-retaining bolt for cracks in area of retaining balls and threads. **NONE ALLOWED.** If crack is visible, replace bolt.

- (c) If corrosion or cracks are found, replace special self-retaining bolt.

- (3) Release retaining balls by holding in retaining pin. Using corrosion preventive compound, Item 114, Appendix D, apply to special self-retaining bolt as follows:

- (a) Work corrosion preventive compound through one retaining ball hole until excess is seen at opposite ball hole. Rotate bolt 180° and repeat procedure.

- (b) While working retaining pin, spray corrosion preventive compound, Item 114, Appendix D, into hole in bolthead and hole in end of special self-retaining bolt.

- (c) Using cleaning cloth, Item 102, Appendix D, wipe off excess corrosion preventive compound from special

self-retaining bolt prior to installation.

- (d) Using corrosion preventive compound, Item 109, Appendix D, apply to exterior retaining ball area and threads.

### NOTE

Position control rod according to identification plates on rod end (Figure 6-4-1-7, Sheet 3, Detail G, View A, and Sheet 7, Detail R, View A and Detail S, View A).

- (4) Install control rod between applicable primary servo and bellcrank, or walking beam.

### WARNING

### FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of special self-retaining bolts are critical characteristics.

### NOTE

- Add washers, as necessary, to make sure nut is bottomed against control rod and not on special self-retaining bolts.
  - To install special self-retaining bolt, insert 1/8-inch Allen wrench in hole in bolthead. Push on pin to release retaining balls at nut end of bolt. Keep pressure on pin while installing self-retaining bolt.
- (5) Install special self-retaining bolt and countersunk or flat washers. Make sure countersunk side of washer is toward bolt-head. Add washers, as required (maximum of two additional washers under countersunk washer), to provide correct grip length and cotter pin position (Figure

6-4-1-7, Sheet 3, Detail G, and Sheet 7, Detail R and Detail S).

- (6) Check special self-retaining bolt for free axial motion and that impedance feature is not restrained in part. If special self-retaining bolt binds and impedance feature is depressed, change washer stackup.

- (7) Install nut.

- (8) **TORQUE NUT TO 600 INCH-POUNDS. IF SLOT IN NUT DOES NOT LINE UP WITH COTTER PIN HOLE, SET TORQUE WRENCH TO 900 INCH-POUNDS AND TIGHTEN NUT UNTIL SLOT IN NUT AND COTTER PIN HOLE LINE UP. DO NOT EXCEED MAXIMUM TORQUE.**

- (9) Install cotter pin.

### NOTE

If unable to torque bolthead, use crowfoot attachment to torque nut, without disturbing cotter pin and without restraining bolthead.

- (10) **APPLY TIGHTENING TORQUE (PROOF LOAD) OF 400 - 420 INCH-POUNDS IN TIGHTENING DIRECTION.** If bolt rotates, perform the following:

- (a) Remove cotter pin, nut, special self-retaining bolt, and washer(s).
- (b) Repeat step aa. (5), adjusting washer thicknesses or adding washers.
- (c) Repeat step aa. (6) through step aa. (10).

- (11) Apply torque stripe (PARA 2-6-5).

- ab. Connect oil inlet and outlet lines, and if installed, rotor brake hose to main transmission as follows:

- (1) **RTR BRAKE** Connect rotor brake hose as follows (Figure 6-4-1-2, Detail A):

- (a) Remove protective cap and plug from rotor brake hose and rotor brake.
- (b) Connect rotor brake hose to elbow on rotor brake. Remove tag.
- (c) **TORQUE B-NUT TO 215 - 245 INCH-POUNDS.**
- (2) Connect oil drain line to oil scupper. Install clamp and oil drain line on bracket on oil scupper and secure with screw and nut (Figure 6-4-1-2, Detail B).
- (3) Remove protective caps and plugs from oil inlet and oil outlet lines, and fittings.
- (4) Connect oil inlet and oil outlet lines to fittings (Figure 6-4-1-2, Detail A).
- (5) **TORQUE B-NUT ON OIL INLET LINE TO 800 - 900 INCH-POUNDS.**
- (6) **TORQUE B-NUT ON OIL OUTLET LINE TO 500 - 700 INCH-POUNDS.**

ac. Install gust lock lever (PARA 6-4-8).

ad. Install tail rotor drive shaft section I (PARA 6-4-38).

ae. Install heating and ventilation system bleed-air tube (PARA 13-4-4).

af. Connect forward support tubes to elastomeric gimbals on input modules (PARA 4-4-6).

ag. Connect engine output shafts to input module flanges (PARA 4-4-6).

ah. Connect engine load-demand push/pull cable supports to accessory modules as follows (Figure 6-4-1-1, Sheet 2, Detail F):

- (1) Using primer, Item 258, Appendix D, apply to support boltholes on accessory modules.
- (2) Position supports on accessory modules.
- (3) Add washers as required to fill gaps between supports and accessory modules.

(4) Install bolts, washers, and nuts securing supports to accessory modules.

(5) **TORQUE BOLTS TO 118 - 131 INCH-POUNDS.**

ai. Connect lines and wiring to main transmission as follows:

(1) Connect hydraulic pressure line and hydraulic return line at quick-disconnects on hydraulic pump modules (Figure 6-4-1-1, Sheet 1, Detail A).

(2) Connect electrical connectors to hydraulic pump modules as follows:

#### NOTE

Electrical connector on side of hydraulic pump module should be capped. It is not used in this installation.

(a) Remove protective caps and plugs from electrical connectors.

(b) Inspect and treat electrical connectors (PARA 1-7-20).

(c) Connect electrical connector, P447 (No. 1 hydraulic pump module), and P432 (No. 2 hydraulic pump module).

(d) Using lockwire, Item 196, Appendix D, lockwire electrical connectors.

(3) Connect drain lines to seal drain elbows.

(4) Connect drain lines to hydraulic pump module bleed relief valves.

(5) Connect electrical connectors to main generators as follows:

(a) Remove protective caps and plugs from electrical connectors.

(b) Inspect and treat electrical connectors (PARA 1-7-20).

| No. 1 Main Generator | No. 2 Main Generator |
|----------------------|----------------------|
| P251 to J251         | P250 to J250         |
| P278 to J278         | P277 to J277         |

| Terminal | Electrical Lead No. |
|----------|---------------------|
| T-1      | X1A6A               |
| T-2      | X2A6B               |
| T-3      | X3A6C               |
| G        | X4A6N               |

- (c) Remove tags.
- (6) Connect electrical leads to main generators as follows (Figure 6-4-1-1, Sheet 1, Detail C or Detail D):
  - (a) Inspect electrical leads in main generator harness for chafing. If chafed, repair electrical leads.
  - (b) If installed, inspect rubber terminal covers for cracks. If cracked, replace rubber terminal covers.

**NOTE**

Do not reinstall terminal block cover. Terminal block cover will be replaced with rubber terminal covers.

- (c) If required, install rubber terminal covers on electrical leads.
- (d) If installing No. 1 main generator, connect electrical leads to terminals and secure with washers, lockwashers, and nuts as follows:

| Terminal | Electrical Lead No. |
|----------|---------------------|
| T-1      | X101A6A             |
| T-2      | X102A6B             |
| T-3      | X103A6C             |
| G        | X104A6N             |

- (e) If installing No. 2 main generator, connect electrical leads to terminals and secure with washers, lockwashers, and nuts as follows:

- (f) Remove tags.
- (g) Push rubber terminal covers firmly over terminal connections.
- (h) Install electrical leads with clamp to end of main generator and secure with screw and washer.
- (i) Arrange electrical lead bundle with most of slack between clamp and feed-through (Figure 6-4-1-1, Sheet 2, Detail E).
- (j) If not already done, using tiedown straps, Item 334, Appendix D, secure electrical lead bundle below clamp at 4 to 6-inch intervals.
- (k) If not already done, using spiral wrap, Item 315, Appendix D, on left side (No. 1 main generator) only, wrap backup hydraulic pump module wiring from 2 or 3 inches above to 2 or 3 inches below point adjacent to main generator electrical lead bundle. Using tiedown straps, Item 334, Appendix D, secure wrap.

- (7) Connect electrical connectors under front engine supports as follows (Figure 6-4-1-1, Sheet 1, Detail B):
  - (a) Remove protective caps and plugs from electrical connectors.
  - (b) Inspect and treat electrical connectors (PARA 1-7-20).
  - (c) Connect electrical connectors, P400 (right side), and P401 (left side), to brackets under forward support tube.



- aj. Using sealing compound, Item 292, Appendix D, apply to boltheads, nuts, and exposed threads.
- ak. Reposition cables and reconnect rod ends of collective bias tube to mixer (PARA 4-4-27).
- al. Remove protective covering from upper wall perimeter of main module housing.
- am. If required, install swashplate (PARA 5-4-22).
- an. If removed, install fire extinguisher line on right firewall (PARA 12-4-5).
- ao. Install left and right main rotor pylon air inlet fairings (PARA 2-4-106).
- ap. Install main rotor pylon transmission cover (PARA 2-4-107).
- aq. Inspect and clean accessory module oil screens (PARA 6-4-16).
- ar. Inspect and clean chip detectors, and if installed, chip detector strainers (PARA 6-4-15, PARA 6-4-17, and PARA 6-4-18).
- as. Perform operational check of chip detector system (PARA 6-2-2).
- at. If removed, install main rotor blades (PARA 5-4-31).
- au. If disconnected, connect deice harness to main rotor deice junction box (PARA 12-4-26).
- av. Service main transmission (PARA 1-3-12).
- aw. Make sure area is clean and free of foreign material.
- ax. Install soundproofing panels, rear of main transmission drip pan, to cabin ceiling (PARA 2-4-69).
- ay. Install main transmission drip pan (PARA 2-4-70).
- az. Close engine cowlings.
- ba. Close main rotor pylon sliding cover.
- bb. Install oil cooler compartment access doors (PARA 2-4-113).
- bc. Perform main rotor system rig check (PARA 11-4-4).
- bd. Perform main transmission functional test (PARA 6-3-4).
- be. Perform torque stabilization check as follows:
  - (1) On helicopters with engine output shaft, 70361-08014-041, perform torque stabilization check after first ground run and again after 3 to 13 flight hours (PARA 4-4-6).
  - (1) On helicopters with engine output shafts, 70361-08003-042/-043, perform torque stabilization check after first ground run, after 1 to 3 flight hours, and after 25 to 30 flight hours (PARA 4-4-6).

#### 6-4-1.2 Reseat Dowel Pin.

##### NOTE

This procedure applies to the left aft dowel pin located at the leg of the aft bellcrank support. Reseating of any other dowel pin on main transmission is not authorized.

- a. Using protective covering, protect adjacent flight control surfaces, firewalls, hydraulic lines and wiring harnesses.
- b. **701214-SUBQ** Remove bolts, washers, and nuts securing retaining angle to leg of aft bellcrank support. Remove retaining angle (Figure 6-4-1-7, Sheet 5, Detail M).
- c. Using 2 - 5 pound rawhide mallet and brass drift, seat left aft dowel pin as follows:
  - (1) If left aft dowel pin, 70351-08404-102, 70351-08405-102, or NAS607-12-14P, is installed, seat dowel pin until flush with, or below surface of, leg of aft bellcrank support.
  - (2) **70583-701213** If left aft dowel pin, 70351-08404-104, is installed, seat dowel



pin to a maximum of 0.248-inch above surface of leg of aft bellcrank support leg. ■

- (3) **701214-SUBQ** If left aft dowel pin, 70351-08404-104, is installed, seat dowel pin to a maximum of 0.128-inch above surface of leg of aft bellcrank support leg. ■

- d. **701214-SUBQ** Install retaining angle as follows: ■

**WARNING**

**FSCAP**

Verification of retaining angle installation is a critical characteristic.

**NOTE**

Bolts securing retaining angle to leg of aft bellcrank support are installed with boltheads facing down.

- (1) Install retaining angle to leg of aft bellcrank support and secure with bolts, washers, and nuts.
- (2) Using feeler gage, measure clearance between left aft dowel pin and retaining angle. **MINIMUM CLEARANCE IS 0.010-INCH.**
- (3) **TORQUE NUTS TO 12 - 20 INCH-POUNDS.**
- (4) Install cotter pin.



6-4-2 MAIN TRANSMISSION TAIL TAKEOFF FLANGE.

PARAGRAPH BREAKDOWN

|         |                                               | PAGE    |
|---------|-----------------------------------------------|---------|
| 6-4-2.1 | Replace Main Transmission Tail Takeoff Flange | 6-4-2-2 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Inside Micrometer
- Pin, 0.072-inch diameter
- Powertrain Repairer’s Toolkit
- Tail Drive Flange Wrench, 70700-20687-102
- Torque Reactor, 70700-20688-041
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 100 - 500 ft lbs

Materials/Parts

- Abrasive Cloth, Item 1A, Appendix D
- Abrasive Cloth, Item 9, Appendix D

- Abrasive Cloth, Item 10, Appendix D
- Abrasive Mat, Item 14, Appendix D
- Lockwire, Item 198, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Primer, Item 258, Appendix D

References

- Appendix D
- PARA 1-6-4
- PARA 1-6-18
- PARA 1-7-49
- PARA 2-6-5
- PARA 6-4-1
- PARA 6-4-38
- PARA 6-4-70

### 6-4-2.1 Replace Main Transmission Tail Takeoff Flange.

#### 6-4-2.1.1 Remove.

#### CAUTION

Damage to transmission and drive shaft will result if rotor blades are not tied down prior to performing maintenance.

- a. Tie down main and tail rotor blades (PARA 1-6-4).
- b. If not already done, disengage gust lock (PARA 1-6-18).
- c. Open oil cooler compartment access doors.
- d. Remove tail rotor drive shaft section I (PARA 6-4-38). Keep tail rotor drive shaft flexible coupling with tail rotor drive shaft section I.
- e. **RTR BRAKE** Remove rotor brake assembly (PARA 6-4-70). 
- f. On helicopters with main transmission tail takeoff flange, 70351-08159, remove flange as follows (Figure 6-4-2-1, Sheet 1, Detail A):
  - (1) Remove bolts from retaining nut on main transmission tail takeoff flange.
  - (2) Unscrew and remove retaining nut.
  - (3) Pull main transmission tail takeoff flange from pinion shaft.
- g. On helicopters with main transmission tail takeoff flange, 70351-48149, remove flange as follows (Figure 6-4-2-1, Sheet 1, Detail A):
  - (1) Using torque reactor and tail drive flange wrench, remove retaining nut.
  - (2) Pull main transmission tail takeoff flange from pinion shaft.
- h. On helicopters with main transmission tail takeoff flange, 70351-28159, remove flange as follows (Figure 6-4-2-1, Sheet 2, Detail A):

- (1) Using torque reactor and tail drive flange wrench, remove retaining nut.
- (2) Pull main transmission tail takeoff flange and rotor brake disc or gust lock flange from pinion shaft.
- (3) Remove packing from pinion shaft. Discard packing.

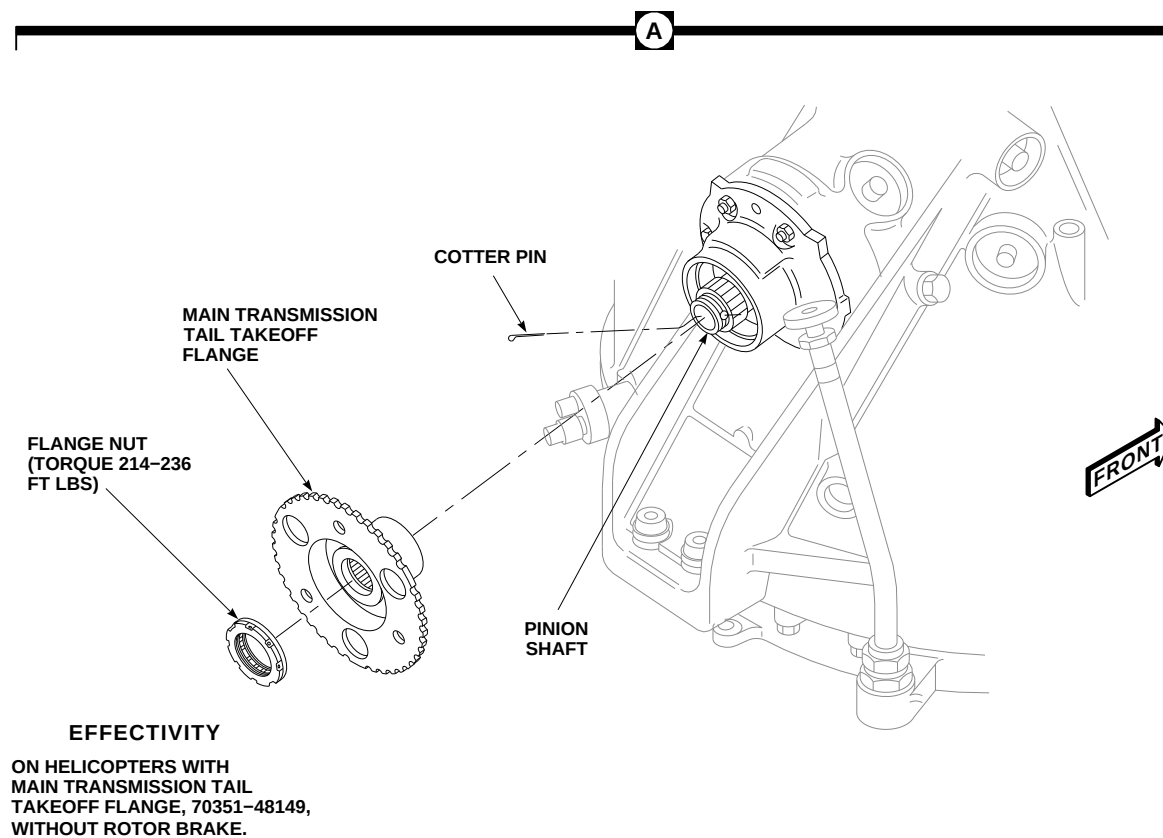
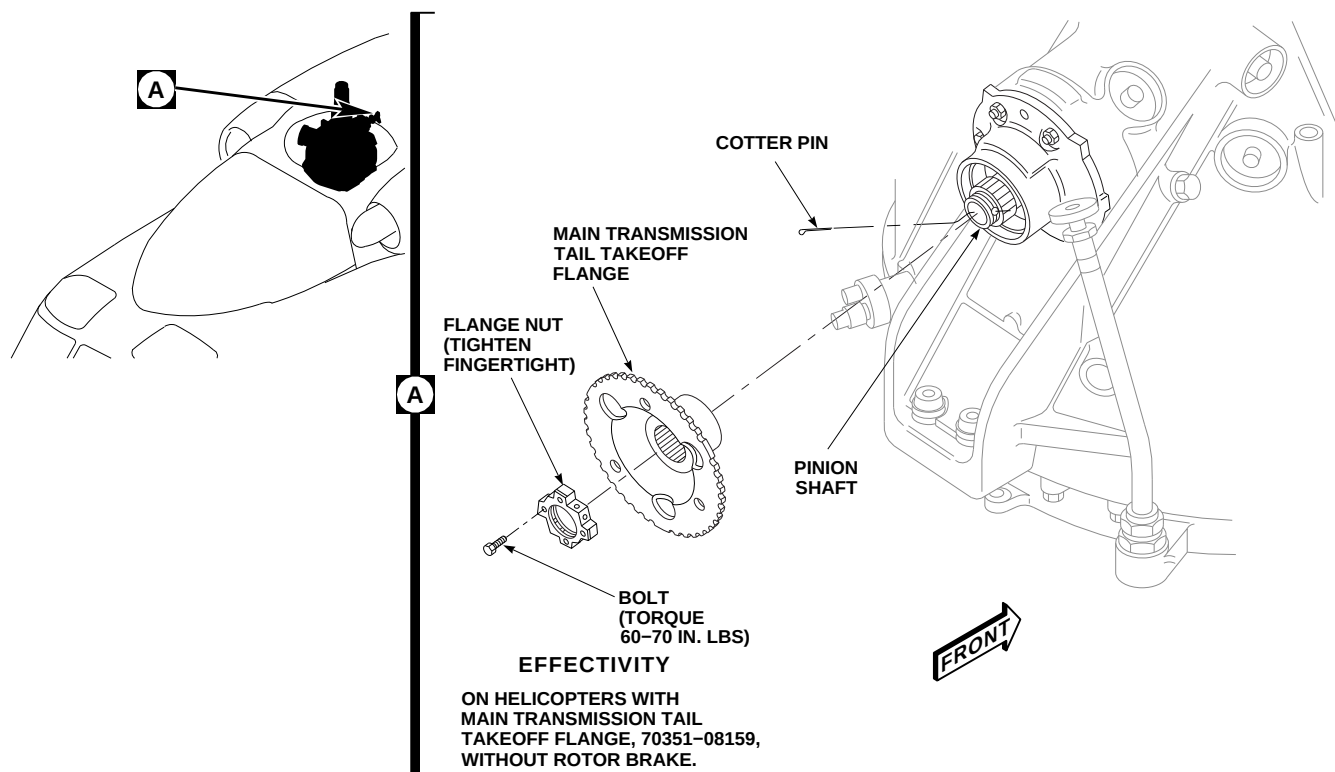
#### NOTE

If rotor brake disc is frozen to main transmission tail takeoff flange, replace rotor brake disc.

- (4) Remove bolts and nuts securing rotor brake disc or gust lock flange to main transmission tail takeoff flange. Slide rotor brake disc or gust lock flange off main transmission tail takeoff flange.

#### 6-4-2.1.2 Inspect/Repair.

- a. Inspect chrome-plated seal journal area of main transmission tail takeoff flange for nicks or scratches (Figure 6-4-2-2, Sheet 1 and Sheet 2). **NONE ALLOWED.** If nicked or scratched, replace tail takeoff flange. Wear pattern caused by seal is not cause for replacement, unless wear causes seal to leak.
- b. Check present dimension of damaged area. Compare reading with minimum repair limits.
- c. Check depth of damaged area to determine whether tail takeoff flange minimum dimension can be maintained after repair.
- d. Inspect tail takeoff flange for scratches. **FINE SCRATCHES NEED NOT BE POLISHED IF NOT THROUGH PAINT.**
- e. Inspect splines for corrosion and wear. Evidence of corrosion of main transmission splines is not cause for removal of main module. **NO WEAR INTO BASE METAL THROUGH PLATING IS PERMITTED (NO FRETTING OR SPALLING).** If wear exceeds limits, replace main module (PARA 6-4-1).
- f. Check internal spline dimension as follows:



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FIGURE 6-4-2-1. REPLACE MAIN TRANSMISSION TAIL TAKEOFF FLANGE (SHEET 1 OF 2)

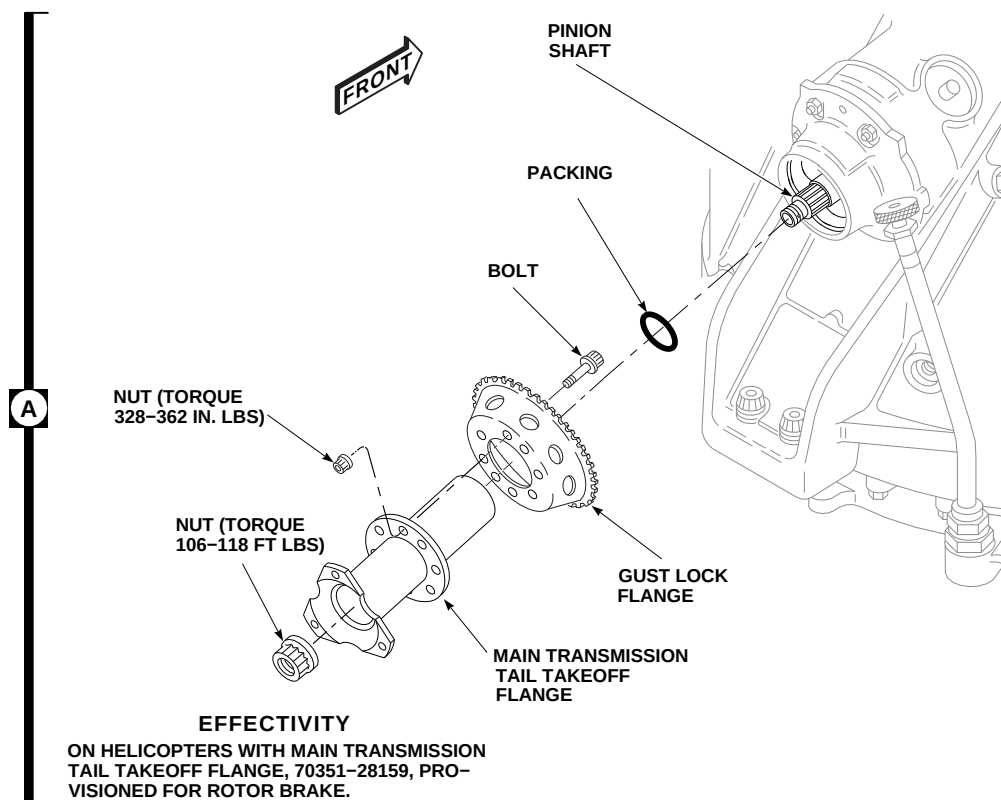
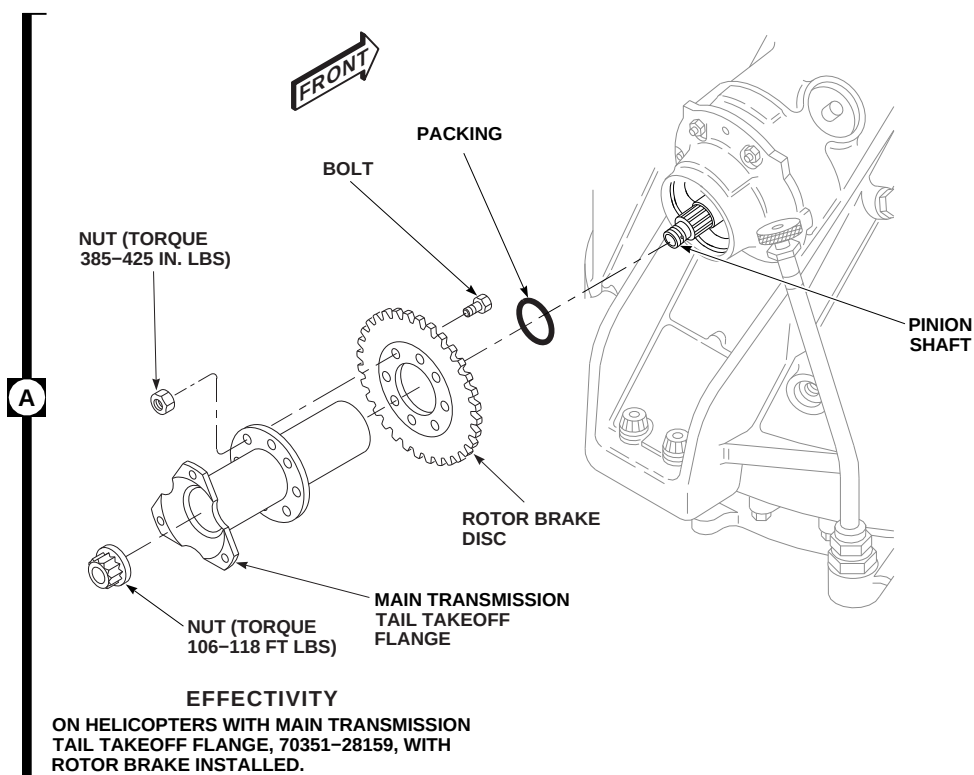
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FIGURE 6-4-2-1. REPLACE MAIN TRANSMISSION TAIL TAKEOFF FLANGE (SHEET 2 OF 2)

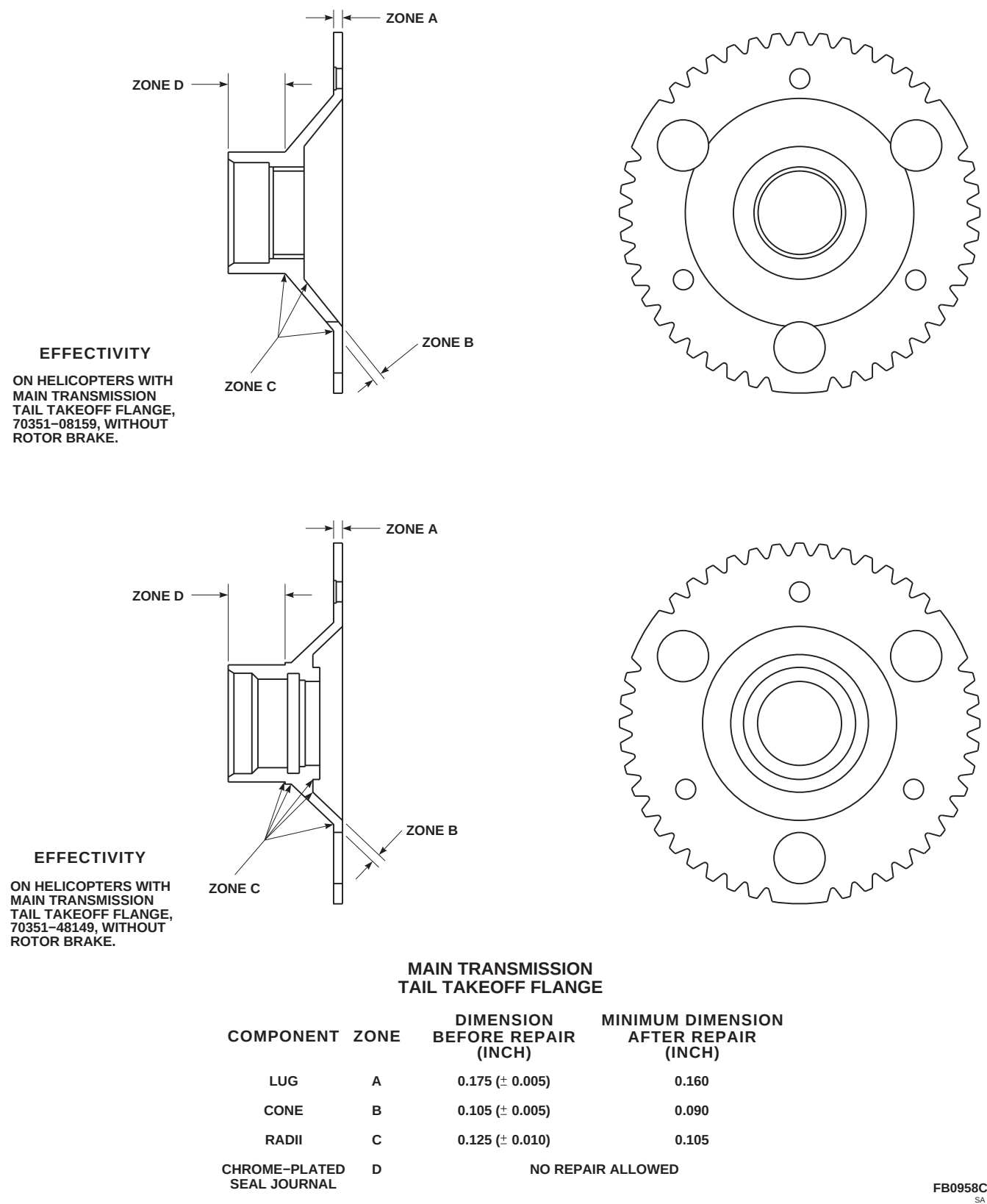


FIGURE 6-4-2-2. INSPECT MAIN TRANSMISSION TAIL TAKEOFF FLANGE (SHEET 1 OF 2)

- (1) Secure 0.072-inch diameter pin between internal splines.
  - (2) Secure second 0.072-inch diameter pin between internal splines directly opposite first pin.
  - (3) Using inside micrometer, measure inside diameter (ID) between two pins. **ID MUST NOT EXCEED 1.736-INCHES.** If ID is beyond limits, replace tail takeoff flange.
  - (4) Repeat step f. (1) through step f. (3) with pins located in four different positions.
- g. Inspect retaining nut for corrosion. **REPLACE RETAINING NUT IF:**
- (1) **PITS ARE 0.030-INCH DEEP.**
  - (2) **PITTING COVERS MORE THAN 10% OF MATING SURFACE.**
  - (3) **THREADS SHOW SIGNS OF CORROSION.**
- h. Minor corrosion may be removed from retaining nut using abrasive cloth, Item 10, Appendix D, abrasive cloth, Item 1A, Appendix D, or abrasive mat, Item 14, Appendix D.

#### 6-4-2.1.3 Repair.

##### NOTE

Repair minor damage, such as nicks, scratches, or scoring by polishing.

- a. Before starting repair on tail takeoff flange, determine whether polishing out damage would result in a dimension below repairable limits (Figure 6-4-2-2). If damage repair will exceed permitted limits, replace tail takeoff flange.

##### NOTE

Rework of main transmission tail takeoff flange may be done on helicopter.

- b. If damage can be repaired, remove damaged tail takeoff flange (PARA 6-4-2.1.1).

- c. Using abrasive cloth, Item 9, Appendix D, blend smooth. Remove only what is necessary so tail takeoff flange will stay within limits. Blend radius on tail takeoff flange to + 1/2-inch or greater.

##### NOTE

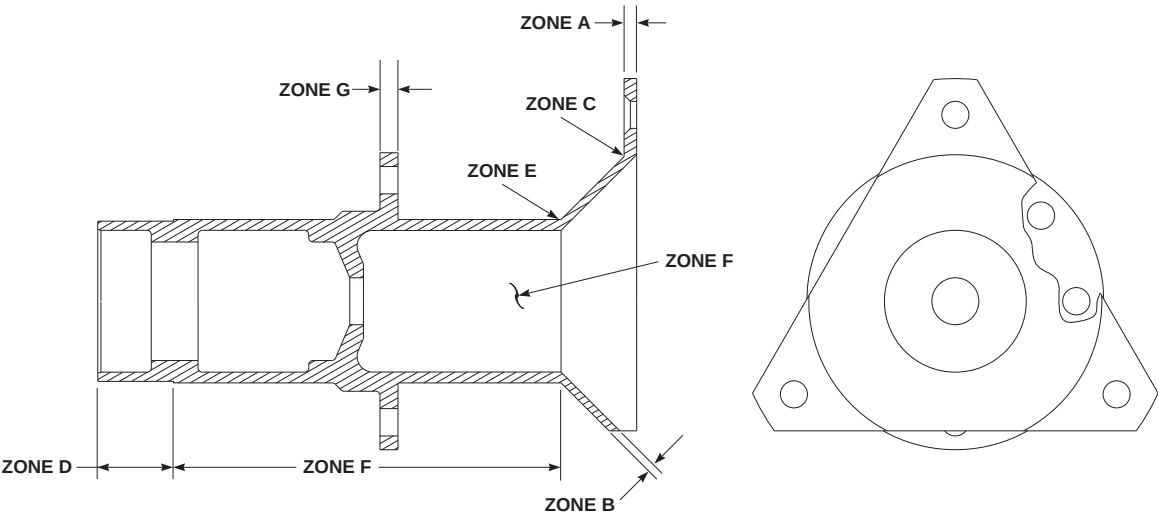
Polish only in damaged area and not entire circumference, unless damage extends around it.

- d. Using abrasive cloth, Item 10, Appendix D, or similar polishing cloth, blend reworked area to remove all traces of abrasive marks and match surface smoothness.
- e. If gear teeth have nicks, using abrasive cloth, Item 9, Appendix D, blend smooth. **NICKS, DENTS, AND SCRATCHES MUST NOT EXCEED 0.010-INCH. NO RAISED AREAS ARE TO REMAIN AFTER POLISHING.**
- f. Touch up paint on reworked areas (PARA 2-6-5). Do not paint around boltholes.

#### 6-4-2.1.4 Install.

- a. On helicopters with main transmission tail takeoff flange, 70351-08159, install as follows (Figure 6-4-2-1, Sheet 1, Detail A):
  - (1) Slide main transmission tail takeoff flange on pinion shaft.
  - (2) Apply primer, Item 258, Appendix D, on inner side of retaining nut.
  - (3) **INSTALL RETAINING NUT AND TIGHTEN FINGERTIGHT. BACK OFF ENOUGH TO LINE UP COTTER PIN HOLE.** Insert cotter pin with head inside pinion shaft.
  - (4) Install bolts on retaining nut.
  - (5) **TORQUE TO 60 - 70 INCH-POUNDS.**
  - (6) Using lockwire, Item 198, Appendix D, lockwire bolts.
- b. On helicopters with main transmission tail takeoff flange, 70351-48149, install as follows (Figure 6-4-2-1, Sheet 1, Detail A):





| MAIN TRANSMISSION<br>TAIL TAKEOFF FLANGE |      |                                      |                                             |  |
|------------------------------------------|------|--------------------------------------|---------------------------------------------|--|
| COMPONENT                                | ZONE | DIMENSION<br>BEFORE REPAIR<br>(INCH) | MINIMUM DIMENSION<br>AFTER REPAIR<br>(INCH) |  |
| LUG                                      | A    | 0.175 ( ±0.005)                      | 0.160                                       |  |
| CONE                                     | B    | 0.105 ( ±0.005)                      | 0.090                                       |  |
| RADII                                    | C    | 0.375 ( ±0.010)                      | 0.350                                       |  |
| CHROME-PLATED<br>SEAL JOURNAL            | D    | NO REPAIR ALLOWED                    |                                             |  |
| RADII                                    | E    | 0.500 ( ±0.010)                      | 0.450                                       |  |
| ID, OD                                   | F    | NICKS OR SCRATCHES UP TO 0.010 DEEP  |                                             |  |
| FLANGE                                   | G    | 0.250 ( ±0.010)                      | 0.235                                       |  |

**EFFECTIVITY**

ON HELICOPTERS WITH MAIN TRANSMISSION  
TAIL TAKEOFF FLANGE, 70351-28159, WITH  
ROTOR BRAKE INSTALLED OR PROVISIONED  
FOR ROTOR BRAKE.

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FIGURE 6-4-2-2. INSPECT MAIN TRANSMISSION TAIL TAKEOFF FLANGE (SHEET 2 OF 2)

- (1) Slide main transmission tail takeoff flange on pinion shaft.
  - (2) Apply primer, Item 258, Appendix D, on inner side of retaining nut.
  - (3) Thread retaining nut onto pinion shaft. Make sure retaining nut can be fully run on by hand.
  - (4) Using torque reactor and tail drive flange wrench, **TORQUE RETAINING NUT TO 214 - 236 FOOT-POUNDS.**
  - (5) If required, align cotter pin hole by turning retaining nut in tightening direction. Install cotter pin with head inside pinion shaft.
- c. On helicopters with rotor brake provisioned or installed, install main transmission tail takeoff flange, 70351-28159, as follows (Figure 6-4-2-1, Sheet 2, Detail A):
- (1) Position rotor brake disc or gust lock flange on main transmission tail takeoff flange and align boltholes.
  - (2) Install bolts and nuts connecting main transmission tail takeoff flange to rotor brake disc or gust lock flange.
  - (3) **TORQUE NUTS SECURING ROTOR BRAKE DISC TO 385 - 425 INCH-POUNDS.**
  - (4) **TORQUE NUTS SECURING GUST LOCK FLANGE TO 328 - 362 INCH-POUNDS.**
  - (5) Using lubricating oil, Item 205, Appendix D, coat packing and install on pinion shaft.
  - (6) Install main transmission tail takeoff flange, with rotor brake disc or gust lock flange installed, on pinion shaft.
  - (7) Measure drag torque of retaining nut as follows:

**NOTE**

Retaining nut may be reused only if it meets minimum drag torque requirement of 32 inch-pounds.

- (a) With retaining nut and bolt not installed on helicopter, and with no axial load being applied to retaining nut, thread retaining nut onto mating bolt so that at least two threads of bolt extend past end of retaining nut.
  - (b) Measure torque required to back off retaining nut. **RETAINING NUT SHALL BACK OFF BOLT WITH A MINIMUM TORQUE OF 32 INCH-POUNDS.** If torque is beyond limits, replace retaining nut.
- (8) Install retaining nut as follows:
- (a) Thread retaining nut onto pinion shaft.
  - (b) Using torque reactor and tail drive flange wrench, **TORQUE RETAINING NUT TO 107 - 118 FOOT-POUNDS.**
- d. Install tail rotor drive shaft section I (PARA 6-4-38).
- e. **RTR BRAKE** Install rotor brake assembly (PARA 6-4-70). 
- f. Make sure area is clean and free of foreign material.
- g. Close oil cooler compartment access doors.
- h. Remove tiedowns securing main and tail rotor blades.
- i. If required, engage gust lock (PARA 1-6-18). 
- j. **RTR BRAKE** Perform rotor brake nut torque stabilization check (PARA 1-7-49). 

6-4-3    MAIN TRANSMISSION OUTPUT SHAFT SEAL.

PARAGRAPH BREAKDOWN

|         |                                             | PAGE    |
|---------|---------------------------------------------|---------|
| 6-4-3.1 | Replace Main Transmission Output Shaft Seal | 6-4-3-2 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Ballpeen Hammer, 8-Ounce
- Barstock, 4-inches-square
- Drill
- Drill Bit, 5⁄32-inch diameter
- Nylon Wedge, Locally-Made, Figure H-260 , Appendix H
- Powertrain Repairer’s Toolkit
- Screws, No. 8 Self-Tapping, 2-inches long
- Seal Installation Tool, Locally-Made, Figure H-157, Appendix H
- Seal Puller, Locally-Made, Figure H-160, Appendix H

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Grease, Item 149, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D
- Lubricating Oil, Item 208, Appendix D
- Machinery Towel, Item 211, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- Appendix H
- PARA 1-3-12
- PARA 6-4-2

### 6-4-3.1 Replace Main Transmission Output Shaft Seal.

#### 6-4-3.1.1 Remove.

#### NOTE

- Checking amount of leakage from main transmission output shaft seal can prevent premature seal replacement before leakage is over recommended allowable limits. Leakage is gaged by measuring drop in oil level as shown on oil level gage (dipstick). Total leakage must not be over  $\frac{1}{8}$ -inch in any one flight. Allow at least 2 hours for oil to drain back before checking. If over  $\frac{1}{8}$ -inch, replace seal.

- On helicopters with main transmission, 70351-48000 and 70351-58000, for extended flights over 3 hours, the allowable leakage is to be compatible with a 10 hour flight. Loss rate of 0.050-inch in 1 hour,  $\frac{1}{4}$ -inch in 5 hours and  $\frac{1}{2}$ -inch in 10 hours as indicated on oil level gage (dipstick). If loss of oil exceeds the above criteria, replace seal.

- Remove main transmission tail takeoff flange (PARA 6-4-2).
- Remove retaining ring (Figure 6-4-3-1, Detail A).

#### NOTE

Self-tapping screws may be used as an alternate to remove seal if seal puller is not available. If seal puller is not available, proceed to step h.

- Locally make seal puller (Figure H-160, Appendix H).
- Locally make nylon wedge (Figure H-260, Appendix H).
- Insert nylon wedge between seal puller and pinion shaft to protect main transmission housing (Figure 6-4-3-1, Detail B).

#### CAUTION

Damage to main transmission will result if too much force is applied while using ballpeen hammer and seal puller. Main module housing will crack. Use care when using ballpeen hammer and seal puller.

- Using 8-ounce ballpeen hammer, gently tap end of seal puller in an outward direction, away from main transmission.
- Moving seal puller and nylon wedge in a clockwise direction, repeat step e. and step f. until seal is removed.
- If seal puller is not available, remove seal from main transmission as follows:

#### CAUTION

- Damage to main transmission will result if self-tapping screws completely penetrate seal. Screws will damage bearing inside main transmission. Install screws so that only outer face of seal is penetrated.

- Damage to main transmission will result if metal chips or shavings enter main transmission by drilling or punching holes in seal. Metal chips or shavings will contaminate main transmission oil. Do not drill or punch holes in seal.

- Using drill with  $\frac{5}{32}$ -inch diameter drill bit, drill holes in barstock to align  $\frac{1}{8}$ -inch holes in seal.
- Using 2-inch long, No. 8 self-tapping screws, insert through barstock and into holes of seal.
- TIGHTEN SCREWS UNTIL BARSTOCK PLATE IS SEATED SNUG AGAINST PINION SHAFT AND SEAL**

### 6-4-3-2

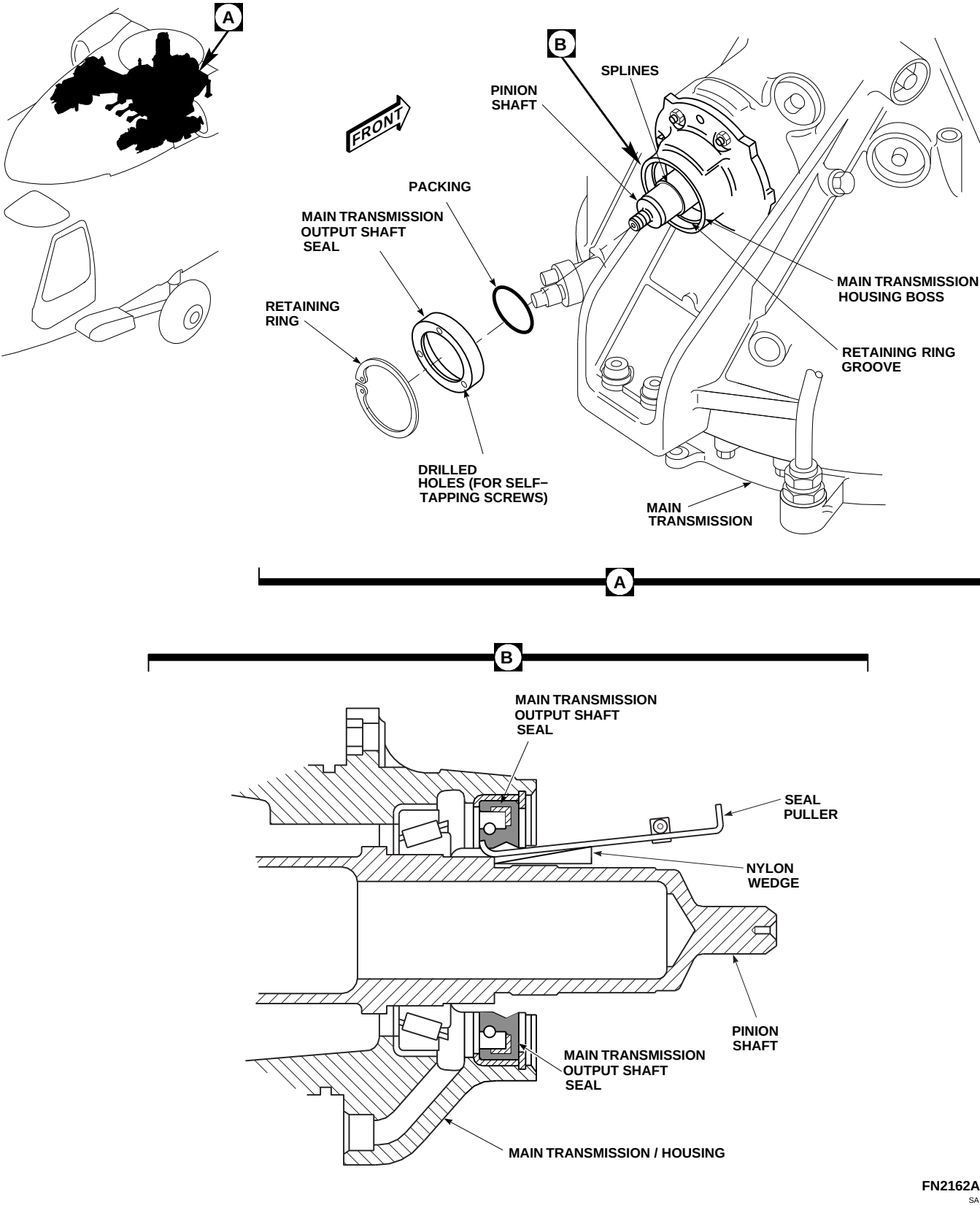


FIGURE 6-4-3-1. REPLACE MAIN TRANSMISSION OUTPUT SHAFT SEAL

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**IS REMOVED FROM MAIN TRANSMISSION. EXERCISE CAUTION NOT TO DAMAGE PINION SHAFT.**

- i. Remove packing from pinion shaft. Discard packing.

**6-4-3.1.2 Install.**

- a. Install packing on pinion shaft in groove behind splines (Figure 6-4-3-1, Detail A).
- b. Prior to installing main transmission output shaft seal, perform the following:
  - (1) Using grease, Item 149, Appendix D, coat inside diameter of seal.
  - (2) Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean mating surface of seal.
  - (3) Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, coat outside diameter of seal and mating surface.
- c. Locally make seal installation tool (Figure H-157, Appendix H).

**CAUTION**

Be careful not to cock or distort seal.

- d. Using seal installation tool, install new seal into main transmission. Make sure seal is fully seated and beyond retaining ring groove.

- e. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to retaining ring and install in main transmission housing while wet.
- f. Apply sealing compound, Item 292, Appendix D, to main transmission housing boss to prevent water entrapment and corrosion above retaining ring groove.
- g. Using epoxy primer coating kit, Item 135, Appendix D, touch up sealing compound with epoxy primer.
- h. Install main transmission tail takeoff flange (PARA 6-4-2).
- i. Service main transmission (PARA 1-3-12).
- j. Perform check of seal as follows:
  - (1) With qualified person at controls, run up helicopter.
  - (2) Check seal for leaks.

6-4-4 MAIN TRANSMISSION BREATHER PLUG.

PARAGRAPH BREAKDOWN

|                                                 | PAGE    |
|-------------------------------------------------|---------|
| 6-4-4.1 Replace Main Transmission Breather Plug | 6-4-4-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Scraper, Locally-Made
- Powertrain Repairer’s Toolkit

Materials/Parts

- Lubricating Oil, Item 205, Appendix D
- Machinery Towel, Item 211, Appendix D

- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 6-4-18
- PARA 6-4-25

6-4-4.1 Replace Main Transmission Breather Plug.

6-4-4.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Open oil cooler compartment access doors.
- If main transmission breather plug is being replaced due to oil leakage, perform the following:
  - Check main transmission for proper oil level.
  - Check main transmission oil filler cap for evidence of debris or ice.
  - Check input module return tubes for proper torque (PARA 6-4-25).

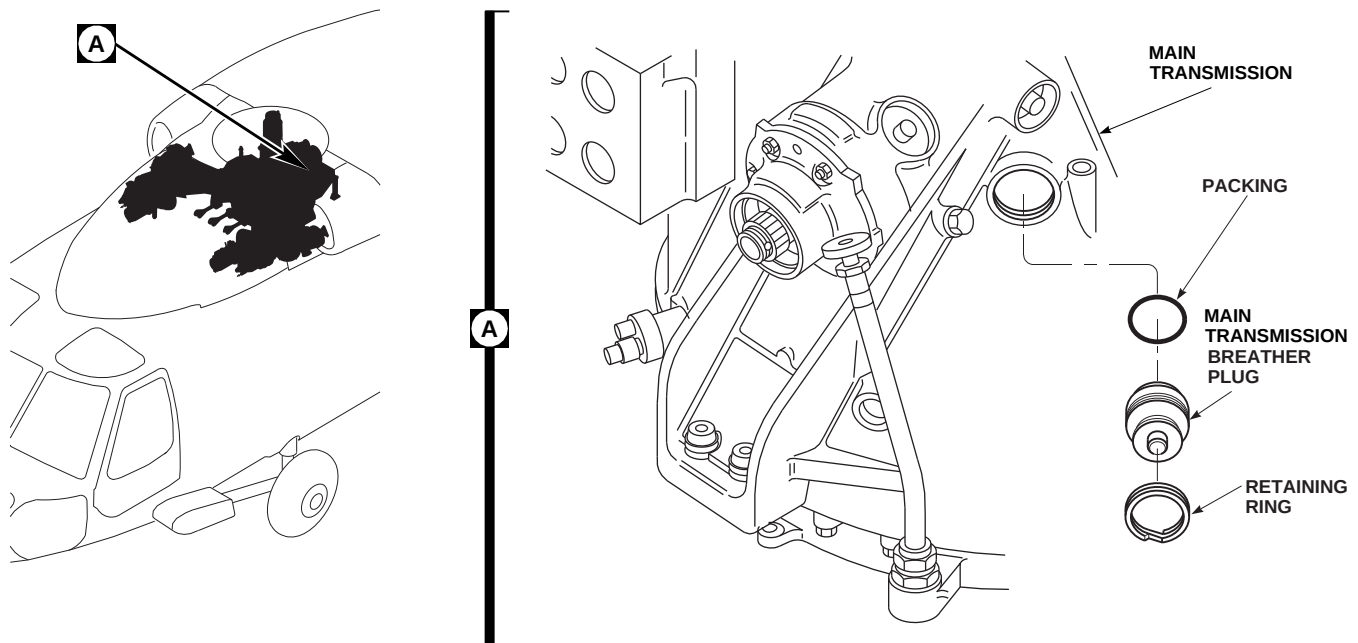
- Check main transmission chip detectors for evidence of leaks (PARA 6-4-18).

- Using nonmetallic scraper, remove sealing compound from main transmission breather plug.

NOTE

Do not remove screws from main transmission breather plug.

- Remove retaining ring securing main transmission breather plug to main transmission (Figure 6-4-4-1, Detail A).
- Install bolt in center of main transmission breather plug.
- Grasp bolt and remove main transmission breather plug from main transmission.

FN2167  
SA**FIGURE 6-4-4-1. REPLACE MAIN TRANSMISSION BREATHER PLUG**

- h. Remove packing from main transmission breather plug. Discard packing.
- i. Remove bolt from main transmission breather plug.
- c. Install main transmission breather plug in main transmission with hole at 6 o'clock position.
- d. Install retaining ring to secure main transmission breather plug to main transmission.

**6-4-4.1.2. Install.**

- a. Using lubricating oil, Item 205, Appendix D, lubricate packing.
- b. Install packing on main transmission breather plug (Figure 6-4-4-1, Detail A).
- e. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean main transmission breather plug and surrounding surface.
- f. Apply a bead of sealing compound, Item 292, Appendix D, around main transmission breather plug.
- g. Close oil cooler compartment access doors.

**NOTE**

Main transmission breather plug must be installed with hole at bottom (6 o'clock) position to allow oil to drain.

**6-4-4-2**



6-4-5 SWASHPLATE GUIDE AND MAIN ROTOR SHAFT SEAL.

PARAGRAPH BREAKDOWN

|                                                                                                                                                                                                                                                                      | PAGE                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6-4-5.1 Replace Swashplate Guide and Main Rotor Shaft Seal                                                                                                                                                                                                           | 6-4-5-1                                                                                                                                                                                         |
| <hr/>                                                                                                                                                                                                                                                                |                                                                                                                                                                                                 |
| <b>INITIAL SETUP</b>                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>Corrosion Resistant Coating, Item 118, Appendix D</li> </ul>                                                                                             |
| <b>Tools</b>                                                                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>Epoxy Primer Coating Kit, Item 135, Appendix D</li> </ul>                                                                                                |
| <ul style="list-style-type: none"> <li>Aircraft Mechanic’s Toolkit</li> <li>Chalk</li> <li>Grease Pencil</li> <li>Nonmetallic Drift, Locally-Made</li> <li>Powertrain Repairer’s Toolkit</li> <li>Rawhide Mallet</li> <li>Torque Wrench, 30 - 150 in. lbs</li> </ul> | <ul style="list-style-type: none"> <li>Grease, Item 149, Appendix D</li> <li>Sealing Compound, Item 292, Appendix D</li> </ul>                                                                  |
| <b>Materials/Parts</b>                                                                                                                                                                                                                                               | <b>References</b>                                                                                                                                                                               |
| <ul style="list-style-type: none"> <li>Abrasive Cloth, Item 8, Appendix D</li> <li>Assembly Fluid, Item 79, Appendix D</li> </ul>                                                                                                                                    | <ul style="list-style-type: none"> <li>Appendix D</li> <li>Appropriate MTF</li> <li>PARA 1-3-12</li> <li>PARA 1-6-16</li> <li>PARA 5-4-22</li> <li>PARA 11-4-4</li> <li>PARA 11-4-87</li> </ul> |
| <hr/>                                                                                                                                                                                                                                                                |                                                                                                                                                                                                 |

6-4-5.1 Replace Swashplate Guide and Main Rotor Shaft Seal.

6-4-5.1.1 Remove.

- a. Perform primary servo rig check (PARA 11-4-87).
- b. Turn off all electrical power (PARA 1-6-16).
- c. Remove swashplate (PARA 5-4-22).
- d. If same swashplate guide is going to be reused, using chalk or grease pencil, match mark swashplate guide and main transmission.

**CAUTION**

Damage to main module will result if foreign material enters main transmission. Use extreme care to keep foreign material out of main module.

- e. Remove bolts and washers securing swashplate guide and spacer to main transmission. Remove swashplate guide and packing (Figure 6-4-5-1, Detail A). Discard packing.
- f. Remove spacer and remove packing. Discard packing.
- g. Remove retaining ring from swashplate guide.
- h. Turn swashplate guide over. Using rawhide mallet and nonmetallic drift, tap out main rotor shaft seal.

#### 6-4-5.1.2 Inspect/Repair.

- a. Inspect swashplate guide in AREA A and AREA B for wear and scoring in chrome-plating (Figure 6-4-5-1, Detail B). **WEAR AND SCORING SHALL NOT PENETRATE INTO BASE METAL.**
- b. Inspect AREA A of swashplate guide chrome surface for peeling, chipping, and flaking as follows:
  - (1) **DAMAGE SHALL NOT EXCEED 1/4-INCH DIAMETER.**
  - (2) **NO MORE THAN 2 AREAS OF SURFACE DAMAGE WITHIN 2-INCHES OF EACH OTHER ARE ALLOWED.**
  - (3) **NO MORE THAN 20 TOTAL AREAS OF SURFACE DAMAGE.**
  - (4) If damage is beyond limits or if damage penetrates into base metal, replace swashplate guide.
- c. If surface damage in AREA A is within limits, using abrasive cloth, Item 8, Appendix D, gently smooth edge of damage.

- d. Inspect AREA B of swashplate guide chrome-plated surface for peeling, chipping, or flaking. **NONE ALLOWED.** If damage is found, replace swashplate guide.
- e. Inspect AREA C of swashplate guide for nicks, dents, or scratches. Minor damage may be blended smooth using abrasive cloth, Item 8, Appendix D. **DEPTH OF DAMAGE OR REWORK MUST NOT EXCEED 0.040-INCH.** If damage or rework is beyond limits, replace swashplate guide.

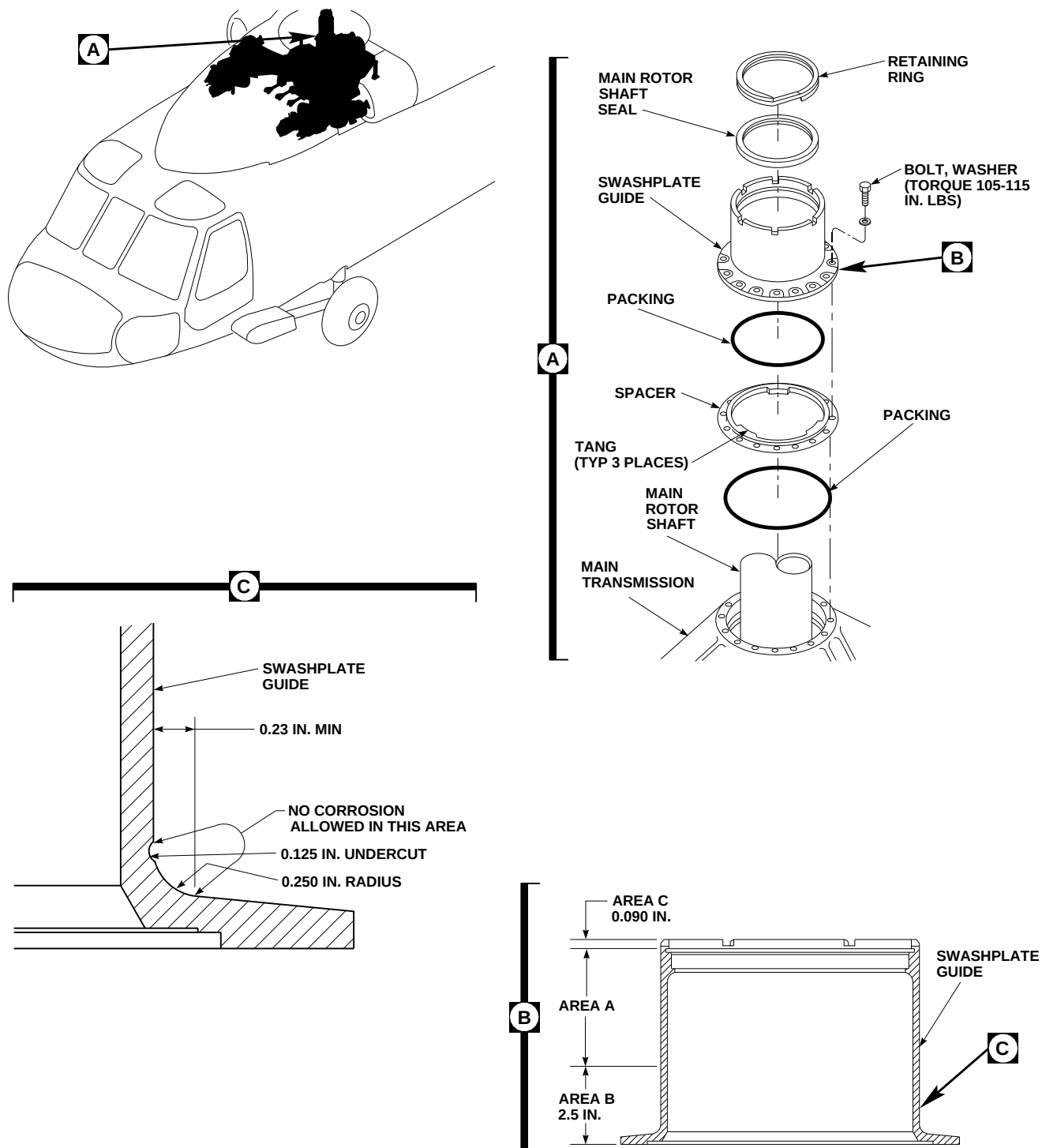
#### NOTE

Perform all blending of corrosion with abrasive cloth, Item 8, Appendix D.

- f. Inspect swashplate guide for corrosion as follows:
  - (1) **NO CORROSION ALLOWED IN CHROME PLATE, 1/4-INCH RADIUS, OR 1/8-INCH UNDERCUT** (Figure 6-4-5-1, Detail C).
  - (2) **SURFACE CORROSION IN AREA C MAY BE BLENDED OUT UP TO 0.020-INCH DEEP OVER NOT MORE THAN 20% OF SURFACE AREA.**
  - (3) **SURFACE CORROSION ON NONCHROME-PLATED AREA OF FLANGE MAY BE BLENDED OUT UP TO 0.010-INCH DEEP OVER NOT MORE THAN 10% OF SURFACE AREA.**
  - (4) If damage or rework is beyond limits, replace swashplate guide.
- g. Using corrosion resistant coating, Item 118, Appendix D, apply to all reworked areas.

#### 6-4-5.1.3 Install.

- a. Using rawhide mallet and nonmetallic drift, install main rotor shaft seal in swashplate guide. Tap main rotor shaft seal evenly and carefully until retaining ring can be installed in slot on swashplate guide (Figure 6-4-5-1, Detail A).



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SA

FIGURE 6-4-5-1. REPLACE SWASHPLATE GUIDE AND MAIN ROTOR SHAFT SEAL

- b. Install retaining ring in swashplate guide.
- c. Install packing on swashplate guide. Using assembly fluid, Item 79, Appendix D, hold packing in place.
- d. Install packing in main transmission. Using assembly fluid, Item 79, Appendix D, hold packing in place.

**CAUTION**

Make sure tangs of spacer engage main rotor shaft bearing outer slots.

- e. Install spacer with flat side facing main transmission.
- f. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to spacer mating surface on main transmission, outboard of packing groove.
- g. Using grease, Item 149, Appendix D, pack inside diameter of main rotor shaft seal.
- h. Using grease, Item 149, Appendix D, apply a light coat to mating surface of main rotor shaft seal on main rotor shaft.
- i. If same swashplate guide is being installed, line up match marks on swashplate guide and main transmission.
- j. Install swashplate guide and secure swashplate guide and spacer with bolts and washers.
- k. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**
- l. Apply sealing compound, Item 292, Appendix D, to seal area around joint of washer and bolt.
- m. Apply sealing compound, Item 292, Appendix D, to seal assembly.
- n. Remove excess sealing compound.
- o. Install swashplate (PARA 5-4-22).
- p. Service main transmission (PARA 1-3-12).
- q. Perform main rotor system rig check (PARA 11-4-4).
- r. Perform maintenance test flight (Appropriate MTF).

6-4-6 MAIN ROTOR SHAFT PLUG.

PARAGRAPH BREAKDOWN

|         |                               | PAGE    |
|---------|-------------------------------|---------|
| 6-4-6.1 | Replace Main Rotor Shaft Plug | 6-4-6-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Hammer
- Heat Lamps
- Powertrain Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Weight, 1 - 3 lbs
- Wood Block, 2-inches × 3½-inches × 55-inches

Materials/Parts

- Adhesive, Item 32, Appendix D
- Lubricating Oil, Item 205, Appendix D

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Primer, Item 253, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-3-12
- PARA 2-4-70
- PARA 5-4-28

6-4-6.1 Replace Main Rotor Shaft Plug.

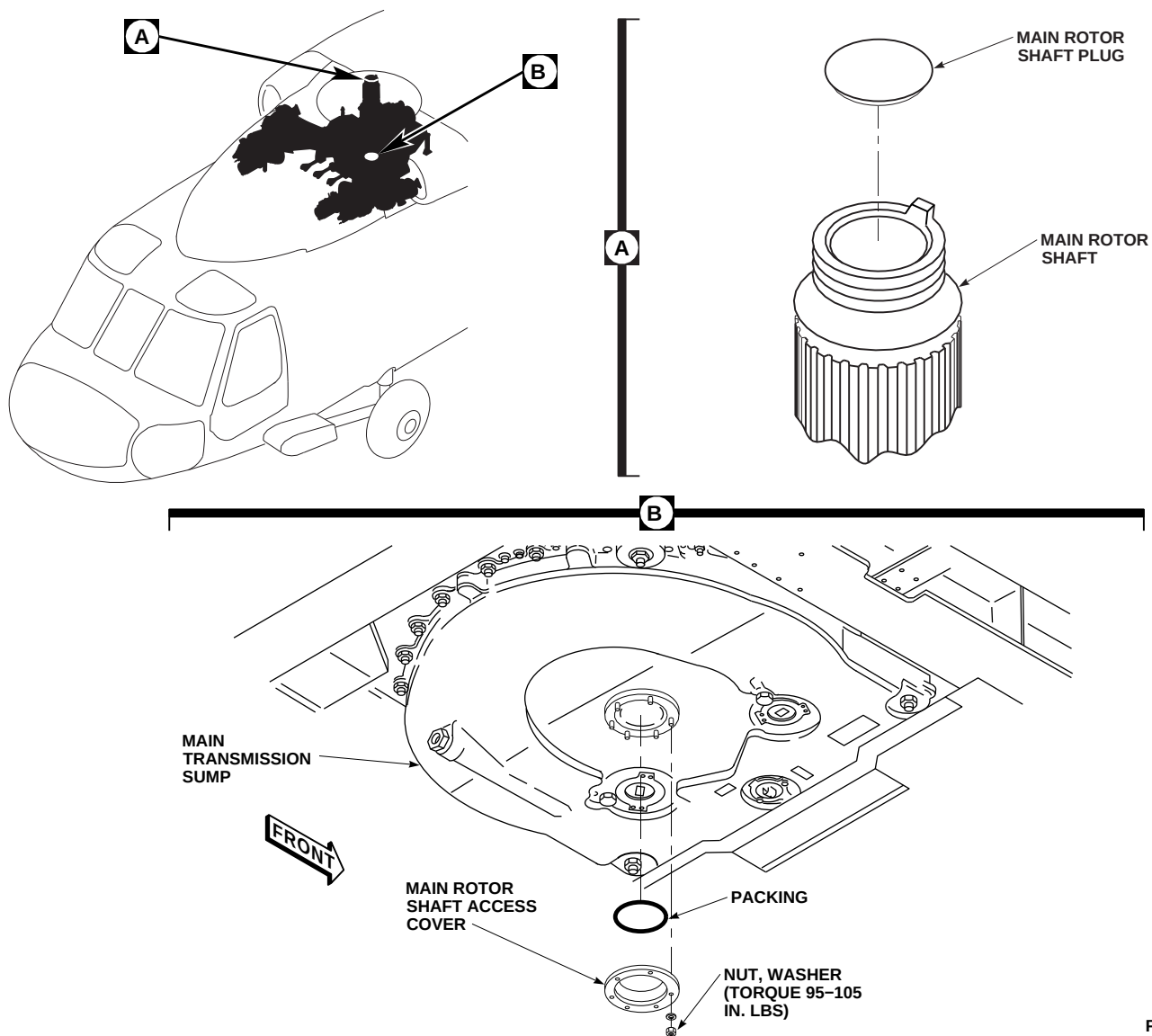
NOTE

This procedure applies to helicopters without blade deice distributor and slip-ring installed.

6-4-6.1.1. Remove.

- Remove main transmission drip pan (PARA 2-4-70).
- Drain main transmission (PARA 1-3-12).

- Remove bifilar cover (PARA 5-4-28).
- Remove nuts, washers, and main rotor shaft access cover from main transmission sump. Discard packing (Figure 6-4-6-1, Detail B).
- Position wood block inside main rotor shaft. Tap end of wood block with hammer until main rotor shaft plug at top of main rotor shaft is unseated (Figure 6-4-6-1, Detail A).
- Remove main rotor shaft plug from inside main rotor shaft extension.

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SA**FIGURE 6-4-6-1. REPLACE MAIN ROTOR SHAFT PLUG****6-4-6.1.2. Install.**

- Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean areas to be bonded.
- Apply a thin coat of primer, Item 253, Appendix D, to bonding surfaces. Allow to cure for a minimum of 1 hour at 70° - 75°F (21° - 23°C).
- Apply adhesive, Item 32, Appendix D, to bonding surfaces.
- Install main rotor shaft plug onto main rotor shaft (Figure 6-4-6-1, Detail A).
- Use suitable weight (1 - 3 pounds) to secure main rotor shaft plug in place.
- Allow adhesive to cure for 24 hours at 70° - 75°F (21° - 23°C), or using heat lamps, for 2 hours at 110° - 130°F (43° - 54°C).
- Using sealing compound, Item 292, Appendix D, seal edges of main rotor shaft plug.

**6-4-6-2**

- h. Install new packing, lubricated with lubricating oil, Item 205, Appendix D, on main rotor shaft access cover (Figure 6-4-6-1, Detail B).
- i. Install main rotor shaft access cover on main transmission sump and secure with washers and nuts.
- j. **TORQUE NUTS TO 95 - 105 INCH-POUNDS.**
- k. Install bifilar cover (PARA 5-4-28).
- l. Service main transmission (PARA 1-3-12).
- m. Install main transmission drip pan (PARA 2-4-70).





6-4-7 MAIN ROTOR SHAFT INSPECT/REPAIR.

PARAGRAPH BREAKDOWN

|                                         | PAGE    |
|-----------------------------------------|---------|
| 6-4-7.1 Inspect/Repair Main Rotor Shaft | 6-4-7-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Powertrain Repairer’s Toolkit

Materials/Parts

- Abrasive Cloth, Item 1A, Appendix D
- Abrasive Cloth, Item 10, Appendix D
- Adhesive, Item 34, Appendix D
- Brush, Acid, Item 84, Appendix D
- Cheesecloth, Item 90, Appendix D
- Corrosion Compound, Item 108, Appendix D
- Corrosion Preventive Compound, Item 112, Appendix D

- Epoxy Coating Kit, Item 126A, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- PARA 1-6-18
- PARA 5-4-1
- PARA 5-4-30
- PARA 6-4-1
- PARA 6-4-5
- PARA 6-4-6
- PARA 12-4-26

6-4-7.1 Inspect/Repair Main Rotor Shaft.

6-4-7.1.1 Remove.

- a. If not already done, engage gust lock (PARA 1-6-18).
- b. If blade deice kit is installed, remove main rotor blade deice distributor (PARA 12-4-26).

- c. If blade deice kit is not installed, perform the following:
  - (1) Remove bifilar cover (PARA 5-4-30).
  - (2) Remove main rotor shaft plug (PARA 6-4-6).

- d. Remove upper main rotor shaft nut (PARA 5-4-1).

### NOTE

- If damage cannot be reached due to installation of main rotor shaft extension or swashplate guide, removal of main rotor head and swashplate guide may be required.
  - Main rotor head will be removed when swashplate guide is removed.
- e. If required, remove swashplate guide (PARA 6-4-5).

#### 6-4-7.1.2 Inspect/Repair.

### CAUTION

Damage to main transmission will result if foreign material enters main transmission. Do not allow foreign material to enter main transmission.

- a. Mask off bore of main transmission where main rotor shaft passes through.
- b. Inspect main rotor shaft surface for nicks, dents, scratches, and tool marks (Figure 6-4-7-1, Detail A). **NONE ALLOWED.** If damage is found, replace main module (PARA 6-4-1).
- c. Inspect main rotor shaft areas excluding lower split cone surface and upper tapered surface for corrosion. **NONE ALLOWED.** If corrosion is found, replace main module (PARA 6-4-1).
- d. Inspect main rotor shaft lower split cone surface and upper tapered surface for corrosion. **NONE ALLOWED.** If corrosion is found, repair as follows:
  - (1) Using acid brush, Item 84, Appendix D, apply corrosion compound, Item 108, Appendix D, to corroded areas. Allow to cure for 30 seconds. Using machinery towel, Item 211, Appendix D, wipe dry.

- (2) Using abrasive cloth, Item 10, Appendix D, blend any corrosion remaining after application of corrosion compound per step d. (1).

- (3) **REPAIR IN EXPOSED AREA OF UPPER TAPERED SURFACE MAY BE UP TO 100% OF AREA DEFINED IN FIGURE 6-4-7-1, DETAIL B, UP TO 0.010-INCH DEEP.** If repairs exceed limits, replace main module (PARA 6-4-1).

- (4) **ON REMAINING UPPER TAPERED SURFACE AND LOWER SPLIT CONE SURFACE, NO SINGLE REPAIRED AREA SHALL EXCEED 1-SQUARE-INCH. TOTAL REPAIRED AREA SHALL NOT EXCEED 20 % OF SURFACE AND SHALL NOT EXCEED 0.010-INCH DEEP.** If repairs exceed limits, replace main module (PARA 6-4-1).

- (5) Using corrosion preventive compound, Item 112, Appendix D, apply thin coat to reworked areas.

- e. Using methyl ethyl ketone, Item 217, Appendix D, clean adhesive on area below main rotor shaft splines.

- f. Inspect adhesive on area below splines of main rotor shaft for damage extending into base metal. **NONE ALLOWED.** If damage extends into base metal, replace main transmission (PARA 6-4-1).

- g. If damage does not extend into base metal, perform the following:

- (1) Using abrasive cloth, Item 1A, Appendix D, blend out epoxy damage.
- (2) Using cheesecloth, Item 90, Appendix D, clean repaired area.
- (3) Apply adhesive, Item 34, Appendix D, to reworked areas. Allow adhesive to dry for 24 hours at room temperature.
- (4) Using epoxy coating kit, Item 126A, Appendix D, apply one coat of epoxy coating.

**6-4-7.1.3 Install.**

- a. Remove masking from bore of main transmission.
- b. If required, install swashplate guide (PARA 6-4-5).
- c. Install upper main rotor shaft nut (PARA 5-4-1).
- d. If blade deice kit is installed, install main rotor blade deice distributor (PARA 12-4-26).
- e. If blade deice kit is not installed, perform the following:
  - (1) Install main rotor shaft plug (PARA 6-4-6).
  - (2) Install bifilar cover (PARA 5-4-30).
- f. If required, disengage gust lock (PARA 1-6-18).

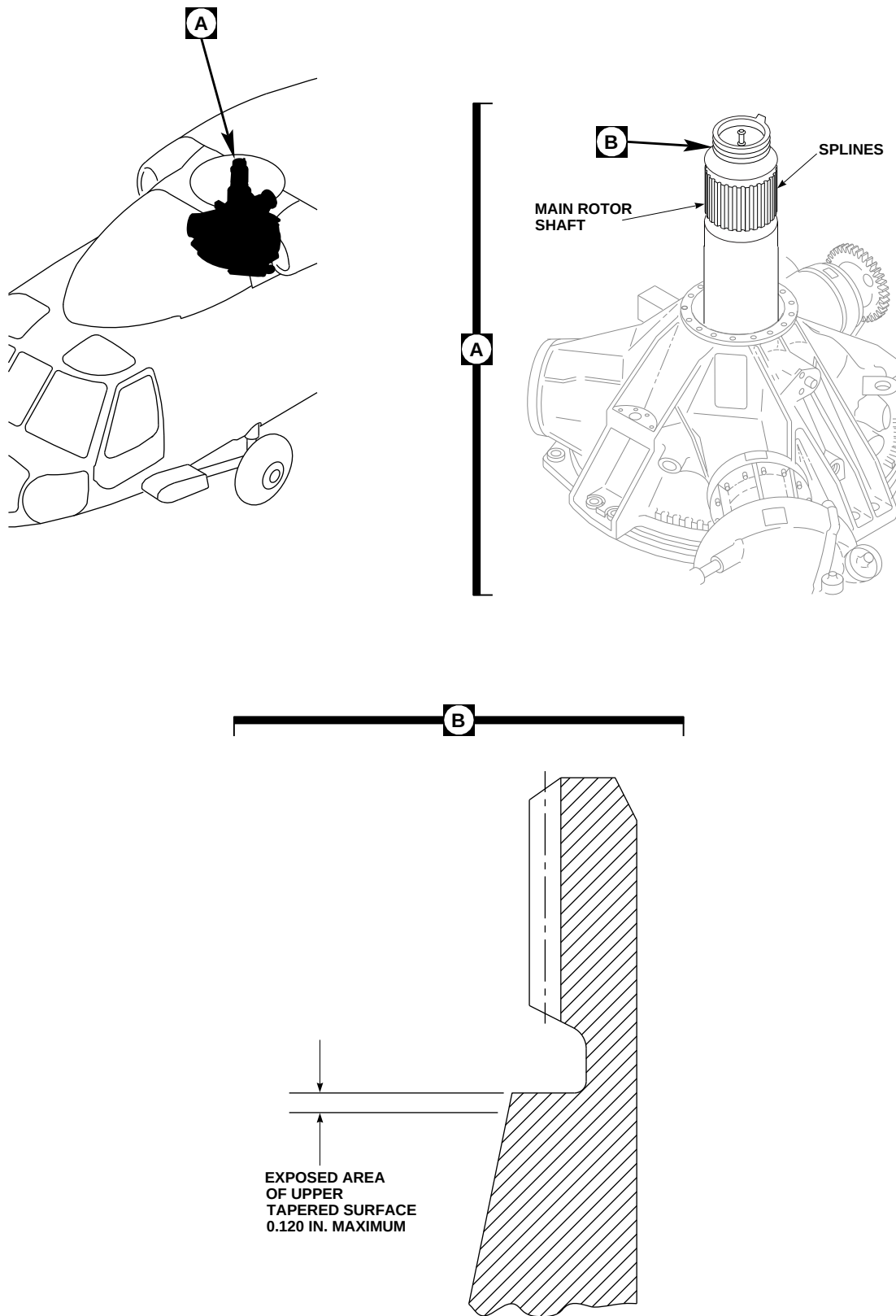


FIGURE 6-4-7-1. INSPECT/REPAIR MAIN ROTOR SHAFT

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6-4-8 GUST LOCK ROD AND LEVER.

PARAGRAPH BREAKDOWN

|         |                                 | PAGE    |
|---------|---------------------------------|---------|
| 6-4-8.1 | Replace Gust Lock Rod and Lever | 6-4-8-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drill
- Drill Bit, 0.072-inch diameter
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Vise, Soft-Jawed

Materials/Parts

- Adhesive, Item 37, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- PARA 1-6-4
- PARA 1-6-16
- PARA 2-4-70
- PARA 6-4-9

6-4-8.1 Replace Gust Lock Rod and Lever.

6-4-8.1.1 Remove.



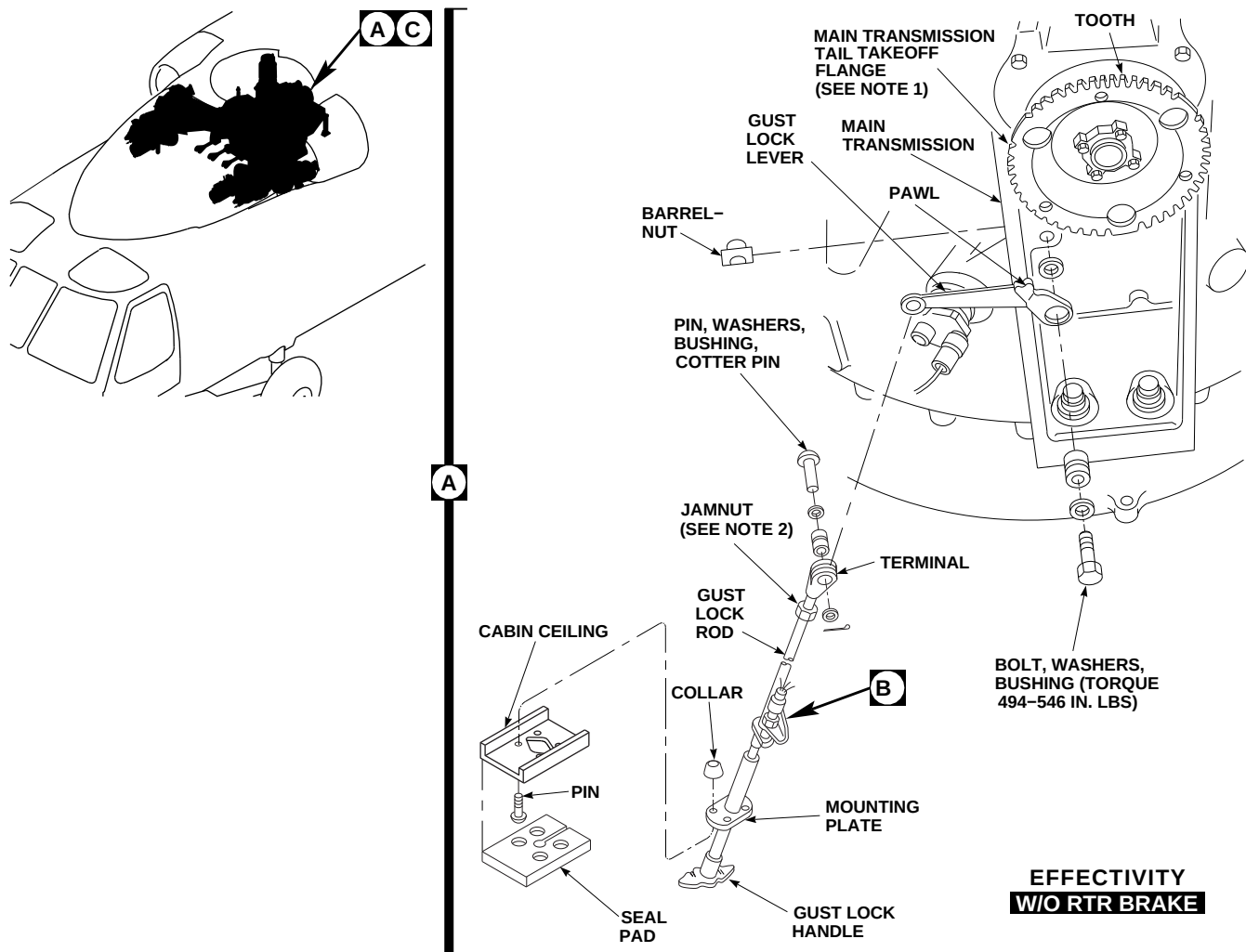
- a. Turn off all electrical power (PARA 1-6-16).



Injury to personnel and damage to equipment will result if gust lock rod and gust lock lever are removed without main and tail rotor blades being secure. Make sure main and tail rotor blades are secured before removing gust lock rod and gust lock lever.

Damage to main transmission and tail rotor drive shaft will result if main and tail rotor blades are not tied down prior to performing maintenance.

- b. Tie down main and tail rotor blades (PARA 1-6-4).
- c. Open oil cooler compartment access doors.
- d. Remove cotter pin, pin, washers, and bushing from terminal. Disconnect upper end of gust lock rod from gust lock lever (Figure 6-4-8-1, Sheet 1, Detail A or Sheet 2, Detail C).



## NOTES

1. TAIL ROTOR DRIVE SHAFT SECTION I REMOVED FOR CLARITY.
2. TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP INCREASE IN TORQUE IS FELT.

B

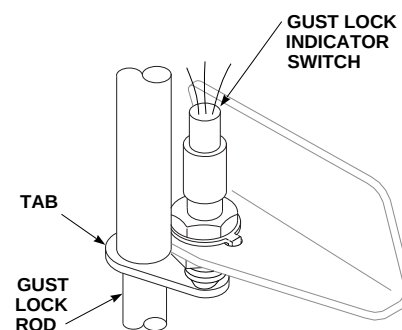
GUST LOCK ROD  
INSTALLED AT GUST LOCK  
INDICATOR SWITCHFN1958 1A  
SA

FIGURE 6-4-8-1. REPLACE GUST LOCK ROD AND LEVER (SHEET 1 OF 2)

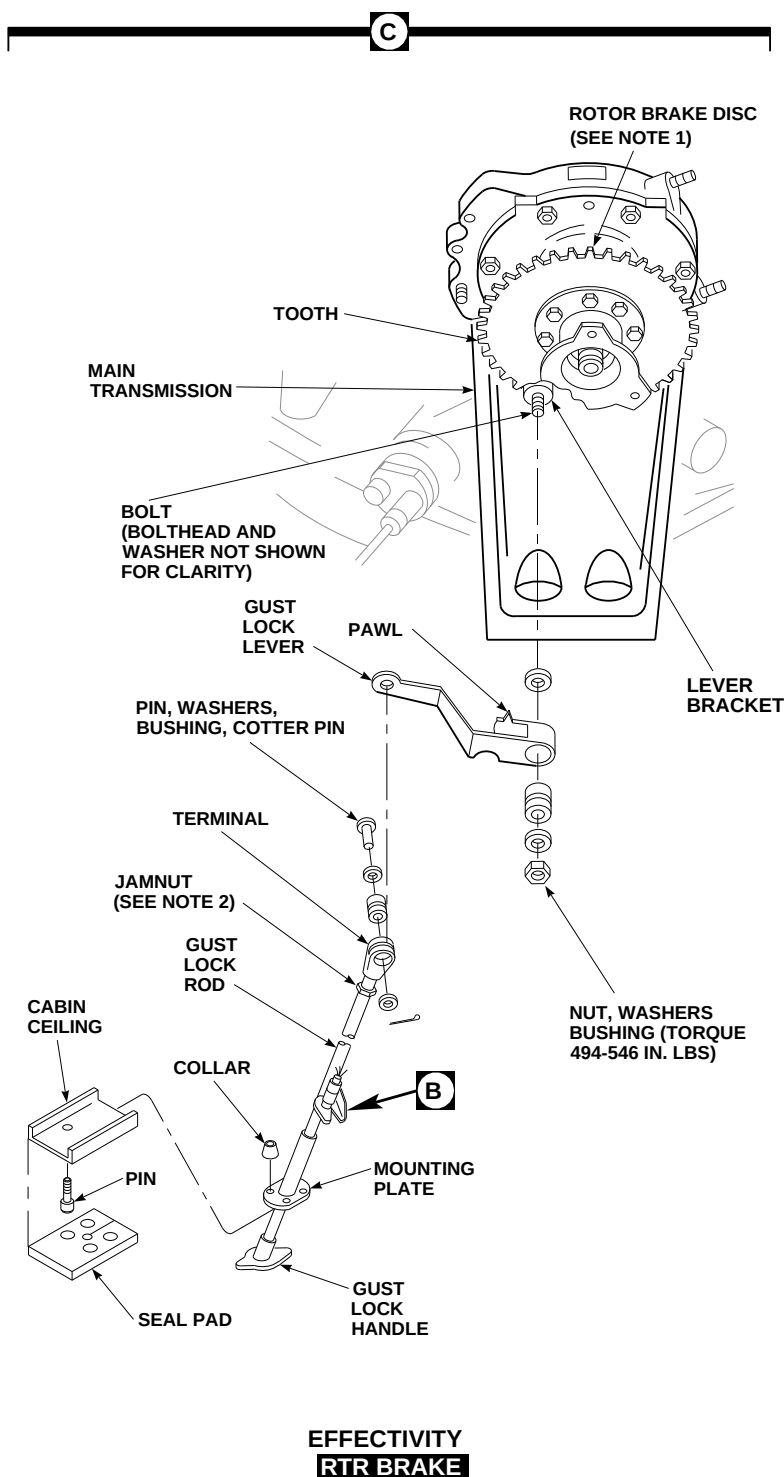
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FIGURE 6-4-8-1. REPLACE GUST LOCK ROD AND LEVER (SHEET 2 OF 2)

- e. Remove main transmission drip pan (PARA 2-4-70).
- f. Remove seal pad above gust lock handle.
- g. Using drill with 0.072-inch diameter drill bit, unfasten gust lock rod by drilling out pin and collar from cabin ceiling and mounting plate. Remove gust lock rod by pulling gust lock rod up and rotating it 90° so gust lock handle will pass through slot in cabin ceiling.
- h. Loosen jamnut and remove terminal from gust lock rod.
- i. Remove gust lock lever as follows:
  - (1) **W/O RTR BRAKE** Remove bolt, washers, and gust lock lever with bushing from main transmission. Remove bushing from gust lock lever (Figure 6-4-8-1, Sheet 1, Detail A).
  - (2) **RTR BRAKE** Remove nut, washers, gust lock lever with bushing, and bolt from lever bracket. Remove bushing from gust lock lever (Figure 6-4-8-1, Sheet 2, Detail C).

#### 6-4-8.1.2 Install.

- a. **W/O RTR BRAKE** Check breakaway torque of barrel-nut in main transmission securing gust lock lever as follows:

#### CAUTION

Damage to recess in airframe or beam will result if barrel-nut is incorrectly installed. To prevent damage to recess, make sure barrel-nut is installed with barrel (round side) toward bolthead.

#### NOTE

Make sure that only barrel-nuts, RMLH2577, are used. These barrel-nuts are identified with the letters "EN" or "SPS" stamped on an end (additional marking may also include "LH2577").

- (1) Remove barrel-nut from main transmission (Figure 6-4-8-1, Sheet 1, Detail A).
- (2) Place barrel-nut in soft-jawed vise.
- (3) Install bolt into barrel-nut until two threads extend beyond barrel-nut.
- (4) Using torque wrench, back out bolt.
- (5) **MINIMUM BREAKAWAY TORQUE IS 18 INCH-POUNDS.** Proceed as follows:

- (a) If breakaway torque is 18 inch-pounds or more, bolt and barrel-nut may be reused. Install barrel-nut in main transmission.
- (b) If breakaway torque is less than 18 inch-pounds, replace bolt and barrel-nut. Repeat steps a. (2) through a. (5).

- b. **W/O RTR BRAKE** Install gust lock lever as follows:

- (1) Install barrel-nut in main transmission.
- (2) Install bushing in gust lock lever and position gust lock lever and washer against main transmission.
- (3) Install bolt and washer.
- (4) **TORQUE BOLT TO 494 - 546 INCH-POUNDS.**

- c. **RTR BRAKE** Install gust lock lever as follows (Figure 6-4-8-1, Sheet 2, Detail C):

- (1) Install bolt and washer in lever bracket.
- (2) Position bushing in gust lock lever and position gust lock lever and washer on bolt against lever bracket.
- (3) Install washer and nut.
- (4) **TORQUE NUT TO 494 - 546 INCH-POUNDS.**



- d. Install jamnut and terminal on upper end of gust lock rod.
- e. **TIGHTEN JAMNUT HANDTIGHT.**
- f. Install gust lock rod as follows:
  - (1) Lower gust lock handle through slot and turn 90° so mounting plate holes line up with holes in cabin ceiling, and tab on gust lock rod lines up with gust lock indicator switch (Figure 6-4-8-1, Sheet 1, Detail B).
  - (2) Secure lower end of gust lock rod in cabin ceiling with pins and collars (Figure 6-4-8-1, Sheet 1, Detail A or Sheet 2, Detail C).
- g. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean seal pad mounting surface.
- h. Using adhesive, Item 37, Appendix D, bond seal pad on cabin ceiling above gust lock handle.
- i. **W/O RTR BRAKE** Adjust gust lock lever as follows (Figure 6-4-8-1, Sheet 1, Detail A):
  - (1) Mesh tooth on main transmission tail takeoff flange with pawl of gust lock lever.
  - (2) Loosen jamnut and adjust terminal so that when gust lock handle is engaged (in), holes in terminal and gust lock lever are lined up.
  - (3) Install pin, washer, and bushing through terminal and gust lock lever.
- (4) Make sure gust lock lever is fully engaged.
- j. **RTR BRAKE** Adjust gust lock lever as follows (Figure 6-4-8-1, Sheet 2, Detail C):
  - (1) Mesh tooth on rotor brake disc with pawl of gust lock lever.
  - (2) Loosen jamnut and adjust terminal so that when gust lock handle is engaged (in), holes in terminal and gust lock lever are lined up.
  - (3) Install pin, washers, and bushing through gust lock lever and terminal.
  - (4) Make sure gust lock lever is fully engaged.
- k. **TIGHTEN TERMINAL JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT** (Figure 6-4-8-1, Sheet 1, Detail A or Sheet 2, Detail C).
- l. Using lockwire, Item 197, Appendix D, lockwire jamnut.
- m. Secure pin attaching terminal to gust lock lever with washer and cotter pin.
- n. Adjust gust lock indicator switch (PARA 6-4-9).
- o. Install main transmission drip pan (PARA 2-4-70).
- p. Remove tiedowns securing main and tail rotor blades.



6-4-9 GUST LOCK INDICATOR SWITCH.

PARAGRAPH BREAKDOWN

|         |                                    | PAGE    |
|---------|------------------------------------|---------|
| 6-4-9.1 | Replace Gust Lock Indicator Switch | 6-4-9-1 |

INITIAL SETUP

Tools

- Electrical Repairer’s Toolkit

Materials/Parts

- Lockwire, Item 196, Appendix D
- Tags

References

- Appendix D
- Appendix F
- PARA 1-6-4
- PARA 1-6-16

6-4-9.1 Replace Gust Lock Indicator Switch.

6-4-9.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).

CAUTION

Damage to main transmission and tail rotor drive shaft will result if main and tail rotor blades are not tied down prior to performing maintenance.

- Tie down main and tail rotor blades (PARA 1-6-4).
- Open oil cooler compartment access doors.

CAUTION

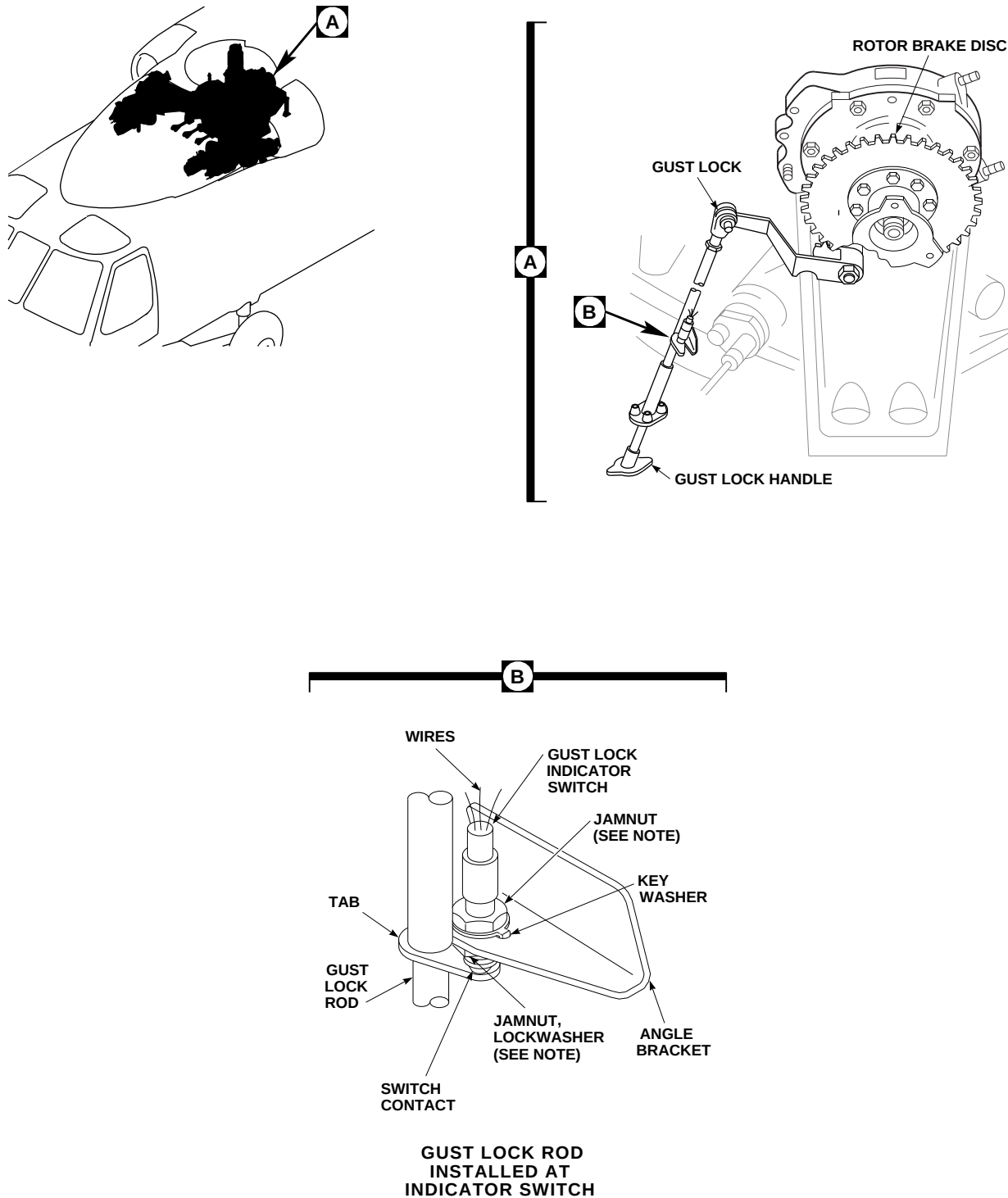
Damage to gust lock will result if operated improperly. To prevent accidental gust lock operation and damage, gust

lock handle button must be out when gust lock is fully engaged or disengaged.

- Fully disengage gust lock (Figure 6-4-9-1, Detail A).
- Measure and record the distance between tab and switch contact (Figure 6-4-9-1, Detail B).
- Tag wires and disconnect wire splice connections between gust lock indicator switch and harness assembly.
- Remove jamnuts, lockwasher, key washer, and gust lock indicator switch from angle bracket.

6-4-9.1.2 Install.

- Assemble gust lock indicator switch with upper jamnut and key washer in position on angle bracket (Figure 6-4-9-1, Detail B).
- Install lockwasher and lower jamnut on indicator switch assembly, and position switch contact with tab to recorded measurement.

**NOTE**

TIGHTEN JAMNUT 1/6 TO 1/3 TURN  
PAST POINT WHERE RISE IN TORQUE IS  
FELT.

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**FIGURE 6-4-9-1. REPLACE GUST LOCK INDICATOR SWITCH**

- c. **TIGHTEN JAMNUTS  $\frac{1}{6}$  TO  $\frac{1}{3}$  TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- d. Connect gust lock switch wiring to harness assembly using wire splice connections (Appendix F). Remove tags.
- e. Turn on all electrical power (PARA 1-6-16).
- f. Actuate gust lock in both engaged and disengaged positions, checking GUST LOCK indicator light on caution/advisory panel for proper operation.
- g. If required, adjust gust lock switch (PARA 6-4-9.1.3).
- h. Lockwire, Item 196, Appendix D, jamnuts.
- i. Turn off all electrical power (PARA 1-6-16).
- j. Make sure area is clean and free of foreign material.
- k. Close oil cooler compartment access doors.
- l. Remove tiedowns securing main and tail rotor blades.

#### 6-4-9.1.3 Adjust.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Open oil cooler compartment access doors.



Damage to main transmission and tail rotor drive shaft will result if main and tail rotor blades are not tied down prior to performing maintenance.

- c. Tie down main and tail rotor blades (PARA 1-6-4).
- d. Turn on all electrical power (PARA 1-6-16).



Damage to gust lock will result if operated improperly. To prevent accidental gust lock operation and damage, gust lock handle button must be out when gust lock is fully engaged or disengaged.

- e. Engage gust lock (Figure 6-4-9-1, Detail A).
- f. Loosen jamnuts and position gust lock indicator switch up or down on angle bracket until GUST LOCK indicator light on caution/advisory panel lights up when tab actuates switch contact (Figure 6-4-9-1, Detail B).
- g. **TIGHTEN GUST LOCK INDICATOR SWITCH IN POSITION WITH JAMNUTS. TIGHTEN JAMNUTS  $\frac{1}{6}$  TO  $\frac{1}{3}$  TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- h. Actuate gust lock in both engaged and disengaged positions, checking GUST LOCK indicator light on caution/advisory panel for proper operation.
- i. If necessary, realign gust lock indicator switch.
- j. Lockwire, Item 196, Appendix D, jamnuts.
- k. Turn off all electrical power (PARA 1-6-16).
- l. Make sure area is clean and free of foreign material.
- m. Close oil cooler compartment access doors.
- n. Remove tiedowns securing main and tail rotor blades.



6-4-10 LEVER BRACKET **RTR BRAKE** .

PARAGRAPH BREAKDOWN

|                                | PAGE     |
|--------------------------------|----------|
| 6-4-10.1 Replace Lever Bracket | 6-4-10-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Vise, Soft-Jawed

References

- PARA 1-6-4
- PARA 1-6-16

- PARA 6-4-2
- PARA 6-4-8
- PARA 6-4-38
- PARA 6-4-70

6-4-10.1 Replace Lever Bracket.

6-4-10.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).



Damage to main transmission and tail rotor drive shaft will result if main and tail rotor blades are not tied down prior to performing maintenance.

- Tie down main and tail rotor blades (PARA 1-6-4).
- Remove tail rotor drive shaft section I (PARA 6-4-38).
- Remove rotor brake assembly (PARA 6-4-70).

- Remove main transmission tail takeoff flange (PARA 6-4-2).
- Remove gust lock lever (PARA 6-4-8).
- Remove bolt and washer securing lever bracket and shim to main transmission. Remove lever bracket and shim. Do not discard shim (Figure 6-4-10-1, Detail A).
- Remove bolt and washer from lever bracket.
- Remove and save barrel-nut from main transmission.

6-4-10.1.2. Install.

- Check breakaway torque of barrel-nut in main transmission that secures gust lock lever as follows:

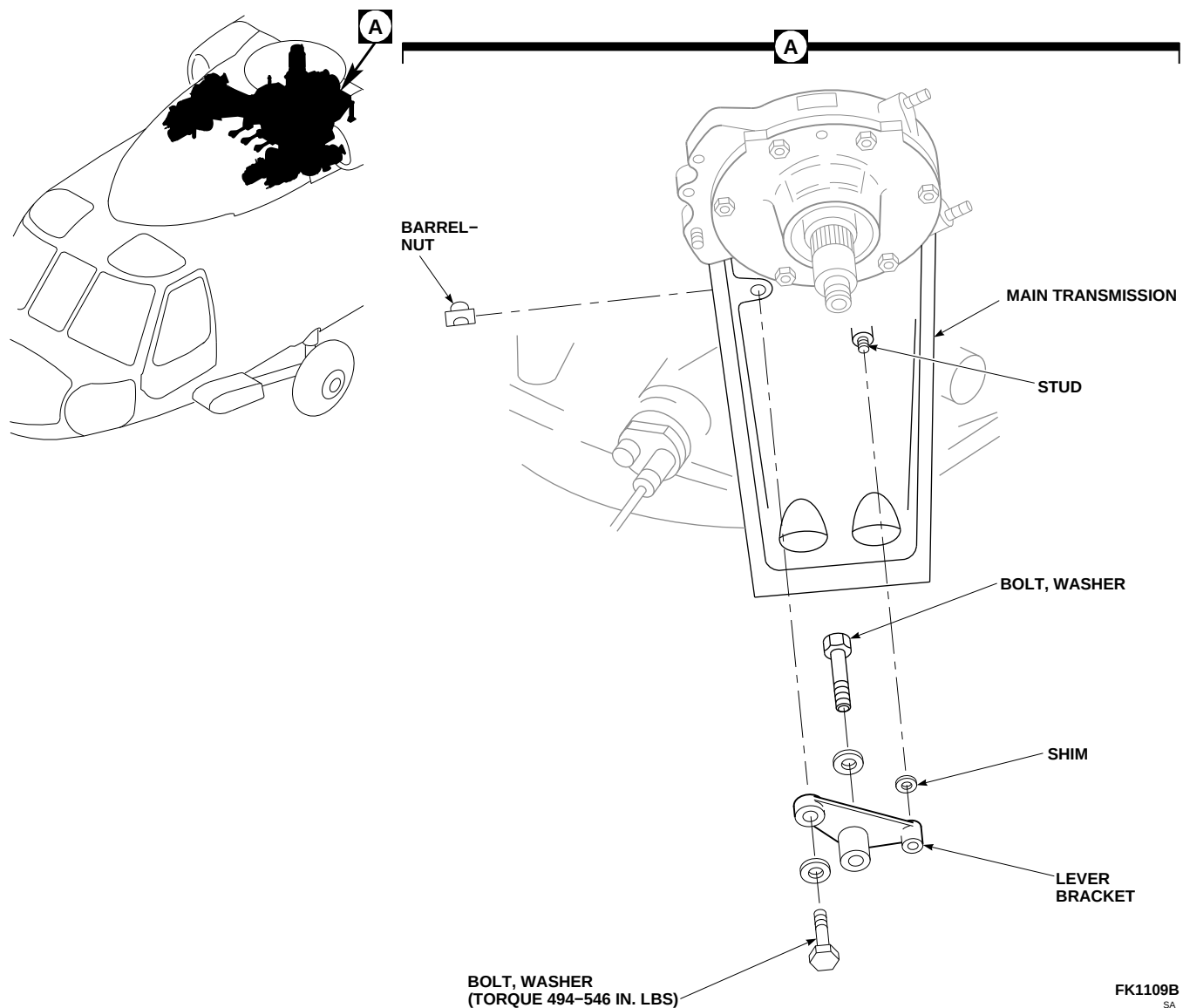


FIGURE 6-4-10-1. REPLACE LEVER BRACKET

**CAUTION**

Damage to recess in airframe or beam will result if barrel-nut is incorrectly installed. To prevent damage to recess, make sure barrel-nut is installed with barrel (round side) toward bolthead.

**NOTE**

Make sure that only barrel-nuts, RMLH2577, are used. These barrel-nuts are identified with the letters "EN" or

"SPS" stamped on an end (additional marking may also include "LH2577").

- (1) Place barrel-nut in soft-jawed vise.
- (2) Install bolt into barrel-nut until two threads extend beyond barrel-nut.
- (3) Using torque wrench, back out bolt.
- (4) **MINIMUM BREAKAWAY TORQUE IS 18 INCH-POUNDS.** Proceed as follows:
  - (a) If breakaway torque is 18 inch-pounds or more, bolt and barrel-nut



- may be reused. Install barrel-nut in main transmission.
- (b) If breakaway torque is less than 18 inch-pounds, replace bolt and barrel-nut. Repeat steps a. (2) through a. (5).
- b. Install bolt and washer in center hole of lever bracket with threads pointing toward rear of helicopter (Figure 6-4-10-1, Detail A).
- c. Install barrel-nut in main transmission.
- d. Position shim and lever bracket on stud of main transmission.
- e. Install bolt and washer.
- f. **TORQUE BOLT TO 494 - 546 INCH-POUNDS.**
- g. Install gust lock lever (PARA 6-4-8).
- h. Install main transmission tail takeoff flange (PARA 6-4-2).
- i. Install rotor brake assembly (PARA 6-4-70).
- j. Install tail rotor drive shaft section I (PARA 6-4-38).
- k. Remove tiedowns securing main and tail rotor blades.



6-4-11 MAIN MODULE HOUSING REPAIR.

PARAGRAPH BREAKDOWN

|                                     | PAGE     |
|-------------------------------------|----------|
| 6-4-11.1 Repair Main Module Housing | 6-4-11-2 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Beaker (10 ml)
- Beaker (200 ml)
- Beaker (600 ml)
- Container, Plastic, 1-Pint
- Container, Plastic, 2-Gallon
- Nonmetallic Scraper, Locally-Made
- Powertrain Repairer’s Toolkit
- Protection, Eye
- Vessel(s), with equivalent measuring accuracy of Beakers (Alternate for Beakers)

Materials/Parts

- Abrasive Cloth, Item 6, Appendix D
- Abrasive Cloth, Item 10, Appendix D
- Acetic Acid, Item 24, Appendix D

- Applicator, Item 78, Appendix D
- Chromium Trioxide, Technical, Item 91, Appendix D
- Corrosion Preventive Compound, Item 113, Appendix D
- Distilled Water, Item 123, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Coating Kit, Item 126A, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Ethylene Glycol, Item 137, Appendix D
- Machinery Towel, Item 211, Appendix D
- Nitric Acid, Item 220, Appendix D
- Paint, Polyurethane, Item 224, Appendix D
- Paint Remover, Item 231, Appendix D
- Pressure Sensitive Adhesive Tape, Item 250, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D

**6-4-11.1 Repair Main Module Housing.**

126A, Appendix D, or polyurethane paint, Item 224, Appendix D.

**6-4-11.1.1 General.****NOTE**

- Minor nicks, gouges, and scratches on surface of main module housing may be repaired. Before starting repair, measure depth of damage to determine whether blending out damage will result in main module still being within repair limits. Make sure to measure depth of damage from an area which has not been previously repaired. If limits will be exceeded, replace main module.
- Minor nicks, gouges, scratches, and corrosion within snapping groove of jet or breather plug may be repaired. Before starting repair, measure depth of damage to determine whether blending out damage will result in main module still being within repair limits. Make sure to measure depth of damage from an area which has not been previously repaired. If limits will be exceeded, replace main module.
- Repair of damaged area may be done with main module installed.

- All repairs shall be recorded in helicopter component historical records.

- a. Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean any oil or grease from damaged area.

**NOTE**

Housings coated with epoxy coating are glossy silver. Housings coated with polyurethane paint are matte surface medium-gray.

- b. Fine scratches that remove only paint may be blended smooth using abrasive cloth, Item 10, Appendix D. After blending, touch up blended area as required using epoxy coating kit, Item

**6-4-11.1.2 Repair Limits.**

- a. Main module housing repair limits are as follows:

- (1) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM ALL OVER OUTER CAST SURFACES AND ADJACENT RIBS OF MAIN MODULE IS 0.030-INCH** (Figure 6-4-11-1, Sheet 1).

- (2) **MAXIMUM DEPTH OF MATERIAL REMOVAL ON MOUNTING FEET AND MOUNTING FEET RIBS IS 0.030-INCH** (Figure 6-4-11-1, Sheet 1, Detail A).

- (a) **NO MORE THAN TWO REPAIRS ALLOWED PER MOUNTING FOOT.**

- (b) **REPAIRS SHALL NOT EXCEED 10% OF TOTAL SURFACE AREA OF MOUNTING FOOT.**

- (c) **NO MATERIAL REMOVAL FROM BUSHING BORE ALLOWED.**

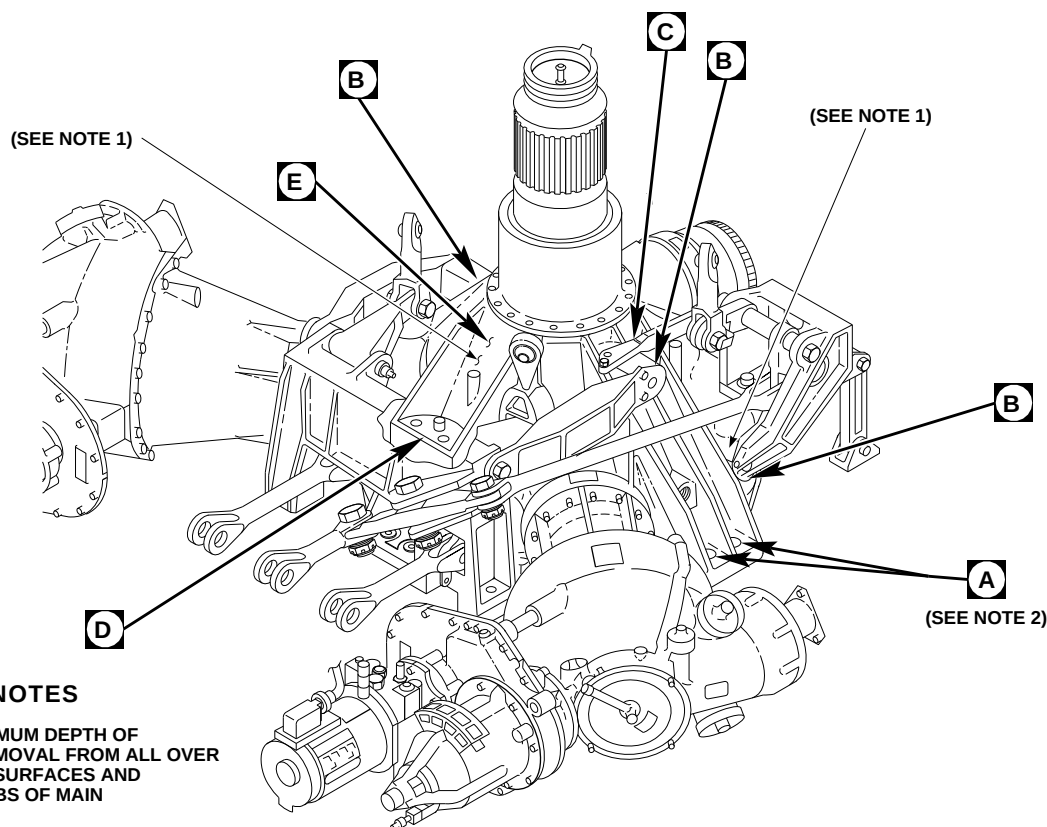
- (d) **SPOT-FACE CORNER MAY BE BLENDED TO ¼-INCH RADIUS AS LONG AS REPAIR DOES NOT EXTEND MORE THAN 0.030-INCH INTO FLAT AREA ADJACENT TO BUSHING OR RIB.**

- (e) **SHARP CORNERS MAY BE BLENDED TO A 0.050-INCH RADIUS.**

- (3) **MAXIMUM DEPTH OF MATERIAL REMOVAL ON FLAT PORTIONS OF BRIDGE PADS IS 0.040-INCH** (Figure 6-4-11-1, Sheet 2, Detail B).

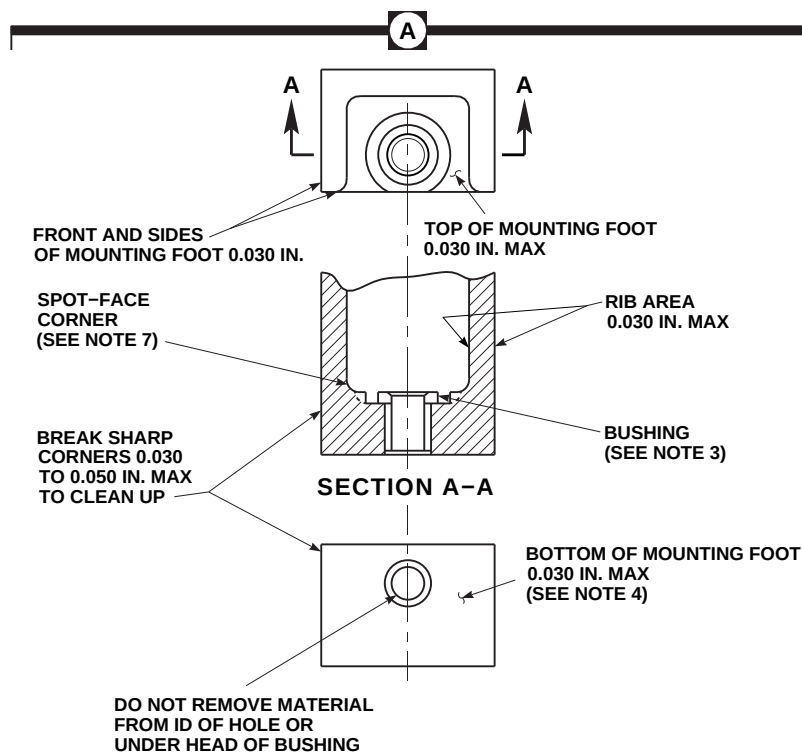
- (a) **NO MORE THAN ONE REPAIR PER ½-INCH-SQUARE AREA WITH A MAXIMUM REPAIR OF 25% OF PAD.**

**6-4-11-2 Change 11**



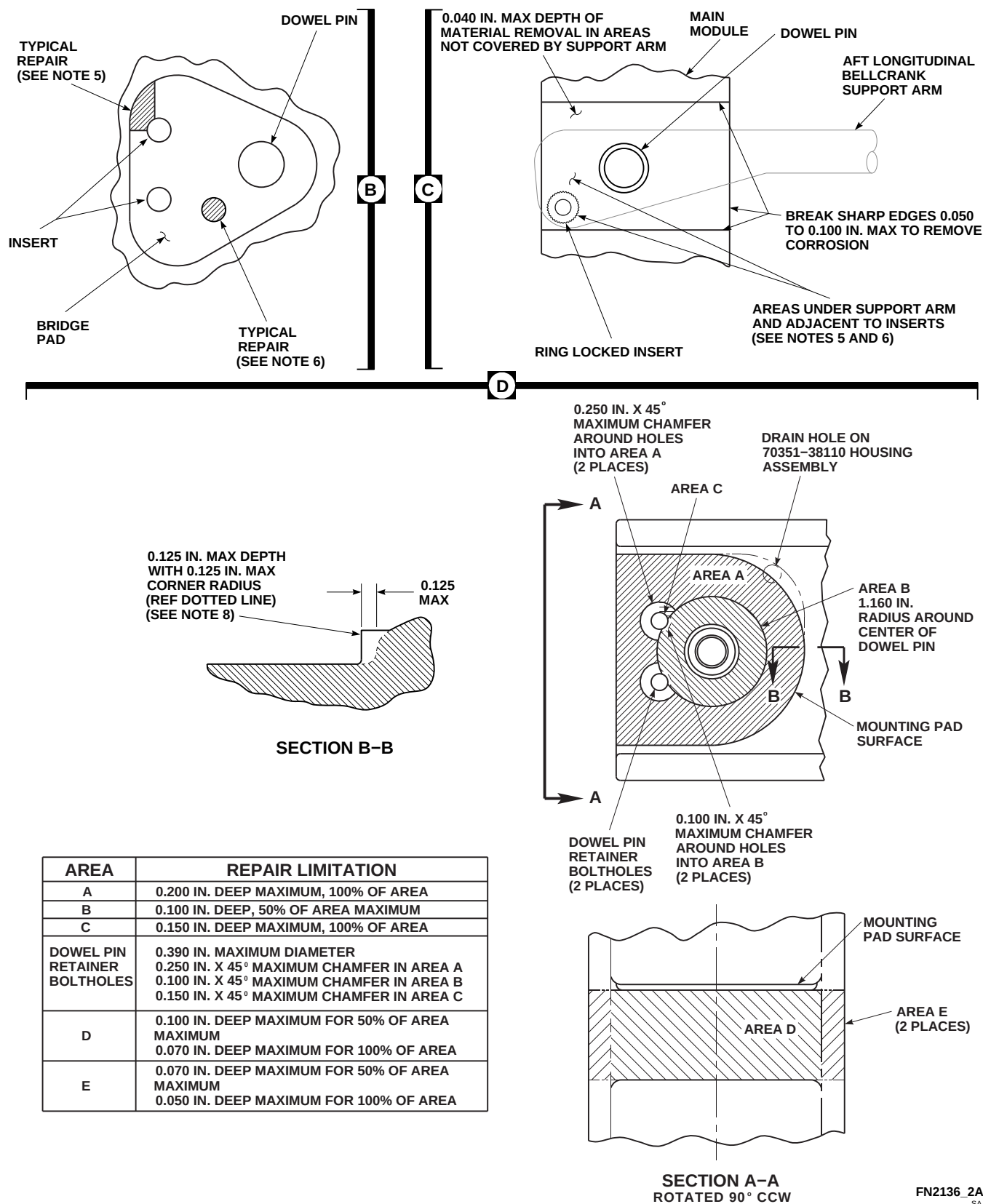
## NOTES

1. 0.030 IN. MAXIMUM DEPTH OF MATERIAL REMOVAL FROM ALL OVER OUTER CAST SURFACES AND ADJACENT RIBS OF MAIN MODULE.
2. DO NOT REMOVE MATERIAL FROM MOUNTING BOLTHOLE BUSHING. 0.030 IN. MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SURFACE AROUND MOUNTING BOLTS, NO MORE THAN 2 REPAIRS SHALL BE MADE PER MOUNTING FOOT.
3. REMOVE BUSHING TO REPAIR CORROSION AT DEPOT LEVEL ONLY.
4. DO NOT REMOVE MORE THAN 10% OF TOTAL SURFACE AREA OF MOUNTING FOOT.
5. 0.015 IN. MAXIMUM DEPTH OF MATERIAL REMOVAL NEAR PAD INSERTS UP TO 90° SEGMENT.
6. 0.040 IN. MAXIMUM DEPTH OF MATERIAL REMOVAL ON FLAT PORTION OF BRIDGE PADS. ONLY ONE REPAIR PER 1/2-INCH-SQUARE AREA WITH A MAXIMUM REPAIR OF 25% OF PAD.
7. TO REMOVE CORROSION, SPOT-FACE CORNERS MAY BE BLENDED 360° WITH A FILLET RADIUS UP TO 0.250 IN. (SHOWN AS DOTTED LINE). RADIUS MAY NOT EXTEND MORE THAN 0.030 IN. INTO TOP OF FLAT AREA ADJACENT TO BUSHING OF RIB.
8. MAINTAIN A 0.125 IN. CORNER RADIUS FOR BLENDING.
9. ZONES 1 AND 2 ARE THE GUST LOCK BOLTHOLE.
10. ZONES 3, 4, AND 5 ARE THE BARREL-NUT HOLE.
11. ZONE 6 IS THE GUST LOCK BOLTHOLE CONICAL SURFACE.
12. ZONE 7 IS THE BARREL-NUT HOLE CONICAL SURFACE.



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**FIGURE 6-4-11-1. REPAIR MAIN MODULE HOUSING (SHEET 1 OF 4)**



**FIGURE 6-4-11-1. REPAIR MAIN MODULE HOUSING (SHEET 2 OF 4)**

**6-4-11-4      Change 11**

This Document contains technical data subject to ITAR. See WARNING and classifications on first page.

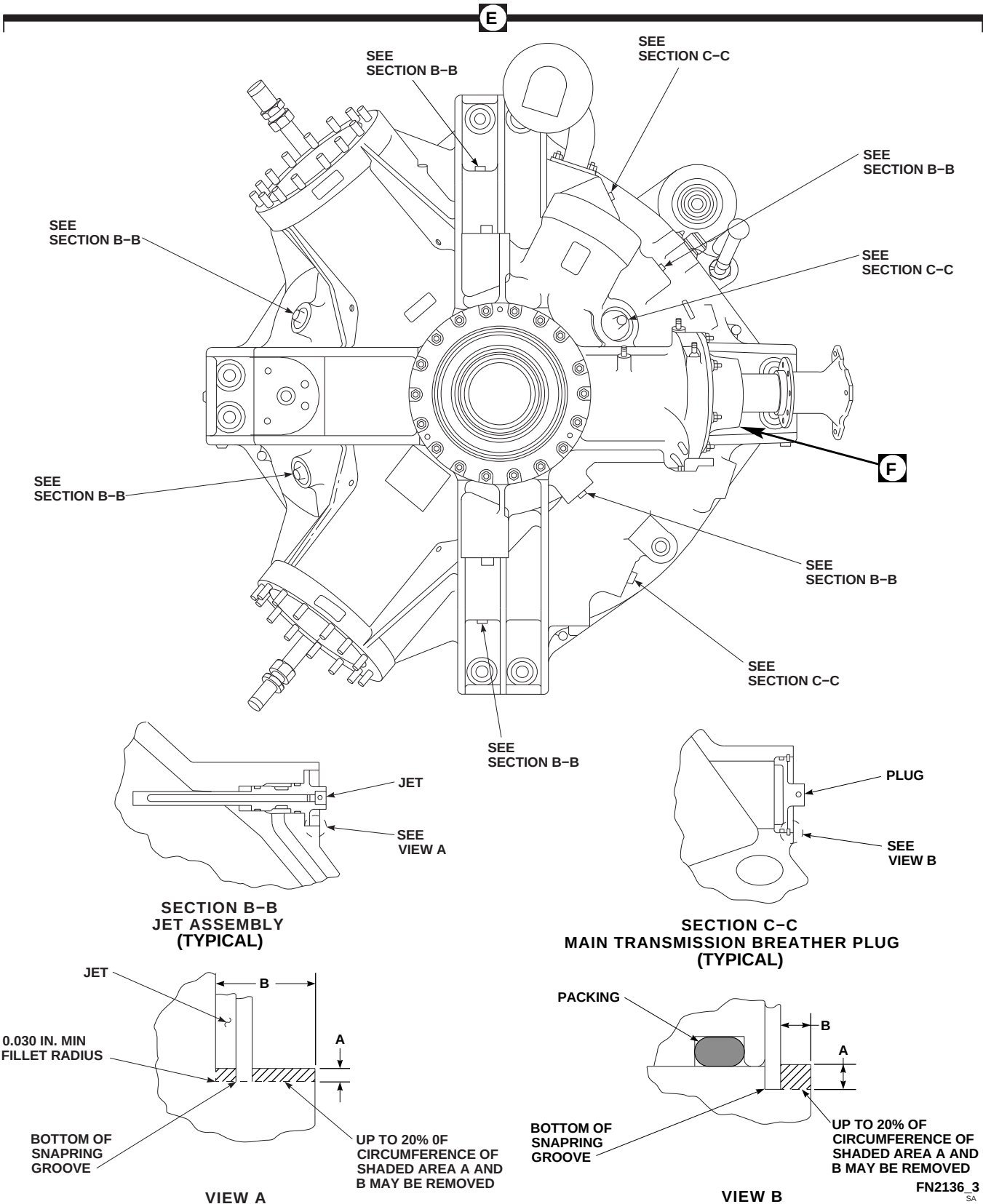
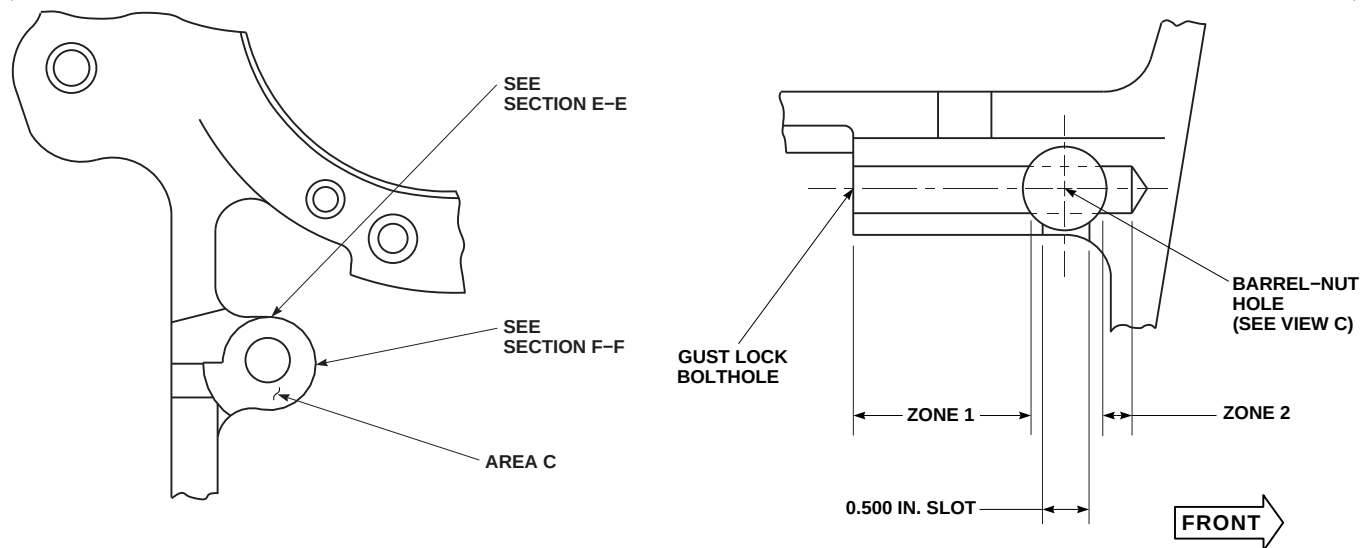
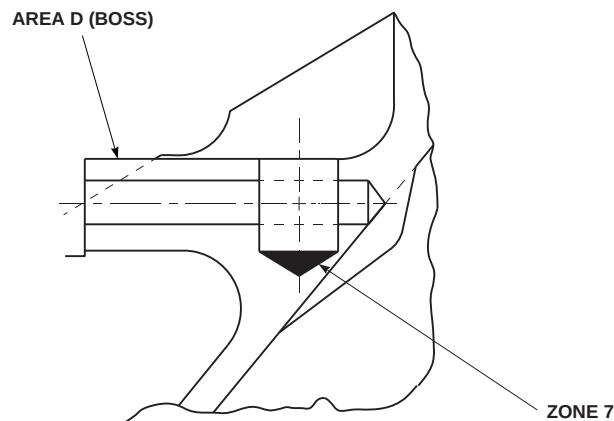


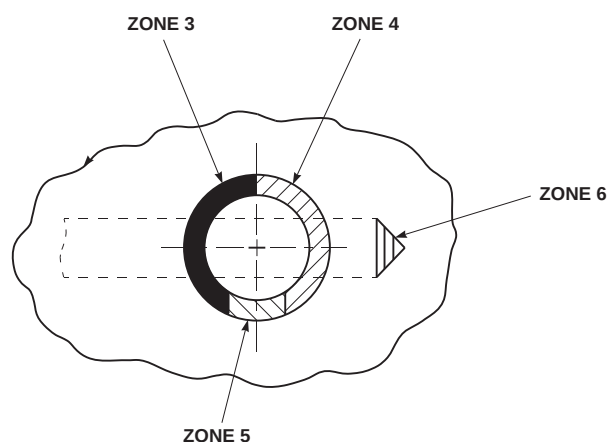
FIGURE 6-4-11-1. REPAIR MAIN MODULE HOUSING (SHEET 3 OF 4)

**F**

**SECTION E-E  
GUST LOCK BOLTHOLE AND  
BARREL-NUT HOLE HOUSING,  
TOP VIEW**



**SECTION F-F  
GUST LOCK BOLTHOLE AND  
BARREL-NUT HOLE HOUSING,  
SIDE VIEW**



**VIEW C**

| AREA | REPAIR LIMITATIONS     |                             |                           |
|------|------------------------|-----------------------------|---------------------------|
| C    | 0.040 IN. DEEP MAXIMUM |                             |                           |
| D    | ZONE / LOCATION        | MAX. DEPTH MATERIAL REMOVAL | MAX. PCT OF AREA REPAIRED |
|      | 1 (SEE NOTE 9)         | 0.015 IN.                   | 25%                       |
|      | 2 (SEE NOTE 9)         | 0.100 IN.                   | 100%                      |
|      | 3 (SEE NOTE 10)        | 0.030 IN.                   | 25%                       |
|      | 4 (SEE NOTE 10)        | 0.060 IN.                   | 100%                      |
|      | 5 (SEE NOTE 10)        | 0.100 IN.                   | 100%                      |
|      | 6 (SEE NOTE 11)        | 0.030 IN.                   | 100%                      |
|      | 7 (SEE NOTE 12)        | 0.060 IN.                   | 100%                      |
|      | 0.500 SLOT, FWD        | 0.080 IN.                   | 100%                      |
|      | 0.500 SLOT, AFT        | 0.050 IN.                   | 100%                      |

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**FIGURE 6-4-11-1. REPAIR MAIN MODULE HOUSING (SHEET 4 OF 4)**



- (b) **MAXIMUM DEPTH OF MATERIAL REMOVAL NEAR PAD INSERTS (UP TO A 90° SEGMENT) IS 0.015-INCH.**
- (4) **FORWARD BELLCRANK SUPPORT MOUNTING PAD AREA MAY BE BLENDED TO LIMITS SHOWN IN FIGURE 6-4-11-1, SHEET 2, DETAIL D.**
- (5) **SPOT-FACE CORNER OF MOUNTING PAD MAY BE BLENDED TO LIMITS SHOWN IN FIGURE 6-4-11-1, SHEET 2, SECTION B-B.**
- (6) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM AFT LONGITUDINAL BELLCRANK SUPPORT ARM MOUNTING PAD AREA NOT COVERED BY SUPPORT ARM IS 0.040-INCH** (Figure 6-4-11-1, Sheet 2, Detail C).
  - (a) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM AREA UNDER SUPPORT ARM IS 0.040-INCH. NO MORE THAN ONE REPAIR PER ½-INCH-SQUARE AREA WITH A MAXIMUM REPAIR OF 25% OF PAD AREA.**
  - (b) **MAXIMUM DEPTH OF MATERIAL REMOVAL NEAR RING LOCKED INSERT (UP TO A 90° SEGMENT) IS 0.015-INCH.**
  - (c) **SHARP EDGES MAY BE BLENDED TO A 0.100-INCH RADIUS.**
- (7) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM JET / BREATHER PLUG ASSEMBLIES IS 20% OF CIRCUMFERENCE OF SHADED AREAS "A" AND "B"** (Figure 6-4-11-1, Sheet 3, Detail E).
- (8) **REFER TO FIGURE 6-4-11-1, SHEET 4, DETAIL F, FOR DAMAGE LIMITS**

**TO GUST LOCK BOLTHOLE AND BARREL-NUT HOLE.**

**6-4-11.1.3 Repair Procedure.**

- a. Using paint remover, Item 231, Appendix D, remove paint from damaged area.
- b. Using nonmetallic scraper, remove sealing compound from damaged area.
- c. Using abrasive cloth, Item 6, Appendix D, carefully blend damaged area smooth. Remove minimum amount of material required to blend out damaged area.
- d. Using abrasive cloth, Item 10, Appendix D, blend reworked area to remove all traces of abrasive cloth.
- e. Remove displaced metal caused by abrasive blending by using magnesium-etching procedure as follows:

**CAUTION**

- Magnesium-etchant measurements are to be as accurate as possible to values measured in a milliliter beaker or vessel with equivalent mixing accuracy.
- Magnesium-etchant formulas shall be prepared and stored in unbreakable 1-pint plastic containers and safely stored in designated areas not subjected to extreme heat or cold.
- Before preparing magnesium-etchant, make sure storage area does not contain premixed magnesium-etchant formula. The correct stock bottle for this repair shall be marked MAGNESIUM-ETCHANT.
- (1) Prepare magnesium-etchant solution in clean unbreakable plastic container by mixing the following:

| MATERIAL                              | QTY             |
|---------------------------------------|-----------------|
| Ethylene Glycol, Item 137, Appendix D | 600 milliliters |
| Acetic Acid, Item 24, Appendix D      | 200 milliliters |
| Nitric Acid, Item 220, Appendix D     | 10 milliliters  |
| Distilled Water, Item 123, Appendix D | 190 milliliters |

- (2) Using dry-cleaning solvent, Item 124, Appendix D, clean area to be etched.
- (3) Using pressure sensitive adhesive tape, Item 250, Appendix D, mask area.
- (4) Pour a small amount of magnesium-etchant from stock bottle into 1-pint plastic container.

### CAUTION

Damage to components will result if magnesium-etchant is allowed to come in contact with areas not being etched. Extreme caution must be used to protect those areas not being etched.

- (5) Using applicator, Item 78, Appendix D, apply magnesium-etchant to reworked area. Swab area for about 10 - 15 seconds.
- (6) Neutralize magnesium-etchant immediately by flooding area with generous amount of clean water.

### WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel

debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- (7) Using clean, dry, compressed air, blow dry area thoroughly.
  - (8) After use, do not return unused magnesium-etchant to stock bottle. Dispose of used magnesium-etchant per local instructions.
  - (9) Identify contents of stock bottle as MAGNESIUM-ETCHANT, before placing bottle in storage area.
  - (10) Prepare technical chromium trioxide solution in clean 2-gallon plastic container by mixing 24 ounces of technical chromium trioxide, Item 91, Appendix D, with 1 gallon of water.
  - (11) Using technical chromium trioxide solution, treat bare surface of housing.
- f. Apply sealing compound, Item 292, Appendix D, to touch up reworked areas.
  - g. After sealing compound has dried, using epoxy primer coating kit, Item 135, Appendix D, apply one coat of epoxy primer.

### NOTE

Housings coated with epoxy coating are glossy silver. Housings coated with polyurethane paint are matte surface medium-gray.

- h. Using epoxy coating kit, Item 126A, Appendix D, or polyurethane paint, Item 224, Appendix D, apply one coat of epoxy primer, depending on original paint finish.
- i. Using corrosion preventive compound, Item 113, Appendix D, apply one coat.
- j. Allow corrosion preventive compound to cure for 30 minutes.

- k. Using corrosion preventive compound, Item 113, Appendix D, apply a second coat.
- l. Make sure area is clean and free of foreign material.
- m. Record brief description of main module repairs in helicopter component historical records.



6-4-12 INPUT MODULE HOUSING REPAIR.

PARAGRAPH BREAKDOWN

|                                      | PAGE     |
|--------------------------------------|----------|
| 6-4-12.1 Repair Input Module Housing | 6-4-12-2 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Beaker (10 ml)
- Beaker (200 ml)
- Beaker (600 ml)
- Container, Plastic, 1-Pint
- Container, Plastic, 2-Gallon
- Nonmetallic Scraper, Locally-Made
- Powertrain Repairer’s Toolkit
- Protection, Eye
- Vessel(s), with equivalent measuring accuracy of Beakers (Alternate for Beakers)

Materials/Parts

- Abrasive Cloth, Item 6, Appendix D
- Abrasive Cloth, Item 10, Appendix D
- Acetic Acid, Item 24, Appendix D

- Applicator, Item 78, Appendix D
- Chromium Trioxide, Technical, Item 91, Appendix D
- Corrosion Preventive Compound, Item 113, Appendix D
- Distilled Water, Item 123, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Coating Kit, Item 126A, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Ethylene Glycol, Item 137, Appendix D
- Machinery Towel, Item 211, Appendix D
- Nitric Acid, Item 220, Appendix D
- Paint, Polyurethane, Item 224, Appendix D
- Paint Remover, Item 231, Appendix D
- Pressure Sensitive Adhesive Tape, Item 250, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D

**6-4-12.1 Repair Input Module Housing.****6-4-12.1.1 General.****NOTE**

- Minor nicks, gouges, and scratches on surface of input module housing may be repaired. Before starting repair, measure depth of damage to determine whether blending out damage will result in input module still being within repair limits. Make sure to measure depth of damage from an area which has not been previously repaired. If limits will be exceeded, replace input module.
  - Minor nicks, gouges, scratches, and corrosion within snapping groove of jet or breather plug may be repaired. Before starting repair, measure depth of damage to determine whether blending out damage will result in input module still being within repair limits. Make sure to measure depth of damage from an area which has not been previously repaired. If limits will be exceeded, replace input module.
  - Repair of damaged area may be done with input module installed.
  - All repairs shall be recorded in helicopter component historical records.
  - Repair of both input module housings is the same.
- a. Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean any oil or grease from damaged area.

**NOTE**

Housings coated with epoxy coating are glossy silver. Housings coated with polyurethane paint are matte surface medium-gray.

- b. Fine scratches that remove only paint may be blended smooth using abrasive cloth, Item 10,

Appendix D. After blending, touch up blended area as required using epoxy coating kit, Item 126A, Appendix D, or polyurethane paint, Item 224, Appendix D.

**6-4-12.1.2 Repair Limits.**

- a. Input module housing limits are as follows:

- (1) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM ALL OVER OUTER CAST SURFACES OF INPUT MODULE INPUT HOUSING IS 0.020-INCH.**
  - (a) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM PINION HOUSING IS 0.020-INCH** (Figure 6-4-12-1, Detail A).
  - (b) **NO MATERIAL REMOVAL ALLOWED ON PILOT DIAMETER.**
- (2) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM JET ASSEMBLY IS 20% OF SNAPPING GROOVE BOTTOM CIRCUMFERENCE** (Figure 6-4-12-1, Detail B).
- (3) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM ALL OVER OUTER CAST SURFACES OF INPUT MODULE CENTER HOUSING IS 0.040-INCH.**
- (4) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM ALL OVER OUTER CAST SURFACES OF INPUT MODULE OUTPUT HOUSING IS 0.030-INCH.**

**NOTE**

Output housing part number is stamped on housing.

- (5) **OUTPUT HOUSING RIBS, 70351-08019-003, CAN BE BLENDED TO MINIMUM 0.420-INCH RIB THICKNESS. OUTPUT HOUSING RIBS, 70351-08019-001 AND 70351-08019-002, CAN BE BLENDED TO MINIMUM 0.170-INCH THICKNESS.**

- (a) NO MORE THAN TWO REPAIRS ALLOWED PER 4-INCH-SQUARE AREA.
- (b) TOTAL LENGTH OF ALL BLENDED AREAS NOT TO EXCEED 5-INCHES ALONG CIRCUMFERENCE (INCLUDING RIBS) AT ANY SECTION OF HOUSING.

**NOTE**

Only 30% of the total number of holes in bolt pattern may be repaired.

- (6) MAXIMUM DEPTH OF MATERIAL REMOVAL FROM OUTPUT HOUSING TO CENTER HOUSING MOUNTING FLANGE BOLTHOLES IS 0.015-INCH. MAXIMUM DIAMETER NOT TO EXCEED 0.440-INCH (Figure 6-4-12-1, Detail C).
  - (a) MAXIMUM DEPTH OF MATERIAL REMOVAL FROM MOUNTING SURFACE UNDER FASTENER IS 0.015-INCH. REPAIR SHALL NOT COVER MORE THAN 25% OF SURFACE AREA UNDER FASTENER.
  - (b) MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SPOT-FACE SURFACE NOT COVERED BY FASTENER IS 0.030-INCH.
  - (c) MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SHARP EDGE AROUND SPOT-FACE SURFACE (NOT INCLUDING AREA UNDER FASTENER) IS 0.060-INCH.
  - (d) CORNER OF SPOT-FACE MAY BE BLENDED TO 0.100-INCH RADIUS AS LONG AS DEPTH INTO SPOT-FACE DOES NOT EXCEED 0.030-INCH.

**NOTE**

Only 30% of the total number of holes in bolt pattern may be repaired.

- (7) MAXIMUM DEPTH OF MATERIAL REMOVAL FROM OUTPUT HOUSING TO MAIN MODULE MOUNTING FLANGE BOLTHOLES IS 0.015-INCH. MAXIMUM DIAMETER NOT TO EXCEED 0.440-INCH (Figure 6-4-12-1, Detail D).
  - (a) MAXIMUM DEPTH OF MATERIAL REMOVAL FROM MOUNTING SURFACE UNDER FASTENER IS 0.015-INCH. REPAIR SHALL NOT COVER MORE THAN 25% OF SURFACE AREA UNDER FASTENER.
  - (b) MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SPOT-FACE SURFACE NOT COVERED BY FASTENER IS 0.025-INCH.
  - (c) CORNER OF SPOT-FACE MAY BE BLENDED TO 0.100-INCH RADIUS AS LONG AS DEPTH INTO SPOT-FACE DOES NOT EXCEED 0.025-INCH.
  - (d) SHARP EDGES MAY BE BLENDED TO A 0.100-INCH RADIUS.

**6-4-12.1.3 Repair Procedure.**

- a. Using paint remover, Item 231, Appendix D, remove paint from damaged area.
- b. Using nonmetallic scraper, remove sealing compound from damaged area.
- c. Using abrasive cloth, Item 6, Appendix D, carefully blend damaged area smooth. Remove minimum amount of material required to blend out damaged area.
- d. Using abrasive cloth, Item 10, Appendix D, blend reworked area to remove all traces of abrasive cloth.
- e. Remove displaced metal caused by abrasive blending by using magnesium-etching procedure as follows:

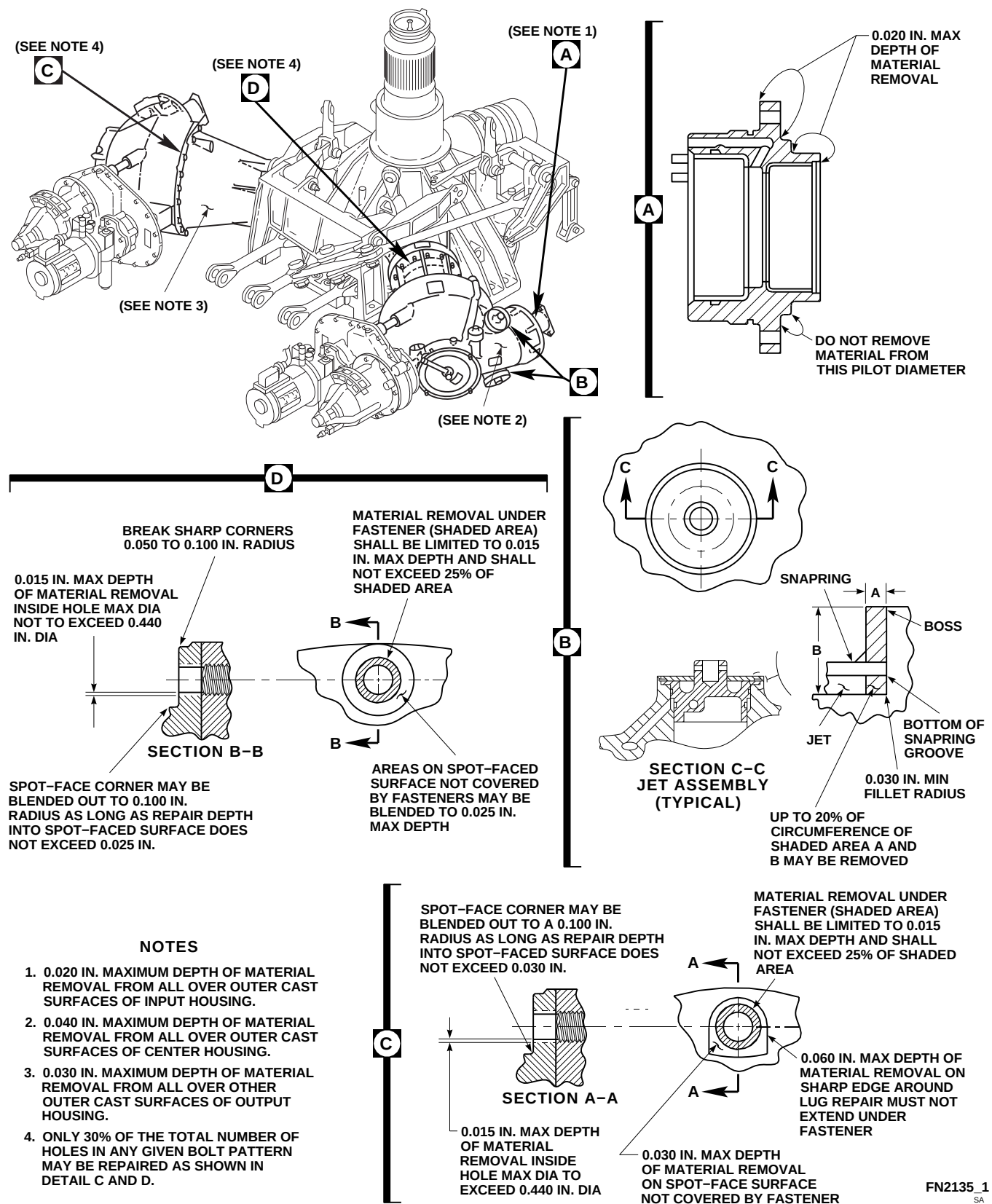


FIGURE 6-4-12-1. REPAIR INPUT MODULE HOUSING



CAUTION

- Magnesium-etchant measurements are to be as accurate as possible to values measured in a milliliter beaker or vessel with equivalent mixing accuracy.
  - Magnesium-etchant formulas shall be prepared and stored in unbreakable 1-pint plastic containers and safely stored in designated areas not subjected to extreme heat or cold.
  - Before preparing magnesium-etchant, make sure storage area does not contain premixed magnesium-etchant formula. The correct stock bottle for this repair shall be marked MAGNESIUM-ETCHANT.
- (1) Prepare magnesium-etchant solution in clean unbreakable plastic container by mixing the following:

| MATERIAL                              | QTY             |
|---------------------------------------|-----------------|
| Ethylene Glycol, Item 137, Appendix D | 600 milliliters |
| Acetic Acid, Item 24, Appendix D      | 200 milliliters |
| Nitric Acid, Item 220, Appendix D     | 10 milliliters  |
| Distilled Water, Item 123, Appendix D | 190 milliliters |

- (2) Using dry-cleaning solvent, Item 124, Appendix D, clean area to be etched.
- (3) Using pressure sensitive adhesive tape, Item 250, Appendix D, mask area.
- (4) Pour a small amount of magnesium-etchant from stock bottle into 1-pint plastic container.

CAUTION

- Damage to components will result if magnesium-etchant is allowed to come in contact with areas not being etched. Extreme caution shall be used to protect those areas not being etched.
- (5) Using applicator, Item 78, Appendix D, apply magnesium-etchant to reworked area. Swab area for about 10 - 15 seconds.
- (6) Neutralize magnesium-etchant immediately by flooding area with generous amount of clean water.

WARNING

- Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.
- (7) Using clean, dry, compressed air, blow dry area thoroughly.
- (8) After use, do not return unused magnesium-etchant to stock bottle. Dispose of used magnesium-etchant per local instructions.
- (9) Identify contents of stock bottle as MAGNESIUM-ETCHANT, before placing bottle in storage area.
- (10) Prepare technical chromium trioxide solution in clean 2-gallon plastic container by mixing 24 ounces of technical chromium trioxide, Item 91, Appendix D, with 1 gallon of water.
- (11) Using technical chromium trioxide solution, treat bare surface of housing.

- f. Apply sealing compound, Item 292, Appendix D, to touch up reworked areas.
- g. After sealing compound has dried, using epoxy primer coating kit, Item 135, Appendix D, apply one coat of epoxy primer.

**NOTE**

Housings coated with epoxy coating are glossy silver. Housings coated with polyurethane paint are matte surface medium-gray.

- h. Using epoxy coating kit, Item 126A, Appendix D, or polyurethane paint, Item 224, Appendix D, apply one coat of epoxy primer, depending on original paint finish.

- i. Using corrosion preventive compound, Item 113, Appendix D, apply one coat.
- j. Allow corrosion preventive compound to cure for 30 minutes.
- k. Using corrosion preventive compound, Item 113, Appendix D, apply a second coat.
- l. Make sure area is clean and free of foreign material.
- m. Record brief description of input module repairs in helicopter component historical records.

6-4-13 ACCESSORY MODULE HOUSING REPAIR.

PARAGRAPH BREAKDOWN

|                                          | PAGE     |
|------------------------------------------|----------|
| 6-4-13.1 Repair Accessory Module Housing | 6-4-13-2 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Beaker (10 ml)
- Beaker (200 ml)
- Beaker (600 ml)
- Container, Plastic, 1-Pint
- Container, Plastic, 2-Gallon
- Nonmetallic Scraper, Locally-Made
- Powertrain Repairer’s Toolkit
- Protection, Eye
- Vessel(s), with equivalent measuring accuracy of Beakers (Alternate for Beakers)

Materials/Parts

- Abrasive Cloth, Item 6, Appendix D
- Abrasive Cloth, Item 10, Appendix D
- Acetic Acid, Item 24, Appendix D

- Applicator, Item 78, Appendix D
- Chromium Trioxide, Technical, Item 91, Appendix D
- Corrosion Preventive Compound, Item 113, Appendix D
- Distilled Water, Item 123, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Coating Kit, Item 126A, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Ethylene Glycol, Item 137, Appendix D
- Machinery Towel, Item 211, Appendix D
- Nitric Acid, Item 220, Appendix D
- Paint, Polyurethane, Item 224, Appendix D
- Paint Remover, Item 231, Appendix D
- Pressure Sensitive Adhesive Tape, Item 250, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D

**6-4-13.1 Repair Accessory Module Housing.****6-4-13.1.1 General.****NOTE**

- Minor nicks, gouges, and scratches on surface of accessory module housing may be repaired. Before starting repair, measure depth of damage to determine whether blending out damage will result in accessory module still being within repair limits. Make sure to measure depth of damage from an area which has not been previously repaired. If limits will be exceeded, replace accessory module.
  - Repair of damaged area may be done with accessory module installed.
  - All repairs shall be recorded in helicopter component historical records.
  - Repair of both accessory module housings is the same.
- a. Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean any oil or grease from damaged area.

**NOTE**

Housings coated with epoxy coating are glossy silver. Housings coated with polyurethane paint are matte surface medium-gray.

- b. Fine scratches that remove only paint may be blended smooth using abrasive cloth, Item 10, Appendix D. After blending, touch up as required using epoxy coating kit, Item 126A, Appendix D, or polyurethane paint, Item 224, Appendix D.

**6-4-13.1.2 Repair Limits.**

- a. Accessory module housing repair limits are as follows:
- (1) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM ALL OVER**

**OUTER CAST SURFACES IS 0.060-INCH** (Figure 6-4-13-1).

**NOTE**

- Only 30% of the total number of holes in bolt pattern may be repaired.
  - Boltholes with bushings are not repairable.
- (2) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM MOUNTING FLANGE BOLTHOLES IS 0.015-INCH. MAXIMUM DIAMETER NOT TO EXCEED 0.315-INCH** (Figure 6-4-13-1, Detail A).
- (a) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM MOUNTING SURFACE UNDER FASTENER IS 0.015-INCH. REPAIR SHALL NOT COVER MORE THAN 25% OF SURFACE AREA UNDER FASTENER.**
- (b) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SPOT-FACE SURFACE NOT COVERED BY FASTENER IS 0.030-INCH.**
- (c) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SHARP EDGE AROUND SPOT-FACE SURFACE (NOT INCLUDING AREA UNDER FASTENER) IS 0.060-INCH.**
- (d) **CORNER OF SPOT-FACE MAY BE BLENDED TO 0.100-INCH RADIUS AS LONG AS DEPTH INTO SPOT-FACE DOES NOT EXCEED 0.030-INCH.**

**6-4-13.1.3 Repair Procedure.**

- a. Using paint remover, Item 231, Appendix D, remove paint from damaged area.
- b. Using nonmetallic scraper, remove sealing compound from damaged area.

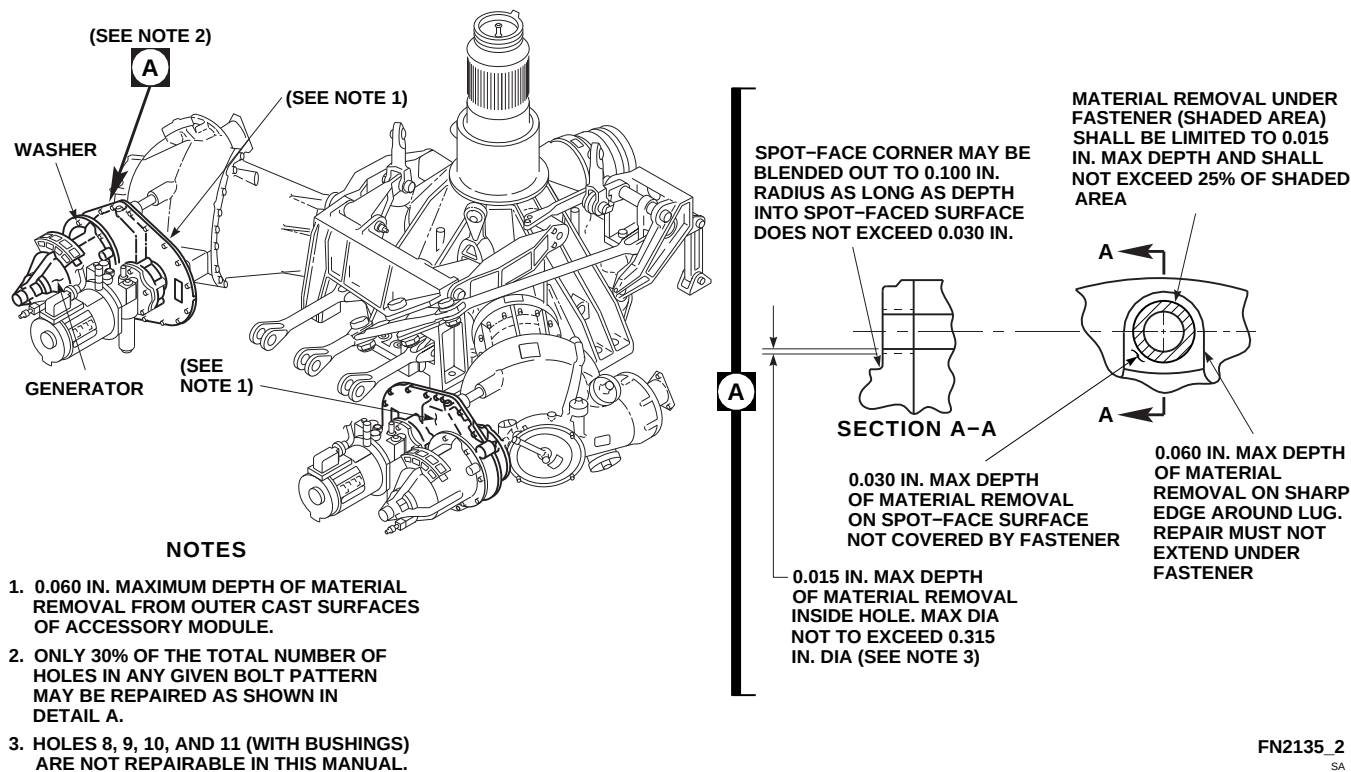


FIGURE 6-4-13-1. REPAIR ACCESSORY MODULE HOUSING

- c. Using abrasive cloth, Item 6, Appendix D, carefully blend damaged area smooth. Remove minimum amount of material required to blend out damaged area.
- d. Using abrasive cloth, Item 10, Appendix D, blend reworked area to remove all traces of abrasive cloth.
- e. Remove displaced metal caused by abrasive blending by using magnesium-etching procedure as follows:

**CAUTION**

- Magnesium-etchant measurements are to be as accurate as possible to values measured in a milliliter beaker or vessel with equivalent mixing accuracy.
- Magnesium-etchant formulas shall be prepared and stored in unbreakable 1-pint plastic containers and safely

stored in designated areas not subjected to extreme heat or cold.

- Before preparing magnesium-etchant, make sure storage area does not contain premixed magnesium-etchant formula. The correct stock bottle for this repair shall be marked MAGNESIUM-ETCHANT.

- (1) Prepare magnesium-etchant solution in clean unbreakable plastic container by mixing the following:

| MATERIAL                              | QTY             |
|---------------------------------------|-----------------|
| Ethylene Glycol, Item 137, Appendix D | 600 milliliters |
| Acetic Acid, Item 24, Appendix D      | 200 milliliters |
| Nitric Acid, Item 220, Appendix D     | 10 milliliters  |

**MATERIAL****QTY**

Ethylene Glycol,  
Item 137, Appendix  
D 600 milliliters

Distilled Water, Item  
123, Appendix D 190 milliliters

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- (2) Using dry-cleaning solvent, Item 124, Appendix D, clean area to be etched.
- (3) Using pressure sensitive adhesive tape, Item 250, Appendix D, mask area.
- (4) Pour a small amount of magnesium-etchant from stock bottle into 1-pint plastic container.

**CAUTION**

Damage to components will result if magnesium-etchant is allowed to come in contact with areas not being etched. Extreme caution shall be used to protect those areas not being etched.

- (5) Using applicator, Item 78, Appendix D, apply magnesium-etchant to reworked area. Swab area for about 10 - 15 seconds.
- (6) Neutralize magnesium-etchant immediately by flooding area with generous amount of clean water.

**WARNING**

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure

all personnel are wearing safety glasses when compressed air is being used.

- (7) Using clean, dry, compressed air, blow dry area thoroughly.
  - (8) After use, do not return unused magnesium-etchant to stock bottle. Dispose of used magnesium-etchant per local instructions.
  - (9) Identify contents of stock bottle as MAGNESIUM-ETCHANT, before placing bottle in storage area.
  - (10) Prepare technical chromium trioxide solution in clean 2-gallon plastic container by mixing 24 ounces of technical chromium trioxide, Item 91, Appendix D, with 1 gallon of water.
  - (11) Using technical chromium trioxide solution, treat bare surface of housing.
- f. Apply sealing compound, Item 292, Appendix D, to touch up reworked areas.
  - g. After sealing compound has dried, using epoxy primer coating kit, Item 135, Appendix D, apply one coat of epoxy primer.

**NOTE**

Housings coated with epoxy coating are glossy silver. Housings coated with polyurethane paint are matte surface medium-gray.

- h. Apply one coat of epoxy coating kit, Item 126A, Appendix D, or polyurethane paint, Item 224, Appendix D, apply one coat of epoxy primer depending on original paint finish.
- i. Using corrosion preventive compound, Item 113, Appendix D, apply one coat.
- j. Allow corrosion preventive compound to cure for 30 minutes.
- k. Using corrosion preventive compound, Item 113, Appendix D, apply a second coat.

- l. Make sure area is clean and free of foreign material.
- m. Record brief description of accessory module repairs in helicopter component historical records.





6-4-14 MAIN, INPUT, AND ACCESSORY MODULE HOUSING WATER ENTRAPMENT AREA CORROSION PROTECTION.

PARAGRAPH BREAKDOWN

|          |                                                | PAGE     |
|----------|------------------------------------------------|----------|
| 6-4-14.1 | Corrosion Protection of Water Entrapment Areas | 6-4-14-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Powertrain Repairer’s Toolkit

Materials/Parts

- Corrosion Preventive Compound, Item 114, Appendix D

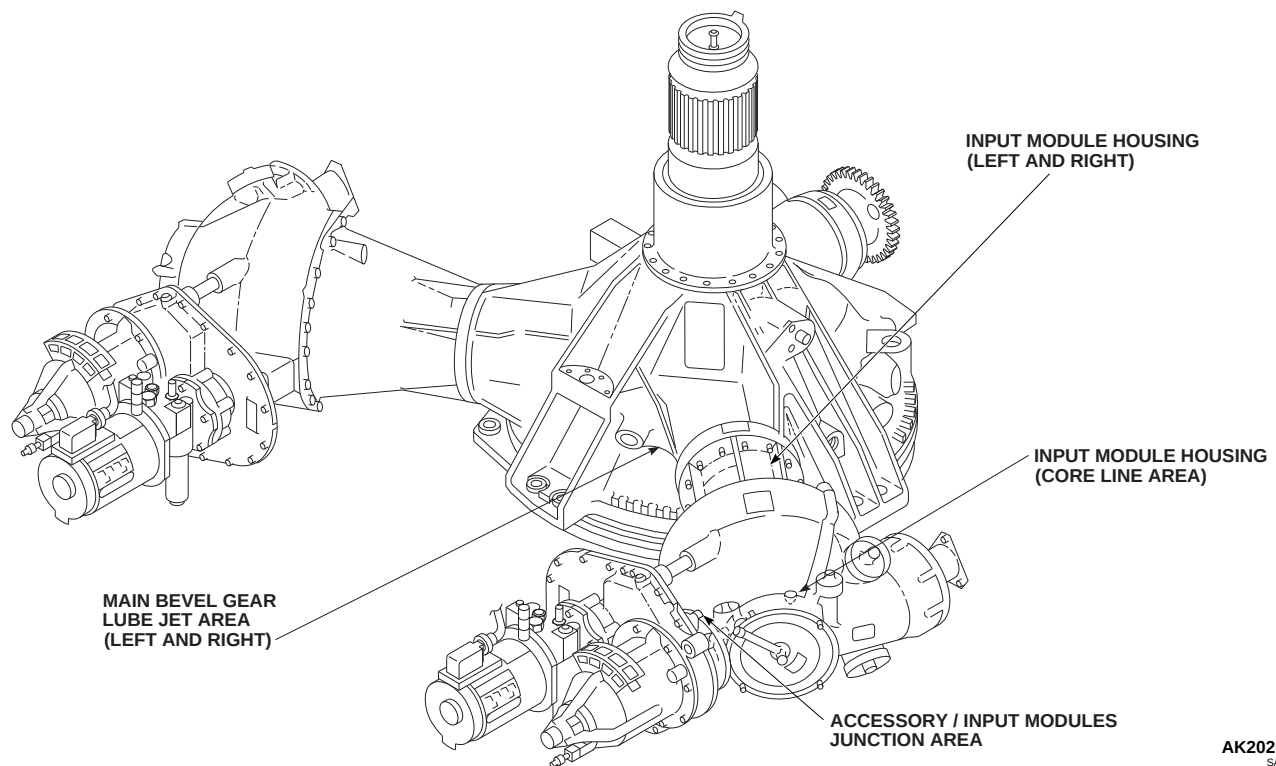
- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint, Polyurethane, Item 228, Appendix D

References

- Appendix D

6-4-14.1 Corrosion Protection of Water Entrapment Areas.

- a. Using corrosion preventive compound, Item 114, Appendix D, clean water entrapment areas (Figure 6-4-14-1), and allow to dry.
- b. Using methyl ethyl ketone, Item 217, Appendix D, repeat cleaning procedure, and allow to dry.
- c. Using polyurethane paint, Item 228, Appendix D, apply one coat to cleaned areas.



**FIGURE 6-4-14-1. CORROSION PROTECTION OF WATER ENTRAPMENT AREAS**

6-4-15 ACCESSORY MODULE CHIP DETECTOR.

PARAGRAPH BREAKDOWN

|          |                                        | PAGE     |          |                            | PAGE     |
|----------|----------------------------------------|----------|----------|----------------------------|----------|
| 6-4-15.1 | Replace Accessory Module Chip Detector | 6-4-15-1 | 6-4-15.2 | Replace Self-Closing Valve | 6-4-15-4 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 2-Gallon
- Powertrain Repairer’s Toolkit
- Protection, Eye

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Lockwire, Item 198, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D

- Lubricating Oil, Item 208, Appendix D
- Protective Caps and Plugs
- Tape, Item 317C, Appendix D
- Tiedown Strap, Item 329, Appendix D

References

- Appendix D
- PARA 1-3-12
- PARA 1-6-16
- PARA 1-7-20
- PARA 6-2-2
- PARA 6-3-6

6-4-15.1 Replace Accessory Module Chip Detector.

NOTE

Replacement of left or right accessory module chip detector is the same, except as noted.

6-4-15.1.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Pull out CHIP DET circuit breaker (DC ESNTL BUS - copilot’s side upper console circuit breaker panel).

- c. Open main rotor pylon sliding cover.
- d. Disconnect electrical connector, P405 (right side), or P406 (left side), from accessory module chip detector (Figure 6-4-15-1, Detail A).
- e. Install protective caps and plugs on electrical connectors.
- f. Remove end of accessory module chip detector from bracket.
- g. Remove tape or tiedown straps from harness (Figure 6-4-15-1, Detail B).

- h. Remove screw, washer, and nut securing clamp(s) and harness to bracket on accessory module. If accessory module chip detector is going to be replaced, remove clamp(s) from harness. Keep all attaching hardware.

### WARNING

Injury to personnel will result if precautions are not taken when removing accessory module chip detector. To prevent injury, personnel must wear eye protection and make sure transmission oil is cool before removing accessory module chip detector.

- i. Remove accessory module chip detector and packings from self-closing valve in accessory module. Discard packings (Figure 6-4-15-1, Detail C).
- j. Install protective plugs in self-closing valve.

#### 6-4-15.1.2 Inspect.

- a. Inspect accessory module chip detector for chips (PARA 6-3-6).
- b. If pin or washer cannot be located, perform main transmission chip detectors particle inspections (PARA 6-3-6).

### CAUTION

Do not immerse accessory module chip detector in dry-cleaning solvent.

- c. Using dry-cleaning solvent, Item 124, Appendix D, clean magnetic plug of accessory module chip detector. If removed, clean sleeve in dry-cleaning solvent.

### WARNING

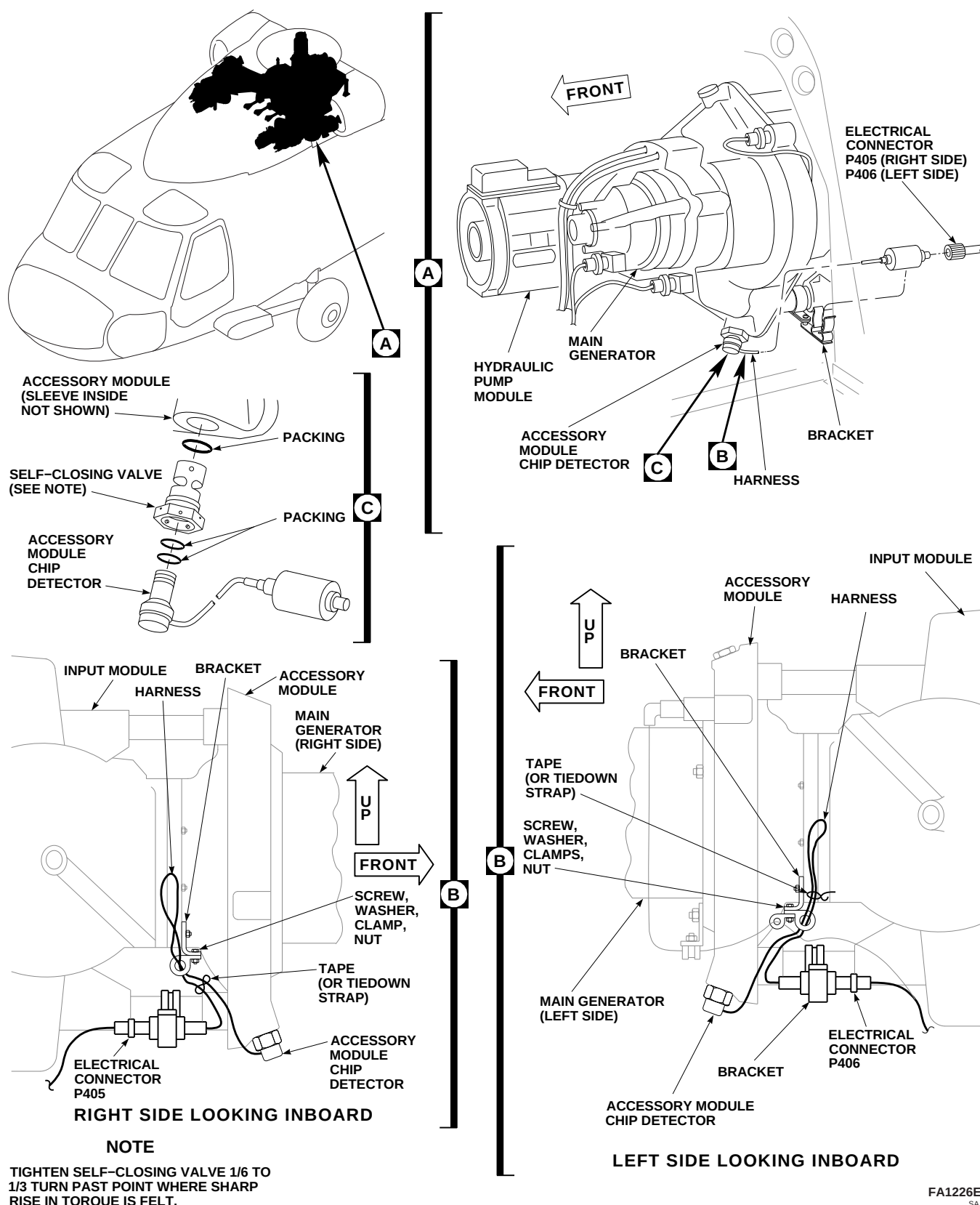
Injury to personnel will result if precautions are not taken while using com-

pressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- d. Using clean, dry, compressed air, blow dry accessory module chip detector.

#### 6-4-15.1.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Remove protective plugs from self-closing valve.
- c. Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packings to be installed on accessory module chip detector.
- d. Perform operational check of chip detector system (PARA 6-2-2).
- e. Install packings on accessory module chip detector, and install in self-closing valve (Figure 6-4-15-1, Detail C).
- f. Remove protective caps and plugs from electrical connectors.
- g. Inspect and treat electrical connectors (PARA 1-7-20).
- h. Install end of accessory module chip detector in bracket (Figure 6-4-15-1, Detail A).
- i. Connect electrical connector, P405 (right side), or P406 (left side), to accessory module chip detector.
- j. If installing new accessory module chip detector, install clamp(s) on harness (Figure 6-4-15-1, Detail B).
- k. Route harness and secure to bracket on accessory module with clamp(s), screw, washer, and nut.



**FIGURE 6-4-15-1. REPLACE ACCESSORY MODULE CHIP DETECTOR AND SELF-CLOSING VALVE**

**NOTE**

Tiedown straps may be used as an alternate for tape.

- l. Using tape, Item 317C, Appendix D, or tiedown straps, Item 329, Appendix D, secure slack in harness.
- m. If necessary, service main transmission (PARA 1-3-12).
- n. Make sure area is clean and free of foreign material.
- o. Close main rotor pylon sliding cover.
- p. Push in CHIP DET circuit breaker (DC ESNTL BUS - copilot's side upper console circuit breaker panel).
- q. Perform operational check of both accessory module chip detector electrical circuits (PARA 6-2-2).

**6-4-15.2 Replace Self-Closing Valve.****NOTE**

Replacement of left or right self-closing valve is the same.

**6-4-15.2.1 Remove.**

- a. If not already removed, remove accessory module chip detector (PARA 6-4-15.1.1), or protective plug from hole.
- b. Position 2-gallon container underneath self-closing valve before removing to catch oil.
- c. Remove self-closing valve and packing from accessory module. Discard packing (Figure 6-4-15-1, Detail C).
- d. Install protective plug in accessory module.

**6-4-15.2.2 Inspect.****NOTE**

Leakage of lubricating oil through center of self-closing valve indicates

loose internal hardware. This disables temperature sensor portion of accessory module chip detector.

- a. Visually inspect self-closing valve for leakage from center of self-closing valve. **NONE ALLOWED.** If any leakage is found, replace self-closing valve.
- b. Check inside of self-closing valve for presence and security of contact pin located in center of valve. **NO MISSING PIN OR LOOSENESS ALLOWED.** If contact pin is loose or missing, locate pin and washer and replace self-closing valve.
- c. Inspect self-closing valve for metal particles (PARA 6-3-6).

**CAUTION**

Do not immerse self-closing valve in dry-cleaning solvent.

- d. Using dry-cleaning solvent, Item 124, Appendix D, clean self-closing valve.

**WARNING**

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- e. Using clean, dry, compressed air, blow dry self-closing valve.

**6-4-15.2.3 Install.**

- a. Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing.

- b. Remove protective plug from accessory module.
- c. Install packing on self-closing valve, and install self-closing valve in accessory module (Figure 6-4-15-1, Detail C).
- d. **TIGHTEN SELF-CLOSING VALVE  $\frac{1}{6}$  TO  $\frac{1}{3}$  TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- e. Using lockwire, Item 198, Appendix D, lockwire self-closing valve to accessory module.
- f. Install protective plug in self-closing valve.
- g. Install accessory module chip detector (PARA 6-4-15.1.3).





6-4-16 ACCESSORY MODULE OIL SCREEN.

PARAGRAPH BREAKDOWN

|                                              | PAGE     |
|----------------------------------------------|----------|
| 6-4-16.1 Replace Accessory Module Oil Screen | 6-4-16-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Scraper, Locally-Made
- Powertrain Repairer’s Toolkit
- Protection, Eye

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D

- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D
- Lubricating Oil, Item 208, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 6-3-6

6-4-16.1 Replace Accessory Module Oil Screen.

NOTE

Replacement of the left or right accessory module oil screen is the same.

6-4-16.1.1. Remove.

- Open main rotor pylon sliding cover.
- Using nonmetallic scraper, remove sealing compound from both ends of lower oil return tube.

WARNING

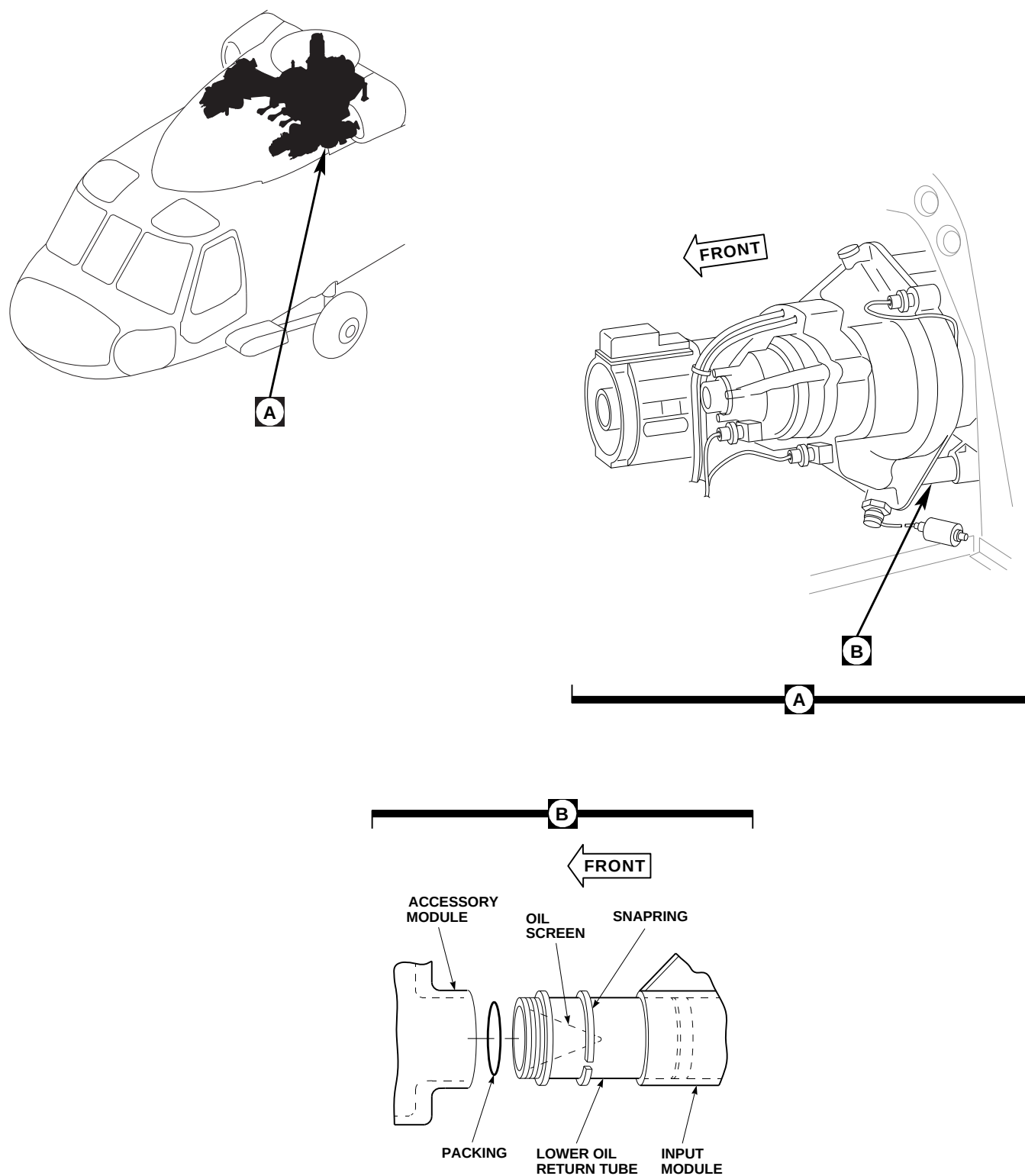
Injury to personnel will result if precautions are not taken when removing oil

screen. To prevent injury, personnel must wear eye protection and make sure transmission oil is cool before removing oil screen.

- Unlock snapping on lower oil return tube on input module side between accessory and input module, and slide to front along lower oil return tube (Figure 6-4-16-1, Detail B).
- Push lower oil return tube into input module to expose cone-shaped oil screen. Remove and discard packing.
- Remove oil screen and inspect for metal particles (PARA 6-3-6).
- Using dry-cleaning solvent, Item 124, Appendix D, clean oil screen.

**6-4-16.1.2. Install.**

- a. Insert oil screen into lower oil return tube (Figure 6-4-16-1, Detail B).
- b. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to snapping.
- c. Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing.
- d. Seat lower oil return tube with packing in accessory module. Slide snapping to rear along lower oil return tube and lock in groove.
- e. Apply sealing compound, Item 292, Appendix D, to lower oil return tube at both ends to prevent water entrapment and corrosion above snapping groove.
- f. Make sure area is clean and free of foreign material.
- g. Close main rotor pylon sliding cover.



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FIGURE 6-4-16-1. REPLACE ACCESSORY MODULE OIL SCREEN

6-4-16-3/(6-4-16-4 Blank)



6-4-17 INPUT MODULE CHIP DETECTOR.

PARAGRAPH BREAKDOWN

|          |                                    | PAGE     |          |                            | PAGE     |
|----------|------------------------------------|----------|----------|----------------------------|----------|
| 6-4-17.1 | Replace Input Module Chip Detector | 6-4-17-1 | 6-4-17.2 | Replace Self-Closing Valve | 6-4-17-5 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Borescope Kit
- Container, 2-Gallon
- Light Source
- Powertrain Repairer’s Toolkit
- Protection, Eye
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Lockwire, Item 197, Appendix D
- Lockwire, Item 198, Appendix D

- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D
- Lubricating Oil, Item 208, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-3-12
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-70
- PARA 6-2-2
- PARA 6-3-6
- PARA 6-4-1

6-4-17.1 Replace Input Module Chip Detector.

NOTE

Replacement of left or right input module chip detector is the same, except as noted.

6-4-17.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).

- Remove main transmission drip pan (PARA 2-4-70).
- Disconnect electrical connector, P403 (left side) or P404 (right side), from input module chip detector (Figure 6-4-17-1, Detail B).
- Install protective caps and plugs on electrical connectors.

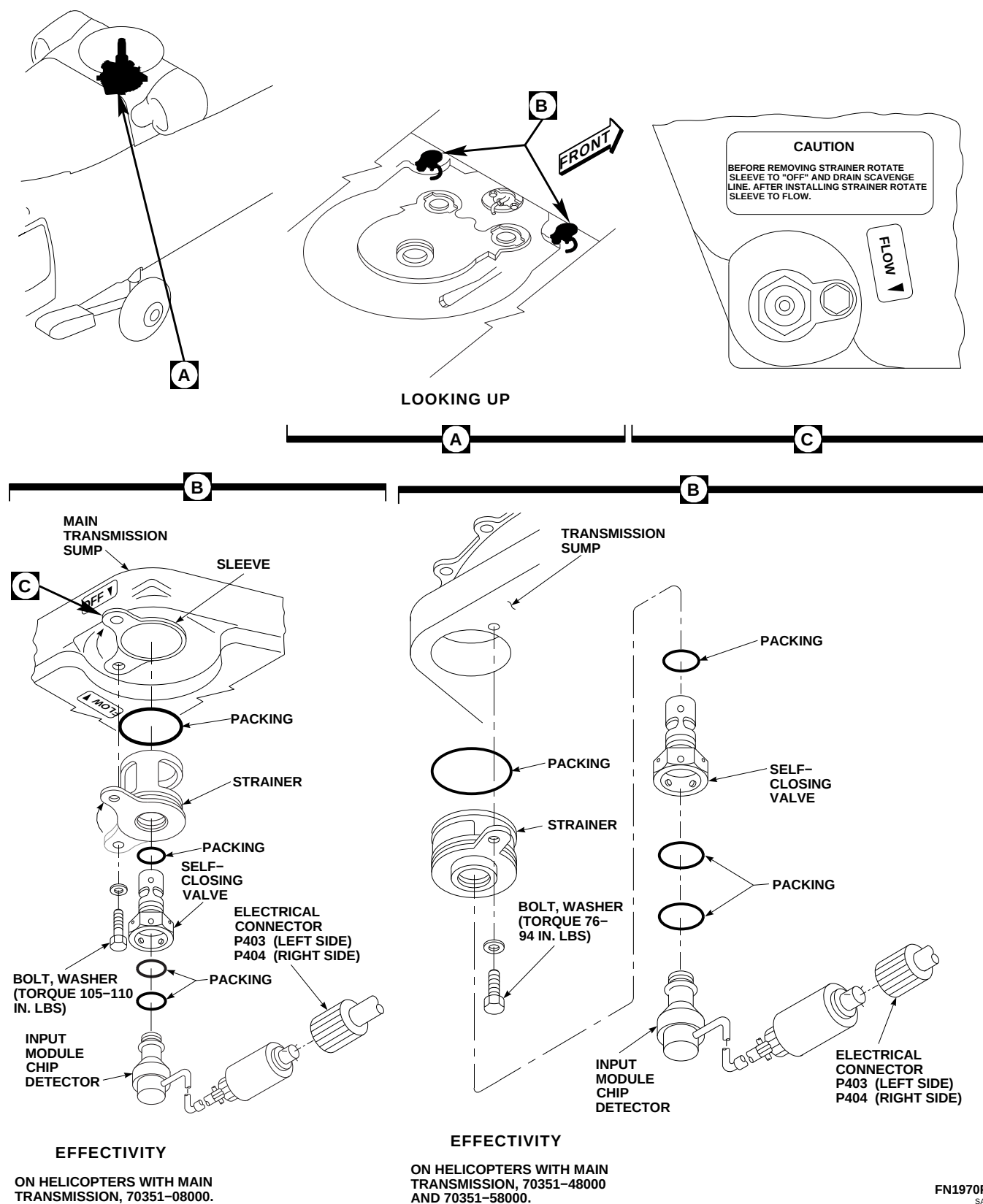


FIGURE 6-4-17-1. REPLACE INPUT MODULE CHIP DETECTOR

- e. On helicopters with main transmission, 70351-08000, remove input module chip detector, self-closing valve, and strainer as follows:

**WARNING**

Injury to personnel will result if precautions are not taken when removing input module chip detector and strainer. To prevent injury, personnel must wear eye protection and make sure transmission oil is cool before removing input module chip detector and strainer.

- (1) Remove bolt and washer securing strainer and sleeve to main transmission sump. Turn strainer and sleeve OFF (Figure 6-4-17-1, Detail B and Detail C).
- (2) Remove input module chip detector from self-closing valve by pushing in, turning, and pulling out (Figure 6-4-17-1, Detail B).
- (3) Remove packings from input module chip detector. Discard packings.

**NOTE**

When removing strainer, about 1-quart of lubrication oil will drain out.

- (4) Position 2-gallon container, or larger, underneath strainer to catch oil.
  - (5) Remove strainer and packing leaving sleeve in main transmission sump. Discard packing.
  - (6) Install protective plug in main transmission sump.
  - (7) Remove self-closing valve and packing from strainer. Discard packing.
- f. On helicopters with main transmission, 70351-48000 and 70351-58000, remove input module chip detector, self-closing valve, and strainer as follows:

**WARNING**

Injury to personnel will result if precautions are not taken when removing input module chip detector and strainer. To prevent injury, personnel must wear eye protection and make sure transmission oil is cool before removing input module chip detector and strainer.

- (1) Position 2-gallon container underneath input module chip detector to catch oil.
- (2) Remove input module chip detector from self-closing valve by pushing in, turning, and pulling out.
- (3) Remove packings from input module chip detector. Discard packings.

**NOTE**

About 1-gallon of lubrication oil will drain out when self-closing valve is removed.

- (4) Remove self-closing valve and packing from strainer. Discard packing.
- (5) Remove bolt and washer and remove strainer and packing from main transmission sump. Discard packing.
- (6) Install protective plug in main transmission sump.

**6-4-17.1.2 Inspect.**

- a. Inspect input module chip detector and strainer for metal particles (PARA 6-3-6).
- b. On helicopters with main transmission, 70351-08000, using borescope kit, inspect openings in sleeve and main transmission sump cored hole. **MAXIMUM OPENING OVERLAP IS 0.030-INCH.** If limit is exceeded, replace main transmission (PARA 6-4-1).
- c. On helicopters with main transmission, 70351-08000, using borescope kit and a light source,

inspect scavenge passage between sleeve window and main transmission oil pump for obstructions. **NONE ALLOWED.** If scavenge passage is not free of obstructions, replace main transmission (PARA 6-4-1).

### CAUTION

Do not immerse input module chip detector or strainer in dry-cleaning solvent.

- d. Using dry-cleaning solvent, Item 124, Appendix D, clean input module chip detector and strainer.

### WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- e. Using clean, dry, compressed air, blow dry input module chip detector and strainer.

#### 6-4-17.1.3 Install.

- a. On helicopters with main transmission, 70351-08000, install input module chip detector and strainer as follows:
  - (1) Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing and install on strainer.
  - (2) Remove protective plug from input module.
  - (3) Install strainer and packing in sleeve in main transmission sump.

- (4) Install self-closing valve (PARA 6-4-17.2.3).
- (5) Check input module chip detector electrical circuit (PARA 6-2-2).
- (6) Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packings and install on input module chip detector.
- (7) Install input module chip detector and packings in self-closing valve, push in and turn. Turn sleeve to FLOW (Figure 6-4-17-1, Detail B and Detail C).
- (8) Install bolt and washer.
- (9) **TORQUE BOLT TO 105 - 110 INCH-POUNDS.**
- (10) Using lockwire, Item 197, Appendix D, lockwire bolt to self-closing valve.
- b. On helicopters with main transmission, 70351-48000 and 70351-58000, install input module chip detector and strainer as follows:
  - (1) Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing and install on strainer.
  - (2) Remove protective plug from main transmission sump.
  - (3) Install strainer in main transmission sump (Figure 6-4-17-1, Detail B).
  - (4) Install strainer retaining bolt and washer in main transmission sump.
  - (5) **TORQUE BOLT TO 76 - 94 INCH-POUNDS.**
  - (6) Install self-closing valve (PARA 6-4-17.2.3).
  - (7) Using lockwire, Item 197, Appendix D, lockwire bolt to self-closing valve.
  - (8) Perform operational check of chip detector system (PARA 6-2-2).



- (9) Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packings and install on input module chip detector.
- (10) Install input module chip detector in self-closing valve, push in and turn.
- c. Remove protective caps and plugs from electrical connectors.
- d. Inspect and treat electrical connectors (PARA 1-7-20).
- e. Connect electrical connector, P403 (left side) or P404 (right side), to input module chip detector.
- f. Using lockwire, Item 197, Appendix D, lockwire electrical connector.
- g. Check main transmission for proper oil level. If necessary, service main transmission (PARA 1-3-12).
- h. Make sure area is clean and free of foreign material.
- i. Install main transmission drip pan (PARA 2-4-70).

#### 6-4-17.2 Replace Self-Closing Valve.

##### NOTE

Replacement of left or right self-closing valve is the same.

##### 6-4-17.2.1 Remove.

- a. If not already removed, remove input module chip detector (PARA 6-4-17.1.1), or protective plug from hole.
- b. Position 2-gallon container underneath self-closing valve before removing to catch oil.
- c. Remove self-closing valve and packing from input module. Discard packing (Figure 6-4-17-1, Detail B).
- d. Install protective plug in input module.

#### 6-4-17.2.2 Inspect.

##### NOTE

Leakage of lubricating oil through center of self-closing valve indicates loose internal hardware. This disables temperature sensor portion of input module chip detector.

- a. Visually inspect self-closing valve for leakage, from center of valve. **NONE ALLOWED.** If any leakage is found, replace self-closing valve.
- b. Check inside of self-closing valve for presence and security of contact pin located in center of valve. **NO MISSING PIN OR LOOSENESS ALLOWED.** If contact pin is loose or missing, locate pin and washer and replace self-closing valve.
- c. Inspect self-closing valve for metal particles (PARA 6-3-6).

##### CAUTION

Do not immerse self-closing valve in dry-cleaning solvent.

- d. Using dry-cleaning solvent, Item 124, Appendix D, clean self-closing valve.

##### WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- e. Using clean, dry, compressed air, blow dry self-closing valve.

**6-4-17.2.3 Install.**

- a. Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing.
- b. Remove protective plug from input module.
- c. Install packing on self-closing valve and install self-closing valve in input module (Figure 6-4-17-1, Detail B).
- d. **TIGHTEN SELF-CLOSING VALVE  $\frac{1}{6}$  TO  $\frac{1}{3}$  TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- e. Using lockwire, Item 198, Appendix D, lockwire self-closing valve to input module.
- f. Install protective plug in self-closing valve.
- g. Install input module chip detector (PARA 6-4-17.1.3).

6-4-18 MAIN MODULE SUMP CHIP DETECTOR/TEMPERATURE SENSOR.

PARAGRAPH BREAKDOWN

|                                                                     | PAGE     |
|---------------------------------------------------------------------|----------|
| 6-4-18.1 Replace Main Module Sump Chip Detector/ Temperature Sensor | 6-4-18-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 2-Gallon
- Powertrain Repairer’s Toolkit
- Protection, Eye
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D

- Lubricating Oil, Item 208, Appendix D
- Protective Caps and Plugs
- Tiedown Strap, Item 329, Appendix D

References

- Appendix D
- PARA 1-3-12
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-70
- PARA 6-2-2
- PARA 6-3-5
- PARA 6-3-6

6-4-18.1 Replace Main Module Sump Chip Detector/Temperature Sensor.

6-4-18.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove main transmission drip pan (PARA 2-4-70).
- Disconnect electrical connector, P412, from main module sump chip detector/temperature sensor (Figure 6-4-18-1, Sheet 1, Detail B).

- Install protective caps and plugs on electrical connectors.
- Remove tiedown straps from harness (Figure 6-4-18-1, Sheet 2, Detail C).
- Remove screw(s), lockwasher(s), and washer(s) securing clamp(s) and harness to main module. If main module sump chip detector/temperature sensor is going to be replaced, remove from harness. Keep all attaching hardware.

- g. Remove bolts and washers securing main module sump chip detector/temperature sensor and strainer from sleeve (Figure 6-4-18-1, Sheet 1, Detail B).
- h. Turn sleeve in main transmission sump to OFF position as indicated by arrow (Figure 6-4-18-1, Sheet 3, Detail D).
- i. Position 2-gallon container underneath main module sump chip detector/temperature sensor.

### WARNING

Injury to personnel will result if precautions are not taken when removing main module sump chip detector/temperature sensor. To prevent injury, personnel must wear eye protection and make sure transmission oil is cool before removing.

### NOTE

When removing main module sump chip detector/temperature sensor and strainer, about 1 pint of oil will drain out.

- j. Remove main module sump chip detector/temperature sensor and strainer, and packing, from sleeve in main transmission sump. Discard packing (Figure 6-4-18-1, Sheet 1, Detail B).
- k. Install protective plug in main transmission sump.

#### 6-4-18.1.2 Inspect.

- a. Inspect main module sump chip detector/temperature sensor and strainer for metal particles (PARA 6-3-6).
- b. Using dry-cleaning solvent, Item 124, Appendix D, clean main module sump chip detector/temperature sensor, magnetic element, and strainer, if necessary.

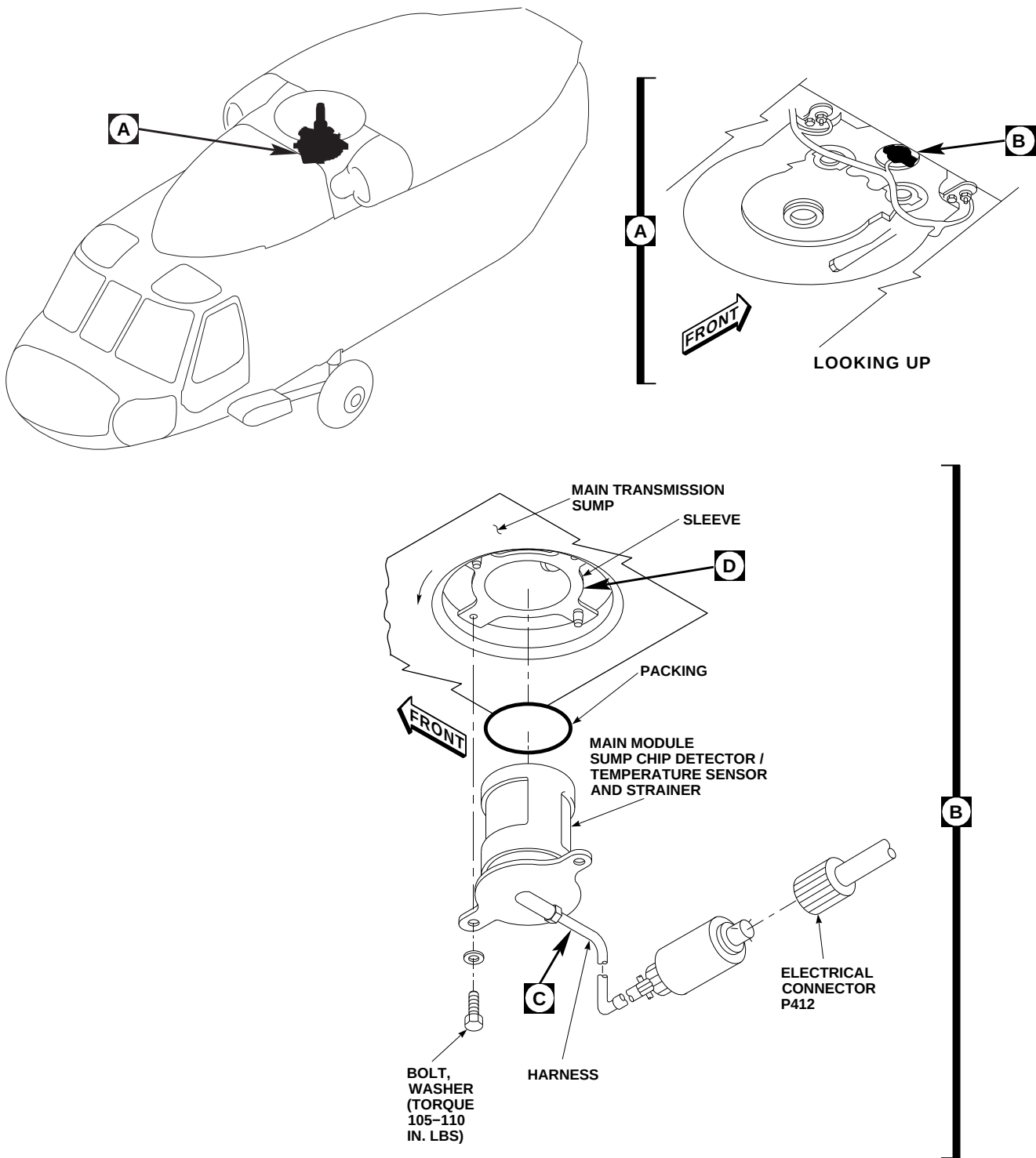
### WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- c. Using clean, dry, compressed air, blow dry main module sump chip detector/temperature sensor, magnetic element, and strainer.
- d. Check main module sump chip detector/temperature sensor electrical circuit (PARA 6-2-2).

#### 6-4-18.1.3 Install.

- a. Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing.
- b. Remove protective plug from main transmission sump.
- c. Install packing in groove on main module sump chip detector/temperature sensor and strainer. Insert main module sump chip detector/temperature sensor into sleeve in main transmission sump (Figure 6-4-18-1, Sheet 1, Detail B).
- d. Turn sleeve in main transmission sump to FLOW position as indicated by arrow (Figure 6-4-18-1, Sheet 3, Detail D), and install bolts and washers (Figure 6-4-18-1, Sheet 1, Detail B).
- e. **TORQUE BOLTS TO 105 - 110 INCH-POUNDS.**
- f. Remove protective caps and plugs from electrical connectors.
- g. Inspect and treat electrical connectors (PARA 1-7-20).



NOTE

POSITION LOOP IN HARNESS TO ALLOW ROTATION OF CHIP DETECTOR FOR REMOVAL.

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FIGURE 6-4-18-1. REPLACE MAIN MODULE SUMP CHIP DETECTOR/TEMPERATURE SENSOR (SHEET 1 OF 3)

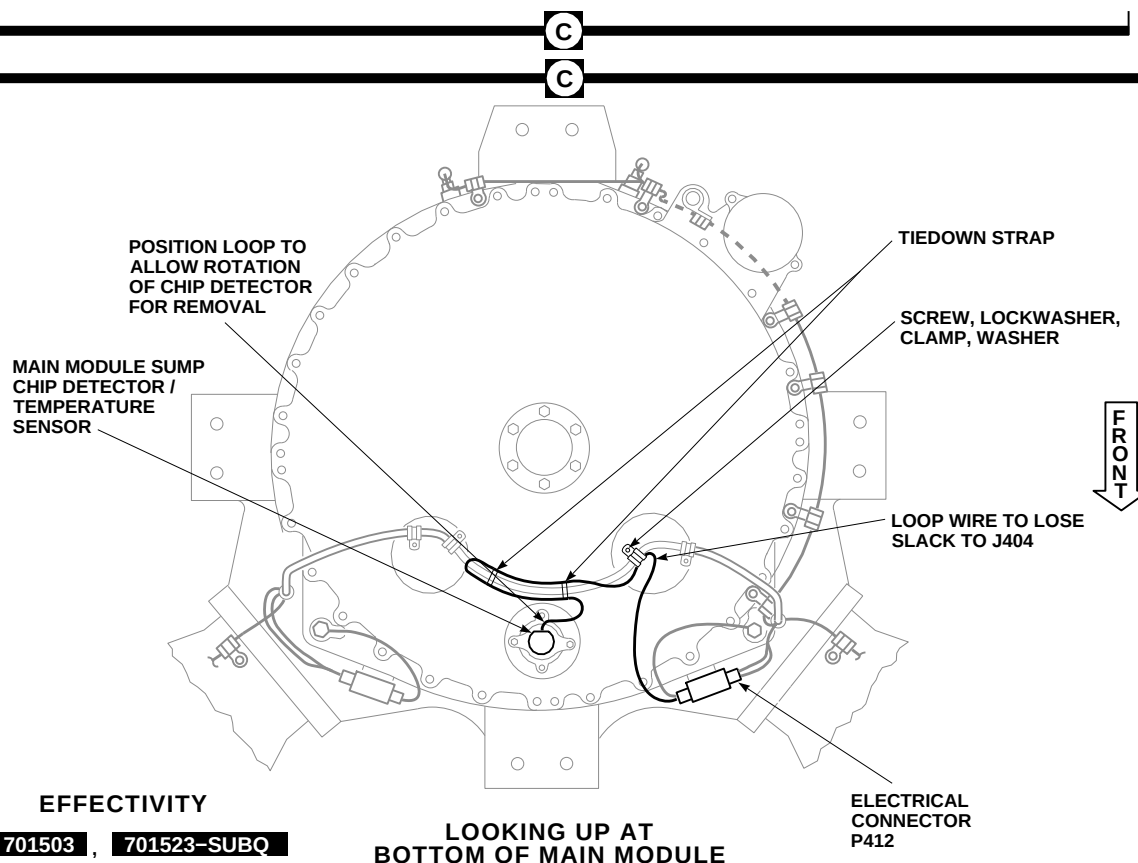
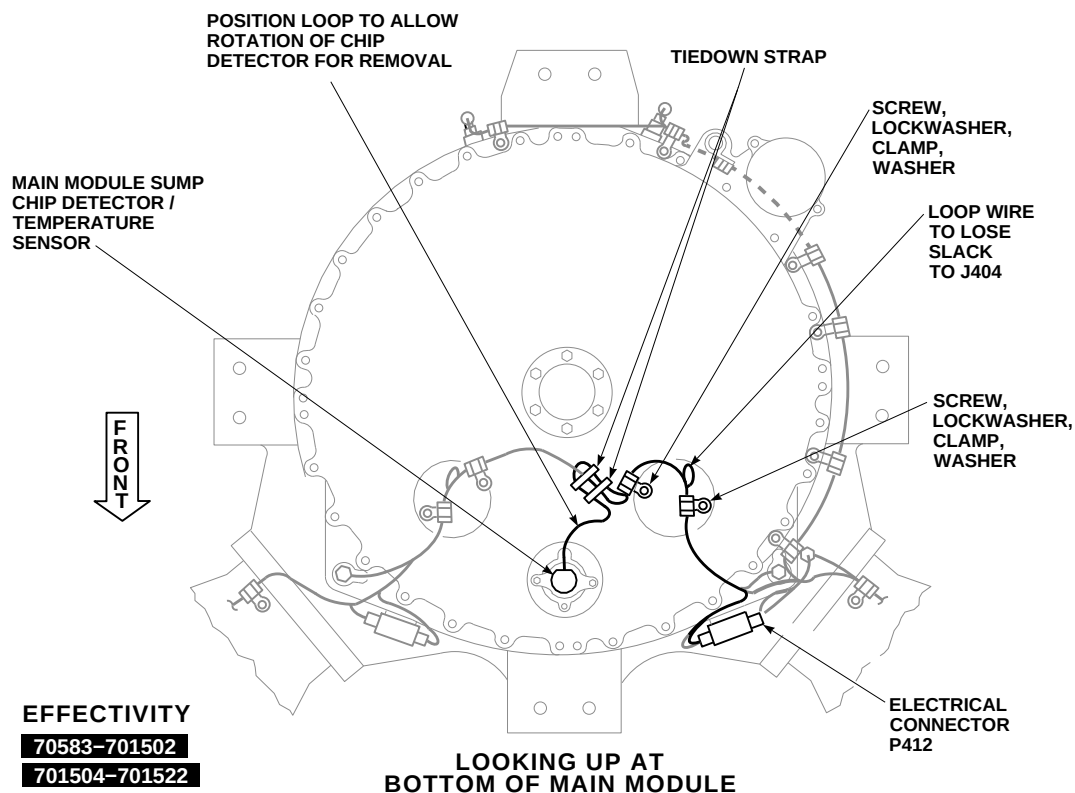
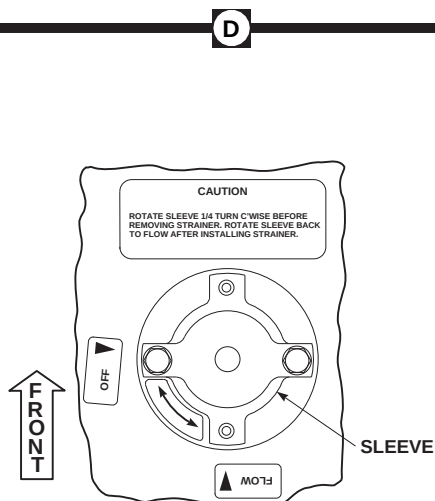
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FIGURE 6-4-18-1. REPLACE MAIN MODULE SUMP CHIP DETECTOR/TEMPERATURE SENSOR (SHEET 2 OF 3)



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**FIGURE 6-4-18-1. REPLACE MAIN MODULE SUMP CHIP DETECTOR/TEMPERATURE SENSOR (SHEET 3 OF 3)**

- |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>h. Connect electrical connector, P412, to main module sump chip detector/temperature sensor.</p> <p>i. If installing new main module sump chip detector/temperature sensor, install clamp(s) on harness.</p> | <p>l. Check oil filter for bypass condition. If in bypass, refer to PARA 6-3-5.</p> <p>m. Check main transmission for proper oil level. If necessary, service main transmission (PARA 1-3-12).</p> <p>n. Make sure area is clean and free of foreign material.</p> <p>o. Install main transmission drip pan (PARA 2-4-70).</p> <p>p. After main transmission chip detectors, oil strainer screens, and filters have been checked, perform the following:</p> <p>(1) Refer to PARA 6-3-6 to determine serviceability of main transmission.</p> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**NOTE**

Position loop in harness to allow rotation of chip detector for removal.

- j. Route harness and secure to main module with screw(s), lockwasher(s), washer(s), and clamp(s) (Figure 6-4-18-1, Sheet 2, Detail C).
- k. Using tiedown straps, Item 329, Appendix D, secure slack in harness.

- (2) Either continue main transmission in service, perform main transmission serviceability inspection (PARA 6-3-5),

or replace module, as indicated in PARA 6-3-6.



6-4-19 MAIN TRANSMISSION OIL FILTER.

PARAGRAPH BREAKDOWN

|          |                                           | PAGE     |          |                | PAGE     |
|----------|-------------------------------------------|----------|----------|----------------|----------|
| 6-4-19.1 | Replace Main Transmis-<br>sion Oil Filter | 6-4-19-1 | 6-4-19.2 | Replace Sleeve | 6-4-19-9 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Quart
- Extension, 6-inch long, ¼-inch drive
- Nonmetallic Drift
- Positioner Tool, Locally-Made, Figure H-257, Appendix H
- Powertrain Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D
- Lubricating Oil, Item 208, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- Appendix H
- PARA 1-3-12
- PARA 2-4-70

6-4-19.1 Replace Main Transmission Oil Filter.

6-4-19.1.1 Remove.

NOTE

On helicopters with main transmission sumps, 70351-38140 and 70351-48140,

- make sure helicopter is level or slightly nose down when replacing main transmission oil filter.
- Remove main transmission drip pan (PARA 2-4-70).

**NOTE**

On helicopters with main transmission sumps, 70351-38140 and 70351-48140, main transmission oil filter is installed on main transmission sump with three bolts.

- b. Remove bolts and washers securing main transmission oil filter to main transmission sump (Figure 6-4-19-1, Sheet 1, Detail B).

**NOTE**

- On helicopters with main transmission sumps, 70351-08140, main transmission oil filter will turn with sleeve.
  - On helicopters with main transmission sumps, 70351-38140 and 70351-48140, no sleeve is installed in main transmission sump.
- c. On helicopters with main transmission sumps, 70351-08140, turn main transmission oil filter ¼-turn clockwise.
  - d. On helicopters with main transmission sumps, 70351-38140 and 70351-48140, turn main transmission oil filter ½-turn.
  - e. Position 1-quart container to catch residual oil.

**NOTE**

On helicopters with main transmission sumps, 70351-38140 and 70351-48140, valve may or may not remain in main transmission sump. If valve remains in main transmission sump, it is not necessary to remove valve.

- f. Pull main transmission oil filter out of main transmission sump and catch any oil with 1-quart container.
- g. On main transmission oil filters, 70351-08143-101 or -104 (from main transmission sump, 70351-08140), depress spring clips (-101 only), and remove valve and packing from filter housing (Figure 6-4-19-1, Sheet 2, Detail C). Discard packing.

- h. On main transmission oil filter, 70351-38801-101 (from main transmission sump, 70351-38140 or -48140), if valve remained with main transmission oil filter, remove valve and packing from filter housing (Figure 6-4-19-1, Sheet 2, Detail D). Discard packing.
- i. Remove first and second stage filter elements and packings. Discard packings (Figure 6-4-19-1, Sheet 2, Detail C or Detail D).
- j. If required, remove pressure indicator as follows (Figure 6-4-19-1, Sheet 2, Detail E):
  - (1) Remove retaining ring securing pressure indicator to filter housing.
  - (2) Push pressure indicator into filter housing and remove.
  - (3) Remove packing and packing retainer from pressure indicator. Discard packing and packing retainer.
- k. On helicopters with main transmission sump, 70351-08140, if required, replace sleeve (PARA 6-4-19.2).

**6-4-19.1.2 Install.**

- a. Using dry-cleaning solvent, Item 124, Appendix D, clean filter housing.
- b. If required, install pressure indicator as follows (Figure 6-4-19-1, Sheet 2, Detail E):
  - (1) Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing and packing retainer and install on pressure indicator.
  - (2) Install pressure indicator in filter housing.
  - (3) Using steady finger pressure, rotate pressure indicator during installation. If resistance suddenly increases, then returns to normal during installation, remove pressure indicator and inspect for cut packing or packing retainer. If necessary, replace packing and/or packing retainer.

### NOTE

Make sure retaining ring secures pressure indicator in position.

- (4) Install retaining ring in groove in pressure indicator on bottom of filter housing. Push pressure indicator toward filter housing.
- c. On filter housing of main transmission oil filter, 70351-08143-101, reset pressure indicator as follows (Figure 6-4-19-1, Sheet 2, Detail E):
  - (1) Using 6-inch long, 1/4-inch drive extension, press spring inside filter housing down and hold.
  - (2) Push pressure indicator up and hold.
  - (3) Release spring, then release pressure indicator.
- d. On filter housing of main transmission oil filter, 70351-38801-101 or 70351-081431-104, reset pressure indicator as follows (Figure 6-4-19-1, Sheet 2, Detail E):
  - (1) Turn filter housing upside down.
  - (2) Press pressure indicator into filter housing.
- e. On helicopters with main transmission oil filter, 70351-08143-101, assemble and install main transmission oil filter in main transmission sump, 70351-08140, as follows (Figure 6-4-19-1, Sheet 2, Detail C):

### CAUTION

Damage to main transmission will result if incorrect filter element is installed. Incorrect filter element will allow chips to enter main transmission. Make sure correct filter elements are used when installing main transmission oil filters.

### NOTE

First and second stage oil filter elements are throwaway-type and must be replaced.

- (1) Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing and install in first stage filter element.
- (2) Install first stage filter element in filter housing.
- (3) Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing and install in second stage filter element.
- (4) Install second stage filter in first stage filter element.
- (5) Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packings and install on valve.

### NOTE

If installed, make sure tips of spring clips are seated in filter housing groove.

- (6) Depress spring clips and place valve in filter housing.
- (7) Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing and install on filter housing (Figure 6-4-19-1, Sheet 1, Detail B).
- (8) Push main transmission oil filter into sleeve.

### CAUTION

Damage to main transmission will result if arrow on sleeve does not line up with FLOW position arrow.

- (9) Locally make positioner tool (Figure H-257, Appendix H).
- (10) Using positioner tool, turn sleeve counterclockwise, until arrow on sleeve lines up with FLOW position arrow.
- (11) Install bolts and washers.

**(12) TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

- f. On helicopters with main transmission oil filter, 70351-08143-104, assemble and install main transmission oil filter in main transmission sump, 70351-08140, as follows (Figure 6-4-19-1, Sheet 2, Detail C):

- (1) Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packings and install in first stage filter element.
- (2) Install first stage filter element in filter housing.
- (3) Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing and install in second stage filter element.

**CAUTION**

Second stage filter element must be installed in correct orientation or gear box oil blockage and pressure drop will occur. Top of second stage filter element is the end where packing is installed and top end is inserted into valve.

- (4) Install second stage filter element in first stage filter element.
- (5) Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing and install on valve.
- (6) Install valve in filter housing.
- (7) Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing and install on filter housing (Figure 6-4-19-1, Sheet 1, Detail B).
- (8) Install main transmission oil filter into sleeve in main transmission sump.

**CAUTION**

Damage to main transmission will result if arrow on sleeve does not line up with FLOW position arrow.

- (9) Locally make positioner tool (Figure H-257, Appendix H).
- (10) Using positioner tool, turn sleeve counterclockwise, until arrow on sleeve lines up with FLOW position arrow.
- (11) Install bolts and washers.
- (12) **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

**NOTE**

On helicopters with main transmission sumps, 70351-38140 and 70351-48140, if valve remained in main transmission sump, proceed to step h.

- g. On main transmission oil filter, 70351-38801-101 (from main transmission sump, 70351-38140 or -48140), if valve remained with main transmission oil filter, perform the following (Figure 6-4-19-1, Sheet 2, Detail D):

**CAUTION**

Damage to main transmission will result if incorrect filter element is installed. Incorrect filter element will allow chips to enter main transmission. Make sure correct filter elements are used when installing main transmission oil filters.

**NOTE**

First and second stage oil filter elements are throwaway-type and must be replaced.

- (1) Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packings and install in first stage filter element.

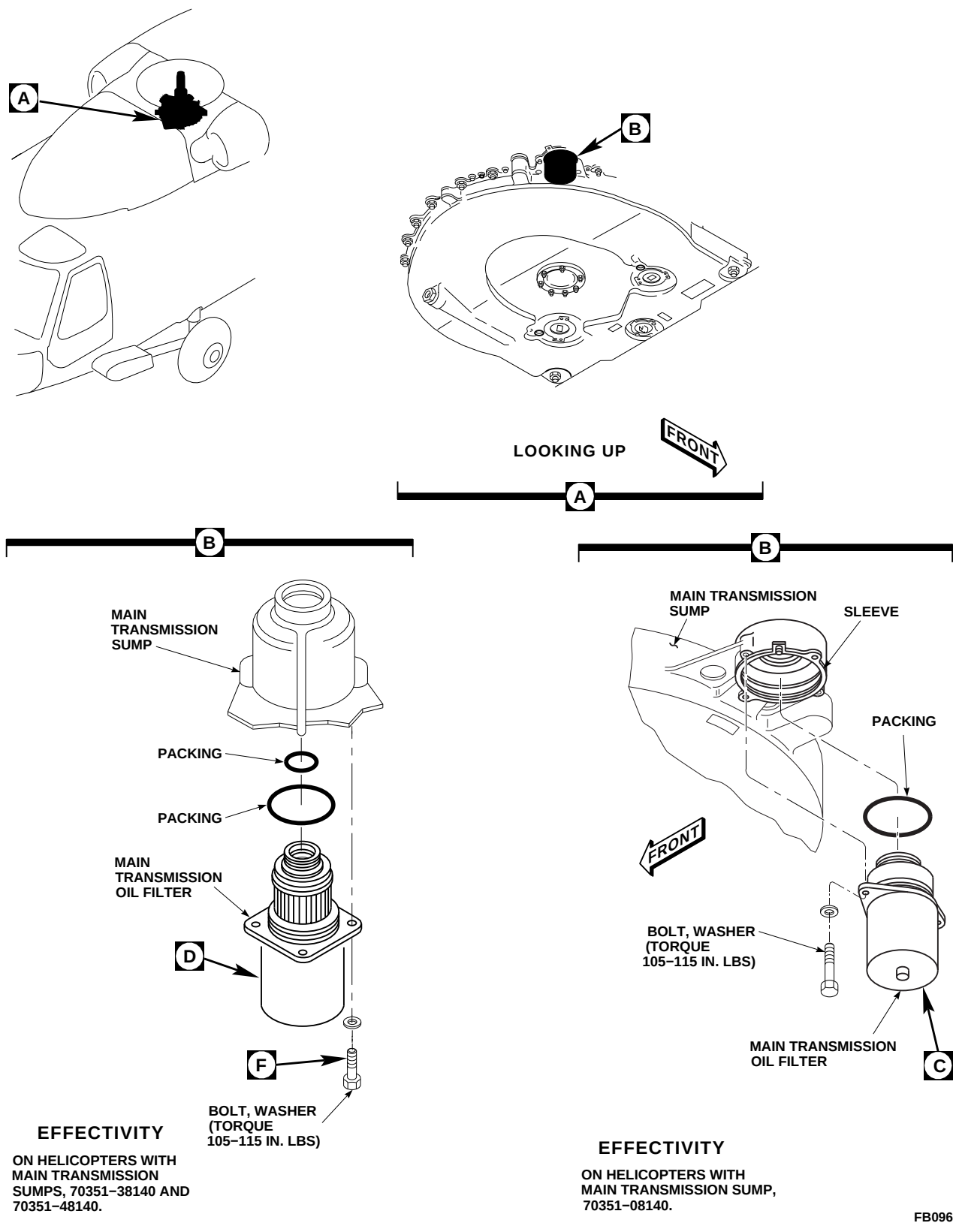


FIGURE 6-4-19-1. REPLACE MAIN TRANSMISSION OIL FILTER (SHEET 1 OF 3)

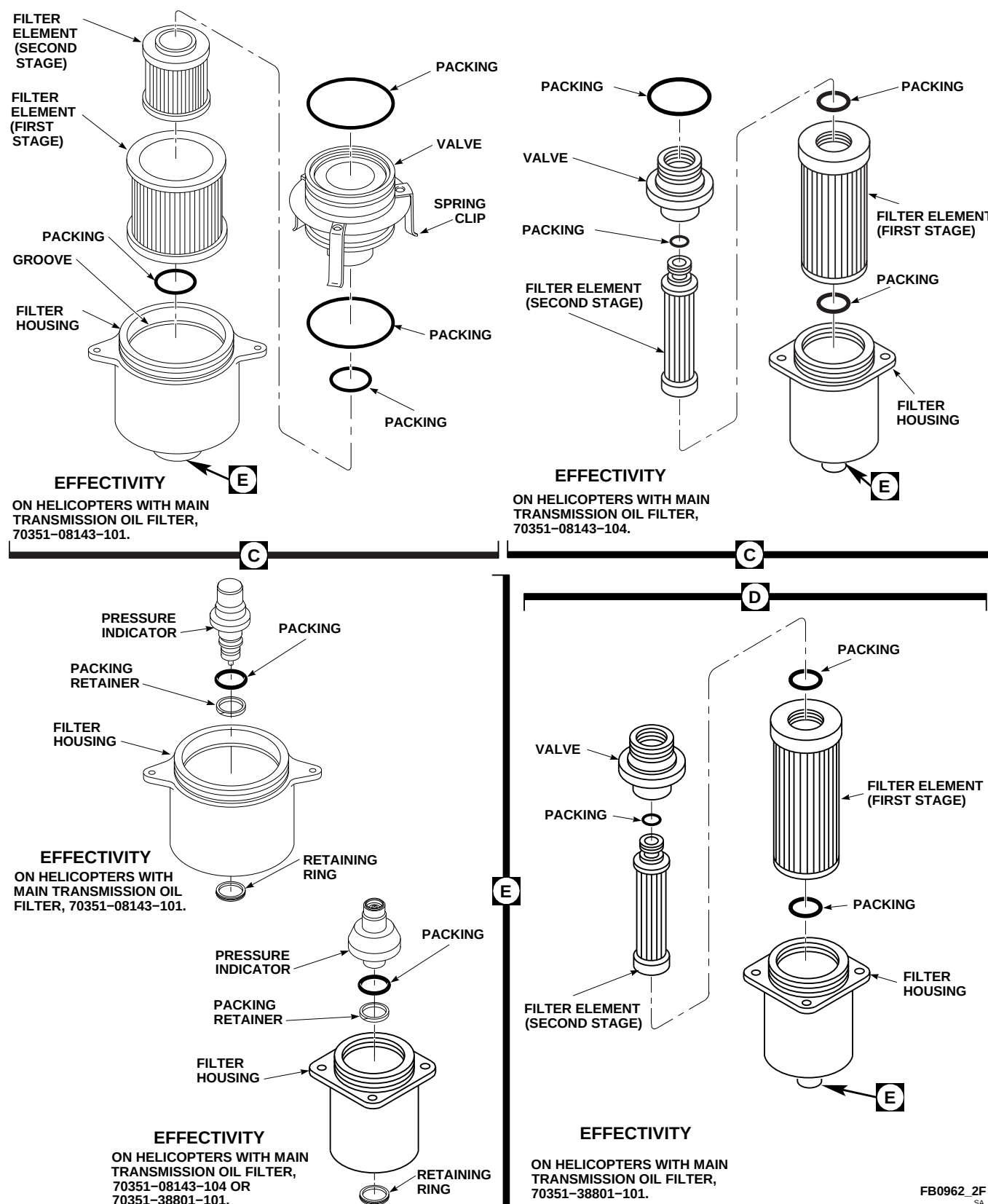
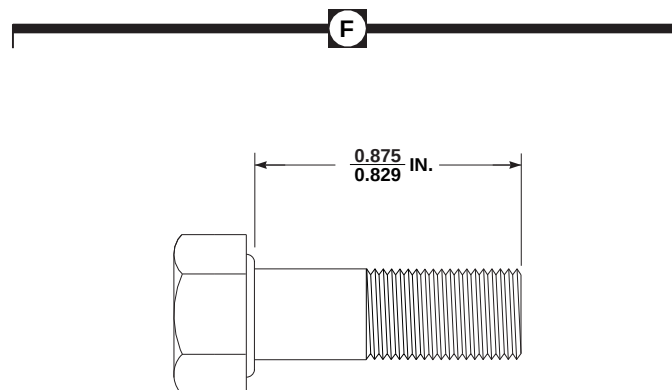


FIGURE 6-4-19-1. REPLACE MAIN TRANSMISSION OIL FILTER (SHEET 2 OF 3)



**BOLT, MS21094-4006**

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**FIGURE 6-4-19-1. REPLACE MAIN TRANSMISSION OIL FILTER (SHEET 3 OF 3)**

- (2) Install first stage filter element in filter housing.
- (6) Install valve on filter housing (Figure 6-4-19-1, Sheet 2, Detail D).

- (3) Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing and install in second stage filter element.



Second stage filter element must be installed in correct orientation or gear box oil blockage and pressure drop will occur. Top of second stage filter element is end where packing is installed and top end is inserted into valve.

- (4) Install second stage filter element in first stage filter element.
- (5) Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packings and install on filter housing (Figure 6-4-19-1, Sheet 1, Detail B).

### WARNING

Injury to personnel and damage to equipment may result if incorrect bolt is used to secure main transmission oil filter. Incorrect bolt length can result in complete loss of hot transmission fluid, causing possible injury to personnel and damage to main transmission. Make sure bolt length is verified prior to installing main transmission oil filter.

### NOTE

Bolt part number, MS21094-4006, is not identified on bolt.

- (7) Verify bolt length by measuring from bottom face of bolthead to bottom face of threaded end (Figure 6-4-19-1, Sheet 3, Detail F). **LENGTH MUST BE**

**BETWEEN 0.829 - 0.875-INCH.** If bolt is not within limits, replace bolt.

- (8) Install assembled main transmission oil filter in main transmission sump with bolts and washers (Figure 6-4-19-1, Sheet 1, Detail B).

- (9) **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

- h. On main transmission oil filter, 70351-38801-101, if valve remained in main transmission sump, 70351-38140 or -48140, perform the following (Figure 6-4-19-1, Sheet 2, Detail D):

### CAUTION

Damage to main transmission will result if incorrect filter element is installed. Incorrect filter element will allow chips to enter main transmission. Make sure correct filter elements are used when installing main transmission oil filter.

### NOTE

First and second stage oil filter elements are throwaway-type and must be replaced.

- (1) Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing and install on second stage filter element.

### CAUTION

Second stage filter element must be installed in correct orientation or gear box oil blockage and pressure drop will occur. Top of second stage filter element is end where packing is installed and top end is inserted into valve.

- (2) Install second stage filter element on valve in main transmission sump.

- (3) Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packings and install in first stage filter element.
- (4) Install first stage filter element on valve in main transmission sump.
- (5) Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packings and install on filter housing (Figure 6-4-19-1, Sheet 1, Detail B).

### WARNING

Injury to personnel and damage to equipment may result if incorrect bolt is used to secure main transmission oil filter. Incorrect bolt length can result in complete loss of hot transmission fluid causing possible injury to personnel and damage to main transmission. Make sure bolt length is verified prior to installing main transmission oil filter.

### NOTE

Bolt part number, MS21094-4006, is not identified on bolt.

- (6) Verify bolt length by measuring from bottom face of bolthead to bottom face of threaded end (Figure 6-4-19-1, Sheet 3, Detail F). **LENGTH MUST BE BETWEEN 0.829 - 0.875-INCH.** If bolt is not within limits, replace bolt.
- (7) Install filter housing, with packings, in main transmission sump with bolts and washers (Figure 6-4-19-1, Sheet 1, Detail B).
- (8) **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**
- i. Check main transmission for proper oil level. If necessary, service main transmission (PARA 1-3-12).
- j. Make sure area is clean and free of foreign material.



- k. Perform check of main transmission oil filter as follows:
  - (1) With qualified person at controls, run up helicopter.
  - (2) Check main transmission oil filter and pressure indicator for leaks.
- l. Install main transmission drip pan (PARA 2-4-70).
- m. Make an entry in helicopter logbook noting when main transmission oil filter was replaced.

## 6-4-19.2 Replace Sleeve.

### 6-4-19.2.1 Remove.

#### NOTE

This procedure applies to helicopters with main transmission sump, 70351-08140.

- a. If necessary, remove main transmission oil filter (PARA 6-4-19.1.1).
- b. Turn sleeve so dowel pin lines up with hole in sleeve (Figure 6-4-19-2, Detail B).

#### CAUTION

Damage to main transmission sump will occur if care is not taken when removing dowel pin. When removing dowel pin, be careful not to damage main transmission sump.

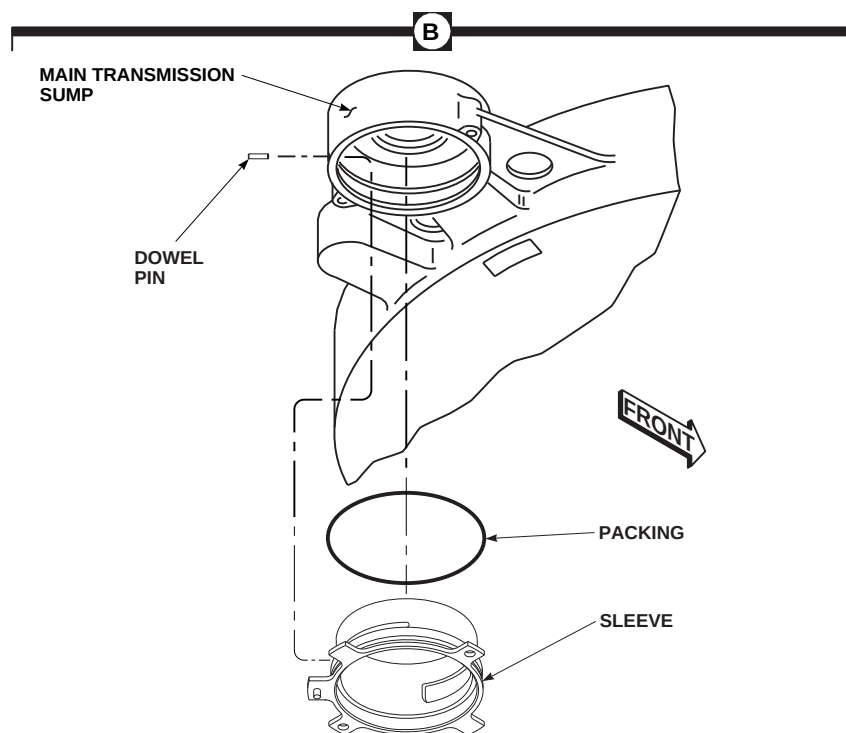
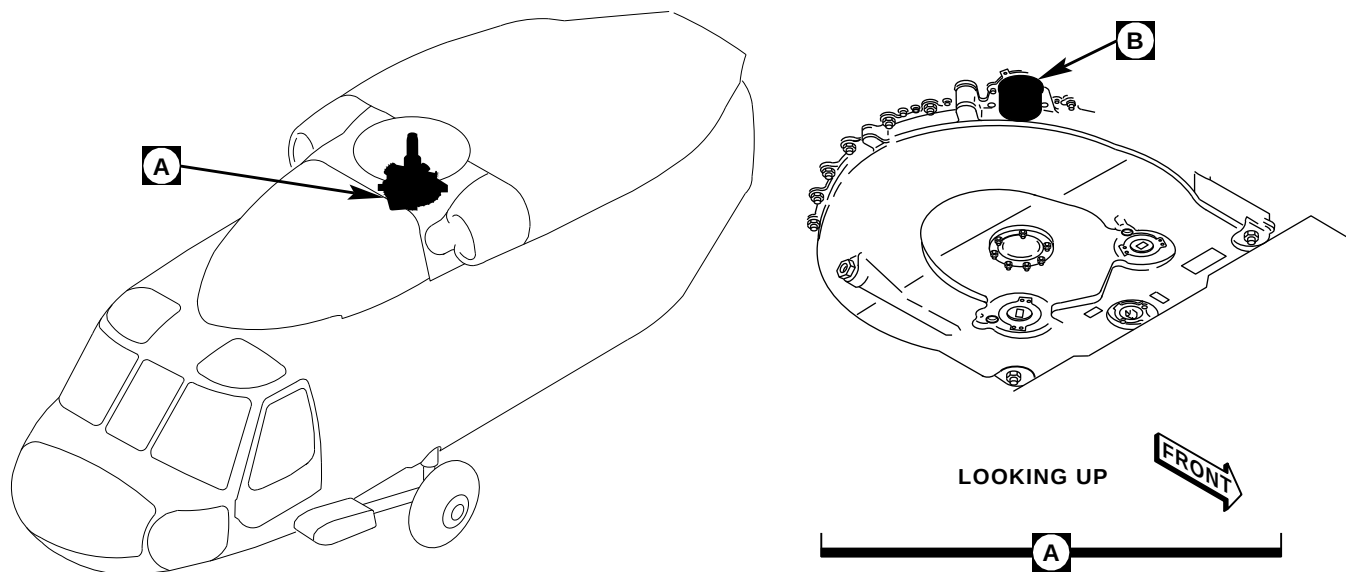
- c. Using nonmetallic drift, remove dowel pin by pushing through main transmission sump.
- d. Remove sleeve and packing from main transmission sump. Discard packing.

### 6-4-19.2.2 Inspect.

Visually inspect inside diameter (ID) of sleeve for size and damage. **ID IS NOT TO EXCEED 0.1245 - 0.1250-INCH. NO DAMAGE ALLOWED.** If ID is beyond limits, or damage is found, replace sleeve.

### 6-4-19.2.3 Install.

- a. Using lubricating oil, Item 205, lubricate packing and install on sleeve (Figure 6-4-19-2, Detail B).
- b. Install sleeve in main transmission sump, aligning hole in sleeve with hole in main transmission sump.
- c. Install new dowel pin with outside diameter of 0.1256 - 0.1261-inch.
- d. Apply coat of sealing compound, Item 292, to dowel pin.
- e. Using nonmetallic drift, install dowel pin until dowel pin is flush to 0.010-inch below with main transmission sump.
- f. Remove excess sealing compound from inside sleeve.
- g. Install main transmission oil filter (PARA 6-4-19.1.2).



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FIGURE 6-4-19-2. REPLACE SLEEVE

## 6-4-20 MAIN TRANSMISSION OIL PUMP.

## PARAGRAPH BREAKDOWN

|          |                                    | PAGE     |
|----------|------------------------------------|----------|
| 6-4-20.1 | Replace Main Transmission Oil Pump | 6-4-20-1 |

## INITIAL SETUP

## Tools

- Aircraft Mechanic's Toolkit
- Bolt, NAS563-65 (2-inch minimum thread length)
- Container, 1-Quart
- Oil Pump Installation Tool, Locally-Made, Figure H-45, Appendix H
- Powertrain Repairer's Toolkit
- Rawhide Mallet
- Torque Wrench, 30 - 150 in. lbs

## Materials/Parts

- Assembly Fluid, Item 79, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D
- Lubricating Oil, Item 208, Appendix D

## References

- Appendix D
- Appendix H
- PARA 1-3-12
- PARA 2-4-70
- PARA 6-2-2

### 6-4-20.1 Replace Main Transmission Oil Pump.

## NOTE

Replacement of both main transmission oil pumps is the same.

**6-4-20.1.1. Remove.**

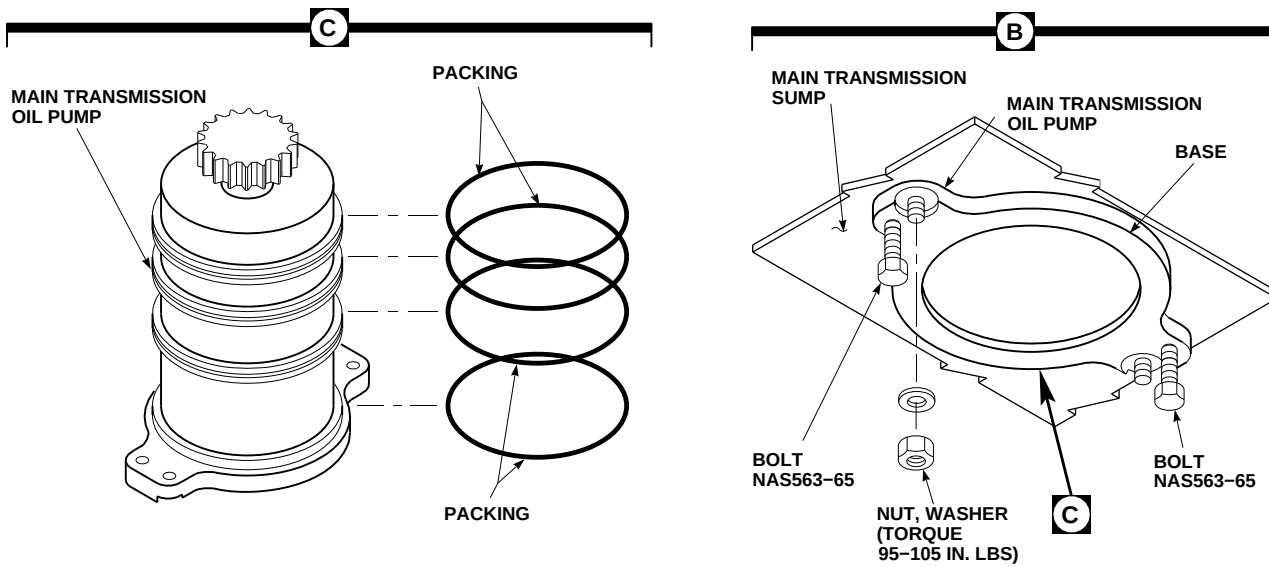
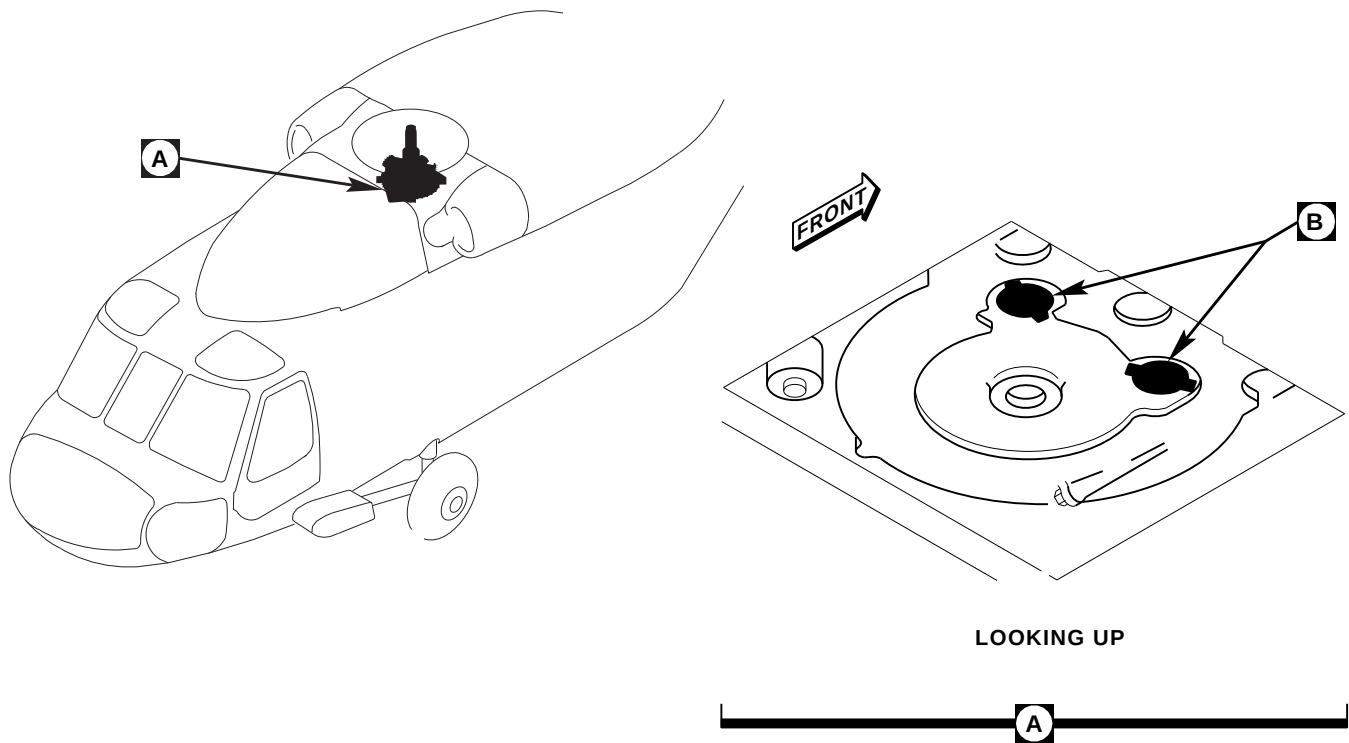
- a. Remove main transmission drip pan (PARA 2-4-70).
- b. Drain main transmission (PARA 1-3-12).
- c. Remove nuts and washers securing main transmission oil pump (Figure 6-4-20-1, Detail B).

- d. Position 1-quart container to catch any residual oil.

## NOTE

- Bolts are used as jacking bolts and, when tightened, push main transmission oil pump away from main transmission sump.
  - Minimum threaded length of bolts is 2-inches (NAS563-65 or longer).
- e. Install two bolts, NAS563, with at least a 2-inch thread (NAS563-65, or longer) in threaded holes of base.

**6-4-20-1**



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FIGURE 6-4-20-1. REPLACE MAIN TRANSMISSION OIL PUMP

6-4-20-2

- f. **TIGHTEN BOLTS EVENLY TO DRAW MAIN TRANSMISSION OIL PUMP AWAY FROM MAIN TRANSMISSION SUMP.**
- g. Remove main transmission oil pump and packings. Discard packings (Figure 6-4-20-1, Detail C).
- h. Remove jacking bolts from main transmission oil pump.

#### 6-4-20.1.2. Install.

- a. Locally make oil pump installation tool (Figure H-45, Appendix H).
- b. Using assembly fluid, Item 79, Appendix D, and lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packings.
- c. Install packings on main transmission oil pump (Figure 6-4-20-1, Detail C).



Damage to main transmission oil pump housing will result if attaching nuts are

used to draw main transmission oil pump into main transmission sump. Turn rotor head to align splines and install main transmission oil pump.

- d. Push main transmission oil pump into main transmission sump. Using oil pump installation tool and rawhide mallet, seat main transmission oil pump. Tap around tube with rawhide mallet, being careful not to cock main transmission oil pump or damage packings.
- e. Install washers and nuts on main transmission oil pump (Figure 6-4-20-1, Detail B).
- f. **TORQUE NUTS TO 95 - 105 INCH-POUNDS.**
- g. Service main transmission (PARA 1-3-12).
- h. Perform operational check (PARA 6-2-2).
- i. Install main transmission drip pan (PARA 2-4-70).



6-4-21 MAIN TRANSMISSION OIL PRESSURE SENSOR.

PARAGRAPH BREAKDOWN

|                                                        | PAGE     |
|--------------------------------------------------------|----------|
| 6-4-21.1 Replace Main Transmission Oil Pressure Sensor | 6-4-21-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Quart
- Nonmetallic Scraper, Locally-Made
- Powertrain Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 197, Appendix D
- Lubricating Oil, Item 205, Appendix D

- Lubricating Oil, Item 206, Appendix D
- Lubricating Oil, Item 208, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-3-12
- PARA 1-6-16
- PARA 1-7-20
- PARA 6-3-4

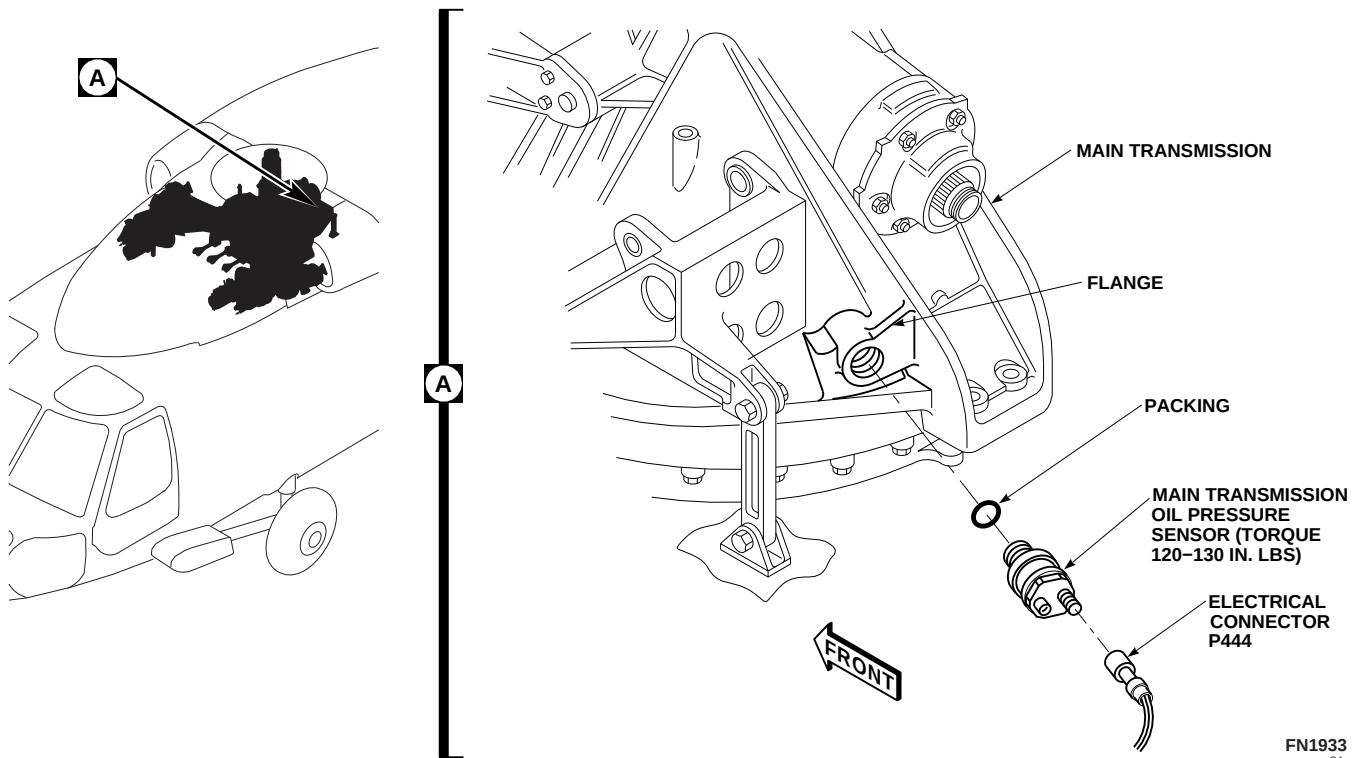
6-4-21.1 Replace Main Transmission Oil Pressure Sensor.

6-4-21.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Open oil cooler compartment access doors.
- Disconnect electrical connector, P444, from main transmission oil pressure sensor (Figure 6-4-21-1, Detail A).
- Install protective caps and plugs on electrical connectors.



- Damage to threaded port will result if excessive torque is used to remove main transmission oil pressure sensor. Make sure all sealing compound is removed prior to removing main transmission oil pressure sensor.
- Using nonmetallic scraper, remove sealing compound from area around mating flange.
  - Position 1-quart container to catch any residual oil.

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SA**FIGURE 6-4-21-1. REPLACE MAIN TRANSMISSION OIL PRESSURE SENSOR**

- g. Remove main transmission oil pressure sensor from mating flange of main transmission. Discard packing.

**6-4-21.1.2. Install.**

- Turn off all electrical power (PARA 1-6-16).
- Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing.
- Install packing on main transmission oil pressure sensor (Figure 6-4-21-1, Detail A).
- Install main transmission oil pressure sensor in mating flange of main transmission.

**CAUTION**

Damage to main transmission oil pressure sensor will result if overtightened. Oil pressure readings will be effected. Do not overtighten main transmission oil pressure sensor.

- e. **TORQUE MAIN TRANSMISSION OIL PRESSURE SENSOR TO 120 - 130 INCH-POUNDS.**

- Lockwire, Item 197, Appendix D, main transmission oil pressure sensor.
- Apply sealing compound, Item 292, Appendix D, to seal area around mating flange.
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connector (PARA 1-7-20).
- Connect electrical connector, P444, to main transmission oil pressure sensor.
- Check main transmission for proper oil level. If necessary, service main transmission (PARA 1-3-12).
- Make sure area is clean and free of foreign material.
- Close oil cooler compartment access doors.

**6-4-21-2**



- n. Perform main transmission functional test (PARA 6-3-4).



6-4-22 MAIN TRANSMISSION OIL PUMP PRESSURE REGULATING VALVES.

PARAGRAPH BREAKDOWN

|          |                                                                       | PAGE     |
|----------|-----------------------------------------------------------------------|----------|
| 6-4-22.1 | Replace Main Transmis-<br>sion Oil Pump Pressure<br>Regulating Valves | 6-4-22-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Quart
- Nonmetallic Scraper, Locally-Made
- Powertrain Repairer’s Toolkit
- Screwdriver, Broad-Tipped
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Lockwire, Item 196, Appendix D
- Lockwire, Item 197, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D

- Lubricating Oil, Item 208, Appendix D
- Protective Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-3-12
- PARA 1-6-16
- PARA 2-4-70
- PARA 6-2-2
- PARA 6-3-4
- PARA 6-3-6
- PARA 6-3-7

6-4-22.1 Replace Main Transmission Oil Pump Pressure Regulating Valves.

NOTE

Replacement of both main transmission oil pump pressure regulating valves is the same.

6-4-22.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove main transmission drip pan (PARA 2-4-70).
- Drain main transmission (PARA 1-3-12).

- d. Position 1-quart container to catch any residual oil.
- e. Using nonmetallic scraper, remove sealing compound from around main transmission oil pump pressure regulating valve.
- f. Unscrew and remove main transmission oil pump pressure regulating valve and packings from main transmission sump. Discard packings (Figure 6-4-22-1, Detail B).
- g. Install protective plug in main transmission sump.

#### 6-4-22.1.2. Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packings.
- c. Install packings on main transmission oil pump pressure regulating valve (Figure 6-4-22-1, Detail B).
- d. Remove protective plug from main transmission sump.
- e. Screw main transmission oil pump pressure regulating valve into main transmission sump until it is bottomed.
- f. **TORQUE MAIN TRANSMISSION OIL PUMP PRESSURE REGULATING VALVE TO 280 - 305 INCH-POUNDS.**
- g. Lockwire, Item 197, Appendix D, main transmission oil pump pressure regulating valve to main transmission sump.
- h. Apply sealing compound, Item 292, Appendix D, to seal area around main transmission oil pump pressure regulating valve and main transmission sump interface.
- i. Make sure area is clean and free of foreign material.
- j. Service main transmission (PARA 1-3-12).
- k. Perform main transmission functional test (PARA 6-3-4).
- l. Install main transmission drip pan (PARA 2-4-70).

#### 6-4-22.1.3. Adjust.

##### NOTE

- Normal oil pressure, stabilized after warmup, is 50 - 55 psi.
- Adjust oil pressure when engine is shut down (main transmission is 0% Nr).
- Isolate cause of pressure drops prior to adjusting pressure (PARA 6-2-2).

- a. Remove main transmission drip pan (PARA 2-4-70).
- b. If oil pressure, after warmup, is above normal operating pressure, perform the following:

##### NOTE

One complete counterclockwise turn of both screws will lower oil pressure 3 - 4 psi.

- (1) Using broad-tipped screwdriver, turn both screws counterclockwise an equal amount (Figure 6-4-22-2, Detail A).
- (2) After adjustment, screws may be adjusted up to ¼ turn counterclockwise maximum, to align slot with lockwire holes.
- (3) Lockwire, Item 196, Appendix D, screws.
- (4) Ground run engine 100% Nr and record oil pressure (PARA 6-3-4).
- (5) If oil pressure will not stabilize at 50 - 55 psi after main transmission oil pump pressure regulating valve adjustment, replace both main transmission oil pump pressure regulating valves.
- c. If oil pressure, after warmup, is below normal operating pressure, perform the following:
  - (1) Check chip detectors, strainers, and screens for contamination in main, input, and accessory modules (PARA 6-3-6).

#### 6-4-22-2

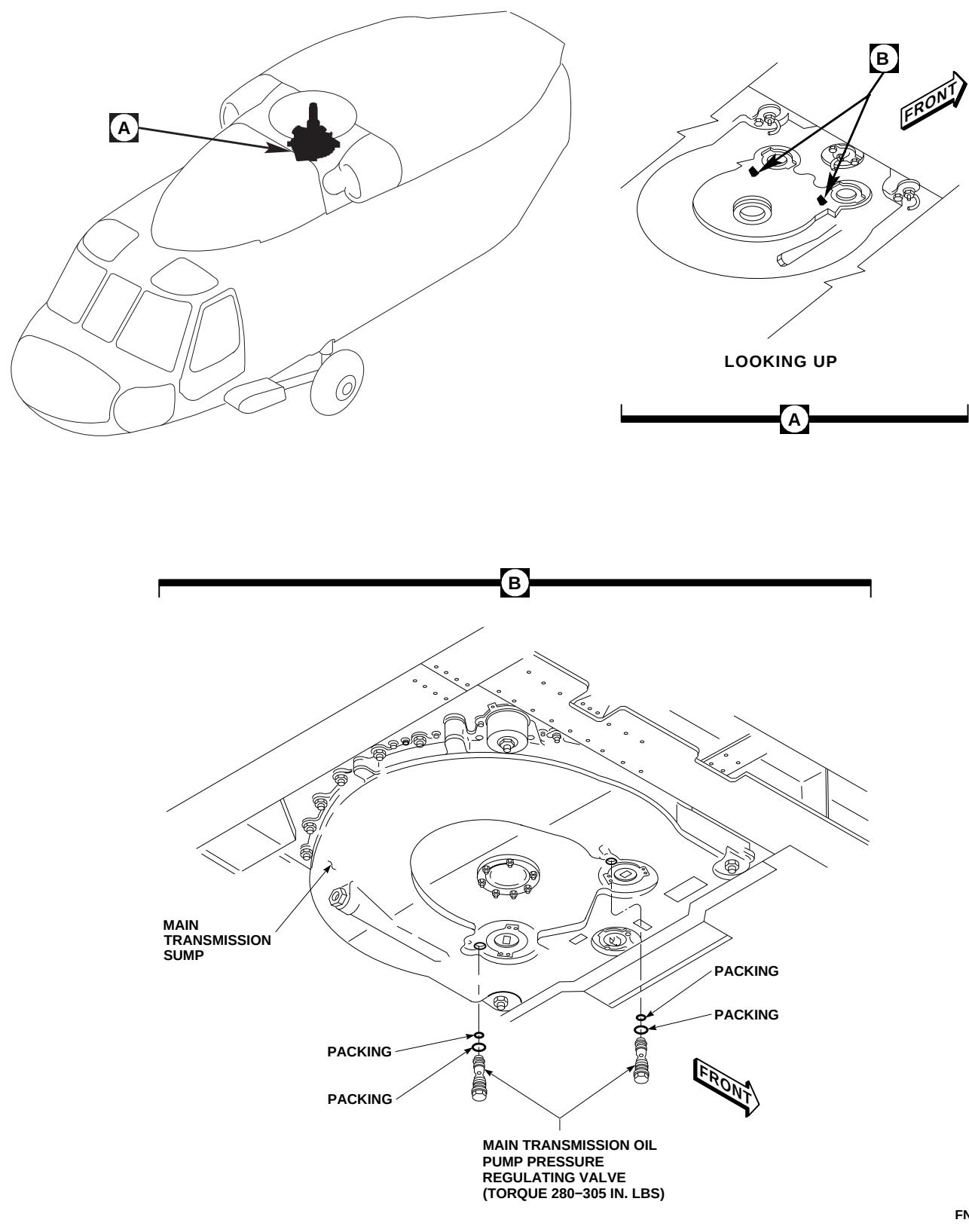
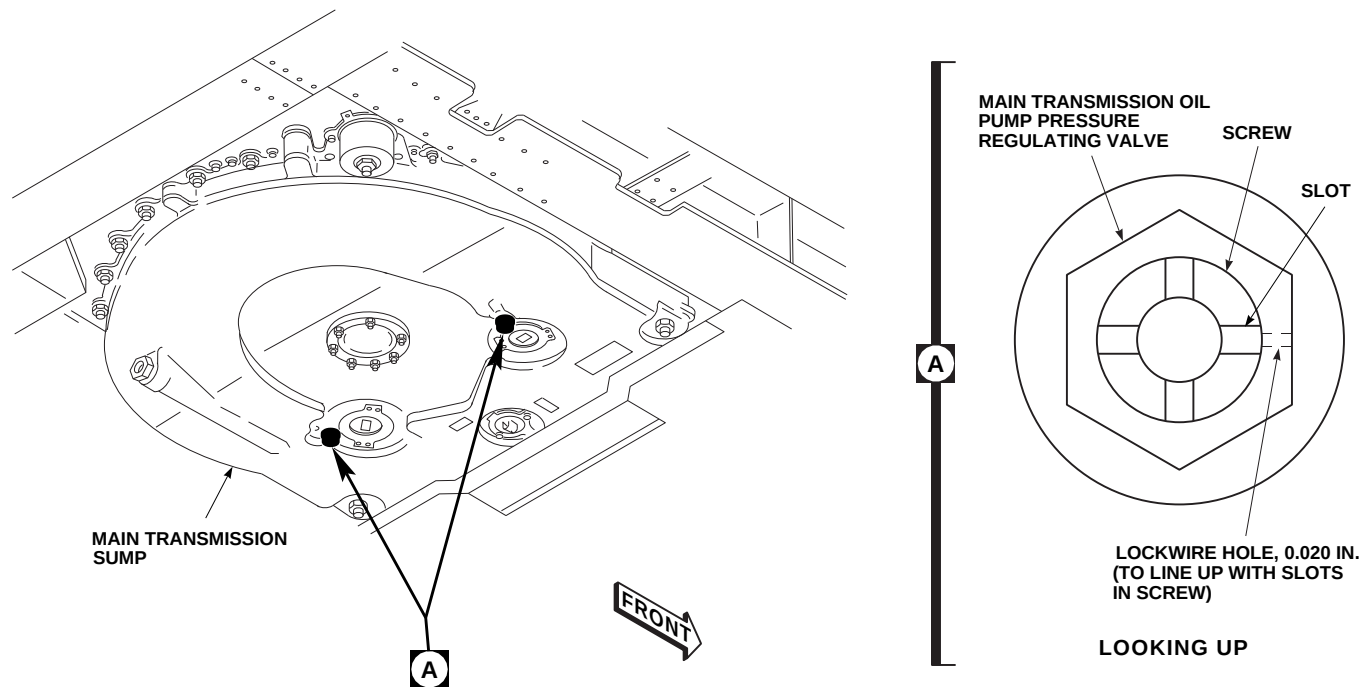


FIGURE 6-4-22-1. REPLACE MAIN TRANSMISSION OIL PUMP PRESSURE REGULATING VALVES

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**FIGURE 6-4-22-2. ADJUST MAIN TRANSMISSION OIL PUMP PRESSURE REGULATING VALVES**

- (2) Check oil filter for contamination (PARA 6-3-7).

**NOTE**

Adjust main transmission oil pump pressure regulating valves when engine is shut down (main transmission is 0% Nr).

- d. If no contamination is found, using broad-tipped screwdriver, turn both screws clockwise an equal amount.

**NOTE**

One complete clockwise turn of both screws will increase oil pressure 3 - 4 psi.

- e. After adjustment, screws may be adjusted up to 1/4 turn counterclockwise maximum, to align slot with lockwire holes.
- f. Lockwire, Item 196, Appendix D, screws.
- g. Ground run engine 100% Nr and record oil pressure (PARA 6-3-4).
- h. If oil pressure will not stabilize at 50 - 55 psi after main transmission oil pump pressure regulating valve adjustment, replace both main transmission oil pump pressure regulating valves.
- i. Install main transmission drip pan (PARA 2-4-70).

6-4-23    MAIN TRANSMISSION OIL GAGE TUBE ASSEMBLY.

PARAGRAPH BREAKDOWN

|                                                              | PAGE     |
|--------------------------------------------------------------|----------|
| 6-4-23.1    Replace Main Transmission Oil Gage Tube Assembly | 6-4-23-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Powertrain Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Lubricating Oil, Item 205, Appendix D

- Lubricating Oil, Item 206, Appendix D
- Lubricating Oil, Item 208, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Plug

References

- Appendix D
- PARA 1-6-16

6-4-23.1    Replace Main Transmission Oil Gage Tube Assembly.

6-4-23.1.1.    Remove.

- Turn off all electrical power (PARA 1-6-16).
- Open oil cooler compartment access doors.
- Push down and twist oil level gage to remove from oil gage tube (Figure 6-4-23-1, Detail A).
- Remove fitting and packing from oil gage tube. Discard packing.
- Using machinery towel, Item 211, Appendix D, wet with dry-cleaning solvent, Item 124, Appendix D, clean B-nut/fitting connection area at base of oil gage tube and main transmission sump.

- Remove nut, bolt, and clamp from bracket on main transmission.
- Loosen B-nut from fitting on main transmission and remove oil gage tube.
- Install protective plug on fitting on main transmission.
- Keep oil level gage, clamp, and hardware for installation.

6-4-23.1.2.    Install.

- Turn off all electrical power (PARA 1-6-16).
- Remove protective plug from fitting and clean threads using machinery towel, Item 211, Appendix D, wet with dry-cleaning solvent, Item 124, Appendix D. Use care not to contaminate oil reservoir.

- c. Position and install oil gage tube onto fitting (Figure 6-4-23-1, Detail A).
- d. Position and assemble clamp to bracket on main transmission and secure with bolt and nut.
- e. **TORQUE NUT TO 105 - 115 INCH-POUNDS.**
- f. **TIGHTEN B-NUT. TORQUE B-NUT TO 460 - 500 INCH-POUNDS.**
- g. Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing.
- h. Install packing on fitting.

- i. Install fitting with packing into oil gage tube.

**NOTE**

Lockwire not required for fitting.

- j. **TORQUE FITTING TO 30 - 108 INCH-POUNDS.**
- k. Install oil level gage into oil gage tube. Push down and twist oil level gage to lock into fitting.
- l. Make sure area is clean and free of foreign material.
- m. Close oil cooler compartment access doors.



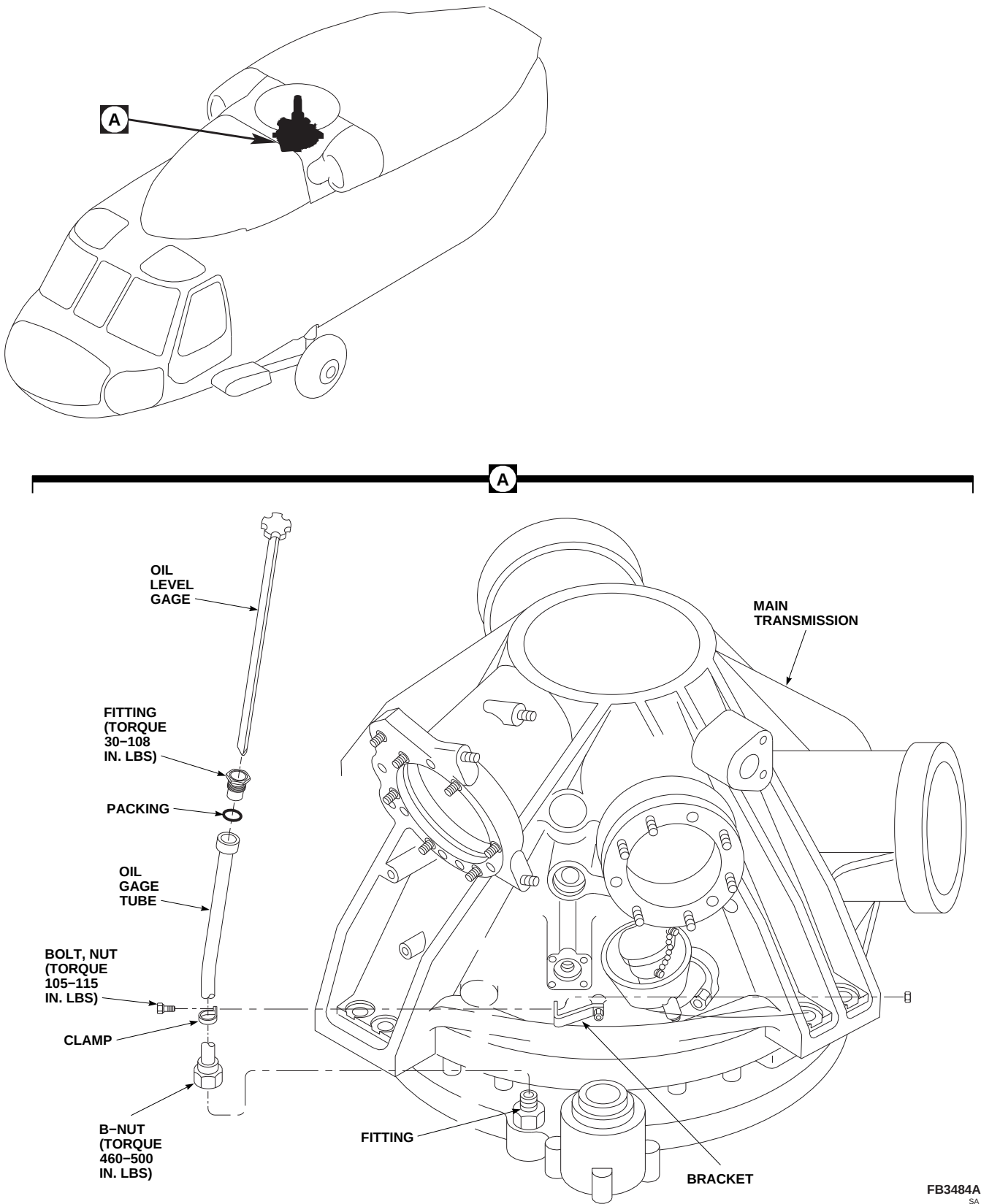


FIGURE 6-4-23-1. REPLACE MAIN TRANSMISSION OIL GAGE TUBE ASSEMBLY

6-4-23-3/(6-4-23-4 Blank)



6-4-24 MAIN TRANSMISSION OIL TEMPERATURE SWITCH.

PARAGRAPH BREAKDOWN

|          |                                                          | PAGE     |
|----------|----------------------------------------------------------|----------|
| 6-4-24.1 | Replace Main Transmis-<br>sion Oil Temperature<br>Switch | 6-4-24-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Scraper, Locally-Made
- Powertrain Repairer’s Toolkit

Materials/Parts

- Lockwire, Item 197, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D

- Lubricating Oil, Item 208, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 6-3-4

6-4-24.1 Replace Main Transmission Oil Temperature Switch.

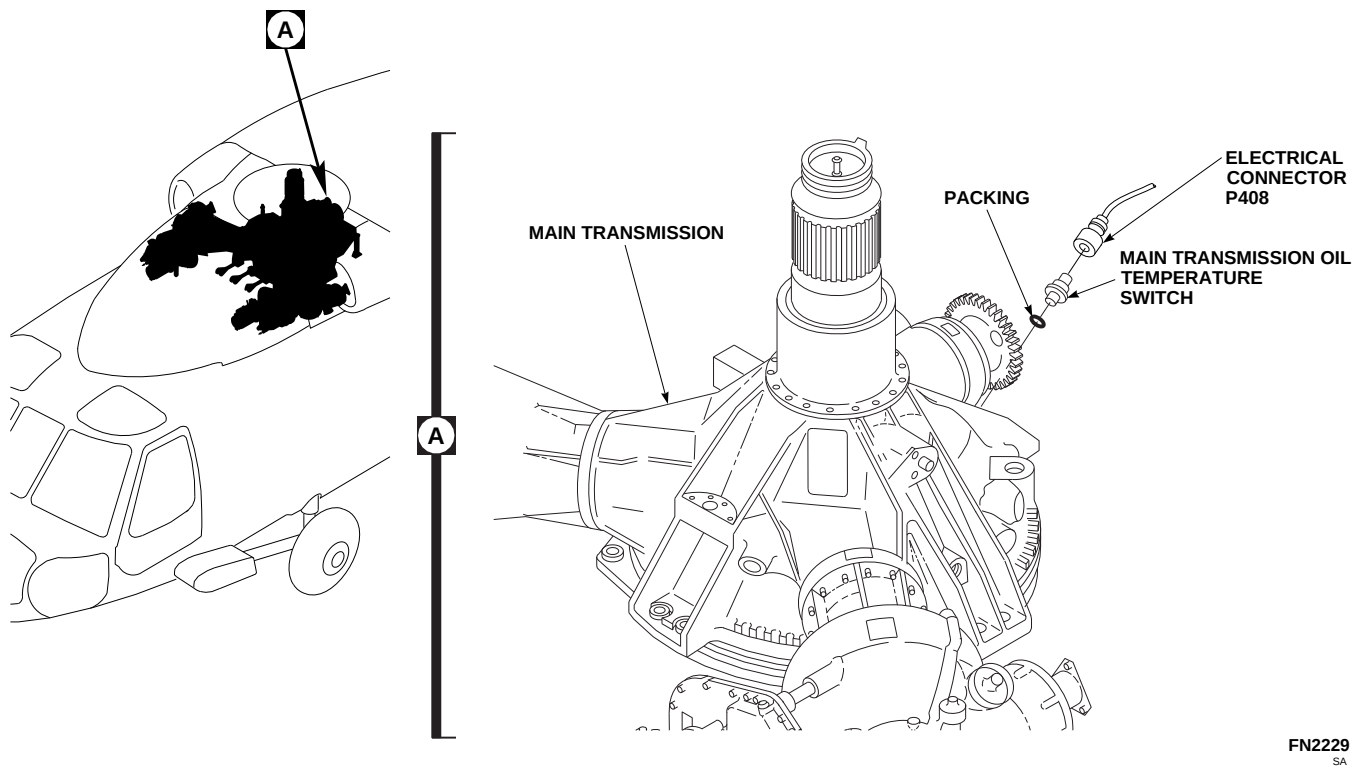
6-4-24.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Open oil cooler compartment access doors.
- Disconnect electrical connector, P408, from main transmission oil temperature switch (Figure 6-4-24-1, Detail A).
- Install protective caps and plugs on electrical connectors.
- Using nonmetallic scraper, remove sealing compound from around mating flange.

- Remove main transmission oil temperature switch and packing from mating flange of main transmission. Discard packing.

6-4-24.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing.
- Install packing on main transmission oil temperature switch (Figure 6-4-24-1, Detail A).
- Install main transmission oil temperature switch in mating flange of main transmission.
- Lockwire, Item 197, Appendix D, main transmission oil temperature switch.

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SA**FIGURE 6-4-24-1. REPLACE MAIN TRANSMISSION OIL TEMPERATURE SWITCH**

- f. Apply sealing compound, Item 292, Appendix D, to seal area around mating flange.
- g. Remove protective caps and plugs from electrical connectors.
- h. Inspect and treat electrical connector (PARA 1-7-20).
- i. Connect electrical connector, P408, to main transmission oil temperature switch.
- j. Make sure area is clean and free of foreign material.
- k. Close oil cooler compartment access doors.
- l. Perform main transmission functional test (PARA 6-3-4).

6-4-25 INPUT MODULE.

PARAGRAPH BREAKDOWN

|                               | PAGE     |
|-------------------------------|----------|
| 6-4-25.1 Replace Input Module | 6-4-25-2 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Hoist, Capacity to Support Input Module (Alternate for Cabin Maintenance Crane)
- Hoisting Sling, 70700-20320-042
- Jacking Bolts, 1/4-28
- Nonmetallic Scraper, Locally-Made
- Powertrain Repairer’s Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Universal Hoist Sling Assembly, 70700-20320-045
- Wrench, 1 3⁄8-inch, Open End

Materials/Parts

- Chemical Conversion Coating, Item 118, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Lockwire, Item 197, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D

- Lubricating Oil, Item 208, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint, Polyurethane, Item 224, Appendix D
- Petrolatum, Item 234, Appendix D
- Primer, Item 258, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D
- Tape, Item 317C, Appendix D
- Tiedown Strap, Item 329, Appendix D

References

- Appendix D
- PARA 1-3-12
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-106
- PARA 2-6-5
- PARA 4-4-1
- PARA 4-4-4
- PARA 4-4-6
- PARA 6-2-2

- PARA 6-3-4
- PARA 6-3-8
- PARA 6-4-15
- PARA 6-4-16
- PARA 6-4-17
- PARA 6-4-18
- PARA 6-4-28
- PARA 7-2-1
- PARA 7-4-12
- PARA 9-2-1
- PARA 9-4-16
- PARA 18-4-1

### 6-4-25.1 Replace Input Module.

#### NOTE

- Replacement of left or right input modules is identical, except as noted.
- Input module with accessory module installed, but without hydraulic pump module and main generator installed, weighs 140 pounds.

#### 6-4-25.1.1 Remove.

- a. If cabin maintenance crane will be used, assemble and erect crane (PARA 18-4-1).
- b. Turn off all electrical power (PARA 1-6-16).
- c. Open main rotor pylon sliding cover.
- d. Open engine cowlings.
- e. Remove engine air inlets (PARA 4-4-4).
- f. Remove main rotor pylon air inlet fairings (PARA 2-4-106).
- g. Disconnect engine output shaft from input module flange (PARA 4-4-6).
- h. Disconnect forward support tube from elastomeric gimbal on input module (PARA 4-4-6).

#### NOTE

When removing right input module, both hydraulic pump module and main

generator must be removed in order to clear refill handpump. When removing left input module, only hydraulic pump module must be removed.

- i. Remove hydraulic pump module (PARA 7-4-12).
- j. If removing right input module, remove main generator (PARA 9-4-16).
- k. Keep oil tubes between main generator and accessory module with accessory module.
- l. Disconnect electrical connector, P405 (right side), or P406 (left side), from accessory module chip detector (Figure 6-4-25-1, Sheet 1, Detail A).
- m. Install protective caps and plugs on electrical connectors.
- n. Remove end of accessory module chip detector from bracket.
- o. Remove tape or tiedown straps from harness (Figure 6-4-25-1, Sheet 1, Detail B).
- p. Remove screw, washer, and nut that secures clamp(s) and harness to bracket on accessory module.
- q. On helicopters with main transmission, 70351-08000 or 70084-35001, if removing right accessory module, disconnect electrical connector from rotor speed (Nr) sensor as follows:

### 6-4-25-2 Change 8

- (1) Disconnect electrical connector, P443, from back side of rotor speed (Nr) sensor on right accessory module (Figure 6-4-25-1, Sheet 1, Detail A).
  - (2) Install protective caps and plugs on electrical connectors.
- r. On helicopters with main transmission, 70084-35007, 70351-48000, or 70351-58000, disconnect electrical connector from rotor speed (Nr) sensor as follows:
- (1) Disconnect electrical connector, P407 (left side), or P443 (right side), from back side of rotor speed (Nr) sensor on accessory module (Figure 6-4-25-1, Sheet 1, Detail A).
  - (2) Install protective caps and plugs on electrical connectors.

**NOTE**

Right accessory module does not have a low oil pressure switch.

- s. On left accessory module, disconnect electrical connector, P409, from low oil pressure switch.
- t. Install protective caps and plugs on electrical connectors.
- u. Disconnect engine load-demand push/pull cable supports from accessory module by removing bolts, washers, and nuts. Retain bolts, washers, and nuts with engine load demand push/pull cable supports.
- v. Attach straps of hoisting sling to input module. Place one strap around accessory module mounting flange and connect other strap to lifting bracket (Figure 6-4-25-1, Sheet 2, Detail C).

**WARNING**

- Hoist cable may break, causing injury to personnel or damage to equipment, if hoist is used to free a sticking

component being removed from helicopter. Do not use too much force when removing a component.

- Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment if hook is allowed to contact pulley when removing a component from helicopter. Do not allow hook assembly to contact pulley.
- w. Attach hoisting sling to hoist or cabin maintenance crane.
  - x. Adjust length of straps for even distribution of weight before lifting.
  - y. Raise hoist and take up slack.
  - z. Remove nuts and washers securing input module to main module.

**WARNING**

Injury to personnel and damage to equipment will result if caution is not taken when hoisting input module. Keep area clear when hoisting input module.

**CAUTION**

Damage to quill shaft will result if quill shaft is not removed from input module, or main module while removing input module. Quill shaft may fall damaging splines. Remove quill shaft while removing input module.

**NOTE**

Care must be taken not to burr studs while removing input module.

- aa. Carefully pull input module from main module.
- ab. Lift input module clear of helicopter and place on stand. If input module does not pull off main module easily, it may be jacked out as follows:

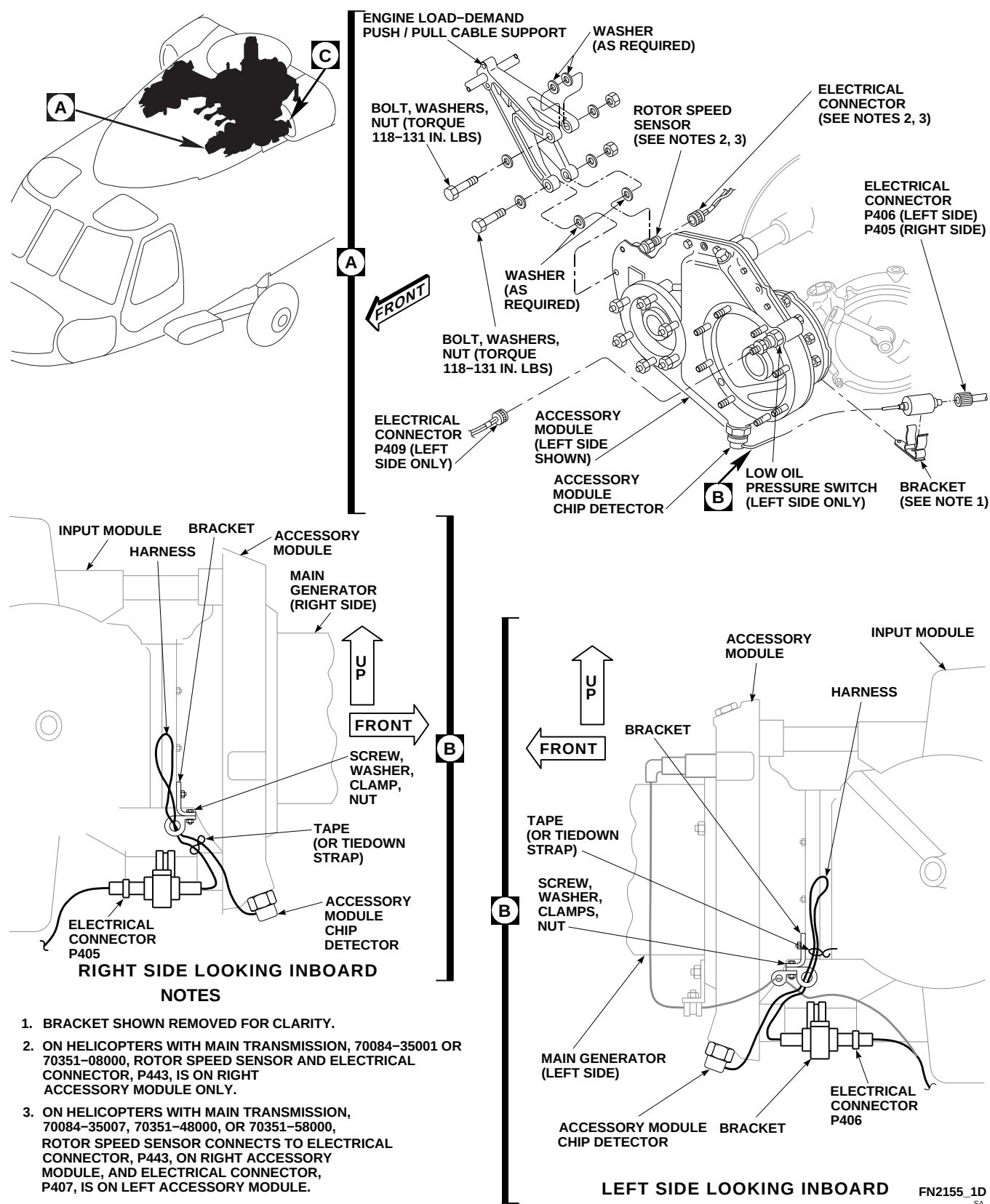


FIGURE 6-4-25-1. REPLACE INPUT MODULE (SHEET 1 OF 2)



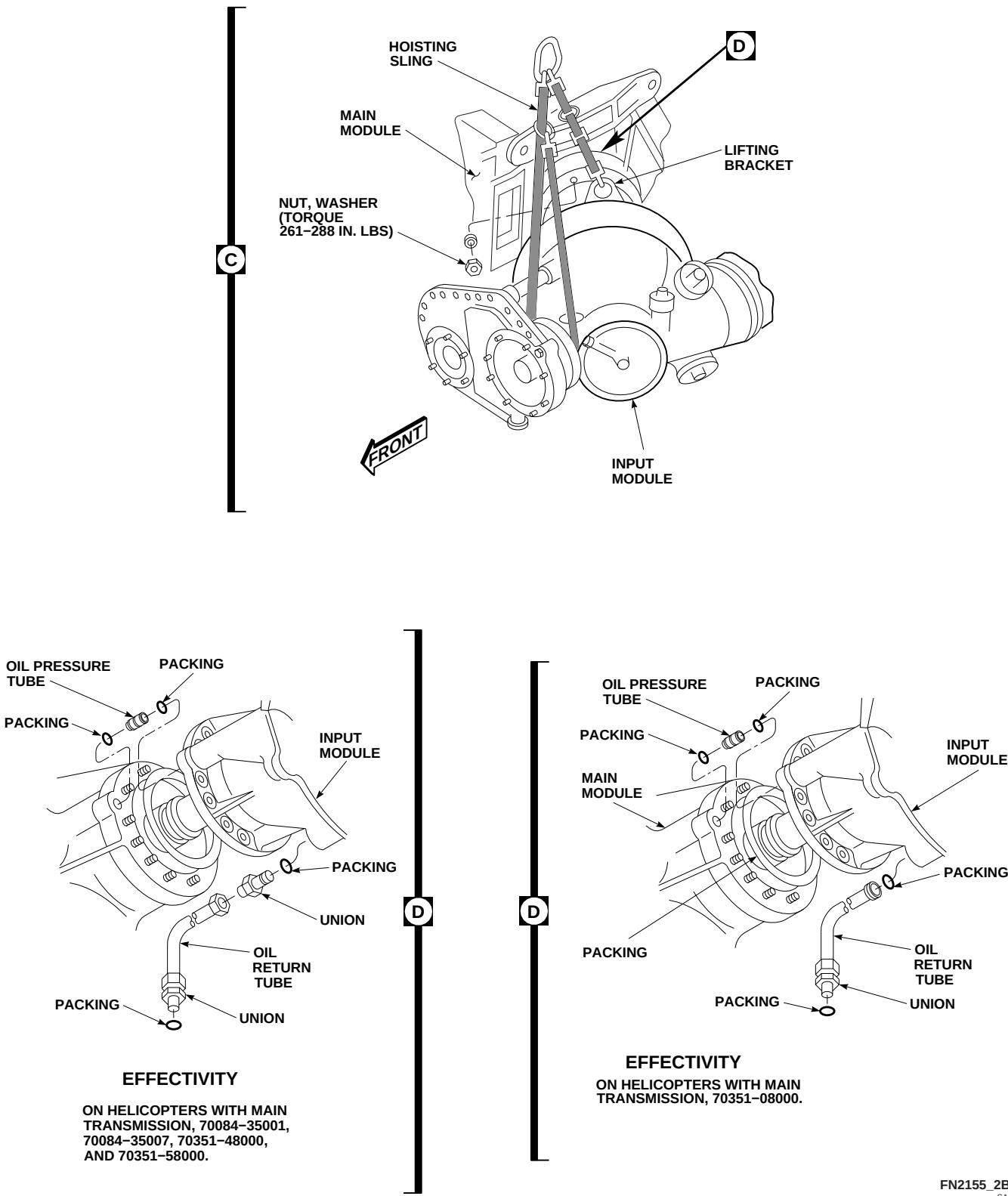


FIGURE 6-4-25-1. REPLACE INPUT MODULE (SHEET 2 OF 2)

- (1) Install 1/4-28 jacking bolts in three jacking holes in input module.

**NOTE**

Be careful not to cock input module.

- (2) Slowly and equally thread in bolts to push input module away from main module.
- (3) Make sure oil return tube stays with main module.
- (4) Cover port of oil return tube.
- (5) Remove jacking bolts from input module.
- (6) Remove hoisting sling.

- ac. Using nonmetallic scraper, remove sealing compound.
- ad. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean primer off mating surface and hardware.
- ae. Inspect input pinion for axial movement (PARA 6-3-8).

**6-4-25.1.2 Input Module Teardown/Buildup.****NOTE**

If replacement input module will be installed, remove following parts from old input module and install on new input module.

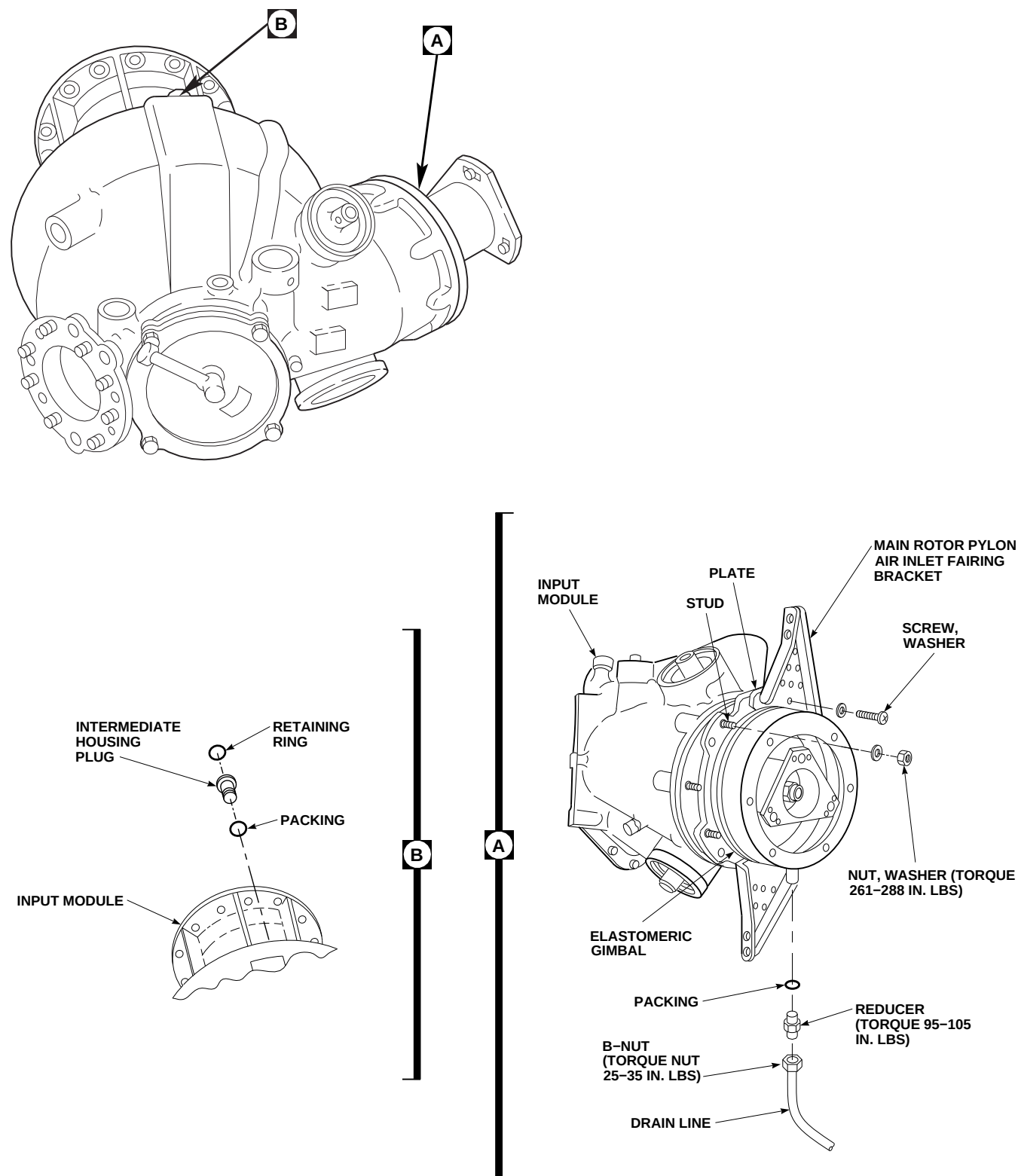
- a. Remove screws and washers securing main rotor pylon air inlet fairing bracket to plates (Figure 6-4-25-2, Detail A).
- b. Remove nuts and washers securing lifting bracket. Remove lifting bracket (Figure 6-4-25-1, Sheet 2, Detail C).
- c. Disconnect drain line from reducer (Figure 6-4-25-2, Detail A).
- d. Install protective cap on drain line.

- e. Remove reducer and packing from elastomeric gimbal. Discard packing.
- f. Install protective plug in elastomeric gimbal.

**NOTE**

When removing elastomeric gimbal, do not remove two nuts on short studs. These nuts are part of input module and removal is not required to remove elastomeric gimbal.

- g. Remove elastomeric gimbal and attaching plates. Remove six nuts and washers in equal steps, otherwise they will not clear studs before hitting housing. If new input module is not readily available for installation, cover opening in input module.
- h. Remove retaining ring, intermediate housing plug, and packing from input module. Discard packing (Figure 6-4-25-2, Detail B).
- i. Remove accessory module (PARA 6-4-28).
- j. Using clean machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean quill shaft.
- k. Visually inspect quill shaft and splines for damage. **NONE ALLOWED.** If damaged, replace quill shaft.
- l. Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate quill shaft.
- m. Install quill shaft in input module.
- n. Install accessory module (PARA 6-4-28).
- o. If required, install lifting bracket on input module split half at 12 o'clock position and secure with nuts and washers (Figure 6-4-25-1, Sheet 2, Detail C).
- p. **TORQUE NUTS TO 261 - 288 INCH-POUNDS.**
- q. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean mounting surfaces on elastomeric gimbal and input module.



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FIGURE 6-4-25-2. TEARDOWN/BUILDUP INPUT MODULE

- r. Apply coat of primer, Item 258, Appendix D, to elastomeric gimbal mounting surface on input module.

### NOTE

When installing elastomeric gimbal, do not install washers under nuts which secure plates.

- s. Install elastomeric gimbal and plates and secure with nuts and washers (Figure 6-4-25-2, Detail A). Start all nuts together and turn down on studs two or three threads at a time.
- t. **TORQUE NUTS TO 261 - 288 INCH-POUNDS.**
- u. Using petrolatum, Item 234, Appendix D, lubricate packing.
- v. Install packing on reducer. Install reducer in elastomeric gimbal.
- w. **TORQUE REDUCER TO 95 - 105 INCH-POUNDS.**
- x. Connect drain line to reducer.
- y. **TORQUE B-NUT TO 25 - 35 INCH-POUNDS.**
- z. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to all exposed nuts and studs securing elastomeric gimbal to input module.
- aa. Apply sealing compound, Item 292, Appendix D, to cover mounting hardware of elastomeric gimbal.
- ab. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer over sealing compound.
- ac. Using polyurethane paint, Item 224, Appendix D, apply one coat over epoxy primer.

- ad. Install main rotor pylon air inlet fairing bracket on plates and secure with screws and washers.

### CAUTION

Damage to input module will result if all shipping plugs are not removed. Make sure all shipping plugs are removed.

- ae. Remove shipping plugs from replacement input module.
- af. Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, coat packing. Install packing on intermediate housing plug (Figure 6-4-25-2, Detail B).
- ag. Install intermediate housing plug into replacement input module and secure with retaining ring.
- ah. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to retaining ring. Allow to dry.
- ai. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to retaining ring groove in intermediate housing. Install retaining ring while primer is still wet. Apply primer to face of installed retaining ring. Allow to dry.
- aj. Apply sealing compound, Item 292, Appendix D, around area of retaining ring.
- ak. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer over sealing compound.
- al. Using polyurethane paint, Item 224, Appendix D, apply one coat over epoxy primer.

### 6-4-25.1.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).

**CAUTION**

Damage to input module will result if all shipping plugs are not removed. Make sure shipping plugs are removed from all port openings.

- b. Attach straps of hoisting sling, to input module. Position one strap around accessory module attachment flange. Connect other strap to input module lifting bracket (Figure 6-4-25-1, Sheet 2, Detail C).
- c. Attach hoisting sling to hoist or cabin maintenance crane.
- d. Adjust lengths of straps for even distribution of weight.
- e. On helicopters with main transmission, 70084-08000, remove oil return tube from main transmission sump (Figure 6-4-25-1, Sheet 2, Detail D).
- f. On helicopters with main transmission, 70084-35001, 70084-35007, 70351-48000, and 70351-58000, remove oil return tube from union and remove union from input module (Figure 6-4-25-1, Sheet 2, Detail D).
- g. Remove packings from oil pressure tube, oil return tube, and union, if applicable. Discard packings.
- h. Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, coat replacement packings.
- i. On helicopters with main transmission, 70084-08000, install packings on oil return tube.
- j. On helicopters with main transmission, 70084-35001, 70084-35007, 70351-48000, and 70351-58000, install packings on oil return tube and union.
- k. Install oil return tube in main transmission sump.
- l. On helicopters with main transmission, 70084-35001, 70084-35007, 70351-48000, and 70351-58000, install union in input module.

- m. Install packings on oil pressure tube and insert tube in main module.
- n. Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, coat packings of oil pressure tube and place packings in groove on input module.

**WARNING**

Injury to personnel and damage to equipment will result if caution is not taken when hoisting input module. Keep area clear when hoisting input module.

**NOTE**

Care must be taken not to burr mounting studs while installing input module.

- o. Lift input module into place. Make sure oil pressure tube and oil return tube packings are in position and seated.
- p. Apply coat of sealing compound, Item 292, Appendix D, to area between Rosan lockring and mating housing counterbore.
- q. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean mounting surface on main and input modules.
- r. Using chemical conversion coating, Item 118, Appendix D, touch up magnesium surfaces.

**WARNING**

Injury to personnel and damage to equipment will result if transmission oil is contaminated. To avoid contaminating transmission oil system, do not apply primer to bore diameters that are sealing surfaces for packings.

- s. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to input module

mounting surface and fastener holes. Do not apply primer to packing or packing groove.

### NOTE

- Install input module while primer is still wet.
- When changing input module to opposite side, make sure that boltholes of elastomeric gimbal and forward support line up.
- t. Guide quill shaft into main module while installing input module onto mounting flange. If necessary, turn input flange to line up splines. Also guide oil return tube into input module. If necessary, loosen nut on oil return tube to line up.
- u. Install nuts and washers (Figure 6-4-25-1, Sheet 2, Detail C).
- v. **TORQUE NUTS TO 261 - 288 INCH-POUNDS.**
- w. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to all exposed washers, nuts, and studs securing input module to main module.
- x. Apply sealing compound, Item 292, Appendix D, to seal area around joint of washer and nut, threads of bolts to nut, and seam between input module and main module.
- y. Remove extra sealing compound.
- z. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer over sealing compound.
- aa. Using polyurethane paint, Item 224, Appendix D, apply one coat over epoxy primer (PARA 2-6-5).

### NOTE

To tighten or loosen nut on oil return tube, using 1<sup>3</sup>/<sub>8</sub>-inch open-end wrench, hold hex on fitting in main transmission

sump. End of fitting in main transmission sump is not threaded.

- ab. **HOLD HEX ON UNION AND TIGHTEN NUT ON OIL RETURN TUBE.**
- ac. Remove hoisting sling.
- ad. Inspect and clean accessory, input, and main module oil return screens, chip detectors, and chip detector strainers (PARA 6-4-15, PARA 6-4-16, PARA 6-4-17, and PARA 6-4-18).
- ae. Check chip detector electrical circuit (PARA 6-2-2).
- af. Connect forward support tube to elastomeric gimbal on input module (PARA 4-4-6).
- ag. Connect engine output drive shaft to input module flange (PARA 4-4-6).
- ah. If right input module was removed, install main generator (PARA 9-4-16).
- ai. Install hydraulic pump module (PARA 7-4-12).
- aj. Remove protective caps and plugs from electrical connectors.
- ak. Inspect and treat electrical connectors (PARA 1-7-20).
- al. Connect electrical connector, P405 (right side), or P406 (left side), to accessory module chip detector (Figure 6-4-25-1, Sheet 1, Detail A).
- am. Route harness and secure to bracket on accessory module with clamp(s), screw, washer, and nut (Figure 6-4-25-1, Sheet 1, Detail B).

### NOTE

Tiedown straps may be used as an alternate for tape.

- an. Secure slack in harness with tape, Item 317C, Appendix D, or tiedown straps, Item 329, Appendix D.
- ao. On helicopters with main transmission, 70351-08000 or 70084-35001, connect electrical connector to rotor speed (Nr) sensor as follows:

- (1) Remove protective caps and plugs from electrical connectors.
  - (2) Inspect and treat electrical connectors (PARA 1-7-20).
  - (3) Connect electrical connector, P443, to back side of rotor speed (Nr) sensor on right accessory module (Figure 6-4-25-1, Sheet 1, Detail A).
  - (4) Using lockwire, Item 197, Appendix D, lockwire electrical connector.
- ap. On helicopters with main transmission, 70084-35007, 70351-48000, or 70351-58000, connect electrical connector to rotor speed (Nr) sensor as follows:
- (1) Remove protective caps and plugs from electrical connectors.
  - (2) Inspect and treat electrical connectors (PARA 1-7-20).
  - (3) Connect electrical connector, P407 (left side), or P443 (right side), to back side of rotor speed (Nr) sensor on accessory module (Figure 6-4-25-1, Sheet 1, Detail A).
  - (4) Using lockwire, Item 197, Appendix D, lockwire electrical connectors.
- aq. On left accessory module, connect electrical connector, P409, to low oil pressure switch.
- ar. Using lockwire, Item 197, Appendix D, lockwire electrical connector.
- as. Connect engine load-demand push/pull cable support to accessory module as follows:
- (1) Apply coat of primer, Item 258, Appendix D, to boltholes of engine load-demand push/pull cable support on accessory module.
  - (2) Position engine load-demand push/pull cable support on accessory module.
  - (3) Add washers as required to fill gap between engine load-demand push/pull cable support and accessory module.
  - (4) Bolt engine load-demand push/pull cable support to accessory module.
  - (5) **TORQUE TO 118 - 131 INCH-POUNDS.**
- at. Using polyurethane paint, Item 224, Appendix D, apply one coat to any water entrapment areas of input module.
- au. Install main rotor pylon air inlet fairings (PARA 2-4-106).
- av. Install engine air inlets (PARA 4-4-4).
- aw. Service main transmission (PARA 1-3-12).
- ax. Make sure area is clean and free of foreign material.
- ay. Close engine cowlings.
- az. Close main rotor pylon sliding cover.
- ba. Perform main transmission functional test (PARA 6-3-4).
- bb. Perform operational check of AC electrical system (PARA 9-2-1) and hydraulic system (PARA 7-2-1).
- bc. On helicopters with engine output shafts, 70361-08004-042/-043, perform torque stabilization check after first ground run, after 1 to 3 flight hours, and after 25 to 30 flight hours (PARA 4-4-1).
- bd. On helicopters with engine output shaft, 70361-08014-041, perform torque stabilization check after first ground run and again after 3 to 13 flight hours (PARA 4-4-1).





6-4-26 INPUT MODULE OIL SEAL.

PARAGRAPH BREAKDOWN

|                                        | PAGE     |
|----------------------------------------|----------|
| 6-4-26.1 Replace Input Module Oil Seal | 6-4-26-2 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Picks
- Pliers
- Powertrain Repairer’s Toolkit
- Screws, MS35206-245 (or equivalent)
- Scribes
- Seal Installation Tool, Locally-Made, Figure H-157, Appendix H
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Cloth, Cleaning, Item 102A, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Lockwire, Item 197, Appendix D

- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D
- Lubricating Oil, Item 208, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint, Polyurethane, Item 224, Appendix D
- Primer, Item 258, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- Appendix H
- PARA 2-6-5
- PARA 4-4-1
- PARA 6-3-4
- PARA 6-3-8
- PARA 6-4-27

**6-4-26.1 Replace Input Module Oil Seal.****NOTE**

- Replacement of left or right input module oil seal is the same.
- Prior to installation of new input module oil seal, replacement packings must be preswelled with lubricating oil, Item 206, Appendix D, for a minimum of 48 hours.
- Input module oil seal, 70968-08502-101, has two parts. When replacing seal, both parts are replaced.

**6-4-26.1.1 Remove.**

- Remove applicable engine (PARA 4-4-1).
- Inspect input pinion for axial movement (PARA 6-3-8).
- Remove screws and washers securing main rotor pylon air inlet fairing bracket to main rotor pylon air inlet fairing (Figure 6-4-26-1, Detail A).
- Disconnect drain line from reducer.
- Remove nuts and washers securing elastomeric gimbal to input module. Remove nuts in equal steps, otherwise they will not clear studs before hitting housing. Note location of washers. Remove elastomeric gimbal from input module.
- Remove and inspect input module input flange (PARA 6-4-27).
- Remove retaining ring.
- Remove rear spacer.
- Thread three screws, MS35206-245, or equivalent, into three holes in rear half of input module oil seal. Using screws, pull out rear half from pinion gear and housing.

- Remove front half of input module oil seal from pinion gear as follows.

- Using two scribes or picks, find flats on front half of input module oil seal and gently work front half off rear packing. Discard rear packing.

- Remove front half from pinion gear.

- If front half of input module oil seal does not have flats or is difficult to remove, perform alternate remove method as follows:

- Using lockwire, Item 197, Appendix D, wrap lockwire around lip of front half, and twist, leaving enough lockwire to grip with pliers.

- Using a second piece of lockwire, wrap around lip of front half, 180° from the first wire, and twist, leaving enough lockwire to grip with pliers.

- Using two pairs of pliers, grip both wires and pull, removing front half of input module oil seal from pinion gear. Remove and discard rear packing.

- Remove front packing from pinion gear. Discard front packing.

**6-4-26.1.2 Install.****NOTE**

Prior to installation of new input module oil seal, replacement packings must be preswelled with lubricating oil, Item 206, Appendix D, for a minimum of 48 hours.

- Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, coat front packing and install on pinion gear. Make sure front packing is installed in groove (Figure 6-4-26-1, Detail A).

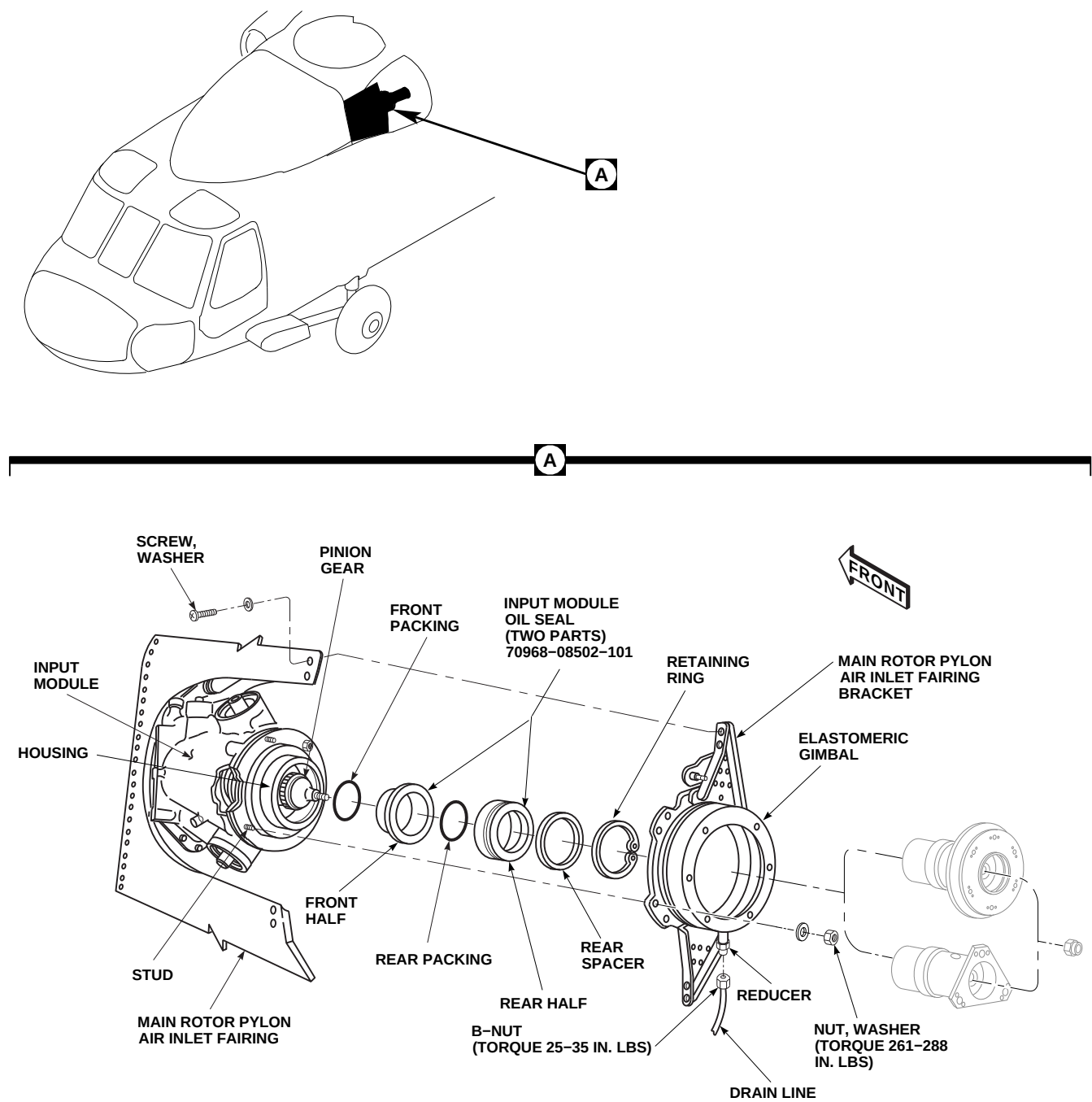


FIGURE 6-4-26-1. REPLACE INPUT MODULE OIL SEAL

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**CAUTION**

- Leave input module oil seal in shipping container until ready for use to prevent possible contamination. Any type of foreign material which makes contact with sealing surface can cause leakage.
  - Handle input module oil seal with extreme care.
  - Do not use thinners, cleaning agents, or water on input module oil seal.
  - Damage to input module oil seal will result if mating surfaces come in contact with skin or if mating surfaces are not lubricated during assembly. Oil from skin will damage carbon seal material. Do not touch mating surfaces of input module oil seal. Touching mating surfaces can cause leakage.
- b. Using cleaning cloth, Item 102A, Appendix D, wipe faces of input module oil seal (front and rear halves).
  - c. Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate front and rear packings, area on both halves of input module oil seal that comes in contact with packings and mating surfaces of both halves of input module oil seal.

**CAUTION**

Do not hammer front half of input module oil seal into place.

- d. Install front half of input module oil seal on pinion gear with flange toward rear. Seat front half against bearing.
- e. Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, coat rear packing and install on pinion gear. Make sure rear packing is installed in groove.

- f. Locally make seal installation tool (Figure H-157, Appendix H).

**CAUTION**

Do not hammer rear half of input module oil seal into place.

**NOTE**

Make sure rear half of input module oil seal does not "bang" into position after packing squeezes past liner.

- g. Install rear half of input module oil seal in housing with metallic surface toward rear. Using seal installation tool, seat input module oil seal. Make sure input module oil seal is seated.
- h. Install rear spacer.
- i. Install retaining ring.
- j. Install input module input flange (PARA 6-4-27).
- k. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean mounting surfaces on elastomeric gimbal and input module.
- l. Using primer, Item 258, Appendix D, apply a coat to mating surface of input module.
- m. Install elastomeric gimbal onto input module with nuts and washers while primer is still wet. Start all nuts together and run them down on studs two or three threads at a time.
- n. **TORQUE NUTS TO 261 - 288 INCH-POUNDS.**
- o. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to all exposed nuts and studs securing elastomeric gimbal to input module.
- p. Using sealing compound, Item 292, Appendix D, cover mounting hardware of elastomeric gimbal.

- q. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer over sealing compound.
- r. Using polyurethane paint, Item 224, Appendix D, apply one coat over epoxy primer (PARA 2-6-5).
- s. Connect drain line to reducer.
- t. **TORQUE B-NUT TO 25 - 35 INCH-POUNDS.**
- u. Install main rotor pylon air inlet fairing bracket to main rotor pylon air inlet fairing and secure with screws and washers.
- v. Install engine (PARA 4-4-1).
- w. Perform main transmission functional test (PARA 6-3-4).



6-4-27 INPUT MODULE INPUT FLANGE.

PARAGRAPH BREAKDOWN

|                                            | PAGE     |
|--------------------------------------------|----------|
| 6-4-27.1 Replace Input Module Input Flange | 6-4-27-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Flange Puller, Locally-Made, Figure H-50, Appendix H
- Magnifying Glass
- Powertrain Repairer’s Toolkit
- Torque Reactor, Locally-Made, Figure H-181, Appendix H
- Torque Reactor Plate, Locally-Made, Figure H-259, Appendix H
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Abrasive Cloth, Item 7, Appendix D
- Abrasive Cloth, Item 8, Appendix D
- Abrasive Cloth, Item 10, Appendix D
- Sealing Compound, Item 268, Appendix D

References

- Appendix D
- Appendix H
- PARA 1-3-12
- PARA 2-6-5
- PARA 4-4-1
- PARA 6-3-8

6-4-27.1 Replace Input Module Input Flange.

NOTE

Replacement of left or right input module input flange is the same.

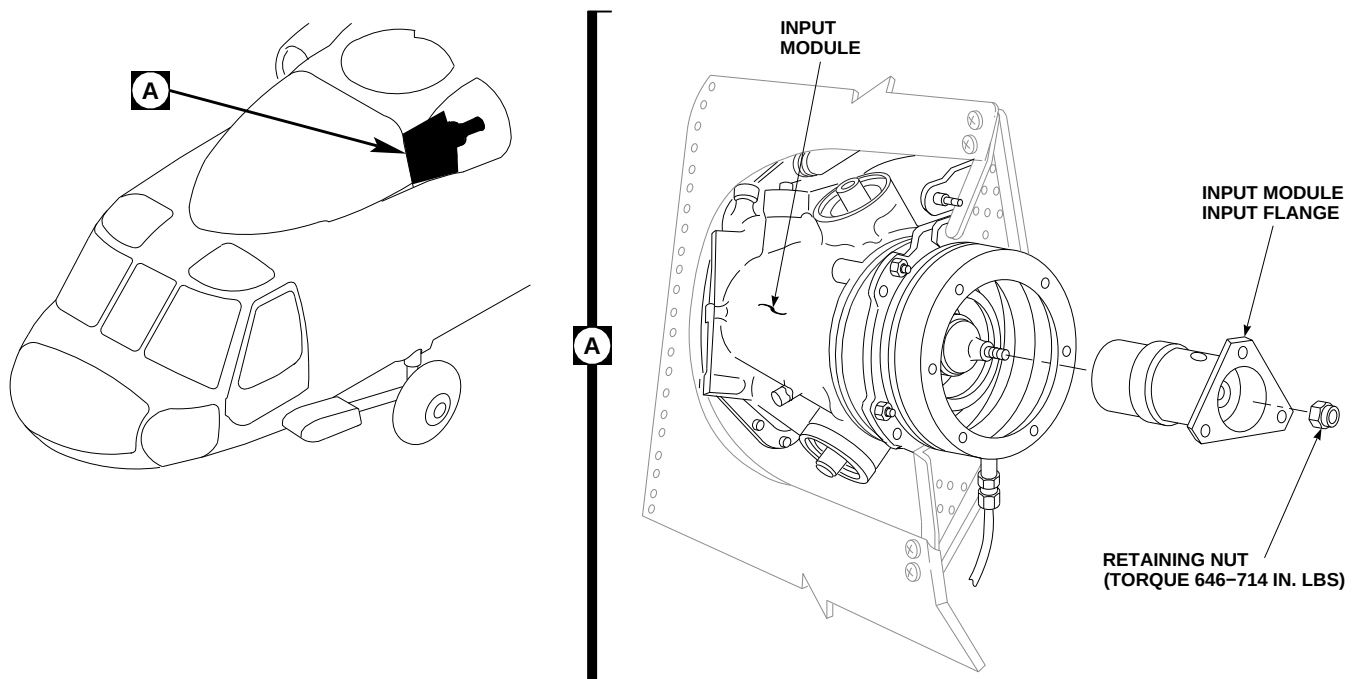
6-4-27.1.1 Remove.

NOTE

Input module input flange, 70351-08206-102, is preferred and should be installed if available. If the -102 input

module input flange is not available, input module input flange, 70351-08206-041, may be used until the -102 input module input flange becomes available.

- Remove applicable engine (PARA 4-4-1).
- Inspect input pinion for axial movement (PARA 6-3-8).
- Locally make torque reactor (Figure H-181, Appendix H), or torque reactor plate (Figure H-259, Appendix H).

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**FIGURE 6-4-27-1. REPLACE INPUT MODULE INPUT FLANGE, 70351-08206-102**

- d. Using torque reactor or torque reactor plate, hold input module input flange or input flange/coupling and remove retaining nut securing input flange or input flange/coupling to input module (Figure 6-4-27-1, Detail A or Figure 6-4-27-2, Detail A).
- e. Remove torque reactor or torque reactor plate from input module input flange.
- f. Remove input flange or input flange/coupling from input module. If input flange or input flange/coupling is difficult to remove, perform the following:
  - (1) On helicopters with input flange/coupling 70351-08255-102, remove gimbal (PARA 6-4-26).
  - (2) Locally make flange puller (Figure H-50, Appendix H).
  - (3) Carefully attach flange puller to input module input flange using bolts from input module input flange.
  - (4) Slowly screw in main puller bolt until input module input flange is pulled free from input shaft.
- g. Discard packing.

#### 6-4-27.1.2 Inspect.

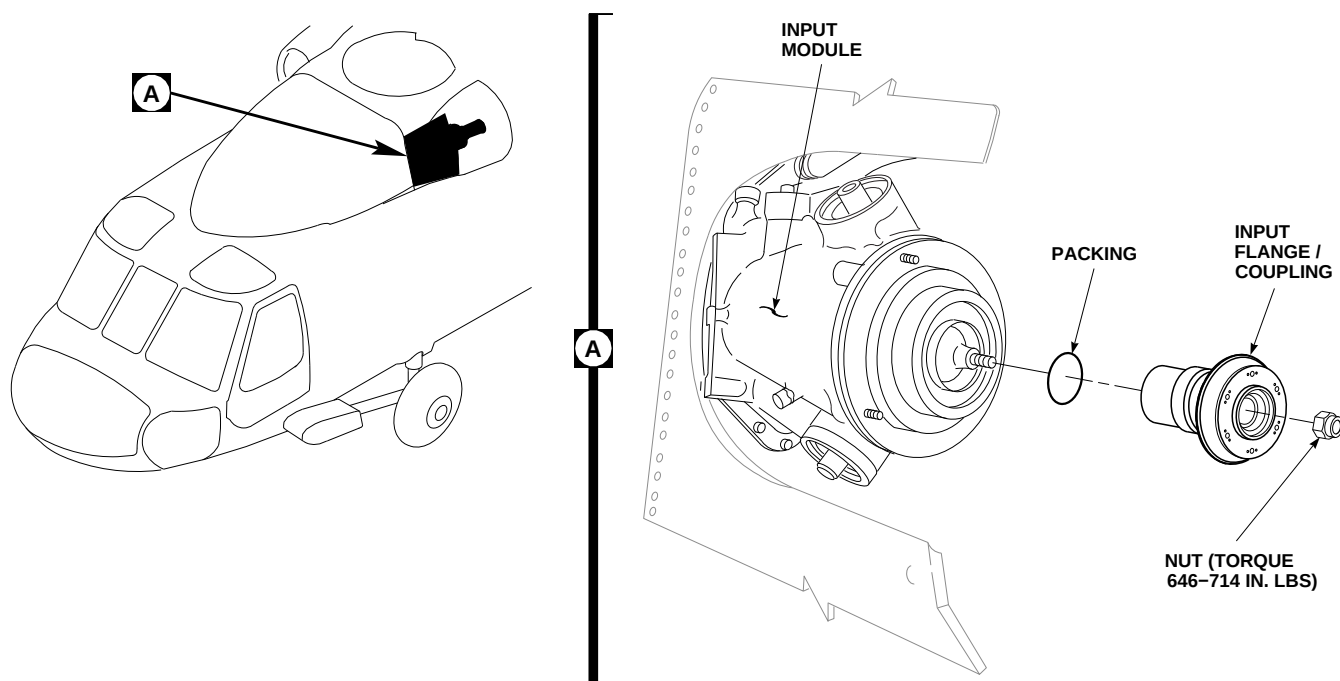
- a. Inspect input flange/coupling, 70351-08255-102, as follows (Figure 6-4-27-3, Sheet 1, and Figure 6-4-27-3, Sheet 2):

#### NOTE

Flange/coupling is made of titanium alloy, so bare metal may have some discoloration and variation in appearance (non-uniform color, from straw to blue, staining, handling marks, etc). Flange/coupling is a welded construction and weld seams are visible. Discoloration as described, or visible weld seams are not reason for rejection.

- (1) Using magnifying glass, visually inspect flange/coupling for cracks. **NONE AL-**





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**FIGURE 6-4-27-2. REPLACE INPUT MODULE INPUT FLANGE/COUPLING, 70351-08255-102**

**LOWED.** If cracks are found, replace flange/coupling.

- (2) Coupling is coated with green epoxy primer on surfaces that are external when coupling is assembled into helicopter. Chipping and flaking of green epoxy primer is acceptable and may be touched up. The coupling has a ring of material on external surface above internal spline for balance correction. This ring will typically have an area where material was removed and coated with green epoxy primer during manufacturing.
- (3) Using magnifying glass, visually inspect flange/coupling diaphragm profile area for dents or wavy appearance. **NONE ALLOWED.** If dents or wavy appearance or any damage is found in diaphragm profile area, replace flange/coupling.
- (4) Using magnifying glass, visually inspect flange/coupling diaphragm profile surface for scratches or nicks into base metal.

**NONE ALLOWED.** If scratches or nicks into base metal of diaphragm profile surface is found, replace flange/coupling.

#### NOTE

There are four vent slots equally spaced in diaphragm rim. These slots are normal and not cause for rejection.

- (5) Scratches and nicks to a maximum of 0.015-inch in diaphragm rim, hub, or flanges (all surfaces other than diaphragm profile) are acceptable. If scratches and nicks exceed limit, replace flange/coupling.
- (6) Inspect bolted mounting flange face and axial stop face for fretting. Light fretting, up to 0.0005-inch deep, is acceptable and may be polished out using abrasive cloth, Item 8, Appendix D.
- (7) Inspect for raised edges or otherwise proud material on mounting flange face,

flange pilot outside diameter (OD), spline end pilot inside diameter (ID), and spline end face. Blend and polish raised areas using abrasive cloth, Item 7, Appendix D, and abrasive cloth, Item 8, Appendix D. **NO BLENDING OR POLISHING IS ALLOWED IN THE DIAPHRAGM PROFILE AREA.**

- (8) Internal spline area is coated with Ever-lube #620C dry-film lubricant. Light marking of the coating on the spline teeth is acceptable.
- (9) Inspect flange/coupling nutplates for missing rivets, rivets above surface of flange face, or missing locknuts. **NONE ALLOWED.** If rivets are missing, or locknuts or rivets are above surface of flange face, replace flange/coupling.
- (10) Check running torque on nutplate locknuts. If running torque is below limit on any locknut, replace flange/coupling.
- (11) Inspect axial stop face for nicks or scratches. **NICKS OR SCRATCHES UP TO 0.010-INCH OVER 10% OF AREA IS ALLOWED.** Remove only raised edges.
- (12) Measure inside diameter (ID) of pinion pilot bore. If ID exceeds 0.5287-inch, replace flange/coupling.
- (13) Measure outside diameter (OD) of flange pilot. If OD is less than 2.4762-inches, replace flange/coupling.
- (14) Inspect flange pilot outside diameter (OD) for nicks and scratches. **NICKS AND SCRATCHES UP TO 0.005-INCH DEEP OVER LESS THAN 30% OF ENTIRE CIRCUMFERENCE ALLOWED. NO MORE THAN 10% ALLOWED IN ANY ONE 90° QUADRANT SECTION.** If nicks and scratches are beyond limits, replace flange/coupling.
- (15) Measure inside diameter (ID) of pilot face. If ID exceeds 2.3664-inches, replace flange/coupling.

- (16) Measure outside diameter (OD) of pilot face. If OD less than 2.580-inches, replace flange/coupling.
- (17) Measure wear of spline. **NO MORE THAN 0.002-INCH WEAR STEP IN SPLINES ALLOWED.** If spline wear step is greater than 0.002-inches, replace flange/coupling.

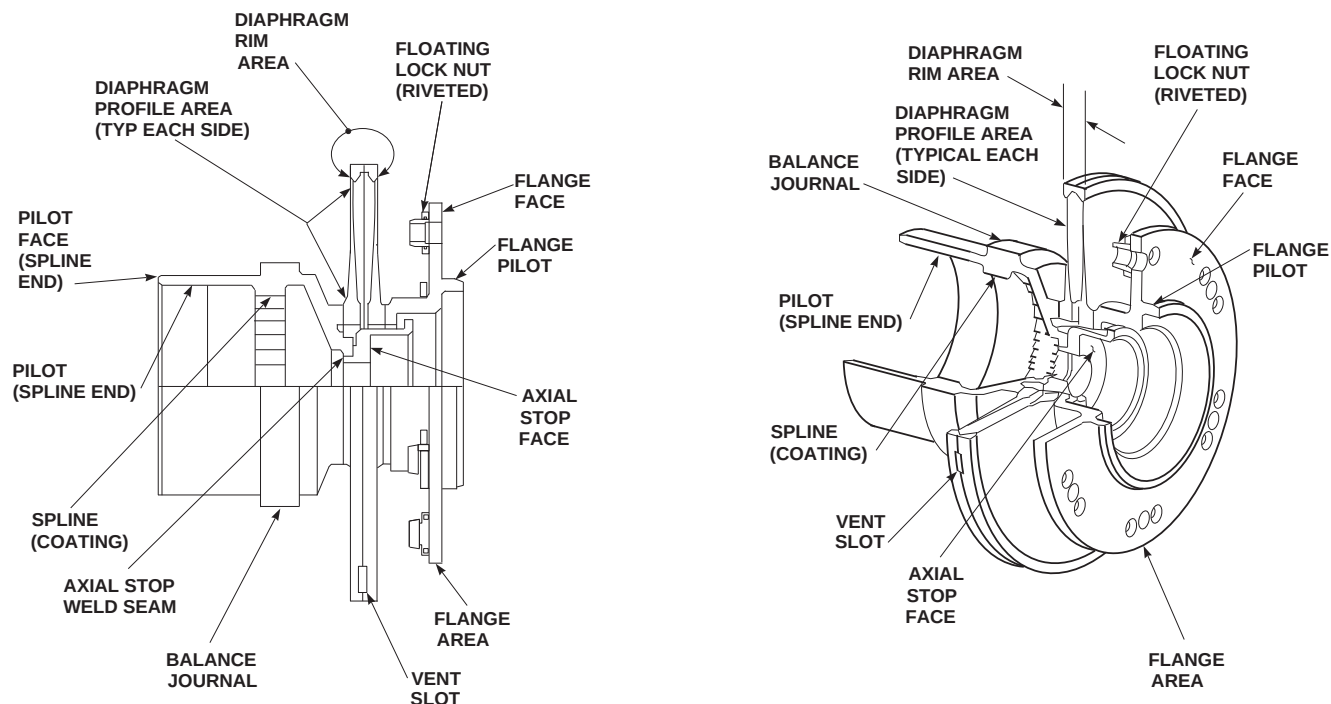
b. Inspect input flange 70351-08206-102 or 70351-08206-041 as follows:

- (1) Check dimension of damaged areas. Compare with minimum repair limits (Figure 6-4-27-4).
- (2) Inspect to determine whether minimum dimension of input module input flange can be maintained after repair.
- (3) Inspect input module input flange for fretting in contact area of washer. **MAXIMUM FRETTING DEPTH IS 0.0005-INCH.** If fretting is beyond limits, replace input module input flange.

#### NOTE

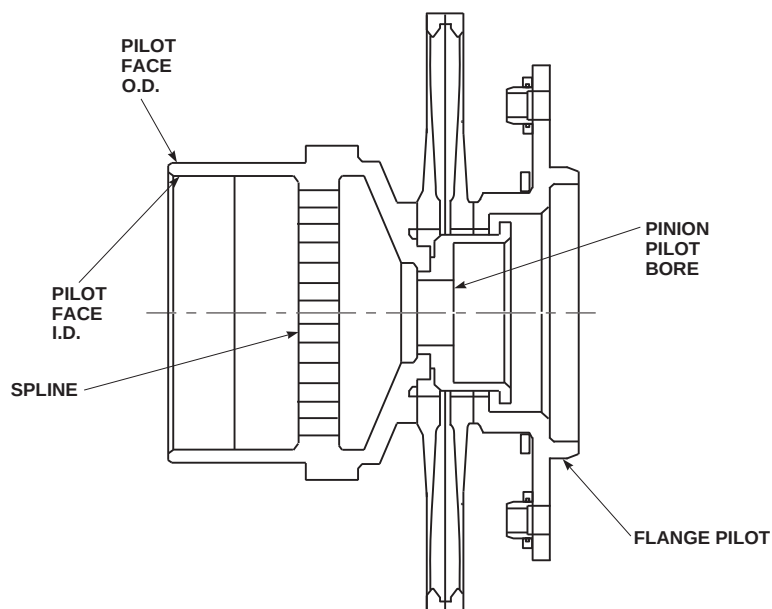
If input module flange, 70351-08206-041 with nutplates is used, replace with input module flange, 70351-08206-102, if available, if input module flange, 70351-08206-041 is used, after torque has stabilized, place a drop of sealing compound, Item 268, Appendix D, to threads of bolts adjacent to nutplates. Make sure compound seeps into nutplates.

- (4) On input module input flange, 70351-08206-041, inspect rivets on nutplate for proper installation, looseness, and fretting on heads. If rivets are loose, not flush with flange surface, or fretting is found on heads, replace input module input flange.
- (5) On input module input flange, 70351-08206-041, thread bolts into nutplates. If bolts can be fully installed by hand, replace input module input flange.



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FIGURE 6-4-27-3. INSPECT INPUT FLANGE/COUPLING (SHEET 1 OF 2)



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FIGURE 6-4-27-3. INSPECT INPUT FLANGE/COUPLING (SHEET 2 OF 2)

## 6-4-27.1.3 Repair.

**NOTE**

- Repair is for input flange, 70351-08206-102 and 70351-08206-041, only.
  - No balance checks are required after reworking input module input flange per this procedure.
  - Repair may be done with input module input flange installed on helicopter.
- a. Before starting repair on input module input flange, determine whether polishing out damage would result in a diameter below repairable limits (Figure 6-4-27-4). If damage repair is greater than permitted limits, replace input module input flange.
  - b. If necessary, remove damaged input module input flange (PARA 6-4-27.1.1).
  - c. Using abrasive cloth, Item 10, Appendix D, polish out minor nicks, scratches, and scoring.
  - d. Using abrasive cloth, Item 7, Appendix D, blend out damage. Remove only what is necessary so input module input flange will stay within limits. Blend damaged area to a minimum 1/2-inch radius.
  - e. Polish reworked area to remove all traces of abrasion marks. Using abrasive cloth, Item 10, Appendix D, match surface smoothness.

## 6-4-27.1.4 Install.

**CAUTION**

- Damage to input module internal bearing retaining clips will result if input module input flange is seated by tapping with mallet. Use only slight hand pressure and torque on retaining nut to seat input flange.
- Input module flange/coupling, 70351-08255-102, must be installed with elas-

tomeric gimbal, 70361-08015-102 (PARA 6-4-25) and engine output drive shaft, 70361-08014-041 (PARA 4-4-6). Input module flange, 70351-08206-102, or 70351-08206-041, with gimbal 70361-08001-101, 70361-08001-102, or 70361-08001-103, may be used with drive shaft 70361-08004-042 or -043.

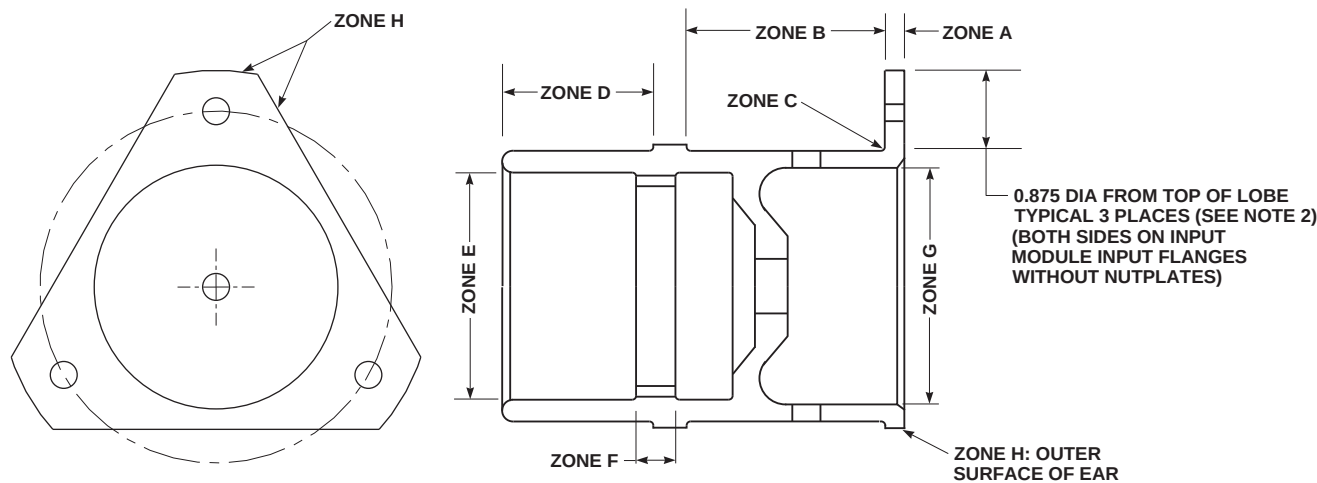
**NOTE**

- When using input module flange/coupling, 70351-08255-102, verify flange/coupling is properly marked, "For H-60/S-70 only". Dash numbers other than -102 are not interchangeable.
  - If input module flange, 70351-08206-041 with nutplates is used, replace with input module flange, 70351-08206-102, if available, if input module flange, 70351-08206-041 is used, after torque has stabilized, place a drop of sealing compound, Item 268, Appendix D, to threads of bolts adjacent to nutplates. Make sure compound seeps into nutplates.
- a. On helicopters with input flange/coupling 70351-08255-102, install packing on pinion gear (Figure 6-4-27-2, Detail A).
  - b. Slide input module input flange or input flange/coupling into input module oil seal until seated (Figure 6-4-27-1, Detail A or Figure 6-4-27-2, Detail A).
  - c. Install retaining nut.

**NOTE**

Do not remove torque reactor or torque reactor plate until retaining nut is torqued.

- d. Using torque reactor or torque reactor plate, hold input module input flange or input flange/coupling and check drag torque of retaining nut. **MINIMUM DRAG TORQUE IS 18 INCH-POUNDS.** If drag torque is beyond limits, replace retaining nut.



| NOTES                                                   | COMPONENT | ZONE | DIMENSION                      | REPAIR LIMIT       |
|---------------------------------------------------------|-----------|------|--------------------------------|--------------------|
|                                                         |           |      | BEFORE REPAIR                  |                    |
| 1. ALL DIMENSIONS ARE IN INCHES, UNLESS NOTED.          | LENGTH    | A    | (THICK FLANGE) 0.210 (± 0.005) | 0.200 (SEE NOTE 2) |
|                                                         | OD        | B    | 2.600 (± 0.010)                | 2.580              |
|                                                         | RADII     | C    | 0.125 (± 0.010)                | 0.105              |
|                                                         | OD        | D    | 2.600 (± 0.010)                | 2.580              |
| 2. MAXIMUM FRETTING DEPTH IS 0.0005.                    | ID        | E    | 2.3654 (± 0.0005)              | 2.3664             |
|                                                         | SPLINE    | F    | 2.0349 (± 0.0009)              | (SEE NOTE 3)       |
| 3. NOT MORE THAN 0.002 WEAR STEP IN SPLINES ALLOWED.    | ID        | G    | 2.360 (± 0.010)                | 2.380              |
|                                                         | EAR       | H    | N/A                            | (SEE NOTE 4)       |
| 4. BLEND 0.020 MAXIMUM DEPTH WITH 0.250 MINIMUM RADIUS. |           |      |                                |                    |

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FIGURE 6-4-27-4. REPAIR INPUT MODULE INPUT FLANGE

- e. TORQUE RETAINING NUT TO 646 - 714 INCH-POUNDS.

f. Remove torque reactor or torque reactor plate from input module input flange.

g. Apply torque stripe (PARA 2-6-5).

h. Inspect input pinion (PARA 6-3-8).
- i. On helicopters with input flange/coupling 70351-08255-102, install gimbal (PARA 6-4-26).

j. Service main transmission (PARA 1-3-12).

k. Install engine (PARA 4-4-1).



6-4-28    ACCESSORY MODULE.

PARAGRAPH BREAKDOWN

|          |                          | PAGE     |
|----------|--------------------------|----------|
| 6-4-28.1 | Replace Accessory Module | 6-4-28-2 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Jacking Bolt, 1/4-28
- Nonmetallic Scraper, Locally-Made
- Powertrain Repairer’s Toolkit
- Soft Mallet
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Chemical Conversion Coating, Item 118, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Coating Kit, Item 126A, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Lockwire, Item 197, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D
- Lubricating Oil, Item 208, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

- Paint, Polyurethane, Item 224, Appendix D
- Primer, Item 258, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D
- Tape, Item 317C, Appendix D
- Tiedown Strap, Item 329, Appendix D

References

- Appendix D
- PARA 1-3-8
- PARA 1-3-12
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-5
- PARA 6-2-2
- PARA 6-3-4
- PARA 6-3-6
- PARA 6-3-7
- PARA 6-4-13
- PARA 6-4-15
- PARA 6-4-16

- PARA 6-4-17
- PARA 6-4-18
- PARA 6-4-30
- PARA 6-4-75
- PARA 6-4-76
- PARA 7-4-12
- PARA 9-4-16

### 6-4-28.1 Replace Accessory Module.

#### NOTE

- Replacement of left or right accessory module is the same, except as noted.
- Accessory module without the hydraulic pump module and main generator on weighs about 28 pounds.

#### 6-4-28.1.1. Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Open main rotor pylon sliding cover.
- c. Remove hydraulic pump module (PARA 7-4-12).
- d. Remove main generator (PARA 9-4-16).
- e. Disconnect electrical connector, P405 (right side), or P406 (left side), from accessory module chip detector (Figure 6-4-28-1, Sheet 1, Detail A).
- f. Install protective caps and plugs on electrical connectors.
- g. Remove end of accessory module chip detector from bracket.
- h. Remove tape or tiedown straps from harness (Figure 6-4-28-1, Sheet 2, Detail B).
- i. Remove screw, washer, and nut securing clamp(s) and harness to bracket on accessory module.
- j. On helicopters with main transmission, 70351-08000 or 70084-35001, if removing right ac-

cessory module, disconnect electrical connector from rotor speed (Nr) sensor as follows (Figure 6-4-28-1, Sheet 1, Detail A):

- (1) Disconnect electrical connector, P443, from back side of rotor speed (Nr) sensor on right accessory module.
  - (2) Install protective caps and plugs on electrical connectors.
- k. On helicopters with main transmission, 70084-35007, 70351-48000, or 70351-58000, disconnect electrical connector from rotor speed (Nr) sensor as follows (Figure 6-4-28-1, Sheet 1, Detail A):
    - (1) Disconnect electrical connector, P407 (left side), or P443 (right side), from back side of rotor speed (Nr) sensor on accessory module.
    - (2) Install protective caps and plugs on electrical connectors.
- #### NOTE
- Right accessory module does not have a low oil pressure switch.
- l. On left accessory module, disconnect electrical connector, P409, from low oil pressure switch.
  - m. Install protective caps and plugs on electrical connectors.
  - n. Disconnect engine load-demand push/pull cable supports from accessory module by removing bolts, washers, and nuts. Retain bolts, washers, and nuts with engine load demand push/pull cable supports.

### 6-4-28-2



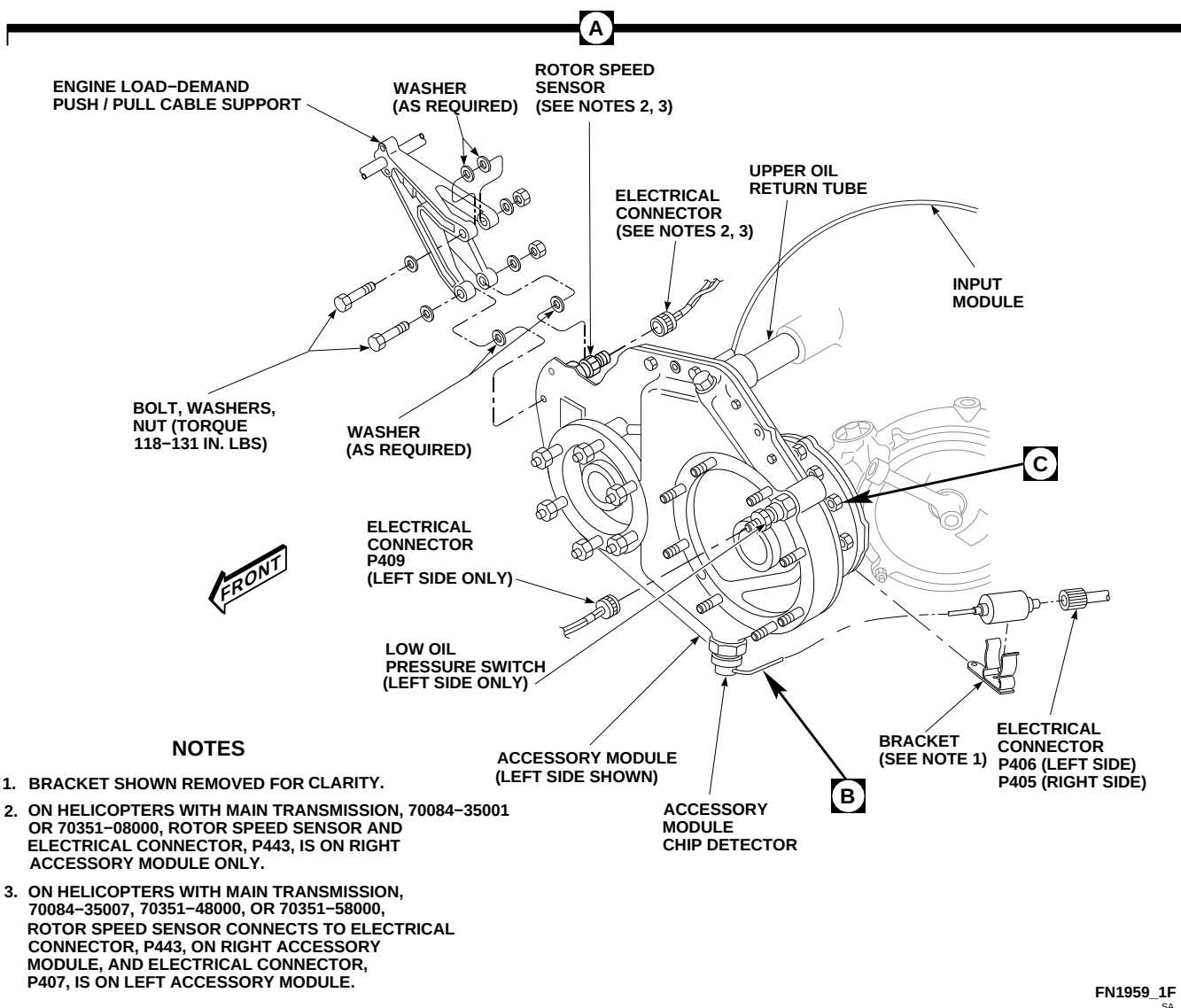
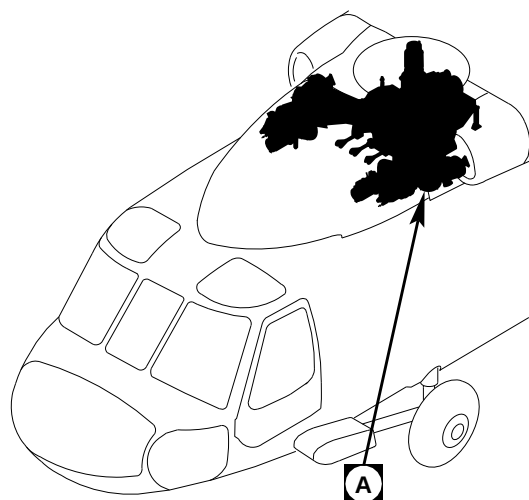


FIGURE 6-4-28-1. REPLACE ACCESSORY MODULE (SHEET 1 OF 2)

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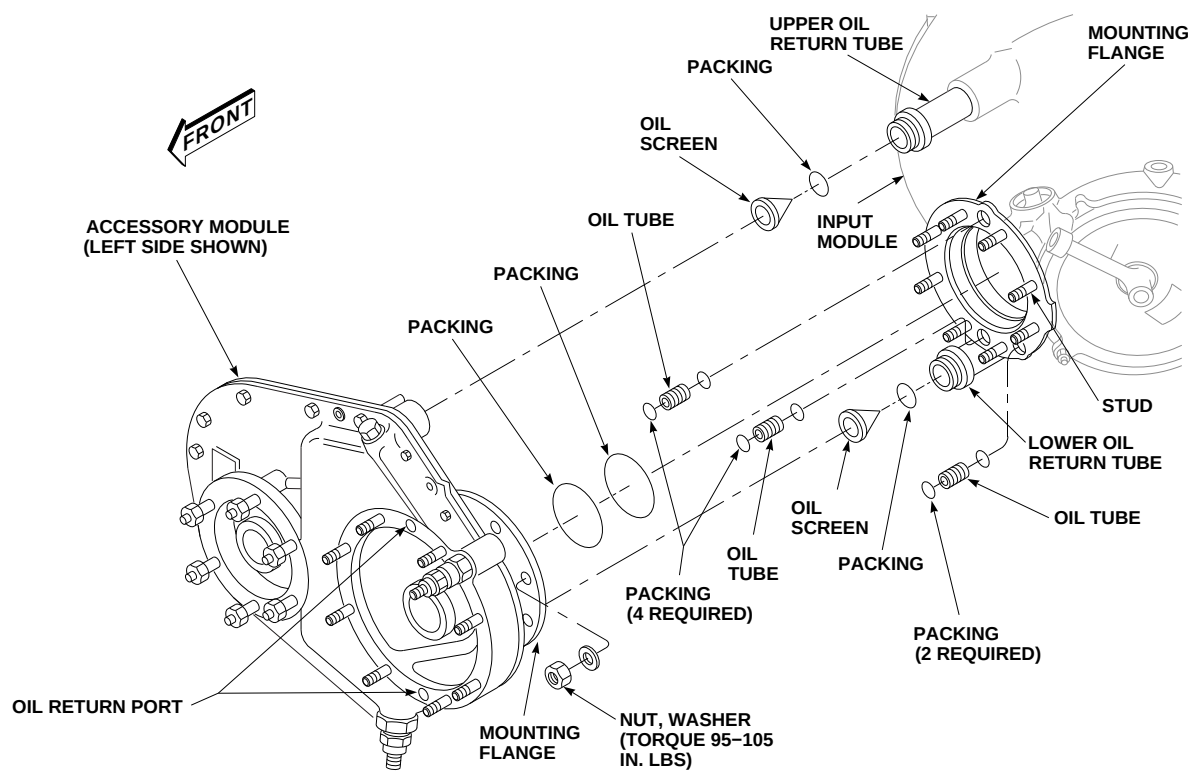
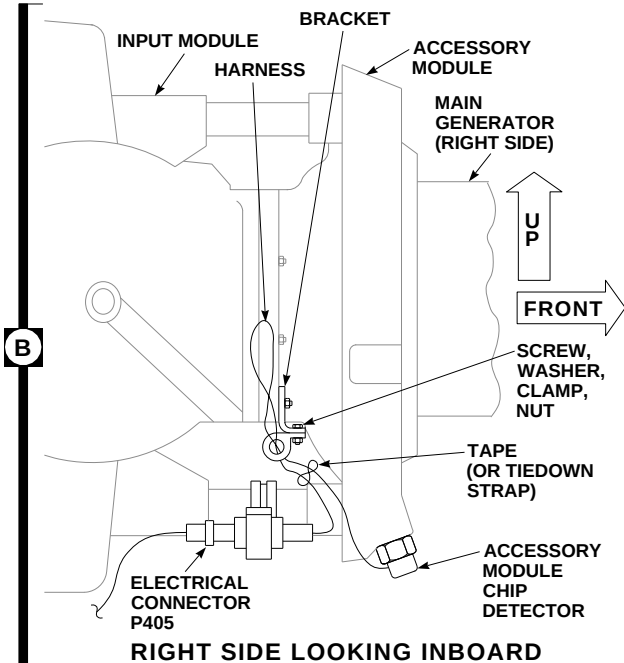
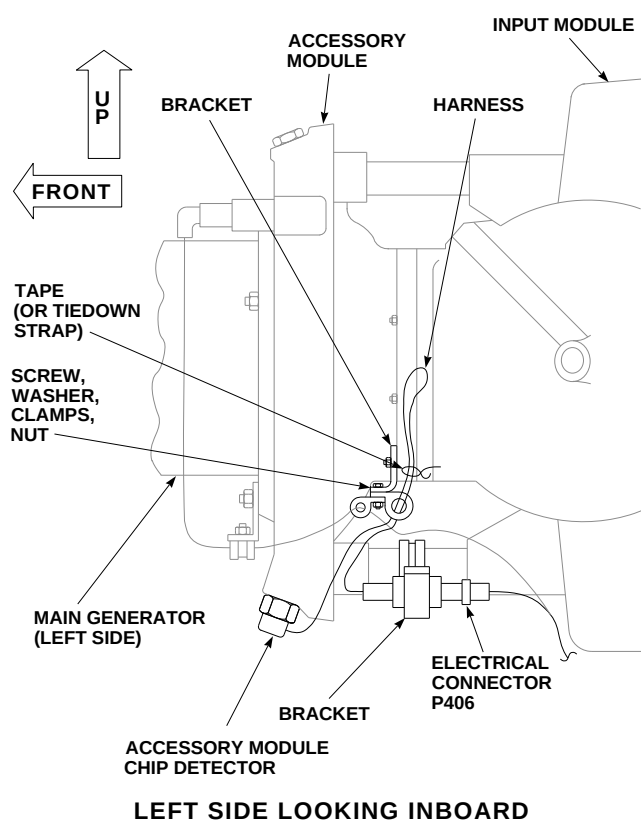


FIGURE 6-4-28-1. REPLACE ACCESSORY MODULE (SHEET 2 OF 2)

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6-4-28-4

- o. Using nonmetallic scraper, remove sealing compound from upper and lower oil return tubes.
- p. Remove retaining rings from upper and lower oil return tubes on accessory module.
- q. Remove nuts and washers from mounting flange of accessory module. Remove accessory module and packings. Discard packings (Figure 6-4-28-1, Sheet 2, Detail C).

**NOTE**

When removing accessory module, oil return tube will slide free of accessory module and stay with input module.

- r. Lift accessory module. If accessory module does not pull off input module easily, it may be jacked out as follows:
  - (1) Install 1/4-28 jacking bolt in jacking hole in mounting flange on input module.

**NOTE**

Be careful not to cock accessory module.

- (2) Slowly thread in jacking bolt, and at same time gently tap opposite side of accessory module using soft mallet. Be careful not to cock accessory module.
- s. Check accessory module shim on mounting flange of accessory module for separation and torn areas. **ACCESSORY MODULE SHIM MAY BE REUSED AS LONG AS IT DOES NOT HAVE ANY MAJOR DAMAGE SUCH AS CUTS, RAISED OR DEPRESSED AREAS, OR ANY OTHER DAMAGE THAT PREVENTS ACCESSORY MODULE FROM MATING PROPERLY WITH MOUNTING FLANGE ON INPUT MODULE.** If damaged beyond limits, replace accessory module shim (PARA 6-4-30).

- t. Check oil screens for metal particle contamination. Compare any particles found with accept/reject table (PARA 6-3-6).
- u. Carefully package and return any foreign material with accessory module. If necessary, replace oil screens.
- v. Keep oil screens with input module.
- w. Remove oil tubes between accessory module and input module. Discard packings. Keep oil tubes with input module.
- x. If accessory module is going to be replaced, remove engine load-demand push/pull cable support from new accessory module as follows (Figure 6-4-28-1, Sheet 1, Detail A):
  - (1) Remove both bolts, washers, and nuts from engine load-demand push/pull cable support on new accessory module to be installed.
  - (2) Install these bolts, washers, and nuts in accessory module just removed (old accessory module).
  - (3) **TORQUE NUTS TO 118 - 131 INCH-POUNDS.**
- y. Using nonmetallic scraper, remove sealing compound.
- z. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean primer off mating surface and hardware.
- aa. If new accessory module is not readily available for installation, cover opening in input module.

**6-4-28.1.2. Inspect.**

Visually inspect accessory module housing for nicks, gouges, scratches, and corrosion. If necessary, repair (PARA 6-4-13), or replace accessory module housing.

**6-4-28.1.3. Accessory Module Teardown/Buildup.****NOTE**

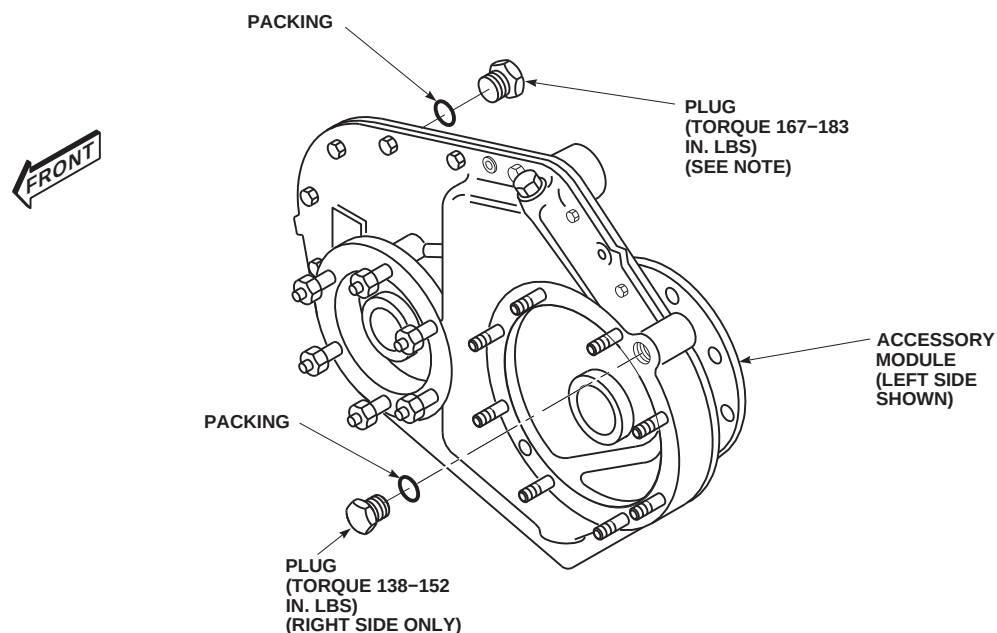
Perform the following if a replacement accessory module is going to be installed.

- a. Install protective caps and plugs as items are removed.
- b. On left accessory module, perform the following:
  - (1) Remove low oil pressure switch (PARA 6-4-76).
  - (2) On helicopters with main transmission, 70084-08000 or 70084-35001, remove plug and packing. Retain plug for installation on replacement module. Discard packing (Figure 6-4-28-2).
  - (3) Remove and inspect accessory module chip detector (PARA 6-4-15).
- c. On right accessory module, perform the following:
  - (1) Remove plug and packing. Retain plug for installation on replacement module. Discard packing.
  - (2) Remove rotor speed (Nr) sensor (PARA 6-4-75).
  - (3) Remove and inspect accessory module chip detector (PARA 6-4-15).
- d. Remove protective caps and plugs as items are installed.
- e. On replacement left accessory module, perform the following:

- (1) Install packing on plug and install in accessory module.
- (2) **TORQUE PLUG TO 167 - 183 INCH-POUNDS.**
- (3) Lockwire, Item 197, Appendix D, plug.
- (4) Install accessory module chip detector (PARA 6-4-15).
- (5) Install low oil pressure switch (PARA 6-4-76).
- f. On replacement right accessory module, perform the following:
  - (1) Install rotor speed (Nr) sensor (PARA 6-4-75).
  - (2) Install packing on plug and install in front of accessory module.
  - (3) **TORQUE PLUG TO 138 - 152 INCH-POUNDS.**
  - (4) Lockwire, Item 197, Appendix D, plug.
  - (5) Install accessory module chip detector (PARA 6-4-15).

**6-4-28.1.4. Install.**

- a. Remove oil screens and packings from oil return tubes (Figure 6-4-28-1, Sheet 2, Detail C). Discard packings.
- b. Using dry-cleaning solvent, Item 124, Appendix D, clean screens.
- c. Lubricate packings with lubricating oil, Item 205, Item 206, or Item 208, Appendix D.
- d. Install oil screens and packings in upper and lower oil return tubes, with point of oil screen in oil return tube.



**NOTE**

ON HELICOPTERS WITH MAIN MODULE, 70084-35001 OR 70351-08000, PLUG IS INSTALLED ON LEFT ACCESSORY MODULE ONLY.

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**FIGURE 6-4-28-2. TEARDOWN/BUILDUP ACCESSORY MODULE**

**NOTE**

Oil return ports for main generator may or may not be drilled in accessory gear box. Oil tube bores will always be drilled in accessory gear box to accept oil tubes and packings. Make sure oil tubes and packings are installed between input and accessory gear box, whether oil return ports are drilled or not, to prevent leakage from input module oil return ports.

- e. Place packings on oil tubes.
- f. Install oil tubes in input module.
- g. Place packings on accessory module.
- h. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean mounting surfaces on input and accessory module.

- i. Apply a coat of sealing compound, Item 292, Appendix D, between Rosan lockring and mating housing counterbore.
- j. Using chemical conversion coating, Item 118, Appendix D, touch up magnesium surfaces.

**WARNING**

Injury to personnel and damage to equipment will result if transmission oil is contaminated. To avoid contaminating transmission oil system, do not apply primer to bore diameters that are sealing surfaces for packings.

**NOTE**

Install accessory module while primer is wet.

- k. Apply a coat of primer, Item 258, Appendix D, to accessory module mounting surface and

fastener holes on input module. Do not apply primer to packing or packing grooves.

**CAUTION**

Damage to accessory module will result if it is not carefully installed on input module. Be careful not to drop accessory module or burr stud threads when installing accessory module.

- l. Place accessory module on mounting flange of input module. Make sure oil return tubes and oil tubes are properly seated.
- m. Install washers and nuts.
- n. **TORQUE NUTS TO 95 - 105 INCH-POUNDS.**
- o. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to all exposed washers, nuts, and studs securing accessory module to input module.
- p. Touch up paint on exposed boltheads, nuts, and threads, as necessary (PARA 2-6-5).
- q. Apply sealing compound, Item 292, Appendix D, to seal area around joint of washer and nut, thread of studs to nut, and seam between accessory module and input module.
- r. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, remove excess sealing compound.
- s. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer over sealing compound.

**NOTE**

Housings coated with epoxy coating are glossy silver. Housings coated with polyurethane paint are matte surface medium-gray.

- t. Using epoxy coating kit, Item 126A, Appendix D, apply one coat of epoxy coating, or one coat

of polyurethane paint, Item 224, Appendix D, depending on original paint finish.

- u. Install main generator (PARA 9-4-16).
- v. Install hydraulic pump module (PARA 7-4-12).
- w. Inspect and clean accessory, input, and main module oil return screens, chip detectors, and chip detector strainers (PARA 6-4-15, PARA 6-4-16, PARA 6-4-17, and PARA 6-4-18).
- x. Check accessory module chip detector electrical circuit (PARA 6-2-2).
- y. Inspect and clean main transmission oil filter (PARA 6-3-7).
- z. Remove protective caps and plugs from electrical connectors.
- aa. Inspect and treat electrical connectors (PARA 1-7-20).
- ab. Install end of accessory module chip detector in bracket.
- ac. Connect electrical connector, P405 (right side), or P406 (left side), to accessory module chip detector (Figure 6-4-28-1, Sheet 1, Detail A).
- ad. Route harness and secure to bracket on accessory module with clamp(s), screw, washer, and nut (Figure 6-4-28-1, Sheet 2, Detail B).

**NOTE**

Tiedown straps may be used as an alternate for tape.

- ae. Secure slack in harness with tape, Item 317C, Appendix D, or tiedown straps, Item 329, Appendix D.
- af. On helicopters with main transmission, 70351-08000 or 70084-35001, if right accessory module was removed, connect electrical connector to rotor speed (Nr) sensor as follows:

- (1) Remove protective caps and plugs from electrical connectors.

- (2) Inspect and treat electrical connectors (PARA 1-7-20).
  - (3) Connect electrical connector, P443, to back side of rotor speed (Nr) sensor on right accessory module.
  - (4) Lockwire, Item 197, Appendix D, electrical connector.
- ag. On helicopters with main transmission, 70084-35007, 70351-48000, and 70351-58000, connect electrical connector to rotor speed (Nr) sensor as follows:
- (1) Remove protective caps and plugs from electrical connectors.
  - (2) Inspect and treat electrical connectors (PARA 1-7-20).
  - (3) Connect electrical connector, P407 (left side), or P443 (right side), to back side of rotor speed (Nr) sensor on accessory module.
  - (4) Lockwire, Item 197, Appendix D, electrical connectors.
- ah. Remove protective caps and plugs from electrical connectors.
- ai. Inspect and treat electrical connectors (PARA 1-7-20).
- aj. On left accessory module, connect electrical connector, P409, to low oil pressure switch over main generator.
- ak. If electrical connector is not self-locking, lockwire, Item 197, Appendix D, electrical connector.
- al. Connect engine load-demand push/pull cable support to accessory module as follows:
- (1) Apply a coat of primer, Item 258, Appendix D, to boltholes of engine load-demand push/pull cable support on accessory module.
  - (2) Position engine load-demand push/pull cable support on accessory module.
  - (3) Add washers as required to fill gaps between engine load-demand push/pull cable support and accessory module.
  - (4) Bolt engine load-demand push/pull cable support to accessory module.
  - (5) **TORQUE TO 118 - 131 INCH-POUNDS.**
- am. Service main transmission (PARA 1-3-12).
- an. Check hydraulic fluid level in hydraulic pump module. Service as necessary (PARA 1-3-8).
- ao. Make sure area is clean and free of foreign material.
- ap. Close main rotor pylon sliding cover.
- aq. Perform main transmission functional test (PARA 6-3-4).





6-4-29 ACCESSORY MODULE SPLINE ADAPTER.

PARAGRAPH BREAKDOWN

|                                                  | PAGE     |
|--------------------------------------------------|----------|
| 6-4-29.1 Replace Accessory Module Spline Adapter | 6-4-29-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Mallet
- Nonmetallic Drift, Locally-Made
- Spline Adapter Removal Tool, Locally-Made, Figure H-261 , Appendix H

Materials/Parts

- Lubricating Oil, Item 202, Appendix D

References

- Appendix D
- Appendix H
- PARA 6-4-25
- PARA 9-4-16

6-4-29.1 Replace Accessory Module Spline Adapter.

NOTE

Replacement of right and left accessory module spline adapters is the same.

6-4-29.1.1. Remove.

- Remove main generator (PARA 9-4-16).
- Locally make spline adapter removal tool (Figure H-261 , Appendix H).
- Install spline adapter removal tool into accessory module spline adapter, making sure splines engage each other.
- Tighten nut of spline adapter removal tool and remove accessory module spline adapter (Figure 6-4-29-1, Detail A).

- Remove spline adapter removal tool by loosening nut and removing from accessory module spline adapter.

6-4-29.1.2. Inspect.

- Inspect accessory module spline adapter for wear, cracks, or discoloration due to heat. **NONE ALLOWED.** If damaged, replace accessory module spline adapter.
- Inspect accessory module spline adapter for missing pieces. **DIMENSION OF MISSING PIECES SHALL NOT EXCEED 0.10-INCH. IF ANY PIECE OF ACCESSORY MODULE SPLINE ADAPTER WITH DIMENSION GREATER THAN 0.010-INCH IS MISSING, CHECK OIL RETURN TUBES ON INPUT MODULE AND CHIP DETECTOR SCREENS FOR MISSING PIECES. IF MISSING PIECES CANNOT BE LOCATED, REPLACE INPUT MODULE (PARA 6-4-25), AND SPLINE ADAPTER.**

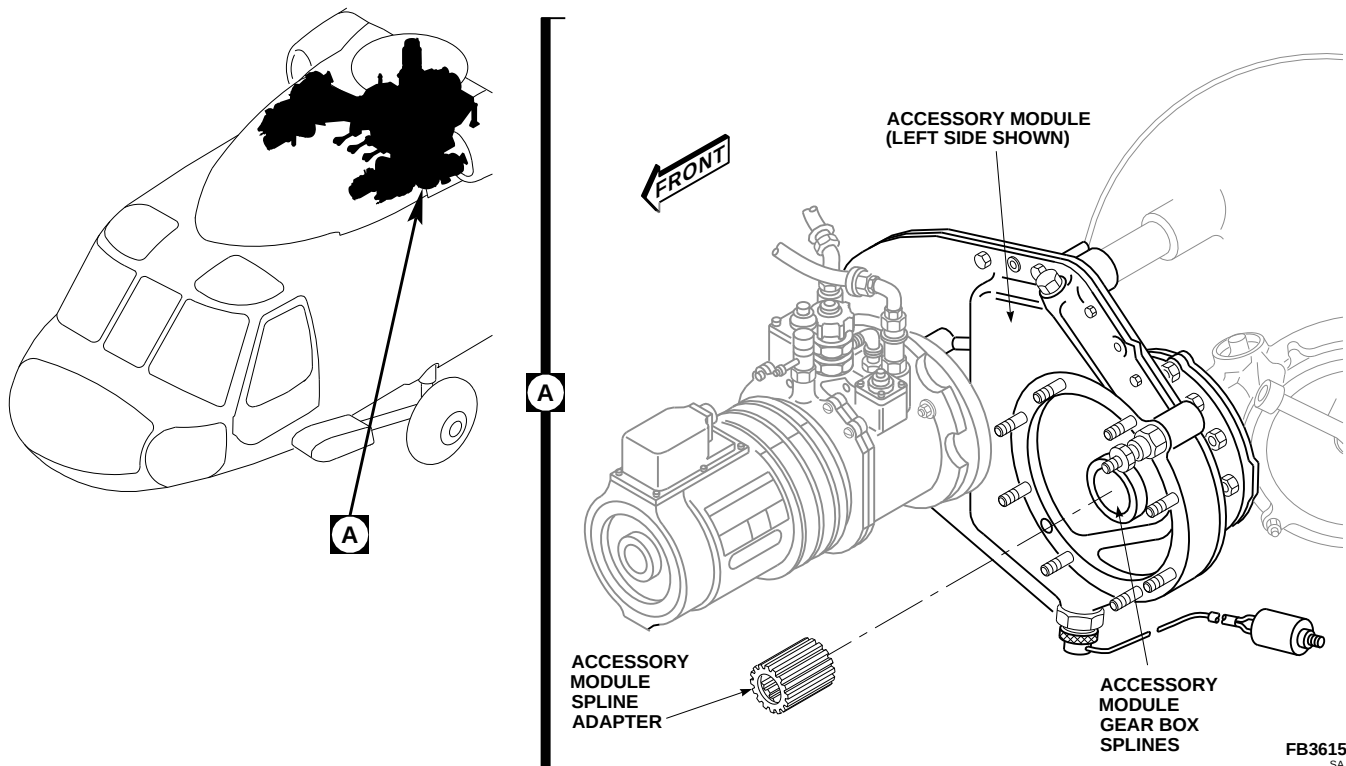


FIGURE 6-4-29-1. REPLACE ACCESSORY MODULE SPLINE ADAPTER

#### 6-4-29.1.3. Install.

- a. Check accessory module gear box splines to make sure they are free and clean of foreign material.

#### CAUTION

- Do not open bag containing spline adapter until ready for use. Torlon (green) spline adapters will absorb moisture and increase in size.

- Spline adapter is a press-fit and is difficult to install. Do not use excessive force to install spline adapter.
- b. Lightly coat outside surface of spline adapter with lubricating oil, Item 202, Appendix D.
  - c. Using mallet and nonmetallic drift, carefully tap accessory module spline adapter (tapered end first) into accessory module gear box splines until seated (Figure 6-4-29-1, Detail A).
  - d. Install main generator (PARA 9-4-16).

6-4-30 ACCESSORY MODULE SHIM.

PARAGRAPH BREAKDOWN

|          |                               | PAGE     |
|----------|-------------------------------|----------|
| 6-4-30.1 | Replace Accessory Module Shim | 6-4-30-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Scraper, Locally-Made
- Pliers

- Powertrain Repairer’s Toolkit

Materials/Parts

- Epoxy Primer Coating Kit, Item 135, Appendix D

References

- Appendix D
- PARA 6-4-28

6-4-30.1 Replace Accessory Module Shim.

NOTE

Replacement of left and right accessory module shims is the same.

6-4-30.1.1. Remove.

- Remove accessory module (PARA 6-4-28).
- Insert nonmetallic scraper between mounting flange on accessory module and accessory module shim. Using nonmetallic scraper as a wedge, gently pry accessory module shim away from mounting flange until head of pin is above surface of accessory module shim (Figure 6-4-30-1, Detail A).
- Using pliers, grasp head of pin and pull pin out.

- Repeat step b. and step c. for remaining pin and remove accessory module shim.

6-4-30.1.2. Install.

- Measure and record thickness of accessory module shim in three places 120° apart. Record average of three measurements.
- Peel new accessory module shim to average thickness ± 0.001-inch of average measurement recorded in step a.
- Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to mounting flange on accessory module.
- Position accessory module shim on mounting flange and secure with pins (Figure 6-4-30-1, Detail A).
- Install accessory module (PARA 6-4-28).

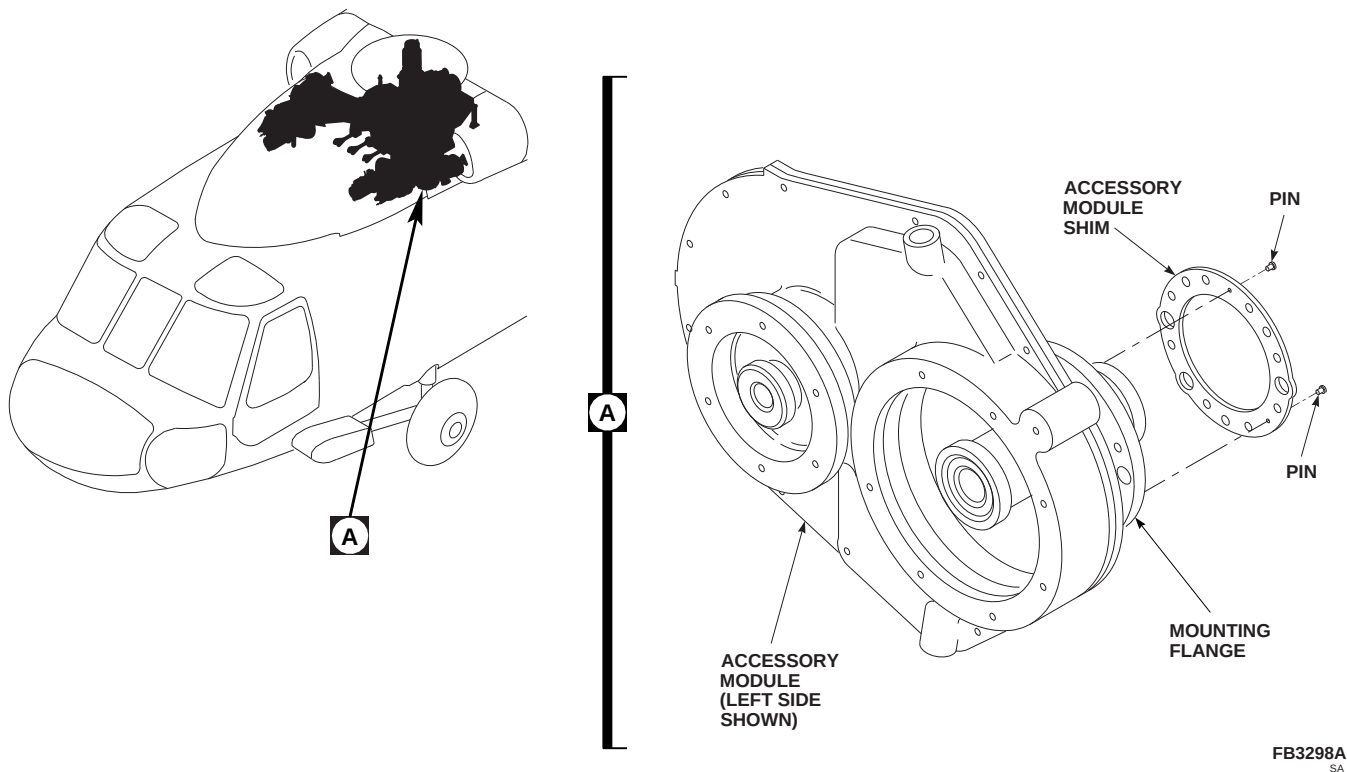


FIGURE 6-4-30-1. REPLACE ACCESSORY MODULE SHIM

6-4-31 INTERMEDIATE GEAR BOX.

PARAGRAPH BREAKDOWN

|                                        | PAGE     |
|----------------------------------------|----------|
| 6-4-31.1 Replace Intermediate Gear Box | 6-4-31-2 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drill
- Drill Bit, 1¼-inch diameter
- Feeler Gage
- Fluorescent Penetrant Inspection Kit
- Nonmetallic Scraper, Locally-Made
- Nonpermanent Ink Marker
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Wood Blocks, to Support Drive Shaft

Materials/Parts

- Brush, Acid, Item 84, Appendix D
- Corrosion Preventive Compound, Item 113, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Gear Box Shim, 70219-06000-106, Locally-Made, Figure H-199, Appendix H
- Gear Box Shim, 70219-06000-107, Locally-Made, Figure H-199, Appendix H

- Gear Box Shim, 70219-06000-108, Locally-Made, Figure H-199, Appendix H
- Gear Box Shim, 70219-06000-110, Locally-Made, Figure H-199, Appendix H
- Gear Box Shim, 70219-06000-111, Locally-Made, Figure H-199, Appendix H
- Gear Box Spacer, 70219-06000-109, Locally-Made, Figure H-200, Appendix H
- Machinery Towel, Item 211, Appendix D
- Masking Tape, 3-inches, Item 215, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Molybdenum Disulfide, Item 219, Appendix D
- Paint, Polyurethane, Item 228, Appendix D
- Primer, Item 255, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- Appendix H
- Appropriate Flight Manual
- Appropriate MTF
- ASTM E1417

- PARA 1-3-13
- PARA 1-6-16
- PARA 1-6-18
- PARA 1-7-20
- PARA 2-4-139
- PARA 6-2-2

### 6-4-31.1 Replace Intermediate Gear Box.

#### NOTE

Intermediate gear box weighs 32 pounds.

#### 6-4-31.1.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Drain intermediate gear box (PARA 1-3-13).

#### NOTE

Insulation sleeve on electrical connector is used to prevent water intrusion.

- c. Pull insulation sleeve back away from electrical connector, P613 (Figure 6-4-31-1, Sheet 1, View A).
- d. Disconnect electrical connector, P613, from intermediate gear box chip detector/temperature sensor (Figure 6-4-31-1, Sheet 1, Detail B).
- e. Install protective caps and plugs on electrical connectors.

#### CAUTION

Damage to tail rotor drive shaft sections III and IV will result if handled improperly. Handle tail rotor drive shaft sections III and IV carefully to prevent from being dented, scratched, or bent.

#### NOTE

Tail rotor drive shaft sections III and IV coupling bolts, self-locking nuts, and

viscous damper bearing assembly attachment bolts and nuts can be reused, provided they show no visual evidence of damage.

- f. Using machinery towels, Item 211, Appendix D, or wood blocks, firmly support rear end of tail rotor drive shaft section III and lower end of tail rotor drive shaft section IV.

#### WARNING

Injury to personnel and damage to equipment will result if tail rotor is not secured when tail rotor drive shaft sections III and IV are disconnected. Secure tail rotor to prevent tail rotor drive shaft sections III and IV and tail rotor from turning.

- g. Secure tail rotor blades to tail rotor pylon to keep from freewheeling.
- h. If not already done, engage gust lock (PARA 1-6-18).
- i. Disconnect tail rotor drive shaft section III to input flange of intermediate gear box by removing bolts, spherical washers, shims, and nuts (Figure 6-4-31-1, Sheet 2, Detail C).
- j. Disconnect tail rotor drive shaft section IV to output flange of intermediate gear box by removing bolts, spherical washers, shims, and nuts.
- k. Remove bolts securing intermediate gear box to supports (Figure 6-4-31-1, Sheet 3, Detail F).
- l. Remove intermediate gear box.

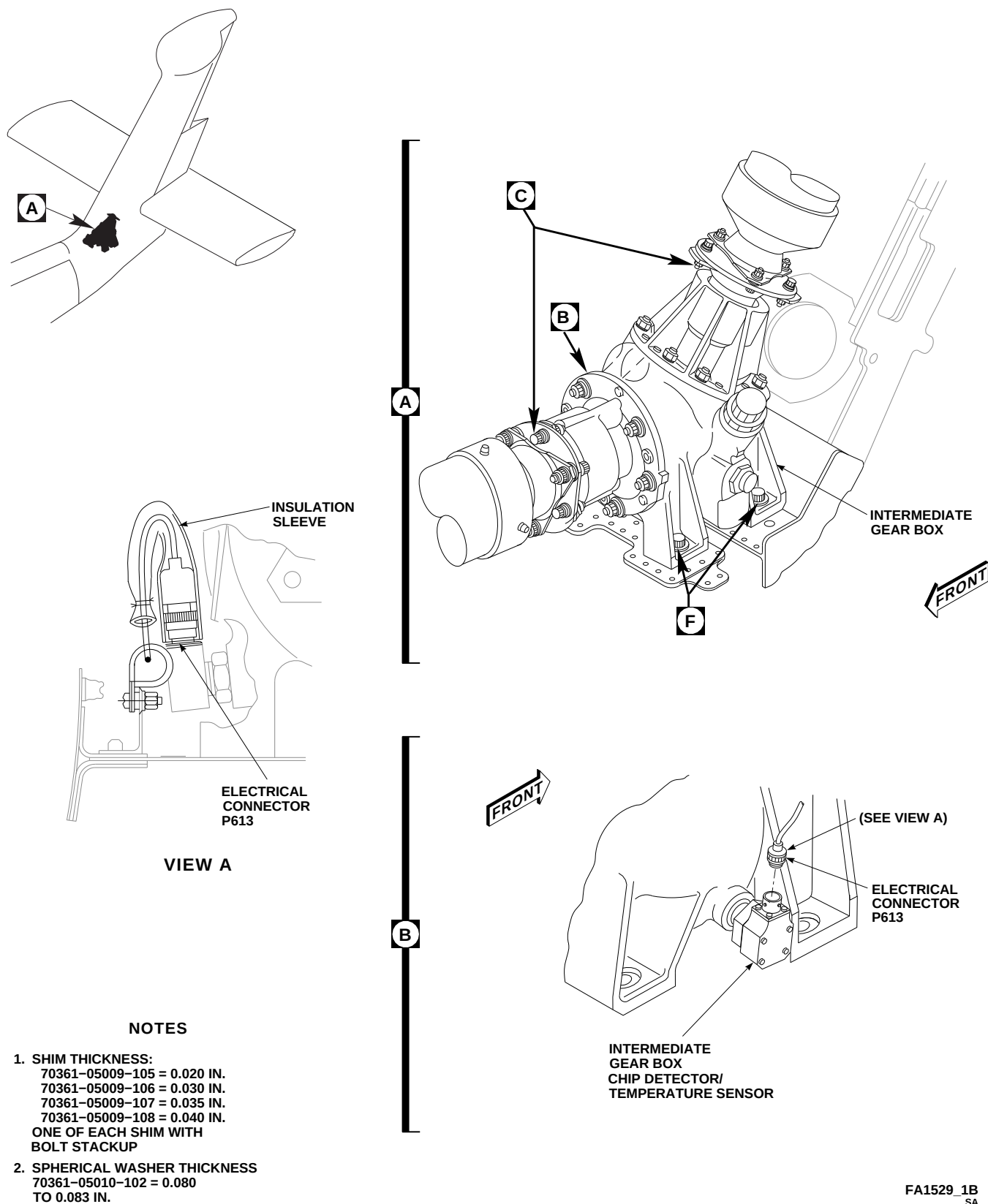


FIGURE 6-4-31-1. REPLACE INTERMEDIATE GEAR BOX (SHEET 1 OF 3)

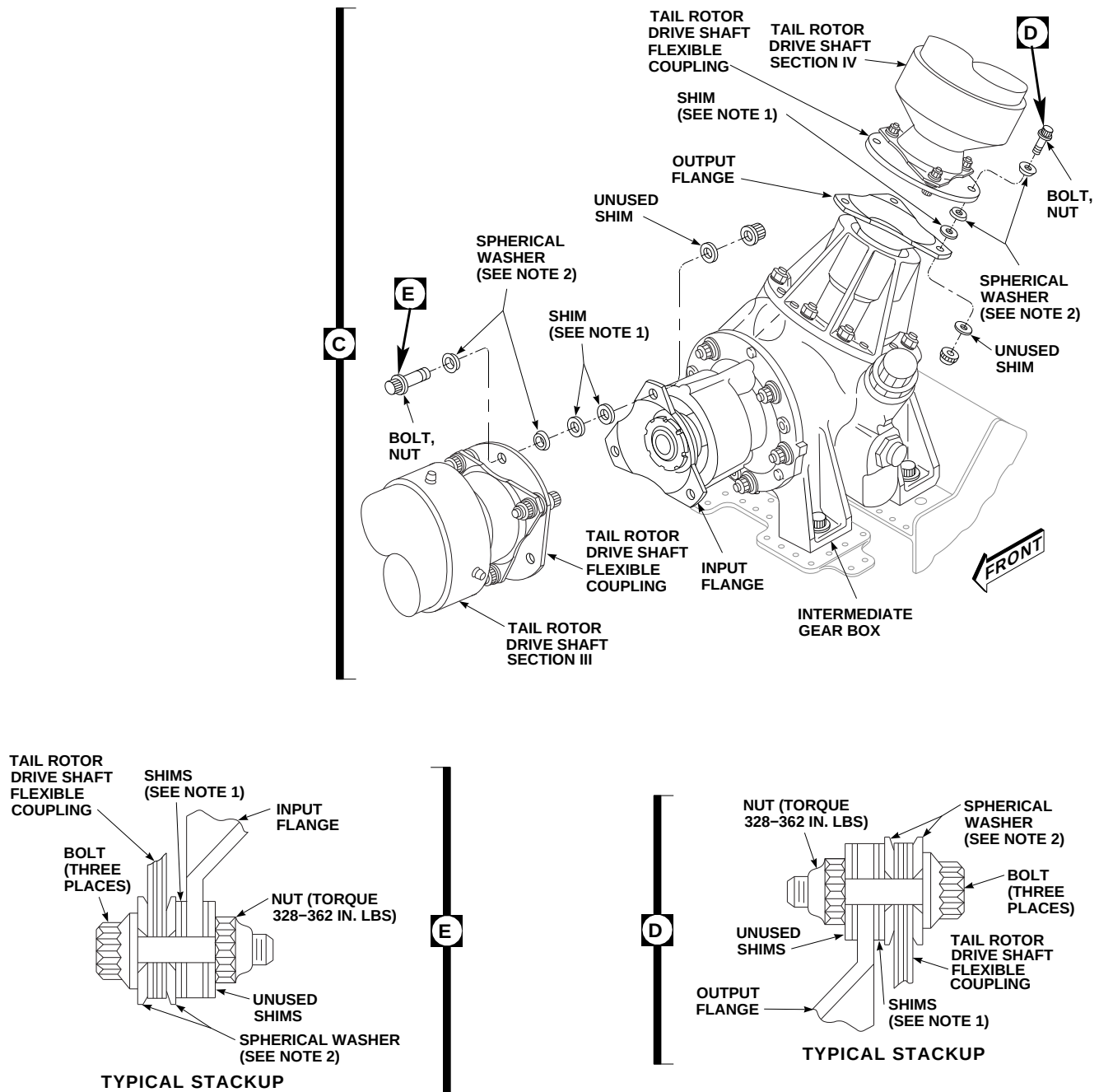
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FIGURE 6-4-31-1. REPLACE INTERMEDIATE GEAR BOX (SHEET 2 OF 3)



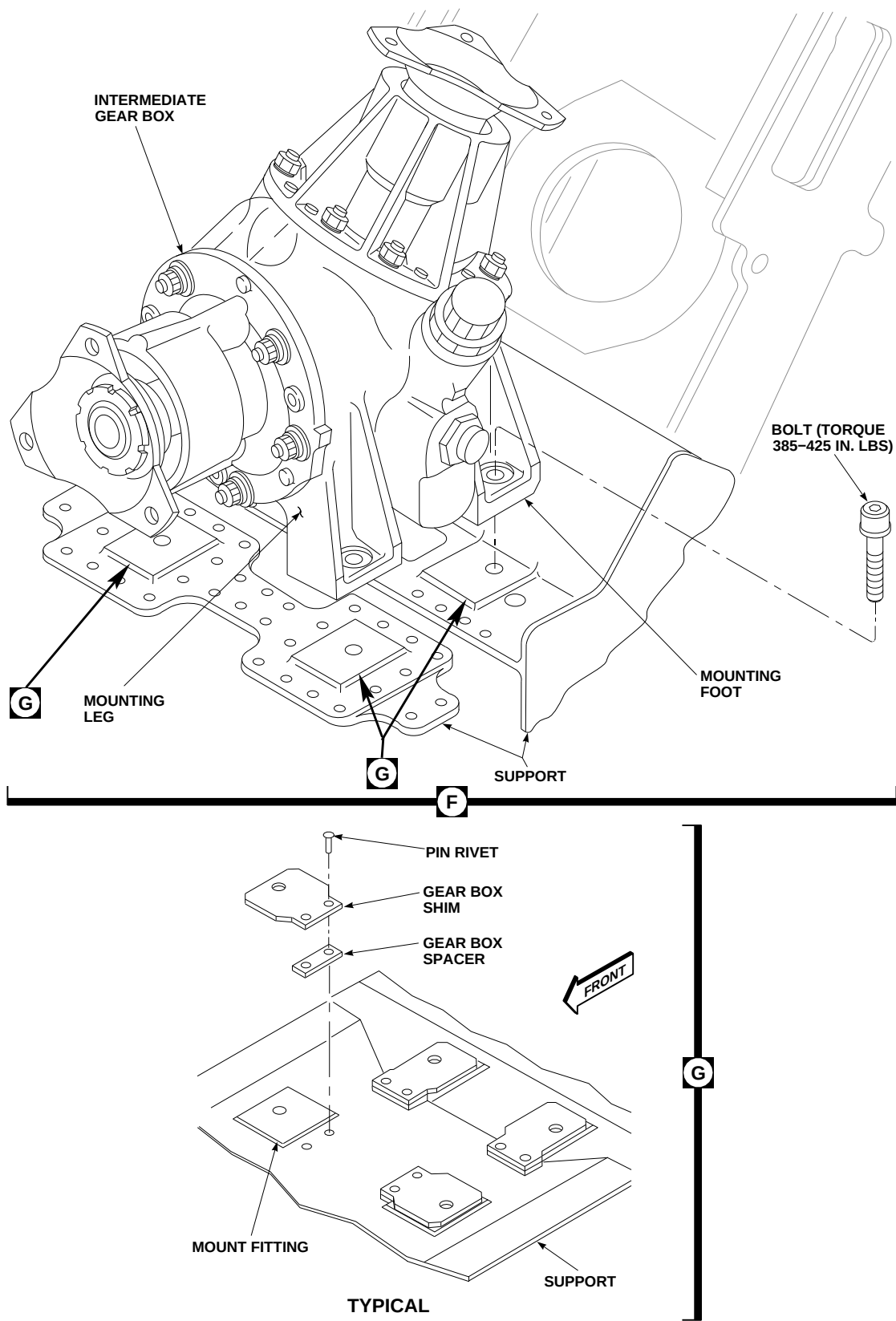


FIGURE 6-4-31-1. REPLACE INTERMEDIATE GEAR BOX (SHEET 3 OF 3)

**6-4-31.1.2 Inspect.**

**CAUTION**

Damage to barrel-nuts will result if mounting bolts are forced into them. Threads will be stripped. Do not attempt to force mounting bolts into barrel-nuts.

**NOTE**

Make sure that only barrel-nuts, RMLH2577, are used. These barrel-nuts are identified with letters "EN" or "SPS" stamped on an end (additional marking may also include "LH2577").

- a. Thread bolts into front mounting holes until gap between mounting surface and underside of bolthead is 0.669 - 0.741-inch.
- b. Thread bolts into rear mounting holes until gap between mounting surface and underside of bolthead is 0.619 - 0.691-inch.
- c. Using torque wrench, back out bolt.
- d. **MINIMUM BREAKAWAY TORQUE IS 9.5 INCH-POUNDS.** If breakaway torque is less than 9.5 inch-pounds, replace barrel-nuts.

**6-4-31.1.3 Install/Repair.**

- a. Check intermediate gear box for mounting gap as follows:
  - (1) Using nonmetallic scraper, remove any sealing compound from mounting surfaces of intermediate gear box and supports.
  - (2) Place intermediate gear box on supports and install mounting bolts. **TIGHTEN BOLTS AS REQUIRED UNTIL BOLT-HEAD MAKES LIGHT CONTACT WITH MOUNTING FEET. DO NOT EXCEED 100 INCH-POUNDS** (Figure 6-4-31-1, Sheet 3, Detail F).
  - (3) Using feeler gage, check for gaps between mounting surfaces. **MAXIMUM GAP IS**

**0.030-INCH.** If gap exceeds 0.030-inch under any mounting foot, install gear box shim to bring gap within limits as follows:

- (a) Remove bolts and remove intermediate gear box from supports.

**NOTE**

If a gear box shim was previously installed on a mount fitting which now requires a thicker gear box shim, existing shim must be removed.

- (b) If necessary, remove pin rivets securing gear box shim to mount fitting and support (Figure 6-4-31-1, Sheet 3, Detail G).
- (c) If necessary, using nonmetallic scraper, remove original gear box shim and gear box spacer from mount fitting.
- (d) Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, remove old sealing compound from mount fitting.

**NOTE**

- Use only one gear box shim, one gear box spacer, and two rivets on each mount fitting.
- Replacement gear box shim thickness must be original gear box shim thickness plus thickness required for correction.
  - (e) Locally make replacement gear box shims (Figure H-199, Appendix H) and gear box spacer (Figure H-200, Appendix H).
  - (f) Position replacement gear box shim and gear box spacer on mount fitting.
  - (g) Using suitable nonpermanent ink marker, mark location of fastener holes on gear box shim.

- (h) Using drill with  $1\frac{1}{64}$ -inch diameter drill bit, drill holes in gear box shim.
- (i) Apply coat of primer, Item 255, Appendix D, to all gear box shim and gear box spacer surfaces. Allow to cure.
- (j) Using acid brush, Item 84, Appendix D, apply brush-coat of sealing compound, Item 292, Appendix D, to mount fitting.
- (k) Install gear box shim and gear box spacer on mounting fitting and support using pin rivets.
- b. Position intermediate gear box on supports and install bolts (Figure 6-4-31-1, Sheet 3, Detail F).
- c. **TORQUE BOLTS TO 385 - 425 INCH-POUNDS.**
- d. Using feeler gage, measure gap between mounting feet of intermediate gear box and supports. **NONE ALLOWED.** If gap is found, perform the following:
  - (1) Remove bolts and remove intermediate gear box from supports.
  - (2) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect mounting legs of intermediate gear box for cracks (ASTM E1417). **NONE ALLOWED.** If cracks are found in mounting feet, replace intermediate gear box.
  - (3) If no cracks are found, repeat step a. through step d.
- e. Remove intermediate gear box from supports.
- f. Using molybdenum disulfide, Item 219, Appendix D, coat surface of supports.

- g. Apply coat of sealing compound, Item 292, Appendix D, to mounting feet.
- h. Apply small bead of sealing compound, Item 292, Appendix D, to inner rim of mounting boltholes on top side of mounting feet.
- i. Place intermediate gear box on supports.
- j. Using epoxy primer coating kit, Item 135, Appendix D, apply to bolts, and install bolts.

**WARNING**

**FSCAP**

Verification of minimum torque of bolts is a critical characteristic.

- k. **TORQUE BOLTS TO 385 - 425 INCH-POUNDS.**
- l. Form a fillet with squeezed out sealing compound around base of mounting feet.
- m. Using 3-inch masking tape, Item 215, Appendix D, mask off adjacent areas to prevent overspray.
- n. Using corrosion preventive compound, Item 113, Appendix D, apply to entire area of mounting feet.
- o. Allow corrosion preventive compound to cure for 10 to 30 minutes.
- p. Using corrosion preventive compound, Item 113, Appendix D, apply second coat.
- q. Inspect every three days and using corrosion preventive compound, Item 113, Appendix D, reapply two coats as necessary until bolt torque stabilizes.
- r. Connect tail rotor drive shaft section III as follows (Figure 6-4-31-1, Sheet 2, Detail C and Detail E):

**WARNING**

Injury to personnel and damage to equipment will result if tail rotor drive shaft section III is bolted incorrectly to adjacent flange. Three adjacent flange bolts will be connected to tail rotor drive shaft flexible coupling only. Three drive shaft bolts will be connected to tail rotor drive shaft flexible coupling only. Make sure drive shaft section III flanges are not bolted directly through tail rotor drive shaft flexible coupling to adjacent flange.

**CAUTION**

Damage to drive shaft will result if handled improperly. Handle tail rotor drive shaft section III carefully so they will not get dented, scratched, or bent. To prevent possible damage to drive shaft, place tail rotor drive shaft section III on padded area.

**NOTE**

- When installing bolts in tail rotor drive shaft flexible couplings, make sure bolt-heads are against tail rotor drive shaft flexible coupling.
  - Do not lockwire bolts with drilled heads installed on tail rotor drive shaft flexible couplings.
  - Do not turn tail rotor drive shaft section III from original position, as gap may change. If tail rotor drive shaft section III is moved, recheck gap.
- (1) Support tail rotor drive shaft section III in position with input flange of intermediate gear box.

- (2) Determine shim thickness at opposite end of tail rotor drive shaft section III as follows:

**NOTE**

- Spherical washers are to be installed on each side of tail rotor drive shaft flexible coupling.
- Spherical side of washer must face tail rotor drive shaft flexible coupling.
  - (a) Support tail rotor drive shaft section III in installed position with three bolts and spherical washers. Do not install nuts at this time.
  - (b) Using feeler gage, measure gap at three attachment points.
  - (c) Determine average gap by adding readings and dividing by 3.

**Example****Measure Gap:**

Gap                      A = 0.125

B = 0.142

C = 0.137

Total                      = 0.404

**Divide by 3:**  $0.404 \div 3 = 0.1346$ -inch average gap.

**Round off:** 0.135-inch shim combination.

- (3) Select shim combination resulting from computation above. Make three stackups for each end of tail rotor drive shaft section III. **MAXIMUM GAP IS 0.125-INCH AT EACH END OF TAIL ROTOR DRIVE SHAFT SECTION III.**

### NOTE

- Shim thickness must be the same for all three bolt locations at each end of tail rotor drive shaft section III. Shim combination at one end of tail rotor drive shaft section III does not have to be the same as shim combination at other end of tail rotor drive shaft section III as long as total shim combination adds up closest to average gap.
- Tail rotor drive shaft section III attaching nuts must be torqued within 15 minutes after applying primer to bolts.
- (4) Using epoxy primer coating kit, Item 135, Appendix D, apply to bolt grip lengths and under boltheads. Do not apply primer to threads.

### NOTE

- Spherical washers are installed on each side of tail rotor drive shaft flexible coupling.
- Spherical side of washer must face tail rotor drive shaft flexible coupling.
- One shim stackup is to be installed on each bolt.
- Shim stackups are to be located between input flange and spherical washer.
- Install unused shims under nut.
- (5) Secure input flange to tail rotor drive shaft flexible coupling with bolts, shims, spherical washers, and nuts.
- (6) Using torque wrench, check drag torque on nuts. **MINIMUM DRAG TORQUE IS 9.5 INCH-POUNDS.** If torque is beyond limits, replace nuts.

### WARNING

#### FSCAP

Verification of minimum torque and torque stabilization check of nuts is a critical characteristic.

- (7) **TORQUE NUTS TO 328 - 362 INCH-POUNDS.**
- (8) Remove machinery towels or wood blocks used in supporting tail rotor drive shaft section III.
- s. Connect tail rotor drive shaft section IV as follows (Figure 6-4-31-1, Sheet 2, Detail C and Detail D):

### WARNING

Injury to personnel and damage to equipment will result if tail rotor drive shaft section IV are bolted incorrectly to adjacent flange. Three adjacent flange bolts will be connected to tail rotor drive shaft flexible coupling only. Three drive shaft bolts will be connected to tail rotor drive shaft flexible coupling only. Make sure drive shaft section IV flanges are not bolted directly through tail rotor drive shaft flexible coupling to adjacent flange.

### CAUTION

Damage to drive shaft will result if handled improperly. Handle tail rotor drive shaft section IV carefully so they will not get dented, scratched, or bent. To prevent possible damage to drive

shaft, place tail rotor drive shaft section IV on padded area.

- (1) Support tail rotor drive shaft section IV in position with output flange of intermediate gear box.
- (2) Determine shim thickness at opposite end of tail rotor drive shaft section III as follows:

#### NOTE

- Spherical washers are to be installed on each side of tail rotor drive shaft flexible coupling.
- Spherical side of washer must face tail rotor drive shaft flexible coupling.
  - (a) Support tail rotor drive shaft section IV in installed position with three bolts and spherical washers. Do not install nuts at this time.
  - (b) Using feeler gage, measure gap at three attachment points.
  - (c) Determine average gap by adding readings and dividing by 3.

#### Example

##### Measure Gap:

Gap                      A = 0.125

B = 0.142

C = 0.137

Total                      = 0.404

**Divide by 3:**  $0.404 \div 3 = 0.1346$ -inch average gap.

**Round off:** 0.135-inch shim combination.

- 
- (3) Select shim combination resulting from computation above. Make three stackups

for end of tail rotor drive shaft section IV. **MAXIMUM GAP IS 0.125-INCH AT END OF TAIL ROTOR DRIVE SHAFT SECTION IV.**

#### NOTE

- Shim thickness must be the same for all three bolt locations at each end of shaft. Shim combination at one end of shaft does not have to be the same as shim combination at other end of shaft as long as total shim combination adds up closest to average gap.
- Tail rotor drive shaft section IV attaching nuts must be torqued within 15 minutes after applying primer to bolts.
- (4) Using epoxy primer coating kit, Item 135, Appendix D, apply to bolt grip lengths and under boltheads. Do not apply primer to threads.

#### NOTE

- Spherical washers are to be installed on each side of tail rotor drive shaft flexible coupling.
- Spherical side of washer must face tail rotor drive shaft flexible coupling.
- One shim stackup is to be installed on each bolt.
- Shim stackup is to be located between output flange and spherical washer.
- Install unused shims under nut.
- (5) Secure output flange to tail rotor drive shaft flexible coupling with bolts, shims, spherical washers, and nuts.
- (6) Using torque wrench, check drag torque on nuts. **MINIMUM DRAG TORQUE IS 9.5 INCH-POUNDS.** If torque is beyond limits, replace nuts.

**WARNING**

**FSCAP**

Verification of minimum torque and torque stabilization check of nuts is a critical characteristic.

- (7) **TORQUE NUTS TO 328 - 362 INCH-POUNDS.**
- (8) Remove machinery towels or wood blocks used in supporting tail rotor drive shaft section IV.

**CAUTION**

Damage to bearings will result if contact occurs with corrosion preventive compounds. Corrosion preventive compounds will penetrate bearing seals and dissolve lubrication grease. Do not spray corrosion preventive compounds in vicinity of tail rotor drive shaft section III bearings. Shield bearings from spray.

- t. Using corrosion preventive compound, Item 113, Appendix D, spray all spherical washers.
- u. Check intermediate gear box chip detector/temperature sensor electrical circuit (PARA 6-2-2).
- v. Remove protective caps and plugs from electrical connectors.
- w. Inspect and treat electrical connectors (PARA 1-7-20).
- x. Connect electrical connector, P613, to intermediate gear box chip detector/temperature sensor.

**NOTE**

Insulation sleeve on electrical connector is used to prevent water intrusion.

- y. Pull insulation sleeve over electrical connector, P613 (Figure 6-4-31-1, Sheet 1, View A).

**WARNING**

Injury to personnel and damage to equipment will result if intermediate gear box is operated with low oil level. Operation with low oil level or without oil will cause loss of tail rotor drive. Lightly shake pylon to observe oil sloshing when checking oil level.

**CAUTION**

Damage to intermediate gear box will result if intermediate gear box is overfilled. Do not overfill intermediate gear box.

- z. Service intermediate gear box (PARA 1-3-13). Do not install intermediate gear box fairing.
- aa. Make sure area is clean and free of foreign material.
- ab. Perform check of intermediate gear box as follows:
  - (1) With qualified person at controls, run up helicopter (Appropriate Flight Manual).
  - (2) Check intermediate gear box for leaks.
- ac. Install intermediate gear box fairing (PARA 2-4-139).
- ad. Check helicopter logbook to make sure tail rotor drive shaft and intermediate gear box mount bolts torque checks are shown as due 9 to 11 flight hours after installation.
- ae. Using polyurethane paint, Item 228, Appendix D, apply one coat to any water entrapment areas of intermediate gear box.
- af. If required, disengage gust lock (PARA 1-6-18).

- ag. Perform maintenance test flight (Appropriate MTF).



6-4-32 INTERMEDIATE GEAR BOX INPUT SEAL AND FLANGE.

PARAGRAPH BREAKDOWN

|                                                              | PAGE     |
|--------------------------------------------------------------|----------|
| 6-4-32.1 Replace Intermediate Gear Box Input Seal and Flange | 6-4-32-2 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Ballpeen Hammer, 8-Ounce
- Nylon Wedge, Locally-Made, Figure H-260, Appendix H
- Powertrain Repairer’s Toolkit
- Screws, Self-Tapping
- Seal Installation Tool, Locally-Made, Figure H-157, Appendix H
- Seal Puller, Locally-Made, Figure H-160, Appendix H
- Tail Drive Flange Wrench, 70700-20687-102
- Torque Reactor, 70700-20688-041
- Torque Wrench, 100 - 500 ft lbs

Materials/Parts

- Abrasive Cloth, Item 1A, Appendix D
- Abrasive Cloth, Item 10, Appendix D
- Abrasive Mat, Item 14, Appendix D

- Corrosion Preventive Compound, Item 113, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Grease, Item 149, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D
- Lubricating Oil, Item 208, Appendix D
- Machinery Towel, Item 211, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- Appendix H
- Appropriate Flight Manual
- PARA 1-3-13
- PARA 1-6-16
- PARA 2-4-139
- PARA 6-3-10
- PARA 6-4-40

### 6-4-32.1 Replace Intermediate Gear Box Input Seal and Flange.

#### 6-4-32.1.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Remove intermediate gear box fairing (PARA 2-4-139).
- c. Remove tail rotor drive shaft section III (PARA 6-4-40).
- d. Drain intermediate gear box (PARA 1-3-13).
- e. Remove cotter pin from retaining nut (Figure 6-4-32-1, Detail A).
- f. Using torque reactor and tail drive flange wrench, remove retaining nut.
- g. Pull intermediate gear box input flange off input shaft.
- h. Remove retaining ring.

#### NOTE

Self-tapping screws may be used as an alternate to remove intermediate gear box input seal if seal puller is not available. If seal puller is not available, proceed to step n.

- i. Locally make seal puller (Figure H-160, Appendix H).
- j. Locally make nylon wedge (Figure H-260, Appendix H).
- k. Insert nylon wedge between seal puller and input shaft to protect intermediate gear box housing (Figure 6-4-32-1, Detail B).

#### CAUTION

Damage to intermediate gear box will result if too much force is applied while using ballpeen hammer and seal puller.

Intermediate gear box housing will crack. Use care when using ballpeen hammer and seal puller.

- l. Using 8-ounce ballpeen hammer, gently tap end of seal puller in outward direction, away from intermediate gear box.
- m. Moving seal puller and nylon wedge in clockwise direction, repeat step k. and step l. until intermediate gear box input seal is removed (Figure 6-4-32-1, Detail A and Detail B).
- n. If seal puller is not available, remove intermediate gear box input seal as follows:

#### WARNING

- Damage to intermediate gear box will result if self-tapping screws completely penetrate intermediate gear box input seal. Screws will damage bearing inside intermediate gear box. Install screws so that only outer face of intermediate gear box input seal is penetrated.
- Damage to intermediate gear box will result if metal chips or shavings enter intermediate gear box by drilling or punching holes in intermediate gear box input seal. Metal chips or shavings will contaminate intermediate gear box oil. Do not drill or punch holes in intermediate gear box input seal.
  - (1) Using three self-tapping screws, thread into holes in intermediate gear box input seal.
  - (2) Carefully pull out intermediate gear box input seal.

#### 6-4-32.1.2 Inspect.

- a. Inspect intermediate gear box input flange (PARA 6-3-10).
- b. Inspect retaining nut for corrosion. **REPLACE RETAINING NUT IF:**

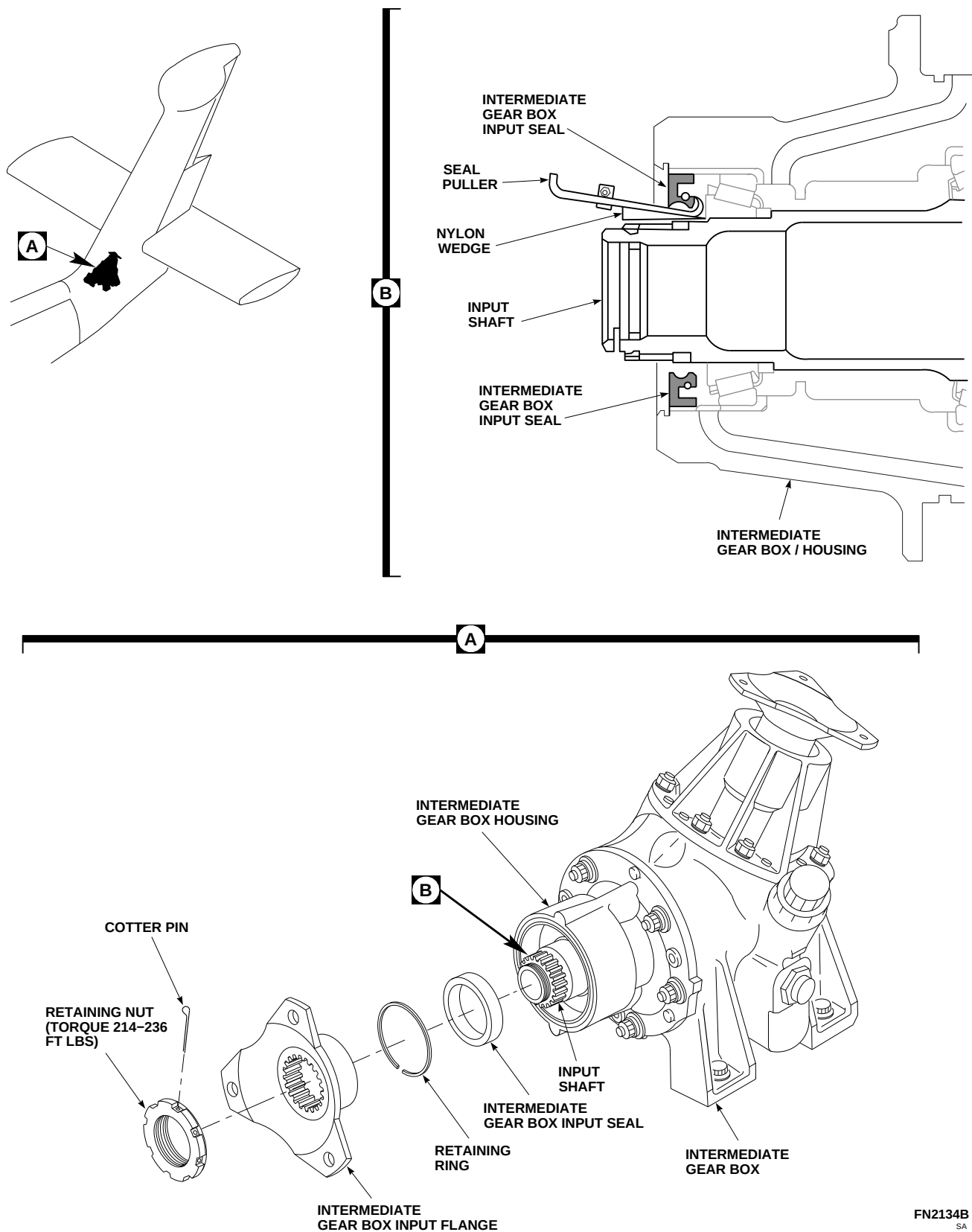


FIGURE 6-4-32-1. REPLACE INTERMEDIATE GEAR BOX INPUT SEAL AND FLANGE

- (1) **PITS ARE 0.030-INCH DEEP.**
- (2) **PITS COVER MORE THAN 10% OF MATING SURFACE.**
- (3) **THREADS SHOW SIGNS OF CORROSION.**

#### 6-4-32.1.3 Repair.

- a. Remove minor corrosion from retaining nut using abrasive cloth, Item 10, Appendix D, abrasive cloth, Item 1A, Appendix D, or abrasive mat, Item 14, Appendix D.

#### NOTE

Do not paint threads or mating surface of retaining nut.

- b. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to flats and exposed face of retaining nut.

#### 6-4-32.1.4 Install.

- a. Prior to installing intermediate gear box input seal, perform the following:
  - (1) Using grease, Item 149, Appendix D, coat inside diameter of intermediate gear box input seal.
  - (2) Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean mating surface of intermediate gear box input seal.
  - (3) Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, coat outside diameter of intermediate gear box input seal and mating surface.
- b. Locally make seal installation tool (Figure H-157, Appendix H).

#### CAUTION

Intermediate gear box input seal and intermediate gear box will be damaged

if incorrectly installed. Intermediate gear box input seal will leak. Make sure intermediate gear box input seal is not cocked or distorted during installation. Make sure intermediate gear box input seal is installed flush to 0.020-inch below retaining ring groove, prior to installing retaining ring.

- c. Using seal installation tool, gently tap new intermediate gear box input seal into intermediate gear box housing (Figure 6-4-32-1, Detail A).
- d. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to retaining ring and install in intermediate gear box housing while wet.
- e. Apply sealing compound, Item 292, Appendix D, to boss of intermediate gear box housing to prevent water entrapment and corrosion above retaining ring groove.
- f. Using epoxy primer coating kit, Item 135, Appendix D, touch up sealing compound.
- g. Slide intermediate gear box input flange onto input shaft.
- h. Using corrosion preventive compound, Item 113, Appendix D, apply over entire retaining nut.
- i. Install retaining nut as follows:
  - (1) Install retaining nut, making sure retaining nut can be fully turned by hand. Make sure bevel side of retaining nut is toward intermediate gear box input seal.
  - (2) Using torque reactor and tail drive flange wrench, **TIGHTEN RETAINING NUT.**
  - (3) **TORQUE RETAINING NUT TO 214 - 236 FOOT-POUNDS.**
  - (4) **INSTALL NEW COTTER PIN. IF REQUIRED TO ALIGN COTTER PIN HOLE, TURN RETAINING NUT IN A TIGHTENING DIRECTION.**

- j. Install tail rotor drive shaft section III (PARA 6-4-40).

**WARNING**

Injury to personnel and damage to equipment will result if intermediate gear box is operated with low oil level. Operation with low oil level or without oil will cause loss of tail rotor drive. Lightly shake pylon to observe oil sloshing when checking oil level.

- k. Service intermediate gear box (PARA 1-3-13).
- l. Make sure area is clean and free of foreign material.
- m. Perform check of intermediate gear box input seal and flange as follows:
  - (1) With qualified person at controls, run up helicopter (Appropriate Flight Manual).
  - (2) Check input seal and flange for leaks.
- n. Install intermediate gear box fairing (PARA 2-4-139).



6-4-33 INTERMEDIATE GEAR BOX OUTPUT SEAL AND FLANGE.

PARAGRAPH BREAKDOWN

|                                                               | PAGE     |
|---------------------------------------------------------------|----------|
| 6-4-33.1 Replace Intermediate Gear Box Output Seal and Flange | 6-4-33-2 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Ballpeen Hammer, 8-Ounce
- Nylon Wedge, Locally-Made, Figure H-260, Appendix H
- Powertrain Repairer’s Toolkit
- Screws, Self-Tapping
- Seal Installation Tool, Locally-Made, Figure H-157, Appendix H
- Seal Puller, Locally-Made, Figure H-160, Appendix H
- Tail Drive Flange Wrench, 70700-20687-102
- Torque Reactor, 70700-20688-041
- Torque Wrench, 100 - 500 ft lbs

Materials/Parts

- Abrasive Cloth, Item 1A, Appendix D
- Abrasive Cloth, Item 10, Appendix D
- Abrasive Mat, Item 14, Appendix D

- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Grease, Item 149, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D
- Lubricating Oil, Item 208, Appendix D
- Machinery Towel, Item 211, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- Appendix H
- Appropriate Flight Manual
- PARA 1-3-13
- PARA 1-6-16
- PARA 2-4-139
- PARA 6-3-10
- PARA 6-4-41

### 6-4-33.1 Replace Intermediate Gear Box Output Seal and Flange.

#### 6-4-33.1.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Remove intermediate gear box fairing (PARA 2-4-139).
- c. Remove tail rotor drive shaft section IV (PARA 6-4-41).
- d. Remove cotter pin from retaining nut (Figure 6-4-33-1, Detail A).
- e. Using torque reactor and tail drive flange wrench, remove retaining nut.
- f. Pull intermediate gear box output flange off output shaft.
- g. Remove retaining ring.

#### NOTE

Self-tapping screws may be used as an alternate to remove intermediate gear box output seal if seal puller is not available. If seal puller is not available, proceed to step m.

- h. Locally make seal puller (Figure H-160, Appendix H).
- i. Locally make nylon wedge (Figure H-260, Appendix H).
- j. Insert nylon wedge between seal puller and output shaft to protect intermediate gear box housing (Figure 6-4-33-1, Detail B).

#### CAUTION

Damage to intermediate gear box will result if too much force is applied while using ballpeen hammer and seal puller. Intermediate gear box housing will crack. Use care when using ballpeen hammer and seal puller.

- k. Using 8-ounce ballpeen hammer, gently tap end of seal puller in an outward direction, away from intermediate gear box.
- l. Moving seal puller and nylon wedge in a clockwise direction, repeat step j. and step k. until intermediate gear box output seal is removed (Figure 6-4-33-1, Detail A and Detail B).
- m. If seal puller is not available, remove intermediate gear box output seal as follows:

#### WARNING

- Damage to intermediate gear box will result if self-tapping screws completely penetrate intermediate gear box output seal. Screws will damage bearing inside intermediate gear box. Install screws so that only outer face of intermediate gear box output seal is penetrated.
  - Damage to intermediate gear box will result if metal chips or shavings enter intermediate gear box by drilling or punching holes in intermediate gear box output seal. Metal chips or shavings will contaminate intermediate gear box oil. Do not drill or punch holes in intermediate gear box output seal.
- (1) Using three self-tapping screws, thread into holes in intermediate gear box output seal.
  - (2) Carefully pull out intermediate gear box output seal.

#### 6-4-33.1.2 Inspect.

- a. Inspect intermediate gear box output flange (PARA 6-3-10).
- b. Inspect retaining nut for corrosion. **REPLACE RETAINING NUT IF:**
  - (1) **PITS ARE 0.030-INCH DEEP.**
  - (2) **PITTED MORE THAN 10% OF MATING SURFACE.**



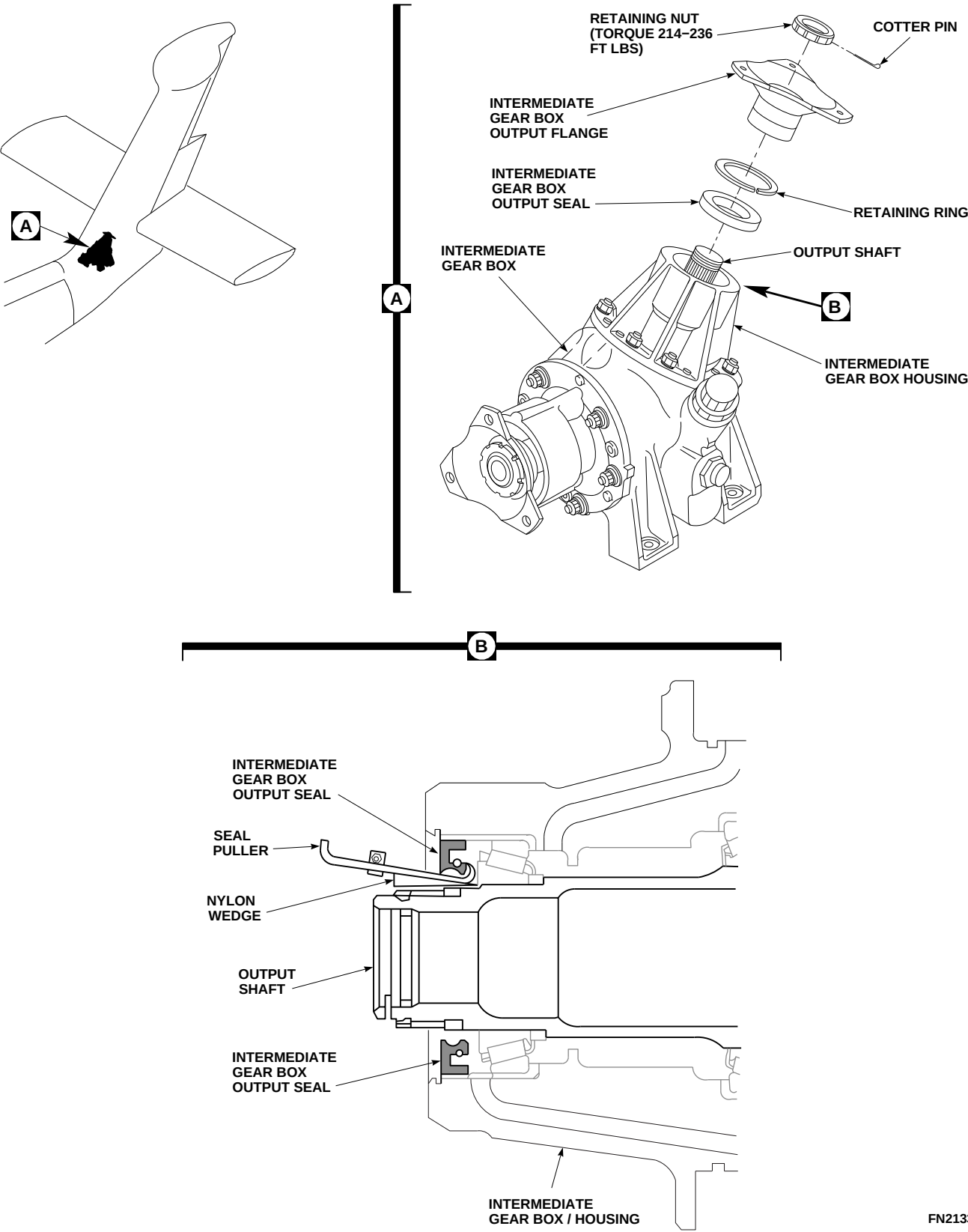


FIGURE 6-4-33-1. REPLACE INTERMEDIATE GEAR BOX OUTPUT SEAL AND FLANGE

(3) **THREADS SHOW SIGNS OF CORROSION.**

**6-4-33.1.3 Repair.**

- a. Remove minor corrosion from retaining nut using abrasive cloth, Item 10, Appendix D, abrasive cloth, Item 1A, Appendix D, or abrasive mat, Item 14, Appendix D.

**NOTE**

Do not paint threads or mating surface of retaining nut.

- b. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to flats and exposed face of retaining nut.

**6-4-33.1.4 Install.**

- a. Prior to installing intermediate gear box output seal, perform the following:
  - (1) Using grease, Item 149, Appendix D, coat inside diameter of intermediate gear box output seal.
  - (2) Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean mating surface of intermediate gear box output seal.
  - (3) Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, coat outside diameter of intermediate gear box output seal and mating surface.
- b. Locally make seal installation tool (Figure H-157, Appendix H).

**CAUTION**

Intermediate gear box output seal and intermediate gear box will be damaged if incorrectly installed. Intermediate gear

box output seal will leak. Make sure intermediate gear box output seal is not cocked or distorted during installation. Make sure intermediate gear box output seal is installed flush to 0.020-inch below retaining ring groove prior to installing retaining ring.

- c. Using seal installation tool, gently tap new intermediate gear box output seal into intermediate gear box housing (Figure 6-4-33-1, Detail A).
- d. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to retaining ring and install in intermediate gear box housing while wet.
- e. Using sealing compound, Item 292, Appendix D, apply to boss of intermediate gear box housing to prevent water entrapment and corrosion above retaining ring groove.
- f. Using epoxy primer coating kit, Item 135, Appendix D, touch up sealing compound.
- g. Slide intermediate gear box output flange on output shaft.
- h. Install retaining nut as follows:
  - (1) Install retaining nut, making sure retaining nut can be fully turned by hand. Make sure bevel side of retaining nut is toward intermediate gear box output seal.
  - (2) Using torque reactor and tail drive flange wrench, **TIGHTEN RETAINING NUT.**
  - (3) **TORQUE RETAINING NUT TO 214 - 236 FOOT-POUNDS.**
  - (4) **INSTALL NEW COTTER PIN. IF REQUIRED TO ALIGN COTTER PIN HOLE, TURN RETAINING NUT IN A TIGHTENING DIRECTION.**
- i. Install tail rotor drive shaft section IV (PARA 6-4-41).

**WARNING**

Injury to personnel and damage to equipment will result if intermediate gear box is operated with low oil level. Operation with low oil level or without oil will cause loss of tail rotor drive. Lightly shake pylon to observe oil sloshing when checking oil level.

j. Service intermediate gear box (PARA 1-3-13).

- k. Make sure area is clean and free of foreign material.
- l. Perform check of intermediate gear box output seal and flange as follows:
  - (1) With qualified person at controls, run up helicopter (Appropriate Flight Manual).
  - (2) Check output seal and flange for leaks.
- m. Install intermediate gear box fairing (PARA 2-4-139).



6-4-34 INTERMEDIATE GEAR BOX SIGHT PLUG.

PARAGRAPH BREAKDOWN

|                                                   | PAGE     |
|---------------------------------------------------|----------|
| 6-4-34.1 Replace Intermediate Gear Box Sight Plug | 6-4-34-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Powertrain Repairer’s Toolkit

Materials/Parts

- Lockwire, Item 197, Appendix D
- Lubricating Oil, Item 205, Appendix D

- Lubricating Oil, Item 206, Appendix D
- Lubricating Oil, Item 208, Appendix D

References

- Appendix D
- PARA 1-3-13
- PARA 1-6-16
- PARA 2-4-139

6-4-34.1 Replace Intermediate Gear Box Sight Plug.

6-4-34.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove intermediate gear box fairing (PARA 2-4-139).
- Drain intermediate gear box (PARA 1-3-13).
- Unscrew and remove intermediate gear box sight plug and packing from intermediate gear box (Figure 6-4-34-1, Detail A). Discard packing.

6-4-34.1.2. Install.

- Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing and install on intermediate gear box sight plug (Figure 6-4-34-1, Detail A).

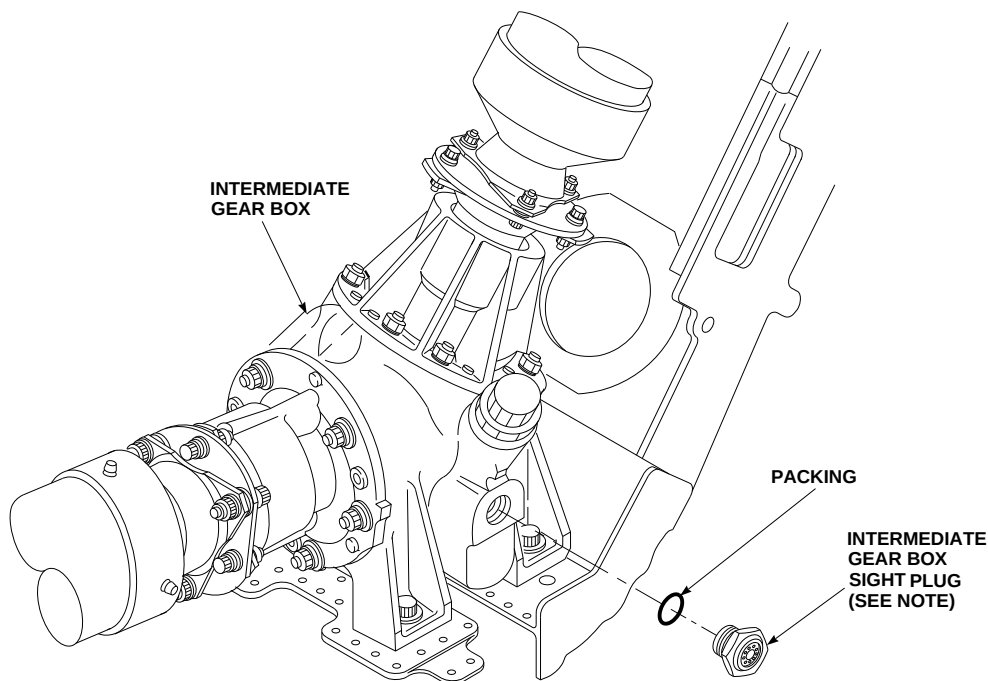
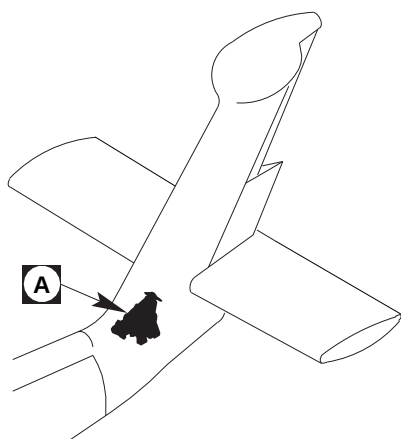
- Screw intermediate gear box sight plug into intermediate gear box.
- TIGHTEN INTERMEDIATE GEAR BOX SIGHT PLUG ¼ TO ⅓ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- Lockwire, Item 197, Appendix D, intermediate gear box sight plug to intermediate gear box.



Injury to personnel and damage to equipment will result if intermediate gear box is operated with low oil level. Operation with low oil level or without oil will cause loss of tail rotor drive. Lightly shake pylon to observe oil sloshing when checking oil level.

- Service intermediate gear box (PARA 1-3-13).

- f. Make sure area is clean and free of foreign material.
- g. Perform check of intermediate gear box sight plug as follows:
  - (1) With qualified person at controls, run up helicopter.
  - (2) Check intermediate gear box sight plug for leaks.
- h. Install intermediate gear box fairing (PARA 2-4-139).



**NOTE**

TIGHTEN  $\frac{1}{6}$  TO  $\frac{1}{3}$   
TURN PAST POINT WHERE  
SHARP RISE IN TORQUE  
IS FELT.

FN2132A  
SA

**FIGURE 6-4-34-1. REPLACE INTERMEDIATE GEAR BOX SIGHT PLUG**

**6-4-34-3/(6-4-34-4 Blank)**





6-4-35 INTERMEDIATE GEAR BOX OIL FILLER PLUG.

PARAGRAPH BREAKDOWN

|                                                        | PAGE     |
|--------------------------------------------------------|----------|
| 6-4-35.1 Replace Intermediate Gear Box Oil Filler Plug | 6-4-35-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Lockwire, Item 197, Appendix D
- Lubricating Oil, Item 205, Appendix D

- Lubricating Oil, Item 206, Appendix D
- Lubricating Oil, Item 208, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-4-139

6-4-35.1 Replace Intermediate Gear Box Oil Filler Plug.

6-4-35.1.1 Remove.

- Remove intermediate gear box fairing (PARA 2-4-139).
- Unscrew and remove intermediate gear box oil filler plug (with cap attached) and packing from intermediate gear box (Figure 6-4-35-1, Detail A). Discard packing.

CAUTION

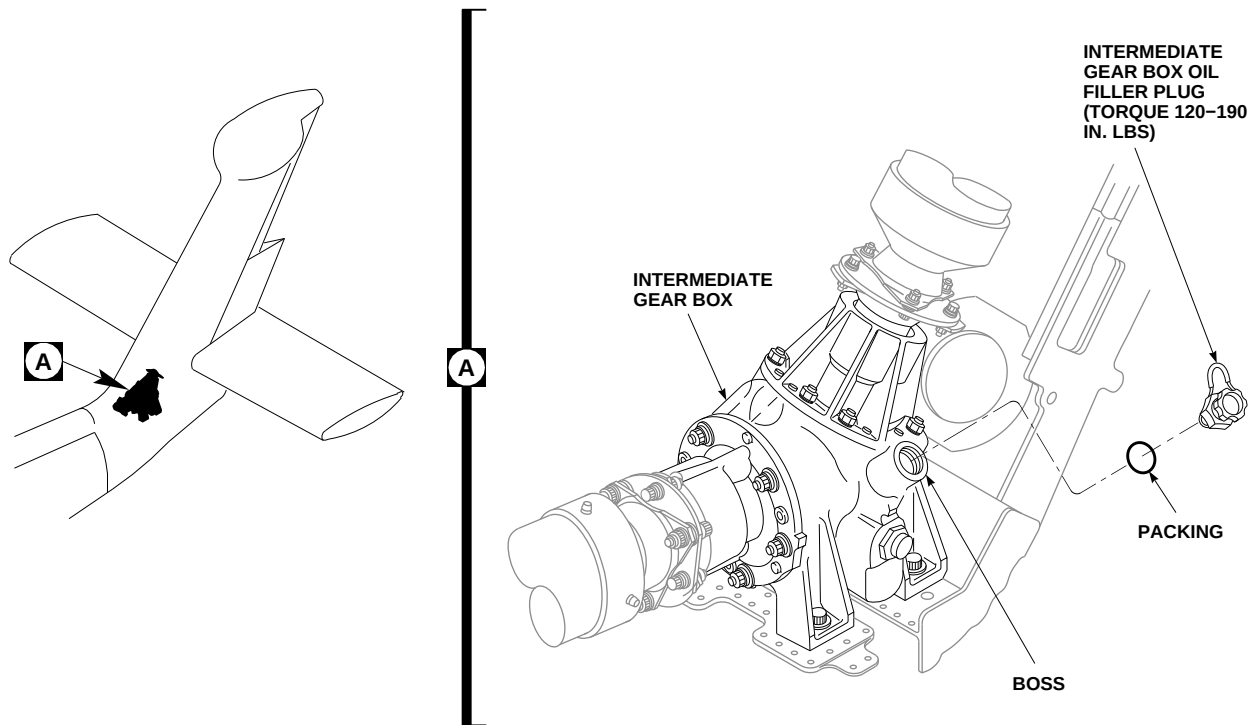
Damage to intermediate gear box will result if sealing compound enters

intermediate gear box. Use extreme care when scraping sealing compound from boss of gear box.

- Using nonmetallic scraper, scrape sealing compound from boss of gear box.

6-4-35.1.2 Install.

- Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing and install on intermediate gear box oil filler plug (Figure 6-4-35-1, Detail A).
- Install intermediate gear box oil filler plug on intermediate gear box.
- TORQUE PLUG TO 120 - 190 INCH-POUNDS.**

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SA**FIGURE 6-4-35-1. REPLACE INTERMEDIATE GEAR BOX OIL FILLER PLUG**

- d. Lockwire, Item 197, Appendix D, intermediate gear box oil filler plug to intermediate gear box.
- e. Apply sealing compound, Item 292, Appendix D, to seal area around seam of intermediate gear box oil filler plug and boss.
- f. Perform check of intermediate gear box oil filler plug as follows:
  - (1) With qualified person at controls, run up helicopter.
  - (2) Check intermediate gear box oil filler plug for leaks.
- g. Install intermediate gear box fairing (PARA 2-4-139).

6-4-36 INTERMEDIATE GEAR BOX HOUSING REPAIR.

PARAGRAPH BREAKDOWN

|                                               | PAGE     |
|-----------------------------------------------|----------|
| 6-4-36.1 Repair Intermediate Gear Box Housing | 6-4-36-2 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Beaker (10 ml)
- Beaker (200 ml)
- Beaker (600 ml)
- Container, Plastic, 1-Pint
- Container, Plastic, 2-Gallon
- Fluorescent Penetrant Inspection Kit
- Powertrain Repairer’s Toolkit
- Protection, Eye
- Vessel(s), with equivalent measuring accuracy of Beakers (Alternate for Beakers)

Materials/Parts

- Abrasive Cloth, Item 6, Appendix D
- Abrasive Cloth, Item 10, Appendix D
- Acetic Acid, Item 24, Appendix D
- Applicator, Item 78, Appendix D

- Chromium Trioxide, Technical, Item 91, Appendix D
- Distilled Water, Item 123, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Coating Kit, Item 126A, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Ethylene Glycol, Item 137, Appendix D
- Machinery Towel, Item 211, Appendix D
- Nitric Acid, Item 220, Appendix D
- Paint, Polyurethane, Item 224, Appendix D
- Paint Remover, Item 231, Appendix D
- Pressure Sensitive Adhesive Tape, Item 250, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 6-4-31

### 6-4-36.1 Repair Intermediate Gear Box Housing.

#### 6-4-36.1.1 General.

##### NOTE

- Minor nicks, gouges, scratches, or corrosion on surface of intermediate gear box housing may be repaired. Before starting repair, measure depth of damage to determine whether blending out damage will result in intermediate gear box still being within repair limits. Make sure to measure depth of damage from an area which has not been previously repaired. If limits will be exceeded, replace intermediate gear box.
  - Repair of damaged area may be done with intermediate gear box installed.
  - The following repair procedures apply to input, center, and output housings (Figure 6-4-36-1).
  - All repairs shall be recorded in helicopter component historical records.
- a. Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean any oil or grease from damaged area.
  - b. Before blending, visually check damaged area for cracks. **NONE ALLOWED.** If cracks are found, replace intermediate gear box (PARA 6-4-31).
  - c. Fine scratches that remove only paint may be blended smooth using abrasive cloth, Item 10, Appendix D. After blending, touch up blended area as required using epoxy coating kit, Item 126A, Appendix D, or polyurethane paint, Item 224, Appendix D.
  - d. Nicks, gouges, scratches, or corrosion on surface of all intermediate gear box housing may be blended out to a ½-inch radius. Only two repairs are allowed in any 4-square-inch area. Refer to repair limits for depths of repair (PARA 6-4-36.1.2).

### 6-4-36.1.2 Repair Limits.

- a. Input housing limits are as follows:

- (1) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM ALL OVER OUTER CAST SURFACE IS 0.035-INCH.**

##### NOTE

Only 30% of the total number of holes in bolt pattern may be repaired.

- (2) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM INSIDE OF BOLT-HOLES IS 0.015-INCH** (Figure 6-4-36-1, Detail A).
- (3) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM MOUNTING SURFACE UNDER FASTENERS (BUT NOT COVERING MORE THAN A 90° SEGMENT) IS 0.015-INCH.**
- (4) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SPOT-FACE SURFACE NOT COVERED BY FASTENER IS 0.030-INCH.**
- (5) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SHARP EDGE AROUND SPOT-FACE SURFACE (NOT INCLUDING AREA UNDER FASTENER) IS 0.060-INCH.**
- (6) **CORNER OF SPOT-FACE MAY BE BLENDED TO 0.100-INCH RADIUS AS LONG AS DEPTH INTO SPOT-FACE DOES NOT EXCEED 0.030-INCH.**

- b. Center housing limits are as follows:

- (1) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM ALL OVER OUTER CAST SURFACE IS 0.030-INCH.**
- (2) **MAXIMUM DEPTH OF MATERIAL REMOVAL ON MOUNTING FEET AND MOUNTING FEET RIBS IS 0.030-INCH** (Figure 6-4-36-1, Detail B).

- (a) **NO MORE THAN TWO REPAIRS ALLOWED PER MOUNTING FOOT.**
- (b) **TOTAL REPAIR SHALL NOT EXCEED 10% OF SURFACE AREA OF MOUNTING FOOT IN ANY 1-SQUARE-INCH AREA OR MORE THAN 30% OF TOTAL BASE OF MOUNTING FOOT.**
- (c) **NO MATERIAL REMOVAL FROM BUSHING BORE ALLOWED.**
- (d) **SPOT-FACE CORNER MAY BE BLENDED TO 0.050 - 0.100-INCH RADIUS AS LONG AS REPAIR DOES NOT GO DEEPER THAN 0.040-INCH INTO SPOT-FACE.**
- (e) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SPOT-FACE SURFACE OF MOUNTING FOOT IS 0.040-INCH.**
- (3) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM OPEN SPOT-FACE SURFACE NOT COVERED BY SIGHT GAGE, CHIP DETECTOR, OR FILLER PLUG IS 0.040-INCH (Figure 6-4-36-1, Detail C).**
- (4) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SURFACE UNDER SIGHT GAGE, CHIP DETECTOR, OR FILLER PLUG IS 0.010-INCH.**
- (5) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SPOT-FACE CORNER AROUND SIGHT GAGE, CHIP DETECTOR, OR FILLER PLUG IS 0.040-INCH WHEN BLENDED TO A 0.050-INCH RADIUS.**
- c. Output housing limits are as follows:
  - (1) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM ALL OVER OUTER CAST SURFACES IS 0.040-INCH.**

## NOTE

Only 30% of the total number of holes in bolt pattern may be repaired.

- (2) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM INSIDE OF BOLT-HOLES IS 0.015-INCH (Figure 6-4-36-1, Detail A).**
- (3) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM MOUNTING SURFACE UNDER FASTENERS (BUT NOT COVERING MORE THAN A 90° SEGMENT) IS 0.015-INCH.**
- (4) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SPOT-FACE SURFACE NOT COVERED BY FASTENER IS 0.030-INCH.**
- (5) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SHARP EDGE AROUND SPOT-FACE SURFACE (NOT INCLUDING AREA UNDER FASTENER) IS 0.060-INCH.**
- (6) **CORNER OF SPOT-FACE MAY BE BLENDED TO 0.100-INCH RADIUS AS LONG AS DEPTH INTO SPOT-FACE DOES NOT EXCEED 0.030-INCH.**

### 6-4-36.1.3 Repair Procedure.

- a. Using paint remover, Item 231, Appendix D, remove paint from damaged area.
- b. Using abrasive cloth, Item 6, Appendix D, carefully blend damaged area smooth. Remove minimum amount of material required to blend out damaged area.
- c. Using abrasive cloth, Item 10, Appendix D, blend reworked area to remove all traces of abrasive cloth.
- d. Remove displaced metal caused by abrasive blending by using magnesium-etching procedure as follows:

**CAUTION**

- Magnesium-etchant measurements are to be as accurate as possible to values measured in a milliliter beaker or vessel with equivalent mixing accuracy.
- Magnesium-etchant formulas shall be prepared and stored in unbreakable 1-pint plastic containers and safely stored in designated areas not subjected to extreme heat or cold.
- Before preparing magnesium-etchant, make sure storage area does not contain premixed magnesium-etchant formula. The correct stock bottle for this repair shall be marked MAGNESIUM-ETCHANT.

- (1) Prepare magnesium-etchant solution in clean unbreakable plastic container by mixing the following:

| MATERIAL                              | QTY             |
|---------------------------------------|-----------------|
| Ethylene Glycol, Item 137, Appendix D | 600 milliliters |
| Acetic Acid, Item 24, Appendix D      | 200 milliliters |
| Nitric Acid, Item 220, Appendix D     | 10 milliliters  |
| Distilled Water, Item 123, Appendix D | 190 milliliters |

- (2) Using dry-cleaning solvent, Item 124, Appendix D, clean area to be etched.
- (3) Using pressure sensitive adhesive tape, Item 250, Appendix D, mask area.
- (4) Pour a small amount of magnesium-etchant from stock bottle into 1-pint plastic container.

**CAUTION**

Damage to components will result if magnesium-etchant is allowed to come in contact with areas not being etched. Extreme caution shall be used to protect those areas not being etched.

- (5) Using applicator, Item 78, Appendix D, apply magnesium-etchant to reworked area. Swab area for about 10 - 15 seconds.
- (6) Neutralize magnesium-etchant immediately by flooding area with generous amount of clean water.

**WARNING**

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- (7) Using clean, dry, compressed air, blow dry area thoroughly.
- (8) After use, do not return unused magnesium-etchant to stock bottle. Dispose of used magnesium-etchant per local instructions.
- (9) Identify contents of stock bottle as MAGNESIUM-ETCHANT, before placing bottle in storage area.
- (10) Prepare technical chromium trioxide solution in clean 2-gallon plastic container by mixing 24 ounces of technical chromium trioxide, Item 91, Appendix D, with 1 gallon of water.
- (11) Using technical chromium trioxide solution, treat bare surface of housing.

- e. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type I, Method C, reworked area for cracks (ASTM E1417). **NONE ALLOWED.** If cracks are found, replace intermediate gear box (PARA 6-4-31).
- f. Apply sealing compound, Item 292, Appendix D, to touch up reworked areas.
- g. After sealing compound has dried, using epoxy primer coating kit, Item 135, Appendix D, apply one coat of epoxy primer.

**NOTE**

Housings coated with epoxy coating are glossy silver. Housings coated with

polyurethane paint are matte surface medium-gray.

- h. Using epoxy coating kit, Item 126A, Appendix D, or polyurethane paint, Item 224, Appendix D, apply one coat of epoxy primer, depending on original paint finish.
- i. Make sure area is clean and free of foreign material.
- j. Record brief description of intermediate gear box repairs in helicopter component historical records.

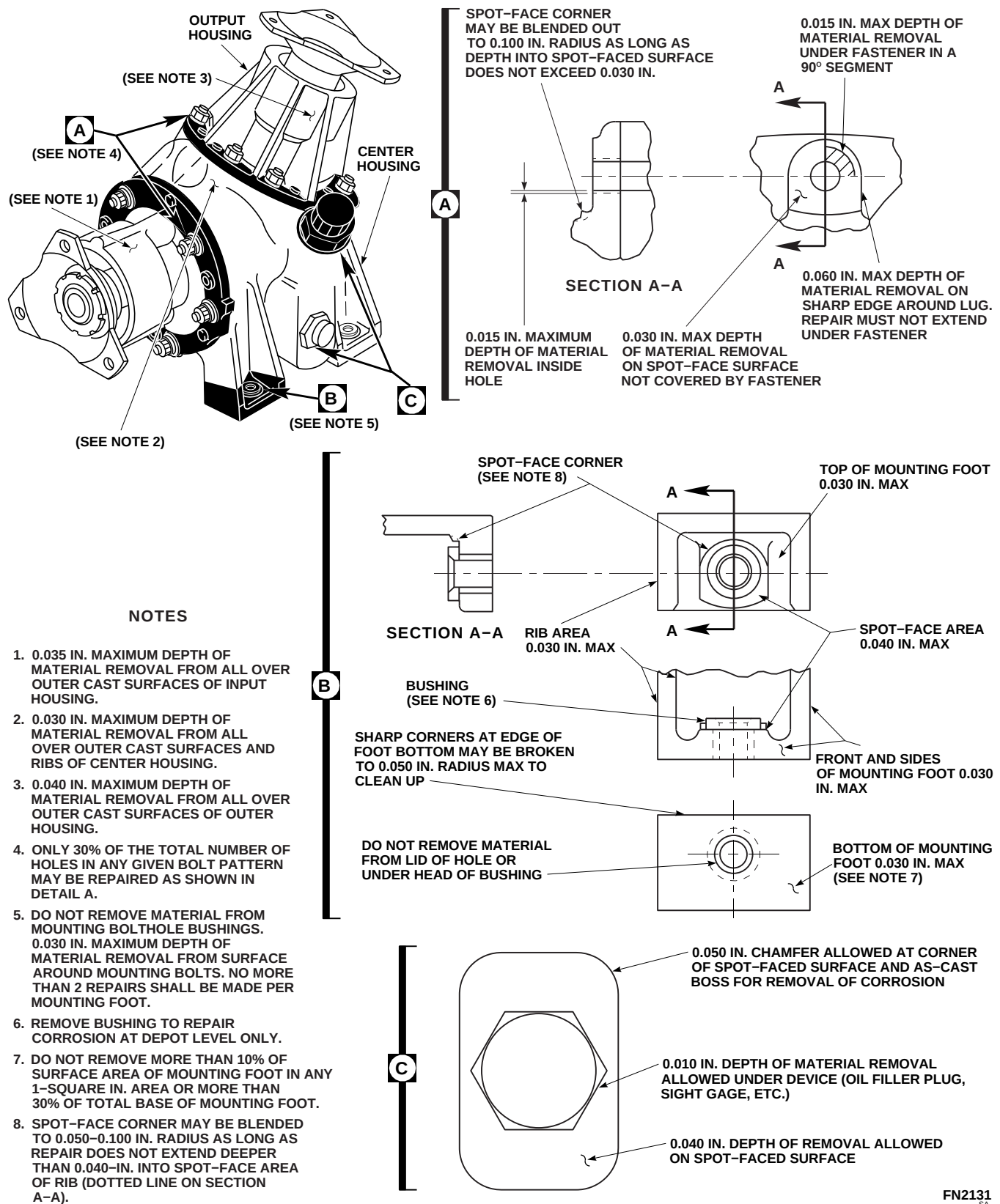
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FIGURE 6-4-36-1. REPAIR INTERMEDIATE GEAR BOX HOUSING



6-4-37 INTERMEDIATE GEAR BOX CHIP DETECTOR/TEMPERATURE SENSOR.

PARAGRAPH BREAKDOWN

|          |                                                                 | PAGE     |          |                            | PAGE     |
|----------|-----------------------------------------------------------------|----------|----------|----------------------------|----------|
| 6-4-37.1 | Replace Intermediate Gear Box Chip Detector/ Temperature Sensor | 6-4-37-1 | 6-4-37.2 | Replace Self-Closing Valve | 6-4-37-4 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Powertrain Repairer’s Toolkit
- Protection, Eye

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Lockwire, Item 198, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D

- Lubricating Oil, Item 208, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-3-13
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-139
- PARA 6-2-2
- PARA 6-3-11

6-4-37.1 Replace Intermediate Gear Box Chip Detector/Temperature Sensor.

6-4-37.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- On copilot’s circuit breaker panel, pull out CHIP DET circuit breaker.
- Remove intermediate gear box fairing (PARA 2-4-139).

NOTE

Insulation sleeve on electrical connector is used to prevent water intrusion.

- Pull insulation sleeve back away from electrical connector, P613 (Figure 6-4-37-1, Detail B, View A).
- Disconnect electrical connector, P613, from intermediate gear box chip detector/temperature sensor (Figure 6-4-37-1, Detail B).

- f. Install protective caps and plugs on electrical connectors.

### WARNING

Injury to personnel will result if precautions are not taken when removing intermediate gear box chip detector/temperature sensor. To prevent injury, personnel must wear eye protection and make sure oil in intermediate gear box is cool before removing intermediate gear box chip detector/temperature sensor.

- g. Remove intermediate gear box chip detector/temperature sensor and packings from self-closing valve in intermediate gear box. Discard packings.
- h. Install protective plug in self-closing valve.

### CAUTION

Do not immerse input module chip detector, self-closing valve, or strainer in dry-cleaning solvent.

- i. Using dry-cleaning solvent, Item 124, Appendix D, clean input module chip detector and self-closing valve.

### WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- j. Using clean, dry, compressed air, blow dry input module chip detector, self-closing valve, and strainer.

#### 6-4-37.1.2 Inspect.

- a. Inspect intermediate gear box chip detector/temperature sensor for chips (PARA 6-3-11).

#### NOTE

Leakage of lubricating oil through center of self-closing valve indicates loose internal hardware. This disables temperature sensor portion of intermediate gear box chip detector/temperature sensor.

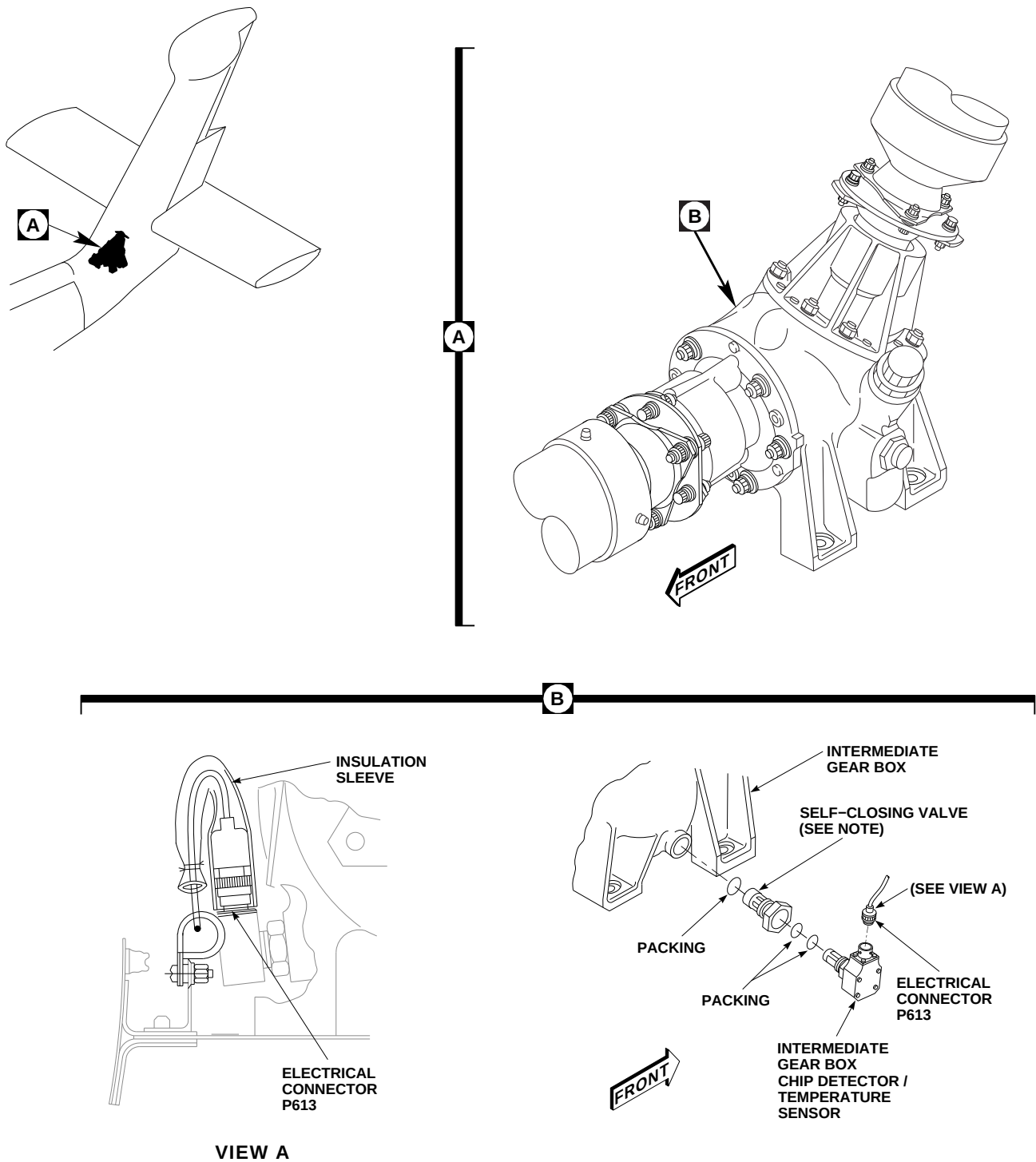
- b. Remove protective plug from self-closing valve.
- c. Visually inspect self-closing valve for leakage from center of self-closing valve. **NONE ALLOWED.** If leakage is found, replace self-closing valve (PARA 6-4-37.2).
- d. Check inside of self closing valve for presence and security of contact pin located in center of valve. **NO MISSING PIN OR LOOSENESS ALLOWED.** If contact pin is loose or missing, locate pin and washer and replace self-closing valve.
- e. If pin or washer cannot be located, perform intermediate gear box chip detector/temperature sensor particle inspections (PARA 6-3-11).
- f. If necessary, replace self-closing valve (PARA 6-4-37.2). If self-closing valve does not need to be replaced, install protective plug in hole.

#### 6-4-37.1.3 Install.

#### NOTE

Intermediate gear box chip detector/temperature sensor and self-closing valve are a matched set and cannot be replaced individually. If replacement chip detector/temperature sensor is being installed, chip detector/temperature sensor and self-closing valve must be replaced as an assembly.

- a. Turn off all electrical power (PARA 1-6-16).



**NOTE**

TIGHTEN SELF-CLOSING VALVE 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.

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**FIGURE 6-4-37-1. REPLACE INTERMEDIATE GEAR BOX CHIP DETECTOR/TEMPERATURE SENSOR AND SELF-CLOSING VALVE**

- b. Remove protective plug from self-closing valve.
- c. Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packings to be installed on intermediate gear box chip detector/temperature sensor.
- d. Check intermediate gear box chip detector/temperature sensor electrical circuit (PARA 6-2-2).
- e. Install packings on chip detector/temperature sensor, and install in self-closing valve (Figure 6-4-37-1, Detail B).
- f. Remove protective caps and plugs from electrical connectors.
- g. Inspect and treat electrical connectors (PARA 1-7-20).
- h. Connect electrical connector, P613, to intermediate gear box chip detector/temperature sensor.

**NOTE**

Insulation sleeve on electrical connector is used to prevent water intrusion.

- i. Pull insulation sleeve over electrical connector, P613 (Figure 6-4-37-1, Detail B, View A).

**WARNING**

Injury to personnel and damage to equipment will result if intermediate gear box is operated with low oil level. Operation with low oil level or without oil will cause loss of tail rotor drive. Lightly shake pylon to observe oil sloshing when checking oil level.

- j. If self-closing valve was replaced, service intermediate gear box (PARA 1-3-13).
- k. Make sure area is clean and free of foreign material.

- l. Perform check of intermediate gear box chip detector/temperature sensor as follows:
  - (1) With qualified person at controls, run up helicopter.
  - (2) Check intermediate gear box chip detector/temperature sensor for leaks.
- m. Install intermediate gear box fairing (PARA 2-4-139).
- n. On copilot's circuit breaker panel, push in CHIP DET circuit breaker.
- o. Perform operational check of both intermediate gear box chip detector/temperature sensor electrical circuits (PARA 6-2-2).

**6-4-37.2 Replace Self-Closing Valve.****6-4-37.2.1 Remove.**

- a. Drain intermediate gear box (PARA 1-3-13).
- b. If not already removed, remove intermediate gear box chip detector/temperature sensor (PARA 6-4-37.1.1) and/or protective plug from hole.
- c. Remove self-closing valve and packing from intermediate gear box. Discard packing (Figure 6-4-37-1, Detail B).
- d. Install protective plug in intermediate gear box.

**6-4-37.2.2 Install.****NOTE**

Intermediate gear box self-closing valve and chip detector/temperature sensor are a matched set and cannot be replaced individually. If replacement self-closing valve is being installed, self-closing valve and chip detector/temperature sensor must be replaced as an assembly.

- a. Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing.
- b. Remove protective plug from intermediate gear box.

- c. Install packing on self-closing valve, and install self-closing valve in intermediate gear box (Figure 6-4-37-1, Detail B).
- d. **TIGHTEN SELF-CLOSING VALVE  $\frac{1}{6}$  TO  $\frac{1}{3}$  TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- e. Using lockwire, Item 198, Appendix D, lockwire self-closing valve to intermediate gear box.
- f. Install protective plug in self-closing valve.
- g. Install intermediate gear box chip detector/temperature sensor (PARA 6-4-37.1.3).



6-4-38 TAIL ROTOR DRIVE SHAFT SECTION I.

PARAGRAPH BREAKDOWN

|          |                                          | PAGE     |
|----------|------------------------------------------|----------|
| 6-4-38.1 | Replace Tail Rotor Drive Shaft Section I | 6-4-38-2 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Feeler Gage
- Powertrain Repairer’s Toolkit
- Straight Edge
- Tail Drive Flange Wrench, 70700-20687-102
- Torque Reactor, 70700-20688-041
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1600 in. lbs
- Torque Wrench, 100 - 500 ft lbs

Materials/Parts

- Abrasive Cloth, Item 2, Appendix D

- Corrosion Preventive Compound, Item 113, Appendix D
- Corrosion Preventive Compound, Item 114, Appendix D
- Corrosion Removing Compound, Item 116, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D
- PARA 1-7-8
- PARA 6-3-15
- PARA 6-4-2
- PARA 6-4-42
- PARA 6-4-50
- TM 1-70-VIB

### 6-4-38.1 Replace Tail Rotor Drive Shaft Section I.

#### 6-4-38.1.1 Remove.

#### WARNING

Injury to personnel and damage to equipment will result if tail rotor is not secured when tail rotor drive shaft section I is disconnected. Make sure tail rotor is secured to prevent tail rotor drive shafts and tail rotor from turning.

- a. Secure tail rotor to prevent tail rotor drive shafts and tail rotor from turning.
- b. Open oil cooler compartment access doors.

#### CAUTION

Damage to tail rotor drive shaft section I will result if handled improperly. Handle tail rotor drive shaft section I carefully to prevent from being dented, scratched, or bent.

#### NOTE

- Tail rotor drive shaft flexible coupling bolts and self-locking nuts can be reused, provided they show no visual evidence of damage.
  - Note location of hardware for proper installation.
- c. On helicopters without rotor brake provisioned or installed, remove tail rotor drive shaft section I as follows (Figure 6-4-38-1, Sheet 1, Detail A):
    - (1) Support tail rotor drive shaft section I at both ends.
    - (2) Disconnect flexible couplings to main transmission tail takeoff flange and oil cooler fan input flange by removing bolts, spherical washers, shims, and nuts.

- (3) Remove tail rotor drive shaft section I.

- d. On helicopters with rotor brake provisioned or installed, remove tail rotor drive shaft section I as follows:

- (1) **RTR BRAKE** Remove shield as follows (Figure 6-4-38-1, Sheet 3, Detail F):

- (a) Remove bolt, washers, and nut from upper bracket.
- (b) Remove nuts and washers from studs securing shield.
- (c) Using machinery towels, Item 211, Appendix D, position on tail rotor drive shaft section I.
- (d) Slide shield over flexible couplings and place it on section of tail rotor drive shaft section I covered with machinery towels.

#### CAUTION

- Damage to tail rotor drive shaft section I will result if assembly is disturbed. Tail rotor drive shaft section I, 70361-23001-047, is a dynamically balanced assembly. Do not disturb hardware attaching adapters and inner flexible couplings to tail rotor drive shaft section I.
  - Damage to flexible coupling will result if tail rotor drive shaft section I is allowed to hang unsupported on one end when removing bolts. Support tail rotor drive shaft section I at both ends when removing. Place tail rotor drive shaft section I on padded area after removal.
- (2) On front and rear ends of tail rotor drive shaft section I, remove one bolt, spherical washers, and nut securing flexible couplings to adapters (Figure 6-4-38-1, Sheet 3, Detail F and Sheet 4, Detail J). Do not mix hardware.



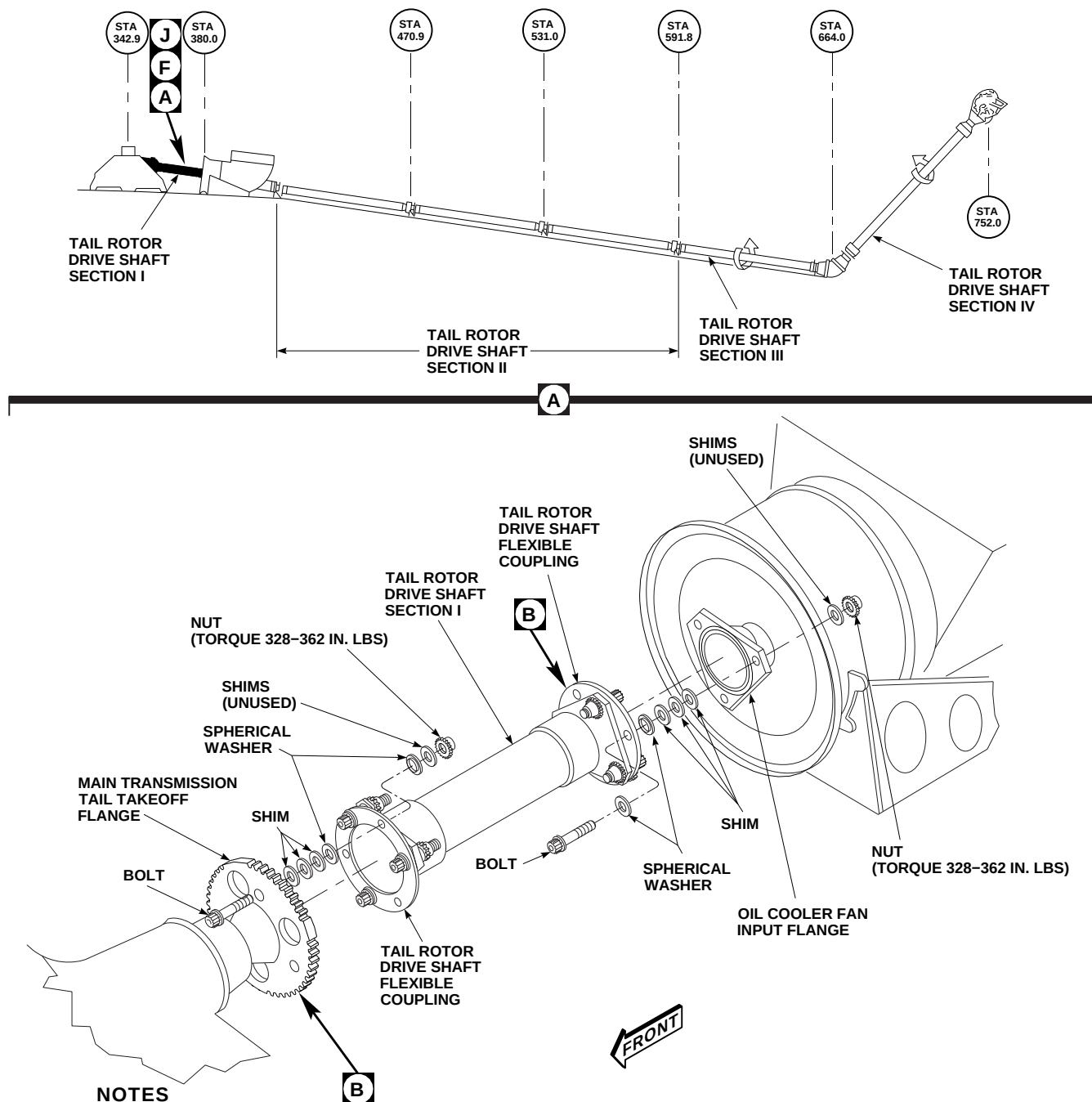
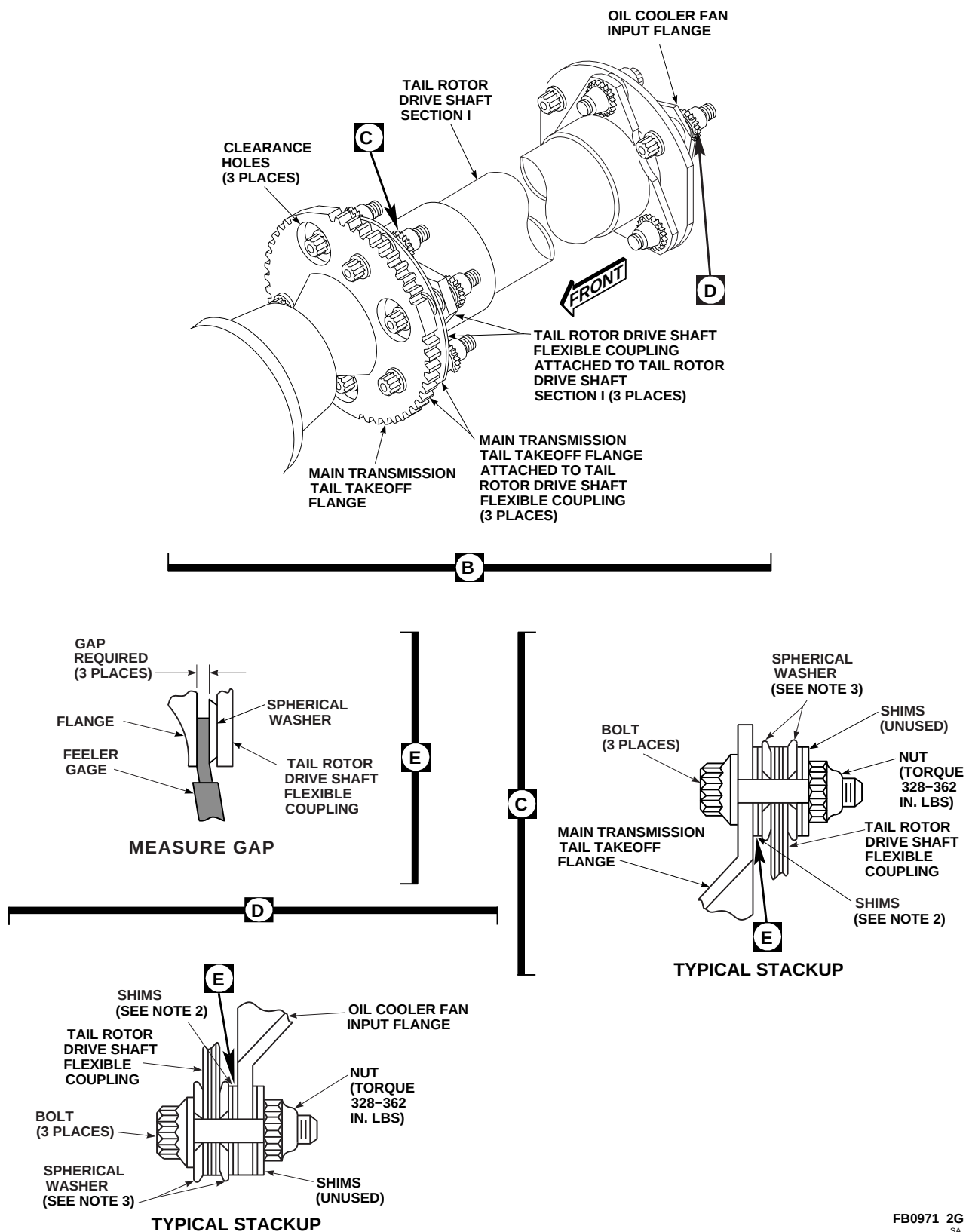


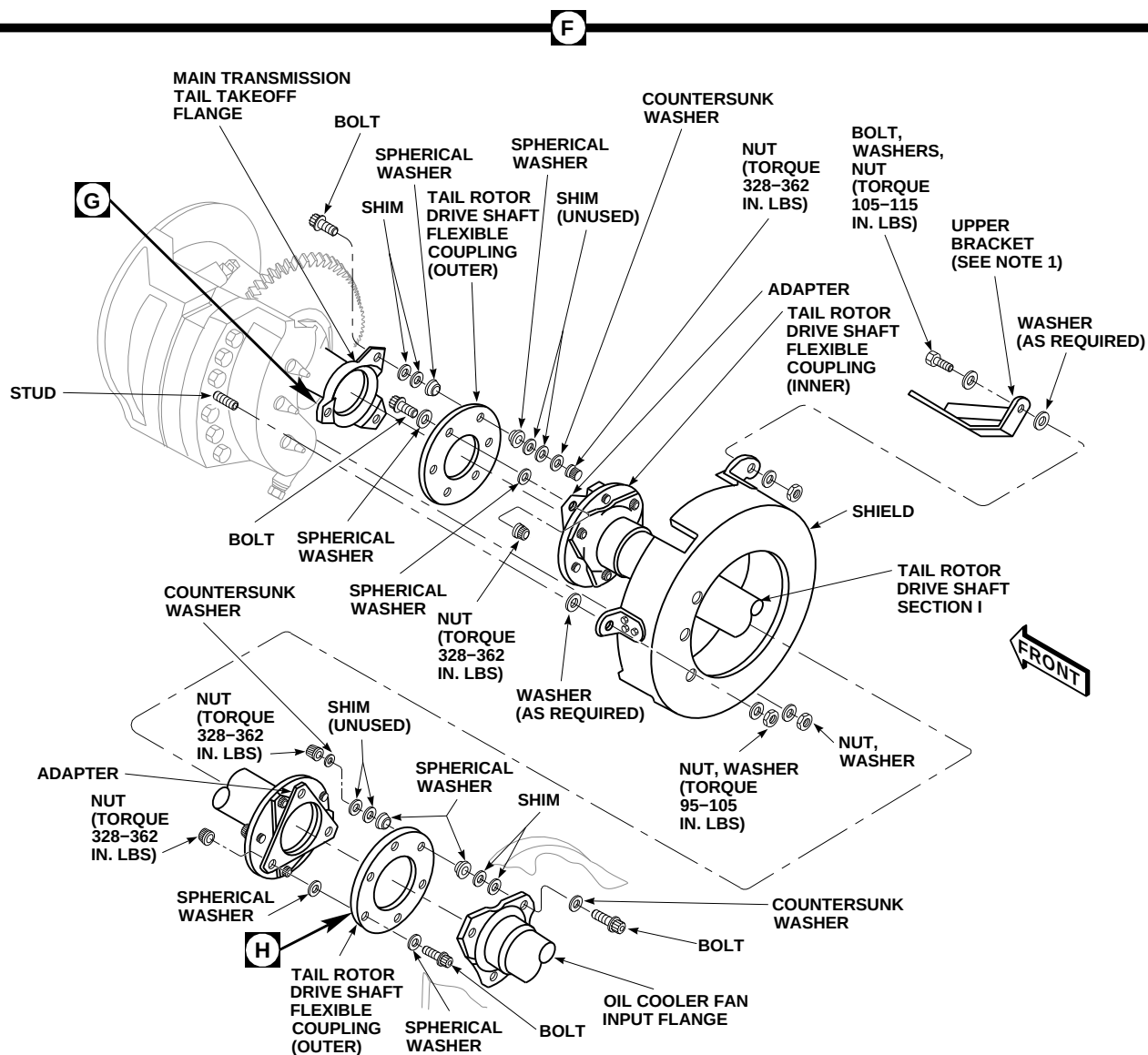
FIGURE 6-4-38-1. REPLACE TAIL ROTOR DRIVE SHAFT SECTION I (SHEET 1 OF 4)

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FIGURE 6-4-38-1. REPLACE TAIL ROTOR DRIVE SHAFT SECTION I (SHEET 2 OF 4)



## EFFECTIVITY

**ON HELICOPTERS WITH ROTOR BRAKE  
INSTALLED OR PROVISIONED.**

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**FIGURE 6-4-38-1. REPLACE TAIL ROTOR DRIVE SHAFT SECTION I (SHEET 3 OF 4)**

## Change 6

**6-4-38-5**



- (3) Support tail rotor drive shaft section I and remove bolts, spherical washers, shims, countersunk washers, and nuts connecting outer flexible couplings to main transmission tail takeoff flange and oil cooler fan input flange.
- (4) Remove tail rotor drive shaft section I. Place tail rotor drive shaft section I on padded area.
- (5) **RTR BRAKE** Remove shield and machinery towels from tail rotor drive shaft section I. 
- (6) If outer flexible couplings are being removed, remove remaining bolts, spherical washers, and nuts securing outer flexible coupling to adapter.

#### 6-4-38.1.2 Inspect.

#### NOTE

If any one disc making up tail rotor drive shaft flexible coupling stackup is damaged, entire flexible coupling must be replaced. Flexible coupling discs are a matched set. Discs cannot be replaced individually. If any one disc is damaged beyond limits, entire flexible coupling must be replaced. Two or more discs between bolts on flexible coupling flanges may separate or buckle. This is acceptable, unless there is a sharp bend or crease in disc, or if lamination spread (gap between two individual discs) is over 0.030-inch. To determine spread, measure gap between disc laminations with feeler gage.

- a. Inspect outer flexible couplings for damage. If any one disc making up stackup of outer flexible coupling is damaged, replace entire outer flexible coupling (PARA 6-4-42).
- b. Inspect inner flexible couplings for damage. If inner flexible coupling is damaged, replace tail rotor drive shaft section I.

#### NOTE

Corrosion preventive compound may turn reddish-brown over time. This is normal and should not be mistaken for corrosion.

- c. Visually inspect spherical washers of flexible coupling for evidence of corrosion, damage, or surface roughness. **NONE ALLOWED.** If corrosion, damage, or surface roughness is found, replace spherical washers.
- d. Inspect adapter boltholes of tail rotor drive shaft section I for elongation. **NONE ALLOWED.** If elongated, replace tail rotor drive shaft section I.
- e. If spherical washer, 70361-05010-103, is unavailable, washer, 70361-05010-102, may be reworked as follows:
  - (1) Using corrosion removing compound, Item 116, Appendix D, clean washer.
  - (2) If corrosion is still found, using abrasive cloth, Item 2, Appendix D, remove corrosion. Make sure all corrosion is removed.
  - (3) **MINIMUM THICKNESS AFTER BLENDING IS 0.080-INCH THICK.** If minimum thickness is under limit, replace washers.
  - (4) Using corrosion preventive compound, Item 114, Appendix D, apply to washer surfaces.
  - (5) Using corrosion preventive compound, Item 113, Appendix D, apply to hardware assembly. Allow to dry.
- f. With retaining nut of main transmission tail takeoff flange exposed, perform torque check as follows:
  - (1) Remove cotter pin from retaining nut, if installed.

**NOTE**

Check torque in tightening direction only.

- (2) Using torque reactor and tail drive flange wrench, **CHECK TORQUE AT 840 INCH-POUNDS. IF TORQUE IS AT OR BELOW 840 INCH-POUNDS, REMOVE MAIN TRANSMISSION TAIL TAKEOFF FLANGE AND INSPECT SPLINES AND MATING SURFACES FOR WEAR (PARA 6-4-2).**
- (3) Using torque reactor and tail drive flange wrench, **CHECK TORQUE AT 1344 INCH-POUNDS. IF TORQUE IS AT 1344 INCH-POUNDS OR MORE, NO FURTHER ACTION IS REQUIRED. IF TORQUE IS BETWEEN 840 INCH-POUNDS AND 1344 INCH-POUNDS, TORQUE RETAINING NUT TO 1344 INCH-POUNDS.**

g. With retaining nut of oil cooler fan input flange exposed, perform torque check as follows:

- (1) Remove cotter pin from retaining nut.

**NOTE**

Check torque in tightening direction only.

- (2) Using torque reactor and tail drive flange wrench, check torque of retaining nut. **IF TORQUE IS 214 - 236 FOOT-POUNDS, INSTALL NEW COTTER PIN. IF REQUIRED TO ALIGN COTTER PIN HOLE, TURN RETAINING NUT IN TIGHTENING DIRECTION.** Torque check is complete.
- (3) **IF TORQUE IS LESS THAN 214 FOOT-POUNDS, INSPECT RETAIN-**

**ING NUT AND OIL COOLER FAN INPUT FLANGE FOR BINDING AND CORROSION. REPLACE RETAINING NUT IF:**

- (a) **PITS ARE 0.030-INCH DEEP.**
- (b) **PITTED MORE THAN 10% OF MATING SURFACE.**
- (c) **THREADS SHOW SIGNS OF CORROSION.**
- (d) **RETAINING NUT CAN NOT BE FULLY RUN ON BY HAND.**
- (4) **IF RETAINING NUT IS NOT BOUND OR CORRODED, USING TORQUE REACTOR AND TAIL DRIVE FLANGE WRENCH, TORQUE RETAINING NUT TO 214 - 236 FOOT-POUNDS. INSTALL COTTER PIN. IF REQUIRED TO ALIGN COTTER PIN HOLE, TURN RETAINING NUT IN TIGHTENING DIRECTION.**

h. Inspect oil cooler fan input flange (PARA 6-4-50).

**6-4-38.1.3 Install.**

- a. Using torque wrench, check drag torque on nuts as follows:
  - (1) With nut and bolt not installed, thread nut on mating bolts so that two threads of bolt extend past end of nut.
  - (2) Measure torque required to back off nut. **MINIMUM DRAG TORQUE IS 9.5 INCH-POUNDS.** If torque is beyond limits, replace nuts.

**CAUTION**

- Damage to powertrain components will result if tail rotor drive shaft section I is unbalanced. Mixing of washers will cause an unbalanced tail rotor drive shaft section I. Do not mix spherical washers, 70361-05010-102, with washers, 70361-05010-103.
- Damage to tail rotor drive shaft section I will result if handled improperly. Handle tail rotor drive shaft section I carefully to prevent from being dented, scratched, or bent. To prevent possible damage, place tail rotor drive shaft section I on padded area.

**NOTE**

- When installing bolts in flexible couplings, make sure boltheads are against main transmission tail takeoff flange.
  - Do not lockwire bolts with drilled heads installed on flexible couplings.
- b. On helicopters without rotor brake provisioned or installed, install tail rotor drive shaft section I as follows:
- (1) Remove flexible couplings from old tail rotor drive shaft section I and install on new tail rotor drive shaft section I (PARA 6-4-42).
  - (2) Support tail rotor drive shaft section I in position (Figure 6-4-38-1, Sheet 1, Detail A).

**NOTE**

- Spherical washers are to be installed on each side of flexible coupling.
  - Spherical side of washer must face flexible coupling.
- (3) Support tail rotor drive shaft section I (front end) in installed position with three

bolts and spherical washers. Install nuts. **TIGHTEN NUTS ONLY ENOUGH TO PREVENT AXIAL MOVEMENT** (Figure 6-4-38-1, Sheet 2, Detail B and Detail C).

- (4) Support opposite end tail rotor drive shaft section I (rear end) in installed position with three bolts and spherical washers. Do not install nuts at this time (Figure 6-4-38-1, Sheet 2, Detail B and Detail D).
- c. On helicopters with rotor brake provisioned or installed, install tail rotor drive shaft section I as follows (Figure 6-4-38-1, Sheet 3, Detail F and Sheet 4, Detail J):

**NOTE**

- Spherical washers are to be installed on each side of flexible coupling.
  - Spherical side of washer must face flexible coupling.
- (1) If removed, install outer flexible couplings on adapters with two bolts, spherical washers, and nuts. **DO NOT TIGHTEN NUTS NOW.**
  - (2) **RTR BRAKE** Position machinery towels, Item 211, Appendix D, on tail rotor drive shaft section I and slide shield onto tail rotor drive shaft section I. Be careful not to scratch tail rotor drive shaft section I with shield. ◀

**CAUTION**

Damage to flexible coupling will result if tail rotor drive shaft section I is allowed to hang unsupported on one end when installing bolts. Support tail rotor drive shaft section I at both ends during installation.

**NOTE**

- When installing tail rotor drive shaft section I, bolts securing outer flexible



coupling to main transmission tail takeoff flange are installed with bolt-heads facing front.

- When installing tail rotor drive shaft section I, short bolts, SS5025-06-009N, must be installed on front end.
  - Spherical washers are to be installed on each side of flexible coupling.
  - Spherical side of washer must face flexible coupling.
- (3) Support tail rotor drive shaft section I (front end) in installed position with three bolts and spherical washers. Install countersunk washers and nuts (Figure 6-4-38-1, Sheet 3, Detail F and Sheet 4, Detail G and Detail J).
  - (4) Support rear end of tail rotor drive shaft section I in installed position.

#### NOTE

When installing tail rotor drive shaft section I, bolts securing outer flexible coupling to oil cooler fan input flange are installed with bolt-heads facing rear.

- (5) Support opposite end tail rotor drive shaft section I (rear end) in installed position with three bolts and spherical washers

(Figure 6-4-38-1, Sheet 3, Detail F and Sheet 4, Detail H and Detail J).

- (6) Install remaining bolts, spherical washers, and nuts securing flexible coupling to adapters. **TIGHTEN ALL INSTALLED NUTS, BUT DO NOT TORQUE NOW** (Figure 6-4-38-1, Sheet 3, Detail F and Sheet 4, Detail G, Detail H, and Detail J).

- d. Determine shim stackup as follows:

#### NOTE

Do not turn tail rotor drive shaft section I from original position, as gap may change. If tail rotor drive shaft section I is moved, recheck gap.

- (1) Support tail rotor drive shaft section I in installed position with spherical washers on each side of flexible coupling.
- (2) Using feeler gage, measure gap at three attachment points (Figure 6-4-38-1, Sheet 2, Detail E).
- (3) Determine average gap by adding readings and dividing by 3.
- (4) Divide average gap by 2 to determine best shim combination measurement for both ends of tail rotor drive shaft section I.



Example

Measure Gap:

|       |                  |
|-------|------------------|
| Gap   | A = 0.125        |
|       | B = 0.128        |
|       | <u>C = 0.121</u> |
| Total | = 0.374          |

Divide by 3:  $0.374 \div 3 = 0.1246$ -inch average gap.

Divide by 2:  $0.1246 \div 2 = 0.0623$ .

Round off: 0.062-inch shim combination.

NOTE

Due to weight of tail rotor drive shaft section I, shim calculation may result in excess deflection of tail rotor drive flexible coupling. In an effort to center tail rotor drive shaft section I and reduce maximum amount of gap, shims should be installed at each end as equally as possible. In situations where reducing the maximum amount of gap requires unequal amount of shims from one end to the other, it is permissible for shims on opposite ends to be unequal up to 0.020-inch.

- e. Select shim combination resulting from computation above. Make three stackups for each end of tail rotor drive shaft section I. **MAXIMUM GAP IS 0.125-INCH AT EACH END OF TAIL ROTOR DRIVE SHAFT SECTION I.**

NOTE

- Shim thickness must be the same for all three bolt locations at each end of tail

rotor drive shaft section I. Shim combination at one end does not have to be same as shim combination at other end, as long as total shim combination adds up closest to average gap. Store unused shims under nut to maintain balance of tail rotor drive shaft section I.

- Hardware must not be mixed.
- Bolts must be installed within 15 minutes after applying epoxy primer to bolts.
- f. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to bolt grip lengths and under boltheads. Do not apply primer to threads.

NOTE

- Spherical washers are to be installed on each side of flexible coupling.
- Spherical side of washer must face flexible coupling.
- One shim stackup is to be installed on each bolt.
- Shim stackup is to be located between flange and spherical washer.
- Install unused shims under nut.
- g. Secure oil cooler fan input flange to outer flexible coupling with bolts, countersunk washers (if applicable), shims, spherical washers, and nuts (Figure 6-4-38-1, Sheet 2, Detail D and Sheet 4, Detail H).
- h. Secure main transmission tail takeoff flange to outer flexible coupling with bolts, countersunk washers (if applicable), shims, spherical washers, and nuts (Figure 6-4-38-1, Sheet 2, Detail C, and Sheet 4, Detail G).

**WARNING****FSCAP**

Verification of minimum torque and torque stabilization check of nuts is a critical characteristic.

- i. **TORQUE NUTS TO 328 - 362 INCH-POUNDS.**
- j. Perform tail rotor drive shaft shimming inspection as follows:

**NOTE**

- Shims used for tail rotor drive shaft section I are installed between the mating flange and spherical washer. Unused shims are installed under nut (Figure 6-4-38-1, Sheet 2, Detail C and Detail D and Sheet 4, Detail G and Detail H).
- Flanges at both ends of tail rotor drive shaft section I use equal amount of shims.
  - (1) Inspect for proper shimming of flexible couplings. Make sure same amount of shims are used between spherical washer and adjacent flange. There should be same amount of unused shims between spherical washer and nut.
  - (2) Make sure all spherical washers have spherical side facing flexible coupling.
  - (3) Using feeler gage and straight edge placed along flange surface, inspect for deflection in flexible coupling. **NO MORE THAN 0.015-INCH DEFLECTION IN FLEX-**

**IBLE COUPLING IS ALLOWED**  
(Figure 6-4-38-2).

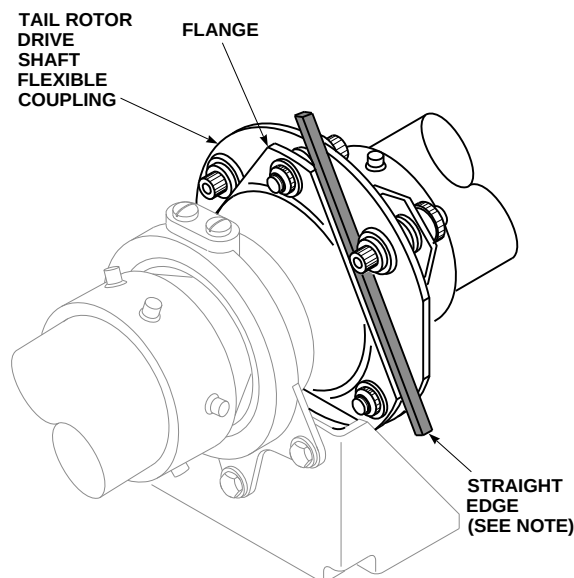
- k. If requirements of tail rotor drive shaft shimming inspection are not met using shim measurements and calculations in step d., adjust shims as follows:

Adjusted Shim Thickness = Calculated Shim Thickness (step d. (4)) - Deflection of Tail Rotor Drive Shaft Flexible Coupling at Lower End (step j. (3))

**Example**

If shim thickness at each end is calculated to be 0.110-inch and deflection at lower end measured with straight edge 0.020-inch (0.005-inch greater than 0.015-inch allowable), then actual shim thickness = 0.110-inch - 0.020-inch = 0.090-inch.

- l. Using corrosion preventive compound, Item 113, Appendix D, spray all spherical washers.
- m. **RTR BRAKE** Secure shield as follows (Figure 6-4-38-1, Sheet 3, Detail F):
  - (1) Slide shield onto rotor brake studs and bracket.
  - (2) Install washers on studs as required to shim shield. Place unused washers under nuts.
  - (3) **TORQUE NUTS TO 95 - 105 INCH-POUNDS.**
  - (4) Install bolt with washers as required to shim shield on upper bracket. Place unused washers under bolthead. Install nut.
  - (5) **TORQUE NUT TO 105 - 115 INCH-POUNDS.**



**NOTE**  
 INSERT FEELER GAGE BETWEEN TAIL ROTOR DRIVE SHAFT FLEXIBLE COUPLING AND STRAIGHT EDGE. MAXIMUM DEFLECTION NOT TO EXCEED 0.015 IN.

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**FIGURE 6-4-38-2. INSPECT TAIL ROTOR DRIVE SHAFT SHIMMING**

**WARNING**

Injury to personnel and damage to equipment will result if oil cooler spline wear inspection is not performed on oil cooler fans, 70361-03005-103 through 70361-03005-106, every time tail rotor drive shaft is disconnected or replaced from main transmission to STA 531.0. Excessive spline wear will result in loss of tail rotor drive. Make sure spline wear indicator is installed and oil cooler spline wear inspection is performed.

- n. On helicopters with oil cooler fan, 70361-03005-103 through 70361-03005-106, installed, perform oil cooler spline wear inspection (PARA 6-3-15).

**WARNING**

Injury to personnel and damage to equipment will result if vibration check

is not performed every time tail rotor drive shaft is disconnected or replaced from main transmission to STA 531.0. Excessive vibration indicates improper installation of oil cooler which will cause oil cooler fan drive shaft splines and oil cooler fan exit flange splines to wear, resulting in loss of tail rotor drive. Make sure vibration check is performed.

- o. Perform oil cooler vibration check (TM 1-70-VIB).
- p. Check helicopter logbook to make sure tail rotor drive shaft torque check is shown as due 9 to 11 flight hours after installation (PARA 1-7-8).
- q. Make sure area is clean and free of foreign material.
- r. Close oil cooler compartment access doors.



6-4-39 TAIL ROTOR DRIVE SHAFT SECTION II.

PARAGRAPH BREAKDOWN

|                                                    | PAGE     |
|----------------------------------------------------|----------|
| 6-4-39.1 Replace Tail Rotor Drive Shaft Section II | 6-4-39-2 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Feeler Gage
- Powertrain Repairer’s Toolkit
- Tail Drive Flange Wrench, 70700-20687-102
- Torque Reactor, 70700-20688-041
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 100 - 500 ft lbs

Materials/Parts

- Abrasive Cloth, Item 2, Appendix D
- Corrosion Preventive Compound, Item 113, Appendix D

- Corrosion Preventive Compound, Item 114, Appendix D
- Corrosion Removing Compound, Item 116, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D

References

- Appendix D
- PARA 1-7-8
- PARA 2-4-112
- PARA 2-4-116
- PARA 2-5-19
- PARA 2-6-5
- PARA 6-3-15
- PARA 6-4-42
- TM 1-70-VIB

### 6-4-39.1 Replace Tail Rotor Drive Shaft Section II.

#### NOTE

Tail rotor drive shaft section II has three sections. Sections are identified as front, middle, and rear. Replacement of the three sections is the same.

#### 6-4-39.1.1 Remove.

#### WARNING

Injury to personnel and damage to equipment will result if tail rotor is not secured when tail rotor drive shaft section II is disconnected. Make sure tail rotor is secured to prevent tail rotor drive shafts and tail rotor from turning.

- a. Secure tail rotor to prevent tail rotor drive shafts and tail rotor from turning.
- b. Gain access to applicable section(s) (front, middle, and rear) of tail rotor drive shaft section II as follows:
  - (1) Front section:
    - (a) Open APU access doors.
    - (b) Remove APU panel (PARA 2-4-112).
  - (2) Middle section:
    - (a) Remove APU panel (PARA 2-4-112).
    - (b) Remove main rotor pylon rear fairing (PARA 2-4-116).
    - (c) Open front tail rotor drive shaft cover.
  - (3) Rear section - Open front and rear tail rotor drive shaft covers.

#### CAUTION

Damage to tail rotor drive shaft section II will result if handled improperly. Handle tail rotor drive shaft section II carefully to prevent from being dented, scratched, or bent. To prevent possible damage, place tail rotor drive shaft section II on padded area.

#### NOTE

- Tail rotor drive shaft flexible coupling bolts, self-locking nuts, and viscous damper bearing assembly attachment bolts and nuts can be reused, provided they show no visual evidence of damage.
  - Note location of hardware for proper installation.
  - Forward end of tail rotor drive shaft flexible coupling comes off with section of tail rotor drive shaft section II being removed. Rear end of tail rotor drive shaft flexible coupling remains on helicopter.
- c. Support section and disconnect tail rotor drive shaft flexible coupling at forward end by removing bolts, spherical washers, shims, and nuts (Figure 6-4-39-1, Detail B).
  - d. Disconnect rear end of section from tail rotor drive shaft flexible coupling by removing bolts, spherical washers, shims, and nuts.
  - e. Remove bolts, washers, and nuts securing viscous damper bearing assembly to airframe support (Figure 6-4-39-1, Detail A).
  - f. Remove section of tail rotor drive shaft section II. Place tail rotor drive shaft section II on padded area.

#### 6-4-39.1.2 Inspect.

- a. Inspect airframe supports for visible cracks and distortion. **NONE ALLOWED.** If cracks and

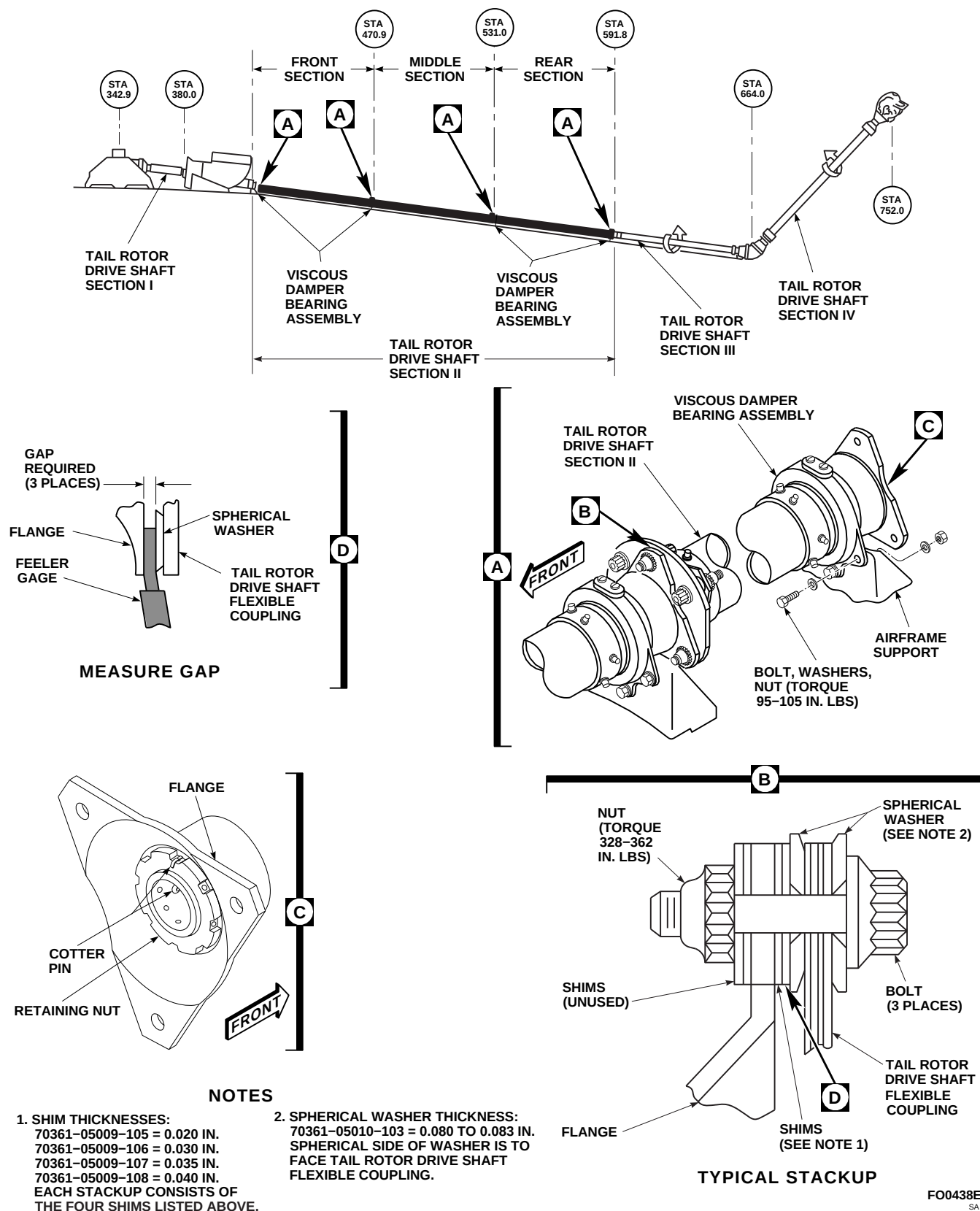


FIGURE 6-4-39-1. REPLACE TAIL ROTOR DRIVE SHAFT SECTION II

distortion are found, replace airframe support (PARA 2-5-19).

- b. Inspect tail rotor drive shaft flexible coupling (PARA 6-4-42).

#### NOTE

Corrosion preventive compound may turn reddish-brown over time. This is normal and should not be mistaken for corrosion.

- c. Visually inspect spherical washers of tail rotor drive shaft flexible coupling for evidence of corrosion, damage, or surface roughness. **NONE ALLOWED.** If corrosion, damage, or surface roughness is found, replace spherical washers.
- d. Inspect tail rotor drive shaft section II flange boltholes for elongation. **NONE ALLOWED.** If boltholes are elongated, replace tail rotor drive shaft section II.
- e. If spherical washer, 70361-05010-103, is unavailable, washer, 70361-05010-102, may be reworked as follows:
- (1) Using corrosion removing compound, Item 116, Appendix D, clean washer.
  - (2) If corrosion is still found, using abrasive cloth, Item 2, Appendix D, remove corrosion. Make sure all corrosion is removed.
  - (3) **MINIMUM THICKNESS AFTER BLENDING IS 0.080-INCH THICK.** If minimum thickness is under limit, replace washers.
  - (4) Using corrosion preventive compound, Item 114, Appendix D, apply to washer surfaces.
  - (5) Using corrosion preventive compound, Item 113, Appendix D, apply to hardware assembly. Allow to dry.
- f. With retaining nut of oil cooler drive shaft and tail rotor drive shaft section II flange exposed, perform torque check as follows:

- (1) Remove cotter pin from retaining nut (Figure 6-4-39-1, Detail C).

#### NOTE

- Torque reactor will be attached to drive shaft flange to react against torquing of retaining nut.
  - Check torque in tightening direction while holding drive shaft with torque reactor.
- (2) Using torque reactor and tail drive flange wrench, check torque of retaining nut. **TORQUE SHALL BE 214 - 236 FOOT-POUNDS.**
  - (3) **IF TORQUE IS 214 - 236 FOOT-POUNDS, TORQUE CHECK IS COMPLETE.** Remove torque reactor and install new cotter pin as follows:

#### NOTE

Cotter pin is installed with head against inside diameter of retaining nut.

- (a) Install new cotter pin. If required to align cotter pin hole, turn retaining nut in tightening direction.
- (b) Fold over rear end of cotter pin against retaining nut, then fold front end (flange against retaining nut mating surface) flush with retaining nut.
- (c) Apply torque stripe (PARA 2-6-5).
- (4) **IF TORQUE IS LESS THAN 214 FOOT-POUNDS, INSPECT RETAINING NUT AND SECTION II DRIVE SHAFT FLANGE FOR BINDING AND CORROSION. REPLACE RETAINING NUT IF:**
  - (a) **PITS ARE 0.030-INCH DEEP.**
  - (b) **PITTED MORE THAN 10% OF MATING SURFACE.**



- (c) **THREADS SHOW SIGNS OF CORROSION.**
- (d) **RETAINING NUT CAN NOT BE FULLY RUN ON BY HAND.**
- (5) **IF RETAINING NUT IS NOT BOUND OR CORRODED, USING TORQUE REACTOR AND TAIL DRIVE FLANGE WRENCH, TORQUE RETAINING NUT TO 214 - 236 FOOT-POUNDS. TORQUE CHECK IS COMPLETE.** Remove torque reactor and install cotter pin per step f. (3).

### 6-4-39.1.3 Install.

#### WARNING

Injury to personnel and damage to equipment will result if section of tail rotor drive shaft section II is bolted incorrectly to adjacent flange. Three adjacent flange bolts will be connected to tail rotor drive shaft flexible coupling only. Three drive shaft bolts will be connected to tail rotor drive shaft flexible coupling only. Make sure flange of section is not bolted directly through tail rotor drive shaft flexible coupling to adjacent flange.

#### CAUTION

Damage to tail rotor drive shaft section II will result if handled improperly. Handle tail rotor drive shaft section II carefully to prevent from being dented, scratched, or bent. To prevent possible damage, place tail rotor drive shaft section II on padded area.

#### NOTE

- When installing bolts in tail rotor drive shaft flexible couplings, make sure bolt-heads are against tail rotor drive shaft flexible coupling.

- Do not lockwire bolts with drilled heads installed on tail rotor drive shaft flexible couplings.
- Installation of all three sections (front, middle, and rear) is the same. For any section removed or disconnected, install both ends using the following procedure.
  - a. Remove tail rotor drive shaft flexible couplings from old section and install on new section (PARA 6-4-42).
  - b. Support front end of tail rotor drive shaft. Attach viscous damper bearing assembly to air-frame support with bolts, washers, and nuts (Figure 6-4-39-1, Detail A).
  - c. Using torque wrench, check drag torque on nuts. **MINIMUM DRAG TORQUE IS 3.5 INCH-POUNDS.** If torque is beyond limits, replace nuts.
  - d. **TORQUE NUTS TO 95 - 105 INCH-POUNDS.**
  - e. Center bearing in viscous damper bearing assembly.
  - f. Determine shim thickness at opposite end of section as follows:

#### NOTE

- Do not turn section from original position, as gap may change. If section is moved, recheck gap.
- Spherical washers are to be installed on each side of tail rotor drive shaft flexible coupling.
- Spherical side of washer must face tail rotor drive shaft flexible coupling.
  - (1) Support section in installed position with three bolts and spherical washers on each side of tail rotor drive shaft flexible coupling.
  - (2) Using feeler gage, measure gap at three attachment points between spherical

washers and flange (Figure 6-4-39-1, Detail D).

- (3) Determine average gap by adding readings and dividing by 3.

### Example

#### Measure Gap:

|       |                  |
|-------|------------------|
| Gap   | A = 0.062        |
|       | B = 0.071        |
|       | <u>C = 0.068</u> |
| Total | = 0.201          |

**Divide by 3:**  $0.201 \div 3 = 0.067$ -inch average gap.

- g. Select shim combination resulting from computation above. Make three stackups. **MAXIMUM GAP IS 0.125-INCH.**

### NOTE

- Shim thickness must be the same for all three bolt locations at each end of section. Shim combination at one end does not have to be the same as shim combination at other end, as long as total shim combination adds up closest to average gap.
  - Nuts must be torqued within 15 minutes after applying epoxy primer to bolts.
- h. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to bolt grip lengths and under boltheads. Do not apply primer to threads.

### NOTE

- Spherical washers are to be installed on each side of tail rotor drive shaft flexible coupling.
- Spherical side of washer must face tail rotor drive shaft flexible coupling.

- One shim stackup is to be installed on each bolt.
- Shim stackup is to be located between flange and spherical washer.
- Install unused shims under nut.

- i. Secure flange to tail rotor drive shaft flexible coupling with bolts, shims, spherical washers, and nuts (Figure 6-4-39-1, Detail B).
- j. Using torque wrench, check drag torque on nuts. **MINIMUM DRAG TORQUE IS 9.5 INCH-POUNDS.** If torque is beyond limits, replace nuts.

## WARNING

### FSCAP

Verification of minimum torque and torque stabilization check of nuts is a critical characteristic.

- k. **TORQUE NUTS TO 328 - 362 INCH-POUNDS.**
- l. Check helicopter logbook to make sure tail rotor drive shaft torque check is shown as due 9 to 11 flight hours after installation (PARA 1-7-8).

## CAUTION

Damage to bearing will result if contact occurs with corrosion preventive compounds. Corrosion preventive compounds can penetrate seals of bearing and dissolve lubrication grease. Do not spray corrosion preventive compounds in vicinity of bearing. Shield bearing from spray.

- m. Using corrosion preventive compound, Item 113, Appendix D, spray all spherical washers.

**WARNING**

Injury to personnel and damage to equipment will result if oil cooler spline wear inspection is not performed on oil cooler fans, 70361-03005-103 through 70361-03005-106, every time tail rotor drive shaft is disconnected or replaced from main transmission to STA 531.0. Excessive spline wear will result in loss of tail rotor drive. Make sure spline wear indicator is installed and oil cooler spline wear inspection is performed.

**NOTE**

Spline wear inspection is not required when tail rotor drive shaft is disconnected or replaced aft of STA 531.0.

- n. On helicopters with oil cooler fans, 70361-03005-103 through 70361-03005-106, installed, perform oil cooler spline wear inspection, if required (PARA 6-3-15).

**WARNING**

Injury to personnel and damage to equipment will result if vibration check is not performed any time a tail rotor drive shaft flexible coupling is disconnected or replaced from main transmission up to and including STA 470.9. Excessive vibration indicates improper installation of oil cooler and/or tail rotor drive shaft flexible coupling, which will

cause oil cooler fan drive shaft splines and oil cooler fan exit flange splines to wear, resulting in loss of tail rotor drive. Make sure vibration check is performed.

**NOTE**

Vibration check is not required when tail rotor drive shaft is disconnected or replaced at STA 531.0 and aft.

- o. If tail rotor drive shaft flexible coupling was disconnected or replaced from main transmission up to and including STA 470.9, perform oil cooler vibration check (TM 1-70-VIB).
- p. Make sure area is clean and free of foreign material.
- q. Close-up access to applicable section(s) (front, middle, and rear) of tail rotor drive shaft section II as follows:
  - (1) Front section:
    - (a) Close APU access doors.
    - (b) Install APU panel (PARA 2-4-112).
  - (2) Middle section:
    - (a) Install APU panel (PARA 2-4-112).
    - (b) Install main rotor pylon rear fairing (PARA 2-4-116).
    - (c) Close front tail rotor drive shaft cover.
  - (3) Rear section- Close front and rear tail rotor drive shaft covers.



6-4-40 TAIL ROTOR DRIVE SHAFT SECTION III.

PARAGRAPH BREAKDOWN

|          |                                            | PAGE     |
|----------|--------------------------------------------|----------|
| 6-4-40.1 | Replace Tail Rotor Drive Shaft Section III | 6-4-40-2 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Feeler Gage
- Powertrain Repairer’s Toolkit
- Straight Edge
- Tail Drive Flange Wrench, 70700-20687-102
- Torque Reactor, 70700-20688-041
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 100 - 500 ft lbs

Materials/Parts

- Abrasive Cloth, Item 2, Appendix D
- Corrosion Preventive Compound, Item 113, Appendix D
- Corrosion Preventive Compound, Item 114, Appendix D
- Corrosion Removing Compound, Item 116, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D

References

- Appendix D
- PARA 1-7-8
- PARA 2-5-19
- PARA 6-4-42

### 6-4-40.1 Replace Tail Rotor Drive Shaft Section III.

#### 6-4-40.1.1 Remove.

#### WARNING

Injury to personnel and damage to equipment will result if tail rotor is not secured when tail rotor drive shaft section III is disconnected. Make sure tail rotor is secured to prevent tail rotor drive shafts and tail rotor from turning.

- a. Secure tail rotor to prevent tail rotor drive shafts and tail rotor from turning.
- b. Open rear tail rotor drive shaft cover.

#### CAUTION

- Damage may occur if tools contact electrical harnesses during removal of tail rotor drive shaft section III. Do not allow tools to contact electrical harnesses.
- Damage to tail rotor drive shaft section III will result if handled improperly. Handle tail rotor drive shaft section III carefully to prevent from being dented, scratched, or bent. To prevent possible damage, place tail rotor drive shaft section III on padded area.

#### NOTE

- Tail rotor drive shaft flexible coupling bolts, self-locking nuts, and viscous damper bearing assembly attachment bolts and nuts can be reused, provided they show no visual evidence of damage.
- Note location of hardware for proper installation.
- c. Support rear end of tail rotor drive shaft section III and remove bolts, spherical washers, shims,

and nuts connecting tail rotor drive shaft flexible coupling on tail rotor drive shaft section III to intermediate gear box input flange (Figure 6-4-40-1, Detail A and Detail C).

- d. Support front end of tail rotor drive shaft section III and remove bolts, spherical washers, shims, and nuts connecting tail rotor drive shaft flexible coupling on tail rotor drive shaft section III to rear section of tail rotor drive shaft section II (Figure 6-4-40-1, Detail A and Detail B).
- e. Remove tail rotor drive shaft section III with tail rotor drive shaft flexible couplings. Place tail rotor drive shaft section III on padded area.

#### 6-4-40.1.2 Inspect.

- a. Inspect airframe supports for visible cracks and distortion. **NONE ALLOWED.** If cracks and distortion are found, replace airframe support (PARA 2-5-19).
- b. Inspect tail rotor drive shaft flexible coupling (PARA 6-4-42).

#### NOTE

Corrosion preventive compound may turn reddish-brown over time. This is normal and should not be mistaken for corrosion.

- c. Visually inspect spherical washers of tail rotor drive shaft flexible coupling for evidence of corrosion, damage, or surface roughness. **NONE ALLOWED.** If corrosion, damage, or surface roughness is found, replace spherical washers.
- d. Inspect flange boltholes of tail rotor drive shaft section III for elongation. **NONE ALLOWED.** If boltholes are elongated, replace tail rotor drive shaft section III.
- e. If spherical washer, 70361-05010-103, is unavailable, washer, 70361-05010-102, may be reworked as follows:
  - (1) Using corrosion removing compound, Item 116, Appendix D, clean washer.

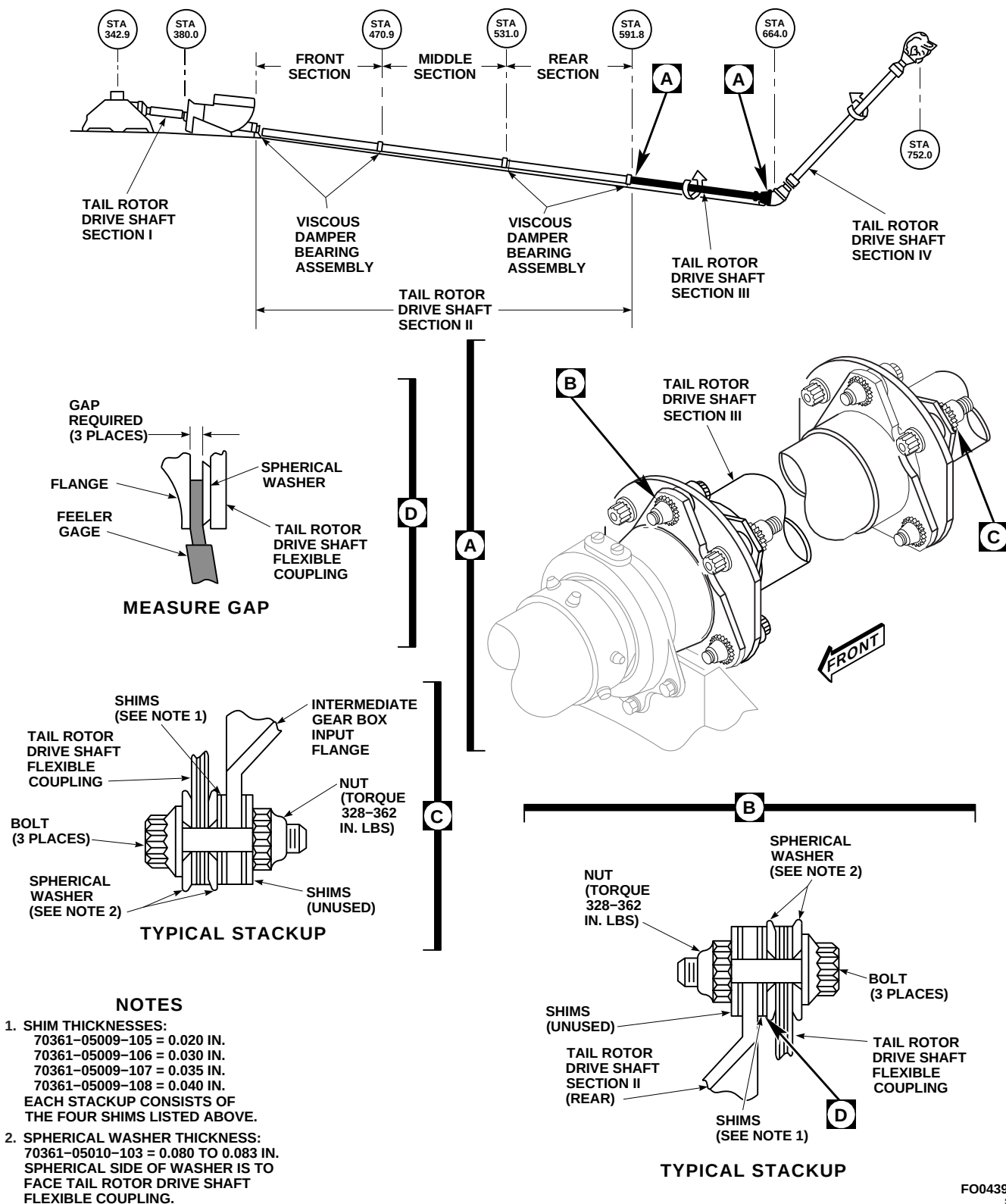


FIGURE 6-4-40-1. REPLACE TAIL ROTOR DRIVE SHAFT SECTION III

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- (2) If corrosion is still found, using abrasive cloth, Item 2, Appendix D, remove corrosion. Make sure all corrosion is removed.
  - (3) **MINIMUM THICKNESS AFTER BLENDING IS 0.080-INCH THICK.** If minimum thickness is under limit, replace washers.
  - (4) Using corrosion preventive compound, Item 114, Appendix D, apply to washer surfaces.
  - (5) Using corrosion preventive compound, Item 113, Appendix D, apply to hardware assembly. Allow to dry.
- f. With retaining nuts of tail rotor drive shaft section II flange and intermediate gear box input flange exposed, perform torque check as follows:

**NOTE**

Torque check for both retaining nuts is the same.

- (1) Remove cotter pin from retaining nut.

**NOTE**

Check torque in tightening direction only.

- (2) Using torque reactor and tail drive flange wrench, check torque of retaining nut. **IF TORQUE IS 214 - 236 FOOT-POUNDS, INSTALL NEW COTTER PIN. IF REQUIRED TO ALIGN COTTER PIN HOLE, TURN RETAINING NUT IN TIGHTENING DIRECTION.** Torque check is complete.
- (3) **IF TORQUE IS LESS THAN 214 FOOT-POUNDS, INSPECT RETAINING NUT AND FLANGE FOR BINDING AND CORROSION. REPLACE RETAINING NUT IF:**
  - (a) **PITS ARE 0.030-INCH DEEP.**
  - (b) **PITTED MORE THAN 10% OF MATING SURFACE.**

- (c) **THREADS SHOW SIGNS OF CORROSION.**
- (d) **RETAINING NUT CAN NOT BE FULLY RUN ON BY HAND.**
- (4) **IF RETAINING NUT IS NOT BOUND OR CORRODED, USING TORQUE REACTOR AND TAIL DRIVE FLANGE WRENCH, TORQUE RETAINING NUT TO 214 - 236 FOOT-POUNDS. INSTALL COTTER PIN. IF REQUIRED TO ALIGN COTTER PIN HOLE, TURN RETAINING NUT IN TIGHTENING DIRECTION.**

**6-4-40.1.3 Install.****WARNING**

Injury to personnel and damage to equipment will result if tail rotor drive shaft section III is bolted incorrectly to adjacent flange. Three adjacent flange bolts will be connected to tail rotor drive shaft flexible coupling only. Three drive shaft bolts will be connected to tail rotor drive shaft flexible coupling only. Make sure section III drive shaft flange is not bolted directly through tail rotor drive shaft flexible coupling to adjacent flange.

**CAUTION**

- Damage may occur if tools contact electrical harnesses during installation of tail rotor drive shaft section III. Do not allow tools to contact electrical harnesses.
- Damage to tail rotor drive shaft section III will result if handled improperly. Handle tail rotor drive shaft section III carefully to prevent from being dented, scratched, or bent. To prevent possible damage, place tail rotor drive shaft section III on padded area.



NOTE

- When installing bolts in tail rotor drive shaft flexible couplings, make sure bolt-heads are against tail rotor drive shaft flexible coupling.
  - Do not lockwire bolts with drilled heads that are installed on tail rotor drive shaft flexible couplings.
- Remove tail rotor drive shaft flexible couplings from old tail rotor drive shaft section III and install on new tail rotor drive shaft section III (PARA 6-4-42).
  - Support tail rotor drive shaft section III in position (Figure 6-4-40-1).

NOTE

- Spherical washers are to be installed on each side of tail rotor drive shaft flexible coupling.
  - Spherical side of washer must face tail rotor drive shaft flexible coupling.
- Support tail rotor drive shaft section III (front end) in installed position with three bolts and spherical washers. Install nuts. **TIGHTEN NUTS ONLY ENOUGH TO PREVENT AXIAL MOVEMENT** (Figure 6-4-40-1, Detail B).
  - Determine shim thickness at opposite end of tail rotor drive shaft section III as follows:

NOTE

- Do not turn tail rotor drive shaft section III from original position, as gap may change. If tail rotor drive shaft section III is moved, recheck gap.

- Spherical washers are to be installed on each side of tail rotor drive shaft flexible coupling.
  - Spherical side of washer must face tail rotor drive shaft flexible coupling.
- Support tail rotor drive shaft section III (rear end) in installed position with three bolts and spherical washers.
  - Using feeler gage, measure gap at three attachment points between spherical washers and flange (Figure 6-4-40-1, Detail D).
  - Determine average gap by adding readings and dividing by 3.
  - Divide average gap by 2 to determine best shim combination measurement for both ends of tail rotor drive shaft section III.

Example

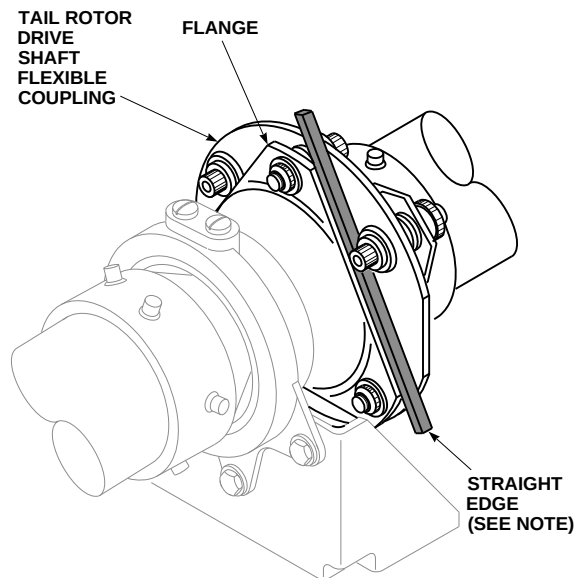
Measure Gap:

|       |                  |
|-------|------------------|
| Gap   | A = 0.125        |
|       | B = 0.142        |
|       | <u>C = 0.137</u> |
| Total | = 0.404          |

Divide by 3:  $0.404 \div 3 = 0.1346$ -inch average gap.

Round off: 0.135-inch average gap.

Divide by 2:  $0.135 \div 2 = 0.068$ -inch shim combination.

**NOTE**

INSERT FEELER GAGE BETWEEN TAIL ROTOR DRIVE SHAFT FLEXIBLE COUPLING AND STRAIGHT EDGE. MAXIMUM DEFLECTION NOT TO EXCEED 0.015 IN.

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**FIGURE 6-4-40-2. INSPECT TAIL ROTOR DRIVE SHAFT SHIMMING**

**NOTE**

Due to weight of tail rotor drive shaft section III, shim calculation may result in excess deflection of tail rotor drive shaft flexible coupling. In an effort to center tail rotor drive shaft section III and reduce maximum amount of gap, shims should be installed at each end as equally as possible. In situations where reducing the maximum amount of gap requires unequal amount of shims from one end to the other, it is permissible for shims on opposite ends to be unequal up to 0.020-inch.

- e. Select shim combination resulting from computation above. Make three stackups for each end of tail rotor drive shaft section III. **MAXIMUM GAP IS 0.125-INCH AT EACH END OF TAIL ROTOR DRIVE SHAFT SECTION III.**

**NOTE**

- Shim thickness must be the same for all three bolt locations at each end of tail

rotor drive shaft section III. Shim combination at one end does not have to be the same as shim combination at other end, as long as total shim combination adds up closest to average gap. Store unused shims under nut to maintain balance of tail rotor drive shaft section III.

- Nuts must be torqued within 15 minutes after applying epoxy primer to bolts.
- f. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to bolt grip lengths and under boltheads. Do not apply primer to threads.

**NOTE**

- Spherical washers are to be installed on each side of tail rotor drive shaft flexible coupling.
- Spherical side of washer must face tail rotor drive shaft flexible coupling.
- One shim stackup is to be installed on each bolt.

- Shim stackup is to be located between flange and spherical washer.
  - Install unused shims under nut.
- g. Secure flange to tail rotor drive shaft flexible coupling with bolts, shims, spherical washers, and nuts (Figure 6-4-40-1, Detail B and Detail C).
- h. Using torque wrench, check drag torque on nuts. **MINIMUM DRAG TORQUE IS 9.5 INCH-POUNDS.** If torque is beyond limits, replace nuts.

**WARNING**

**FSCAP**

Verification of minimum torque and torque stabilization check of nuts is a critical characteristic.

- i. **TORQUE NUTS TO 328 - 362 INCH-POUNDS.**
- j. Perform tail rotor drive shaft shimming inspection as follows:

**NOTE**

- Shims used for tail rotor drive shaft section III are installed between the mating flange and spherical washer. Unused shims are installed under nut (Figure 6-4-40-1, Detail B and Detail C).
- Flanges at both ends of tail rotor drive shaft section III use equal amount of shims.
  - (1) Inspect for proper shimming of tail rotor drive shaft flexible couplings. Make sure same amount of shims are used between spherical washer and adjacent flange. There should be the same amount of unused shims between spherical washer and nut.
  - (2) Make sure all spherical washers have spherical side facing tail rotor drive shaft flexible coupling.

- (3) Using feeler gage and straight edge placed along flange surface, inspect for deflection in tail rotor drive shaft flexible coupling. **NO MORE THAN 0.015-INCH DEFLECTION IN TAIL ROTOR DRIVE SHAFT FLEXIBLE COUPLING IS ALLOWED** (Figure 6-4-40-2).

- k. If requirements of tail rotor drive shaft shimming inspection are not met using shim measurements and calculations in step d., adjust shims as follows:

Adjusted Shim Thickness = Calculated Shim Thickness (step d. (4)) - Deflection of Tail Rotor Drive Shaft Flexible Coupling at Lower End (step j. (3)).

**Example**

If shim thickness at each end is calculated to be 0.110-inch and deflection at lower end measured with straight edge 0.020-inch (0.005-inch greater than 0.015-inch allowable), then actual shim thickness = 0.110-inch - 0.020-inch = 0.090-inch.

- l. Check helicopter logbook to make sure tail rotor drive shaft torque check is shown as due 9 to 11 flight hours after installation (PARA 1-7-8).

**CAUTION**

Damage to bearing will result if contact occurs with corrosion preventive compounds. Corrosion preventive compounds can penetrate seals of bearing and dissolve lubrication grease. Do not spray corrosion preventive compounds in vicinity of bearing. Shield bearing from spray.

- m. Using corrosion preventive compound, Item 113, Appendix D, spray all spherical washers.
- n. Make sure area is clean and free of foreign material.

- o. Close rear tail rotor drive shaft cover.

6-4-41 TAIL ROTOR DRIVE SHAFT SECTION IV.

PARAGRAPH BREAKDOWN

|          |                                           | PAGE     |
|----------|-------------------------------------------|----------|
| 6-4-41.1 | Replace Tail Rotor Drive Shaft Section IV | 6-4-41-2 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Feeler Gage
- Powertrain Repairer’s Toolkit
- Straight Edge
- Tail Drive Flange Wrench, 70700-20687-102
- Torque Reactor, 70700-20688-041
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 100 - 500 ft lbs

Materials/Parts

- Abrasive Cloth, Item 2, Appendix D

- Corrosion Preventive Compound, Item 113, Appendix D
- Corrosion Preventive Compound, Item 114, Appendix D
- Corrosion Removing Compound, Item 116, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D

References

- Appendix D
- PARA 1-7-8
- PARA 2-4-137
- PARA 2-4-139
- PARA 2-5-19
- PARA 6-4-42

### 6-4-41.1 Replace Tail Rotor Drive Shaft Section IV.

#### 6-4-41.1.1 Remove.

#### WARNING

Injury to personnel and damage to equipment will result if tail rotor is not secured when tail rotor drive shaft section IV is disconnected. Make sure tail rotor is secured to prevent tail rotor drive shafts and tail rotor from turning.

- a. Secure tail rotor to prevent tail rotor drive shafts and tail rotor from turning.
- b. Open tail rotor pylon leading edge cover/antenna.
- c. Remove intermediate gear box fairing (PARA 2-4-139).
- d. Remove tail gear box fairing (PARA 2-4-137).
- e. Remove bolts, spherical washers, shims, and nuts connecting tail rotor drive shaft flexible coupling on tail rotor drive shaft section IV to intermediate gear box output flange (Figure 6-4-41-1, Detail A).

#### CAUTION

- Damage may occur if tools contact electrical harnesses during removal of tail rotor drive shaft section IV. Do not allow tools to contact electrical harnesses.
- Damage to tail rotor drive shaft section IV will result if handled improperly. Handle tail rotor drive shaft section IV carefully to prevent from being dented, scratched, or bent. To prevent possible damage, place tail rotor drive shaft section IV on padded area.

#### NOTE

- Tail rotor drive shaft flexible coupling bolts and self-locking nuts can be reused, provided they show no visual evidence of damage.
- Note location of hardware for proper installation.
- f. Remove bolts, spherical washers, shims, and nuts connecting tail rotor drive shaft flexible coupling on tail rotor drive shaft section IV to tail gear box input flange (Figure 6-4-41-1, Detail B).
- g. Remove tail rotor drive shaft section IV with tail rotor drive shaft flexible coupling. Place tail rotor drive shaft section IV on padded area.

#### 6-4-41.1.2 Inspect.

- a. Inspect airframe supports for visible cracks and distortion. **NONE ALLOWED.** If cracks and distortion are found, replace airframe support (PARA 2-5-19).
- b. Inspect tail rotor drive shaft flexible coupling (PARA 6-4-42).

#### NOTE

Corrosion preventive compound may turn reddish-brown over time. This is normal and should not be mistaken for corrosion.

- c. Visually inspect spherical washers of tail rotor drive shaft flexible coupling for evidence of corrosion, damage, or surface roughness. **NONE ALLOWED.** If corrosion, damage, or surface roughness is found, replace spherical washers.
- d. Inspect flange boltholes of tail rotor drive shaft section IV for elongation. **NONE ALLOWED.** If boltholes are elongated, replace tail rotor drive shaft section IV.
- e. If spherical washer, 70361-05010-103, is unavailable, washer, 70361-05010-102, may be reworked as follows:

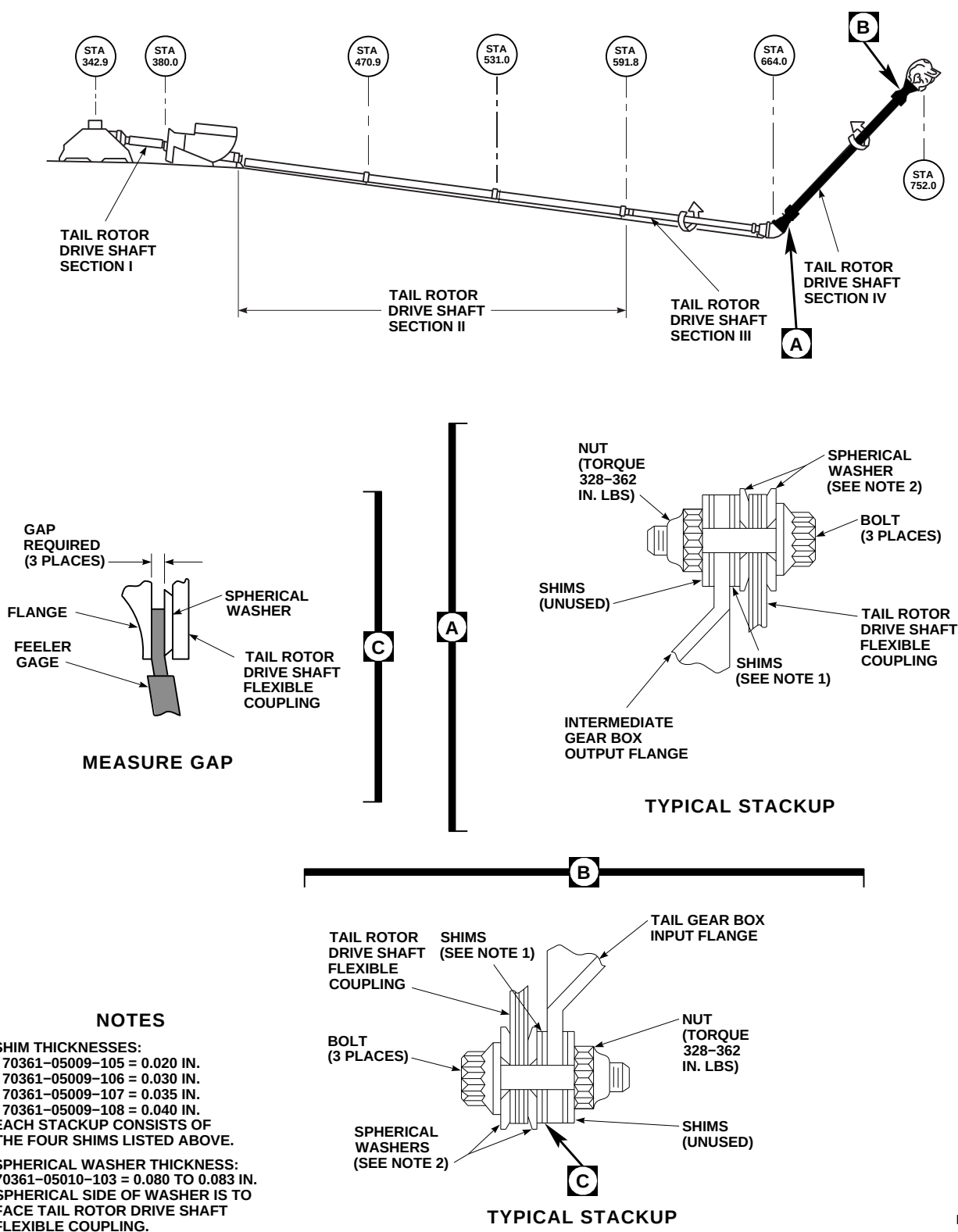


FIGURE 6-4-41-1. REPLACE TAIL ROTOR DRIVE SHAFT SECTION IV

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- (1) Using corrosion removing compound, Item 116, Appendix D, clean washer.
  - (2) If corrosion is still found, using abrasive cloth, Item 2, Appendix D, remove corrosion. Make sure all corrosion is removed.
  - (3) **MINIMUM THICKNESS AFTER BLENDING IS 0.080-INCH THICK.** If minimum thickness is under limit, replace washers.
  - (4) Using corrosion preventive compound, Item 114, Appendix D, apply to washer surfaces.
  - (5) Using corrosion preventive compound, Item 113, Appendix D, apply to hardware assembly. Allow to dry.
- f. With retaining nuts of intermediate gear box output flange and tail gear box input flange exposed, perform torque check as follows:

**NOTE**

Torque check for both retaining nuts is the same.

- (1) Remove cotter pin from retaining nut.

**NOTE**

Check torque in tightening direction only.

- (2) Using torque reactor and tail drive flange wrench, check torque of retaining nut. **IF TORQUE IS 214 - 236 FOOT-POUNDS, INSTALL NEW COTTER PIN. IF REQUIRED TO ALIGN COTTER PIN HOLE, TURN RETAINING NUT IN TIGHTENING DIRECTION.** Torque check is complete.
- (3) **IF TORQUE IS LESS THAN 214 FOOT-POUNDS, INSPECT RETAINING NUT AND FLANGE FOR BINDING AND CORROSION. REPLACE RETAINING NUT IF:**
  - (a) **PITS ARE 0.030-INCH DEEP.**

- (b) **PITTED MORE THAN 10% OF MATING SURFACE.**
- (c) **THREADS SHOW SIGNS OF CORROSION.**
- (d) **RETAINING NUT CAN NOT BE FULLY RUN ON BY HAND.**
- (4) **IF RETAINING NUT IS NOT BOUND OR CORRODED, USING TORQUE REACTOR AND TAIL DRIVE FLANGE WRENCH, TORQUE RETAINING NUT TO 214 - 236 FOOT-POUNDS. INSTALL COTTER PIN. IF REQUIRED TO ALIGN COTTER PIN HOLE, TURN RETAINING NUT IN TIGHTENING DIRECTION.**

**6-4-41.1.3 Install.****WARNING**

Injury to personnel and damage to equipment will result if tail rotor drive shaft section IV is bolted incorrectly to adjacent flange. Three adjacent flange bolts will be connected to tail rotor drive shaft flexible coupling only. Three drive shaft bolts will be connected to tail rotor drive shaft flexible coupling only. Make sure section IV drive shaft flange is not bolted directly through tail rotor drive shaft flexible coupling to adjacent flange.

**CAUTION**

- Damage may occur if tools are allowed to contact electrical harnesses during installation of tail rotor drive shaft section IV. Do not allow tools to contact electrical harnesses.
- Damage to tail rotor drive shaft section IV will result if handled improperly. Handle tail rotor drive shaft section IV



carefully to prevent from being dented, scratched, or bent. To prevent possible damage, place tail rotor drive shaft section IV on padded area.

NOTE

- When installing bolts in tail rotor drive shaft flexible couplings, make sure bolt-heads are against tail rotor drive shaft flexible coupling.
  - Do not lockwire bolts with drilled heads that are installed on tail rotor drive shaft flexible couplings.
- a. Remove tail rotor drive shaft flexible couplings from old tail rotor drive shaft section IV and install on new tail rotor drive shaft section IV (PARA 6-4-42).
  - b. Support tail rotor drive shaft section IV in position (Figure 6-4-41-1).

NOTE

- Spherical washers are to be installed on each side of tail rotor drive shaft flexible coupling.
  - Spherical side of washer must face tail rotor drive shaft flexible coupling.
- c. Support tail rotor drive shaft section IV in installed position (at intermediate gear box end) with three bolts and spherical washers. Install nuts. **TIGHTEN NUTS ONLY ENOUGH TO PREVENT AXIAL MOVEMENT** (Figure 6-4-41-1, Detail A).
  - d. Have assistant support tail rotor drive shaft section IV to keep tail rotor drive shaft flexible coupling from bending under weight of tail rotor drive shaft section IV.
  - e. Determine shim thickness at opposite end of tail rotor drive shaft section IV as follows:

NOTE

- Do not turn tail rotor drive shaft section IV from original position, as gap may

change. If tail rotor drive shaft section IV is moved, recheck gap.

- Spherical washers are to be installed on each side of tail rotor drive shaft flexible coupling.
  - Spherical side of washer must face tail rotor drive shaft flexible coupling.
- (1) Support tail rotor drive shaft section IV in installed position with three bolts and spherical washers.
  - (2) Using feeler gage, measure gap at three attachment points between spherical washers and flange (Figure 6-4-41-1, Detail C).
  - (3) Determine average gap by adding readings and dividing by 3.
  - (4) Divide average gap by 2 to determine best shim combination measurement for both ends of tail rotor drive shaft section IV.

Example

Measure Gap:

|       |                  |
|-------|------------------|
| Gap   | A = 0.125        |
|       | B = 0.142        |
|       | <u>C = 0.137</u> |
| Total | = 0.404          |

**Divide by 3:**  $0.404 \div 3 = 0.1346$ -inch average gap.

**Round off:** 0.135-inch average gap.

**Divide by 2:**  $0.135 \div 2 = 0.068$ -inch shim combination.

**NOTE**

Due to weight of tail rotor drive shaft section IV, shim calculation may result in excess deflection of tail rotor drive flexible coupling. In an effort to center tail rotor drive shaft section IV and reduce maximum amount of gap, shims should be installed at each end as equally as possible. In situations where reducing the maximum amount of gap requires unequal amount of shims from one end to the other, it is permissible for shims on opposite ends to be unequal up to 0.020-inch.

- f. Select shim combination resulting from computation above. Make three stackups for each end of tail rotor drive shaft section IV. **MAXIMUM GAP IS 0.125-INCH AT EACH END OF TAIL ROTOR DRIVE SHAFT SECTION IV.**

**NOTE**

- Shim thickness must be the same for all three bolt locations at each end of tail rotor drive shaft section IV. Shim combination at one end does not have to be the same as shim combination at other end, as long as total shim combination adds up closest to average gap. Store unused shims under nut to maintain balance of tail rotor drive shaft section IV.
  - Nuts must be torqued within 15 minutes after applying epoxy primer to bolts.
- g. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to bolt grip lengths and under boltheads. Do not apply primer to threads.

**NOTE**

- Spherical washers are to be installed on each side of tail rotor drive shaft flexible coupling.
- Spherical side of washer must face tail rotor drive shaft flexible coupling.

- One shim stackup is to be installed on each bolt.
  - Shim stackup is to be located between flange and spherical washer.
  - Install unused shims under nut.
- h. Secure flange to tail rotor drive shaft flexible coupling with bolts, shims, spherical washers, and nut (Figure 6-4-41-1, Detail A and Detail B).
- i. Using torque wrench, check drag torque on nuts. **MINIMUM DRAG TORQUE IS 9.5 INCH-POUNDS.** If torque is beyond limits, replace nuts.

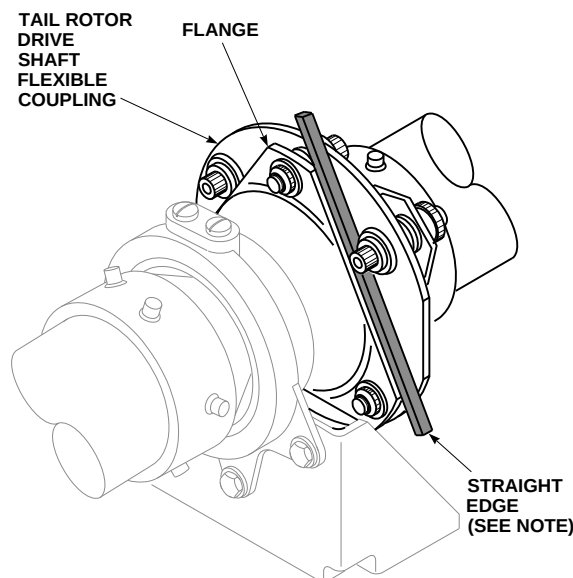
**WARNING****FSCAP**

Verification of minimum torque and torque stabilization check of nuts is a critical characteristic.

- j. **TORQUE NUTS TO 328 - 362 INCH-POUNDS.**
- k. Perform tail rotor drive shaft shimming inspection as follows:

**NOTE**

- Shims used for tail rotor drive shaft section IV are installed between mating flange and spherical washer. Unused shims are installed under nut (Figure 6-4-41-1, Detail A and Detail B).
  - Flanges at both ends of tail rotor drive shaft section IV use equal amount of shims.
- (1) Inspect for proper shimming of tail rotor drive shaft flexible couplings. Make sure same amount of shims are used between spherical washer and adjacent flange. There should be the same amount of unused shims between spherical washer and nut.



**NOTE**  
INSERT FEELER GAGE BETWEEN TAIL ROTOR DRIVE SHAFT FLEXIBLE COUPLING AND STRAIGHT EDGE. MAXIMUM DEFLECTION NOT TO EXCEED 0.015 IN.

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**FIGURE 6-4-41-2. INSPECT TAIL ROTOR DRIVE SHAFT SHIMMING**

- (2) Make sure all spherical washers have spherical side facing tail rotor drive shaft flexible coupling.
  - (3) Using feeler gage and straight edge placed along flange surface, inspect for deflection in tail rotor drive shaft flexible coupling. **NO MORE THAN 0.015-INCH DEFLECTION IN TAIL ROTOR DRIVE SHAFT FLEXIBLE COUPLING IS ALLOWED** (Figure 6-4-41-2).
- l. If the requirements of tail rotor drive shaft shimming inspection are not met using shim measurements and calculations in step e., adjust shims as follows:
 

Adjusted Shim Thickness = Calculated Shim Thickness (step e. (4)) - Deflection of Tail Rotor Drive Shaft Flexible Coupling at Lower End (step k. (3)).
  - m. Check helicopter logbook to make sure tail rotor drive shaft torque check is shown as due 9 to 11 flight hours after installation (PARA 1-7-8).
  - n. Using corrosion preventive compound, Item 113, Appendix D, spray all spherical washers.
  - o. Make sure area is clean and free of foreign material.
  - p. Install tail gear box fairing (PARA 2-4-137).
  - q. Install intermediate gear box fairing (PARA 2-4-139).
  - r. Close tail rotor pylon leading edge cover/antenna.

**Example**



6-4-42 TAIL ROTOR DRIVE SHAFT FLEXIBLE COUPLING.

PARAGRAPH BREAKDOWN

|                                                           | PAGE     |
|-----------------------------------------------------------|----------|
| 6-4-42.1 Replace Tail Rotor Drive Shaft Flexible Coupling | 6-4-42-2 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Borescope Kit
- Dial Indicating Depth Gage
- Feeler Gage
- Magnifying Glass, 10X
- Micrometer
- Powertrain Repairer’s Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Abrasive Cloth, Item 2, Appendix D
- Abrasive Cloth, Item 10, Appendix D

- Epoxy Primer Coating Kit, Item 135, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- PARA 1-7-8
- PARA 6-3-15
- PARA 6-4-38
- PARA 6-4-39
- PARA 6-4-40
- PARA 6-4-41
- TM 1-70-VIB

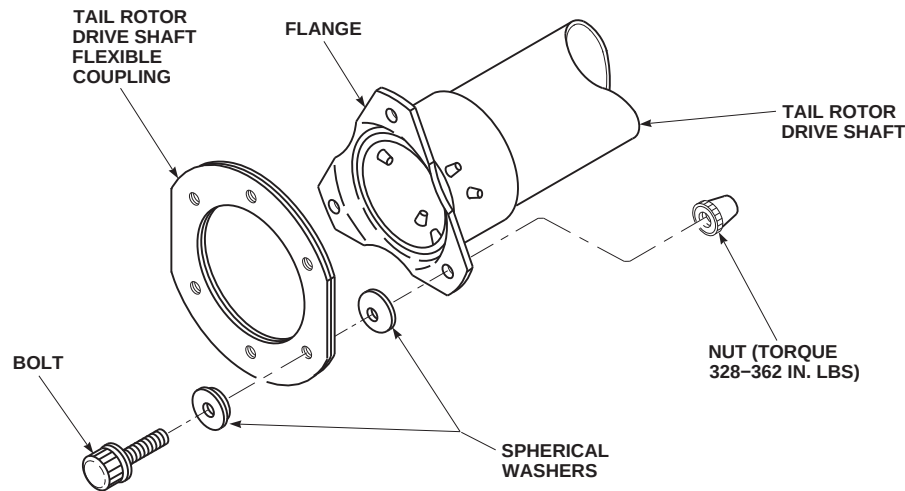
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FIGURE 6-4-42-1. REPLACE TAIL ROTOR DRIVE SHAFT FLEXIBLE COUPLING

**6-4-42.1 Replace Tail Rotor Drive Shaft Flexible Coupling.****NOTE**

Replacement of all tail rotor drive shaft flexible couplings is the same, except as noted.

**6-4-42.1.1 Remove.**

- a. Wipe away any grease or dirt from flexible coupling.

**NOTE**

If flexible coupling is being replaced, proceed to step c.

- b. If flexible coupling is being reused, perform the following:

- (1) Using feeler gage, inspect flexible coupling for buckling. **MAXIMUM GAP**

**OF 0.030-INCH IS PERMITTED BETWEEN LAMINATES.**

- (2) If gap between any two laminates is greater than 0.030-inch, perform the following:

- (a) Break torque on six nuts.
- (b) Retorque nuts in 50 inch-pound increments, in sequence, until each nut is **TORQUED TO 328 - 362 INCH-POUNDS.**
- (c) **IF GAP STILL EXCEEDS 0.030-INCH, REPLACE FLEXIBLE COUPLING.**

- c. Support end of tail rotor drive shaft.
- d. Disconnect tail rotor drive shaft (PARA 6-4-38 (for section I), PARA 6-4-39 (for section II), PARA 6-4-40 (for section III), or PARA, 6-4-41 (for section IV)).

**6-4-42-2 Change 4**

**NOTE**

- There is no longer a requirement that the order of discs alternate between those marked "A" and those marked "B". Flexible coupling may contain any order of "A" and "B" discs. Maintain original orientation and order of discs when disassembling and inspecting flexible coupling.
- If any one disc making up flexible coupling stackup is damaged, entire flexible coupling must be replaced. Flexible coupling discs are a matched set. Discs cannot be replaced individually. If any one disc is damaged beyond limits, entire flexible coupling must be replaced. Two or more discs between bolts on flexible coupling flanges may separate or buckle. This is acceptable, unless there is a sharp bend or crease in disc, or if lamination spread (gap between two individual discs) is over 0.030-inch. To determine spread, measure gap between disc laminations with feeler gage.
- e. Remove bolts, spherical washers, and nuts securing flexible coupling to tail rotor drive shaft (Figure 6-4-42-1).
- f. Remove flexible coupling.

**6-4-42.1.2 Inspect/Repair.****NOTE**

- No more than half number of discs may be blended in same assembled area.
- Fretting of discs occurs in area under spherical washer.
- Red oxide stains on discs are acceptable.
- When inspecting tail rotor drive shaft flexible couplings, pay particular attention to edges around the circumference of spherical washers.
- If flexible coupling is damaged, tail rotor drive shaft may also be damaged.

Before installing new flexible coupling, visually inspect tail rotor drive shaft.

- a. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean flexible coupling.
- b. Using micrometer, check disc stackup. **DISC STACKUP MUST MEASURE 0.140 - 0.151-INCH THICK AND INCLUDE APPROXIMATELY 10 - 14 DISCS.** If stackup measurement is beyond limits, replace flexible coupling.
- c. Using borescope kit, or 10X magnifying glass with adequate lighting, inspect flexible couplings for cracks. **NONE ALLOWED.** If cracks are found, replace flexible coupling.
- d. Using dial indicating depth gage, inspect flexible coupling for corrosion pitting. **CORROSION PITTING SHALL NOT EXCEED 0.0005-INCH DEEP.** If corrosion pitting is less than 0.0005-inch deep, using abrasive cloth, Item 10, Appendix D, blend out corrosion pitting. If corrosion pitting is beyond limits, replace flexible coupling.
- e. Inspect for dents and nicks on edges of individual discs. **NICKS AND DENTS SHALL NOT EXCEED 0.020-INCH DEEP.** If nicks and dents are less than 0.020-inch deep, using abrasive cloth, Item 2, Appendix D, blend out nicks and dents. Using abrasive cloth, Item 10, Appendix D, smooth area. If dents and nicks are beyond limits, replace flexible coupling.
- f. Using dial indicating depth gage, inspect for fretting. **FRETTING SHALL NOT EXCEED 0.0005-INCH DEEP.** If fretting is less than 0.0005-inch deep, using abrasive cloth, Item 10, Appendix D, remove fretting. If fretting is beyond limits, replace flexible coupling.

**6-4-42.1.3 Install.****NOTE**

Nuts must be torqued within 15 minutes after applying epoxy primer to bolts.

- a. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to bolt grip

lengths and under boltheads. Do not apply primer to threads.

### NOTE

- Spherical washers are to be installed on each side of tail rotor drive shaft flexible coupling.
- Spherical side of washer must face flexible coupling.
- b. Secure flexible coupling to tail rotor drive shaft with bolts, spherical washers, and nuts (Figure 6-4-42-1).
- c. Using torque wrench, check drag torque on nuts. **MINIMUM DRAG TORQUE IS 9.5 INCH-POUNDS.** If torque is beyond limits, replace nut(s).

### WARNING

#### FSCAP

Verification of minimum torque and torque stabilization check of nuts is a critical characteristic.

- d. **TORQUE NUTS TO 328 - 362 INCH-POUNDS.**
- e. Connect tail rotor drive shaft (PARA 6-4-38 (for section I), PARA 6-4-39 (for section II), PARA 6-4-40 (for section III), or PARA, 6-4-41 (for section IV)).

### WARNING

Injury to personnel and damage to equipment will result if oil cooler spline wear inspection is not performed on oil cooler fans, 70361-03005-103 through 70361-03005-106, every time tail rotor drive shaft is disconnected or replaced from main transmission to STA 531.0.

Excessive spline wear will result in loss of tail rotor drive. Make sure spline wear indicator is installed and oil cooler spline wear inspection is performed.

### NOTE

Spline wear inspection is not required when tail rotor drive shaft is disconnected or replaced aft of STA 531.0.

- f. On helicopters with oil cooler fans, 70361-03005-103 through 70361-03005-106, installed, perform oil cooler spline wear inspection, if required (PARA 6-3-15).

### WARNING

Injury to personnel and damage to equipment will result if vibration check is not performed any time a flexible coupling is disconnected or replaced from main transmission up to and including STA 470.9. Excessive vibration indicates improper installation of oil cooler and/or flexible coupling, which will cause oil cooler fan drive shaft splines and oil cooler fan exit flange splines to wear, resulting in loss of tail rotor drive. Make sure vibration check is performed.

### NOTE

Vibration check is not required when tail rotor drive shaft is disconnected or replaced at STA 531.0 and aft.

- g. If flexible coupling was disconnected or replaced from main transmission up to and including STA 470.9, perform oil cooler vibration check (TM 1-70-VIB).
- h. Check helicopter logbook to make sure tail rotor drive shaft torque check is shown as due 9 to 11 flight hours after installation (PARA 1-7-8).



6-4-43 TAIL ROTOR DRIVE SHAFT VISCOUS DAMPER BEARING ASSEMBLY.

PARAGRAPH BREAKDOWN

|                                                                         | PAGE     |
|-------------------------------------------------------------------------|----------|
| 6-4-43.1 Replace Tail Rotor Drive Shaft Viscous Damper Bearing Assembly | 6-4-43-2 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Scraper, Locally-Made
- Permanent Ink Marker
- Powertrain Repairer’s Toolkit
- Puller Kit, Mechanical
- Tail Drive Flange Wrench, 70700-20687-102
- Torque Reactor, 70700-20688-041
- Torque Wrench, 100 - 500 ft lbs

Materials/Parts

- Abrasive Cloth, Item 1A, Appendix D
- Abrasive Cloth, Item 8, Appendix D

- Abrasive Cloth, Item 10, Appendix D
- Abrasive Mat, Item 14, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-6-5
- PARA 6-3-15
- PARA 6-4-39
- PARA 6-4-46
- PARA 6-4-50
- TM 1-70-VIB

### 6-4-43.1 Replace Tail Rotor Drive Shaft Viscous Damper Bearing Assembly.

#### NOTE

- Replacement of all tail rotor drive shaft viscous damper bearing assemblies is the same, except as noted.
- If replacing viscous damper bearing assembly on oil cooler drive shaft, refer to PARA 6-4-50.

#### 6-4-43.1.1 Remove.

- Remove applicable section of tail rotor drive shaft section II (PARA 6-4-39).
- Remove cotter pin from retaining nut (Figure 6-4-43-1, Sheet 1, Detail A).
- Using torque reactor and tail drive flange wrench, remove retaining nut.

#### NOTE

Make sure index marks are present on tail rotor drive shaft and flange. If no index marks are present, index marks must be applied.

- If necessary, using permanent ink marker, apply index marks to tail rotor drive shaft and flange.
- Remove flange from tail rotor drive shaft by pulling straight off.
- Using mechanical puller kit, remove viscous damper bearing assembly from tail rotor drive shaft.

#### 6-4-43.1.2 Disassemble.

#### NOTE

Variations in tolerance may allow movement of viscous damper tube assembly within outer housing. Axial and radial movement inside outer housing is not cause for rejection/replacement,

provided the viscous damper tube assembly does not become cocked inside outer housing and does not move past snapping.

- Using nonmetallic scraper, remove sealing compound from around edges of filler neck and from around mating surfaces of inner housing, outer housing, and snapping.
- Remove snapping from inner housing (Figure 6-4-43-1, Sheet 2, Detail B).
- Manually press out inner housing.
- Remove viscous damper tube assembly.

#### 6-4-43.1.3 Inspect/Repair.

- Inspect tail rotor drive shaft for damage (PARA 6-4-46).
- Inspect retaining nut for corrosion as follows:
  - Using abrasive cloth, Item 10, Appendix D, abrasive cloth, Item 1A, Appendix D, or abrasive mat, Item 14, Appendix D, minor corrosion may be blended out.
  - REPLACE RETAINING NUT IF:**
    - PITS ARE 0.030-INCH DEEP.**
    - PITTED MORE THAN 10% OF MATING SURFACE.**
    - THREADS SHOW SIGNS OF CORROSION.**

#### NOTE

Do not paint threads or mating surface of retaining nut.

- Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to flats and exposed face of retaining nut.
- Inspect outer housing for damage or corrosion as follows:

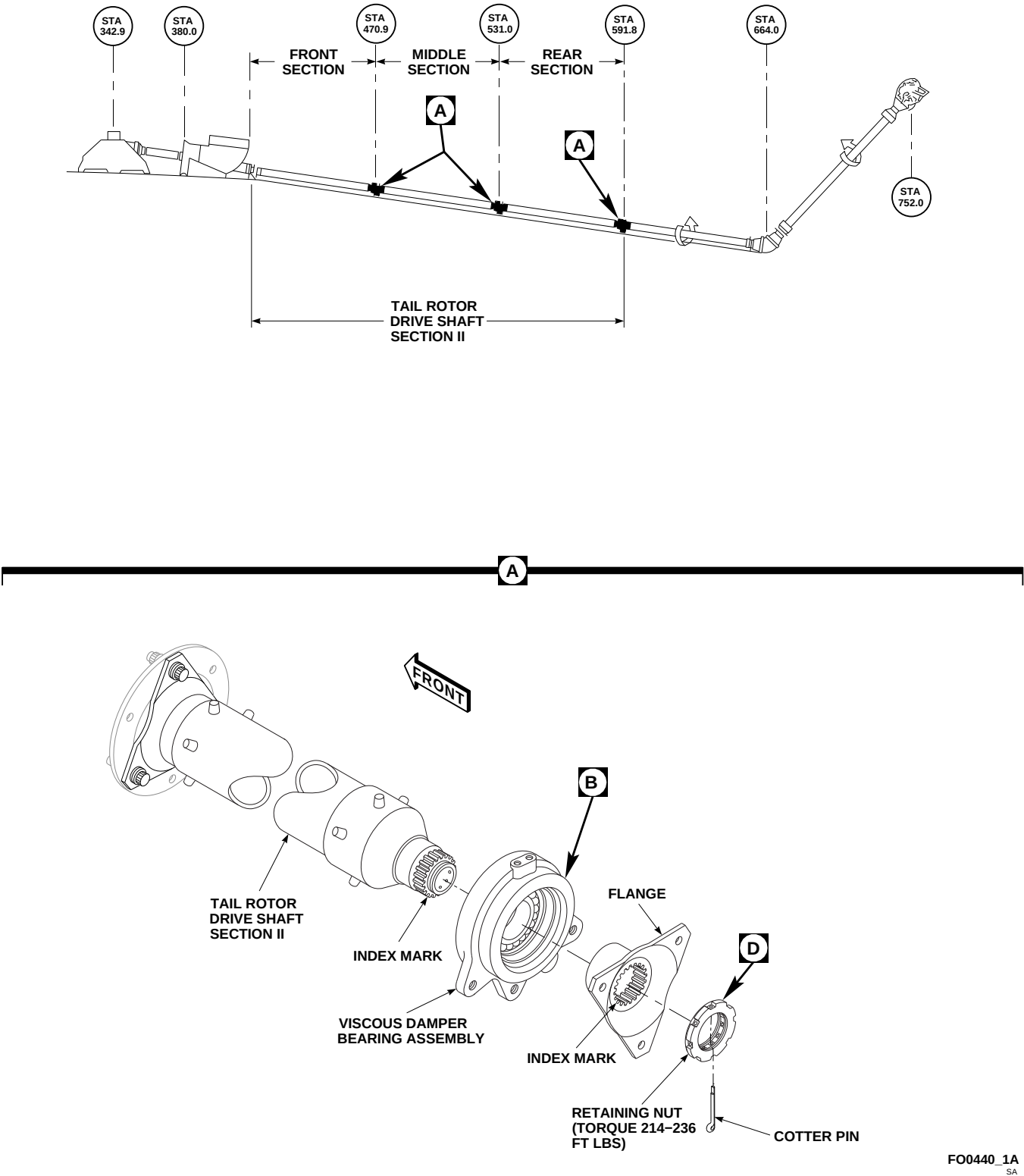


FIGURE 6-4-43-1. REPLACE TAIL ROTOR DRIVE SHAFT VISCOUS DAMPER BEARING ASSEMBLY (SHEET 1 OF 2)

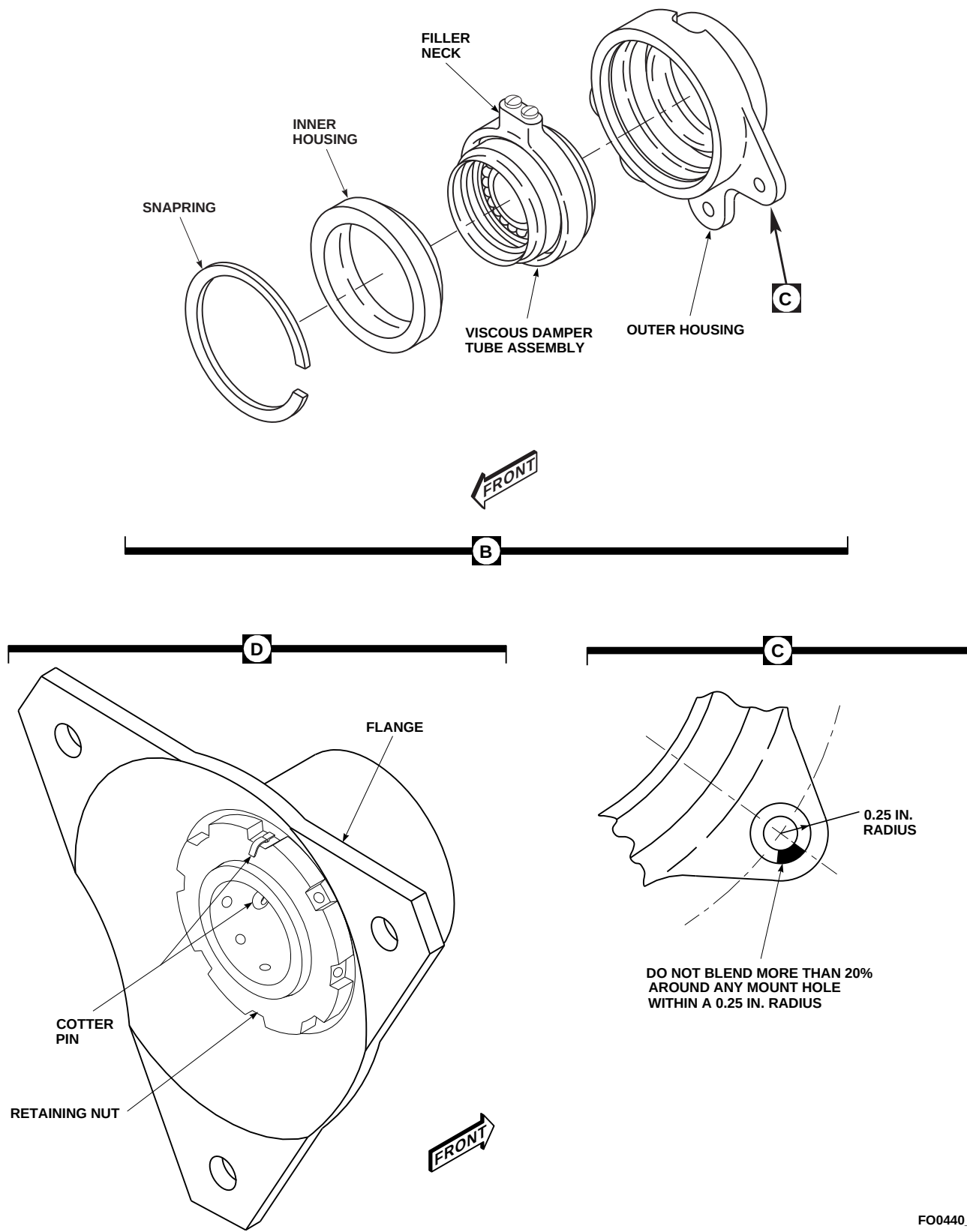


FIGURE 6-4-43-1. REPLACE TAIL ROTOR DRIVE SHAFT VISCOUS DAMPER BEARING ASSEMBLY (SHEET 2 OF 2)

**NOTE**

All blended areas are to be smooth to adjacent surfaces. **NO SHARP EDGES ALLOWED.**

- (1) Using abrasive cloth, Item 8, Appendix D, damage or corrosion on mounting lugs may be blended out up to 0.030-inch deep over not more than 20% of any one lug, and not more than 20% around any hole (Figure 6-4-43-1, Sheet 2, Detail C).
  - (2) Using abrasive cloth, Item 8, Appendix D, damage or corrosion on outer housing may be blended out up to 0.030-inch deep over not more than 20% of total surface area.
  - (3) If damage or corrosion is beyond limits, replace outer housing.
- d. Inspect inner housing for damage or corrosion as follows:

**NOTE**

All blended areas are to be smooth to adjacent surfaces. **NO SHARP EDGES ALLOWED.**

- (1) Using abrasive cloth, Item 8, Appendix D, damage or corrosion on inner housing may be blended out up to 0.030-inch deep over not more than 20% of total surface area.
- (2) If damage or corrosion is beyond limits, replace inner housing.

**6-4-43.1.4 Assemble.**

- a. Install viscous damper tube assembly in outer housing. Make sure viscous damper tube assembly is seated (Figure 6-4-43-1, Sheet 2, Detail B).
- b. Install inner housing and secure with snapping.
- c. Apply  $\frac{1}{16}$  -  $\frac{1}{8}$ -inch bead of sealing compound, Item 292, Appendix D, around edges of filler neck and around mating surfaces of inner housing, outer housing, and snapping.

**6-4-43.1.5 Install.**

- a. Press viscous damper bearing assembly on tail rotor drive shaft section II. Make sure viscous damper bearing assembly is installed in direction shown (Figure 6-4-43-1, Sheet 1, Detail A).

**NOTE**

Make sure rear section of viscous damper bearing assembly is against tapered part of tail rotor drive shaft.

- b. Install flange on tail rotor drive shaft. Make sure flange and tail rotor drive shaft index marks line up.
- c. Install retaining nut as follows:
  - (1) Install retaining nut, making sure retaining nut can be fully run on by hand. Make sure bevel side of retaining nut is toward seal.
  - (2) Using torque reactor and tail drive flange wrench, **TIGHTEN RETAINING NUT.**
  - (3) **TORQUE RETAINING NUT TO 214 - 236 FOOT-POUNDS.**

**NOTE**

Cotter pin is installed with head against inside diameter of retaining nut.

- (4) Remove torque reactor and install new cotter pin (Figure 6-4-43-1, Sheet 1, Detail A and Sheet 2, Detail D). **IF REQUIRED TO ALIGN COTTER PIN HOLE, TURN RETAINING NUT IN A TIGHTENING DIRECTION.**
- (5) Fold over rear end of cotter pin against retaining nut, then fold front end (flange against retaining nut mating surface) flush with retaining nut (Figure 6-4-43-1, Sheet 2, Detail D).
- (6) Apply torque stripe (PARA 2-6-5).
- d. Install tail rotor drive shaft section II as required (PARA 6-4-39).

**WARNING**

Injury to personnel and damage to equipment will result if oil cooler spline wear inspection is not performed on oil cooler fans, 70361-03005-103 through 70361-03005-106, every time tail rotor drive shaft is disconnected or replaced from main transmission up to STA 531.0. Excessive spline wear will result in loss of tail rotor drive. Make sure spline wear indicator is installed and oil cooler spline wear inspection is performed.

**NOTE**

Spline wear inspection is not required when tail rotor drive shaft is disconnected or replaced aft of STA 531.0.

- I** e. On helicopters with oil cooler fans, 70361-03005-103 through 70361-03005-106, installed, perform oil cooler spline wear inspection, if required (PARA 6-3-15).

**WARNING**

Injury to personnel and damage to equipment will result if vibration check is not performed any time a tail rotor drive shaft flexible coupling is disconnected or replaced from main transmission up to and including STA 470.9. Excessive vibration indicates improper installation of oil cooler and/or tail rotor drive shaft flexible coupling, which will cause oil cooler fan drive shaft splines and oil cooler fan exit flange splines to wear, resulting in loss of tail rotor drive. Make sure vibration check is performed.

**NOTE**

Vibration check is not required when tail rotor drive shaft is disconnected or replaced at STA 531.0 and aft.

- f. If tail rotor drive shaft flexible coupling was disconnected or replaced from main transmission up to and including STA 470.9, perform oil cooler vibration check (TM 1-70-VIB).

6-4-44 TAIL ROTOR DRIVE SHAFT BEARING.

PARAGRAPH BREAKDOWN

|                                                 | PAGE     |
|-------------------------------------------------|----------|
| 6-4-44.1 Replace Tail Rotor Drive Shaft Bearing | 6-4-44-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Drift, Locally-Made
- Powertrain Repairer’s Toolkit

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D

- Corrosion Preventive Compound, Item 114, Appendix D

References

- Appendix D
- PARA 6-4-43

6-4-44.1 Replace Tail Rotor Drive Shaft Bearing.

NOTE

Replacement of all tail rotor drive shaft bearings is the same.

DEPTH OF SCRATCHES IS 0.015-INCH. Using abrasive cloth, Item 10, Appendix D, blend out minor scratches.

6-4-44.1.3. Install.



6-4-44.1.1. Remove.

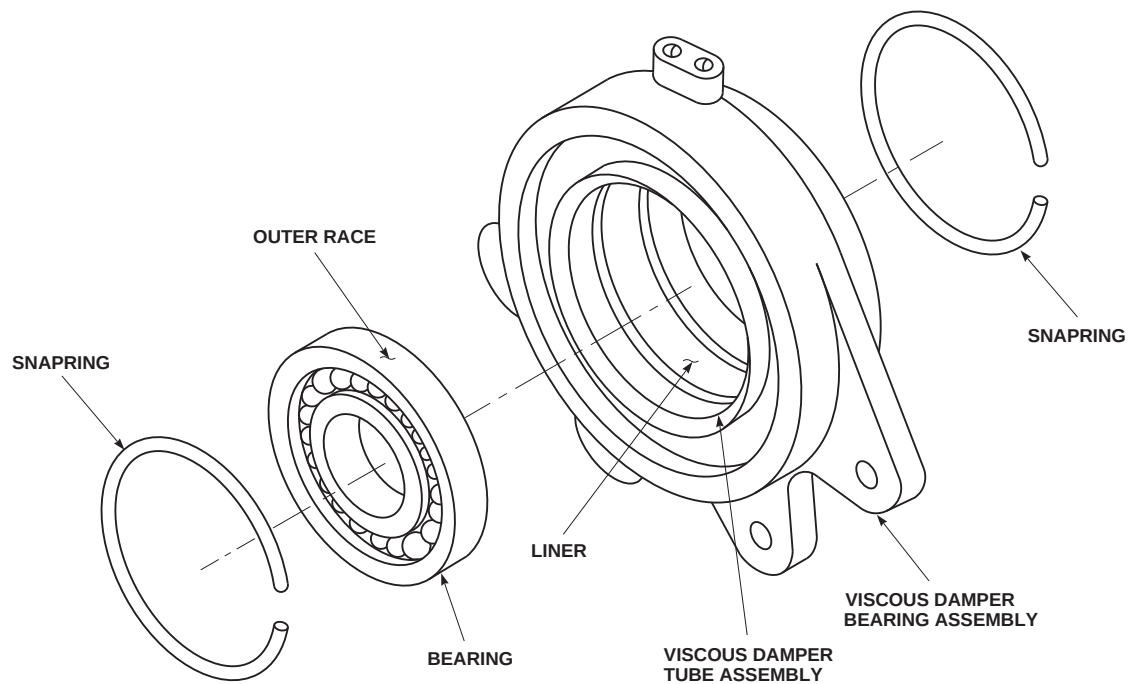
- Remove tail rotor drive shaft viscous damper bearing assembly (PARA 6-4-43).
- Remove snaprings (Figure 6-4-44-1).
- Manually press out bearing from viscous damper tube assembly. If necessary, using nonmetallic drift, tap out gently.

Damage to bearing will result if contact occurs with corrosion preventive compound. Corrosion preventive compound can dissolve lubrication grease in bearing. Do not allow corrosion preventive compound to contact ball bearings. Apply corrosion preventive compound to bearing outer race only.

6-4-44.1.2. Inspect/Repair.

Inspect liner on viscous damper tube assembly for scratches with raised edges and burrs. **MAXIMUM**

- Using corrosion preventive compound, Item 114, Appendix D, apply a light coat to outer race of bearing.

FO0441B  
SA**FIGURE 6-4-44-1. REPLACE TAIL ROTOR DRIVE SHAFT BEARING**

- b. Manually press bearing into viscous damper tube assembly. If necessary, using nonmetallic drift, tap in gently (Figure 6-4-44-1).

**NOTE**

Variations in tolerance may allow movement of viscous damper tube assembly within outer housing. Axial and radial movement inside outer housing is not

cause for rejection/replacement, provided the viscous damper tube assembly does not become cocked inside outer housing and does not move past snapring.

- c. Install snaprings.
- d. Install tail rotor drive shaft viscous damper bearing assembly (PARA 6-4-43).



6-4-45    VISCOUS DAMPER TUBE ASSEMBLY.

PARAGRAPH BREAKDOWN

|                                                  | PAGE     |
|--------------------------------------------------|----------|
| 6-4-45.1    Replace Viscous Damper Tube Assembly | 6-4-45-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Scraper, Locally-Made
- Powertrain Repairer’s Toolkit
- Protection, Eye and Skin

Materials/Parts

- Alcohol, Denatured, Item 70, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-3-16
- PARA 6-3-14
- PARA 6-4-43
- PARA 6-4-44

6-4-45.1    Replace Viscous Damper Tube Assembly.

NOTE

Replacement of all viscous damper tube assemblies is the same.

6-4-45.1.1    Remove.

- Remove tail rotor drive shaft viscous damper bearing assembly (PARA 6-4-43).
- Using nonmetallic scraper, remove sealing compound from edges of filler neck and mating surfaces of inner housing, outer housing, and snapping.
- Remove snapping from inner housing (Figure 6-4-45-1).
- Manually press out inner housing.

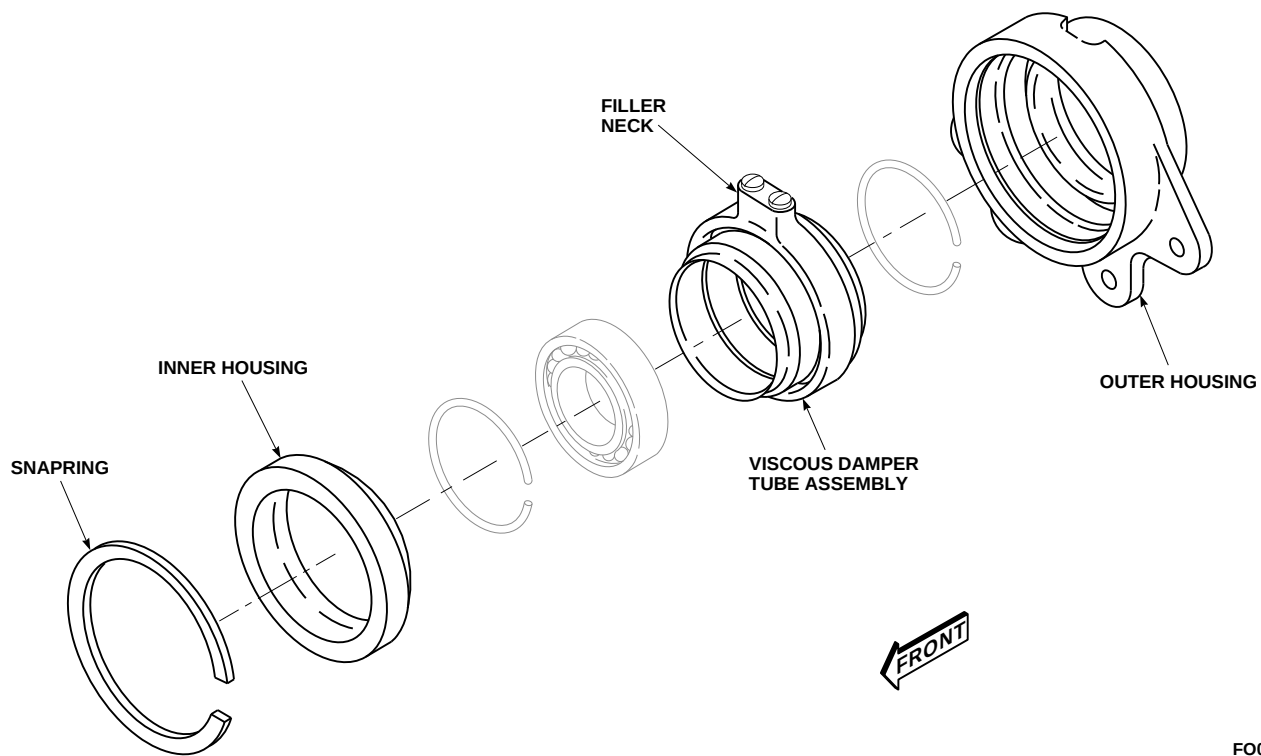
- Remove viscous damper tube assembly.
- Remove bearing from viscous damper tube assembly (PARA 6-4-44).

6-4-45.1.2    Clean.



Injury to personnel will result if care is not taken during flushing. Bearings may contain a white powder (chloride and sodium cyanide), which is a hazardous substance. Use eye protection and rubber gloves during flushing.

- Remove new viscous damper tube assembly from packing material and clean as follows:
  - Flush with hot water.

FO0442C  
SA**FIGURE 6-4-45-1. REPLACE VISCOUS DAMPER TUBE ASSEMBLY**

- (2) Using denatured alcohol, Item 70, Appendix D, rinse viscous damper tube assembly.

**6-4-45.1.3 Install.**

- a. Install tail rotor drive shaft bearing in viscous damper tube assembly (PARA 6-4-44).

**NOTE**

Variations in tolerance may allow movement of viscous damper tube assembly within outer housing. Axial and radial movement inside outer housing is not cause for rejection or replacement, provided the viscous damper tube assembly does not become cocked inside outer housing and does not move past snapping.

- b. Install viscous damper tube assembly on outer housing. Make sure viscous damper tube assembly is seated (Figure 6-4-45-1).  
c. Install inner housing.

- d. Install snapping.

**NOTE**

Flexing of snapping, when installed in groove, is not cause for rejection or replacement of snapping, provided installed snapping meets visual inspection.

- e. Visually inspect snapping for proper installation. **NO PART OF SNAPRING OUTSIDE DIAMETER IS ALLOWED OUTSIDE GROOVE IN VISCOUS DAMPER TUBE.** If any part of snapping is outside groove in viscous damper tube, replace snapping.  
f. Apply  $\frac{1}{16}$  -  $\frac{1}{8}$ -inch bead of sealing compound, Item 292, Appendix D, to edges of filler neck and mating surfaces of inner housing, outer housing, and snapping.  
g. Service tail rotor drive shaft bearing (PARA 1-3-16).  
h. Install tail rotor drive shaft viscous damper bearing assembly (PARA 6-4-43).

**NOTE**

For tail rotor drive shaft bearing leakage limitations, refer to PARA 6-3-14.

- i. Check helicopter logbook to make sure viscous damper bearing assembly is inspected for leak-

age after next flight. If air bubbles are more than 1-inch long, replace viscous damper tube assembly.



6-4-46 TAIL ROTOR DRIVE SHAFT REPAIR.

PARAGRAPH BREAKDOWN

|                                        | PAGE     |
|----------------------------------------|----------|
| 6-4-46.1 Repair Tail Rotor Drive Shaft | 6-4-46-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Caliper Set
- Depth Gage
- Dial Indicating Depth Gage
- Powertrain Repairer’s Toolkit

Materials/Parts

- Abrasive Cloth, Item 1A, Appendix D
- Abrasive Cloth, Item 10, Appendix D

- Primer, Item 258, Appendix D

- Polishing Cloth

References

- Appendix D
- PARA 2-6-5
- PARA 6-4-38
- PARA 6-4-39
- PARA 6-4-40
- PARA 6-4-41

6-4-46.1 Repair Tail Rotor Drive Shaft.

NOTE

No balance checks are required after reworking tail rotor drive shaft per this procedure.

6-4-46.1.1. Repair.

- Before starting repair of tail rotor drive shaft, using caliper set, determine whether polishing out damage would result in a diameter below repairable limits (Figure 6-4-46-1, Sheet 1 and Sheet 2). If damage repair exceeds limits, replace tail rotor drive shaft (PARA 6-4-38 (for section I), PARA 6-4-39 (for section II), PARA

6-4-40 (for section III), or PARA 6-4-41 (for section IV).

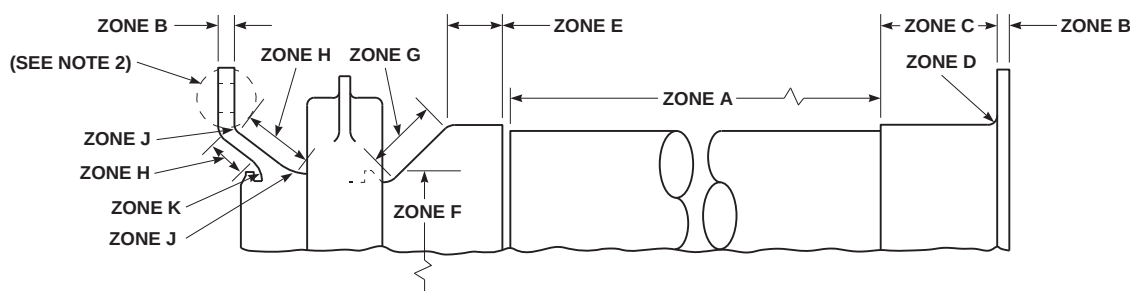
NOTE

Rework of tail rotor drive shafts may be done on helicopter.

- If required, remove damaged section of tail rotor drive shaft (PARA 6-4-38 (for section I), PARA 6-4-39 (for section II), PARA 6-4-40 (for section III), or PARA 6-4-41 (for section IV)).
- Fine scratches that remove only paint may be blended smooth using abrasive cloth, Item 10, Appendix D. After blending, touch up with matching paint (PARA 2-6-5).

- d. Using dial indicating depth gage and calipers, check present dimension of damaged area. Compare reading with minimum repair limits.
- e. Using depth gage, check depth of damaged area to determine whether tail rotor drive shaft minimum dimension after repair can be maintained.
- f. If damage can be repaired, blend smooth using abrasive cloth, Item 1A, Appendix D. Remove only what is necessary so tail rotor drive shaft will stay within limits.
- g. Light corrosion on rivets or on visible part of tail rotor drive shaft inside diameter may be removed using abrasive cloth, Item 1A, Appendix D. Remove only what is necessary so tail rotor drive shaft will stay within limits.
- h. Blend reworked area to remove all traces of abrasive marks and match surface smoothness using abrasive cloth, Item 10, Appendix D, or appropriate polishing cloth.
- i. Clean polished area. Apply a coat of primer, Item 258, Appendix D, to area.
- j. Touch up area with matching paint (PARA 2-6-5).

## ILLUSTRATION FOR TAIL ROTOR DRIVE SHAFT SECTIONS I, II, AND III (TYPICAL)



| COMPONENT                                | ZONE | DIMENSION BEFORE REPAIR (INCH)        | REPAIR LIMIT (INCH)   |
|------------------------------------------|------|---------------------------------------|-----------------------|
| TUBE OD                                  | A    | 3.500 (+0.005-0.000) DIA              | 3.485                 |
| TUBE ID                                  | A    | 3.312 (+0.010) DIA                    | 3.322                 |
| FLANGE                                   |      |                                       |                       |
| LUG                                      | B    | 0.175 ( $\pm 0.005$ )                 | 0.160                 |
| OD                                       | C    | 3.690 ( $\pm 0.010$ ) DIA             | 3.670 DIA             |
| RADIUS                                   | D    | 0.312 ( $\pm 0.010$ ) RADIUS          | (SEE NOTE 3)          |
| STUB SHAFT                               |      |                                       |                       |
| OD                                       | E    | 3.690 ( $\pm 0.010$ ) DIA             | 3.670 DIA             |
| OD                                       | F    | 1.9686 (+0.007- 0.000) DIA            | 1.9676 MIN DIA        |
|                                          | G    | 0.125 ( $\pm 0.010$ ) (WEB THICKNESS) | BLEND TO 0.250 RADIUS |
| COUPLING FLANGE                          |      |                                       |                       |
| CONE                                     | H    | 0.105 ( $\pm 0.005$ )                 | 0.010 MAX DEPTH       |
| RADII                                    | J    | 0.125 ( $\pm 0.010$ ) RADIUS          | BLEND TO 0.250 RADIUS |
|                                          |      |                                       | (SEE NOTE 3)          |
| RADII                                    | K    | 0.190<br>0.45 RADIUS                  | 0.010 MAX DEPTH       |
|                                          |      |                                       | BLEND TO 0.250 RADIUS |
|                                          |      |                                       | (SEE NOTE 3)          |
| OUTER SURFACE OF FLANGE EARS (BOTH ENDS) |      |                                       | 0.010 MAX DEPTH       |
|                                          |      |                                       | BLEND TO 0.250 RADIUS |
|                                          |      |                                       | (SEE NOTE 3)          |

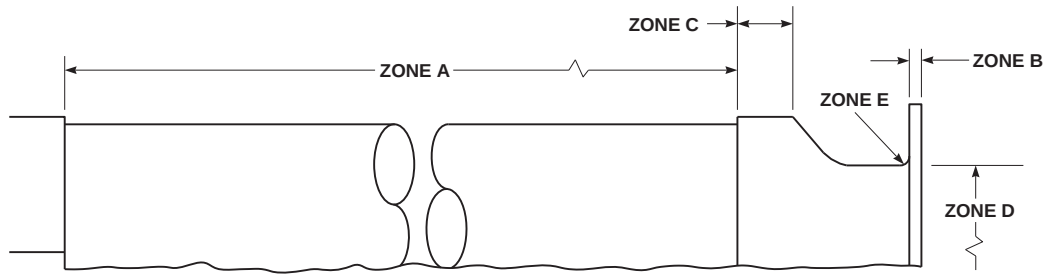
## NOTES

- REPAIR LIMITATIONS
  - REPAIR OF MINOR DAMAGE SUCH AS LIMITED NICKS, SCRATCHES, OR SCORING SHALL BE CLEANED UP BY POLISHING.
  - POLISHING SHALL BE DONE ONLY IN THE DAMAGED AREA AND NOT THE ENTIRE CIRCUMFERENCE UNLESS THE DAMAGE EXTENDS AROUND IT.
  - THE BLEND RADIUS ON SHAFT OD SHALL BE 1/2 INCH OR GREATER.
  - FINE SCRATCHES NEED NOT BE POLISHED IF NOT THROUGH PAINT.
  - AREAS WHERE PAINT HAS BEEN REMOVED ARE TO BE POLISHED WITH CROCUS CLOTH AND REPAINTED.
- DO NOT PAINT AROUND BOLTHOLES.
- BLEND RADIUS SHALL NOT BE LESS THAN 0.010 INCH BELOW THE MINIMUM TOLERANCES SHOWN IN "DIMENSION BEFORE REPAIR" COLUMN.

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FIGURE 6-4-46-1. REPAIR TAIL ROTOR DRIVE SHAFT (SHEET 1 OF 2)

## ILLUSTRATION FOR TAIL ROTOR DRIVE SHAFT SECTION IV



| COMPONENT | ZONE | DIMENSION BEFORE REPAIR (INCH) | REPAIR LIMIT (INCH) |
|-----------|------|--------------------------------|---------------------|
| TUBE OD   | A    | 5.250 (+0.005-0.000) DIA       | 5.235 DIA           |
| TUBE ID   | A    | 5.042 (+0.010) DIA             | 5.052 DIA           |
| FLANGE    | B    | 0.175 ( $\pm 0.005$ )          | 0.160               |
| LUG       | C    | 5.450 ( $\pm 0.010$ ) DIA      | 5.430 DIA           |
| OD        | D    | 3.500 ( $\pm 0.010$ ) DIA      | 3.480 DIA           |
| RADII     | E    | 0.250 ( $\pm 0.010$ ) RADIUS   | (SEE NOTE 3)        |

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FIGURE 6-4-46-1. REPAIR TAIL ROTOR DRIVE SHAFT (SHEET 2 OF 2)



6-4-47 OIL COOLER RADIATOR.

PARAGRAPH BREAKDOWN

|                                                                                                                                                                     | PAGE                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| 6-4-47.1 Replace Oil Cooler Radiator                                                                                                                                | 6-4-47-1                                                                                                       |
| <hr/>                                                                                                                                                               |                                                                                                                |
| INITIAL SETUP                                                                                                                                                       | <ul style="list-style-type: none"> <li>• PARA 1-6-18</li> </ul>                                                |
| Tools                                                                                                                                                               | <ul style="list-style-type: none"> <li>• PARA 2-4-109</li> </ul>                                               |
| <ul style="list-style-type: none"> <li>• Aircraft Mechanic’s Toolkit</li> <li>• Powertrain Repairer’s Toolkit</li> <li>• Torque Wrench, 30 - 150 in. lbs</li> </ul> | <ul style="list-style-type: none"> <li>• PARA 2-4-113</li> <li>• PARA 2-4-115</li> <li>• PARA 2-6-5</li> </ul> |
| Materials/Parts                                                                                                                                                     | <ul style="list-style-type: none"> <li>• PARA 6-3-6</li> </ul>                                                 |
| <ul style="list-style-type: none"> <li>• Abrasive Cloth, Item 10, Appendix D</li> </ul>                                                                             | <ul style="list-style-type: none"> <li>• PARA 6-4-48</li> </ul>                                                |
| References                                                                                                                                                          | <ul style="list-style-type: none"> <li>• PARA 6-4-49</li> </ul>                                                |
| <ul style="list-style-type: none"> <li>• Appendix D</li> <li>• PARA 1-3-12</li> </ul>                                                                               | <ul style="list-style-type: none"> <li>• PARA 7-4-6</li> </ul>                                                 |
| <hr/>                                                                                                                                                               |                                                                                                                |

6-4-47.1 Replace Oil Cooler Radiator.

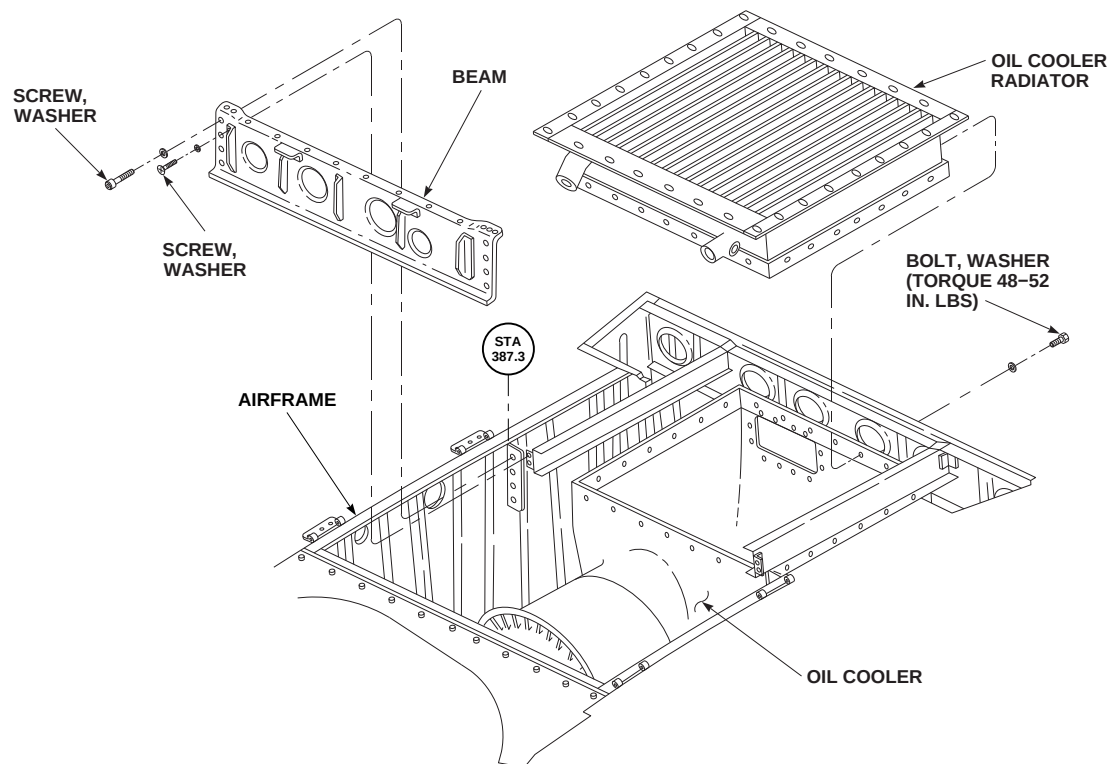
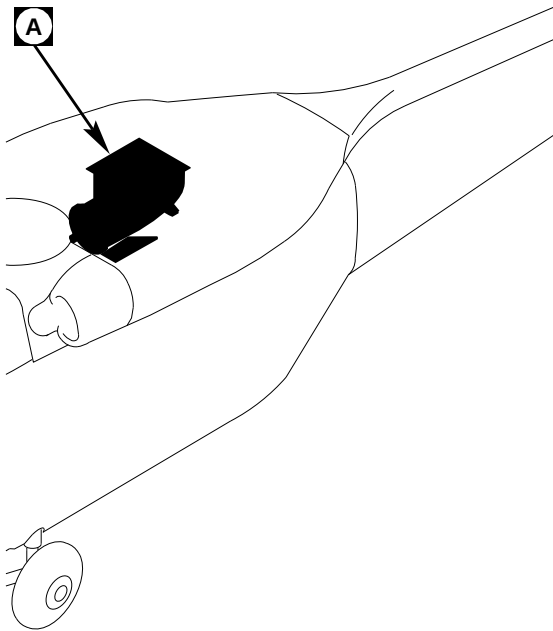
NOTE

Oil cooler radiator weighs about 31 pounds.

6-4-47.1.1 Remove.

- Turn main rotor blades as required to gain access to oil cooler radiator.
- If not already done, engage gust lock (PARA 1-6-18).
- Open APU access doors.
- Remove oil cooler compartment access doors (PARA 2-4-113).

- Remove APU screens (PARA 2-4-109).
- Remove IR jammer panel (PARA 2-4-115).
- Remove pneumatic tubes (PARA 7-4-6).
- Disconnect oil inlet and oil outlet lines from oil cooler radiator (PARA 6-4-49).
- Remove screws and washers securing beam to airframe at STA 387.3. Remove beam (Figure 6-4-47-1, Detail A).
- Remove oil cooler radiator thermostatic valve (PARA 6-4-48).
- Remove bolts and washers securing oil cooler radiator to oil cooler.



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FIGURE 6-4-47-1. REPLACE OIL COOLER RADIATOR

- l. Remove oil cooler radiator.
- m. Drain any remaining oil from oil cooler radiator.
- n. If oil cooler radiator was removed due to clogging, contamination, or blockage, perform the following:
  - (1) Flush oil inlet and oil outlet lines.
  - (2) Inspect all main transmission chip detectors and strainer of main module (PARA 6-3-6).
  - (3) Send oil cooler radiator to approved maintenance facility for cleaning/repair.

#### 6-4-47.1.2 Inspect/Repair.

- a. Nicks and gouges up to 0.100-inch deep (measured from edge of mounting flange) are to be blended smooth with a blend radius of not less than 1/8-inch. An edge distance of not less than 0.120-inch must be maintained at all attachment holes.
- b. Fine scratches that remove only paint may be blended smooth using abrasive cloth, Item 10, Appendix D. After blending, touch up with matching paint (PARA 2-6-5). Fine scratches not through paint need not be polished.



Damage to core of oil cooler radiator will result if attempts are made to straighten core fins. Core fins may break and cause leakage. Do not straighten core fins.

- c. Minor damage to core fins is acceptable, provided there is no leakage or transmission cooling problems reported.

#### 6-4-47.1.3 Install.

- a. Install oil cooler radiator on oil cooler and secure with bolts and washers (Figure 6-4-47-1, Detail A).

- b. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**
- c. Install beam to airframe at STA 387.3 and secure using screws and washers. If necessary, remove outlet fitting to install beam, then install outlet fitting.
- d. Install oil cooler radiator thermostatic valve (PARA 6-4-48).
- e. Connect oil inlet and oil outlet lines to oil cooler radiator (PARA 6-4-49).
- f. Install pneumatic tubes (PARA 7-4-6).
- g. Make sure area is clean and free of foreign material.
- h. Install IR jammer panel (PARA 2-4-115).
- i. Install APU screens (PARA 2-4-109).
- j. Install oil cooler compartment access doors (PARA 2-4-113).
- k. Close APU access doors.
- l. Service main transmission (PARA 1-3-12).
- m. Perform check of oil cooler radiator as follows:
  - (1) With qualified person at controls, run up helicopter.
  - (2) Bring main transmission up to operating temperature and run about 10 minutes. Temperature should remain constant.
  - (3) Shut down helicopter.
  - (4) Check oil cooler and oil cooler radiator for leaks.
  - (5) Check main transmission for proper oil level. If necessary, service main transmission (PARA 1-3-12).
- n. If required, disengage gust lock (PARA 1-6-18).



6-4-48 OIL COOLER RADIATOR THERMOSTATIC VALVE.

PARAGRAPH BREAKDOWN

|                                                         | PAGE     |
|---------------------------------------------------------|----------|
| 6-4-48.1 Replace Oil Cooler Radiator Thermostatic Valve | 6-4-48-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Quart
- Pail
- Powertrain Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-3-12
- PARA 1-6-18
- PARA 2-4-113

6-4-48.1 Replace Oil Cooler Radiator Thermostatic Valve.

6-4-48.1.1 Remove.

- Turn main rotor blades as required to access oil cooler fan and oil cooler radiator.
- If not already done, engage gust lock (PARA 1-6-18).
- Remove oil cooler compartment access doors (PARA 2-4-113).
- Remove screws and washers securing front sections of IR jammer panel and APU screens to beam at STA 387.3 (Figure 6-4-48-1, Detail A).
- Remove screws and washers securing beam at STA 387.3 to airframe.
- Slide beam over oil inlet and oil outlet lines.

- Position 1-quart container under oil cooler radiator thermostatic valve to catch any leaking oil.
- Remove valve and gasket from oil cooler radiator (Figure 6-4-48-1, Detail B). Discard gasket.

6-4-48.1.2 Test.

- Heat small pail of water to 125° - 135°F (52° - 57°C).
- Place oil cooler radiator thermostatic valve in heated water for 5 minutes.
- Remove valve from water and measure quickly (Figure 6-4-48-1, Detail C).
- Heat small pail of water to 155° - 165°F (69° - 74°C).
- Place valve in water for 5 minutes.

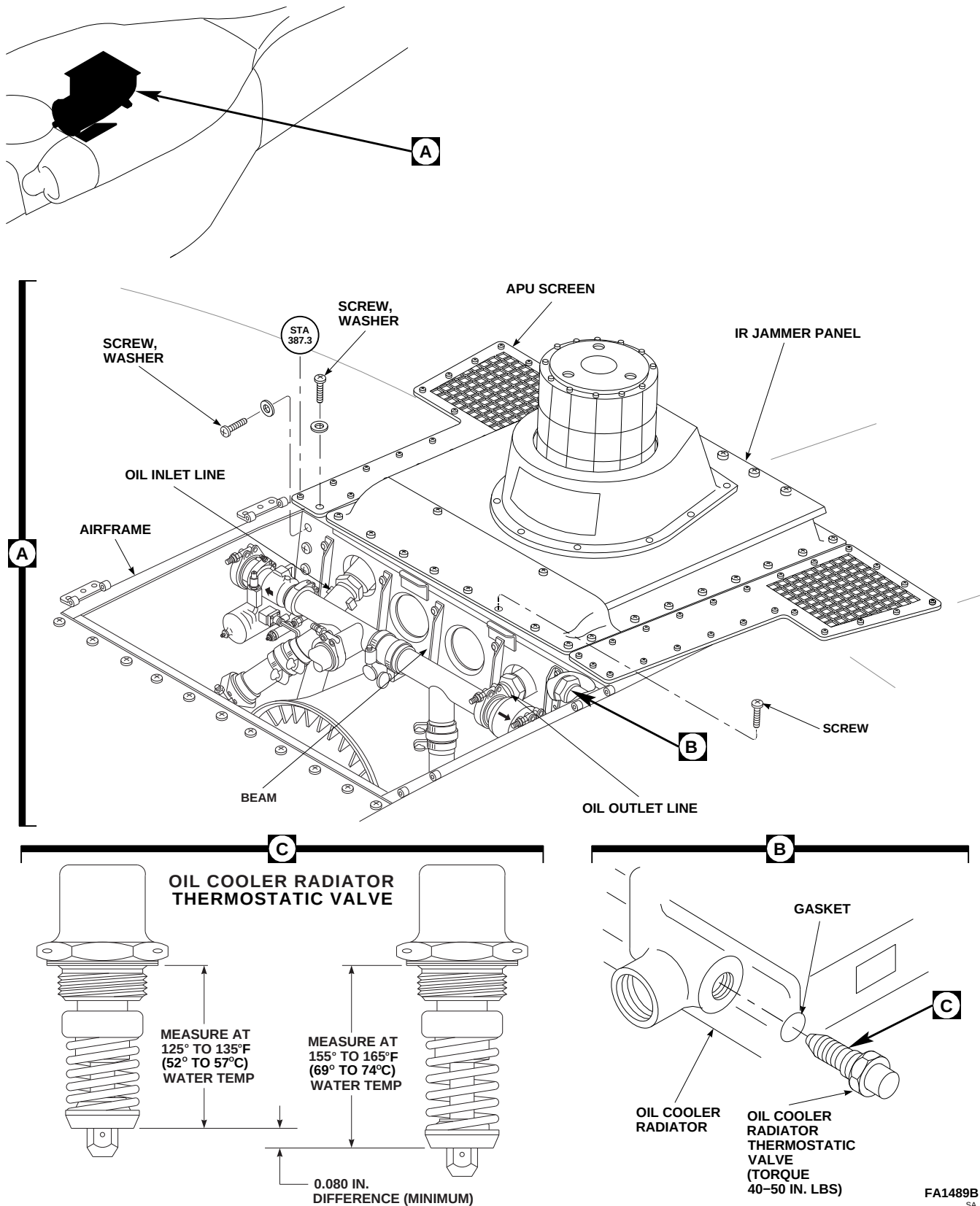


FIGURE 6-4-48-1. REPLACE OIL COOLER RADIATOR THERMOSTATIC VALVE

- f. Remove valve from water and measure quickly.
- g. **MINIMUM DIFFERENCE BETWEEN TWO MEASUREMENTS IS 0.080-INCH.** If valve moves less than 0.080-inch, replace valve.

#### 6-4-48.1.3 Install.

- a. Check valve seat on oil cooler radiator for scoring or wear that would cause leakage.
- b. Install oil cooler radiator thermostatic valve and gasket in oil cooler radiator (Figure 6-4-48-1, Detail B).
- c. **TORQUE VALVE TO 40 - 50 INCH-POUNDS.**
- d. Using lockwire, Item 197, Appendix D, lockwire valve.
- e. Install beam at STA 387.3 to airframe using screws and washers (Figure 6-4-48-1, Detail A).
- f. Make sure area is clean and free of foreign material.

- g. Install screws and washers securing front sections of IR jammer panel and APU screens to beam at STA 387.3.
- h. Install oil cooler compartment access doors (PARA 2-4-113).
- i. Perform check of oil cooler radiator thermostatic valve as follows:
  - (1) With qualified person at controls, run up helicopter.
  - (2) Bring main transmission up to operating temperature and run about 10 minutes.
  - (3) Check for normal oil temperature indication.
  - (4) Shut down helicopter.
  - (5) Check valve for leaks.
  - (6) Check main transmission for proper oil level. If necessary, service main transmission (PARA 1-3-12).
- j. If required, disengage gust lock (PARA 1-6-18).





6-4-49 OIL COOLER RADIATOR LINES AND ELBOWS.

PARAGRAPH BREAKDOWN

|                                                       | PAGE     |
|-------------------------------------------------------|----------|
| 6-4-49.1 Replace Oil Cooler Radiator Lines and Elbows | 6-4-49-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Quart
- Powertrain Repairer’s Toolkit
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1600 in. lbs

Materials/Parts

- Petrolatum, Item 234, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-3-12
- PARA 1-6-16

6-4-49.1 Replace Oil Cooler Radiator Lines and Elbows.

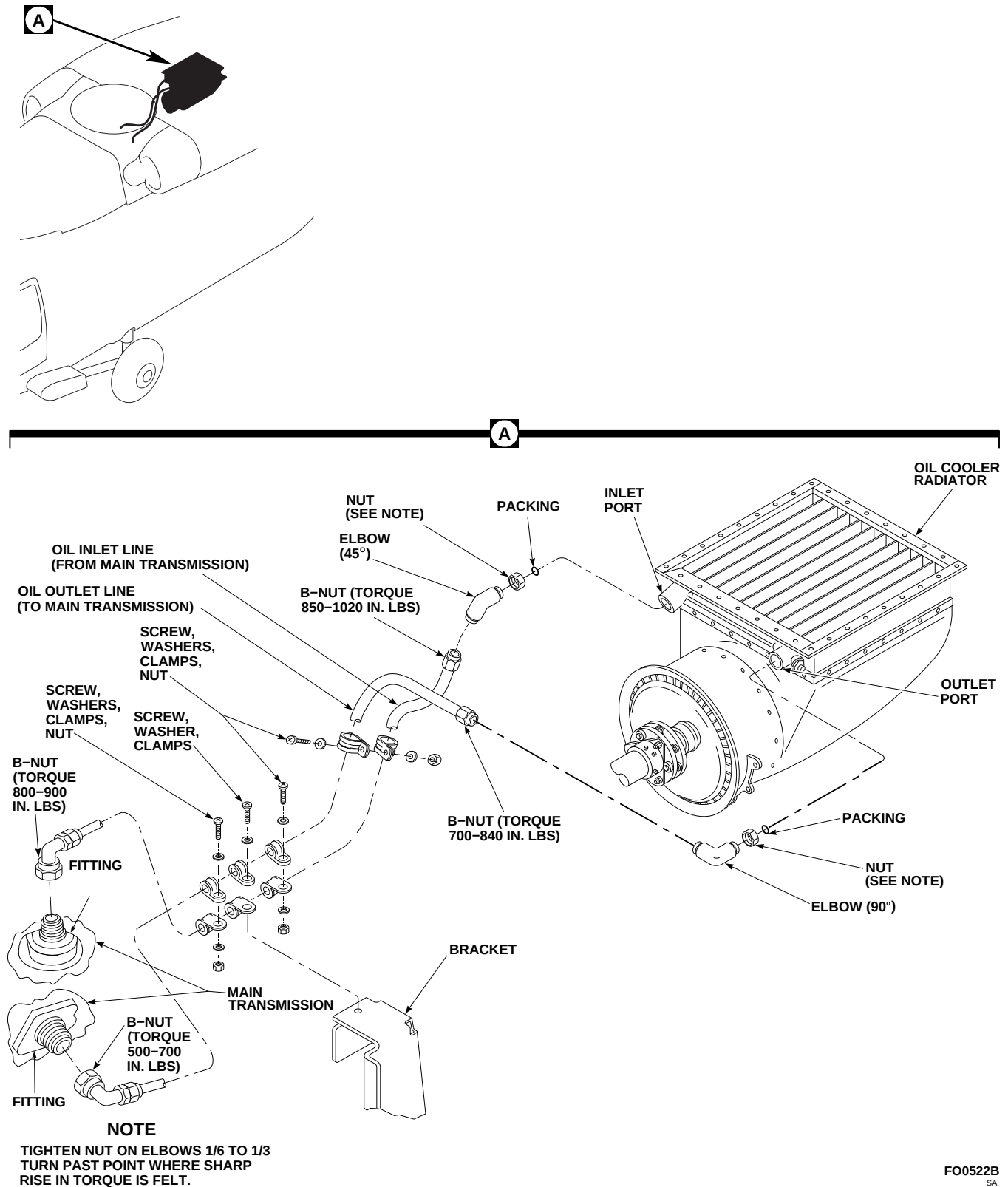
6-4-49.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Turn main rotor blades as required to gain access to oil cooler radiator lines and elbows.
- Open engine cowlings.
- Open oil cooler compartment access doors.
- Remove screws, washers, and nuts securing clamps to oil inlet and oil outlet lines and brackets. Remove clamps (Figure 6-4-49-1, Detail A).
- Position 1-quart container under oil inlet and oil outlet lines to catch oil trapped in line.

- Disconnect oil inlet and oil outlet lines from fittings on main transmission and elbows on oil cooler radiator. Remove lines.
- Loosen nuts and remove elbows from inlet and outlet ports in oil cooler radiator. Cap open fittings, elbows, and plug oil inlet and oil outlet lines.
- Remove nuts and packings from elbows. Discard packings.

6-4-49.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Remove caps from elbows.
- Using petrolatum, Item 234, Appendix D, coat threads of elbows and both packings.



- d. Remove caps from inlet and outlet fittings.



Damage to starter control valve will result if contact occurs between starter control valve and oil cooler inlet elbow. Make sure there is proper clearance between start control valve and oil cooler inlet elbow.

- e. Install nut and packing on 45° elbow. Install elbow in inlet port (Figure 6-4-49-1, Detail A).
- f. Install nut and packing on 90° elbow. Install elbow in outlet port.
- g. **TIGHTEN NUT ON ELBOWS  $\frac{1}{6}$  TO  $\frac{1}{3}$  TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- h. Remove plugs from oil inlet and oil outlet lines.



**FSCAP**

Verification of clearance between oil inlet and oil outlet lines is a critical characteristic.

- i. Connect oil inlet and oil outlet lines. **MAKE SURE THERE IS A 0.120-INCH MINIMUM CLEARANCE BETWEEN OIL INLET AND OIL OUTLET LINES OVER THE ENTIRE LENGTH.**
- j. At fitting on main transmission, **TORQUE B-NUT ON OIL INLET LINE TO 800 - 900 INCH-POUNDS.**

- k. At fitting on main transmission, **TORQUE B-NUT ON OIL OUTLET LINE TO 500 - 700 INCH-POUNDS.**
- l. At 90° elbow on oil cooler radiator, **TORQUE B-NUT ON OIL OUTLET LINE TO 700 - 840 INCH-POUNDS.**
- m. At 45° elbow on oil cooler radiator, **TORQUE B-NUT ON OIL INLET LINE TO 850 - 1020 INCH-POUNDS.**
- n. Install clamps on oil inlet and oil outlet lines with screws, washers, and nuts.
- o. Install clamps on oil inlet and oil outlet lines and attach to bracket with screw and washer.
- p. Service main transmission (PARA 1-3-12).
- q. Make sure area is clean and free of foreign material.
- r. Close oil cooler compartment access doors.
- s. Close engine cowlings.
- t. Perform check of oil inlet and oil outlet lines, and elbows as follows:
  - (1) With qualified person at controls, run up helicopter.
  - (2) Bring main transmission up to operating temperature and run about 10 minutes.
  - (3) Check for normal oil temperature indication.
  - (4) Shut down helicopter.
  - (5) Check oil inlet and oil outlet lines and elbows for leaks.
  - (6) Check main transmission for proper oil level. If necessary, service main transmission (PARA 1-3-12).



6-4-50 OIL COOLER.

PARAGRAPH BREAKDOWN

|          |                                               | PAGE      |          |                                         | PAGE      |
|----------|-----------------------------------------------|-----------|----------|-----------------------------------------|-----------|
| 6-4-50.1 | Replace Oil Cooler                            | 6-4-50-2  | 6-4-50.4 | Input Flange Retaining Nut Torque Check | 6-4-50-20 |
| 6-4-50.2 | Replace Oil Cooler Fan, Drive Shaft, and Duct | 6-4-50-9  |          |                                         |           |
| 6-4-50.3 | Oil Cooler Drive Shaft Runout Check           | 6-4-50-19 |          |                                         |           |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Arbor Press
- Borescope Kit
- Caliper Set
- Depth Gage
- Dial Indicator
- Hoist, Capacity to Support Oil Cooler (Alternate for Cabin Maintenance Crane)
- Hoisting Sling, 70700-20320-042
- Maintenance Trailer
- Micrometer
- Permanent Ink Marker
- Pin, 0.096-inch diameter (2)
- Powertrain Repairer’s Toolkit
- Puller Kit, Mechanical
- Spring Scale

- Tail Drive Flange Wrench, 70700-20687-102
- Torque Reactor, 70700-20688-041
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 100 - 500 ft. lbs
- Transmission Oil Cooler Adapter, 70700-20425-041
- Universal Hoist Sling Assembly, 70700-20320-045

Materials/Parts

- Abrasive Cloth, Item 1A, Appendix D
- Abrasive Cloth, Item 6, Appendix D
- Abrasive Cloth, Item 8, Appendix D
- Abrasive Cloth, Item 10, Appendix D
- Abrasive Mat, Item 14, Appendix D
- Corrosion Preventive Compound, Item 113, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D

- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D
- Polishing Cloth
- Primer, Item 258, Appendix D
- Shim, 70361-03013-104 (2)
- Shim, 70361-03013-105 (2)
- Shim, 70361-03013-106 (1)
- Tiedown Strap, Item 336, Appendix D
- PARA 2-4-113
- PARA 2-4-115
- PARA 2-6-3
- PARA 2-6-5
- PARA 6-3-4
- PARA 6-3-15
- PARA 6-4-17
- PARA 6-4-18
- PARA 6-4-38
- PARA 6-4-39
- PARA 6-4-47
- PARA 6-4-49
- PARA 6-4-51
- PARA 7-4-6
- PARA 18-4-1

## References

- Appendix D
- PARA 1-3-12
- PARA 1-3-20
- PARA 1-3-21
- PARA 2-4-109
- Appropriate Flight Manual
- TM 1-70-VIB

### 6-4-50.1 Replace Oil Cooler.

#### NOTE

Oil cooler weighs 70 pounds.

#### 6-4-50.1.1 Remove.

- a. If cabin maintenance crane will be used, assemble and erect crane (PARA 18-4-1).
- b. Turn main rotor blades as required to gain access to oil cooler radiator lines and elbows.
- c. If necessary, install transmission oil cooler adapter on maintenance trailer.
- d. Remove oil cooler compartment access doors (PARA 2-4-113).
- e. Remove APU screens (PARA 2-4-109).
- f. Remove IR jammer panel (PARA 2-4-115).
- g. Remove oil cooler compartment pneumatic tubes (PARA 7-4-6).
- h. Disconnect oil inlet and oil outlet lines from oil cooler radiator (PARA 6-4-49).
- i. Open APU access doors.
- j. Remove screws and washers securing beam to airframe at STA 387.3. Remove beam from airframe (Figure 6-4-50-1, Detail A).

### 6-4-50-2 Change 9

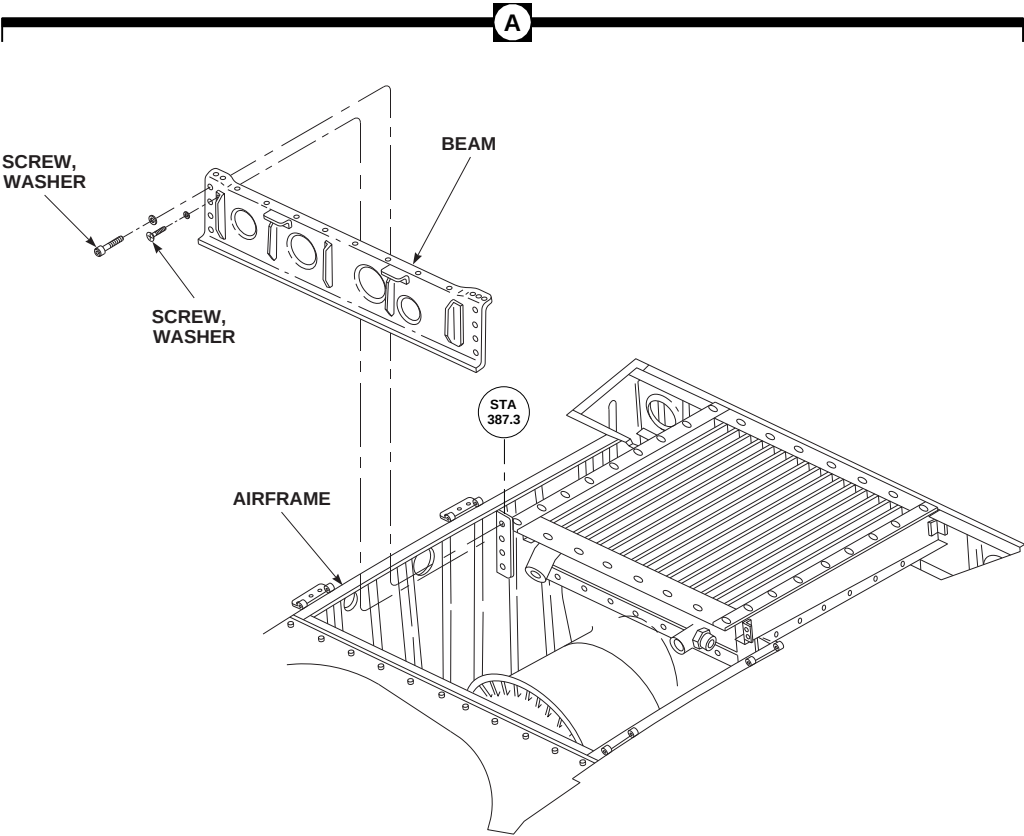
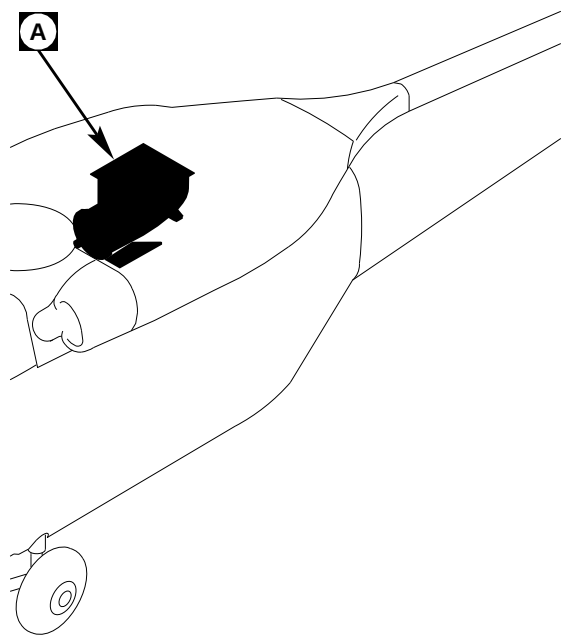


FIGURE 6-4-50-1. REPLACE BEAM AT STA 387.3

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- k. Remove tail rotor drive shaft section I (PARA 6-4-38).
- l. Support front end of tail rotor drive shaft section II.
- m. Disconnect tail rotor drive shaft section II from oil cooler (PARA 6-4-39).
- n. Attach hoisting sling straps to oil cooler (Figure 6-4-50-3, Detail A).
- t. Lower oil cooler onto transmission oil cooler adapter (Figure 6-4-50-2).
- u. Secure oil cooler to transmission oil cooler adapter with bolts, washers, and nuts.
- v. Remove hoisting sling.
- w. If oil cooler is removed due to clogging, contamination, or blockage, perform the following:

### WARNING

- Hoist cable may break, causing injury to personnel or damage to equipment, if hoist is used to free sticking component being removed from helicopter. Do not use excessive force when removing a component.
- Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment, if hook is allowed to contact pulley when removing component from helicopter. Do not allow hook assembly to contact pulley.
- o. Attach hoisting sling to hoist or maintenance crane.
- p. Adjust length of straps.
- q. Raise hoisting sling to remove slack.
- r. Remove bolts, washers, and nuts attaching oil cooler to front and rear supports.
- (1) Flush oil inlet and oil outlet lines.
- (2) Remove and inspect main module sump chip detector/temperature sensor and strainer (PARA 6-4-18).
- (3) Remove and inspect input module chip detectors and strainers (PARA 6-4-17).
- (4) Remove oil cooler radiator (PARA 6-4-47). Send oil cooler radiator to approved maintenance facility for cleaning/repair.
- (5) Install input module chip detectors (PARA 6-4-17) and main module sump chip detector/temperature sensor (PARA 6-4-18).

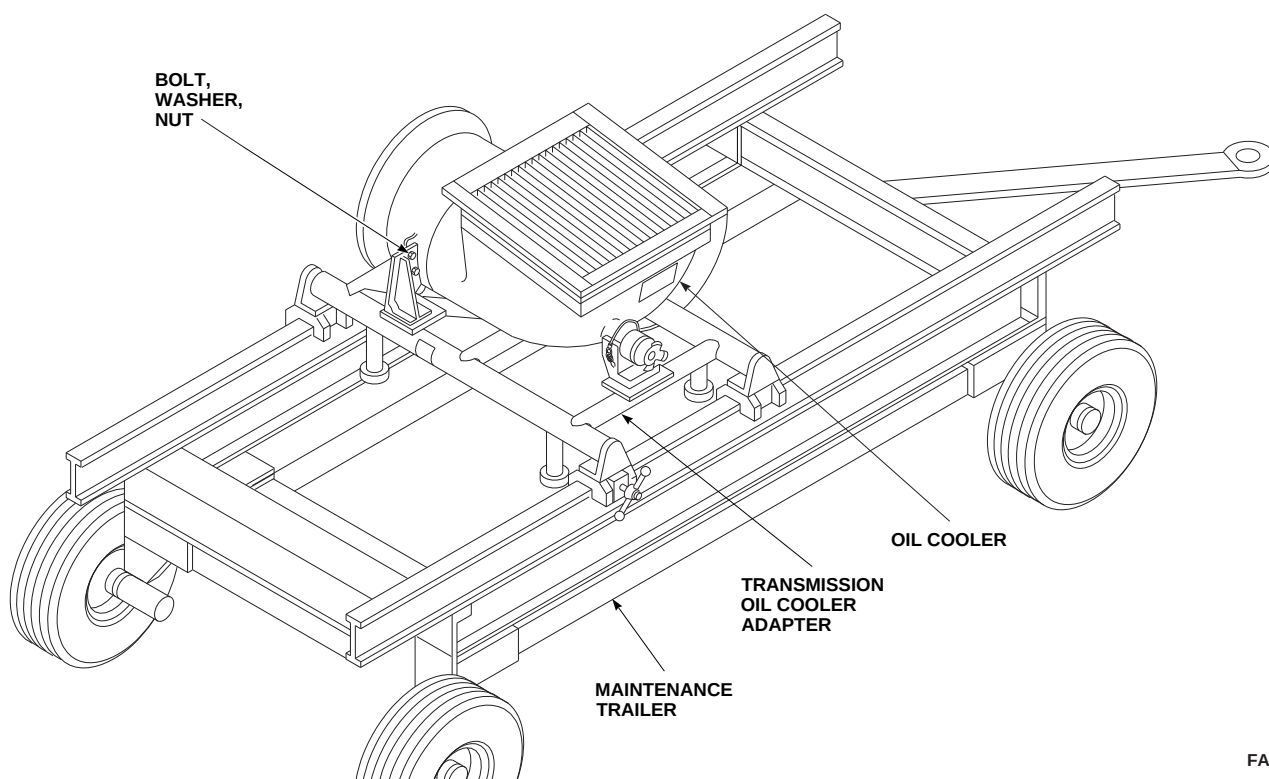
#### 6-4-50.1.2 Inspect.

- a. Perform input flange retaining nut torque check (PARA 6-4-50.4).
- b. Inspect oil cooler drive shaft flange boltholes for elongation. **NONE ALLOWED.** If boltholes are elongated, replace drive shaft (PARA 6-4-50.2).
- c. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to exposed surfaces of retaining nut on oil cooler fan input flange.
- d. Apply corrosion preventive compound, Item 113, Appendix D, to retaining nut on oil cooler fan input flange and area around retaining nut. Do not apply compound to flange mating surfaces.
- s. Lift oil cooler clear of helicopter.

### WARNING

Injury to personnel and damage to equipment will result if caution is not taken when hoisting oil cooler. Keep area clear when hoisting oil cooler.





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**FIGURE 6-4-50-2. POSITION OIL COOLER ON TRANSMISSION OIL COOLER ADAPTER**

### 6-4-50.1.3 Install.

#### **WARNING**

Injury to personnel and damage to equipment will result if oil cooler drive shaft runout check is not performed before oil cooler is installed on helicopter. Failure to properly perform runout check will result in loss of tail rotor drive.

- Perform oil cooler drive shaft runout check (PARA 6-4-50.3).
- Attach hoisting sling straps to oil cooler (Figure 6-4-50-3, Detail A).
- Adjust rear strap so it does not interfere with beams on airframe.
- Attach hoisting sling to hoist or cabin maintenance crane.

- Adjust length of straps.
- Remove bolts, washers, and nuts securing oil cooler to maintenance trailer (Figure 6-4-50-2).

#### **WARNING**

Injury to personnel and damage to equipment will result if caution is not taken when hoisting oil cooler. Keep area clear when hoisting oil cooler.

- Lift oil cooler and position on front and rear supports in helicopter (Figure 6-4-50-3, Detail A).
- Install bolts and washers on front support to line up oil cooler. Do not install bolts in rear supports now.
- TORQUE BOLTS ON FRONT SUPPORT TO 105 - 115 INCH-POUNDS.**
- Remove hoisting sling.

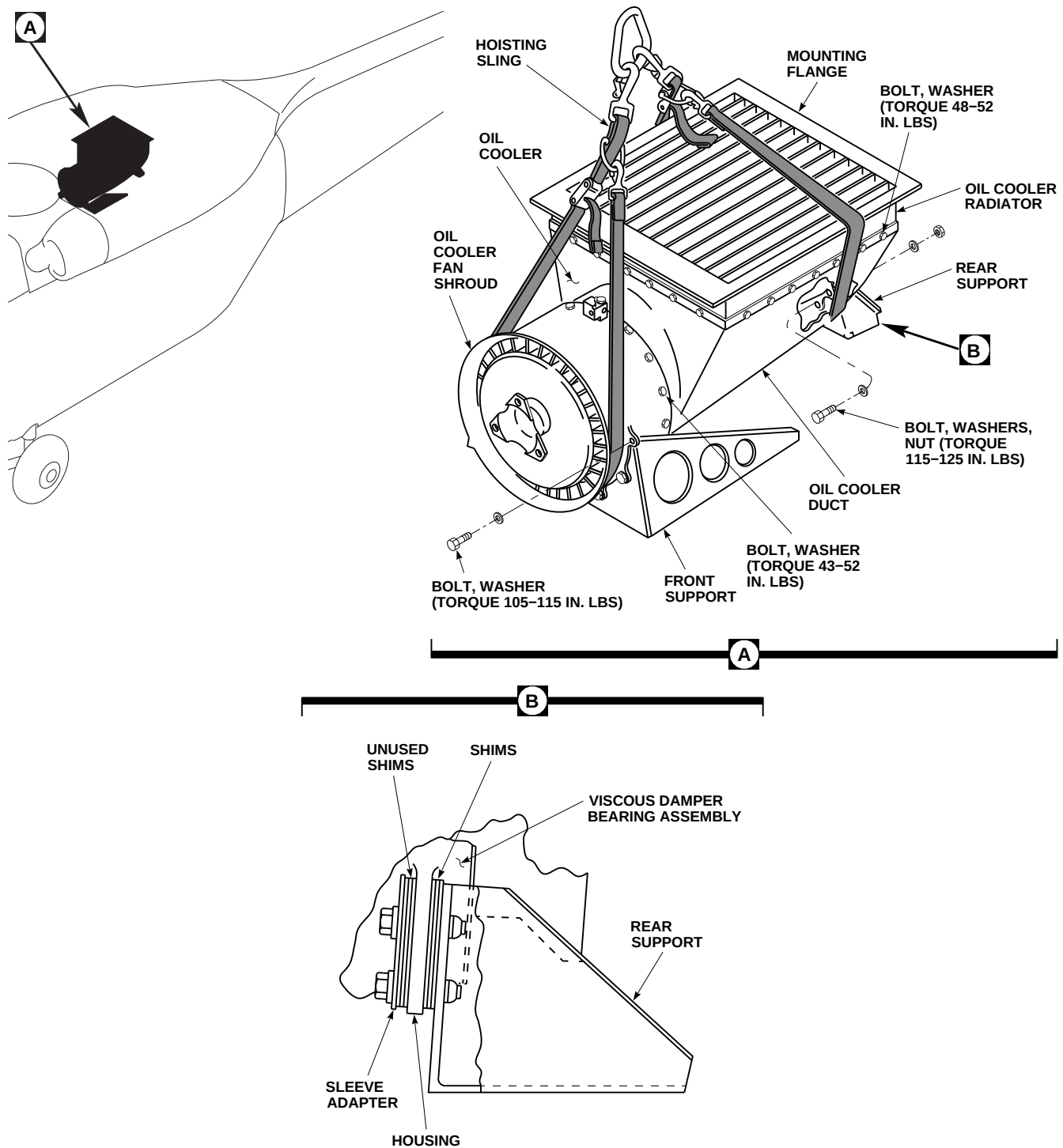
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FIGURE 6-4-50-3. ATTACH HOISTING SLING/REPLACE OIL COOLER

- k. Make sure mounting flange on oil cooler radiator is flush with airframe.
- l. **TIGHTEN FLANGE ATTACHMENT HARDWARE.**
- m. Center bearing in viscous damper bearing assembly.
- n. Perform preload check of oil cooler drive shaft as follows:

**NOTE**

Helicopter fuel cell weight can affect installation preload of oil cooler.

- (1) Using spring scale and tiedown strap, Item 336, Appendix D, align viscous damper bearing assembly with rear support boltholes. **FORCE REQUIRED TO ALIGN BOLTHOLES MUST NOT EXCEED 10 POUNDS.** If more than 10 pounds of force is required, proceed to step o.
- (2) Verify alignment by inserting bolt through viscous damper bearing assembly and rear support on each side of rear support.
- o. If more than 10 pounds of force is required to align viscous damper bearing assembly, perform the following:
  - (1) Remove oil cooler (PARA 6-4-50.1). Do not remove hoisting sling.
  - (2) Loosen all bolts attaching oil cooler fan shroud to oil cooler duct.
  - (3) Loosen all bolts attaching oil cooler radiator to oil cooler duct.
  - (4) Lift oil cooler into place.
  - (5) Repeat step i. through step n.
  - (6) If preload is still beyond limit, defuel helicopter (PARA 1-3-20 or PARA 1-3-21) and repeat step o. (1) through step o. (5).

- (7) If preload is still more than 10 pounds and helicopter has been defueled, replace oil cooler.
- (8) If preload is less than 10 pounds, install and tighten three bolts on each side of oil cooler radiator connection and four bolts around oil cooler fan connection.

**NOTE**

- Oil cooler may be removed from front and rear support to provide easier access to secure oil cooler radiator and oil cooler fan. Recheck preload on installation.
- Make sure oil cooler duct does not interfere with rotation of drive shaft.
- (9) **TORQUE BOLTS ATTACHING OIL COOLER RADIATOR TO OIL COOLER DUCT TO 48 - 52 INCH-POUNDS.**
- (10) **TORQUE BOLTS ATTACHING OIL COOLER FAN SHROUD TO OIL COOLER DUCT TO 43 - 52 INCH-POUNDS.**
- (11) Using lockwire, Item 197, Appendix D, lockwire accelerometer bracket to adjacent bolt and bracket mounting bolt to other adjacent bolt.

- p. Measure gap between rear support and viscous damper bearing assembly (Figure 6-4-50-3, Detail B).
- q. Remove bolts installed to verify alignment.

**NOTE**

A total of five shims are used to assemble rear support, viscous damper bearing assembly, and sleeve adapter. Use two shims, 70361-03013-104, two shims, 70361-03013-105, and one shim, 70361-03013-106, for this assembly.

- r. Select combination of 5 shims (to fill measured gap) and place between rear support and

viscous damper bearing assembly. Place any unused shims between sleeve adapter and viscous damper bearing assembly.

- s. Install sleeve adapter and viscous damper bearing assembly to rear support with bolts, washers, and nuts (Figure 6-4-50-3, Detail A and Detail B).
- t. **TORQUE NUTS TO 115 - 125 INCH-POUNDS.**
- u. Install tail rotor drive shaft section I (PARA 6-4-38).
- v. Connect front end of tail rotor drive shaft section II to exit flange of oil cooler drive shaft (PARA 6-4-39).
- w. Install beam to airframe at STA 387.3 and secure using screws and washers (Figure 6-4-50-1, Detail A).
- x. Connect oil inlet and oil outlet lines to oil cooler radiator (PARA 6-4-49).
- y. Make sure area is clean and free of foreign material.
- z. Install IR jammer panel (PARA 2-4-115).
- aa. Install APU screens (PARA 2-4-109).
- ab. Install oil cooler compartment pneumatic tubes (PARA 7-4-6).
- ac. Make sure area is clean and free of foreign material.
- ad. Install oil cooler compartment access doors (PARA 2-4-113).
- ae. Close APU access doors.
- af. Service main transmission (PARA 1-3-12).

#### NOTE

Spline wear indicator system is not used on oil cooler fan, 70361-03005-107.

- ag. Install oil cooler spline wear indicator (PARA 6-4-51).

#### WARNING

Injury to personnel and damage to equipment will result if oil cooler spline wear inspection is not performed on oil cooler fans, 70361-03005-103 through 70361-03005-106, every time tail rotor drive shaft is disconnected or replaced from main transmission to STA 531.0. Excessive spline wear will result in loss of tail rotor drive. Make sure spline wear indicator is installed and oil cooler spline wear inspection is performed.

- ah. On helicopters with oil cooler fans, 70361-03005-103 through 70361-03005-106, installed, perform spline wear inspection (PARA 6-3-15).
- ai. Perform main transmission functional test (PARA 6-3-4).
- aj. Perform check of oil cooler as follows:
  - (1) With qualified person at controls, run up helicopter (Appropriate Flight Manual).
  - (2) Bring main transmission up to operating temperature and run about 10 minutes.
  - (3) Check for normal oil temperature indication.
  - (4) Shut down helicopter (Appropriate Flight Manual).
  - (5) Check oil cooler for leaks.
  - (6) Check main transmission for proper oil level. If necessary, service main transmission (PARA 1-3-12).

**WARNING**

Injury to personnel and damage to equipment will result if vibration check is not performed any time a tail rotor drive shaft flexible coupling is disconnected or replaced from main transmission up to and including STA 470.9. Excessive vibration indicates improper installation of oil cooler and/or tail rotor drive shaft flexible coupling, which will cause oil cooler fan drive shaft splines and oil cooler fan exit flange splines to wear, resulting in loss of tail rotor drive. Make sure vibration check is performed.

- ak. Perform oil cooler vibration check (TM 1-70-VIB).

## 6-4-50.2 Replace Oil Cooler Fan, Drive Shaft, and Duct.

### 6-4-50.2.1 Remove.

**NOTE**

Rework of oil cooler drive shaft may be performed on helicopter.

- a. If necessary, remove oil cooler (PARA 6-4-50.1).
- b. Remove oil cooler radiator (PARA 6-4-47).
- c. Remove cotter pin from retaining nut on exit flange near viscous damper bearing assembly (Figure 6-4-50-4, Sheet 1).
- d. Using torque reactor and tail drive flange wrench, remove retaining nut from exit flange.

**NOTE**

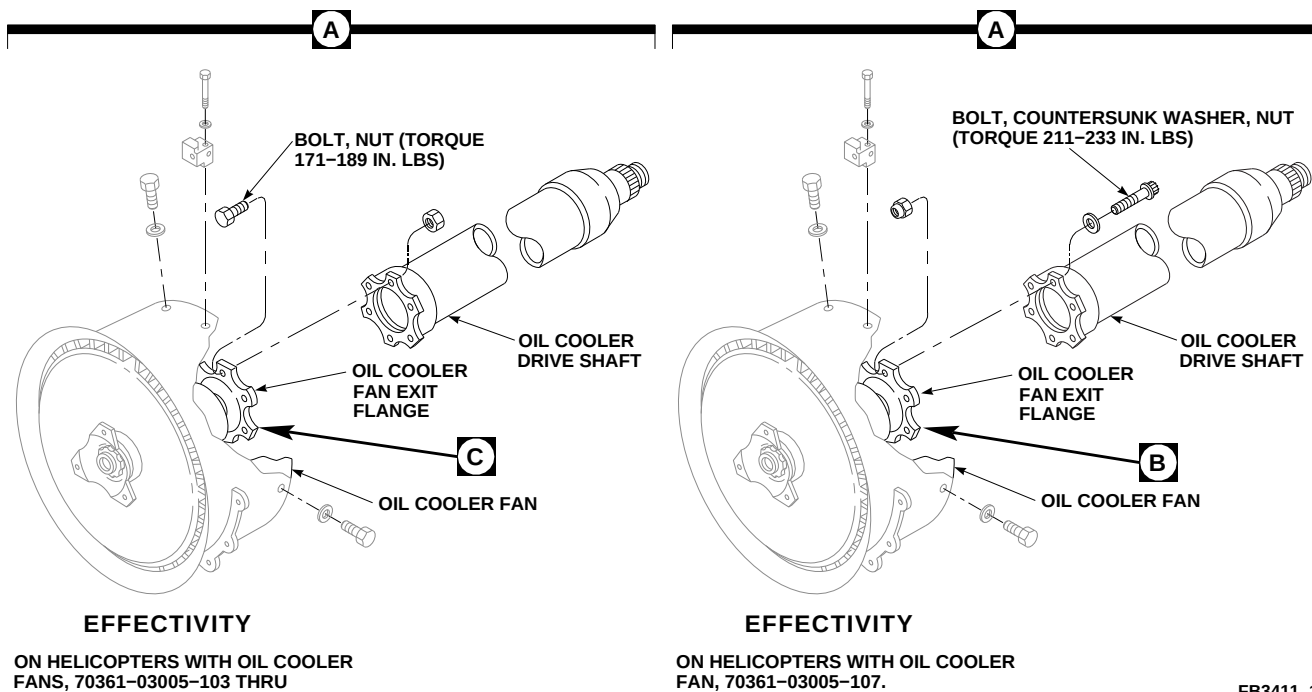
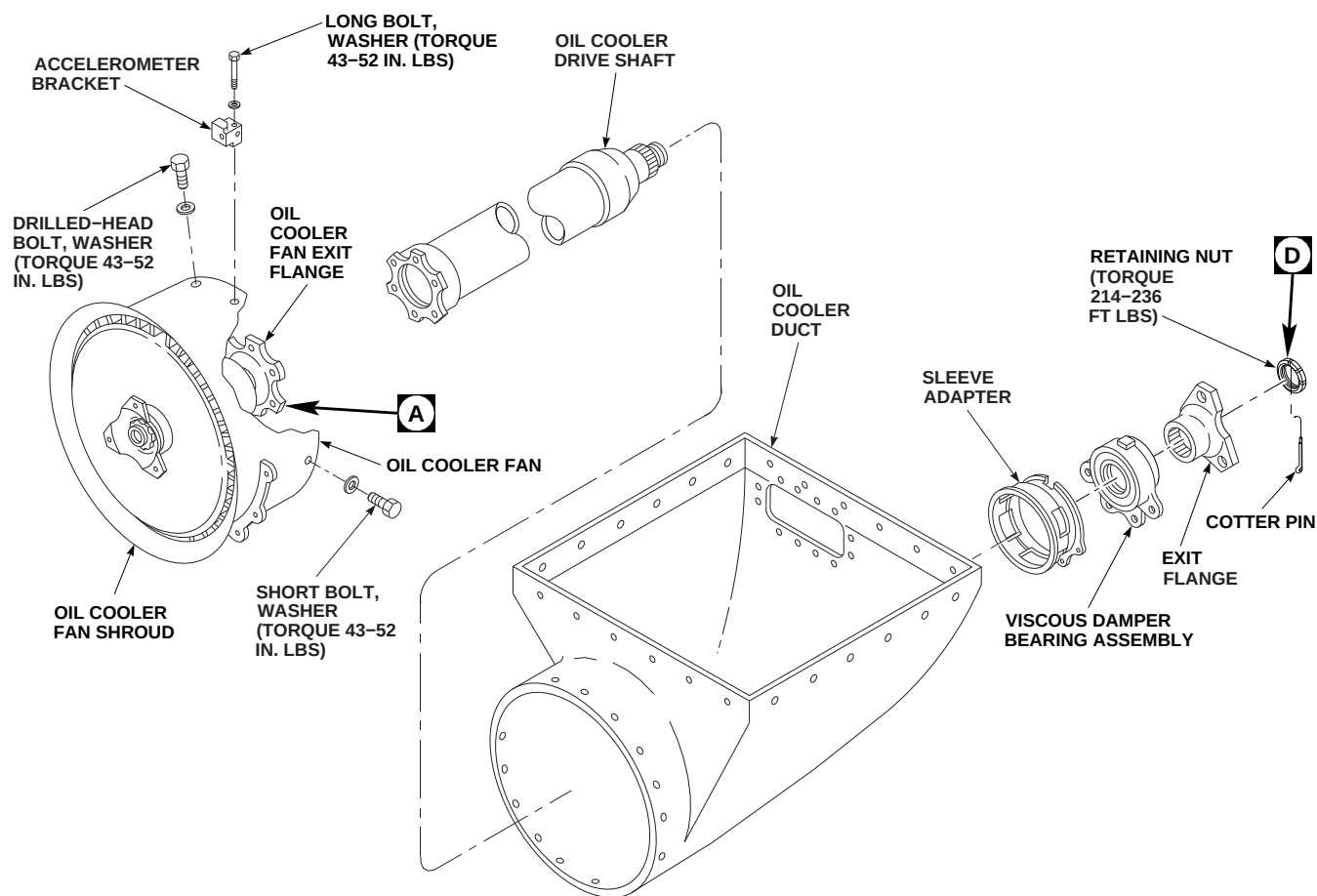
Make sure index marks are present on oil cooler drive shaft and exit flange. If no index marks are present, index marks must be applied.

- e. If necessary, using permanent ink marker, apply index marks to oil cooler drive shaft and exit flange.
- f. Remove exit flange from oil cooler drive shaft by pulling straight off.
- g. Using mechanical puller kit, remove viscous damper bearing assembly from oil cooler drive shaft.
- h. Remove sleeve adapter from oil cooler drive shaft.
- i. Remove long bolt and washer securing accelerometer bracket to oil cooler fan shroud, and remove bracket.
- j. Remove short bolts, drilled-head bolts, and washers securing oil cooler fan shroud to oil cooler duct.
- k. As an assembly, remove oil cooler fan and oil cooler drive shaft from oil cooler duct.
- l. Remove bolts, countersunk washers (if installed), and nuts attaching oil cooler fan exit flange to oil cooler drive shaft (Figure 6-4-50-4, Sheet 1, Detail A). Remove oil cooler drive shaft.

**WARNING**

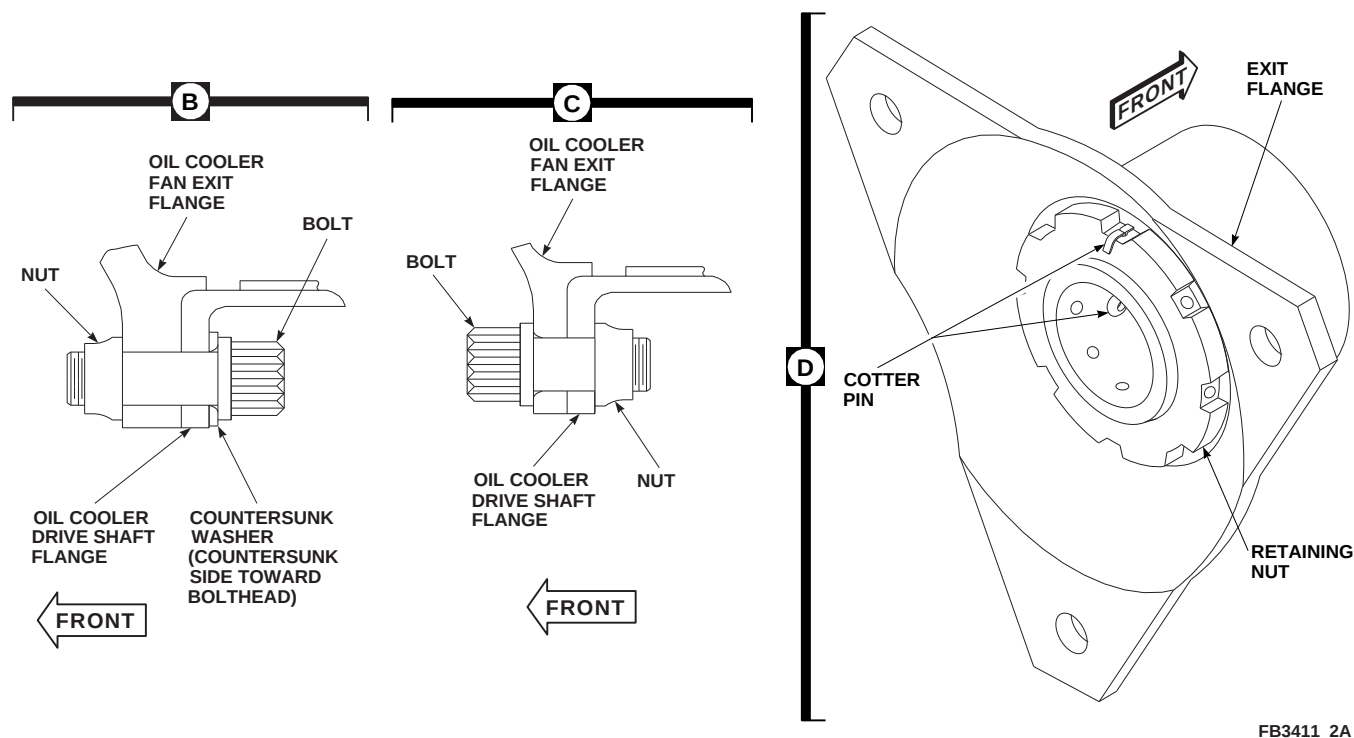
Injury to personnel and damage to equipment will result if retaining nut on exit flange is not torqued properly on oil cooler fans, 70361-03005-103 through 70361-03005-106. Loosening or removal of retaining nut may cause improper torque to be present and is not allowed. On oil cooler fans, 70361-03005-103 through 70361-03005-106, do not loosen or remove retaining nut on oil cooler fan exit flange.

- m. On oil cooler fans, 70361-03005-103 through 70361-03005-106, perform torque check on



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FIGURE 6-4-50-4. REPLACE OIL COOLER FAN, DRIVE SHAFT, AND DUCT (SHEET 1 OF 2)



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FIGURE 6-4-50-4. REPLACE OIL COOLER FAN, DRIVE SHAFT, AND DUCT (SHEET 2 OF 2)

exit flange retaining nut (PARA 6-3-15). Do not install cotter pin now.

#### 6-4-50.2.2 Inspect/Repair.

#### CAUTION

Damage to bearing will result if contact occurs with cleaning solvents and fluids. Cleaning solvents and fluids can penetrate seals of bearing and dissolve lubrication grease. Do not spray on cleaning solvents or fluids in vicinity of bearing. Apply solvents only by wiping.

#### NOTE

No balance checks are required after reworking per this procedure.

- a. Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, thoroughly clean oil cooler drive shaft.

- b. Inspect oil cooler drive shaft and exit flange vi-bropeen index marks for raised or sharp edges. If raised or sharp edges are found, blend using abrasive cloth, Item 10, Appendix D.
- c. Visually inspect oil cooler drive shaft for cracks (Figure 6-4-50-5). **NONE ALLOWED.** If cracks are found, replace oil cooler drive shaft.

#### NOTE

Corrosion between tube and stub shaft, and shaft tube and flange, is identified by flaking or lifting of paint, cracks, or presence of white powder on shaft.

- d. Visually inspect oil cooler drive shaft for corrosion between tube and stub shaft, and tube and flange. **NONE ALLOWED.** If corrosion is found, replace oil cooler drive shaft.
- e. Fine scratches that remove only paint may be blended smooth using abrasive cloth, Item 10, Appendix D. After blending, touch up paint as required (PARA 2-6-5).



- f. Using depth gage and calipers, check dimension of damaged area before rework.
- g. Compare readings with minimum repair limits (Figure 6-4-50-5). **OIL COOLER DRIVE SHAFT MUST NOT BE DAMAGED BEYOND MINIMUM REPAIR LIMITS.** If oil cooler drive shaft is under limits, replace oil cooler drive shaft.
- h. If oil cooler drive shaft minimum dimension after repair can be maintained, repair as follows:
  - (1) Using abrasive cloth, Item 1A, Appendix D, blend damage smooth.
  - (2) Using abrasive cloth, Item 10, Appendix D, or polishing cloth, blend reworked area to remove all traces of abrasive marks and match surface smoothness.
- (2) Secure second 0.096-inch diameter pin between two splines directly opposite first pin.
- (3) Using micrometer, measure outside diameter (OD) of shaft splines over pins. **OD MUST NOT BE LESS THAN 1.9401-INCHES.** If OD exceeds limit, replace oil cooler drive shaft.
- (4) Repeat step i. (1) through step i. (3) with pins located in four different positions.
- j. Inspect threads on end of stub shaft for crossed, stripped, or flattened threads. **NONE ALLOWED.** If threads are damaged, replace oil cooler drive shaft.
- k. On helicopters with oil cooler fans, 70361-03005-103 through 70361-03005-106, inspect inside diameter (ID) of oil cooler fan shaft as follows:
  - (1) Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean oil cooler fan shaft ID of grease and dirt.
  - (2) Using borescope kit, inspect ID of oil cooler fan shaft for missing primer, corrosion, pits, and flaking plating. **NONE ALLOWED.**
  - (3) Inspect ID of oil cooler fan shaft for cracks particularly in area of cotter pin holes. **NONE ALLOWED.**

**CAUTION**

Damage to bearing will result if contact occurs with cleaning solvents and fluids. Cleaning solvents and fluids can penetrate seals of bearing and dissolve lubrication grease. Do not spray on cleaning solvents or fluids in vicinity of bearing. Apply solvents only by wiping.

- (3) Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, thoroughly clean oil cooler drive shaft.
- (4) Touch up paint as required (PARA 2-6-5). Do not apply paint to boltholes or areas 1-inch around boltholes.

**NOTE**

Measurement of stub shaft is only required if wear is visually detected.

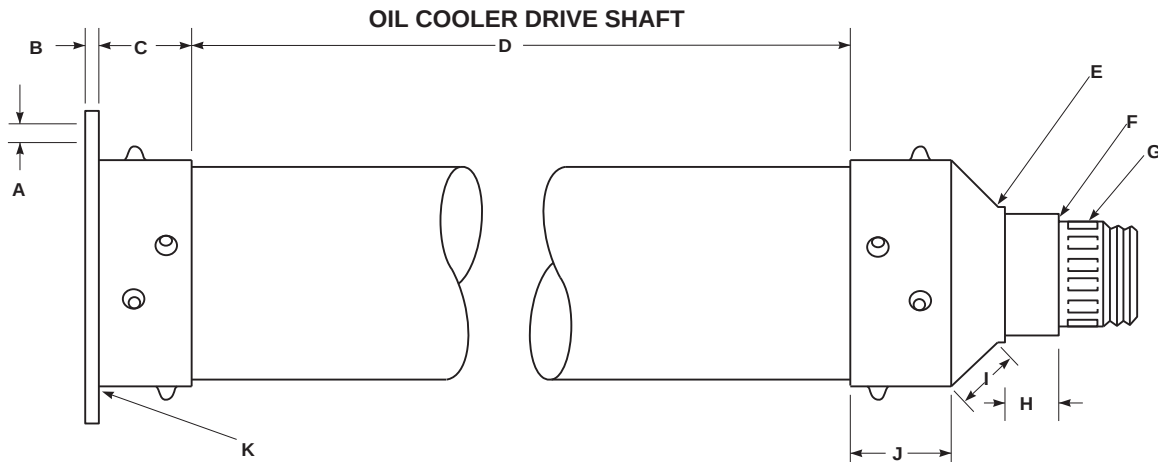
- i. Visually inspect splines on stub shaft for wear. If wear is detected, inspect as follows:
  - (1) Secure 0.096-inch diameter pin between two splines.

**NOTE**

Minor erosion may be noticed on blades of oil cooler fan. This is normal and not a cause for rejection.

- l. Inspect blades of oil cooler fan for nicks, gouges, and cracks. **NO CRACKS ALLOWED IN BLADE/HUB ATTACHMENT. NICKS AND GOUGES MUST NOT EXCEED 0.032-INCH DEEP.** If nicks and gouges do not exceed limit, repair as follows:
  - (1) Using abrasive cloth, Item 6, Appendix D, blend nicks and gouges in blades on oil





| COMPONENT  | ZONE | DIMENSION BEFORE REPAIR (INCH)    | REPAIR LIMIT (INCH)                      |
|------------|------|-----------------------------------|------------------------------------------|
| FLANGE     |      |                                   |                                          |
| HOLE       | A    | +0.322 (+0.005<br>-0.000) DIA     | 0.327 MAX DIA<br>0.0015<br>OUT-OF-ROUND  |
| LUG        | B    | 0.175 (±0.005)                    | 0.160 MIN THICKNESS                      |
| OD         | C    | 3.710 (±0.010)                    | 3.690 MIN DIA                            |
| TUBE       | D    | 3.500 (+0.005<br>-0.000) DIA      | 3.490 MIN DIA                            |
| STUB SHAFT |      |                                   |                                          |
| RADIUS     | E    | 0.250 (±0.005)                    | 0.230 MIN RADIUS                         |
| RADIUS     | F    | 0.032 (±0.007)                    | 0.020 MIN RADIUS                         |
| SPLINES    | G    | 1.9446 (±0.0012)                  | 1.9401 MIN DIA<br>(SEE NOTE 4)           |
| OD         | H    | 1.9686 (+0.007<br>-0.000) DIA     | 1.9682 MIN DIA                           |
|            | I    | 0.125 (±0.010) (WEB<br>THICKNESS) | 0.010 MAX DEPTH<br>BLEND TO 0.250 RADIUS |
| OD         | J    | 3.690 (±0.010) DIA                | 3.670 MIN DIA                            |
| FLANGE     |      |                                   |                                          |
| RADIUS     | K    | 0.125 (±0.010)                    | 0.105 MIN RADIUS<br>(SEE NOTE 3)         |

#### NOTES

- REPAIR LIMITATIONS:
  - REPAIR OF MINOR DAMAGE SUCH AS LIMITED NICKS, SCRATCHES, OR SCORING SHALL BE CLEANED UP BY POLISHING.
  - POLISHING SHALL BE DONE ONLY IN THE DAMAGED AREA AND NOT THE ENTIRE CIRCUMFERENCE UNLESS THE DAMAGE EXTENDS AROUND IT.
  - THE BLEND RADIUS ON SHAFT OD SHALL BE 1/2 IN. OR GREATER.
  - FINE SCRATCHES NEED NOT BE POLISHED IF NOT THROUGH PAINT.
  - AREAS WHERE PAINT HAS BEEN REMOVED ARE TO BE POLISHED WITH CROCUS CLOTH AND REPAINTED.
- DO NOT PAINT AROUND BOLTHOLES.
- BLEND RADIUS SHALL NOT BE LESS THAN 0.010 IN. BELOW THE MINIMUM TOLERANCES SHOWN IN "DIMENSION BEFORE REPAIR" COLUMN.
- MINIMUM DIAMETER OF SPLINES TO BE MEASURED OVER TWO 0.096 IN. DIAMETER PINS.

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FIGURE 6-4-50-5. INSPECT/REPAIR OIL COOLER DRIVE SHAFT

cooler fan that do not exceed 0.032-inch deep to a blend radius of  $\frac{1}{8}$ -inch or greater.

- (2) Using corrosion resistant compound, Item 118, Appendix D, touch up blended area of blades on oil cooler fan.
- m. Inspect bellmouth of oil cooler fan shroud for nicks and gouges. **NICKS AND GOUGES MUST NOT EXCEED 0.062-INCH DEEP.** If nicks and gouges do not exceed limit, repair as follows:
  - (1) Using abrasive cloth, Item 6, Appendix D, blend nicks and gouges in bellmouth that do not exceed 0.062-inch deep to blend radius of  $\frac{1}{8}$ -inch or greater.
  - (2) Using primer, Item 258, Appendix D, prime blended area of bellmouth (PARA 2-6-5).
  - (3) Using matching paint, touch up paint on bellmouth (PARA 2-6-5).
- n. Repair surface damage/corrosion on oil cooler fan shroud as follows:

**CAUTION**

Damage to oil cooler fan bearings and bonded joints will result if contact occurs with cleaning solvents and fluids. Cleaning solvents and fluids can penetrate seals and bonding joints of oil cooler fan bearing and dissolve lubrication grease and adhesives. Do not spray cleaning solvents or fluids in vicinity of oil cooler fan bearings or bonded joints. Apply solvents only by wiping.

**NOTE**

- Oil cooler fan shroud rework may be performed on helicopter.
- No balance checks are required after reworking oil cooler fan shroud per this procedure.

- (1) Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean oil and grease from oil cooler fan shroud.
- (2) Fine scratches that remove only paint may be blended smooth using abrasive cloth, Item 10, Appendix D.
- (3) Check depth of damage area. Blend damage area if damage can be fully repaired and blended without blending deeper than 0.010-inch. Using abrasive cloth, Item 8, Appendix D, or abrasive mat, Item 14, Appendix D, blend repair smooth. Remove only what is required to remove damage.
- (4) Using primer, Item 258, Appendix D, prime blended area of oil cooler fan shroud (PARA 2-6-5).
- (5) Using matching paint, touch up paint on oil cooler fan shroud (PARA 2-6-5).

- o. On oil cooler fans, 70361-03005-107, repair surface damage/corrosion on oil cooler one-piece shaft exit flange as follows:

**CAUTION**

Damage to bearing will result if contact occurs with cleaning solvents and fluids. Cleaning solvents and fluids can penetrate seals of bearing and dissolve lubrication grease. Do not spray on cleaning solvents or fluids in vicinity of bearing. Apply solvents only by wiping.

**NOTE**

- Oil cooler fan must be separated from oil cooler duct and oil cooler radiator.
- No balance checks are required after reworking oil cooler fan one-piece shaft exit flange per this procedure.
- Before starting repair of oil cooler fan one-piece shaft exit flange, determine

whether polishing out damage would result in dimension beyond repairable limit.

- (1) Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean oil and grease from oil cooler fan one-piece shaft exit flange.
  - (2) Using abrasive cloth, Item 10, Appendix D, blend fine scratches.
  - (3) Check depth of damage area. Blend damage area if damage can be fully repaired and blended without exceeding limits (Figure 6-4-50-6). Using abrasive cloth, Item 8, Appendix D, or abrasive mat, Item 14, Appendix D, blend repair smooth. Remove only what is required to remove damage.
- p. Repair surface damage/corrosion on oil cooler fan input flange as follows:



Damage to bearing will result if contact occurs with cleaning solvents and fluids. Cleaning solvents and fluids can penetrate seals of bearing and dissolve lubrication grease. Do not spray on cleaning solvents or fluids in vicinity of bearing. Apply solvents only by wiping.

#### NOTE

- Oil cooler fan must be separated from oil cooler duct and oil cooler radiator.
  - No balance checks are required after reworking oil cooler fan input flange per this procedure.
  - Before starting repair of oil cooler fan input flange, determine whether polishing out damage would result in dimension beyond repairable limits.
- (1) Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning

solvent, Item 124, Appendix D, clean oil and grease from oil cooler fan input flange.

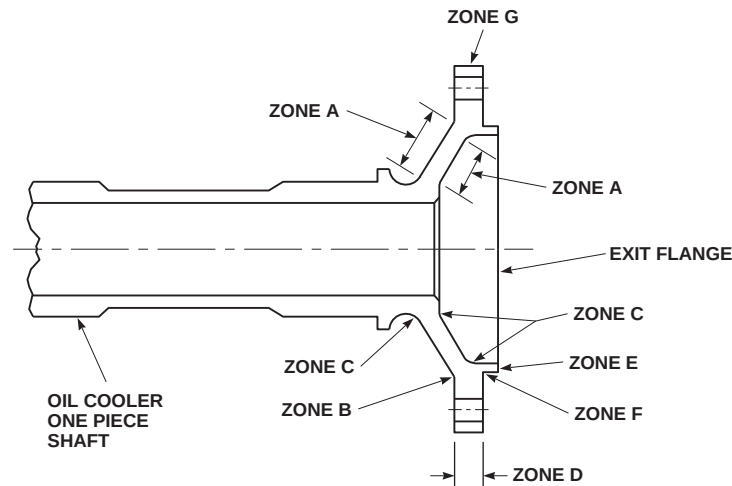
- (2) Using abrasive cloth, Item 10, Appendix D, blend fine scratches.
- (3) Check depth of damage area. Blend damage area if damage can be fully repaired and blended without exceeding limits (Figure 6-4-50-7). Using abrasive cloth, Item 8, Appendix D, or abrasive mat, Item 14, Appendix D, blend repair smooth. Remove only what is required to remove damage.
- (4) Using primer, Item 258, Appendix D, prime blended area of oil cooler fan input flange (PARA 2-6-5).
- (5) Using matching paint, touch up paint on oil cooler fan input flange (PARA 2-6-5).

q. Inspect oil cooler duct as follows:

- (1) Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean oil cooler duct of grease and dirt.
- (2) Inspect oil cooler duct for damage. If necessary, repair oil cooler duct (PARA 2-6-3).
- (3) Inspect vanes inside oil cooler duct for damage. **NONE ALLOWED.** If vanes are damaged, replace oil cooler duct.
- (4) Inspect all nutplates for damage and security. If necessary, replace nutplates.

r. Inspect exit flange retaining nut for corrosion. **REPLACE RETAINING NUT IF:**

- (1) **PITS ARE 0.030-INCH DEEP.**
- (2) **PITTING COVERS MORE THAN 10% OF MATING SURFACE.**
- (3) **THREADS SHOW SIGNS OF CORROSION.**



| COMPONENT                   | ZONE | DIMENSION BEFORE REPAIR (INCH) | REPAIR LIMIT (INCH) |
|-----------------------------|------|--------------------------------|---------------------|
| WEB                         | A    | 0.310/0.280                    | 0.270 MIN           |
| RADIUS                      | B    | 0.125 R ( $\pm 0.010$ )        | (SEE NOTE 3)        |
| RADIUS                      | C    | 0.250 R ( $\pm 0.010$ )        | (SEE NOTE 3)        |
| LUG                         | D    | 0.400 ( $\pm 0.010$ )          | 0.380 (SEE NOTE 2)  |
| FACE                        | E    | N/A                            | (SEE NOTE 4)        |
| RADIUS                      | F    | 0.060 ( $\pm 0.010$ )          | (SEE NOTE 3)        |
| OUTER SURFACE OF FLANGE EAR | G    | N/A                            | (SEE NOTE 4)        |

### EFFECTIVITY

ON HELICOPTERS WITH OIL COOLER FAN,  
70361-03005-107.

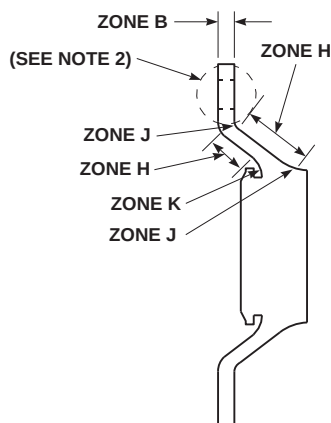
### NOTES

- REPAIR LIMITATIONS:
  - REPAIR OF MINOR DAMAGE SUCH AS LIMITED NICKS, SCRATCHES, OR SCORING SHALL BE CLEANED UP BY POLISHING.
  - POLISHING SHALL BE DONE ONLY IN THE DAMAGED AREA AND NOT THE ENTIRE CIRCUMFERENCE UNLESS THE DAMAGE EXTENDS AROUND IT.
  - THE BLEND RADIUS ON FLANGES SHALL BE 1/2 IN. OR GREATER.
- MAXIMUM FRETTING DEPTH IS 0.0005 IN. POLISH OUT MINOR NICKS, SCRATCHES, FRETTING AND SCORING USING CROCUS CLOTH.
- BLEND RADIUS SHALL NOT BE LESS THAN 0.010 IN. BELOW THE MINIMUM TOLERANCES SHOWN IN THE "DIMENSION BEFORE REPAIR" COLUMN.
- BLEND UP TO 0.020 IN. DEEP IN NO MORE THAN 10% OF THIS TOTAL SURFACE.

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**FIGURE 6-4-50-6. INSPECT/REPAIR OIL COOLER FAN ONE-PIECE SHAFT EXIT FLANGE**

## OIL COOLER INPUT FLANGE



| COMPONENT                                | ZONE | DIMENSION BEFORE REPAIR (INCH) | REPAIR LIMIT (INCH)                                      |
|------------------------------------------|------|--------------------------------|----------------------------------------------------------|
| FLANGE LUG                               | B    | 0.175 ( $\pm 0.005$ )          | 0.160                                                    |
| COUPLING FLANGE CONE                     | H    | 0.105 ( $\pm 0.005$ )          | 0.010 MAX DEPTH<br>BLEND TO 0.250 RADIUS<br>(SEE NOTE 3) |
| RADII                                    | J    | 0.125 ( $\pm 0.010$ ) RADIUS   |                                                          |
| RADII                                    | K    | 0.190<br>0.45 RADIUS           | 0.010 MAX DEPTH<br>BLEND TO 0.250 RADIUS<br>(SEE NOTE 3) |
| OUTER SURFACE OF FLANGE EARS (BOTH ENDS) |      |                                | 0.010 MAX DEPTH<br>BLEND TO 0.250 RADIUS<br>(SEE NOTE 3) |

## NOTES

- REPAIR LIMITATIONS:
  - REPAIR OF MINOR DAMAGE SUCH AS LIMITED NICKS, SCRATCHES, OR SCORING SHALL BE CLEANED UP BY POLISHING.
  - POLISHING SHALL BE DONE ONLY IN THE DAMAGED AREA AND NOT THE ENTIRE CIRCUMFERENCE UNLESS THE DAMAGE EXTENDS AROUND IT.
  - FINE SCRATCHES NEED NOT BE POLISHED IF NOT THROUGH PAINT.
  - AREAS WHERE PAINT HAS BEEN REMOVED ARE TO BE POLISHED WITH CROCUS CLOTH AND REPAINTED.
- DO NOT PAINT AROUND BOLTHOLES.
- BLEND RADIUS SHALL NOT BE LESS THAN 0.010 IN. BELOW THE MINIMUM TOLERANCES SHOWN IN "DIMENSION BEFORE REPAIR" COLUMN.

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FIGURE 6-4-50-7. INSPECT/REPAIR OIL COOLER FAN INPUT FLANGE

- s. Remove minor corrosion from retaining nut using abrasive cloth, Item 10, Appendix D, abrasive cloth, Item 1A, Appendix D, or abrasive mat, Item 14, Appendix D.

### 6-4-50.2.3 Install.

#### WARNING

- Injury to personnel and damage to equipment will result if retaining nut is not torqued properly. Loosening or removal of retaining nut may cause improper torque to be present and is not allowed. Only inspect and repair surfaces that do not require removal or loosening of retaining nut.
- Verify proper installation of attaching hardware, especially countersunk washers. Oil cooler fan, 70361-03005-107, requires different hardware.
- a. On oil cooler fans, 70361-03005-103 through 70361-03005-106, inspect oil cooler fan exit flange and retaining nut (PARA 6-3-15).
- b. Perform breakaway torque check of nuts securing oil cooler fan to oil cooler drive shaft as follows:

#### NOTE

Do not install bolts and nuts on components when performing this torque check.

- (1) Thread bolts into each nut until two threads stick out past end of nut.
- (2) Using torque wrench, back out bolt. **MINIMUM BREAKAWAY TORQUE IS 6.5 INCH-POUNDS.**
- c. If breakaway torque is not within limit, perform the following:
  - (1) Inspect bolts and nuts for evidence of corrosion or wear. **NONE ALLOWED.** If corrosion or wear is found, replace bolts or nuts.

- (2) Repeat step b.

#### NOTE

To prevent improper torquing of exit flange to oil cooler drive shaft, verify proper attaching hardware and proper installation of countersunk washers. Make sure that countersunk washer is installed with countersunk side facing bolthead as shown in Figure 6-4-50-4, Sheet 2, Detail B. Oil cooler fans, 70361-03005-107, have longer attaching bolts and countersunk washers. After assembly, verify that bolt thread extends past self-locking nut.

- d. On helicopters with oil cooler fans, 70361-03005-103 through 70361-03005-106, perform the following (Figure 6-4-50-4, Sheet 1, Detail A and Sheet 2, Detail C):
  - (1) Connect oil cooler fan exit flange to drive shaft with bolts and nuts (bolthead facing front).
  - (2) **TORQUE NUTS TO 171 - 189 INCH-POUNDS.**
- e. On helicopters with oil cooler fan, 70361-03005-107, perform the following (Figure 6-4-50-4, Sheet 1, Detail A and Sheet 2, Detail B):
  - (1) Connect oil cooler fan exit flange to oil cooler drive shaft with bolts, countersunk washers (if installed), and nuts (bolthead facing rear). Make sure countersunk side of washer is toward bolthead.
  - (2) **TORQUE NUTS TO 211 - 233 INCH-POUNDS.**

#### NOTE

When installing oil cooler fan shroud onto oil cooler duct, make sure drilled-head bolts are installed on both sides of accelerometer bracket position.

- f. Install oil cooler fan and oil cooler drive shaft into oil cooler duct using short bolts, drilled-head bolts, and washers (Figure 6-4-50-4, Sheet 1).

- g. **TORQUE SHORT BOLTS AND DRILLED-HEAD BOLTS TO 43 - 52 INCH-POUNDS.**
- h. Install accelerometer bracket with long bolt and washer.
- i. **TORQUE LONG BOLT TO 43 - 52 INCH-POUNDS.**
- j. Using lockwire, Item 197, Appendix D, lockwire accelerometer bracket to adjacent drilled-head bolt and long bolt to other adjacent drilled-head bolt.
- k. Perform oil cooler drive shaft runout check (PARA 6-4-50.3).
- l. Install sleeve adapter on oil cooler drive shaft.
- m. Using arbor press, press viscous damper bearing assembly onto oil cooler drive shaft.
- n. Line up index mark on exit flange and index mark on oil cooler drive shaft.
- o. Slide exit flange on oil cooler drive shaft.

**WARNING**

**FSCAP**

Verification of minimum torque of nut is a critical characteristic.

- p. Install retaining nut as follows:
  - (1) With bevel side of retaining nut toward seal, and making sure retaining nut can be fully turned by hand, install retaining nut.
  - (2) Using torque reactor and tail drive flange wrench, **TIGHTEN RETAINING NUT.**
  - (3) **TORQUE RETAINING NUT TO 214 - 236 FOOT-POUNDS.**

**NOTE**

Cotter pin is installed with head against inside diameter of retaining nut.

- (4) Remove torque reactor and install new cotter pin. **IF REQUIRED TO ALIGN COTTER PIN HOLE, TURN RETAINING NUT IN A TIGHTENING DIRECTION.**
- (5) Fold over rear end of cotter pin against retaining nut, then fold front end (flange against retaining nut mating surface) flush with retaining nut (Figure 6-4-50-4, Sheet 2, Detail D).
- (6) Apply torque stripe (PARA 2-6-5).
- q. Install oil cooler radiator (PARA 6-4-47).
- r. If removed, install oil cooler (PARA 6-4-50.1).

**6-4-50.3 Oil Cooler Drive Shaft Runout Check.**

- a. Remove oil cooler (PARA 6-4-50.1).
- b. Remove bolts and washers securing oil cooler radiator to oil cooler.
- c. Remove oil cooler radiator.
- d. Remove retaining nut, exit flange, and viscous damper bearing assembly (PARA 6-4-50.2).
- e. Attach dial indicator to transmission oil cooler adapter near rear of oil cooler. Make sure dial indicator will not be disturbed during runout check.
- f. Place dial indicator pointer on journal face of oil cooler drive shaft bearing. Make sure dial indicator pointer is perpendicular to centerline of oil cooler drive shaft.

**NOTE**

A clicking or rattling noise when oil cooler fan is rotated is not an indication of possible bearing failure. Actual operating loads are not being simulated by rotating oil cooler fan. Bearing can only be checked by performing the following procedures.

- g. Rotate oil cooler drive shaft by hand through oil cooler fan input flange. Make sure no force

is placed on oil cooler fan input flange that will affect runout reading. **RUNOUT MUST NOT EXCEED 0.010-INCH.**

h. If runout is greater than 0.010-inch, perform the following:

(1) Loosen bolts and nuts attaching oil cooler drive shaft to oil cooler fan input flange.

(2) **TIGHTEN BOLTS AND NUTS UNTIL OIL COOLER DRIVE SHAFT IS SECURE. DO NOT TORQUE BOLTS NOW.**

(3) Repeat runout check. Note point where greatest change in dial indicator reading occurs. Torque nut opposite point of greatest change as follows:

(a) **ON OIL COOLER FANS, 70361-03005-103 THROUGH 70361-03005-106, TORQUE NUTS TO 171 - 189 INCH-POUNDS.**

(b) **ON OIL COOLER FAN, 70361-03005-107, TORQUE NUTS TO 211 - 233 INCH-POUNDS.**

(4) Repeat step h. (3) until all nuts attaching oil cooler drive shaft to oil cooler fan exit flange are torqued.

(5) Repeat runout check. **RUNOUT MUST NOT EXCEED 0.010-INCH.**

i. If runout is still greater than 0.010-inch, perform the following:

(1) Remove bolts and nuts attaching oil cooler drive shaft to oil cooler fan exit flange.

(2) Turn oil cooler drive shaft one bolthole with respect to oil cooler fan exit flange.

(3) Install oil cooler drive shaft to oil cooler fan exit flange with bolts, washers, and nuts. **TIGHTEN BOLTS AND NUTS UNTIL OIL COOLER DRIVE SHAFT IS SECURE. DO NOT TORQUE BOLTS NOW.**

(4) Repeat step h. (3) and step h. (4).

(5) Repeat runout check. **RUNOUT MUST NOT EXCEED 0.010-INCH.** Repeat step i. (1) through step i. (4) until runout is within limit.

(6) If all combinations of boltholes have been used and runout still exceeds 0.010-inch, replace oil cooler drive shaft. Repeat step b. through step g. for new oil cooler drive shaft.

(7) If new oil cooler drive shaft runout still exceeds 0.010-inch, replace oil cooler fan (PARA 6-4-50.2).

j. Install retaining nut, exit flange, and viscous damper bearing assembly (PARA 6-4-50.2).

k. Install oil cooler radiator on oil cooler and secure with bolts and washers.

l. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**

m. Install oil cooler (PARA 6-4-50.1).

#### 6-4-50.4 Input Flange Retaining Nut Torque Check.

##### NOTE

The following procedure is not required on new oil coolers being installed.

a. Remove oil cooler (PARA 6-4-50.1).

b. With retaining nut of oil cooler fan input flange exposed, perform torque check as follows:

(1) Remove cotter pin from retaining nut.

##### NOTE

- Torque reactor will be attached to oil cooler fan input flange to react against torquing of retaining nut.
- Check torque in tightening direction while holding oil cooler fan shaft with torque reactor.



- (2) Using torque reactor and tail drive flange wrench, check torque of retaining nut. **TORQUE MUST BE 214 - 236 FOOT-POUNDS.**
- (3) **IF TORQUE IS 214 - 236 FOOT-POUNDS, TORQUE CHECK IS COMPLETE.** Remove torque reactor and install new cotter pin as follows:

**NOTE**

Cotter pin is installed with head against inside diameter of retaining nut (Figure 6-4-50-5, Sheet 2, Detail B).

- (a) Install new cotter pin. If required to align cotter pin hole, turn retaining nut in tightening direction.
- (b) Fold over rear end of cotter pin against retaining nut, then fold front end (flange against retaining nut mating surface) flush with retaining nut.
- (c) Apply torque stripe (PARA 2-6-5).

- (4) **IF RETAINING NUT TORQUE IS 180 - 213 FOOT-POUNDS, TORQUE RETAINING NUT TO 214 - 236 FOOT-POUNDS. TORQUE CHECK IS COMPLETE.** Remove torque reactor and install cotter pin per step b. (3) (a) through b. (3) (c).
- (5) **IF RETAINING NUT TORQUE IS LESS THAN 180 FOOT-POUNDS, REPLACE OIL COOLER FAN (PARA 6-4-50.2).**

- c. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to exposed surfaces of retaining nut on oil cooler fan input flange retaining.
- d. Apply corrosion preventive compound, Item 113, Appendix D, to retaining nut of oil cooler fan input flange and area around retaining nut. Do not apply compound to flange mating surfaces.
- e. Install oil cooler (PARA 6-4-50.1).



6-4-51 OIL COOLER SPLINE WEAR INDICATOR SYSTEM.

PARAGRAPH BREAKDOWN

|                                                    | PAGE     |
|----------------------------------------------------|----------|
| 6-4-51.1 Spline Wear Indicator System Installation | 6-4-51-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Ruler, 6-inch (or equivalent straight edge)
- Spring Scale
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Adhesive, Item 37, Appendix D
- Brush, Artist, Item 85, Appendix D
- Epoxy Coating Kit, Item 126A, Appendix D

- Machinery Towel, Item 211, Appendix D
- Masking Tape, 1-Inch, Item 212, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint, Polyurethane, Item 225, Appendix D
- Paint, Polyurethane, Item 228, Appendix D
- Tiedown Strap, Item 336, Appendix D

References

- Appendix D
- PARA 2-4-113
- PARA 6-3-15

6-4-51.1 Spline Wear Indicator System Installation.



NOTE

This procedure applies only to helicopters with oil cooler fans, 70361-03005-103 through 70361-03005-106, installed.

- Open front tail rotor drive shaft cover.
- Remove oil cooler compartment access doors (PARA 2-4-113).

Damage to bearing will result if contact occurs with cleaning solvents and fluids. Cleaning solvents and fluids can penetrate seals of bearing and dissolve lubrication grease. Do not spray on cleaning solvents or fluids in vicinity of bearing. Apply solvents only by wiping.

- Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean area on tail rotor drive shaft

near No. 3 viscous damper bearing assembly at STA 531.0 (Figure 6-4-51-1, Sheet 1). Using clean machinery towel, wipe dry.

### NOTE

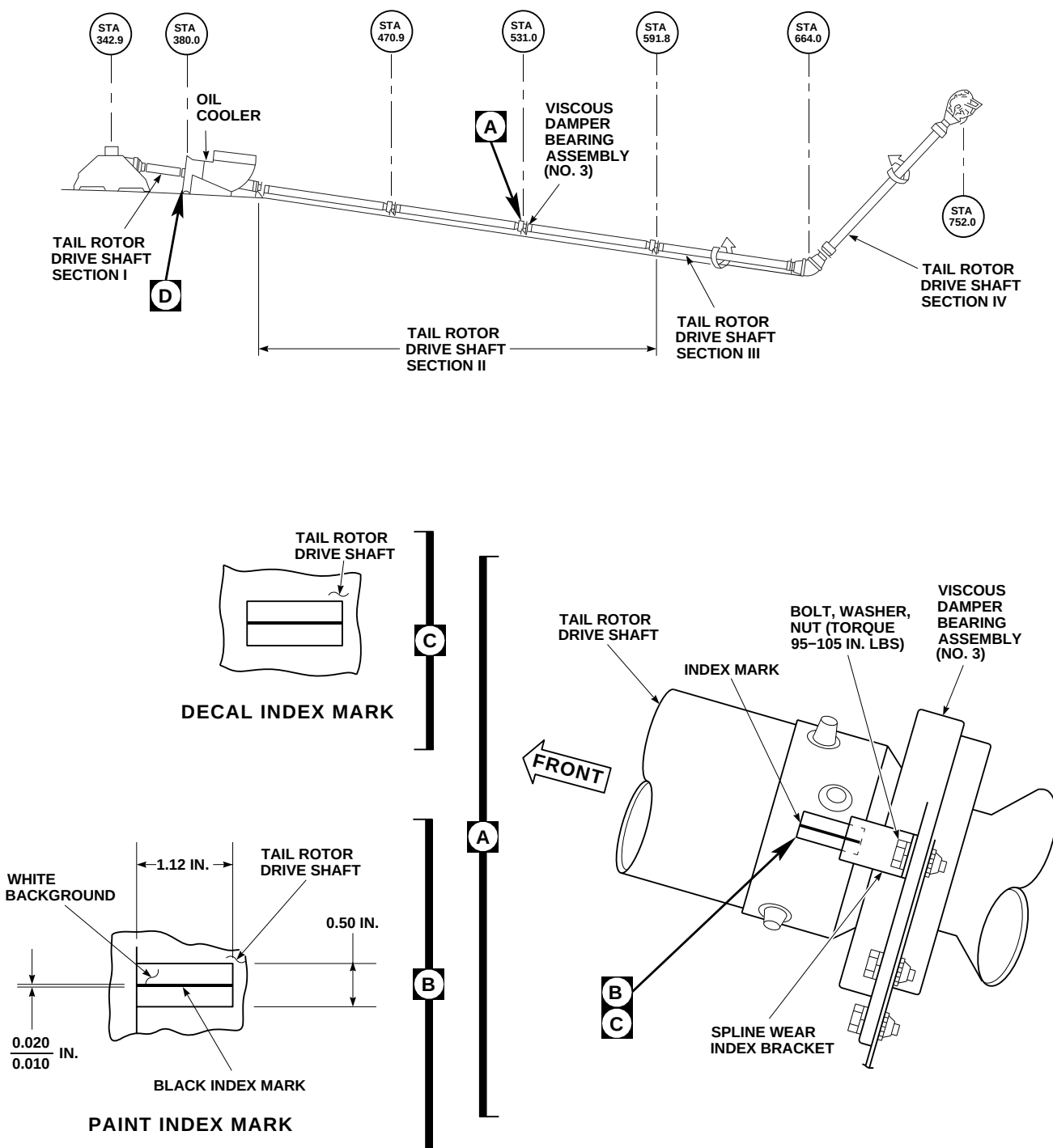
- Location of decal on circumference of tail rotor drive shaft is not important.
  - Black index mark line on decal is to be installed parallel with centerline of tail rotor drive shaft.
- d. To install decal index mark on tail rotor drive shaft, perform the following:
- (1) Using adhesive, Item 37, Appendix D, bond decal to cleaned area on tail rotor drive shaft (Figure 6-4-51-1, Sheet 1, Detail C). Allow to dry.
  - (2) Using artist brush, Item 85, Appendix D, apply a coat of polyurethane paint, Item 228, Appendix D, over decal. Extend coat 1/4-inch maximum beyond edges of decal.

### NOTE

- If decal index mark for tail rotor drive shaft and wear indicator decal for oil cooler fan housing are available, install in place of paint index mark.
  - Location of paint index mark on circumference of tail rotor drive shaft is not important.
  - Black index mark line is to be applied parallel with centerline of tail rotor drive shaft.
- e. To apply paint index mark to tail rotor drive shaft, perform the following:
- (1) Using 1-inch masking tape, Item 212, Appendix D, mask cleaned area on tail rotor

drive shaft to dimensions shown in Figure 6-4-51-1, Sheet 1, Detail B.

- (2) Using polyurethane paint, Item 225, Appendix D, and artist brush, Item 85, Appendix D, paint background inside masked area. Allow to dry.
  - (3) Using 1-inch masking tape, Item 212, Appendix D, mask background area to dimension shown in Figure 6-4-51-1, Sheet 1, Detail B, for black index mark.
  - (4) Using epoxy coating kit, Item 126A, Appendix D, with artist brush, Item 85, Appendix D, paint index mark on background. Allow to dry.
  - (5) Using artist brush, Item 85, Appendix D, apply polyurethane, Item 228, Appendix D, over index mark and background. Allow to dry.
  - (6) Remove masking tape.
- f. Install spline wear index bracket to left side of No. 3 viscous damper bearing assembly at STA 531.0 as follows:
- (1) Remove upper left bolt, washer, and nut from No. 3 viscous damper bearing assembly (Figure 6-4-51-1, Sheet 1, Detail A).
  - (2) Install spline wear index bracket on No. 3 viscous damper bearing assembly and secure with new bolt, washer, and nut.
  - (3) **TORQUE NUT TO 95 - 105 INCH-POUNDS.**
- g. Rotate tail rotor blade in direction of normal rotor rotation until index mark on tail rotor drive shaft aligns with spline wear index bracket slot on No. 3 viscous damper bearing



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FIGURE 6-4-51-1. SPLINE WEAR INDICATOR SYSTEM INSTALLATION (SHEET 1 OF 2)

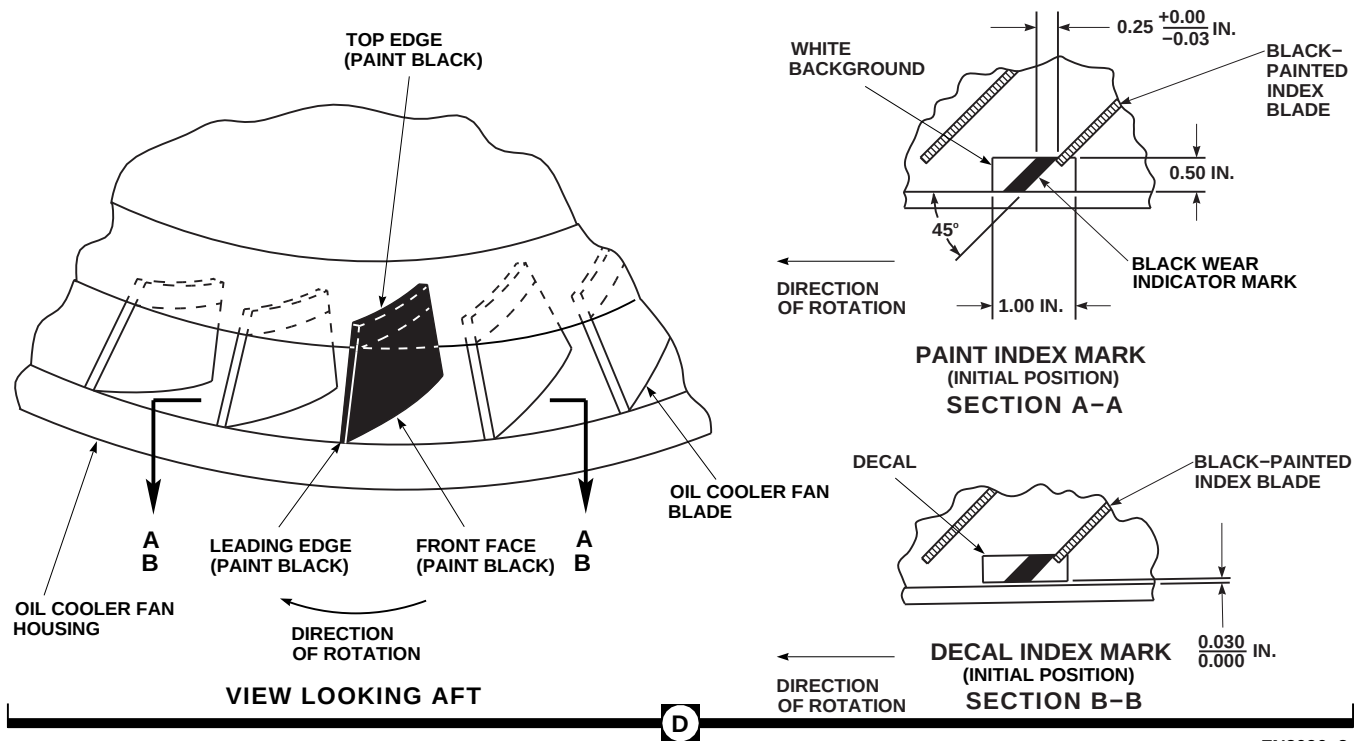
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FIGURE 6-4-51-1. SPLINE WEAR INDICATOR SYSTEM INSTALLATION (SHEET 2 OF 2)

assembly. Use 6-inch ruler (or equivalent straight edge) to line up bracket slot with black index mark line.

### WARNING

Spline wear inspection cannot be accurately performed if alignment of index mark on oil cooler drive shaft and spline wear index bracket on No. 3 viscous damper bearing assembly is disturbed while installing index marks on oil cooler fan. False spline wear readings will result in loss of tail rotor drive. Make sure index mark on tail rotor drive shaft and spline wear index bracket slot remain aligned.

- h. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean area on oil cooler fan housing at approximately 6 o'clock position (Figure 6-4-51-1, Sheet 2, Detail D). Using clean machinery towel, wipe dry.

- i. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean leading edge, top edge and front face of oil cooler fan blade at approximately 6 o'clock position. Using clean machinery towel, wipe dry.
- j. Using epoxy coating kit, Item 126A, Appendix D, with artist brush, Item 85, Appendix D, paint leading edge, top edge, and front face of oil cooler fan blade. Allow to dry.
- k. Using artist brush, Item 85, Appendix D, apply two coats of polyurethane paint, Item 228, Appendix D, over painted area on oil cooler fan blade. Allow to dry.

### NOTE

If decal index mark for oil cooler fan housing is available, install decal in place of paint index mark. Proceed to step m.

- l. To apply paint index mark to oil cooler fan housing, perform the following:

- (1) Using 1-inch masking tape, Item 212, Appendix D, mask cleaned area on oil cooler fan housing to dimensions shown in Figure 6-4-51-1, Sheet 2, Section A-A.
- (2) Using polyurethane paint, Item 225, Appendix D, and artist brush, Item 85, Appendix D, paint background inside masked area. Allow to dry.
- (3) Engage gust lock.
- (4) Using tiedown strap, Item 336, Appendix D, and spring scale, apply a 20 pound load for 5 seconds to tail rotor blade trailing edge. Make sure tail rotor blade is rotated in direction of normal rotation.
- (5) Disengage gust lock with load applied.
- (6) Slowly rotate tail rotor blade in direction of normal rotation until index mark on tail rotor drive shaft aligns with spline wear index bracket slot on No. 3 viscous damper bearing assembly. Use 6-inch ruler (or equivalent straight edge) to line up bracket slot with black index mark line.
- (7) If index mark on tail rotor drive shaft rotates past spline wear index bracket slot on No. 3 viscous damper bearing assembly, continue to rotate blade in direction of normal rotation until index mark is aligned with slot.
- (8) If tail rotor is inadvertently rotated in direction opposite normal rotation, repeat steps l. (3) through l. (7).

**WARNING**

Spline wear inspection cannot be accurately performed if alignment of index mark on oil cooler drive shaft and spline wear index bracket on No. 3 viscous damper bearing assembly is disturbed while installing index marks on oil cooler fan. False spline wear readings

will result in loss of tail rotor drive. Make sure index mark on tail rotor drive shaft and spline wear index bracket slot remain aligned.

**NOTE**

Black-painted oil cooler fan index blade must be at approximately 6 o'clock position.

- (9) Using 1-inch masking tape, Item 212, Appendix D, mask background area on oil cooler fan housing to dimensions shown in Figure 6-4-51-1, Sheet 2, Section A-A. Make sure that masking is properly aligned with black-painted index blade in INITIAL POSITION.
  - (10) Using epoxy coating kit, Item 126A, Appendix D, with artist brush, Item 85, Appendix D, paint wear indicator mark inside masked area. Allow to dry.
  - (11) Remove masking tape.
  - (12) Repeat steps l. (6) through l. (8).
  - (13) Check position of black-painted index fan blade. **BLACK-PAINTED INDEX BLADE MUST BE AT INITIAL POSITION.**
  - (14) If black-painted index blade is at INITIAL POSITION, using artist brush, Item 85, Appendix D, apply two coats of polyurethane paint, Item 228, Appendix D, to black wear indicator mark and white background.
  - (15) If black-painted index blade is not at INITIAL POSITION, perform the following:
    - (a) Remove painted index mark from oil cooler fan housing.
    - (b) Repeat steps l. (1) through l. (15).
- m. To apply decal index mark to oil cooler fan housing, perform the following:

- (1) Engage gust lock.
- (2) Using tiedown strap, Item 336, Appendix D, and spring scale, apply a 20 pound load for 5 seconds to tail rotor blade trailing edge. Make sure tail rotor blade is rotated in direction of normal rotation.
- (3) Disengage gust lock with load applied.
- (4) Slowly rotate tail rotor blade in direction of normal rotation until index mark on tail rotor drive shaft aligns with spline wear index bracket slot on No. 3 viscous damper bearing assembly. Use 6-inch ruler (or equivalent straight edge) to line up bracket slot with black index mark line.
- (5) If index mark on tail rotor drive shaft rotates past spline wear index bracket slot on No. 3 viscous damper bearing assembly, continue to rotate blade in direction of normal rotation until index mark is aligned with slot.
- (6) If tail rotor is inadvertently rotated in direction opposite normal rotation, repeat steps m. (1) through m. (5).

### WARNING

Spline wear inspection cannot be accurately performed if alignment of index mark on oil cooler drive shaft and spline wear index bracket on No. 3 viscous damper bearing assembly is disturbed while installing index marks on oil cooler fan. False spline wear readings will result in loss of tail rotor drive. Make sure index mark on tail rotor drive shaft and spline wear index bracket slot remain aligned.

### NOTE

Black-painted oil cooler fan index blade must be at approximately 6 o'clock position.

- (7) Using adhesive, Item 37, Appendix D, bond decal index mark to cleaned area on oil cooler fan housing. Make sure decal is properly aligned with black-painted index blade in INITIAL POSITION (Figure 6-4-51-1, Sheet 2, Section B-B). Allow to dry.
- (8) Repeat steps m. (4) through m. (6).
- (9) Check position of black-painted index fan blade. **BLACK-PAINTED INDEX BLADE MUST BE AT INITIAL POSITION.**
- (10) If black-painted index blade is at INITIAL POSITION, apply polyurethane paint, Item 228, Appendix D, to decal. Extend coat 1/4-inch maximum beyond edges of decal.
- (11) If black-painted index blade is not at INITIAL POSITION, perform the following:
  - (a) Remove decal from oil cooler fan housing.
  - (b) Repeat steps m. (1) through m. (11).
- n. Install oil cooler compartment access doors (PARA 2-4-113).
- o. Close front tail rotor drive shaft cover.
- p. Perform oil cooler spline wear inspection (PARA 6-3-15).



**6-4-52 TAIL GEAR BOX.****PARAGRAPH BREAKDOWN**

|                                | <b>PAGE</b> |
|--------------------------------|-------------|
| 6-4-52.1 Replace Tail Gear Box | 6-4-52-2    |

**INITIAL SETUP****Tools**

- Aircraft Mechanic's Toolkit
- Airframe Repairer's Toolkit
- Drill
- Drill Bit, No. 29
- Feeler Gage
- Flashlight
- Fluorescent Penetrant Inspection Kit
- Hoist, Capacity to Support Tail Gear Box (Alternate for Tail Rotor Maintenance Crane)
- Hoisting Sling, 70700-20320-042
- Inspection Mirror
- Nonmetallic Scraper, Locally-Made
- Nonpermanent Ink Marker
- Powertrain Repairer's Toolkit
- Tail Rotor Maintenance Crane, 70700-20413-041 (Alternate for Hoist)
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Universal Hoist Sling Assembly, 70700-20320-045

**Materials/Parts**

- Antiseize Compound, Item 74, Appendix D
- Brush, Acid, Item 84, Appendix D
- Corrosion Preventive Compound, Item 113, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Gear Box Shim, 70219-06000-103, Locally-Made, Figure H-199, Appendix H
- Gear Box Shim, 70219-06000-104, Locally-Made, Figure H-199, Appendix H
- Gear Box Shim, 70219-06000-112, Locally-Made, Figure H-199, Appendix H
- Gear Box Spacer, 70219-06000-101, Locally-Made, Figure H-200, Appendix H
- Gear Box Spacer, 70219-06000-102, Locally-Made, Figure H-200, Appendix H
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D
- Masking Tape, 3-inches, Item 215, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Molybdenum Disulfide, Item 219, Appendix D
- Primer, Item 255, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

- Tags

## References

- Appendix D
- Appendix H
- Appropriate MTF
- ASTM E1417
- PARA 1-3-14
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-6-18
- PARA 1-7-8
- PARA 1-7-10
- PARA 1-7-20
- PARA 2-4-137

- PARA 5-4-38
- PARA 6-4-41
- PARA 6-4-57
- PARA 9-4-83
- PARA 9-4-86
- PARA 11-4-6
- PARA 11-4-106
- PARA 11-4-107
- PARA 11-4-113
- PARA 11-4-117
- PARA 11-4-118
- PARA 11-4-119
- PARA 12-4-52
- PARA 18-4-2

### 6-4-52.1 Replace Tail Gear Box.

#### NOTE

Tail gear box with servo installed weighs 125 pounds.

#### 6-4-52.1.1 Remove.

#### NOTE

If tail gear box is being replaced because of contamination, make sure all foreign material from tail gear box chip detector/temperature sensor and oil is carefully packaged and returned with it.

- Turn off all electrical and hydraulic power (PARA 1-6-16 and PARA 1-6-15).
- If not already done, engage gust lock (PARA 1-6-18).

- Disconnect and tag wiring and grounding straps from anticollision and tail incandescent position lights (PARA 9-4-86 and PARA 9-4-83).
- Drain tail gear box (PARA 1-3-14). Do not install tail gear box fairing.
- If tail rotor maintenance crane is going to be used, assemble and erect crane (PARA 18-4-2).
- Remove tail rotor blades (PARA 5-4-38).
- Open tail rotor pylon leading edge cover/antenna.
- Support tail rotor drive shaft section IV and disconnect from tail gear box input flange. Leave flexible coupling with tail rotor drive shaft section IV (PARA 6-4-41).
- Disconnect electrical connector, P610 (1st stage pressure switch), and P609 (2nd stage

### 6-4-52-2 Change 10

pressure switch) from tail rotor servo (Figure 6-4-52-1, Detail A).

- j. Install protective caps and plugs on electrical connectors.

### NOTE

Bracket angles are located on the 2nd stage connector only.

- k. Remove bolts and washers securing 1st and 2nd stage hydraulic connectors and bracket angles to tail rotor servo. Disconnect hydraulic connectors from tail rotor servo.
- l. Disconnect electrical connector, P611, from top of tail pylon (Figure 6-4-52-1, Detail B).
- m. Install protective caps and plugs on electrical connectors.
- n. Disconnect tail rotor pylon flight control cables (PARA 11-4-106).
- o. Disconnect electrical connector, P614, from tail gear box chip detector/temperature sensor.
- p. Install protective caps and plugs on electrical connectors.
- q. Attach hoisting sling to tail gear box.

### WARNING

- Hoist cable may break, causing injury to personnel or damage to equipment, if hoist is used to free sticking component being removed from helicopter. Do not use too much force when removing component.
- Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment if hook is allowed to contact pulley when removing component from helicopter. Do not allow hook assembly to contact pulley.
- r. Attach hoisting sling to hoist or tail rotor maintenance crane. Adjust length of straps. Take up any slack.

- s. If blade deice kit is installed, remove tail rotor deice slipping rotor (PARA 12-4-52).
- t. Remove bolts securing tail gear box to mount fittings.

### WARNING

Injury to personnel and damage to equipment will result if caution is not taken when hoisting tail gear box. Keep area clear when hoisting tail gear box.

- u. Lift tail gear box from mount fittings on tail pylon and place on work surface. Remove hoisting sling.
- v. If new tail gear box is being installed, remove the following components from tail gear box:
  - (1) Tail rotor quadrant spring cylinders (PARA 11-4-118).
  - (2) Tail rotor quadrant support (PARA 11-4-117).
  - (3) Tail rotor quadrant spring cylinder support (PARA 11-4-119).
  - (4) Flight control cable pulleys (PARA 11-4-107).
  - (5) Tail rotor servo and pitch control shaft (PARA 11-4-113).

### 6-4-52.1.2 Inspect.

### NOTE

When inspecting tail gear box fitting, pay particular attention to joining area of fitting to front spar caps in area of high strength fasteners.

- a. Inspect tail gear box fitting for surface cracks around lower barrel-nut holes. **NONE ALLOWED.**
- b. Using inspection mirror and flashlight through access hole in front spar web, inspect inside

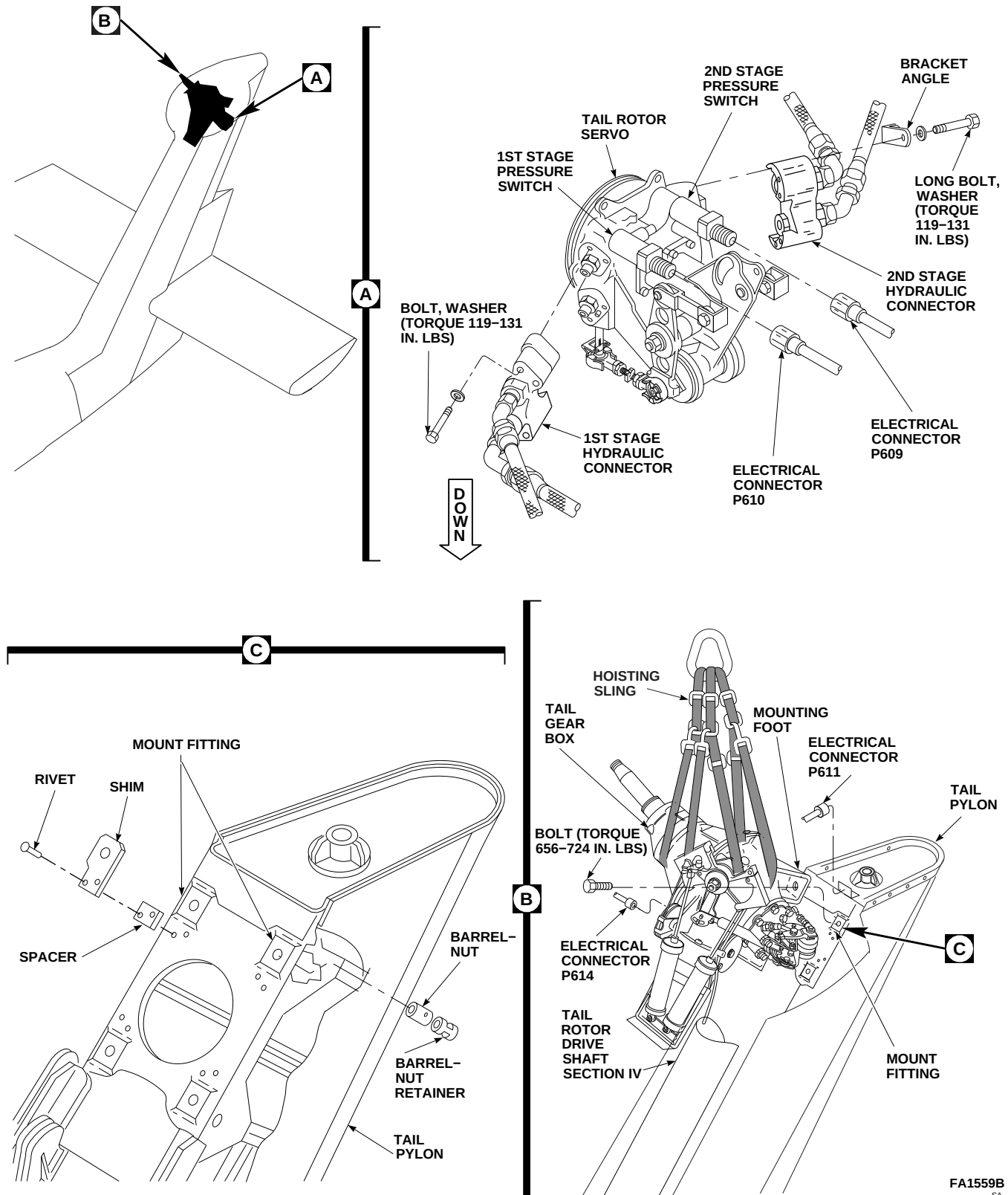
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FIGURE 6-4-52-1. REPLACE TAIL GEAR BOX

surfaces of fitting where attached to front spar cap angles for cracks. **NONE ALLOWED.** One-time flight to approved maintenance facility is allowed.

### 6-4-52.1.3 Install.

- a. If new tail gear box is being installed, install the following components:
  - (1) Tail rotor servo and pitch control shaft (PARA 11-4-113).
  - (2) Tail rotor quadrant spring cylinder support (PARA 11-4-119).
  - (3) Tail rotor quadrant support (PARA 11-4-117).
  - (4) Flight control cable pulleys (PARA 11-4-107).
  - (5) Tail rotor quadrant spring cylinders (PARA 11-4-118).
- b. Check breakaway torque of barrel-nuts in mount fittings that secure tail gear box as follows (Figure 6-4-52-1, Detail C):



Damage to barrel-nut fitting will result if barrel-nut is incorrectly installed. To prevent damage to fitting, make sure barrel-nut is installed with barrel (round side) towards bolthead.

### NOTE

Make sure only barrel-nuts, RMLH2577, are used. These barrel-nuts are identified with letters "EN" or "SPS" stamped on an end (additional marking may also include "LH2577").

- (1) Install bolt into barrel-nut until two threads extend beyond barrel-nut.
- (2) Using torque wrench, back out bolt.

- (3) **MINIMUM BREAKAWAY TORQUE IS 14 INCH-POUNDS.** If breakaway torque is less than 14 inch-pounds, replace barrel-nut.

- c. Using nonmetallic scraper, remove any protective sealing compound from mounting surfaces on tail gear box and from mount fittings on airframe.
- d. Attach hoisting sling to tail gear box (Figure 6-4-52-1, Detail B).



Injury to personnel and damage to equipment will result if caution is not taken when hoisting tail gear box. Keep area clear when hoisting tail gear box.

- e. Lift tail gear box into place on mount fittings. Make sure harness going under tail gear box is in position and not crushed by tail gear box.
- f. Install tail gear box to mount fittings with bolts. **TIGHTEN BOLTS AS REQUIRED UNTIL BOLTHEAD MAKES LIGHT CONTACT WITH TAIL GEAR BOX MOUNTING FEET. DO NOT EXCEED 100 INCH-POUNDS.**
- g. Using feeler gage, check for gaps between mounting feet and mount fittings. **MAXIMUM GAP IS 0.030-INCH.** If gap exceeds 0.030-inch under any mounting foot, install shim to bring gap within limits as follows:

- (1) Remove bolts securing tail gear box and remove tail gear box.

### NOTE

If shim was previously installed on a mounting pad which now requires a thicker shim, existing shim must be removed.

- (2) Remove rivets securing shim to mount fitting (Figure 6-4-52-1, Detail C).

- (3) Using nonmetallic scraper, remove original gear box shim and gear box spacer from mount fitting.
- (4) Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, remove old sealing compound from mounting fitting.

**NOTE**

- Use only one shim, one spacer, and two rivets on each mounting pad.
  - Replacement shim thickness must be original shim thickness plus thickness required for correction.
- (5) Locally make replacement gear box shim (Figure H-199, Appendix H), and gear box spacer (Figure H-200, Appendix H).
  - (6) Position replacement gear box shim and gear box spacer on mount fitting.
  - (7) Using suitable nonpermanent ink marker, mark location of fastener holes on shim.
  - (8) Using drill with No. 29 drill bit, drill holes in shim.
  - (9) Apply coat of primer, Item 255, Appendix D, to all gear box shim and gear box spacer surfaces. Allow to cure.
  - (10) Using acid brush, Item 84, Appendix D, apply brush-coat of sealing compound, Item 292, Appendix D, to mount fitting.
  - (11) Install gear box shim and gear box spacer, 70219-06000-101, on forward mount fittings as required.
  - (12) Install gear box shim and gear box spacer, 70219-06000-102, on aft mount fittings as required.
  - (13) Secure shim and spacer using rivets.
  - (14) Lift tail gear box into place on mount fittings. Make sure harness going under tail

gear box is in position and not crushed by tail gear box.

- h. Install bolts and secure tail gear box to mount fittings (Figure 6-4-52-1, Detail B).
- i. **TORQUE BOLTS TO 656 - 724 INCH-POUNDS.**
- j. Using feeler gage, measure gap between mounting feet and mounting fitting. **NO GAP ALLOWED.** If any gaps exist, perform the following:
  - (1) Remove bolts securing tail gear box and remove tail gear box.
  - (2) Inspect tail gear box housing (PARA 6-4-57).
  - (3) Using fluorescent penetrant inspect kit, fluorescent penetrant inspect mounting legs for cracks (ASTM E1417). **NONE ALLOWED.** If cracks are found in mounting legs, replace tail gear box. If no cracks are found, repeat step f. through step j.
- k. Remove bolts securing tail gear box and remove tail gear box.
- l. Using molybdenum disulfide, Item 219, Appendix D, coat mount fittings.
- m. Apply coat of sealing compound, Item 292, Appendix D, to mounting feet.
- n. Lift tail gear box into place. Make sure harness going under tail gear box is in position and not crushed by tail gear box.

**WARNING****FSCAP**

Verification of minimum torque of bolts is a critical characteristic.

- o. Using antiseize compound, Item 74, Appendix D, apply light coat to bolts.

- p. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to shoulders of bolt and to areas of tail gear box around bolt-holes.
- q. Install bolts while primer is still wet. It may be easier to install top two bolts first.
- r. **TORQUE BOLTS TO 656 - 724 INCH-POUNDS.**
- s. Remove hoisting sling.
- t. Form fillet with squeezed out sealing compound around base of mounting feet.
- u. Using 3-inch masking tape, Item 215, Appendix D, mask off adjacent areas to prevent overspray.
- v. Using corrosion preventive compound, Item 113, Appendix D, apply to entire area of mounting feet.
- w. Allow corrosion preventive compound to cure for 10 to 30 minutes.
- x. Using corrosion preventive compound, Item 113, Appendix D, apply second coat.
- y. Inspect every three days and using corrosion preventive compound, Item 113, Appendix D, reapply two coats as necessary until bolt torque stabilizes.
- z. If blade deice kit is installed, install tail rotor blade deice slipping rotor (PARA 12-4-52).

**WARNING**

Injury to personnel and damage to equipment will result if inboard retention plate is installed without lockwire on bolts and jamnuts. Make sure bolts and jamnuts are lockwired prior to installation of tail rotor blades.

- aa. Check inboard retention plate bolts and jamnuts for lockwire. If required, using lockwire, Item 197, Appendix D, lockwire bolts and jamnuts.

- ab. Check tail rotor drive shaft section IV to tail gear box coupling alignment as follows:

**NOTE**

- Spherical washers are to be installed on each side of tail rotor drive shaft flexible coupling.
  - Spherical side of washer must face tail rotor drive shaft flexible coupling.
- (1) Support tail rotor drive shaft section IV and connect shaft to tail gear box flange using bolts and spherical washers. Do not install nuts or shims.
  - (2) Measure and record gap between flange and spherical washer at the three attachment points. Do not rotate shaft when measuring gaps.
  - (3) Calculate difference between maximum and minimum gaps and record. **MAXIMUM GAP DIFFERENCE MUST NOT EXCEED 0.017-INCH.**
  - (4) If gap difference exceeds limit, remove tail gear box and recheck tail gear box mounting.
  - (5) Support tail rotor drive shaft section IV and remove bolts and washers.

- ac. Connect tail rotor drive shaft section IV to tail gear box (PARA 6-4-41).
- ad. Connect tail rotor pylon flight control cables (PARA 11-4-106).
- ae. Remove protective caps and plugs from electrical connectors.
- af. Inspect and treat electrical connectors (PARA 1-7-20).
- ag. Connect electrical connector, P614, to tail gear box chip detector/temperature sensor.
- ah. Align 1st stage connector with guide pins and tail rotor servo fittings. Secure 1st stage connector to tail rotor servo using bolts and wash-



ers. **TORQUE BOLTS TO 119 - 131 INCH-POUNDS.** Using lockwire, Item 197, Appendix D, lockwire bolts (Figure 6-4-52-1, Detail A).

- ai. Align 2nd stage connector with guide pins and tail rotor servo fittings. Secure 2nd stage connector and bracket angles to tail rotor servo using long bolts and washers. **TORQUE BOLTS TO 119 - 131 INCH-POUNDS.** Using lockwire, Item 197, Appendix D, lockwire bolts.
- aj. Remove protective caps and plugs from electrical connectors.
- ak. Inspect and treat electrical connector (PARA 1-7-20).
- al. Connect electrical connector, P610 (1st stage pressure switch), and P609 (2nd stage pressure switch), to tail rotor servo.
- am. Connect electrical connector, P611, on top of tail pylon (Figure 6-4-52-1, Detail B).
- an. Close tail rotor pylon leading edge cover/antenna.
- ao. Install tail rotor blades (PARA 5-4-38).
- ap. Perform tail rotor system rig check (PARA 11-4-6).

### WARNING

Injury to personnel and damage to equipment will result if tail gear box is

operated with low oil level. Operation with low oil level or without oil will cause loss of tail rotor drive. Lightly shake tail pylon to observe oil sloshing when checking oil level.

- aq. Service tail gear box (PARA 1-3-14). Do not install tail gear box fairing.
- ar. Inspect tail rotor servo valve input linkage (PARA 11-4-113).
- as. Make sure area is clean and free of foreign material.
- at. Install tail gear box fairing (PARA 2-4-137).
- au. Connect wiring and grounding straps to anticollision light and tail incandescent position lights (PARA 9-4-86 and PARA 9-4-83). Remove tags.
- av. Check helicopter logbook to make sure tail gear box mounting bolt torque check, tail rotor drive shaft section IV, and tail rotor cable tension check are shown as due 9 to 11 flight hours after installation (PARA 1-7-8 and PARA 1-7-10).
- aw. If required, disengage gust lock (PARA 1-6-18).
- ax. Perform maintenance test flight (Appropriate MTF).



6-4-53 TAIL GEAR BOX INPUT SEAL AND FLANGE.

PARAGRAPH BREAKDOWN

|                                                      | PAGE     |
|------------------------------------------------------|----------|
| 6-4-53.1 Replace Tail Gear Box Input Seal and Flange | 6-4-53-2 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Ballpeen Hammer, 8-Ounce
- Burnisher (or equivalent)
- Nylon Wedge, Locally-Made, Figure H-260, Appendix H
- Powertrain Repairer’s Toolkit
- Rawhide or Rubber Mallet
- Screws, No. 8 Self-Tapping, 2-inches long
- Seal Installation Tool, Locally-Made, Figure H-157, Appendix H
- Seal Puller, Locally-Made, Figure H-160, Appendix H
- Tail Drive Flange Wrench, 70700-20687-102
- Torque Reactor, 70700-20688-041
- Torque Wrench, 100 - 500 ft lbs

Materials/Parts

- Abrasive Cloth, Item 1A, Appendix D
- Abrasive Cloth, Item 10, Appendix D

- Abrasive Mat, Item 14, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Grease, Item 149, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D
- Lubricating Oil, Item 208, Appendix D
- Machinery Towel, Item 211, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- Appendix H
- PARA 1-3-14
- PARA 1-6-16
- PARA 2-4-137
- PARA 6-3-10
- PARA 6-4-41

### 6-4-53.1 Replace Tail Gear Box Input Seal and Flange.

#### 6-4-53.1.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Drain tail gear box (PARA 1-3-14). Do not install tail gear box fairing.
- c. Remove tail rotor drive shaft section IV (PARA 6-4-41).
- d. Remove cotter pin from retaining nut (Figure 6-4-53-1, Detail A).
- e. Using torque reactor and tail drive flange wrench, remove retaining nut.
- f. Pull tail gear box input flange off input shaft. If necessary, using rawhide or rubber mallet, tap flange gently and evenly.
- g. Remove retaining ring.

#### NOTE

Self-tapping screws may be used as an alternate to remove tail gear box input seal if seal puller is not available. If seal puller is not available, proceed to step n.

- h. Locally make seal puller (Figure H-160, Appendix H).
- i. Locally make nylon wedge (Figure H-260, Appendix H).
- j. Insert nylon wedge between seal puller and input shaft to protect tail gear box housing (Figure 6-4-53-1, Detail B).

#### CAUTION

Damage to tail gear box will result if too much force is applied while using ballpeen hammer and seal puller. Tail gear box housing will crack. Use care when using ballpeen hammer and seal puller.

- k. Using 8-ounce ballpeen hammer, gently tap end of seal puller in an outward direction, away from tail gear box.
- l. Moving seal puller and nylon wedge in a clockwise direction, repeat step j. and step k. until input seal is removed (Figure 6-4-53-1, Detail A and Detail B).
- m. If seal puller is not available, remove input seal as follows:

#### WARNING

- Damage to tail gear box will result if self-tapping screws completely penetrate input seal. Screws will damage bearing inside tail gear box. Install screws so that only outer face of input seal is penetrated.

- Damage to tail gear box will result if metal chips or shavings enter tail gear box by drilling or punching holes in input seal. Metal chips or shavings will contaminate tail gear box oil. Do not drill or punch holes in input seal.

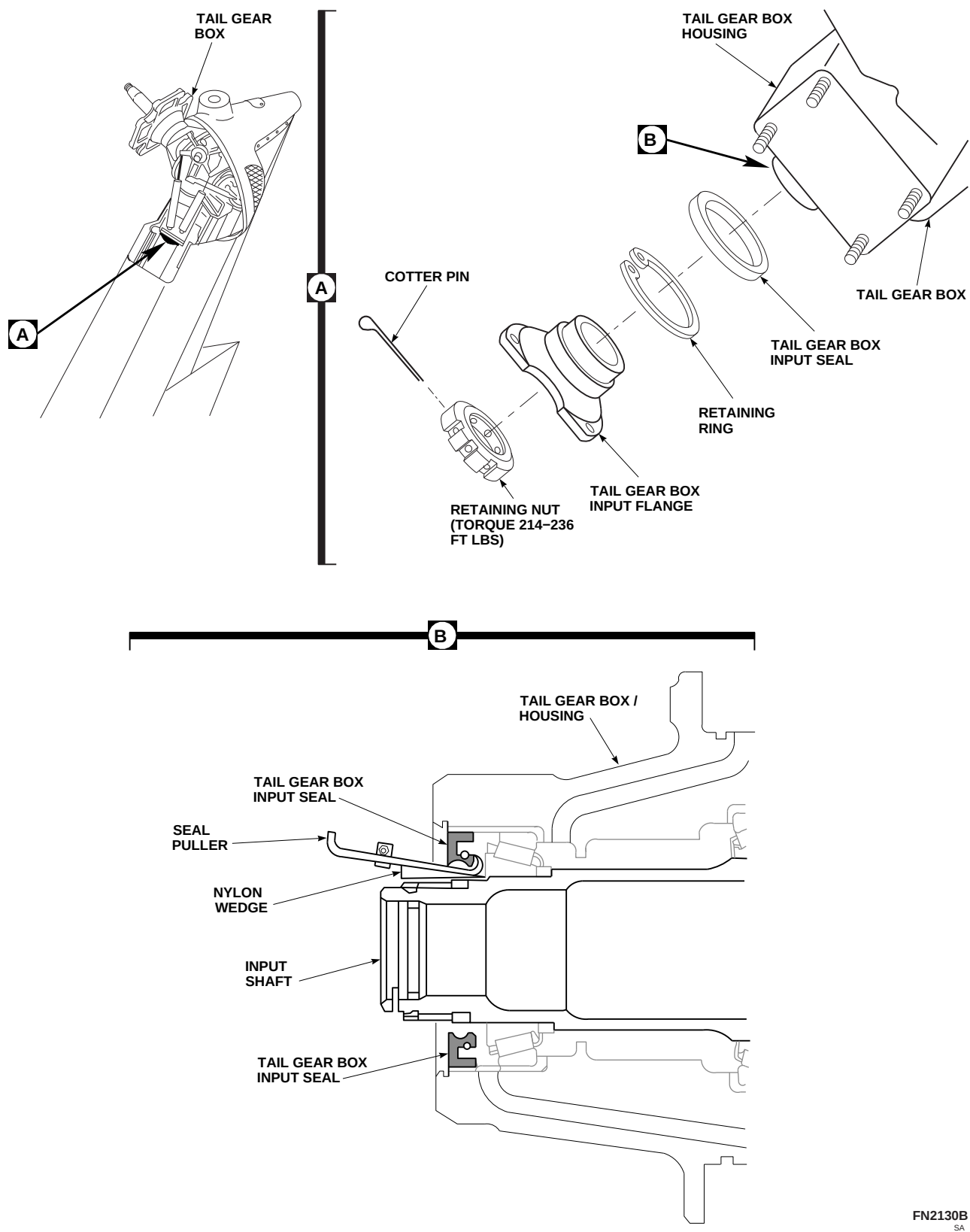
- (1) Using three self-tapping screws, thread into holes in input seal.
- (2) Carefully pull out input seal.

#### 6-4-53.1.2 Inspect/Repair.

#### NOTE

Inspection and repair of tail gear box flange is identical to intermediate gear box input flange repair.

- a. Inspect, and if necessary, repair tail gear box input flange (PARA 6-3-10).
- b. Inspect retaining nut for corrosion. **REPLACE RETAINING NUT IF:**
  - (1) **PITS ARE 0.030-INCH DEEP.**
  - (2) **PITTED MORE THAN 10% OF MATING SURFACE.**



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FIGURE 6-4-53-1. REPLACE TAIL GEAR BOX INPUT SEAL AND FLANGE

(3) **THREADS SHOW SIGNS OF CORROSION.**

- c. Remove minor corrosion from retaining nut using abrasive cloth, Item 10, Appendix D, abrasive cloth, Item 1A, Appendix D, or abrasive mat, Item 14, Appendix D.

**NOTE**

Do not paint threads or mating surface of retaining nut.

- d. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to flats and exposed face of retaining nut.
- e. Visually inspect input shaft for damage such as nicks, scratches, and gouges. Using burnisher, or equivalent, blend out damaged area. **MAXIMUM ALLOWABLE DEPTH AFTER REPAIR IS 0.005-INCH.**

**6-4-53.1.3 Install.**

- a. Prior to installing tail gear box input seal, perform the following:
- (1) Using grease, Item 149, Appendix D, coat inside diameter of input seal.
  - (2) Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean mating surface of input seal.
  - (3) Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, coat outside diameter of input seal and mating surface.
- b. Locally make seal installation tool (Figure H-157, Appendix H).

**CAUTION**

- Input seal and tail gear box will be damaged if incorrectly installed. Input seal

will leak. Make sure input seal is not cocked during installation. Make sure input seal is installed flush to 0.020-inch below retaining ring groove prior to installing retaining ring.

- Be careful not to cock or distort seal.
- c. Using seal installation tool, install new input seal by gently tapping into tail gear box housing (Figure 6-4-53-1, Detail A).
- d. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to retaining ring and install in tail gear box housing while wet.
- e. Apply sealing compound, Item 292, Appendix D, to boss of tail gear box housing to prevent water entrapment and corrosion above retaining ring groove.
- f. Using epoxy primer coating kit, Item 135, Appendix D, touch up sealing compound with epoxy primer.
- g. Slide input flange on input shaft.
- h. Install retaining nut as follows:
- (1) Install retaining nut, making sure retaining nut can be fully run on by hand. Make sure bevel side of retaining nut is toward tail gear box input seal.
  - (2) Using torque reactor and tail drive flange wrench, **TIGHTEN RETAINING NUT.**
  - (3) **TORQUE RETAINING NUT TO 214 - 236 FOOT-POUNDS.**
  - (4) **INSTALL NEW COTTER PIN. IF REQUIRED TO ALIGN COTTER PIN HOLE, TURN RETAINING NUT IN A TIGHTENING DIRECTION.**
- i. Install tail rotor drive shaft section IV (PARA 6-4-41).

**WARNING**

Injury to personnel and damage to equipment will result if tail gear box is operated with low oil level. Operation with low oil level or without oil will cause loss of tail rotor drive. Lightly shake pylon to observe oil sloshing when checking oil level.

- j. Service tail gear box (PARA 1-3-14).
- k. Make sure area is clean and free of foreign material.
- l. Perform check of input seal and flange as follows:
  - (1) With qualified person at controls, run up helicopter.
  - (2) Run up helicopter until tail gear box oil warms up.
  - (3) Shut down helicopter.
  - (4) Check tail gear box input seal and flange for leaks.
- m. Install tail gear box fairing (PARA 2-4-137).



## 6-4-54 INBOARD RETENTION PLATE AND SEALS.

### PARAGRAPH BREAKDOWN

|                                                    | PAGE     |
|----------------------------------------------------|----------|
| 6-4-54.1 Replace Inboard Retention Plate and Seals | 6-4-54-2 |

#### INITIAL SETUP

##### Tools

- Aircraft Mechanic's Toolkit
- Allen Wrench,  $\frac{5}{32}$ -inch
- Ballpeen Hammer, 8-Ounce
- Feeler Gage, Plastic, 0.035-in.
- Fluorescent Penetrant Inspection Kit
- Ink Marker
- Mallet
- Micrometer
- Nonmetallic Drift, Locally-Made
- Nylon Wedge, Locally-Made, Figure H-260, Appendix H
- Powertrain Repairer's Toolkit
- Protection, Eye
- Screws, No. 8 Self-Tapping, 2-inches long
- Seal Installation Tool, Locally-Made, Figure H-157, Appendix H
- Seal Puller, Locally-Made, Figure H-160, Appendix H
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

- Torque Wrench, 150 - 750 in. lbs

##### Materials/Parts

- Abrasive Cloth, Item 1A, Appendix D
- Abrasive Cloth, Item 12, Appendix D
- Abrasive Wheel, Item 22, Appendix D
- Alcohol, Denatured, Item 70, Appendix D
- Assembly Fluid, Item 79, Appendix D
- Damping Fluid, Item 121, Appendix D (or equivalent)
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Grease, Item 149, Appendix D
- Grease, Item 150, Appendix D
- Lockwire, Item 197, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D
- Lubricating Oil, Item 208, Appendix D
- Machinery Towel, Item 211, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D
- Sealing Compound, Item 280, Appendix D
- Sealing Compound, Item 292, Appendix D

- Sealing Primer, Item 306, Appendix D

## References

- Appendix D
- Appendix H
- ASTM E1417
- PARA 1-3-14
- PARA 1-7-9
- PARA 2-4-137
- PARA 2-6-5
- PARA 5-4-38
- PARA 6-4-52
- PARA 12-4-52

### 6-4-54.1 Replace Inboard Retention Plate and Seals.

#### 6-4-54.1.1 Remove.

#### WARNING

#### FSCAP

Exposed surfaces of bevel gear are critical surfaces which must not be damaged during and after removal of inboard retention plate. Use extreme caution when removing and provide protective covering over critical surfaces after removal of retention plate.

- Remove tail gear box fairing (PARA 2-4-137).
- Remove tail rotor blades (PARA 5-4-38).

#### CAUTION

Damage to retention plate will result if tools come in contact when loosening jamnuts. Do not allow tools to come in contact with retention plate when loosening jamnuts.

#### NOTE

It is not necessary to completely remove bolts. Bolts may be loosened enough to unthread nut with bolts and seal housing still attached.

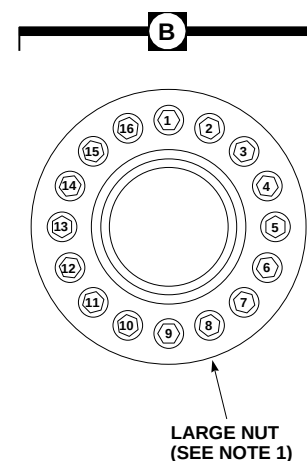
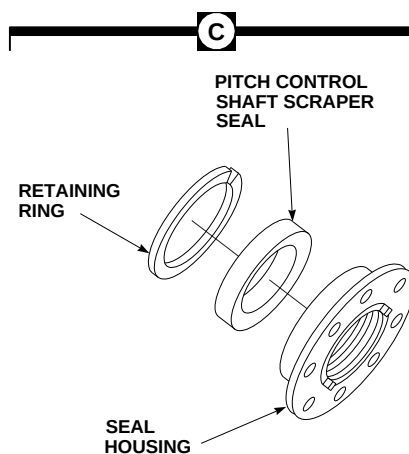
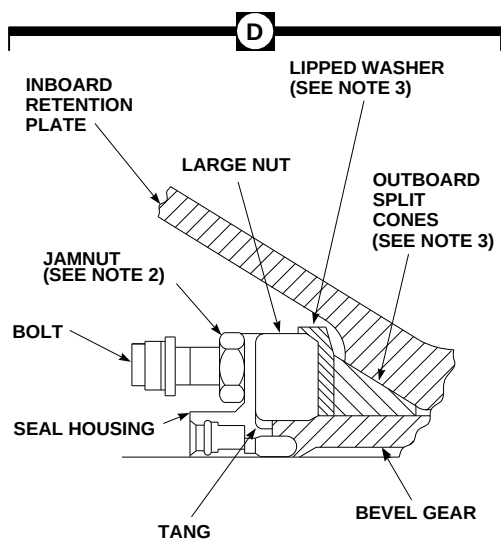
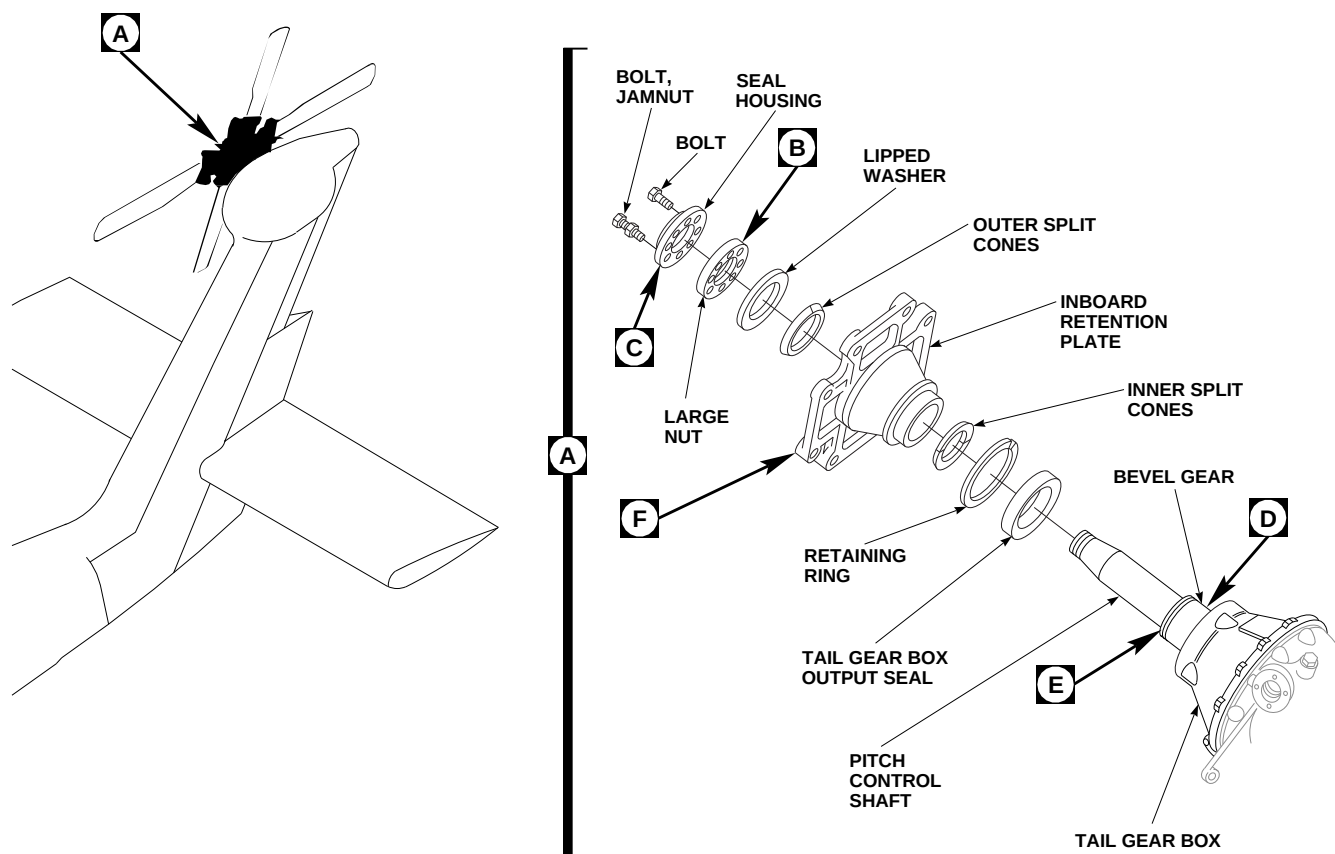
- Loosen both jamnuts (located 180° apart), then loosen all bolts from seal housing (Figure 6-4-54-1, Sheet 1, Detail A).
- As an assembly, remove seal housing, consisting of retaining ring and pitch control shaft scraper seal (Figure 6-4-54-1, Sheet 1, Detail C).
- Remove large nut and lipped washer (Figure 6-4-54-1, Sheet 1, Detail A and Detail D).

#### NOTE

Outer split cones are installed and removed in matched sets. Make sure serial numbers are the same for each cone in a pair.

- Remove outer split cones. Using mallet, gently tap retention plate to loosen outer split cones. If outer split cones do not loosen after tapping with mallet, perform the following:
  - Using damping fluid, Item 121, Appendix D, or equivalent, apply to outer split cones, and wait about 30 minutes.
  - Using mallet, gently tap retention plate.
  - If required, repeat step f. (1) and step f. (2).
- Inspect outer split cones for damage and fretting.
- Slide retention plate off pitch control shaft.





# NOTES

1. TORQUE SEQUENCE: 1, 9, 13, 5, 11, 3, 15, 7, 2, 10, 14, 6, 16, 8, 12, 4.
2. TIGHTEN 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.
3. SPLIT CONE SURFACE SHALL PROTRUDE OUTBOARD FROM INBOARD RETENTION PLATE SURFACE BY APPROXIMATELY 0.10-INCH. LIPPED WASHER SHALL NOT CONTACT INBOARD RETENTION PLATE.

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FIGURE 6-4-54-1. REPLACE INBOARD RETENTION PLATE AND SEALS (SHEET 1 OF 2)

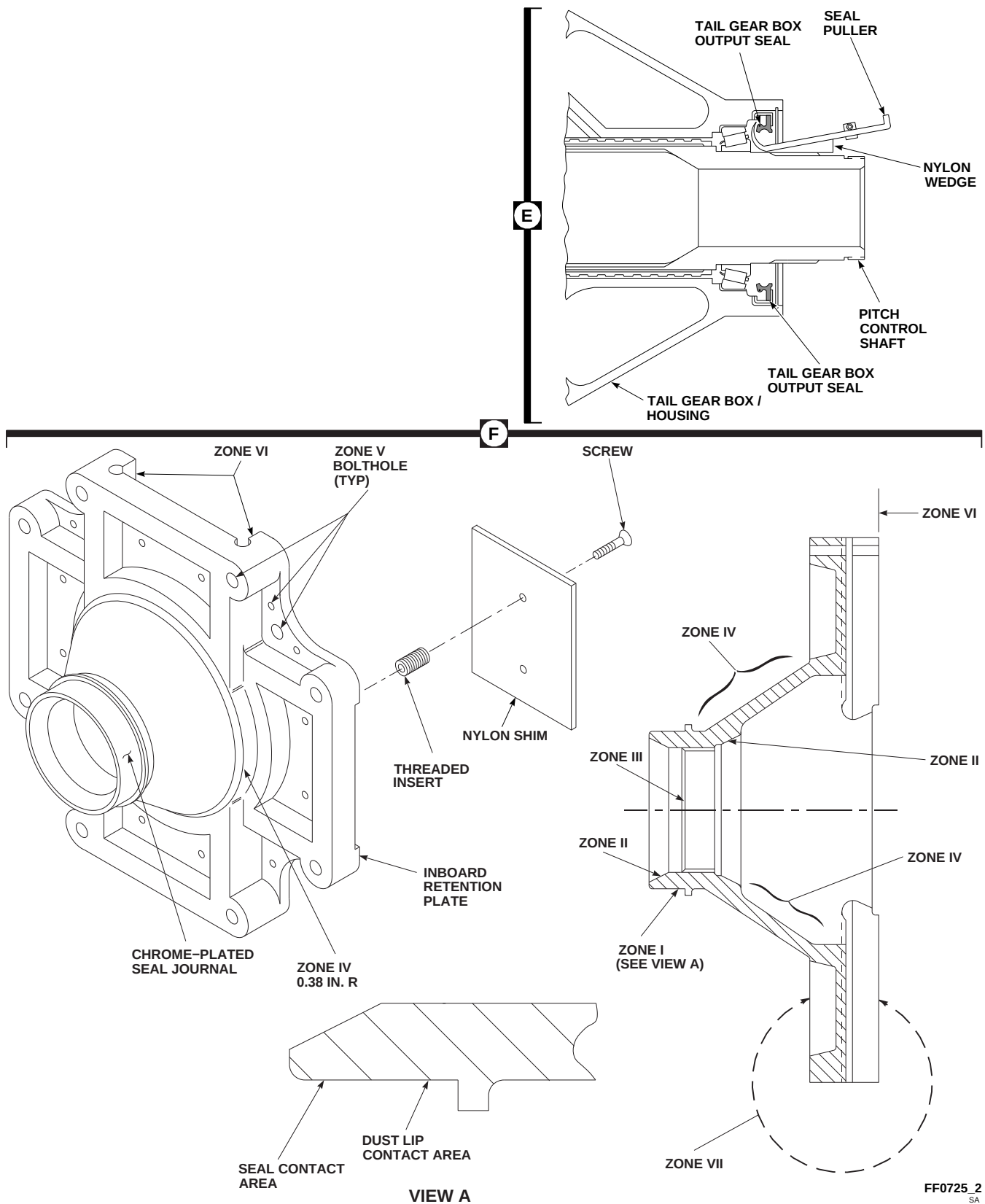


FIGURE 6-4-54-1. REPLACE INBOARD RETENTION PLATE AND SEALS (SHEET 2 OF 2)

- i. Using 1-inch masking tape, Item 212, Appendix D, cover all exposed surfaces of output bevel gear.
- j. If blade deice kit is installed, remove tail rotor blade deice slipping rotor (PARA 12-4-52).
- k. Remove retaining ring.

### NOTE

Self-tapping screws may be used as an alternate to remove tail gear box output seal if seal puller is not available. If seal puller is not available, proceed to step q.

- l. Locally make seal puller (Figure H-160, Appendix H).
- m. Locally make nylon wedge (Figure H-260, Appendix H).
- n. Insert nylon wedge between seal puller and pitch control shaft to protect tail gear box housing (Figure 6-4-54-1, Sheet 2, Detail E).

### CAUTION

Damage to tail gear box will result if too much force is applied while using ballpeen hammer and seal puller. Tail gear box housing will crack. Use care when using ballpeen hammer and seal puller.

- o. Using 8-ounce ballpeen hammer, gently tap end of seal puller in outward direction, away from tail gear box.
- p. Moving seal puller and nylon wedge in clockwise direction, repeat step n. and step o. until tail gear box output seal is removed.
- q. If seal puller is not available, remove tail gear box output seal as follows:

### CAUTION

- Damage to tail gear box will result if self-tapping screws completely penetrate

tail gear box output seal. Screws will damage bearing inside tail gear box. Install screws so that only outer face of tail gear box output seal is penetrated.

- Damage to tail gear box will result if metal chips or shavings enter tail gear box by drilling or punching holes in tail gear box output seal. Metal chips or shavings will contaminate tail gear box oil. Do not drill or punch holes in tail gear box output seal.
  - (1) Using three self-tapping screws, thread screws into holes in tail gear box output seal.
  - (2) Carefully remove tail gear box output seal.
- r. Disassemble seal housing as follows (Figure 6-4-54-1, Sheet 1, Detail C):

### NOTE

Pitch control shaft scraper seal should be replaced whenever tail gear box output seal is replaced.

- (1) Remove retaining ring from seal housing.
- (2) Turn seal housing upside down (boltheads down), and using nonmetallic drift, drive out pitch control shaft scraper seal.

### NOTE

Inner split cones are installed and removed in matched sets. Make sure serial numbers are the same for each cone in a pair.

- s. Check inner split cones for damage and fretting. Remove inner split cones only if they are being replaced (Figure 6-4-54-1, Sheet 1, Detail A).

### 6-4-54.1.2 Inspect/Repair.

- a. Visually inspect inboard retention plate for cracks. **NONE ALLOWED.** If cracks are found, replace inboard retention plate.

- b. If retention plate was removed due to leakage, inspect zone I as follows (Figure 6-4-54-1, Sheet 2, Detail F):

**NOTE**

- Seal contact area is identified by a polished circumferential area or slight groove.
- The seal contacts liner in two places. Wear groove closest to shoulder is where dust lip contacts liner. Wear groove closest to outer edge is where seal contacts liner. These areas are referred to as "seal contact area" (Figure 6-4-54-1, Sheet 2, Detail F, View A).

- (1) Check zone I seal contact area for grooving. **MAXIMUM GROOVE DEPTH IS 0.003-INCH (0.006-INCH ON DIAMETER) IN CHROME-PLATED SEAL CONTACT AREA.** Maximum groove depth is 0.020-inch (0.040-inch on diameter) in dust lip contact area. If grooving exceeds limits, replace inboard retention plate.
- (2) Nicks and scratches in zone I seal contact area that do not penetrate into base metal may be blended out. **NO REPAIR INTO BASE METAL ALLOWED IN SEAL CONTACT AREA.** If damage or repair in seal contact area penetrates into base metal, replace inboard retention plate.
- (3) Nicks and scratches in zone I areas other than seal contact area may be blended out up to 0.010-inch. If damage exceeds limit, replace inboard retention plate.

- c. Inspect/repair zone II inner split cone seats for damage as follows:

- (1) **NO FRACTURES, CRACKS, OR PITS ALLOWED.**
- (2) Using abrasive cloth, Item 12, Appendix D, minor fretting and scratches on inner split cone seats may be blended out up to 0.002-inch. **REPAIR AREA NOT TO**

**EXCEED 10% OF TOTAL SURFACE AREA.**

- (3) If damage exceeds limit, replace inboard retention plate.

- d. Inspect zone III internal spline area for damage as follows:

- (1) **NO DAMAGE MORE THAN UNIFORM WEAR FROM MATING SPLINE ALLOWED. WEAR STEP ON SPLINE IS LIMITED TO 0.002-INCH.**
- (2) **MINOR SCRATCHES UP TO 0.002-INCH DEEP ARE ACCEPTABLE.**
- (3) If damage exceeds limit, replace inboard retention plate.

- e. Inspect nylon shims as follows:

- (1) Remove screws securing nylon shim to inboard retention plate. Remove nylon shims.
- (2) Make sure shim is clean. If necessary, clean shim of foreign matter and residue.
- (3) Using micrometer, inspect nylon shims 1-inch in from edge of each corner. **SHIM THICKNESS MUST BE 0.123 - 0.126-INCH.** If thickness exceeds limits or if surface is not flat, replace nylon shims.

- f. Inspect threaded inserts as follows:

- (1) Thread screws into inserts and check drag torque required to install screws. **DRAG TORQUE MUST BE 2 - 18 INCH-POUNDS.**
- (2) If drag torque exceeds limits, replace screw and clean threaded insert. Recheck drag torque. If drag torque is still not within limits, replace threaded insert as follows:
  - (a) Using  $\frac{5}{32}$ -inch Allen wrench, remove threaded insert.

**WARNING**

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- (b) Using denatured alcohol, Item 70, Appendix D, clean tapped hole in retention plate. Using clean, dry, compressed air, blow dry cleaned area.
  - (c) Inspect tapped hole in inboard retention plate for thread yielding. If more than 10% of thread has yielded, replace inboard retention plate.
  - (d) Using denatured alcohol, Item 70, Appendix D, clean outer threads of threaded insert. Using clean, dry, compressed air, blow dry cleaned area.
  - (e) Using sealing primer, Item 306, Appendix D, apply thin coat to threads of threaded insert and threads of inboard retention plate.
  - (f) Apply thin coat of sealing compound, Item 280, Appendix D, to first two threads of threaded insert.
  - (g) **USING  $\frac{5}{32}$ -INCH ALLEN WRENCH, INSTALL THREADED INSERT FLUSH TO 0.025-INCH BELOW SURFACE WITHIN TAPPED HOLE, TO A DRIVING TORQUE OF 25 - 300 INCH-POUNDS.**
- g. Position nylon shims on retention plate. Thread screws into threaded inserts and note drag

torque required to install screws. **TORQUE SCREWS TO A VALUE 4 - 10 INCH-POUNDS GREATER THAN DRAG TORQUE.**

- h. Visually inspect inboard retention plate for nicks, gouges, fretting, and scoring.
- i. Using abrasive wheel, Item 22, Appendix D, remove raised material and blend out damage. **REPLACE RETENTION PLATE IF:**
  - (1) **CORNER DAMAGE IN ZONE VII AFTER BLENDING IS GREATER THAN 0.100-INCH DEEP, OR 0.040-INCH DEEP IN BOLTHOLE AREAS. MULTIPLE AREAS OF DAMAGE MUST BE AT LEAST 1-INCH APART.**
  - (2) **SURFACE DAMAGE AFTER BLENDING IS GREATER THAN 0.020-INCH DEEP IN OPEN AREAS, OR GREATER THAN 0.020-INCH DEEP IN BOLT-HOLE AREAS. MULTIPLE AREAS OF DAMAGE MUST BE AT LEAST 1-INCH APART.**
  - (3) **SURFACE DAMAGE IN ZONE VI MATING AREAS AFTER BLENDING IS GREATER THAN 0.020-INCH DEEP, OR COVERS MORE THAN 10% OF MATING AREA. BLEND TO REMOVE ANY DAMAGE USING A 0.050-INCH BLEND RADIUS.**
  - (4) **FRETTING IN ZONE VI AREAS THAT MATE WITH OUTBOARD RETENTION PLATE AFTER BLENDING IS GREATER THAN 0.010-INCH DEEP, OR COVERS MORE THAN 10% OF MATING AREA. BLEND TO REMOVE ANY DAMAGE USING A 0.050-INCH BLEND RADIUS.**
  - (5) **BOLTHOLE BORE DAMAGE IN ZONE V EXCEEDS 0.010-INCH DEEP OR COVERS MORE THAN 10% OF BORE SURFACE AREA.**

**CAUTION**

Damage to inboard retention plate will result if care is not taken when inspecting or reworking zone IV areas. Zone IV areas are critical areas of inboard retention plate. Use extreme care when inspecting or reworking all zone IV areas.

- (6) **DAMAGE OR REWORK IN ZONE IV AREAS (INCLUDING 0.38-INCH RADIUS) IS DEEPER THAN 0.0003-INCH. MULTIPLE AREAS OF DAMAGE MUST BE AT LEAST 1-INCH APART.**

- j. Touch up paint on repaired areas (PARA 2-6-5).

**NOTE**

Split cones are installed and removed in matched sets. Make sure serial numbers are the same for each cone in a pair.

- k. Inspect split cones for scratches or pits and repair as follows:
- (1) **USING ABRASIVE CLOTH, ITEM 1A, APPENDIX D, BLEND SCRATCHES OR PITS TO FORM A SMOOTH TRANSITION TO SURROUNDING AREA UP TO 0.005-INCH DEEP NO MORE THAN 10% OF TOTAL SURFACE AREA.**
  - (2) If repairs are beyond limits, replace split cones.
  - l. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect reworked area of repaired cones for cracks (ASTM E1417). **NONE ALLOWED.** If cracks are found, replace split cones.

- m. Visually inspect exposed surfaces of bevel gear as follows:

- (1) Inspect splines on bevel gear for wear and damage. **SCRATCHES UP TO 0.0003-INCH DEEP AND UNIFORM WEAR UP TO 0.0003-INCH FROM MATING SPLINE IS PERMITTED AS LONG AS DAMAGE OR WEAR DOES NOT PENETRATE THROUGH SILVER-PLATE INTO BASE METAL.** If bevel gear splines are damaged or worn beyond limit, replace tail gear box (PARA 6-4-52).
- (2) Inspect split cone journals on bevel gear for damage. **SCRATCHES UP TO 0.0003-INCH DEEP AND UNIFORM WEAR UP TO 0.0003-INCH IS PERMITTED AS LONG AS DAMAGE OR WEAR DOES NOT PENETRATE THROUGH SILVER-PLATE INTO BASE METAL. NO CRACKS, FRACTURES, OR FRETTING THROUGH SILVER-PLATE ALLOWED.** If split cone journals on bevel gear are damaged beyond limits, replace tail gear box (PARA 6-4-52).
- (3) Inspect threads on bevel gear for damage. **NO STRIPPED, CROSSED, OR FLATTENED THREADS ALLOWED.** If threads on bevel gear are damaged, replace tail gear box (PARA 6-4-52).
- n. Visually inspect large nut for nicks and scratches. **IF NICKS OR SCRATCHES GREATER THAN 0.020-INCH DEEP ARE FOUND, REPLACE NUT. NO CROSSED, STRIPPED, FLATTENED, FRACTURED, OR CRACKED THREADS ALLOWED.**
- o. Visually inspect lipped washer for nicks and scratches. **IF DAMAGE IS DEEPER THAN 0.015-INCH OR REPAIR AREA IS GREATER THAN 20% OF SURFACE AREA, REPLACE LIPPED WASHER.**

### 6-4-54.1.3 Install.

#### WARNING

#### FSCAP

Exposed surfaces of bevel gear shaft are critical surfaces which must not be damaged during installation of inboard retention plate. After removal of protective covering, use extreme caution when installing retention plate.

#### CAUTION

- Damage to tail gear box and split cones will result if split cones are not cleaned prior to installation. Split cones will not seat properly. Make sure split cones are clean and free of lubricant before installing split cones on tail gear box.
- Damage to tail gear box will result if incorrect split cone set is used. Inner and outer cone sets are different sizes. Inner cone set, 70358-06343-102, has smaller cross section and is 0.50-inch wide.

#### NOTE

Inner split cones are installed and removed in matched sets. Make sure serial numbers are the same for each cone in a pair.

- a. If inner split cones were removed, using machinery towel, Item 211, Appendix D, thoroughly clean lubricant from inner split cones.

#### NOTE

Make sure not to damage silver-plating on bevel gear.

- b. Remove masking tape and fully clean all exposed surfaces of bevel gear using clean machinery towel, Item 211, Appendix D.

#### NOTE

Inner split cones are installed and removed in matched sets. Make sure serial numbers are the same for each cone in a pair.

- c. Using assembly fluid, Item 79, Appendix D, apply to inner surfaces of inner split cones and install with beveled side facing outboard (Figure 6-4-54-1, Sheet 1). Make sure gaps between inner cones are equal.
- d. If blade deice kit is installed, install tail rotor blade deice slipping rotor (PARA 12-4-52).
- e. Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, apply light coat to pitch control shaft scraper seal.
- f. Assemble seal housing as follows (Figure 6-4-54-1, Sheet 1, Detail C):

#### CAUTION

Tail gear box and pitch control shaft scraper seal will be damaged if incorrectly installed. Make sure scraper seal is not cocked or distorted during installation. Make sure scraper seal is properly seated beyond retaining ring groove before installing retaining ring.

- (1) Using nonmetallic drift, install pitch control shaft scraper seal in seal housing.
- (2) Install retaining ring in groove in seal housing.
- g. Prior to installing tail gear box output seal, perform the following:
  - (1) Using grease, Item 149, Appendix D, coat inside diameter of tail gear box output seal.
  - (2) Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean mating surface of tail gear box output seal.



- (3) Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, coat outside surface of tail gear box output seal and retention plate liner (liner to seal mating surface).
- h. Locally make seal installation tool (Figure H-157, Appendix H).

**CAUTION**

Tail gear box output seal and tail gear box will be damaged if incorrectly installed. Make sure output seal is not cocked or distorted during installation. Make sure output seal is properly seated beyond retaining ring groove prior to installing retaining ring.

- i. Using seal installation tool, install new tail gear box output seal by gently tapping into tail gear box (Figure 6-4-54-1, Sheet 1, Detail A).
- j. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to retaining ring and install in tail gear box while wet.
- k. Apply sealing compound, Item 292, Appendix D, to boss of tail gear box to prevent water entrapment and corrosion above retaining ring groove.
- l. Using epoxy primer coating kit, Item 135, Appendix D, touch up sealing compound.
- m. Using grease, Item 150, Appendix D, apply to splines on bevel gear and inboard retention plate.
- n. Install inboard retention plate.

**CAUTION**

- Retention plate and outer split cones will be damaged if outer split cones are

not clean when installed. Outer split cones will not seat properly. Make sure outer split cones are clean and free of lubricant before installing on tail gear box.

- Damage to tail gear box will result if incorrect split cone set is used. Inner and outer cone sets are different sizes. Outer cone set, 70358-06343-103, has larger cross section and is 0.69-inch wide.
- o. Using machinery towel, Item 211, Appendix D, clean lubricant from outer split cones.

**NOTE**

Make sure outer split cone gaps are positioned 90° from gaps of inner split cones.

- p. Using assembly fluid, Item 79, Appendix D, apply to inner surfaces of outer split cones and install with beveled side facing inboard. Make sure outer split cones are properly seated, and gaps between outer split cones are the same size. If necessary, using nonmetallic drift, tap lightly to seat outer split cones.

**CAUTION**

When correct split cones are used, cone surface should protrude outboard from inboard retention plate surface by approximately 0.10-inch. Lipped washer should not contact inboard retention plate.

- q. Before installing lipped washer, check to make sure that large nut face seats with both sides of washer face.
- r. Install lipped washer on bevel gear with lip outboard from tail gear box (Figure 6-4-54-1, Sheet 1, Detail D).




**CAUTION**

Damage to threads of bevel gear will result if large nut is installed incorrectly. When installing large nut on bevel gear, make sure beveled side of large nut faces inboard and that large nut fits flush with lipped washer.

- s. Install large nut on bevel gear with beveled side of large nut facing inboard and flush with lipped washer.
- t. **TORQUE LARGE NUT HANDTIGHT WITH MINIMUM AMOUNT OF TORQUE THAT WILL ACHIEVE SEAL TANG ENGAGEMENT. DO NOT LOOSEN LARGE NUT.**
- u. As an assembly, install seal housing, pitch control shaft scraper seal, and retaining ring on bevel gear and engage tangs with slots in bevel gear.

**NOTE**

Make sure seal housing does not rotate when aligning boltholes.

- v. Rotate large nut in tightening direction until boltholes line up with holes in seal housing.
- w. Install two bolts with jamnuts 180° apart and located as close as possible to tangs. Install 14 additional bolts securing seal housing to large nut. **INSTALL ALL BOLTS HANDTIGHT.**


**WARNING**
**FSCAP**

Verification of torque stabilization is a critical characteristic.

- x. **USING INK MARKER, MARK ONE OF THE SIXTEEN BOLTS AS A REFERENCE POINT. TORQUE BOLTS TO 20 INCH-**

**POUNDS IN THE FOLLOWING SEQUENCE: 1, 9, 13, 5, 11, 3, 15, 7, 2, 10, 14, 6, 16, 8, 12, AND 4** (Figure 6-4-54-1, Sheet 1, Detail A and Detail B).

- y. **TORQUE BOLTS TO 40 INCH-POUNDS IN THE FOLLOWING SEQUENCE: 1, 9, 13, 5, 11, 3, 15, 7, 2, 10, 14, 6, 16, 8, 12, AND 4. REPEAT TORQUE SEQUENCE UNTIL ALL BOLTS HAVE STABILIZED AND NO MOVEMENT IS FELT AT 40 INCH-POUNDS. DO NOT TIGHTEN JAMNUTS AT THIS TIME.**
- z. Remove bolts securing seal housing to large nut. Remove seal housing.

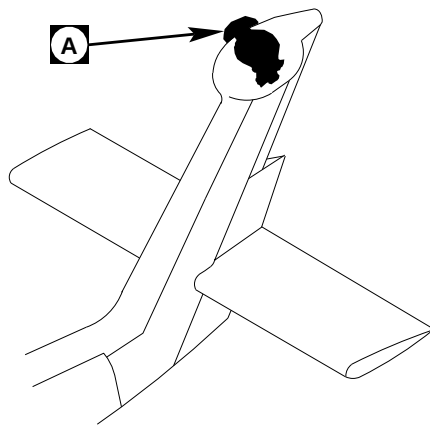
**NOTE**

- Tangs may not be fully engaged in slots until cones are properly seated.
- Make sure nut is fully seated to make sure tangs are engaged in slots in bevel gear.
- To make sure of proper seating, make sure large nut seats flat against lipped washer on both sides.
- aa. **RETIGHTEN LARGE NUT ON BEVEL GEAR HANDTIGHT AND MAKE SURE LARGE NUT IS PROPERLY SEATED AGAINST LIPPED WASHER** (Figure 6-4-54-1, Sheet 1, Detail B and Detail D).
- ab. Repeat step v. and step w.

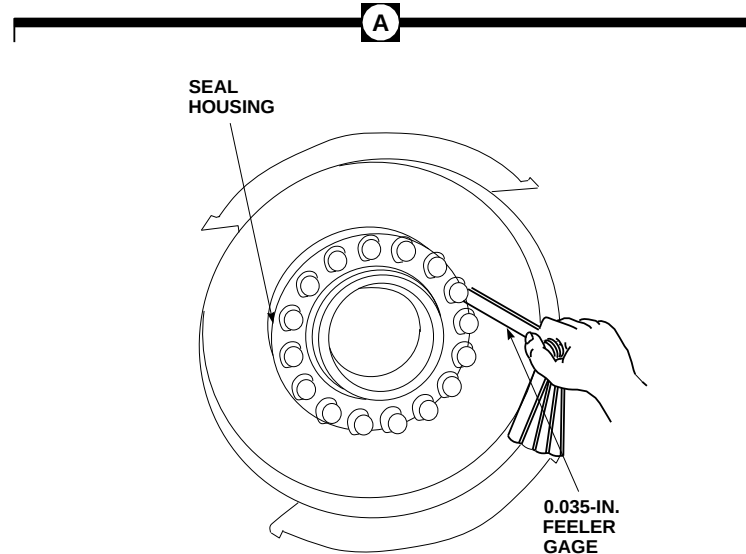

**WARNING**
**FSCAP**

Verification of torque stabilization is a critical characteristic.

- ac. **USING INK MARKER, MARK ONE OF THE SIXTEEN BOLTS AS A REFERENCE POINT. TORQUE BOLTS TO 20 INCH-POUNDS IN THE FOLLOWING**

**NOTE**

FEELER GAGE (0.035 - IN.) SHALL SLIP INTO GAP EASILY WHEN SPLIT CONE SET 70358-06343-103 HAS BEEN INSTALLED ON OUTBOARD END.



**SPLIT CONES (70358-06343-103)  
ON OUTBOARD END**

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**FIGURE 6-4-54-2. GAP CHECK - PROPER SPLIT CONE INSTALLATION**

**SEQUENCE: 1, 9, 13, 5, 11, 3, 15, 7, 2, 10, 14, 6, 16, 8, 12, AND 4** (Figure 6-4-54-1, Sheet 1, Detail A and Detail B). **REPEAT TORQUE SEQUENCE UNTIL ALL BOLTS HAVE STABILIZED AND NO MOVEMENT IS FELT AT 20 INCH-POUNDS. DO NOT TIGHTEN JAMNUTS AT THIS TIME.**

- ad. Verify proper split cone installation as follows (Figure 6-4-54-2, Detail A):

**CAUTION**

To prevent scratching or nicking retention plate, make sure to use plastic feeler gage for gap check.

- (1) Insert plastic feeler gage (0.035-inch) between lipped washer and inboard retention plate. Feeler gage must slip into gap easily when correct split cone set, 70358-06343-103, has been installed on outboard end.

- (2) If plastic feeler gage does not fit in gap, wrong split cone set, 70358-06343-102, has been installed on outboard end. Remove split cones and repeat step c. through step ad. (2)

- ae. **TORQUE BOLTS TO 40 INCH-POUNDS IN THE FOLLOWING SEQUENCE: 1, 9, 13, 5, 11, 3, 15, 7, 2, 10, 14, 6, 16, 8, 12, AND 4** (Figure 6-4-54-1, Sheet 1, Detail B). **REPEAT TORQUE SEQUENCE UNTIL ALL BOLTS HAVE STABILIZED AND NO MOVEMENT IS FELT AT 40 INCH-POUNDS. DO NOT TIGHTEN JAMNUTS AT THIS TIME.**

- af. **TORQUE BOLTS TO 60 INCH-POUNDS IN THE FOLLOWING SEQUENCE: 1, 9, 13, 5, 11, 3, 15, 7, 2, 10, 14, 6, 16, 8, 12, AND 4. REPEAT TORQUE SEQUENCE UNTIL ALL BOLTS HAVE STABILIZED AND NO MOVEMENT IS FELT AT 60 INCH-POUNDS. DO NOT TIGHTEN JAMNUTS AT THIS TIME.**

- ag. **TORQUE BOLTS TO 90 INCH-POUNDS IN THE FOLLOWING SEQUENCE: 1, 9, 13, 5, 11, 3, 15, 7, 2, 10, 14, 6, 16, 8, 12, AND 4. REPEAT TORQUE SEQUENCE UNTIL ALL BOLTS HAVE STABILIZED AND NO MOVEMENT IS FELT AT 90 INCH-POUNDS. DO NOT TIGHTEN JAMNUTS AT THIS TIME.**

**NOTE**

Upon completion of final torque sequence at 120 inch-pounds, bolted connection may be heated up from friction. If possible, repeat 120 inch-pound torque sequence after a 30 minute cool down period.

- ah. **TORQUE BOLTS TO 120 INCH-POUNDS IN THE FOLLOWING SEQUENCE: 1, 9, 13, 5, 11, 3, 15, 7, 2, 10, 14, 6, 16, 8, 12, AND 4. REPEAT TORQUE SEQUENCE UNTIL ALL BOLTS HAVE STABILIZED AND NO MOVEMENT IS FELT AT 120 INCH-POUNDS. DO NOT TIGHTEN JAMNUTS AT THIS TIME.**

**CAUTION**

Damage to retention plate will result if tools come in contact while tightening jamnuts. Do not allow tools to come in contact with retention plate when tightening jamnuts.

- ai. **TIGHTEN JAMNUTS  $\frac{1}{6}$  TO  $\frac{1}{3}$  TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

- aj. Using lockwire, Item 197, Appendix D, lockwire all bolts and jamnuts.
- ak. Install tail rotor blades (PARA 5-4-38).

**WARNING**

Injury to personnel and damage to equipment will result if tail gear box is operated with low oil level. Operation with low oil level or without oil will cause loss of tail rotor drive. Lightly shake pylon to observe oil sloshing when checking oil level.

- al. Service tail gear box (PARA 1-3-14).
- am. Make sure area is clean and free of foreign material.
- an. Check helicopter logbook to make sure retention plate bolt torque check is shown due 9 to 11 flight hours after installation (PARA 1-7-9).
- ao. Perform check of inboard retention plate and seals as follows:
- (1) With qualified person at controls, run up helicopter (Appropriate Flight Manual).
  - (2) Run helicopter until tail gear box oil gets warm.
  - (3) Shut down helicopter (Appropriate Flight Manual).
  - (4) Check inboard retention plate and seals for leaks.
- ap. Install tail gear box fairing (PARA 2-4-137).



6-4-55 TAIL GEAR BOX SIGHT PLUG.

PARAGRAPH BREAKDOWN

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| 6-4-55.1 Replace Tail Gear Box Sight Plug | 6-4-55-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Powertrain Repairer’s Toolkit

Materials/Parts

- Lockwire, Item 197, Appendix D
- Lubricating Oil, Item 205, Appendix D

- Lubricating Oil, Item 206, Appendix D
- Lubricating Oil, Item 208, Appendix D

References

- Appendix D
- PARA 1-3-14
- PARA 1-6-16
- PARA 2-4-137

6-4-55.1 Replace Tail Gear Box Sight Plug.

6-4-55.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Drain tail gear box (PARA 1-3-14).
- Unscrew and remove tail gear box sight plug and packing from tail gear box. Discard packing (Figure 6-4-55-1, Detail B).

6-4-55.1.2 Install.

- Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing and install on tail gear box sight plug (Figure 6-4-55-1, Detail B).
- Install sight plug into tail gear box.
- TIGHTEN SIGHT PLUG 1⁄6 TO 1⁄3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

- Using lockwire, Item 197, Appendix D, lockwire sight plug to tail gear box.



Injury to personnel and damage to equipment will result if tail gear box is operated with low oil level. Operation with low oil level or without oil will cause loss of tail rotor drive. Lightly shake pylon to observe oil sloshing when checking oil level.

- Service tail gear box (PARA 1-3-14). Do not install tail gear box fairing.
- Make sure area is clean and free of foreign material.
- Perform check of sight plug as follows:
  - With qualified person at controls, run up helicopter.

- (2) Run up helicopter until tail gear box oil warms up.
- (3) Shut down helicopter.
- (4) Check tail gear box sight plug for leaks.
- h. Install tail gear box fairing (PARA 2-4-137).

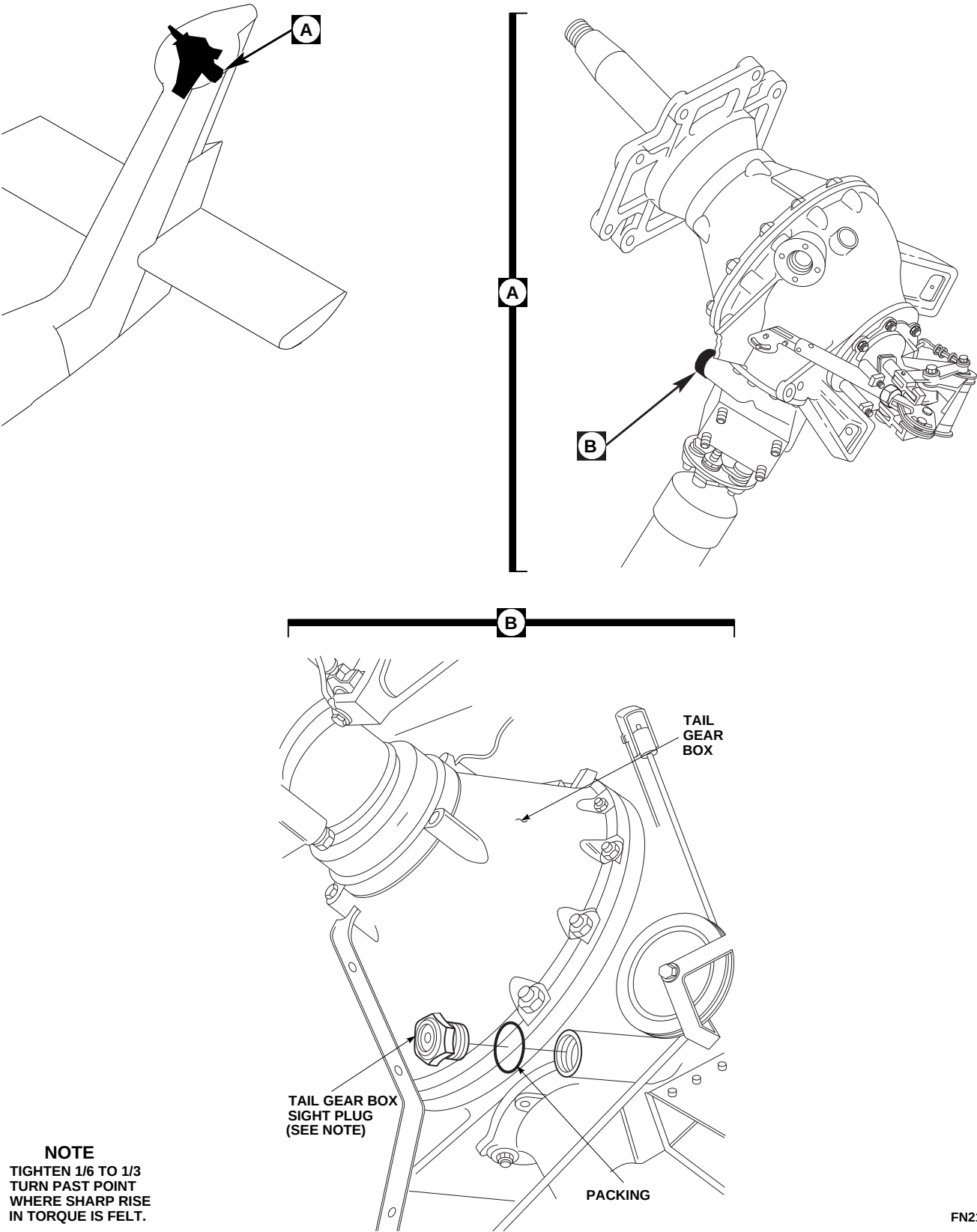


FIGURE 6-4-55-1. REPLACE TAIL GEAR BOX SIGHT PLUG





6-4-56 TAIL GEAR BOX OIL FILLER PLUG.

PARAGRAPH BREAKDOWN

|                                                | PAGE     |
|------------------------------------------------|----------|
| 6-4-56.1 Replace Tail Gear Box Oil Filler Plug | 6-4-56-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Scraper, Locally-Made
- Powertrain Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Lockwire, Item 197, Appendix D
- Lubricating Oil, Item 205, Appendix D

- Lubricating Oil, Item 206, Appendix D
- Lubricating Oil, Item 208, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-4-137

6-4-56.1 Replace Tail Gear Box Oil Filler Plug.

6-4-56.1.1 Remove.

- Remove tail gear box fairing (PARA 2-4-137).
- Unscrew and remove tail gear box oil filler plug (with cap attached) from tail gear box (Figure 6-4-56-1, Detail A). Remove and discard packing.



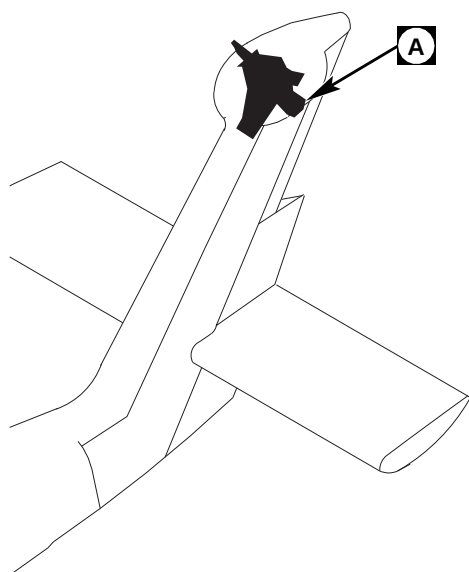
Damage to tail gear box will result if sealing compound is allowed to enter gear box during cleaning. Use extreme

care when scraping sealing compound from gear box boss.

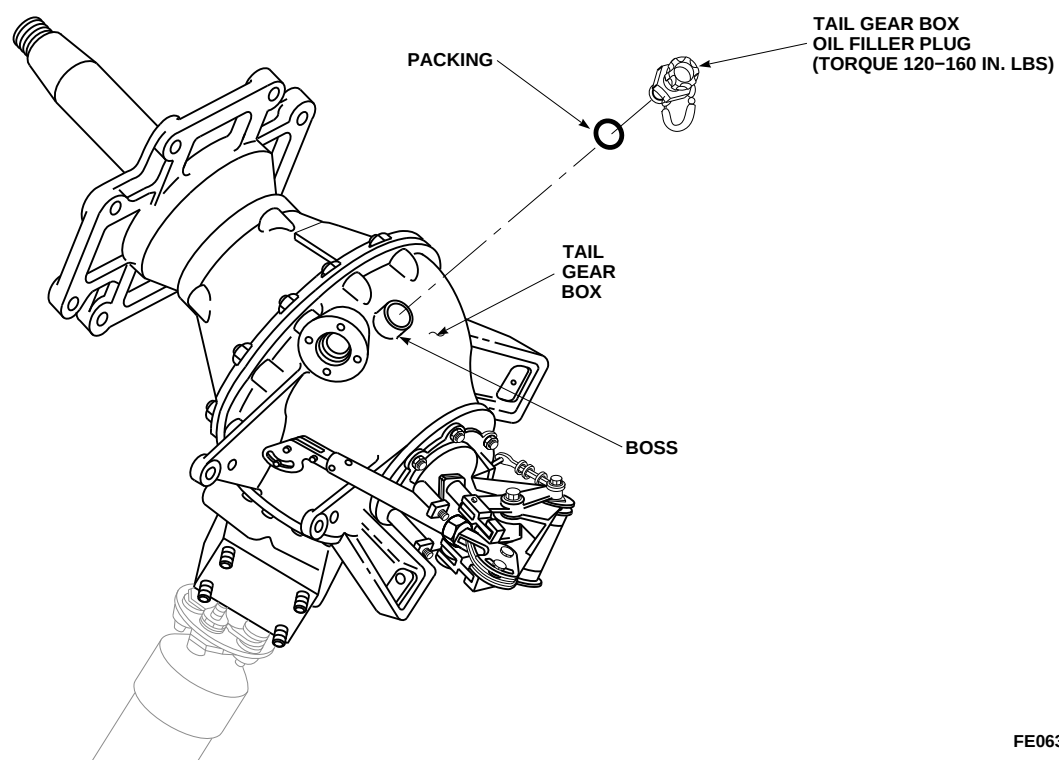
- Using nonmetallic scraper, remove sealing compound from boss of tail gear box.

6-4-56.1.2 Install.

- Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing and install on tail gear box oil filler plug (Figure 6-4-56-1, Detail A).
- Install tail gear box oil filler plug on tail gear box.
- TORQUE TAIL GEAR BOX OIL FILLER PLUG TO 120 - 160 INCH-POUNDS.**



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**FIGURE 6-4-56-1. REPLACE TAIL GEAR BOX OIL FILLER PLUG**

6-4-56-2

Change 5

- d. Lockwire, Item 197, Appendix D, tail gear box oil filler plug to tail gear box.
- e. Apply sealing compound, Item 292, Appendix D, to seal area around base of tail gear box oil filler plug.
- f. Perform check of tail gear box oil filler plug as follows:
  - (1) With qualified person at controls, run up helicopter.
  - (2) Run up helicopter until tail gear box oil warms up.
  - (3) Shut down helicopter.
  - (4) Check tail gear box oil filler plug for leaks.
- g. Install tail gear box fairing (PARA 2-4-137).



6-4-57 TAIL GEAR BOX HOUSING REPAIR.

PARAGRAPH BREAKDOWN

|                                       | PAGE     |
|---------------------------------------|----------|
| 6-4-57.1 Repair Tail Gear Box Housing | 6-4-57-2 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Beaker (10 ml)
- Beaker (200 ml)
- Beaker (600 ml)
- Container, Plastic, 1-Pint
- Container, Plastic, 2-Gallon
- Fluorescent Penetrant Inspection Kit
- Powertrain Repairer’s Toolkit
- Protection, Eye
- Vessel(s), with equivalent measuring accuracy of Beakers (Alternate for Beakers)

Materials/Parts

- Abrasive Cloth, Item 6, Appendix D
- Abrasive Cloth, Item 10, Appendix D
- Acetic Acid, Item 24, Appendix D
- Applicator, Item 78, Appendix D

- Chromium Trioxide, Technical, Item 91, Appendix D
- Distilled Water, Item 123, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Coating Kit, Item 126A, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Ethylene Glycol, Item 137, Appendix D
- Machinery Towel, Item 211, Appendix D
- Nitric Acid, Item 220, Appendix D
- Paint, Polyurethane, Item 224, Appendix D
- Paint Remover, Item 231, Appendix D
- Pressure Sensitive Adhesive Tape, Item 250, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 6-4-52

**6-4-57.1 Repair Tail Gear Box Housing.****6-4-57.1.1 General.****NOTE**

- Minor nicks, gouges, scratches, or corrosion on surface of tail gear box housing may be repaired. Before starting repair, measure depth of damage to determine whether blending out damage will result in tail gear box still being within repair limits. Make sure to measure depth of damage from an area which has not been previously repaired. If limits will be exceeded, replace tail gear box.
- Repair of damaged area may be done with tail gear box installed.
- The following repair procedures apply to input, center, and output housings (Figure 6-4-57-1).
- All repairs shall be recorded in helicopter component historical records.

- a. Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean any oil or grease from damaged area.

**NOTE**

Cracks will appear as radial lines extending outward from center of tail gear box bushing and/or extending down the side of mounting foot.

- b. Before blending, visually check damaged area for cracks. **NONE ALLOWED.** If cracks are found, replace tail gear box (PARA 6-4-52).
- c. Fine scratches that remove only paint may be blended smooth using abrasive cloth, Item 10, Appendix D. After blending, touch up blended area as required using epoxy coating kit, Item 126A, Appendix D, or polyurethane paint, Item 224, appendix D.
- d. Nicks, gouges, scratches, or corrosion on surface of tail gear box housing may be

blended out to a 1/2-inch radius. Only two repairs are allowed in any 4-square-inch area. Refer to PARA 6-4-57.1.2 for depths of repair.

**6-4-57.1.2 Repair Limits.**

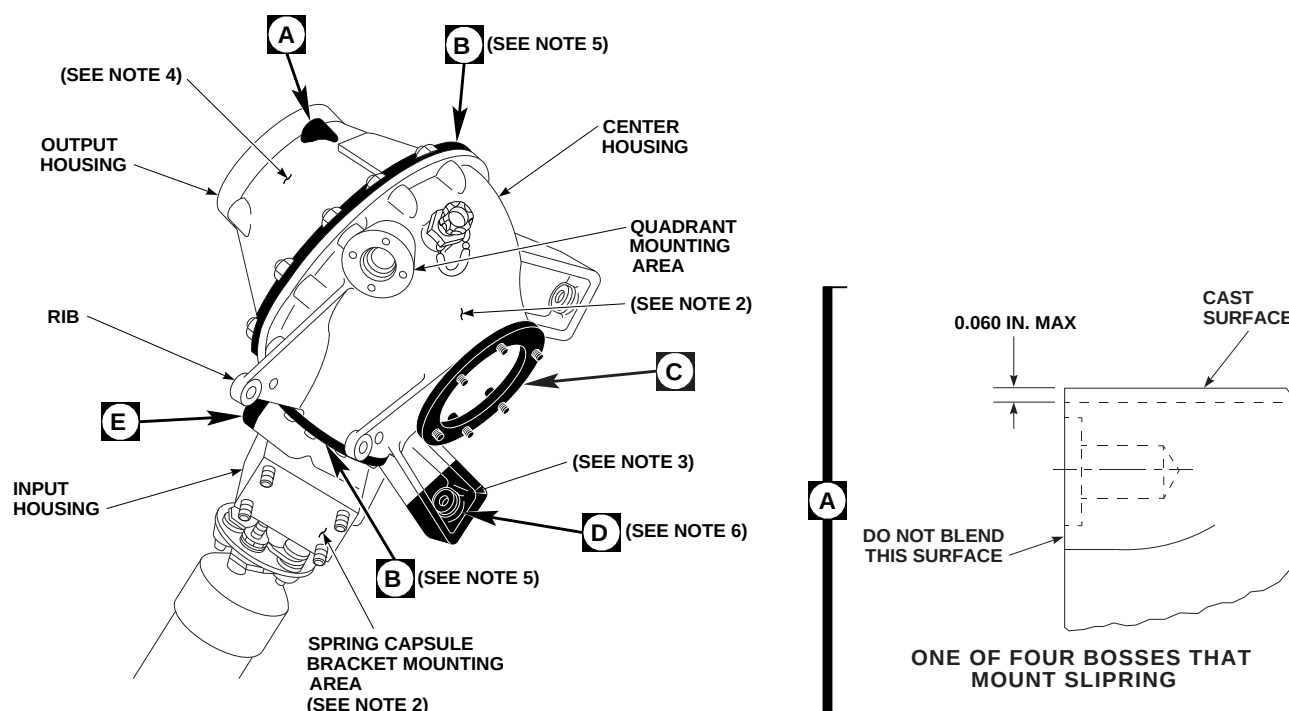
- a. Input housing limits are as follows:

- (1) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM ALL OVER OUTER CAST SURFACES IS 0.040-INCH.**

**NOTE**

Only 30% of the total number of holes in bolt pattern may be repaired.

- (2) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM INSIDE OF BOLT-HOLES IS 0.015-INCH** (Figure 6-4-57-1, Sheet 1, Detail B).
- (3) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM MOUNTING SURFACE UNDER FASTENERS (BUT NOT COVERING MORE THAN A 90° SEGMENT) IS 0.015-INCH.**
- (4) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SPOT-FACE SURFACE NOT COVERED BY FASTENER IS 0.030-INCH.**
- (5) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SHARP EDGE AROUND SPOT-FACE SURFACE (NOT INCLUDING AREA UNDER FASTENER) IS 0.060-INCH.**
- (6) **CORNER OF SPOT-FACE MAY BE BLENDED TO 0.100-INCH RADIUS AS LONG AS DEPTH INTO SPOT-FACE DOES NOT EXCEED 0.030-INCH.**
- (7) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM OPEN SPOT-FACE SURFACE NOT COVERED BY TAIL GEAR BOX SIGHT GAGE OR TAIL GEAR BOX CHIP DETECTOR/TEMPERATURE SENSOR IS 0.040-INCH** (Figure 6-4-57-1, Sheet 3, Detail E).



# NOTES

1. 0.040 IN. MAXIMUM DEPTH OF MATERIAL REMOVAL FROM ALL OVER OUTER CAST SURFACES OF INPUT HOUSING.
2. 0.025 IN. MAXIMUM DEPTH OF MATERIAL REMOVAL FROM ALL OVER OUTER CAST SURFACES, OTHER THAN MOUNTING FEET AND ADJACENT RIBS, OF CENTER HOUSING.
3. 0.015 IN. MAXIMUM DEPTH OF MATERIAL REMOVAL IN AREA OF MOUNTING FEET AND ADJACENT RIBS OF CENTER HOUSING.
4. 0.040 IN. MAXIMUM DEPTH OF MATERIAL REMOVAL ALL OVER OUTER CAST SURFACES, OTHER THAN BOSSES THAT MOUNT SLIPRING, OF OUTPUT HOUSING.
5. ONLY 30% OF THE TOTAL NUMBER OF HOLES IN ANY GIVEN BOLT PATTERN MAY BE REPAIRED AS SHOWN IN DETAIL B.
6. DO NOT REMOVE MATERIAL FROM MOUNTING BOLTHOLE BUSHINGS. 0.015 IN. MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SURFACE AROUND MOUNTING BOLTS. NO MORE THAN 2 REPAIRS SHALL BE MADE PER MOUNTING FOOT.
7. SPOT-FACE CORNER MAY BE BLENDED TO 0.050-0.100 IN. RADIUS AS LONG AS REPAIR DOES NOT EXTEND DEEPER THAN 0.040 IN. INTO SPOTFACE AREA OF RIB (DOTTED LINE ON SECTION A-A).

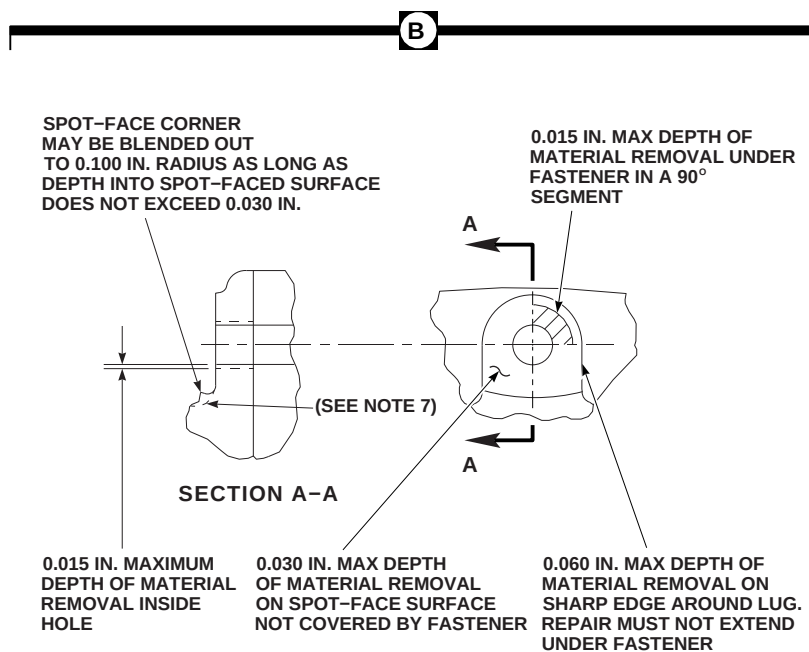
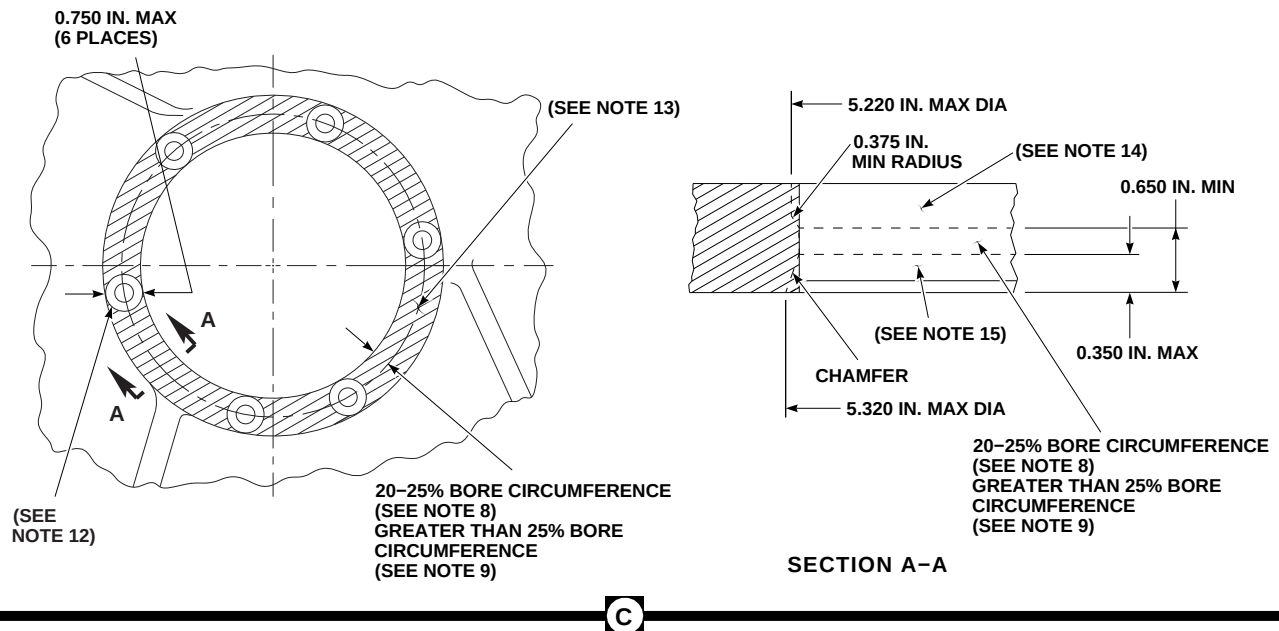


FIGURE 6-4-57-1. REPAIR TAIL GEAR BOX HOUSING (SHEET 1 OF 3)

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## NOTES

8. DEPTH OF REPAIR NOT TO EXCEED 5.124 IN. DIAMETER.
9. REWORK BORE OVERSIZE TO 5.122 / 5.124 IN. DIAMETER TO REMOVE CORROSION MAINTAINING CONCENTRICITY OF 0.002 IN. DIAMETER.
10. REMOVE BUSHING TO REPAIR CORROSION AT DEPOT LEVEL ONLY.
11. DO NOT REMOVE MORE THAN 10% OF SURFACE AREA OF MOUNTING FOOT IN ANY 1-SQUARE IN. AREA OR MORE THAN 30% OF TOTAL BASE OF MOUNTING FOOT.
12. BLEND AREAS OF CORROSION USING 0.250 IN. MINIMUM RADIUS. MAXIMUM DEPTH IS 0.030 IN. MAXIMUM AREA IS 20% AT EACH LUG.
13. BLEND AREAS OF CORROSION USING 0.250 IN. MINIMUM RADIUS. MAXIMUM DEPTH IS 0.060 IN. FROM SURFACE OF LUG. BLENDED AREA SHALL NOT EXCEED 50% BETWEEN LUGS.
14. BLEND WITH A MINIMUM OF 0.375 RADIUS MAINTAINING A 0.650 IN. DIMENSION AND NOT EXCEEDING 5.220 IN. DIAMETER AS SHOWN. DEPTH OF BLENDED AREAS IS NOT TO EXTEND BEYOND THE AREA SPECIFIED.
15. BLEND AREAS OF CORROSION USING 0.375 IN. MINIMUM RADIUS. DEPTH OF BLENDED AREA IS NOT TO EXTEND BEYOND CHAMFER. DIAMETER OF CHAMFER IS NOT TO EXCEED 5.320 IN. DIAMETER AND DEPTH OF CHAMFER NOT TO EXCEED 0.350 IN.

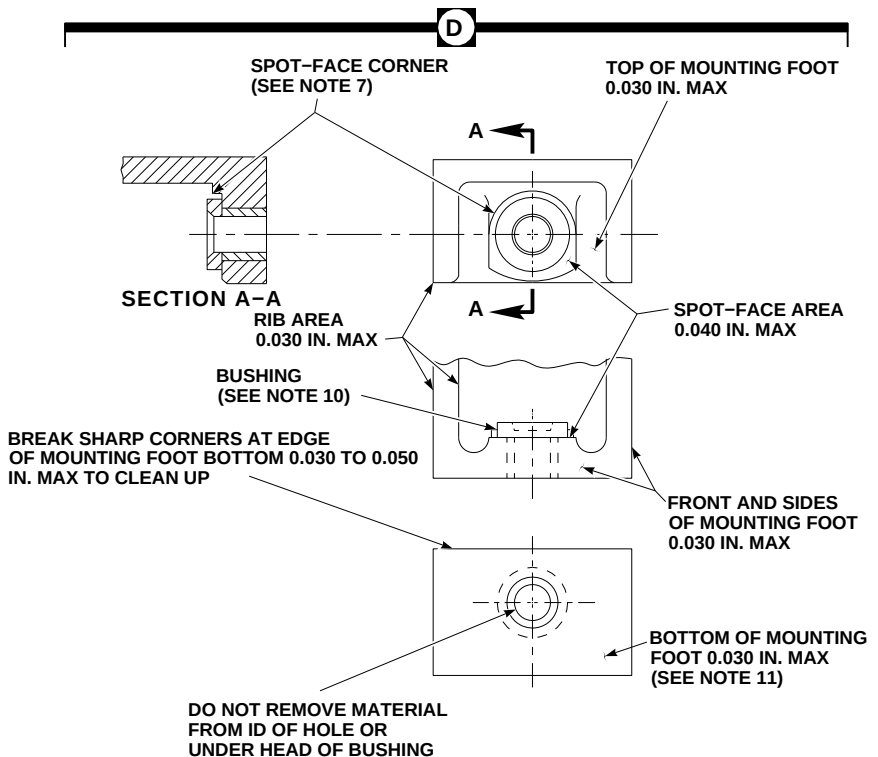
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FIGURE 6-4-57-1. REPAIR TAIL GEAR BOX HOUSING (SHEET 2 OF 3)



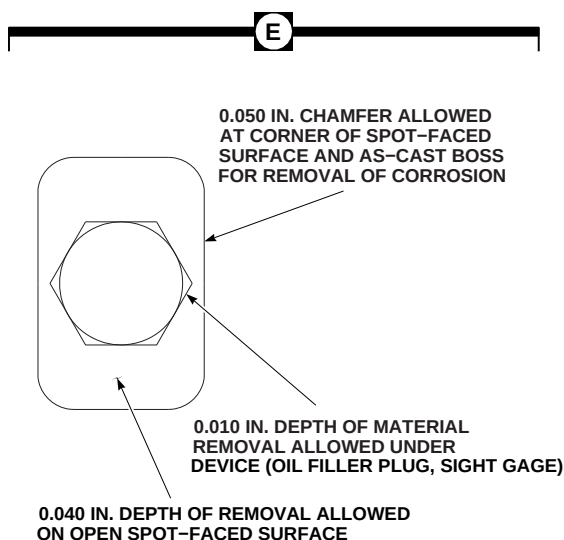
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FIGURE 6-4-57-1. REPAIR TAIL GEAR BOX HOUSING (SHEET 3 OF 3)

- (8) MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SURFACE UNDER TAIL GEAR BOX SIGHT GAGE OR TAIL GEAR BOX CHIP DETECTOR/TEMPERATURE SENSOR IS 0.010-INCH.
  - (9) MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SPOT-FACE CORNER AROUND TAIL GEAR BOX SIGHT GAGE OR TAIL GEAR BOX CHIP DETECTOR/TEMPERATURE SENSOR IS 0.040-INCH WHEN BLENDED TO A 0.050-INCH RADIUS.
  - (10) MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SPRING CAPSULE BRACKET MOUNTING AREA IS 0.040-INCH.
- b. Center housing limits are as follows:
- (1) MAXIMUM DEPTH OF MATERIAL REMOVAL FROM OUTER CAST SURFACES (NOT INCLUDING MOUNTING FEET AND MOUNTING FEET RIBS) IS 0.025-INCH.
  - (2) MAXIMUM DEPTH OF MATERIAL REMOVAL ON MOUNTING FEET AND MOUNTING FEET RIBS IS 0.030-INCH (Figure 6-4-57-1, Sheet 2, Detail D).
    - (a) NO MORE THAN TWO REPAIRS ALLOWED PER MOUNTING FOOT.
    - (b) TOTAL REPAIR SHALL NOT EXCEED 10% OF SURFACE AREA OF MOUNTING FOOT IN ANY 1-SQUARE-INCH AREA OR MORE THAN 30% OF TOTAL BASE OF MOUNTING FOOT.
    - (c) NO MATERIAL REMOVAL FROM BUSHING BORE ALLOWED.
    - (d) SPOT-FACE CORNER MAY BE BLENDED TO 0.050 - 0.100-INCH RADIUS AS LONG AS REPAIR DOES NOT PENETRATE DEEPER THAN 0.040-INCH INTO SPOT-FACE.

(e) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SPOT-FACE SURFACE OF MOUNTING FOOT IS 0.040-INCH.**

(3) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM OPEN SPOT-FACE SURFACE NOT COVERED BY FILLER PLUG IS 0.040-INCH.**

(4) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SURFACE UNDER FILLER PLUG IS 0.010-INCH.**

(5) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SPOT-FACE CORNER AROUND FILLER PLUG IS 0.040-INCH WHEN BLENDED TO A 0.050-INCH RADIUS.**

(6) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM QUADRANT MOUNTING AREA IS 0.025-INCH.**

(7) Refer to Figure 6-4-57-1, Sheet 2, Detail C, for repair limits on tail rotor servo mounting area.

c. Output housing limits are as follows:

(1) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM OUTER CAST SURFACE (EXCEPT SLIPRING MOUNT BOSSES) IS 0.040-INCH.**

#### **NOTE**

Only 30% of the total number of holes in bolt pattern may be repaired.

(2) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM INSIDE OF BOLT-HOLES IS 0.015-INCH** (Figure 6-4-57-1, Sheet 1, Detail B).

(3) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM MOUNTING**

**SURFACE UNDER FASTENERS (BUT NOT COVERING MORE THAN A 90° SEGMENT) IS 0.015-INCH.**

(4) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SPOT-FACE SURFACE NOT COVERED BY FASTENER IS 0.030-INCH.**

(5) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SHARP EDGE AROUND SPOT-FACE SURFACE (NOT INCLUDING AREA UNDER FASTENER) IS 0.060-INCH.**

(6) **CORNER OF SPOT-FACE MAY BE BLENDED TO 0.100-INCH RADIUS AS LONG AS DEPTH INTO SPOT-FACE DOES NOT EXCEED 0.030-INCH.**

(7) **MAXIMUM DEPTH OF REPAIR TO DEICE SLIPRING MOUNTING BOSS CAST SURFACE IS 0.060-INCH** (Figure 6-4-57-1, Sheet 1, Detail A).

(8) **NO MATERIAL REMOVAL FROM SLIPRING MOUNTING SURFACE ALLOWED.**

#### **6-4-57.1.3 Repair Procedure.**

- a. Using paint remover, Item 231, Appendix D, remove paint from damaged area.
- b. Using abrasive cloth, Item 6, Appendix D, carefully blend damaged area smooth. Remove minimum amount of material required to blend out damaged area.
- c. Using abrasive cloth, Item 10, Appendix D, blend reworked area to remove all traces of abrasive cloth.
- d. Remove displaced metal caused by abrasive blending by using magnesium-etching procedure as follows:

CAUTION

- Magnesium-etchant measurements are to be as accurate as possible to values measured in a milliliter beaker or vessel with equivalent mixing accuracy.
  - Magnesium-etchant formulas shall be prepared and stored in unbreakable 1-pint plastic containers and safely stored in designated areas not subjected to extreme heat or cold.
  - Before preparing magnesium-etchant, make sure storage area does not contain premixed magnesium-etchant formula. The correct stock bottle for this repair shall be marked MAGNESIUM-ETCHANT.
- (1) Prepare magnesium-etchant solution in clean unbreakable plastic container by mixing the following:

| MATERIAL                              | QTY             |
|---------------------------------------|-----------------|
| Ethylene Glycol, Item 137, Appendix D | 600 milliliters |
| Acetic Acid, Item 24, Appendix D      | 200 milliliters |
| Nitric Acid, Item 220, Appendix D     | 10 milliliters  |
| Distilled Water, Item 123, Appendix D | 190 milliliters |

- (2) Using dry-cleaning solvent, Item 124, Appendix D, clean area to be etched.
- (3) Using pressure sensitive adhesive tape, Item 250, Appendix D, mask area.
- (4) Pour a small amount of magnesium-etchant from stock bottle into 1-pint plastic container.

CAUTION

- Damage to components will result if magnesium-etchant is allowed to come in contact with areas not being etched. Extreme caution shall be used to protect those areas not being etched.
- (5) Using applicator, Item 78, Appendix D, apply magnesium-etchant to reworked area. Swab area for about 10 - 15 seconds.
- (6) Neutralize magnesium-etchant immediately by flooding area with generous amount of clean water.

WARNING

- Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.
- (7) Using clean, dry, compressed air, blow dry area thoroughly.
- (8) After use, do not return unused magnesium-etchant to stock bottle. Dispose of used magnesium-etchant per local instructions.
- (9) Identify contents of stock bottle as MAGNESIUM-ETCHANT, before placing bottle in storage area.
- (10) Prepare technical chromium trioxide solution in clean 2-gallon plastic container by mixing 24 ounces of technical chromium trioxide, Item 91, Appendix D, with 1 gallon of water.
- (11) Using technical chromium trioxide solution, treat bare surface of housing.

- e. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type I, Method C, reworked area for cracks (ASTM E1417). **NONE ALLOWED.** If cracks are found, replace tail gear box (PARA 6-4-52).
- f. Apply sealing compound, Item 292, Appendix D, to touch up reworked areas.
- g. After sealing compound has dried, using epoxy primer coating kit, Item 135, Appendix D, apply one coat of epoxy primer.

**NOTE**

Housings coated with epoxy coating are glossy silver. Housings coated with

polyurethane paint are matte surface medium-gray.

- h. Using epoxy coating kit, Item 126A, or polyurethane paint, Item 224, Appendix D, apply one coat of epoxy primer, depending on original paint finish.
- i. Make sure area is clean and free of foreign material.
- j. Record brief description of tail gear box repairs in helicopter component historical records.

6-4-58 TAIL GEAR BOX PULLEY GUARDS.

PARAGRAPH BREAKDOWN

|                                              | PAGE     |
|----------------------------------------------|----------|
| 6-4-58.1 Replace Tail Gear Box Pulley Guards | 6-4-58-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drift
- Drill
- Hand Reamer Set, Adjustable

Materials/Parts

- Primer, Item 258, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 11-4-107

6-4-58.1 Replace Tail Gear Box Pulley Guards.

NOTE

Replacement of both tail gear box pulley guards is the same.

6-4-58.1.1. Remove.

- Remove tail gear box pulley (PARA 11-4-107).
- Using drill, drill out rivet. Remove rivet, washer, and spacer (Figure 6-4-58-1, Detail B).

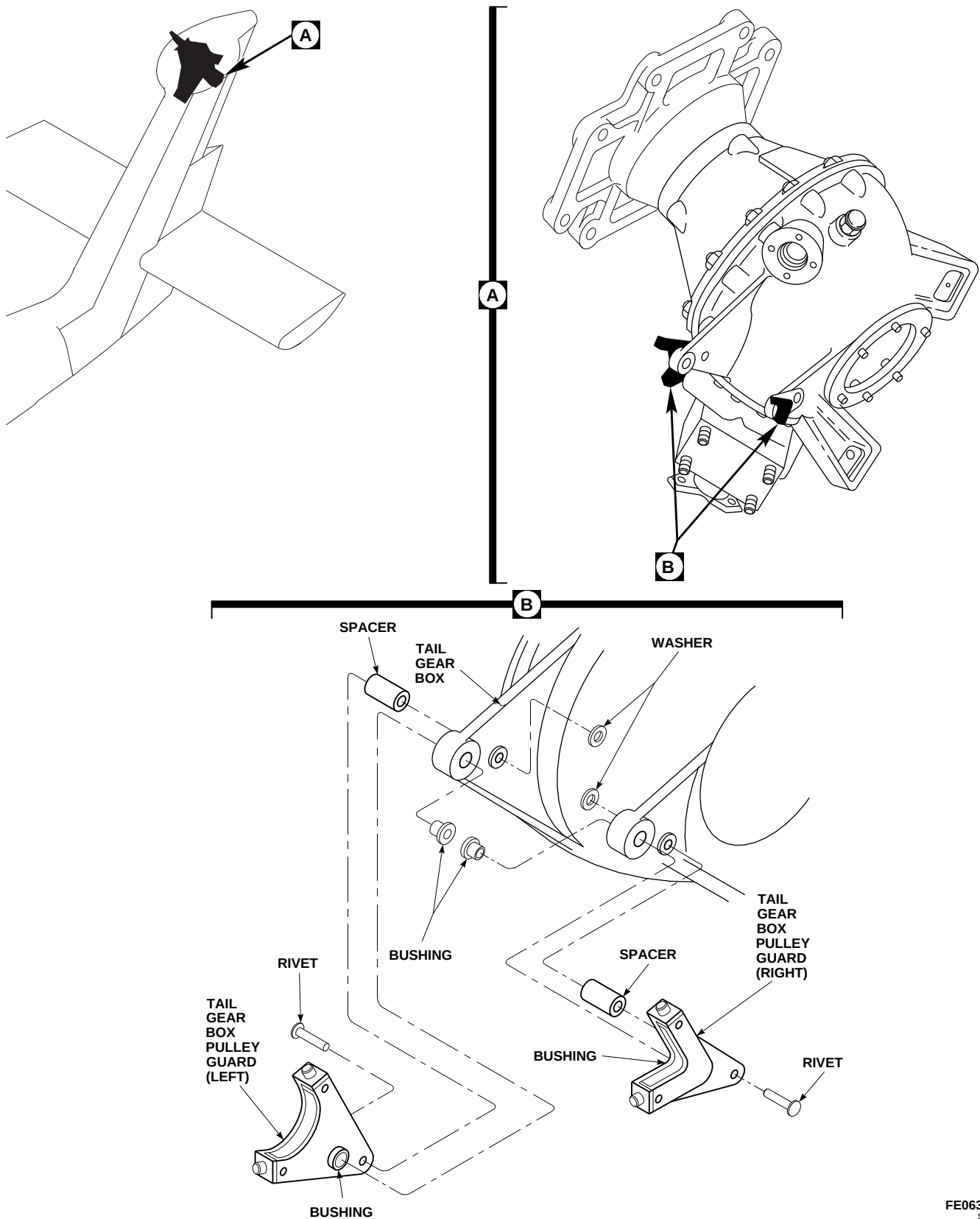
NOTE

There is a 1⁄8-inch gap between bushing ends in center of tail gear box boss.

- Using drift, tap bushing and tail gear box pulley guard from tail gear box. Be careful not to scar hole in tail gear box boss.
- Tap out remaining bushing.

6-4-58.1.2. Install.

- If necessary, using adjustable hand reamer set, ream large hole in tail gear box pulley guard to 0.437 - 0.438-inch.
- Apply a coat of primer, Item 258, Appendix D, to area under shoulder of bushing.
- Press bushing into tail gear box pulley guard while primer is still wet (Figure 6-4-58-1, Detail B).



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FIGURE 6-4-58-1. REPLACE TAIL GEAR BOX PULLEY GUARDS

6-4-58-2

- d. Apply a coat of primer, Item 258, Appendix D, to outer surface of bushing and area where tail gear box pulley guard will contact housing.



Damage to control cable will result if wrong tail gear box pulley guard is installed. Right and left tail gear box pulley guards are not interchangeable. Prior to installing tail gear box pulley guard, make sure that left tail gear box pulley guard is 70400-06670-042, 70400-06670-045, or 70400-06670-047, and right tail gear box pulley guard is 70400-06670-046 or 70400-06670-048 (viewed looking aft).

- e. Press bushing and tail gear box pulley guard into tail gear box while primer is still wet.

- f. Apply a coat of primer, Item 258, Appendix D, to outer surface and area under shoulder of second bushing.

- g. Press bushing into tail gear box while primer is still wet.
- h. Install rivet, spacer, and washer in tail gear box and tail gear box pulley guard.
- i. Using washer with sealing compound, Item 292, Appendix D, seal areas around shoulders of bushings and end of rivet.
- j. Using adjustable hand reamer set, line-ream bushings to 0.312 - 0.313-inch.
- k. Install tail gear box pulley (PARA 11-4-107).





6-4-59 TAIL GEAR BOX CHIP DETECTOR/TEMPERATURE SENSOR.

PARAGRAPH BREAKDOWN

|          |                                                         | PAGE     |          |                            | PAGE     |
|----------|---------------------------------------------------------|----------|----------|----------------------------|----------|
| 6-4-59.1 | Replace Tail Gear Box Chip Detector/ Temperature Sensor | 6-4-59-1 | 6-4-59.2 | Replace Self-Closing Valve | 6-4-59-4 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container
- Protection, Eye

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Lockwire, Item 198, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D

- Lubricating Oil, Item 208, Appendix D
- Protective Caps and Plugs

References

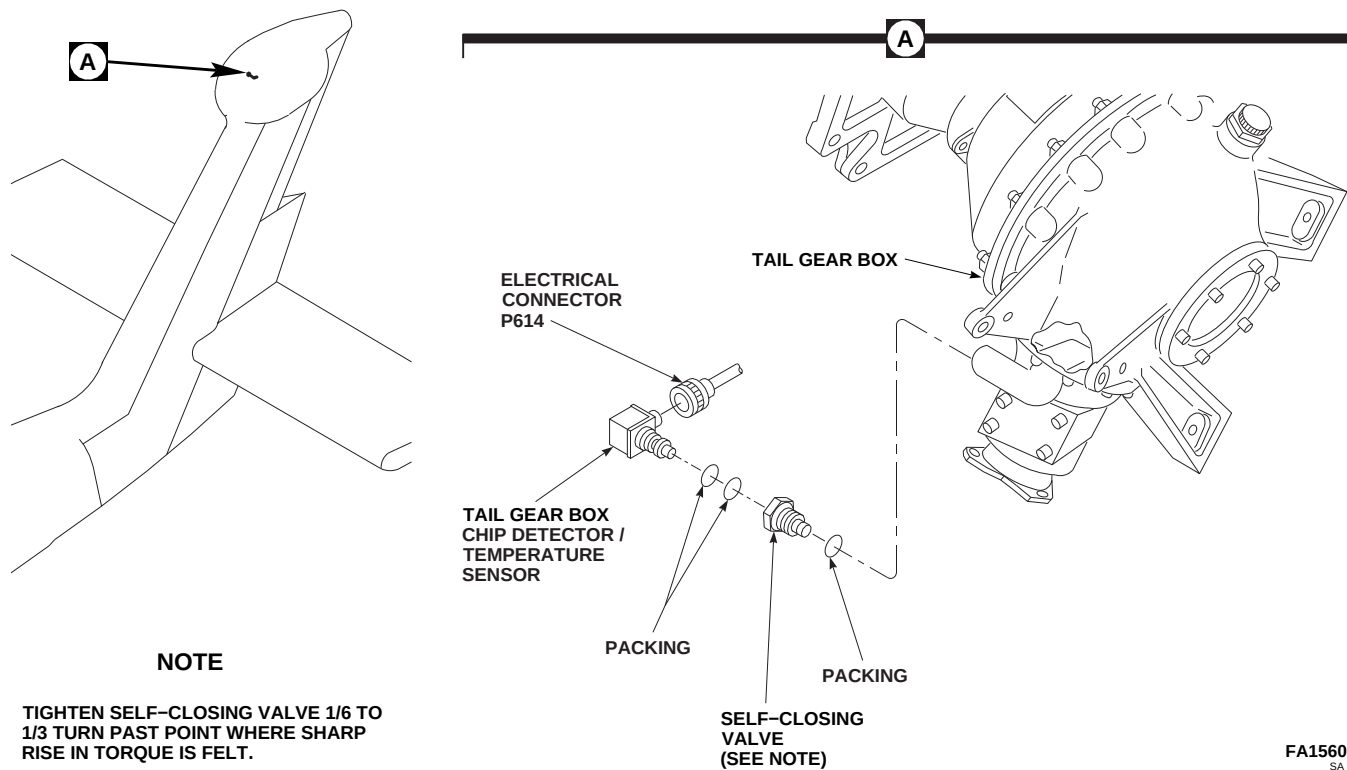
- Appendix D
- PARA 1-3-14
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-137
- PARA 6-2-2
- PARA 6-3-17

6-4-59.1 Replace Tail Gear Box Chip Detector/Temperature Sensor.

6-4-59.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- On copilot’s circuit breaker panel, pull out CHIP DET circuit breaker.

- Remove tail gear box fairing (PARA 2-4-137).
- Disconnect electrical connector, P614, from tail gear box chip detector/temperature sensor (Figure 6-4-59-1, Detail A).
- Install protective caps and plugs on electrical connectors.



**FIGURE 6-4-59-1. REPLACE TAIL GEAR BOX CHIP DETECTOR/TEMPERATURE SENSOR AND SELF-CLOSING VALVE**

**NOTE**

When removing chip detector/temperature sensor, some lubricating oil may drain out.

- f. Place container underneath chip detector/temperature sensor to catch oil.

**WARNING**

Injury to personnel will result if precautions are not taken when removing chip detector/temperature sensor. To prevent injury, personnel must wear eye protection and make sure gear box oil is cool

before removing chip detector/temperature sensor.

- g. Remove chip detector/temperature sensor and packings from self-closing valve in tail gear box. Discard packings.
- h. Install protective plug in self-closing valve.

**CAUTION**

Do not immerse chip detector/temperature sensor in dry-cleaning solvent.

- i. Using dry-cleaning solvent, Item 124, Appendix D, clean chip detector/temperature sensor.

**WARNING**

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- j. Using clean, dry, compressed air, blow dry chip detector/temperature sensor.

**6-4-59.1.2 Inspect.**

- a. Inspect tail gear box chip detector/temperature sensor for chips (PARA 6-3-17).

**NOTE**

Leakage of lubricating oil through center of self-closing valve indicates loose internal hardware. This disables temperature sensor portion of chip detector/temperature sensor.

- b. Remove protective plug from hole and visually inspect self-closing valve for leakage from center of self-closing valve. **NONE ALLOWED.** If any leakage is found, replace self-closing valve (PARA 6-4-59.2).
- c. Inspect inside of self-closing valve for contact pin located in center of self-closing valve and for leakage of oil. If contact pin is loose or missing, or if leakage is found, locate pin and washer and replace self-closing valve (PARA 6-4-59.2).
- d. If pin or washer cannot be located, perform tail gear box chip detector/temperature sensor particle inspections (PARA 6-3-17).
- e. If necessary, replace self-closing valve (PARA 6-4-59.2). If self-closing valve does not need to be replaced, install protective plug in hole.

**6-4-59.1.3 Install.**

**NOTE**

Tail gear box chip detector/temperature sensor and self-closing valve are a matched set and cannot be replaced individually. If replacement chip detector/temperature sensor is being installed, chip detector/temperature sensor and self-closing valve must be replaced as an assembly.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Remove protective plug from self-closing valve.
- c. Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packings to be installed on tail gear box chip detector/temperature sensor.
- d. Check both chip detector/temperature sensor electrical circuits (PARA 6-2-2).
- e. Install packings on chip detector/temperature sensor, and install in self-closing valve (Figure 6-4-59-1, Detail A).
- f. Remove protective caps and plugs from electrical connectors.
- g. Inspect and treat electrical connectors (PARA 1-7-20).
- h. Connect electrical connector, P614, to chip detector/temperature sensor.

**WARNING**

Injury to personnel and damage to equipment will result if gear box is operated with low oil level. Operation with low oil level or without oil will cause loss of tail rotor drive. Lightly shake pylon to observe oil sloshing when checking oil level.

- i. If self-closing valve was replaced, service tail gear box (PARA 1-3-14).

- j. Make sure area is clean and free of foreign material.
- k. Install tail gear box fairing (PARA 2-4-137).
- l. On copilot's circuit breaker panel, push in CHIP DET circuit breaker.
- m. Perform operational check of both chip detector and temperature sensor electrical circuits (PARA 6-2-2).

## 6-4-59.2 Replace Self-Closing Valve.

### 6-4-59.2.1 Remove.

- a. Drain tail gear box (PARA 1-3-14).
- b. If not already removed, remove tail gear box chip detector/temperature sensor (PARA 6-4-59.1.1), or protective plug from hole.
- c. Remove self-closing valve and packing from tail gear box. Discard packing (Figure 6-4-59-1, Detail A).
- d. Install protective plug in tail gear box.

### 6-4-59.2.2 Install.

#### NOTE

Tail gear box self-closing valve and chip detector/temperature sensor are a

matched set and cannot be replaced individually. If replacement self-closing valve is being installed, self-closing valve and chip detector/temperature sensor must be replaced as an assembly.

- a. Using lubricating oil, Item 205, Item 206, or Item 208, Appendix D, lubricate packing.
- b. Remove protective plug from tail gear box.
- c. Install packing on self-closing valve, and install valve in tail gear box (Figure 6-4-59-1, Detail A).
- d. **TIGHTEN SELF-CLOSING VALVE  $\frac{1}{6}$  TO  $\frac{1}{3}$  TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- e. Using lockwire, Item 198, Appendix D, lockwire self-closing valve to tail gear box.
- f. Install protective plug in self-closing valve.
- g. Install tail gear box chip detector/temperature sensor (PARA 6-4-59.1.3).

6-4-60 ROTOR BRAKE MASTER CYLINDER.

PARAGRAPH BREAKDOWN

|                                              | PAGE     |
|----------------------------------------------|----------|
| 6-4-60.1 Replace Rotor Brake Master Cylinder | 6-4-60-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Gallon
- Torque Wrench 30 - 150 in. lbs
- Torque Wrench 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 155, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-6-18
- PARA 6-4-64
- PARA 6-4-68

6-4-60.1 Replace Rotor Brake Master Cylinder.

6-4-60.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- If not already done, engage gust lock and disengage rotor brake (PARA 1-6-18).
- Position 1-gallon container under hydraulic tube connections of rotor brake master cylinder to catch hydraulic fluid.

NOTE

Protective caps or plugs must be installed as parts are disconnected or removed.

- Disconnect pressure tube from tee (Figure 6-4-60-1, Detail A).

- Remove rotor brake pressure gage (PARA 6-4-64).
- Remove tee from rotor brake master cylinder.
- Disconnect supply tube from rotor brake master cylinder.
- Remove bolts and washers securing rotor brake master cylinder to airframe.
- Remove rotor brake master cylinder.
- Remove jamnut, retainer, and packing from tee. Discard packing.

6-4-60.1.2 Install.

- Install rotor brake master cylinder on airframe with bolts and washers (Figure 6-4-60-1, Detail A).

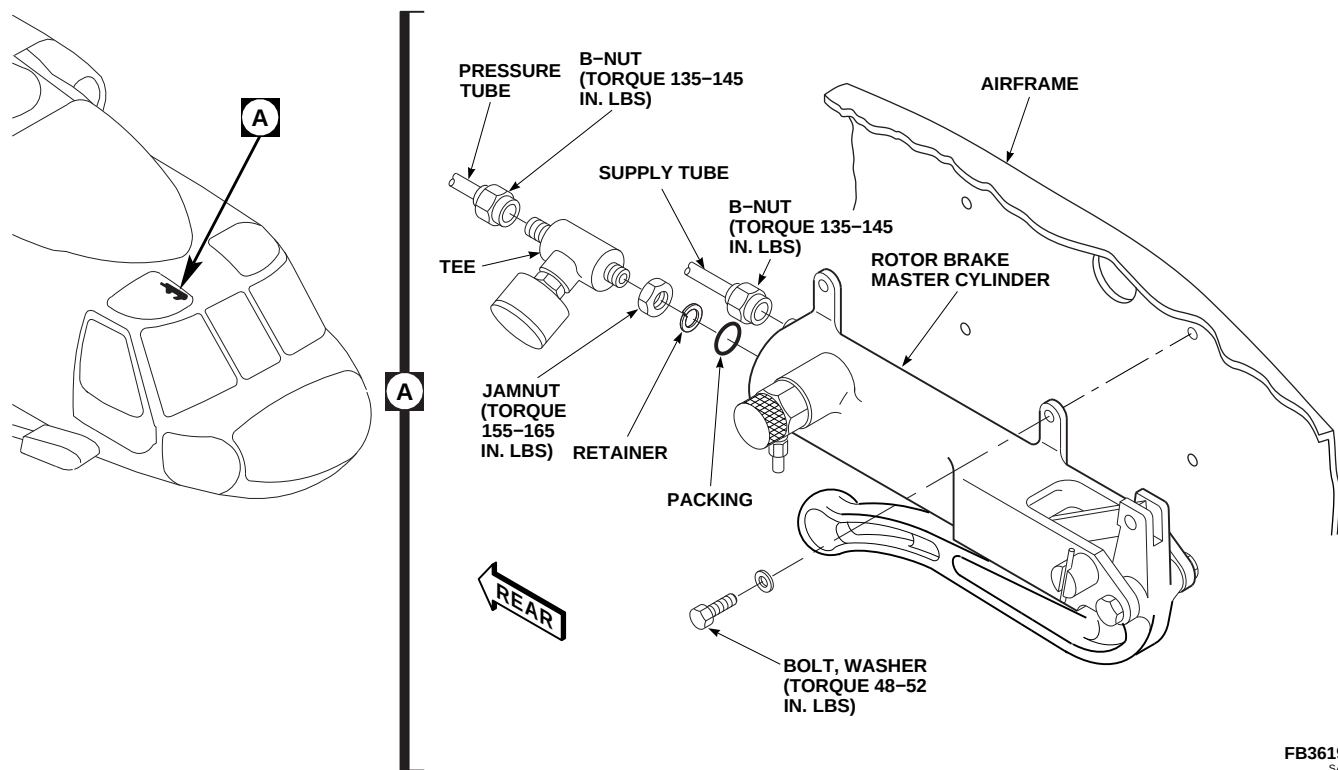


FIGURE 6-4-60-1. REPLACE ROTOR BRAKE MASTER CYLINDER

- b. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**
  - c. Connect supply tube to rotor brake master cylinder.
  - d. **TORQUE B-NUT TO 135 - 145 INCH-POUNDS.**
- NOTE**
- Protective caps or plugs must be removed as parts are installed or connected.
- e. Using hydraulic fluid, Item 155, Appendix D, lubricate packing, retainer, and male threads of tee.
  - f. Install jamnut, retainer, and packing on tee. Turn jamnut and retainer down until packing is pushed firmly against the lower threaded section of tee.
  - g. Install tee into rotor brake master cylinder.
  - h. **WHILE KEEPING JAMNUT FROM TURNING, TIGHTEN TEE AND JAMNUT 1 1/2 TURNS PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
  - i. **POSITION TEE BY TIGHTENING NOT MORE THAN ONE TURN.**
  - j. While maintaining position of tee, **TORQUE JAMNUT TO 155 - 165 INCH-POUNDS.**
  - k. Install rotor brake pressure gage (PARA 6-4-64).
  - l. Connect pressure tube to tee.
  - m. **TORQUE B-NUT TO 135 - 145 INCH-POUNDS.**
  - n. Bleed rotor brake system (PARA 6-4-68).

6-4-61 ROTOR BRAKE ACCUMULATOR.

PARAGRAPH BREAKDOWN

|                                          | PAGE     |
|------------------------------------------|----------|
| 6-4-61.1 Replace Rotor Brake Accumulator | 6-4-61-1 |

---

|                                                                                                                                                                             |                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>INITIAL SETUP</b>                                                                                                                                                        | <ul style="list-style-type: none"> <li>Protective Caps and Plugs</li> </ul>                                                                                        |
| <b>Tools</b> <ul style="list-style-type: none"> <li>Aircraft Mechanic’s Toolkit</li> <li>Torque Wrench 30 - 150 in. lbs</li> <li>Torque Wrench 150 - 750 in. lbs</li> </ul> | <b>References</b> <ul style="list-style-type: none"> <li>Appendix D</li> <li>PARA 1-6-16</li> <li>PARA 1-6-18</li> <li>PARA 1-7-20</li> <li>PARA 6-4-68</li> </ul> |
| <b>Materials/Parts</b> <ul style="list-style-type: none"> <li>Hydraulic Fluid, Item 155, Appendix D</li> <li>Machinery Towel, Item 211, Appendix D</li> </ul>               |                                                                                                                                                                    |

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6-4-61.1 Replace Rotor Brake Accumulator.

NOTE

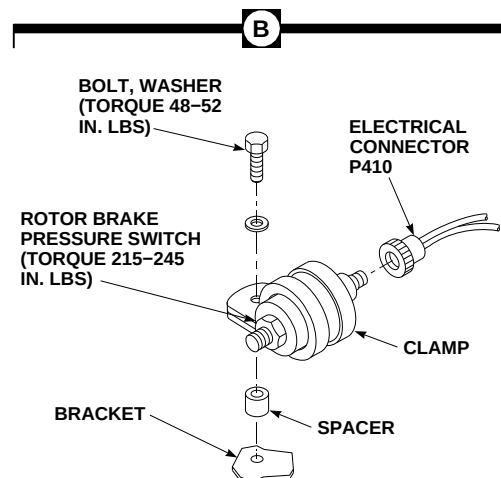
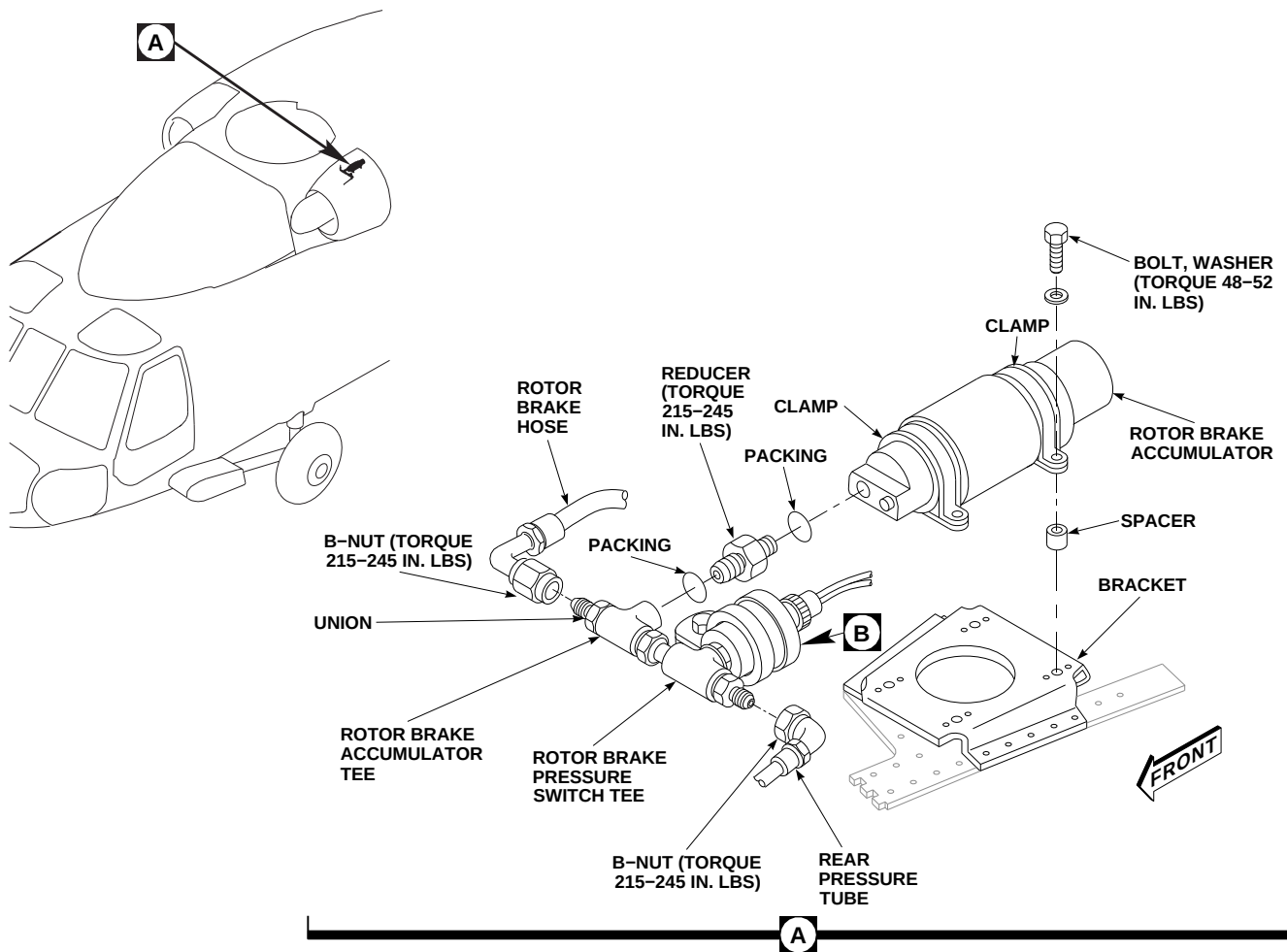
6-4-61.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- If not already done, engage gust lock and disengage rotor brake (PARA 1-6-18).
- Open oil cooler compartment access doors.
- Disconnect electrical connector, P410, from rotor brake pressure switch (Figure 6-4-61-1, Detail B).
- Install protective caps and plugs on electrical connectors.
- Using machinery towels, Item 211, Appendix D, position under rotor brake hose to catch spilled hydraulic fluid.

- Protective caps or plugs must be installed as parts are disconnected or removed.
- Disconnect rotor brake hose from union at rotor brake accumulator tee (Figure 6-4-61-1, Detail A).
- Disconnect rear pressure tube from rotor brake pressure switch tee.

NOTE

- Spacers between rotor brake accumulator and bracket are different lengths. Note location of each spacer as it is removed to make sure it will be installed in same location.
- Remove bolts, washers, clamps, and spacers securing rotor brake accumulator to bracket.



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SA

FIGURE 6-4-61-1. REPLACE ROTOR BRAKE ACCUMULATOR



- j. Remove bolt, washer, clamp, and spacer securing rotor brake pressure switch to bracket (Figure 6-4-61-1, Detail B).
- k. Remove rotor brake accumulator, rotor brake pressure switch, and tee assemblies as one piece (Figure 6-4-61-1, Detail A).
- l. Remove reducer and packings from rotor brake accumulator and rotor brake accumulator tee. Discard packings.

#### 6-4-61.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).

#### NOTE

Protective caps or plugs must be removed as parts are installed or connected.

- b. Using hydraulic fluid, Item 155, Appendix D, lubricate packings and install on reducer (Figure 6-4-61-1, Detail A).
- c. Install reducer on rotor brake accumulator.
- d. **TORQUE REDUCER TO 215 - 245 INCH-POUNDS.**

#### NOTE

Position rotor brake accumulator so bleed screw is accessible for bleeding.

- e. Install rotor brake accumulator with reducer, into rotor brake accumulator tee.
- f. **TORQUE REDUCER TO 215 - 245 INCH-POUNDS.**

#### NOTE

Spacers between rotor brake accumulator and bracket are different lengths.

Make sure spacers are installed in same location from which they were removed.

- g. Install rotor brake accumulator, rotor brake pressure switch, and tees to brackets with bolts, washers, clamps, and spacers. Do not torque bolts now.
- h. Connect rotor brake hose to union in rotor brake accumulator tee.
- i. **TORQUE B-NUT TO 215 - 245 INCH-POUNDS.**
- j. Connect rear pressure tube to rotor brake pressure switch tee.
- k. **TORQUE B-NUT TO 215 - 245 INCH-POUNDS.**
- l. **TORQUE BOLTS SECURING ROTOR BRAKE ACCUMULATOR AND ROTOR BRAKE PRESSURE SWITCH TO BRACKET TO 48 - 52 INCH-POUNDS** (Figure 6-4-61-1, Detail A and Detail B).
- m. Remove protective caps and plugs from electrical connectors.
- n. Inspect and treat electrical connectors (PARA 1-7-20).
- o. Connect electrical connector, P410, to rotor brake pressure switch (Figure 6-4-61-1, Detail B).
- p. Bleed rotor brake system (PARA 6-4-68).
- q. Make sure area is clean and free of foreign material.
- r. Close oil cooler compartment access doors.



6-4-62 ROTOR BRAKE MANIFOLD AND VALVE ASSEMBLY.

PARAGRAPH BREAKDOWN

|                                                          | PAGE     |
|----------------------------------------------------------|----------|
| 6-4-62.1 Replace Rotor Brake Manifold and Valve Assembly | 6-4-62-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-6-18
- PARA 6-4-68

6-4-62.1 Replace Rotor Brake Manifold and Valve Assembly.

6-4-62.1.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. If not already done, engage gust lock and disengage rotor brake (PARA 1-6-18).
- c. Open main rotor pylon sliding cover.
- d. Position machinery towels, Item 211, Appendix D, under rotor brake manifold and valve assembly to catch spilled hydraulic fluid.

NOTE

Protective caps or plugs must be installed as parts are disconnected or removed.

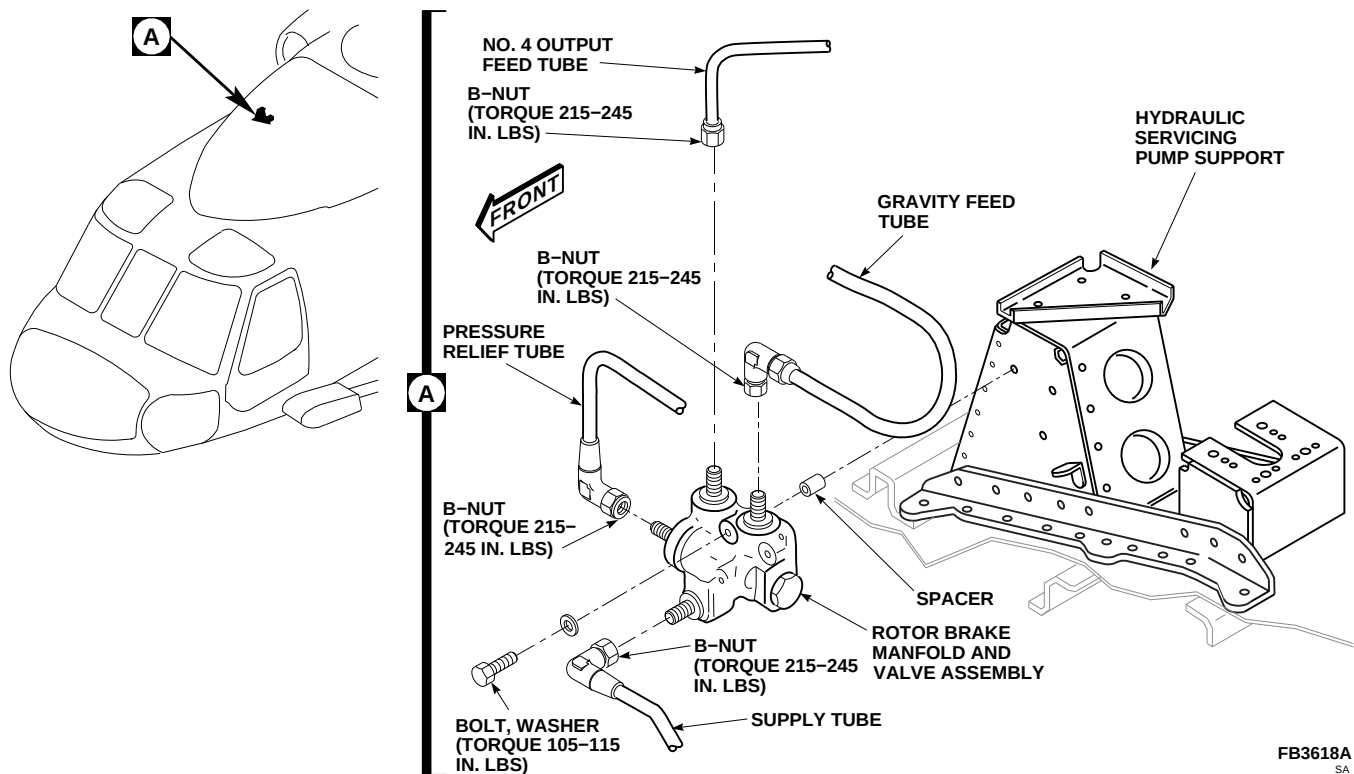
- e. Disconnect supply tube, pressure relief tube, No. 4 output feed tube, and gravity feed tube from manifold and valve assembly (Figure 6-4-62-1, Detail A).
- f. Remove bolts, washers, and spacers securing manifold and valve assembly to hydraulic servicing pump support.
- g. Remove manifold and valve assembly.

6-4-62.1.2 Install.

NOTE

Protective caps or plugs must be removed as parts are installed or connected.

- a. Install rotor brake manifold and valve assembly onto hydraulic servicing pump support

FB3618A  
SA**FIGURE 6-4-62-1. REPLACE ROTOR BRAKE MANFOLD AND VALVE ASSEMBLY**

with bolts, washers, and spacers (Figure 6-4-62-1, Detail A). Do not torque bolts now.

**NOTE**

When installing hydraulic tubes on rotor brake manifold and valve assembly, do not torque B-nuts on tubes until bolts securing manifold and valve assembly are torqued.

- b. Connect supply tube, pressure relief tube, No. 4 output feed tube, and gravity feed tube on manifold and valve assembly.
- c. **TORQUE BOLTS SECURING MANFOLD AND VALVE ASSEMBLY TO 105 - 115 INCH-POUNDS.**
- d. **TORQUE B-NUTS ON HYDRAULIC TUBES TO 215 - 245 INCH-POUNDS.**
- e. Bleed rotor brake system (PARA 6-4-68).
- f. Make sure area is clean and free of foreign material.
- g. Close main rotor pylon sliding cover.

6-4-63 ROTOR BRAKE PRESSURE SWITCH.

PARAGRAPH BREAKDOWN

|                                              | PAGE     |
|----------------------------------------------|----------|
| 6-4-63.1 Replace Rotor Brake Pressure Switch | 6-4-63-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench 30 - 150 in. lbs
- Torque Wrench 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 197, Appendix D

- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-6-18
- PARA 1-7-20
- PARA 6-4-68

6-4-63.1 Replace Rotor Brake Pressure Switch.

6-4-63.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- If not already done, engage gust lock and disengage rotor brake (PARA 1-6-18).
- Open oil cooler compartment access doors.

- Install protective caps and plugs on electrical connectors.
- Remove bolt, washer, and spacer securing clamp to bracket. Remove clamp from pressure switch.
- Remove pressure switch and packing from rotor brake pressure switch tee. Discard packing.

6-4-63.1.2 Install.

NOTE

Protective caps or plugs must be installed as parts are disconnected or removed.

- Disconnect electrical connector, P410, from rotor brake pressure switch (Figure 6-4-63-1, Detail A).

NOTE

Protective caps or plugs must be removed as parts are installed or connected.

- Using hydraulic fluid, Item 155, Appendix D, lubricate packing and install on rotor brake pressure switch (Figure 6-4-63-1, Detail A).

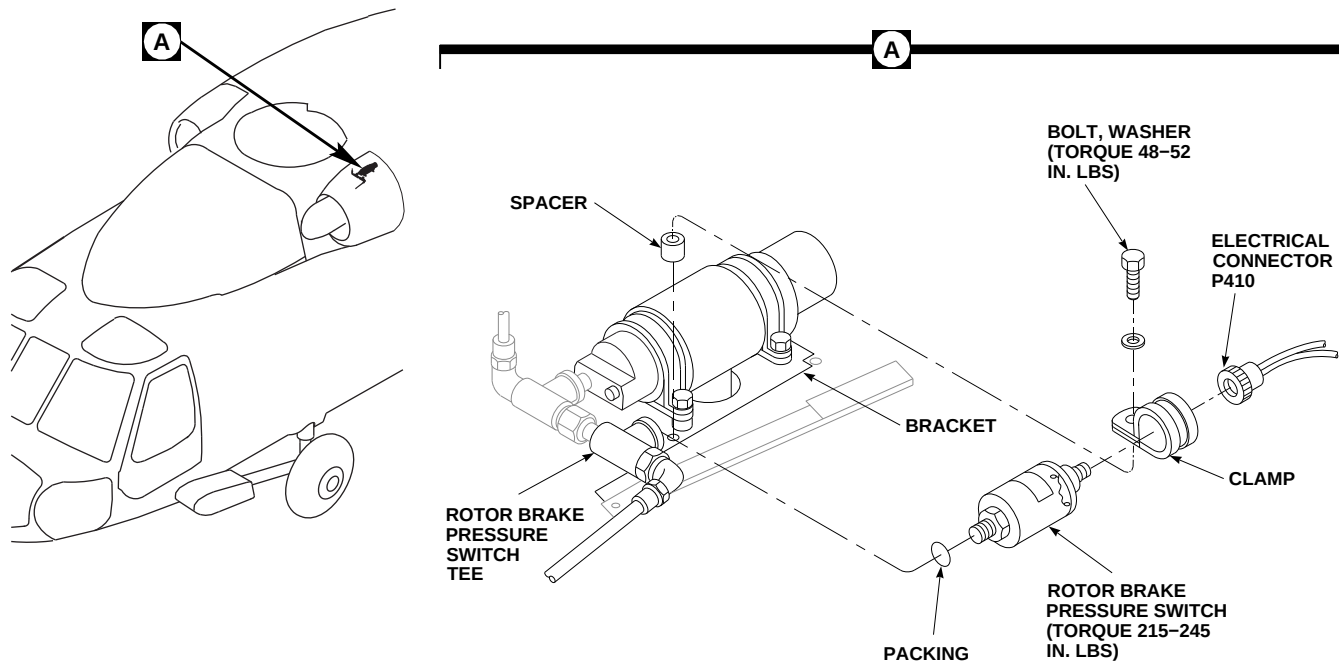
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SA

FIGURE 6-4-63-1. REPLACE ROTOR BRAKE PRESSURE SWITCH

- b. Install pressure switch in rotor brake pressure switch tee.
- c. **TORQUE PRESSURE SWITCH TO 215 - 245 INCH-POUNDS.**
- d. Using lockwire, Item 197, Appendix D, lockwire pressure switch.
- e. Install clamp on pressure switch and install clamp to bracket using bolt, washer, and spacer.
- f. **TORQUE BOLT TO 48 - 52 INCH-POUNDS.**
- g. Remove protective caps and plugs from electrical connector.
- h. Inspect and treat electrical connector (PARA 1-7-20).
- i. Connect electrical connector, P410, to pressure switch.
- j. Bleed rotor brake system (PARA 6-4-68).
- k. Make sure area is clean and free of foreign material.
- l. Close oil cooler compartment access doors.

6-4-64 ROTOR BRAKE PRESSURE GAGE.

PARAGRAPH BREAKDOWN

|                                            | PAGE     |
|--------------------------------------------|----------|
| 6-4-64.1 Replace Rotor Brake Pressure Gage | 6-4-64-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Gallon
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 155, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-6-18
- PARA 6-4-68

6-4-64.1 Replace Rotor Brake Pressure Gage.

6-4-64.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- If not already done, engage gust lock and disengage rotor brake (PARA 1-6-18).

NOTE

Protective caps or plugs must be installed as parts are disconnected or removed.

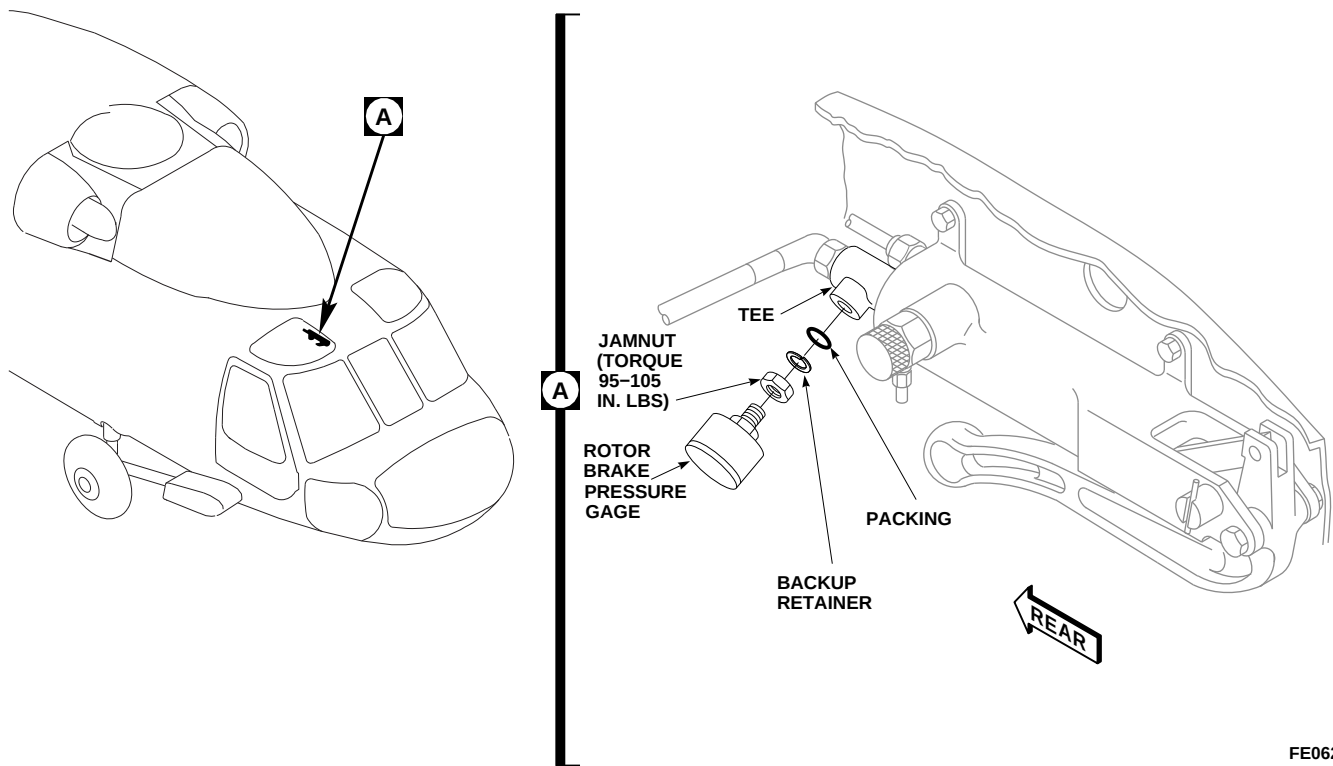
- Position 1-gallon container under tee to catch hydraulic fluid.
- Remove rotor brake pressure gage from tee (Figure 6-4-64-1, Detail A).
- Remove jamnut, backup retainer, and packing from pressure gage. Discard packing.

6-4-64.1.2 Install.

NOTE

Protective caps or plugs must be removed as parts are installed or connected.

- Using hydraulic fluid, Item 155, Appendix D, lubricate packing, backup retainer, and male threads of rotor brake pressure gage.
- Install jamnut, retainer, and packing onto pressure gage (Figure 6-4-64-1, Detail A). Turn jamnut and retainer down until packing is pushed firmly against lower threaded section of pressure gage.
- Install pressure gage into tee. **WHILE HOLDING JAMNUT FROM TURNING, TIGHTEN PRESSURE GAGE AND JAMNUT 1 ½ TURNS PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

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SA**FIGURE 6-4-64-1. REPLACE ROTOR BRAKE PRESSURE GAGE**

- d. **POSITION PRESSURE GAGE BY TIGHTENING NOT MORE THAN ONE TURN.**
- e. **PREVENT PRESSURE GAGE FROM TURNING. TORQUE JAMNUT TO 95 - 105 INCH-POUNDS.**
- f. Bleed rotor brake system (PARA 6-4-68).



6-4-65 ROTOR BRAKE MASTER CYLINDER BLEEDER VALVE.

PARAGRAPH BREAKDOWN

|                                                            | PAGE     |
|------------------------------------------------------------|----------|
| 6-4-65.1 Replace Rotor Brake Master Cylinder Bleeder Valve | 6-4-65-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Gallon

Materials/Parts

- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 197, Appendix D

- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-6-18
- PARA 6-4-68

6-4-65.1 Replace Rotor Brake Master Cylinder Bleeder Valve.

6-4-65.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- If not already done, engage gust lock and disengage rotor brake (PARA 1-6-18).
- Position 1-gallon container under rotor brake master cylinder hydraulic tube connections to catch hydraulic fluid.

NOTE

Protective caps or plugs must be installed as parts are disconnected or removed.

- Remove rotor brake master cylinder bleeder valve and packing from rotor brake master cylinder (Figure 6-4-65-1, Detail A). Discard packing.

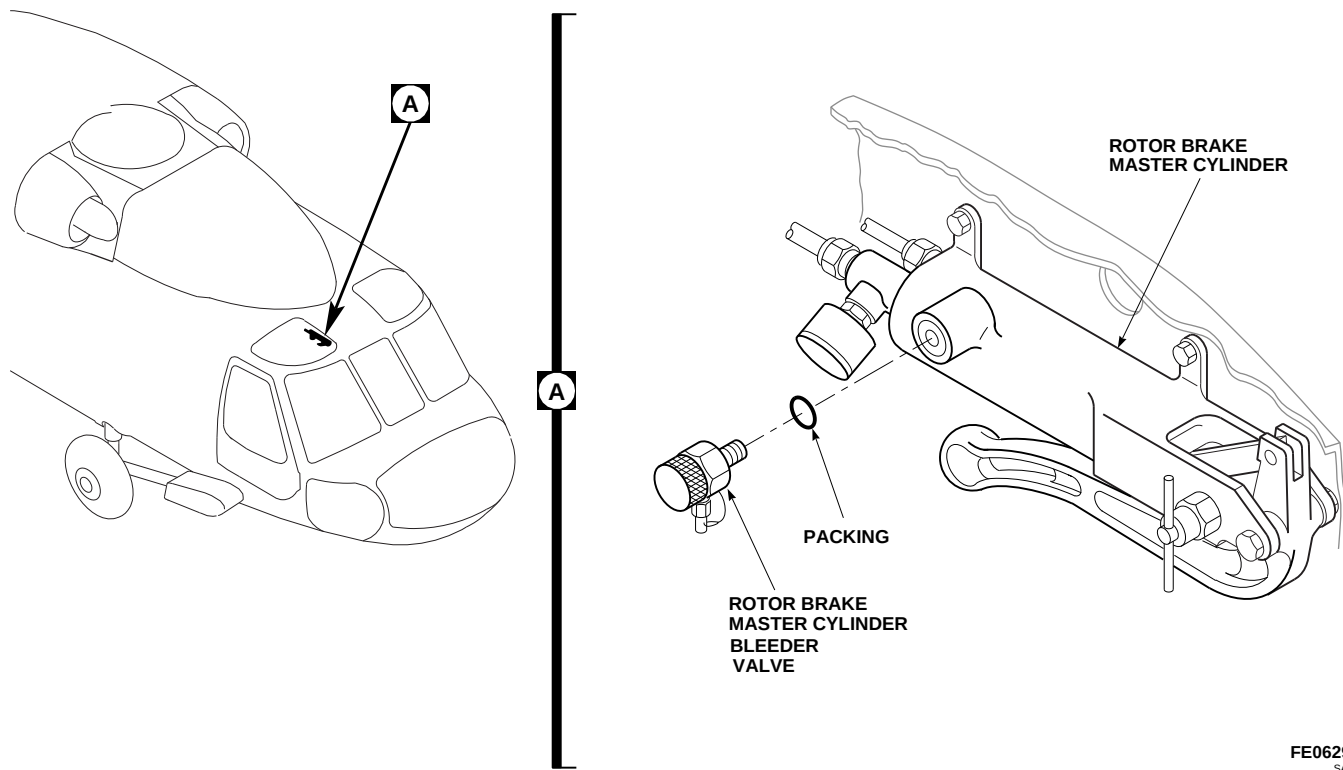
6-4-65.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).

NOTE

Protective caps or plugs must be removed as parts are installed or connected.

- Using hydraulic fluid, Item 155, Appendix D, lubricate packing and install on rotor brake master cylinder bleeder valve (Figure 6-4-65-1, Detail A).
- Install bleeder valve on rotor brake master cylinder.
- Using lockwire, Item 197, Appendix D, lockwire bleeder valve to master cylinder.
- Make sure area is clean and free of foreign material.



**FIGURE 6-4-65-1. REPLACE ROTOR BRAKE MASTER CYLINDER BLEEDER VALVE**

- f. Bleed rotor brake system (PARA 6-4-68).

6-4-66 ROTOR BRAKE PLUNGER.

PARAGRAPH BREAKDOWN

|                                      | PAGE     |
|--------------------------------------|----------|
| 6-4-66.1 Replace Rotor Brake Plunger | 6-4-66-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-6-18

6-4-66.1 Replace Rotor Brake Plunger.

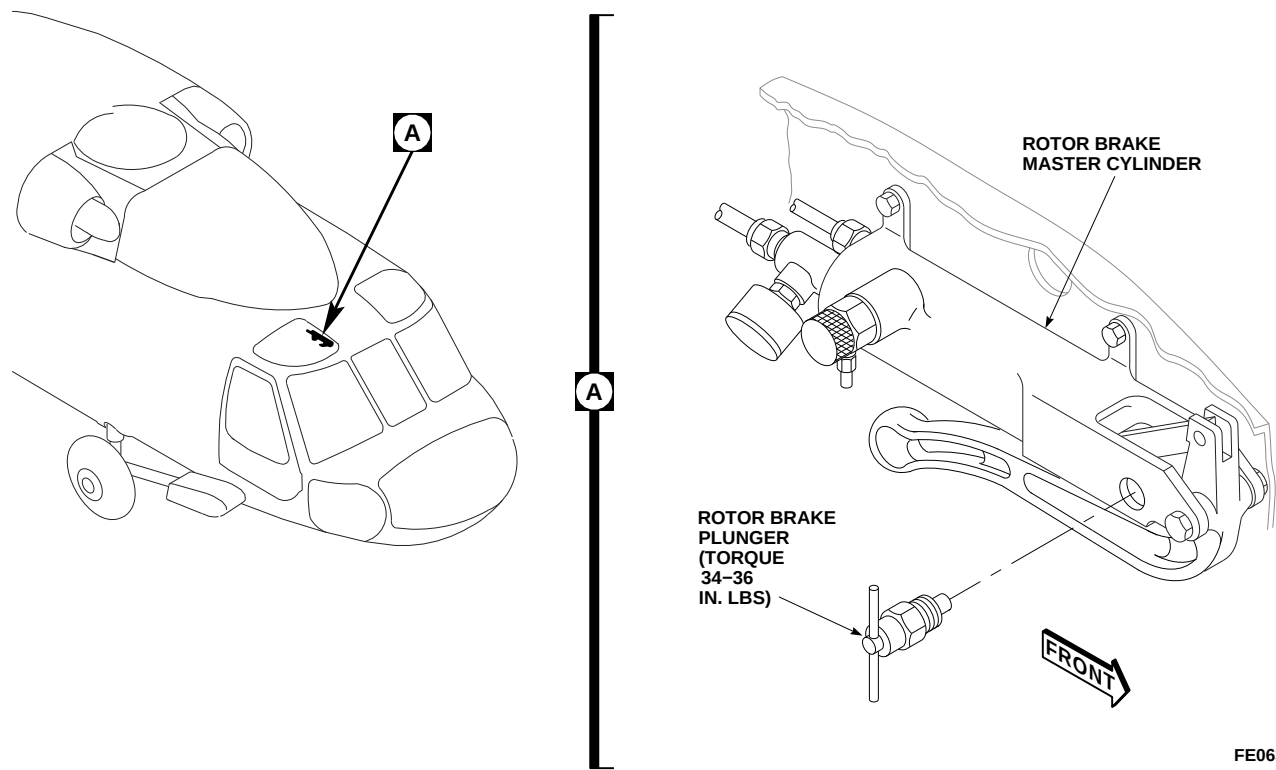
6-4-66.1.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. If not already done, engage gust lock and disengage rotor brake (PARA 1-6-18).
- c. Remove rotor brake plunger from rotor brake master cylinder (Figure 6-4-66-1, Detail A).

6-4-66.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).

- b. Install rotor brake plunger on rotor brake master cylinder (Figure 6-4-66-1, Detail A).
- c. **TORQUE PLUNGER TO 34 - 36 INCH-POUNDS.**
- d. Using lockwire, Item 197, Appendix D, lockwire plunger to master cylinder.
- e. If required, disengage gust lock and engage rotor brake (PARA 1-6-18).

**FIGURE 6-4-66-1. REPLACE ROTOR BRAKE PLUNGER**

6-4-67 ROTOR BRAKE SYSTEM LINES.

PARAGRAPH BREAKDOWN

|                                           | PAGE     |
|-------------------------------------------|----------|
| 6-4-67.1 Replace Rotor Brake System Lines | 6-4-67-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Gallon
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 155, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-6-18
- PARA 6-4-61
- PARA 6-4-63
- PARA 6-4-68
- PARA 7-5-8

6-4-67.1 Replace Rotor Brake System Lines.

NOTE

6-4-67.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- If not already done, engage gust lock and disengage rotor brake (PARA 1-6-18).
- Open main rotor pylon sliding cover.
- Open oil cooler compartment access doors.
- Remove bolt and washer securing channels, spacer, and tube block that hold pressure and supply tubes to overhead structure in cockpit (Figure 6-4-67-1, Sheet 3, Detail D).
- Position 1-gallon container under line connections to catch trapped hydraulic fluid.

- Protective caps or plugs must be installed as parts are disconnected or removed.
- Disconnect and remove supply tube between rotor brake master cylinder and elbow in cabin ceiling (Figure 6-4-67-1, Sheet 1, Detail A). Drain trapped hydraulic fluid into 1-gallon container.
- Disconnect and remove pressure tube between tee and elbow in cabin ceiling.
- Remove screws, washers, and nut securing channels, spacers, and tube blocks that hold pressure and supply tubes to flight controls deck (Figure 6-4-67-1, Sheet 3, Detail E and Detail F).

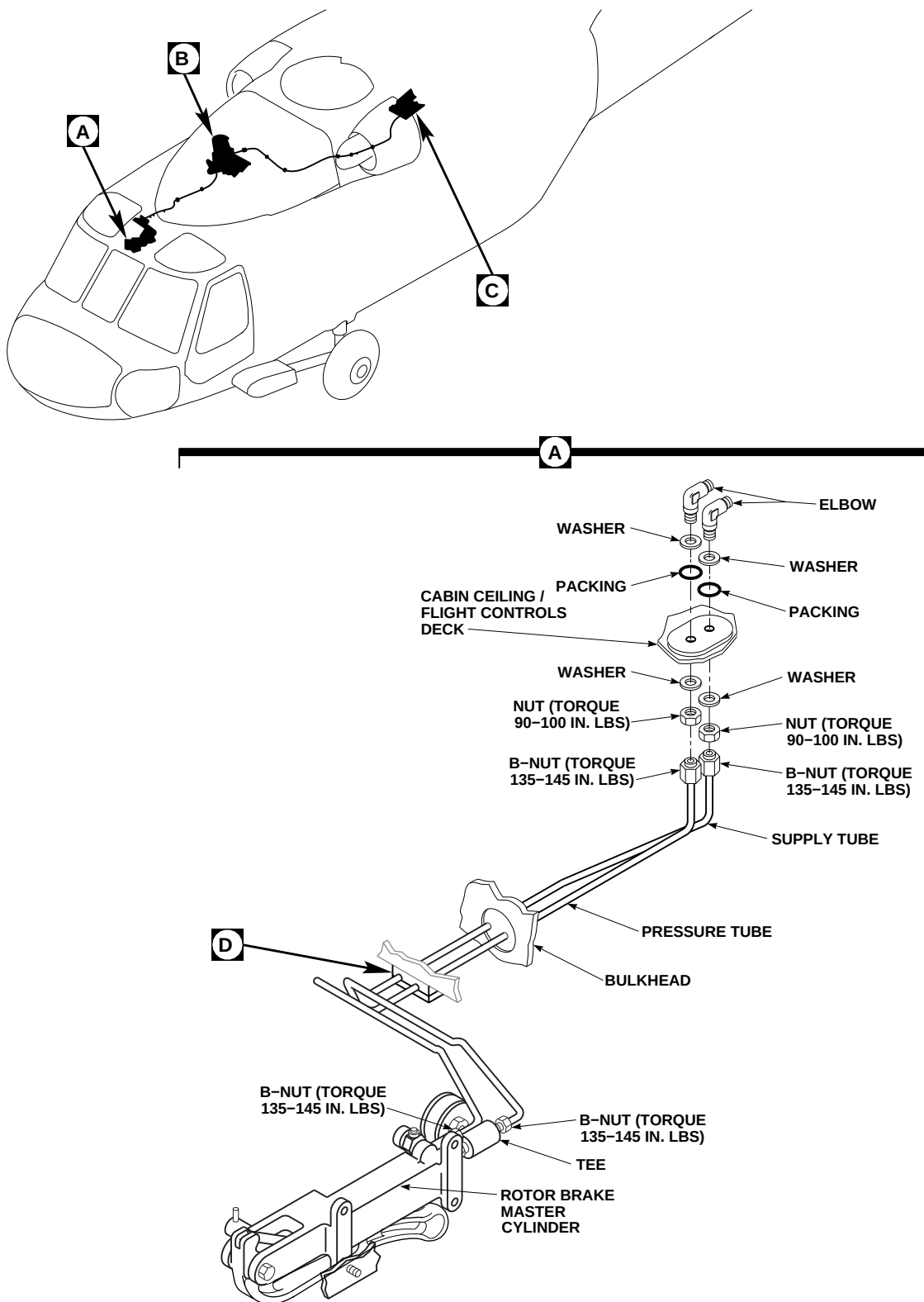
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FIGURE 6-4-67-1. REPLACE ROTOR BRAKE SYSTEM LINES (SHEET 1 OF 3)

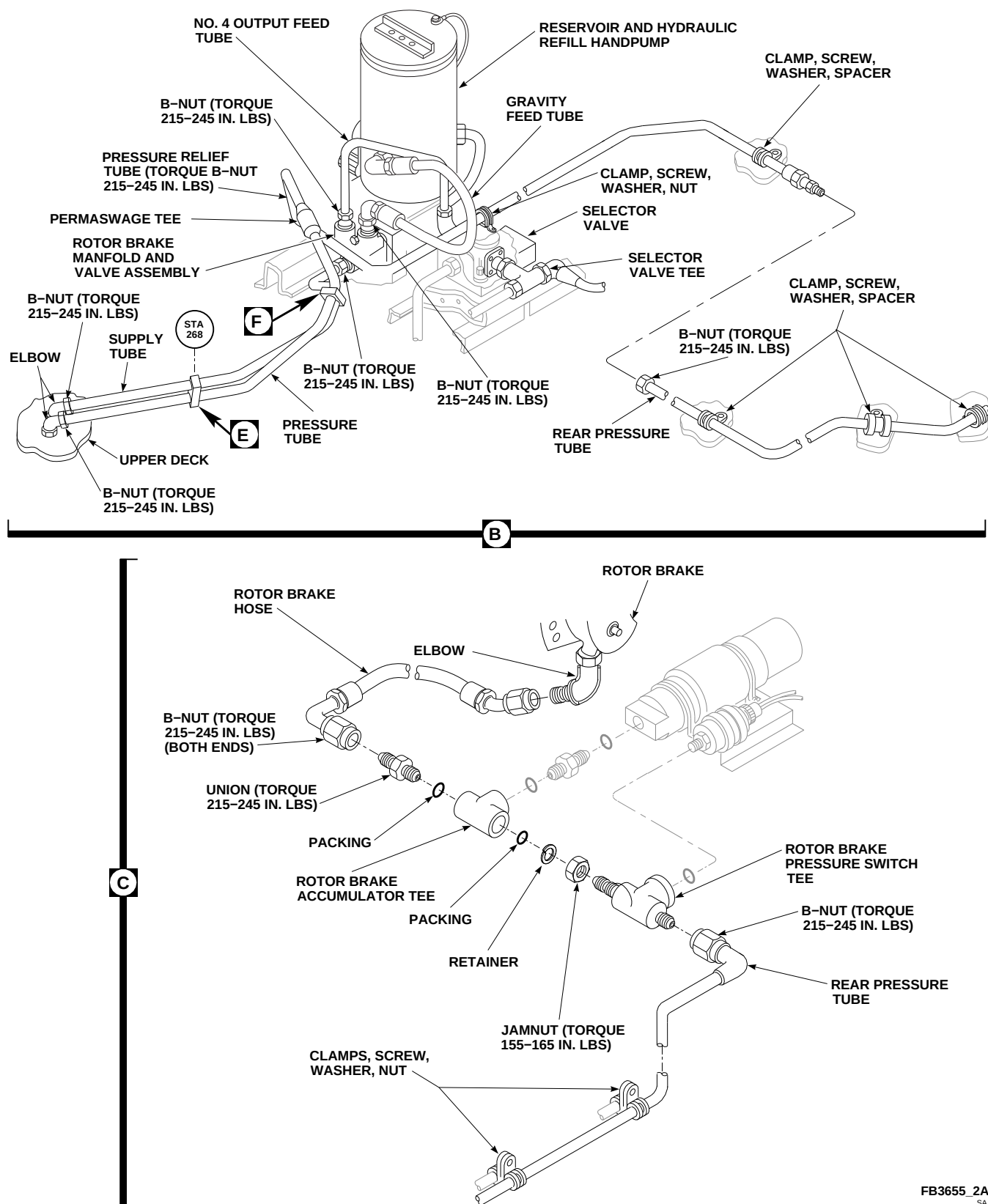
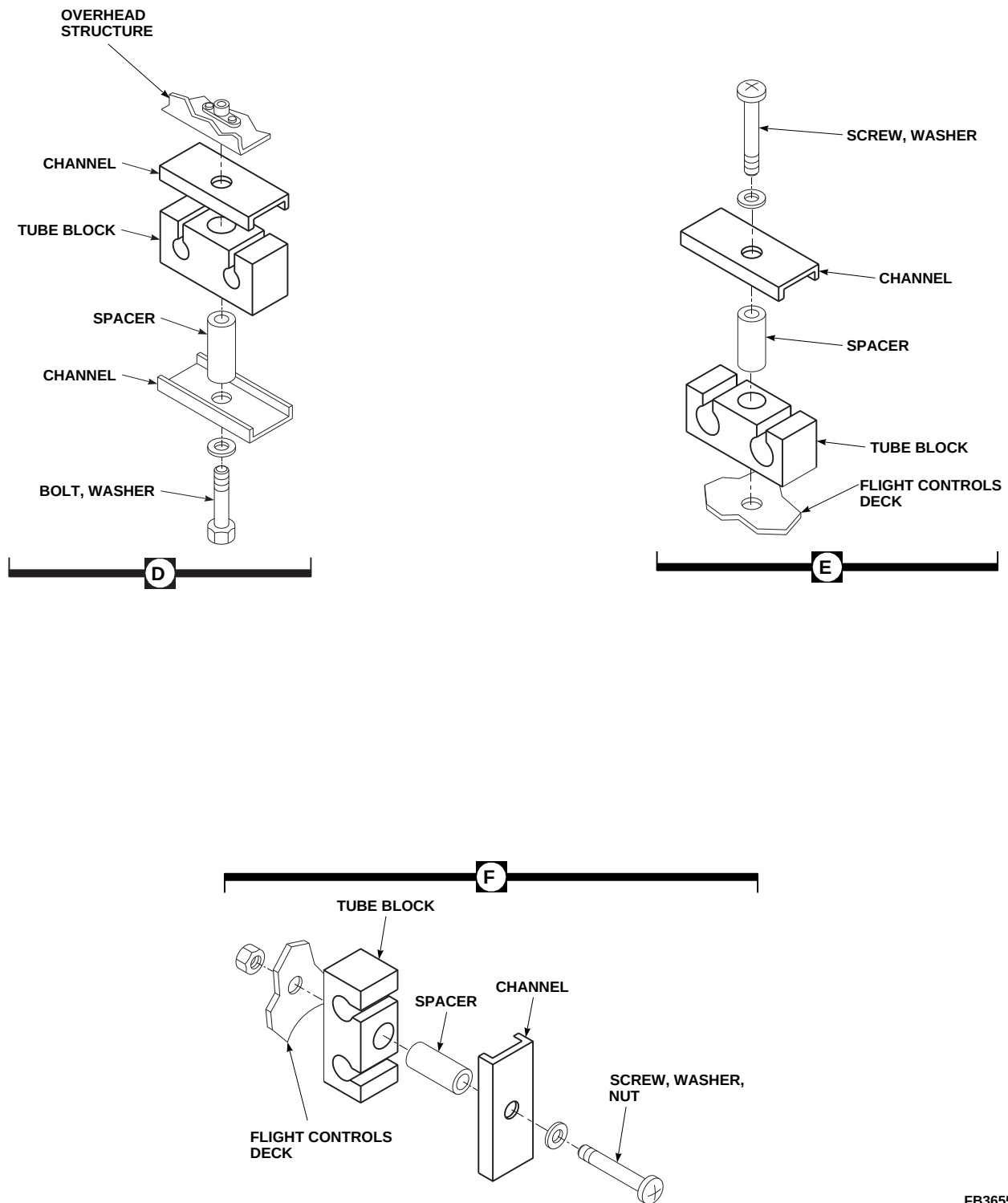


FIGURE 6-4-67-1. REPLACE ROTOR BRAKE SYSTEM LINES (SHEET 2 OF 3)



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SA

FIGURE 6-4-67-1. REPLACE ROTOR BRAKE SYSTEM LINES (SHEET 3 OF 3)



- j. Remove screws, washers, spacers, and nuts securing clamps that hold pressure and rear pressure tubes (Figure 6-4-67-1, Sheet 2, Detail B and Detail C).
- k. Disconnect supply tube between elbow on flight controls deck and rotor brake manifold and valve assembly (Figure 6-4-67-1, Sheet 2, Detail B).
- l. Disconnect pressure tube from elbow on flight controls deck and rear pressure tube.
- m. Disconnect pressure relief tube from rotor brake manifold and valve assembly.
- n. Remove pressure and supply tubes by passing through bulkhead.
- o. Remove nuts and washers from inside cabin, then remove elbows, washers, and packings from flight controls deck (Figure 6-4-67-1, Sheet 1, Detail A). Discard packings.
- p. Disconnect gravity feed tube from reservoir and hydraulic refill handpump, and rotor brake manifold and valve assembly (Figure 6-4-67-1, Sheet 2, Detail B).
- q. Disconnect and remove No. 4 output feed tube between rotor brake manifold and valve assembly, and selector valve.
- r. Disconnect and remove rear pressure tube from rotor brake pressure switch tee (Figure 6-4-67-1, Sheet 2, Detail C).
- s. Remove rotor brake accumulator (PARA 6-4-61).
- t. Remove rotor brake pressure switch (PARA 6-4-63).
- u. Disconnect rotor brake pressure switch tee from rotor brake accumulator tee. Discard packing.
- v. Remove union from rotor brake accumulator tee. Discard packing.
- w. Disconnect and remove rotor brake hose from elbow on rotor brake.

- x. Inspect each tube and hose removed for damage. If damaged, repair or replace tubes or hose as required (PARA 7-5-8).

#### 6-4-67.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Using hydraulic fluid, Item 155, Appendix D, lubricate all threads and packings before installing.

#### NOTE

- For ease of maintenance, do not apply torque until all components have been installed and are positioned correctly.
- Protective caps or plugs must be removed as parts are installed or connected.
- c. Using hydraulic fluid, Item 155, Appendix D, lubricate packing, retainer, and male threads of rotor brake pressure switch tee.
- d. Install jamnut, retainer, and packing on rotor brake pressure switch tee (Figure 6-4-67-1, Sheet 2, Detail C). Turn jamnut and retainer down until packing is pushed firmly against the lower threaded section of rotor brake pressure switch tee.
- e. Install rotor brake pressure switch tee into rotor brake accumulator tee.
- f. **WHILE KEEPING JAMNUT FROM TURNING, TIGHTEN ROTOR BRAKE PRESSURE SWITCH TEE AND JAMNUT 1 ½ TURNS PAST POINT WHERE SHARP RISE IN TORQUE IS FELT. POSITION ROTOR BRAKE PRESSURE SWITCH TEE BY TIGHTENING NOT MORE THAN ONE TURN.**
- g. **PREVENT PRESSURE SWITCH TEE FROM TURNING. TORQUE JAMNUT TO 155 - 165 INCH-POUNDS.**
- h. Install rotor brake pressure switch (PARA 6-4-63).

- i. Install packing on union. Install union in rotor brake accumulator tee.
- j. **TORQUE UNION TO 215 - 245 INCH-POUNDS.**
- k. Install rotor brake accumulator (PARA 6-4-61).

**CAUTION**

Damage to tubes and hoses will result if forced into position. Do not force connections or routing which will strain, kink or twist the tube or hose. Connect tubes and hoses loosely at each end, making sure fittings match squarely before tightening nut.

- l. Install rear pressure tube to rotor brake pressure switch tee.
- m. **TORQUE B-NUT TO 215 - 245 INCH-POUNDS.**
- n. Connect rotor brake hose between union and elbow on rotor brake.
- o. **TORQUE B-NUTS TO 215 - 245 INCH-POUNDS.**
- p. Connect No. 4 output feed tube between rotor brake manifold and valve assembly, and selector valve (Figure 6-4-67-1, Sheet 2, Detail B).
- q. **TORQUE B-NUTS TO 215 - 245 INCH-POUNDS.**
- r. Connect gravity feed tube to reservoir and hydraulic refill handpump, and rotor brake manifold and valve assembly.
- s. **TORQUE B-NUTS TO 215 - 245 INCH-POUNDS.**
- t. Install elbows with washers and packings through flight controls deck and secure with washers and nuts. Position elbows facing rear (Figure 6-4-67-1, Sheet 1, Detail A).
- u. **TORQUE NUTS TO 90 - 100 INCH-POUNDS.**
- v. Connect pressure tube to rear pressure tube and elbow. Connect pressure relief tube to rotor brake manifold and valve assembly (Figure 6-4-67-1, Sheet 2, Detail B).
- w. **TORQUE B-NUTS TO 215 - 245 INCH-POUNDS.**
- x. Connect supply tube between elbow, and rotor brake manifold and valve assembly.
- y. **TORQUE B-NUTS TO 215 - 245 INCH-POUNDS.**
- z. Install tube blocks, spacers, and channels to flight controls deck, and secure with screws, washers, and nut (Figure 6-4-67-1, Sheet 3, Detail E and Detail F).
- aa. Install clamps using screws, washers, spacers, and nuts (Figure 6-4-67-1, Sheet 2, Detail B and Detail C).
- ab. Install pressure tube through bulkhead and connect pressure tube to tee and elbow in cabin ceiling (Figure 6-4-67-1, Sheet 1, Detail A).
- ac. **TORQUE B-NUTS TO 135 - 145 INCH-POUNDS.**
- ad. Install supply tube through bulkhead and connect supply tube to port on master cylinder and elbow in cabin ceiling.
- ae. **TORQUE B-NUTS TO 135 - 145 INCH-POUNDS.**
- af. Install tube block, channels, and spacer on overhead structure with bolt and washer (Figure 6-4-67-1, Sheet 3, Detail D).
- ag. Bleed rotor brake system (PARA 6-4-68).
- ah. Make sure area is clean and free of foreign material.
- ai. Close oil cooler compartment access doors.
- aj. Close main rotor pylon sliding cover.

6-4-68 BLEED ROTOR BRAKE SYSTEM.

PARAGRAPH BREAKDOWN

|                                   | PAGE     |
|-----------------------------------|----------|
| 6-4-68.1 Bleed Rotor Brake System | 6-4-68-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Quart

Materials/Parts

- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D

- Polyurethane Tubing, 1/4-inch
- Polyurethane Tubing, 3/8-inch

References

- Appendix D
- PARA 1-6-16
- PARA 1-6-18
- PARA 6-2-3

6-4-68.1 Bleed Rotor Brake System.

- a. Turn off all electrical power (PARA 1-6-16).
- b. If not already done, engage gust lock and disengage rotor brake (PARA 1-6-18).
- c. Open main rotor pylon sliding cover.
- d. Open oil cooler compartment access doors.

NOTE

To prevent air from getting into rotor brake system during bleeding, make sure the servicing hand pump reservoir is filled with hydraulic fluid, Item 155, Appendix D.

- e. Position servicing selector valve to No. 4 (rotor brake), and press selector valve handle (Figure 6-4-68-1, Detail A).

- f. Remove cap from rotor brake master cylinder bleeder valve on rotor brake master cylinder (Figure 6-4-68-1, Detail B).
- g. Attach 1/4-inch polyurethane tubing to rotor brake master cylinder bleeder valve on rotor brake master cylinder.
- h. Turn reservoir and hydraulic refill handpump handle clockwise to fill rotor brake master cylinder (Figure 6-4-68-1, Detail A).
- i. Position 1-quart container under tubing attached to rotor brake master cylinder bleeder valve on rotor brake master cylinder to catch hydraulic fluid.
- j. Turn knob on rotor brake master cylinder bleeder valve clockwise and allow hydraulic fluid to flow into 1-quart container until fluid is free of air bubbles (Figure 6-4-68-1, Detail B).

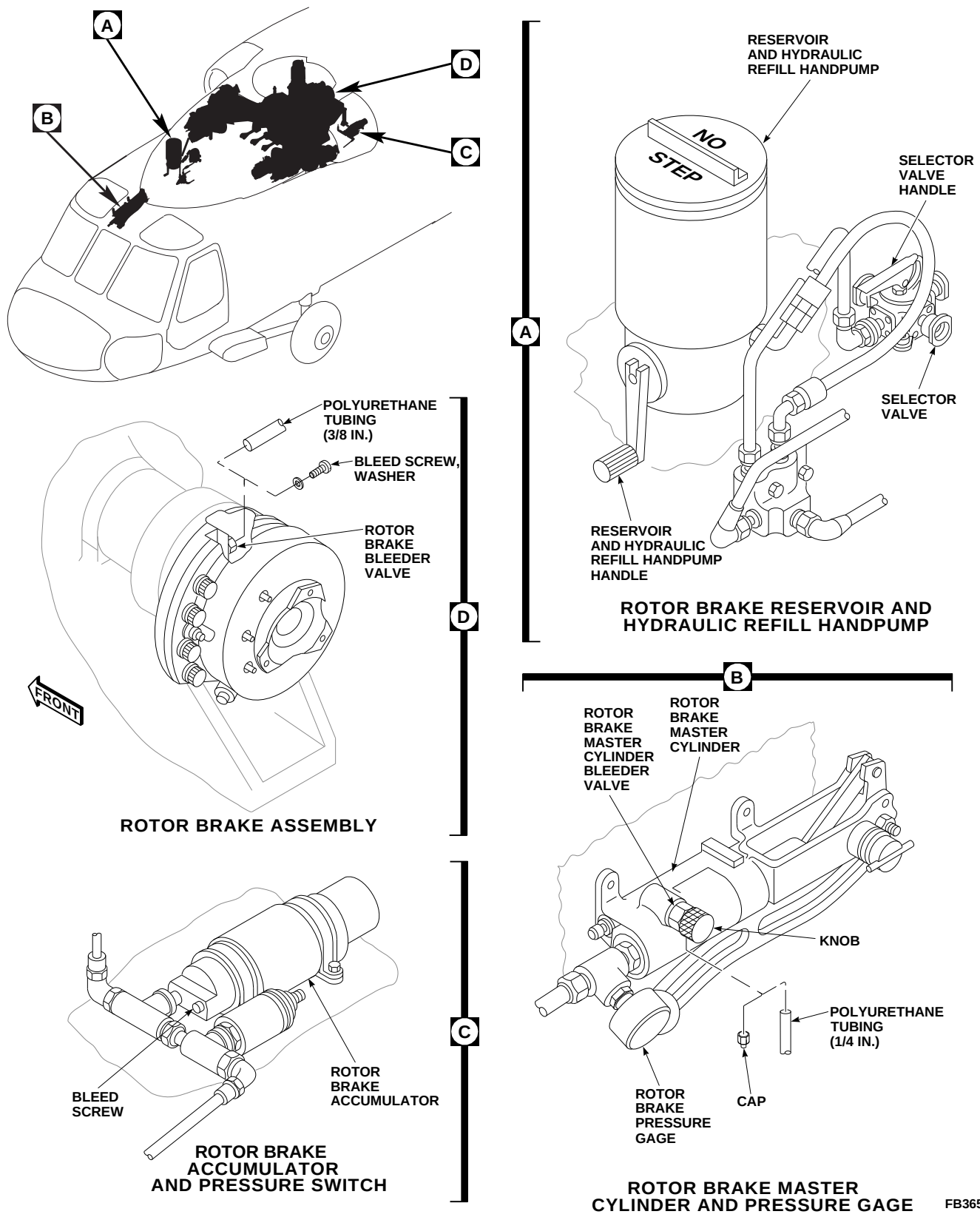


FIGURE 6-4-68-1. BLEED ROTOR BRAKE SYSTEM

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- k. Release knob. Rotor brake master cylinder bleeder valve will close automatically.
- l. Remove tubing from rotor brake master cylinder bleeder valve.
- m. Install cap on rotor brake master cylinder bleeder valve.
- n. Position machinery towel, Item 211, Appendix D, under bleed screw on rotor brake accumulator (Figure 6-4-68-1, Detail C).
- o. Turn reservoir and hydraulic refill handpump clockwise, loosen bleed screw slowly and allow hydraulic fluid to flow until fluid is free of air bubbles (Figure 6-4-68-1, Detail A and Detail C).
- p. **TIGHTEN BLEED SCREW** (Figure 6-4-68-1, Detail C).
- q. Using lockwire, Item 197, Appendix D, lockwire bleed screw to rotor brake accumulator.
- r. Remove bleed screw and washer from rotor brake bleeder valve on rotor brake (Figure 6-4-68-1, Detail D).
- s. Attach  $\frac{3}{8}$ -inch polyurethane tubing to rotor brake bleeder valve on rotor brake.
- t. Position 1-quart container under tubing attached to rotor brake bleeder valve on rotor brake to catch hydraulic fluid.
- u. Turn reservoir and hydraulic refill handpump clockwise, open rotor brake bleeder valve and allow hydraulic fluid to flow into 1-quart container until fluid is free of air (Figure 6-4-68-1, Detail A and Detail D).
- v. Close rotor brake bleeder valve.
- w. Remove tubing from rotor brake bleeder valve.
- x. Install washer and bleed screw in rotor brake bleeder valve (Figure 6-4-68-1, Detail D).
- y. Using lockwire, Item 197, Appendix D, lockwire bleed screw and rotor brake bleeder valve.
- z. Make sure area is clean and free of foreign material.
- aa. Perform operational check of rotor brake system (PARA 6-2-3).
- ab. Close oil cooler compartment access doors.
- ac. Close main rotor pylon sliding cover.
- ad. If required, disengage gust lock and engage rotor brake (PARA 1-6-18).



6-4-69 ROTOR BRAKE TUBING.

PARAGRAPH BREAKDOWN

|                                    | PAGE     |
|------------------------------------|----------|
| 6-4-69.1 Repair Rotor Brake Tubing | 6-4-69-1 |

INITIAL SETUP

- PARA 6-4-68

Tools

- Aircraft Mechanic’s Toolkit

- PARA 7-5-8

References

- PARA 1-6-16
- PARA 1-6-18

6-4-69.1 Repair Rotor Brake Tubing.

- Turn off all electrical power (PARA 1-6-16).
- If not already done, engage gust lock and disengage rotor brake (PARA 1-6-18).
- Open main rotor pylon sliding cover.
- Repair damaged tubing as required (PARA 7-5-8).

- Bleed rotor brake system (PARA 6-4-68).
- Make sure area is clean and free of foreign material.
- Close main rotor pylon sliding cover.





6-4-70 ROTOR BRAKE ASSEMBLY.

PARAGRAPH BREAKDOWN

|                                       | PAGE     |
|---------------------------------------|----------|
| 6-4-70.1 Replace Rotor Brake Assembly | 6-4-70-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- C-Clamp
- Feeler Gage
- Hammer
- Hose, SS48CT1C10100
- Torqueless Grip Installation Plate, Locally-Made, Figure H-245, Appendix H
- Torqueless Grip Installation Tool, Locally-Made, Figure H-244, Appendix H
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1600 in. lbs
- Union, MS21924-6

Materials/Parts

- Cloth, Cleaning, Item 102A, Appendix D
- Corrosion Preventive Compound, Item 109, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- Appendix H
- PARA 1-6-16
- PARA 1-6-18
- PARA 1-7-49
- PARA 6-2-3
- PARA 6-4-68

6-4-70.1 Replace Rotor Brake Assembly.

6-4-70.1.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. If not already done, engage gust lock (PARA 1-6-18).

- c. Open engine cowlings.
- d. Open oil cooler compartment access doors.
- e. Position machinery towels, Item 211, Appendix D, under rotor brake hose to catch spilled hydraulic fluid.

- f. Disconnect rotor brake hose from elbow (Figure 6-4-70-1, Detail A).
  - g. Install protective cap on hose.
  - h. Remove shield on rotor brake as follows:
    - (1) Remove bolt, washers, and nut from upper bracket.
    - (2) Remove nuts and washers from studs securing shield.
    - (3) Position machinery towels, Item 211, Appendix D, on tail rotor drive shaft section I.
    - (4) Slide shield over tail rotor drive shaft flexible couplings and place on cloth covered section of tail rotor drive shaft.
  - i. Loosen jamnut and remove elbow from inlet port.
  - j. Install protective plug on inlet port.
  - k. Remove packing and retainer from elbow on rotor brake. Discard packing.
  - l. Remove bolts, countersunk washers, washers, and nuts securing rotor brake to rotor brake mounting bracket.
  - m. Rotate rotor brake 180° to opposite side of rotor brake disc.
  - n. Loosen clamp securing heater bleed-air tube to crossover tube. Position heater bleed-air tube out of way.
  - o. Remove rotor brake from rotor brake disc, and remove shim from rotor brake mounting bracket.
- c. Remove protective plug from inlet port.
  - d. Install jamnut, retainer, and packing onto elbow. Turn nut and retainer down until packing is pushed firmly against the lower threaded section of elbow (Figure 6-4-70-1, Detail A).
  - e. Install elbow into inlet port on rotor brake.
  - f. **WHILE KEEPING JAMNUT FROM TURNING, TIGHTEN ELBOW AND JAMNUT 1 ½ TURNS PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
  - g. **POSITION ELBOW BY TURNING NO MORE THAN ONE TURN.**
  - h. While maintaining position of elbow, **TORQUE JAMNUT TO 155 - 165 INCH-POUNDS.**
  - i. Inspect rotor brake wear indicator pins to make sure rotor brake pucks are fully retracted (Figure 6-4-70-1, Detail B). Wear indicator pin length with piston fully retracted is 0.80-inch beyond rotor brake housing.

**CAUTION**

To prevent damage to new rotor brake housing, step j. is to be performed only if installing the same rotor brake housing.

- j. If installing same rotor brake housing, perform the following:

**CAUTION**

Damage to rotor brake system will result if torqueless grips are incorrectly adjusted. Make sure torqueless grips are seated correctly.

- (1) Locally make torqueless grip installation tool (Figure H-244, Appendix H).
- (2) Locally make torqueless grip installation plate (Figure H-245, Appendix H).

**6-4-70.1.2 Install.**

- a. Turn off all electrical power (PARA 1-6-16).
- b. Using hydraulic fluid, Item 155, Appendix D, lubricate packing, retainer, and male threads of elbow.

- (3) Using C-clamp, install torqueless grip installation plate to support surface (Figure 6-4-70-1, Detail C).
  - (4) Slide rotor brake housing onto torqueless grip installation plate so rotor brake pucks lay flat.
  - (5) Using corrosion preventive compound, Item 109, Appendix D, apply to grip prior to seating torqueless grips into rotor brake housing. When dry, inspect for complete coverage with no cracks or flaws. If cracks or flaws are found, strip and reapply.
  - (6) Using hammer and torqueless grip installation tool, tap torqueless grips down into rotor brake housing until grips bottom in housing. Repeat for all six pins.
  - (7) If any corrosion preventive compound is removed during torqueless grip installation, reapply corrosion preventive compound, Item 109, Appendix D, to exposed surfaces of the torqueless grips.
  - (8) Using cleaning cloth, Item 102A, Appendix D, wipe off any corrosion preventive compound from wear indicating pins.
- k. Prior to installing rotor brake, conduct the following pressure test without rotor brake bolted to rotor brake mounting bracket:
- (1) Install two bolts in mounting flange of rotor brake housing, used for mounting rotor brake on rotor brake mounting bracket, using extra washers and nuts (Figure 6-4-70-1, Detail A).
  - (2) **TORQUE NUTS TO 684 - 756 INCH-POUNDS.**
  - (3) Remove protective cap from hose.
  - (4) Install union onto hydraulic hose end. Install hose to union. Install end of hose to elbow on rotor brake.
  - (5) **TIGHTEN UNION AND HOSE ENDS.**

- (6) Bleed rotor brake system (PARA 6-4-68).

### NOTE

Rotor brake linings and adjusting pins must be observed for movement while applying 75 psi.

- (7) Slowly apply pressure to 75 psi from rotor brake master cylinder. Observe rotor brake movement; slight movement in brake linings is permitted. Adjusting pins must have no movement.
- (8) Release pressure and remove nuts, washers, and bolts. Disconnect hydraulic hose from rotor brake and remove union from original hose.
- (9) Install protective caps and plugs.
  - l. Mount rotor brake without shim and measure dimensions A and B with rotor brake pucks in retracted position (Figure 6-4-70-1, Detail B). Dimensions A and B are the gaps between rotor brake housing and rotor brake disc. Shim thickness =  $(A - B) \div 2$ .
  - m. Remove rotor brake.

### CAUTION

Damage to rotor brake will result if brake pistons are not retracted prior to installation. Make sure brake pistons are retracted prior to installing.

- n. Install shim on rotor brake mounting bracket (Figure 6-4-70-1, Detail A).
- o. Position rotor brake on rotor brake mounting bracket.
- p. Install rotor brake to rotor brake mounting bracket using bolts, countersunk washers (countersunk side towards bolthead), washers, and nuts.
- q. **TORQUE NUTS TO 912 - 1008 INCH-POUNDS.**

- r. Remove protective cap and plug from rotor brake hose and elbow.
- s. Connect rotor brake hose to elbow on rotor brake.
- t. **TORQUE B-NUT TO 215 - 245 INCH-POUNDS.**
- u. Bleed rotor brake system (PARA 6-4-68).
- v. Perform the following:
  - (1) Apply and release rotor brake by moving rotor brake handle front and rear at least three times.
  - (2) Using feeler gage, verify gap between rotor brake pucks and rotor brake disc on each side of rotor brake disc.
  - (3) **GAP BETWEEN ANY INDIVIDUAL ROTOR BRAKE PUCK AND ROTOR BRAKE DISC MUST BE BETWEEN 0.035-INCH AND 0.075-INCH. TOTAL OF SIX GAPS BETWEEN ROTOR BRAKE PUCKS AND ROTOR BRAKE DISC SHALL NOT EXCEED 0.330-INCH.**
  - (4) If rotor brake puck-to-rotor brake disc gap is not within limits, remove bolts, washers, and nuts securing rotor brake to rotor brake mounting bracket. Peel lamination off shim. Repeat step i. through step v.
- w. **SECURE HEATER BLEED-AIR TUBE TO CROSSOVER TUBE BY TIGHTENING CLAMP.**
- x. Install shield on rotor brake as follows:
  - (1) Slide shield over flex couplings and install on studs of rotor brake.
  - (2) Install nuts and washers. Using washers, shim as required. Install unused washers under nuts.
  - (3) **TORQUE NUTS TO 95 - 105 INCH-POUNDS.**
  - (4) Position shield to upper bracket. Using washers, shim as required. Install unused washers under bolthead.
  - (5) Install bolt, washers, and nut.
  - (6) **TORQUE BOLT TO 95 - 105 INCH-POUNDS.**
  - (7) Remove machinery towel, Item 211, Appendix D, from tail rotor drive shaft section I.
- y. Make sure area is clean and free of foreign material.
- z. Close oil cooler compartment access doors.
- aa. Close engine cowlings.
- ab. Perform operational check of rotor brake system (PARA 6-2-3).
- ac. Perform rotor brake nut torque stabilization check after 5 - 15 rotor brake stops (PARA 1-7-49).

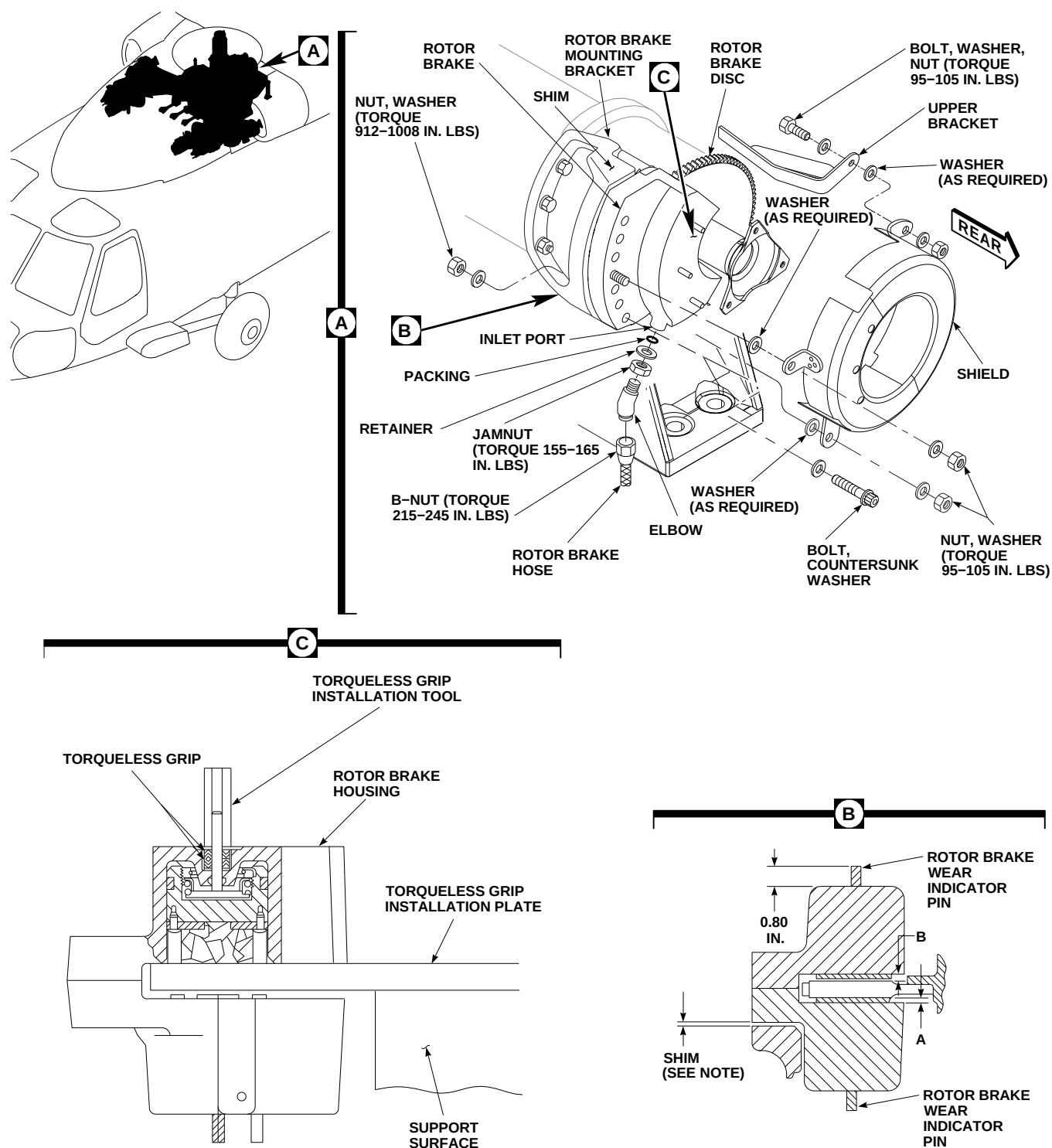


FIGURE 6-4-70-1. REPLACE ROTOR BRAKE ASSEMBLY

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6-4-71 ROTOR BRAKE LININGS (PUCKS).

PARAGRAPH BREAKDOWN

|                                              | PAGE     |
|----------------------------------------------|----------|
| 6-4-71.1 Replace Rotor Brake Linings (Pucks) | 6-4-71-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Brush, Acid, Item 84, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D

- Hydraulic Fluid, Item 155, Appendix D
- Sealing Compound, Item 271A, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 6-2-3
- PARA 6-4-70

6-4-71.1 Replace Rotor Brake Linings (Pucks).

6-4-71.1.1 Inspect.

Measure length of indicator pins extending out from rotor brake housing. **IF ANY INDICATOR PIN EXTENDS 3⁄16-INCH OR LESS FROM ROTOR BRAKE HOUSING, REPLACE ALL ROTOR BRAKE LININGS (PUCKS).**

6-4-71.1.2 Remove.

- Remove rotor brake assembly (PARA 6-4-70).
- Remove bolts securing rotor brake housing halves together (Figure 6-4-71-1, Detail A). Remove and discard packings.

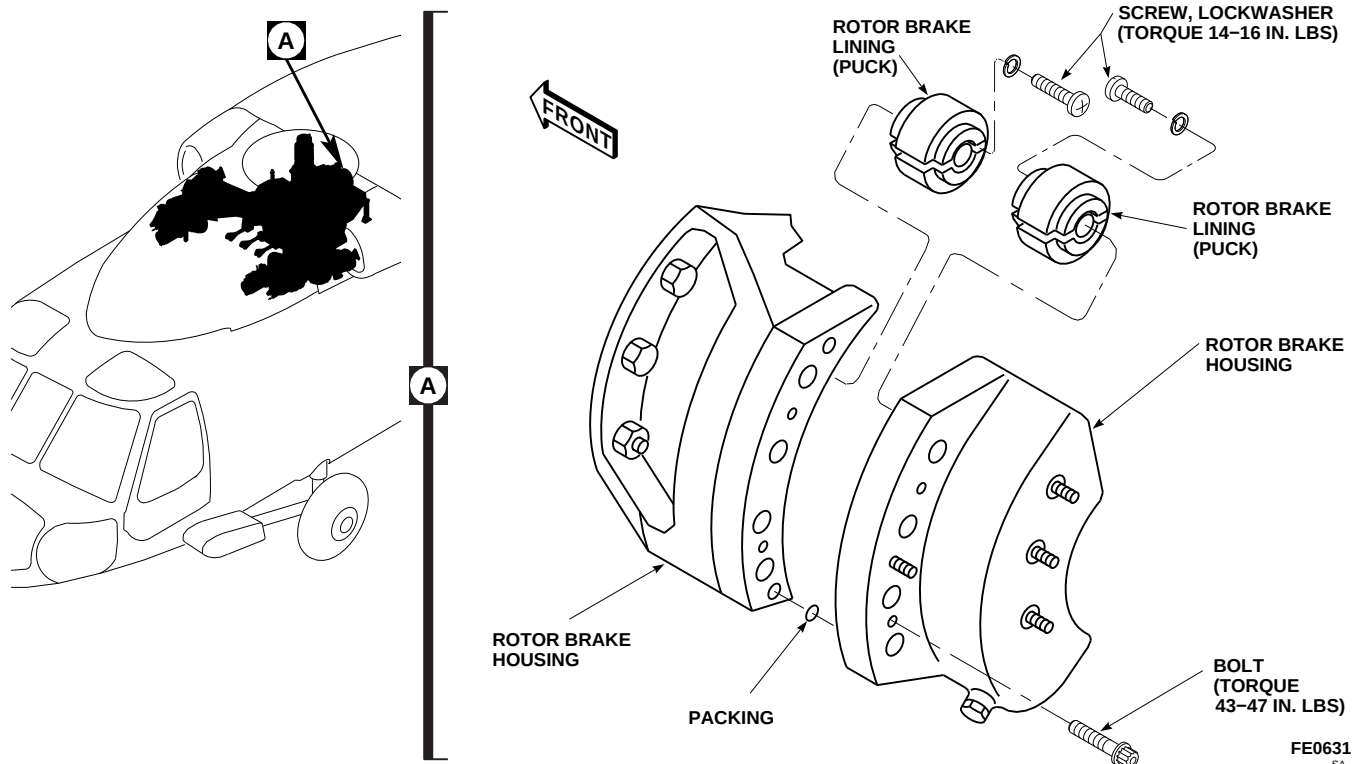
NOTE

When replacing rotor brake linings, replace all six as a set. Do not replace individual linings (pucks).

- Remove screws and lockwashers securing linings (pucks) to pistons. Carefully pry out worn linings (pucks).

6-4-71.1.3 Clean.

- Using acid brush, Item 84, Appendix D, soaked with dry-cleaning solvent, Item 124, Appendix D, clean area around cylinder wall and cylinder surface.
- Make sure area is clean and free of foreign material.



**FIGURE 6-4-71-1. REPLACE ROTOR BRAKE LININGS (PUCKS)**

#### 6-4-71.1.4 Install.

- a. Turn off all electrical power (PARA 1-6-16).

#### NOTE

Make sure screwholes in rotor brake linings (pucks) and piston assembly are aligned.

- b. Install new linings (pucks) into rotor brake housing until piston bottoms (Figure 6-4-71-1, Detail A).
- c. Apply sealing compound, Item 271A, Appendix D, to screw threads and install linings (pucks) to housing with screws and lockwashers.
- d. **TORQUE SCREWS TO 14 - 16 INCH-POUNDS.**

- e. Using hydraulic fluid, Item 155, Appendix D, lubricate packing and install in housing.

#### NOTE

Make sure packing is properly aligned in grooves prior to assembling housing halves together.

- f. Assemble housing halves together and install bolts.
- g. **TORQUE BOLTS TO 43 - 47 INCH-POUNDS.**
- h. Install rotor brake assembly (PARA 6-4-70).
- i. Perform rotor brake break-in procedure (PARA 6-2-3).



6-4-72 ROTOR BRAKE BLEEDER VALVE.

PARAGRAPH BREAKDOWN

|                                            | PAGE     |
|--------------------------------------------|----------|
| 6-4-72.1 Replace Rotor Brake Bleeder Valve | 6-4-72-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container, 1-Gallon
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 155, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-6-18
- PARA 6-4-68

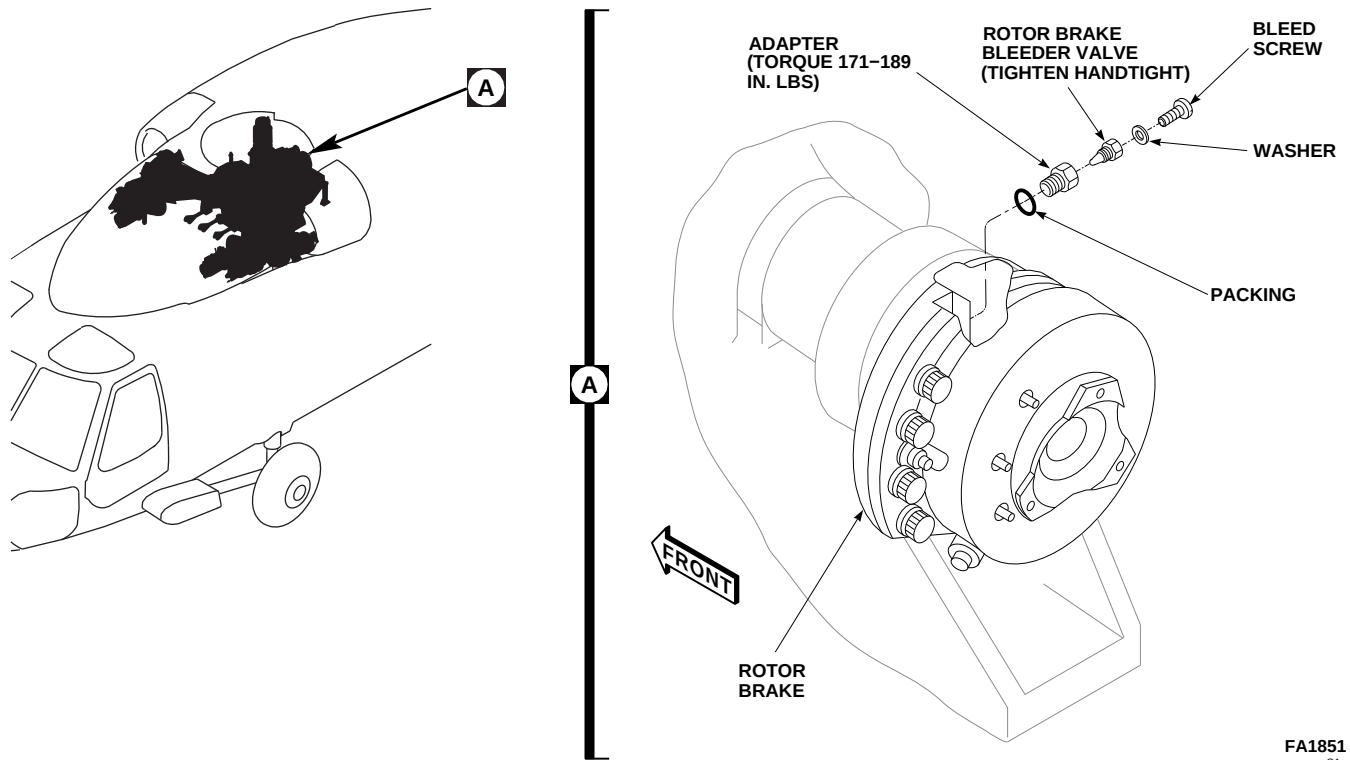
6-4-72.1 Replace Rotor Brake Bleeder Valve.

6-4-72.1.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. If not already done, engage gust lock and disengage rotor brake (PARA 1-6-18).
- c. Open oil cooler compartment access doors.
- d. Remove bleed screw and washer from rotor brake bleeder valve (Figure 6-4-72-1, Detail A). Discard washer.
- e. Position 1-gallon container under bleeder valve, and remove bleeder valve, adapter, and packing. Discard packing.

6-4-72.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Using hydraulic fluid, Item 155, Appendix D, lubricate packing and install on adapter (Figure 6-4-72-1, Detail A).
- c. Install adapter in rotor brake.
- d. **TORQUE ADAPTER TO 171 - 189 INCH-POUNDS.**
- e. Install rotor brake bleeder valve in adapter.
- f. **TIGHTEN ROTOR BRAKE BLEEDER VALVE HANDTIGHT.**
- g. Install bleed screw with new washer in rotor brake bleeder valve.
- h. Bleed rotor brake system (PARA 6-4-68).



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FIGURE 6-4-72-1. REPLACE ROTOR BRAKE BLEEDER VALVE

6-4-73 ROTOR BRAKE DISC.

PARAGRAPH BREAKDOWN

|                                                                                                                                                                                            | PAGE                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| 6-4-73.1 Replace Rotor Brake Disc                                                                                                                                                          | 6-4-73-1                                                                                                              |
| <hr/>                                                                                                                                                                                      |                                                                                                                       |
| INITIAL SETUP                                                                                                                                                                              | <ul style="list-style-type: none"> <li>• PARA 6-3-4</li> </ul>                                                        |
| Tools                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>• PARA 6-3-5</li> </ul>                                                        |
| <ul style="list-style-type: none"> <li>• Aircraft Mechanic’s Toolkit</li> <li>• Torque Wrench, 150 - 750 in. lbs</li> </ul>                                                                | <ul style="list-style-type: none"> <li>• PARA 6-4-2</li> </ul>                                                        |
| References                                                                                                                                                                                 |                                                                                                                       |
| <ul style="list-style-type: none"> <li>• PARA 1-6-16</li> <li>• PARA 6-2-3</li> </ul>                                                                                                      |                                                                                                                       |
| <hr/>                                                                                                                                                                                      |                                                                                                                       |
| 6-4-73.1 Replace Rotor Brake Disc.                                                                                                                                                         |                                                                                                                       |
| 6-4-73.1.1 Remove.                                                                                                                                                                         |                                                                                                                       |
| a. Remove main transmission tail takeoff flange (PARA 6-4-2).                                                                                                                              | b. Install rotor brake disc on main transmission tail takeoff flange with bolts and nuts (Figure 6-4-73-1, Detail A). |
|                                                                                                                                                                                            | c. <b>TORQUE NUTS TO 385 - 425 INCH-POUNDS.</b>                                                                       |
|                                                                                                                                                                                            | d. Install main transmission tail takeoff flange (PARA 6-4-2).                                                        |
|                                                                                                                                                                                            | e. Perform main transmission functional test (PARA 6-3-4).                                                            |
|                                                                                                                                                                                            | f. Perform main transmission serviceability inspection (PARA 6-3-5).                                                  |
|                                                                                                                                                                                            | g. Make sure area is clean and free of foreign material.                                                              |
|                                                                                                                                                                                            | h. Perform operational check of rotor brake system (PARA 6-2-3).                                                      |
| NOTE                                                                                                                                                                                       |                                                                                                                       |
| If rotor brake disc is frozen to main transmission tail takeoff flange, replace rotor brake disc.                                                                                          |                                                                                                                       |
| b. Remove bolts and nuts securing rotor brake disc to main transmission tail takeoff flange. Slide rotor brake disc off main transmission tail takeoff flange (Figure 6-4-73-1, Detail A). |                                                                                                                       |
| 6-4-73.1.2 Install.                                                                                                                                                                        |                                                                                                                       |
| a. Turn off all electrical power (PARA 1-6-16).                                                                                                                                            |                                                                                                                       |

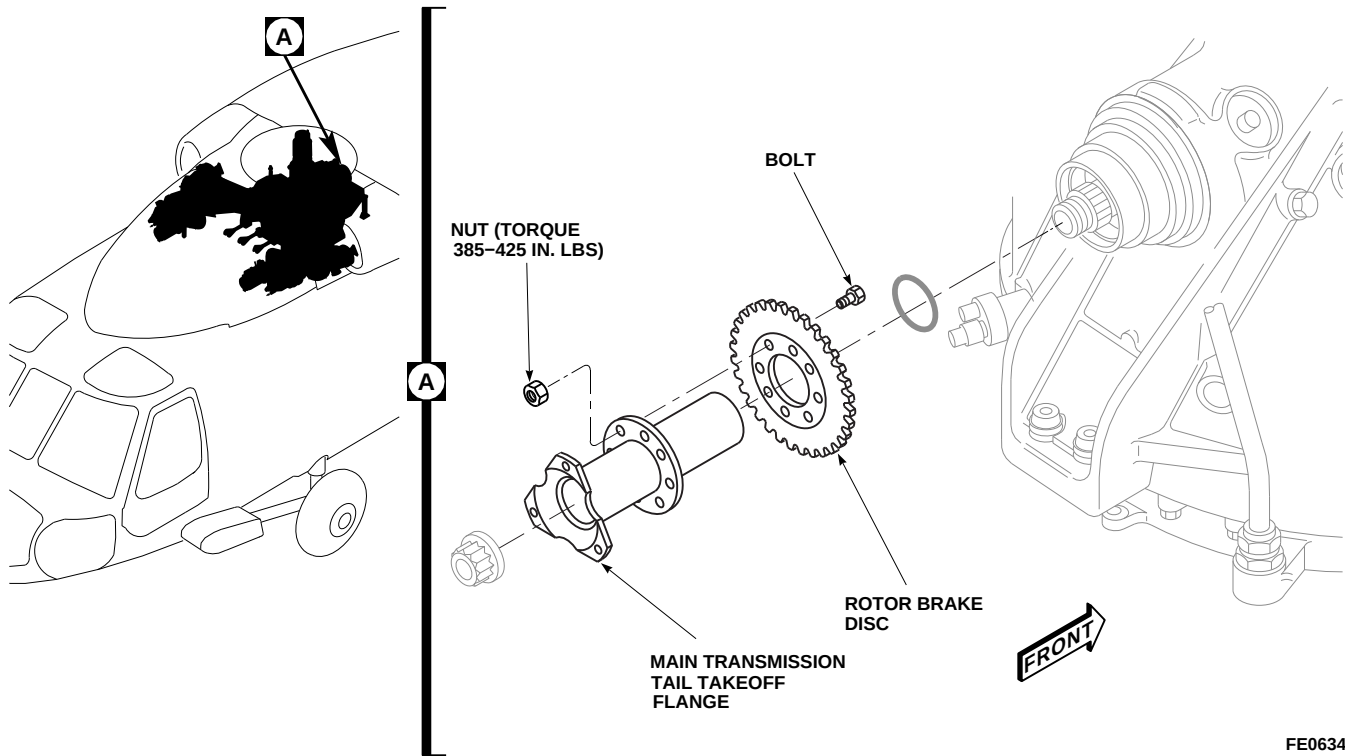


FIGURE 6-4-73-1. REPLACE ROTOR BRAKE DISC

6-4-74 ROTOR BRAKE MOUNTING BRACKET.

PARAGRAPH BREAKDOWN

|                                               | PAGE     |
|-----------------------------------------------|----------|
| 6-4-74.1 Replace Rotor Brake Mounting Bracket | 6-4-74-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Rotor Brake Wrench, 70700-77117-041
- Torque Wrench, 150 - 750 in. lbs

References

- PARA 1-6-16
- PARA 6-4-70

6-4-74.1 Replace Rotor Brake Mounting Bracket.

6-4-74.1.1 Remove.

- Remove rotor brake assembly (PARA 6-4-70).
- Remove bolts, countersunk washers, flat washers, nuts, and rotor brake mounting bracket from stud on main transmission (Figure 6-4-74-1, Detail A).

6-4-74.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Install rotor brake mounting bracket on stud of main transmission (Figure 6-4-74-1, Detail A).

- Install bolts with countersunk washers under boltheads through rotor brake mounting bracket and main transmission.

NOTE

Rotor brake wrench must be used when installing nut, SS5086-06, on lower stud of main transmission.

- Install flat washers and nuts on bolts and on lower stud of main transmission.
- TORQUE NUTS TO 261 - 288 INCH-POUNDS.**
- Install rotor brake assembly (PARA 6-4-70).

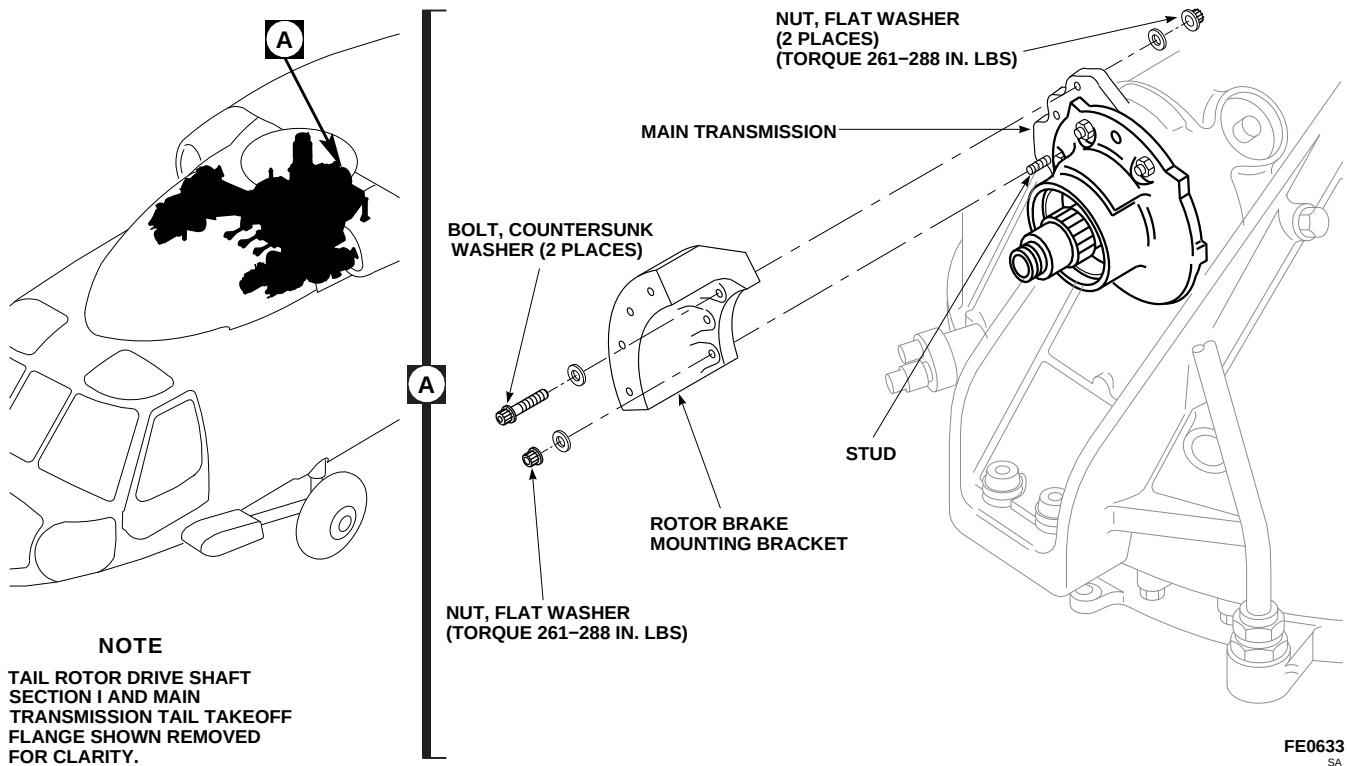


FIGURE 6-4-74-1. REPLACE ROTOR BRAKE MOUNTING BRACKET

6-4-75 ROTOR SPEED (Nr) SENSOR.

PARAGRAPH BREAKDOWN

|                                          | PAGE     |
|------------------------------------------|----------|
| 6-4-75.1 Replace Rotor Speed (Nr) Sensor | 6-4-75-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Protective Caps and Plugs
- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 6-2-2

6-4-75.1 Replace Rotor Speed (Nr) Sensor.

NOTE

- On helicopters with main transmission, 70351-08000 or 70084-35001, rotor speed (Nr) sensor is installed on right accessory module only.
- On helicopters with main transmission, 70084-35007, 70351-48000, or 70351-58000, replacement of both rotor speed (Nr) sensor is the same, except as noted.

6-4-75.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Open main rotor pylon sliding cover.
- On helicopters with main transmission, 70351-08000 or 70084-35001, disconnect electrical connector from rotor speed (Nr) sensor as follows (Figure 6-4-75-1, Detail A):

- Disconnect electrical connector, P443, from rotor speed (Nr) sensor on right accessory module.
- Install protective caps and plugs on electrical connectors.
- On helicopters with main transmission, 70084-35007, 70351-48000, or 70351-58000, disconnect electrical connector from rotor speed (Nr) sensor as follows (Figure 6-4-75-1, Detail A):
  - Disconnect electrical connector, P407 (left side), or P443 (right side), from rotor speed (Nr) sensor on accessory module.
  - Install protective caps and plugs on electrical connectors.
- Remove rotor speed (Nr) sensor and packing from accessory module. Discard packing.
- Install protective plug in accessory module.

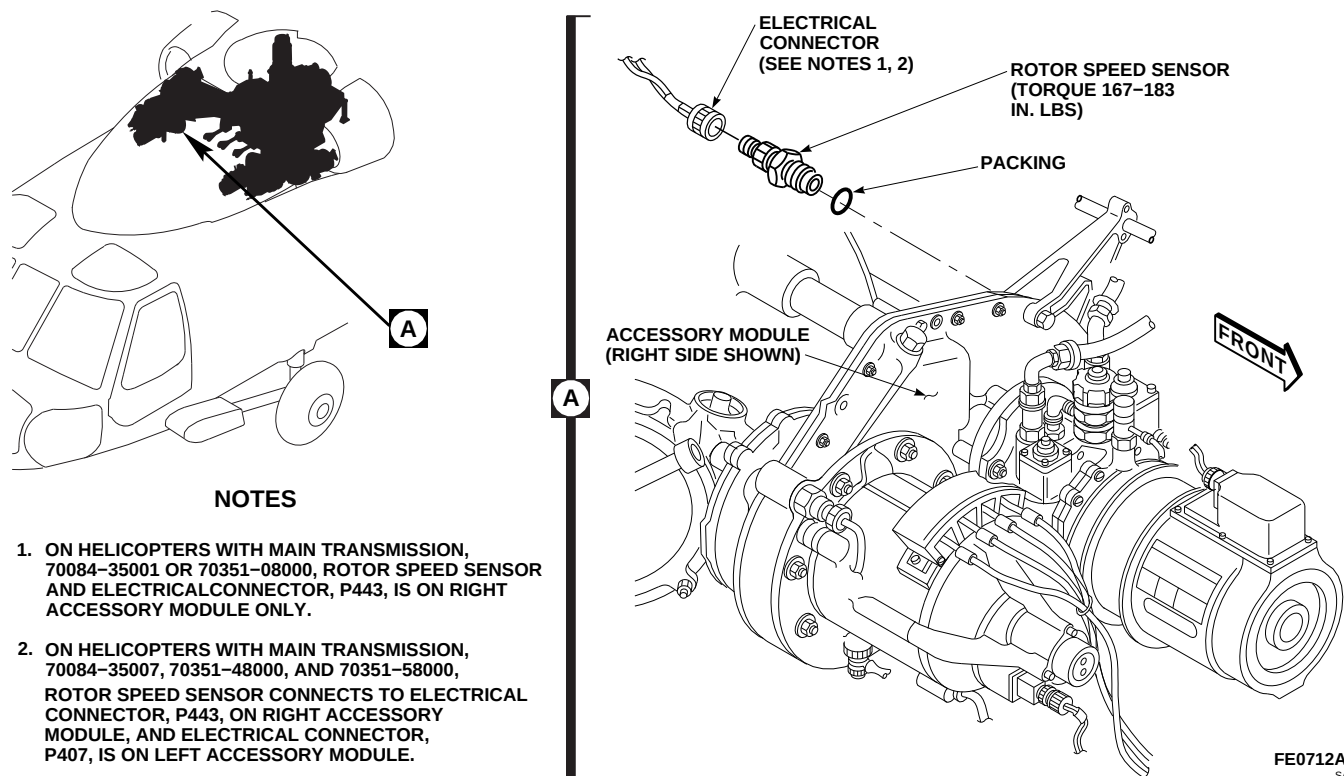


FIGURE 6-4-75-1. REPLACE ROTOR SPEED (NR) SENSOR

**6-4-75.1.2. Install.**

- a. Turn off all electrical power (PARA 1-6-16).
- b. Remove protective plug from accessory module.
- c. Install packing on rotor speed (Nr) sensor and install in accessory module (Figure 6-4-75-1, Detail A).
- d. **TORQUE ROTOR SPEED (Nr) SENSOR TO 167 - 183 INCH-POUNDS.**
- e. On helicopters with main transmission, 70351-08000 or 70084-35001, connect electrical connector to rotor speed (Nr) sensor as follows:
  - (1) Remove protective caps and plugs from electrical connectors.
  - (2) Inspect and treat electrical connectors (PARA 1-7-20).
  - (3) Connect electrical connector, P443, to rotor speed (Nr) sensor on right accessory module.
- f. On helicopters with main transmission, 70084-35007, 70351-48000, or 70351-58000, connect electrical connector to rotor speed (Nr) sensor as follows:
  - (1) Remove protective caps and plugs from electrical connectors.
  - (2) Inspect and treat electrical connectors (PARA 1-7-20).
  - (3) Connect electrical connector, P407 (left side), or P443 (right side), to rotor speed (Nr) sensor on accessory module.
- g. Lockwire, Item 197, Appendix D, electrical connector.
- h. Make sure area is clean and free of foreign material.
- i. Close main rotor pylon sliding cover.

**6-4-75-2**



- j. Perform operational check of instruments and oil warning systems (PARA 6-2-2).



6-4-76 LOW OIL PRESSURE SWITCH.

PARAGRAPH BREAKDOWN

|                                          | PAGE     |
|------------------------------------------|----------|
| 6-4-76.1 Replace Low Oil Pressure Switch | 6-4-76-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Protective Caps and Plugs
- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 6-2-2

6-4-76.1 Replace Low Oil Pressure Switch.

NOTE

Right accessory module does not have a low oil pressure switch.

6-4-76.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Open main rotor pylon sliding cover.
- Disconnect electrical connector, P409, from low oil pressure switch on left accessory module (Figure 6-4-76-1, Detail A).
- Install protective caps and plugs on electrical connectors.
- Remove low oil pressure switch and packing from accessory module. Discard packing.

- Install protective plug in accessory module.

6-4-76.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Remove protective plug from accessory module.
- Install packing on low oil pressure switch and install in accessory module (Figure 6-4-76-1, Detail A).
- TORQUE LOW OIL PRESSURE SWITCH TO 138 - 152 INCH-POUNDS.**
- Remove protective caps and plugs from electrical connectors.
- Inspect and treat electrical connectors (PARA 1-7-20).
- Connect electrical connector, P409, to low oil pressure switch.

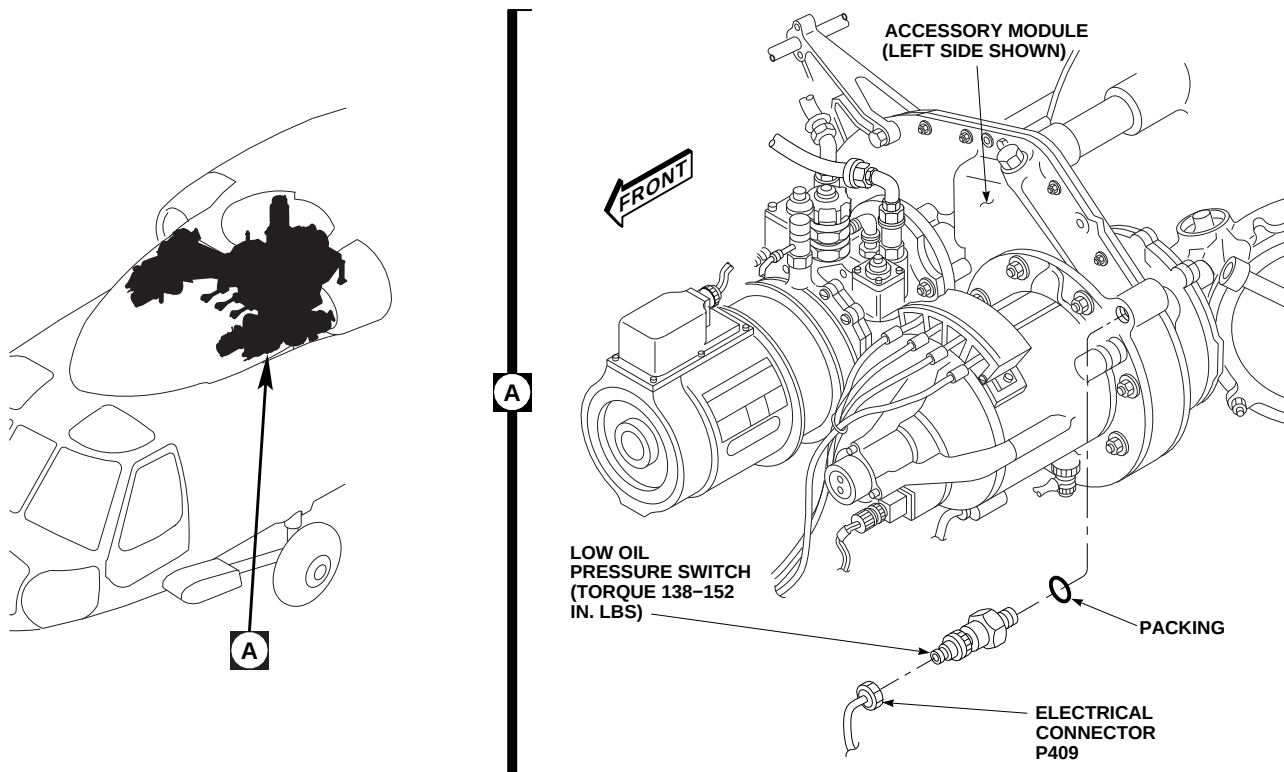
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FIGURE 6-4-76-1. REPLACE LOW OIL PRESSURE SWITCH

- h. Lockwire, Item 197, Appendix D, low oil pressure switch.
- i. Make sure area is clean and free of foreign material.
- j. Close main rotor pylon sliding cover.
- k. Perform operational check of instruments and oil warning systems (PARA 6-2-2).

6-4-77    MAIN ROTOR SHAFT LOWER SEAL.

PARAGRAPH BREAKDOWN

|                                                 | PAGE     |
|-------------------------------------------------|----------|
| 6-4-77.1    Replace Main Rotor Shaft Lower Seal | 6-4-77-1 |

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Powertrain Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lubricating Oil, 206, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 12-4-24

6-4-77.1    Replace Main Rotor Shaft Lower Seal.

6-4-77.1.1    Remove.

- Remove blade deice flange seal and packing (PARA 12-4-24).
- Remove bolts and washers securing liner assembly and retainer to main rotor shaft nut (Figure 6-4-77-1, Detail A).
- Remove liner assembly and retainer from main rotor shaft nut. Remove and discard packing.
- Remove retaining ring and seal from liner.

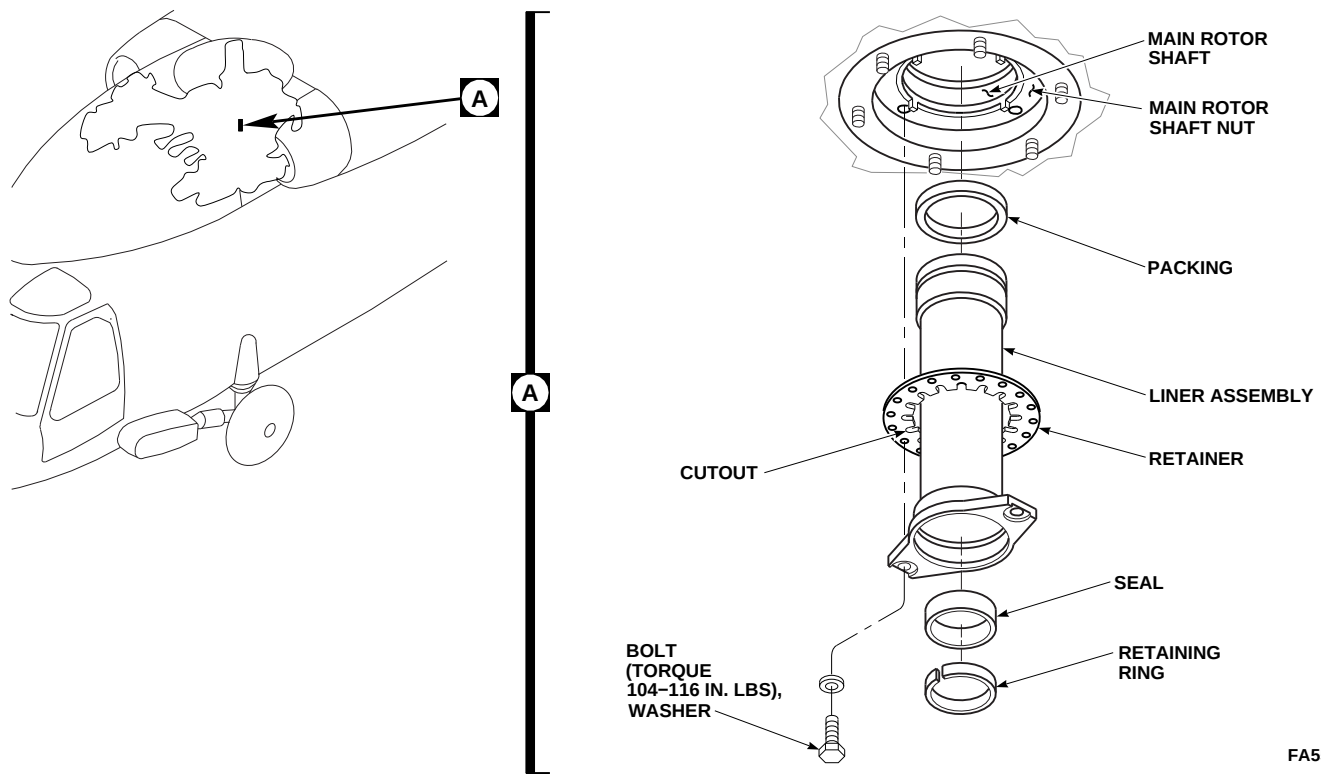
6-4-77.1.2    Install.

- Turn off all electrical power (PARA 1-6-16).

- Install seal, with open side and spring facing up, and install retaining ring in liner assembly (Figure 6-4-77-1, Detail A).
- Lubricate packing with lubricating oil, Item 206, Appendix D, and install on liner assembly.
- Install retainer on liner assembly. Position liner assembly, with retainer, into main rotor shaft.
- Align cutouts in retainer with tangs on main rotor shaft. Secure liner assembly to main rotor shaft nut with bolts and washers. **TORQUE BOLTS TO 104 - 116 INCH-POUNDS.**
- Install blade deice flange seal and packing (PARA 12-4-24).

NOTE

Liner seal side and spring will be facing up when liner assembly is installed in main gear box.



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FIGURE 6-4-77-1. REPLACE MAIN ROTOR SHAFT LOWER SEAL

**SECTION 6-5**

**MAINTENANCE PROCEDURES (BENCH)**

**NOT APPLICABLE**

